

7 – Developmental Testing

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Objective Summary

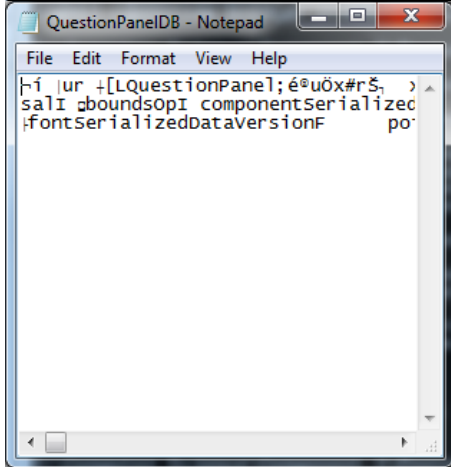
Objective #	Objective Description
	Questions
1	Ability to create Questions
2	Ability to add labels
3	Ability to add text fields
4	Ability to specify the type of data that should be entered into the text field
5	Ability to add radio buttons
6	Ability to add drop-downs
7	Ability to add checkboxes
8	Ability to add password fields
9	Ability to add calendar entry
10	Ability to add a document upload field
11	Ability to delete Questions
12	Ability to sort Questions by difficulty
13	Ability to sort Questions by type
14	Ability to filter Questions by difficulty
15	Ability to filter Questions by type
16	Email addresses, dates, and phone numbers should have format checks
17	Addresses should be looked up via the google maps API to autofill information
18	Passwords should have double entry verification
19	All fields should have a presence check
20	Document upload fields should only allow certain types of files to be uploaded
21	Ability to export a Question to file
22	Ability to import a Question from file
23	Ability to attempt a question
24	Ability to submit a question
	Forms
25	Ability to create forms

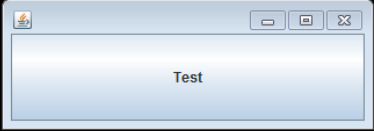
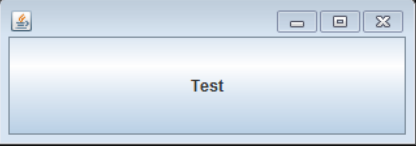
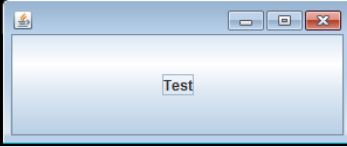
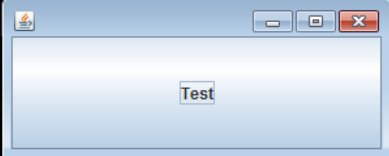
26	Ability to create forms from user weaknesses
27	Ability to sort forms by difficulty
28	Ability to filter forms by difficulty
29	Ability to filter forms by types of questions tested
30	Ability to edit forms
31	Ability to delete forms
32	Ability to export forms to file
33	Ability to import forms from file
34	Ability to attempt a form
35	Ability to submit a form
	User
36	The time taken to complete each question should be stored
37	The number of times that they've failed the validation check for a question should be stored
38	Ability to add users
39	Ability to edit users
40	Ability to delete users
41	Ability to search through users
42	Ability to produce a report on the user's weak areasprogress
43	How much of each form a user has attempted should be tracked.
44	Firstname, lastname, Date of birth and phone number should be stored for each user
45	The attendance for each user should be automatically tracked
46	The system should determine the reason for a user missing a session
	Interface
47	Should be simple
48	Login page which allows users or admins to login
49	There should be a main menu panel
50	There should be a panel which will display the forms to the user and allow them to search through them
51	Allow user to select a form to be attempted
52	Allow form to be submitted
53	There should be a panel which will display the questions to the user and allow them to search through them
54	Allow user to select a question to be attempted
55	Allow question to be submitted
56	Questions that have failed their validation check should be highlighted

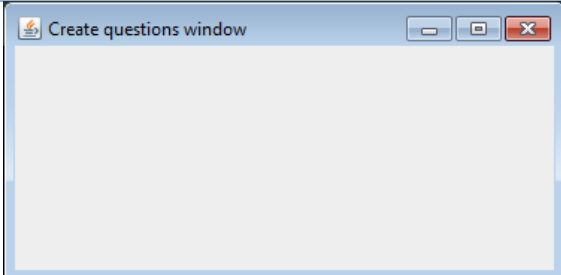
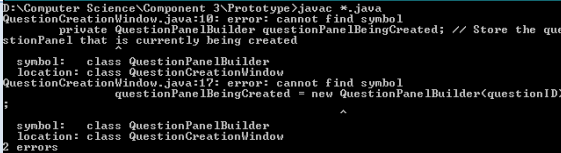
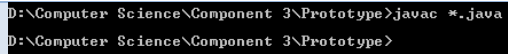
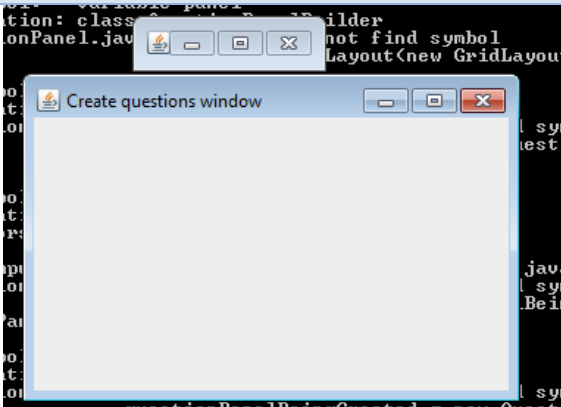
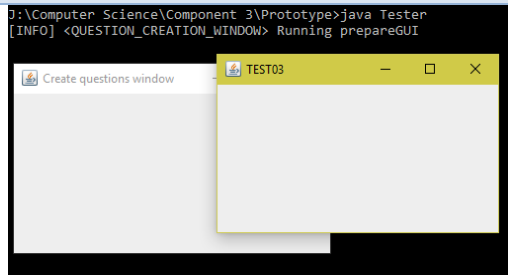
57	The user should be able to create forms by selecting questions from a list
58	There should be a statistics screen which shows the user's progress
59	There should be a context sensitive help button available in each window and panel of the application accessible by a regular user
60	The admin user should have a different main menu
61	The admin user should be able to view information about each user
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63	The admin user should be able to take a register

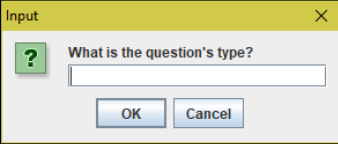
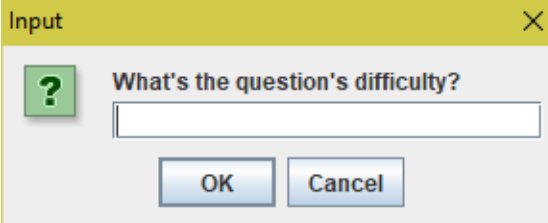
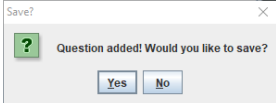
1: Ability to create Questions

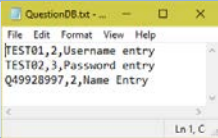
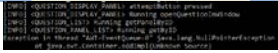
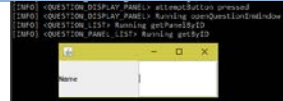
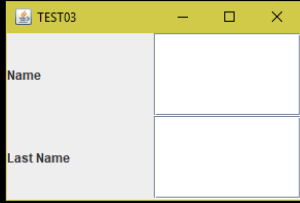
Test	Stage	Result Problem	Evidence	Action and Justification (If Needed)	Evidence of Action (If Needed)
1	Proto	Question successfully created	D:\Computer Science\Component 3\Prototype>java Question Creating question of difficulty 3 and type "Address" Created 3,Address [3, Address] D:\Computer Science\Component 3\Prototype>	No further action needed	
2	Proto	QuestionList class doesn't compile because I put a , rather than <	D:\Computer Science\Component 3\Prototype>javac QuestionList.java QuestionList.java:56: error: ';' expected for (int i = 0; i < 1, nextQuestionLocation; i++) // For each Question in the array ^ QuestionList.java:56: error: illegal start of expression for (int i = 0; i < 1, nextQuestionLocation; i++) // For each Question in the array ^ QuestionList.java:56: error: ')' expected for (int i = 0; i < 1, nextQuestionLocation; i++) // For each Question in the array ^ QuestionList.java:56: error: ';' expected for (int i = 0; i < 1, nextQuestionLocation; i++) // For each Question in the array ^ 4 errors	Add a <	D:\Computer Science\Component 3\Prototype>javac QuestionList.java QuestionList.java:24: error: cannot find symbol BufferedReader br = new BufferedReader(new FileReader(databaseFileName)) ^ database file ^ symbol: class BufferedReader location: class QuestionList 1 error
3	Proto	QuestionList class doesn't compile because I misspelled BufferedReader	D:\Computer Science\Component 3\Prototype>javac QuestionList.java QuestionList.java:24: error: cannot find symbol BufferedReader br = new BufferedReader(new FileReader(databaseFileName)) ^ database file ^ symbol: class BufferedReader location: class QuestionList 1 error	Correct the spelling	D:\Computer Science\Component 3\Prototype>javac QuestionList.java D:\Computer Science\Component 3\Prototype>
4	Proto	Question successfully added to the list and written to file	D:\Computer Science\Component 3\Prototype>java Question Creating question of difficulty 3 and type "Address" Created 3,Address [3, Address] Creating a question list [INFO] <QUESTION_LIST> Running addQuestion [INFO] <QUESTION_LIST> Running writeDatabase UserDB.txt - Notepad File Edit Format View Help 3,Address Ln 1, Col 1		
5	Proto	Question successfully loaded from database	D:\Computer Science\Component 3\Prototype>java Question Creating a question list [INFO] <QUESTION_LIST> Running loadDatabase [INFO] <QUESTION_LIST> Running addQuestion [INFO] <QUESTION_LIST> Running printQuestions 3,Address		
6		Question panels can be added to the Question	D:\Computer Science\Component 3\Prototype>java Tester [INFO] <QUESTION_PANEL_LIST> Running addQuestionPanel Printing the array 0000001		

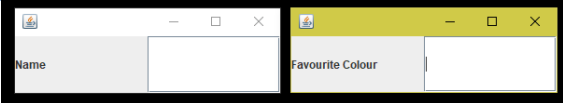
	Proto	panel list.			
7	Proto	Writing the database to file failed because I hadn't created the output folder	<pre>D:\Computer Science\Component 3\Prototype>java Tester [INFO] <QUESTION_PANEL_LIST> Running addQuestionPanel [INFO] <QUESTION_PANEL_LIST> Running writeDatabase [ERROR] <QUESTION_PANEL_LIST> Writing database java.io.FileNotFoundException: questions\QuestionPanelDB.ser (The system cannot find the path specified)</pre>	Make the folder	<pre>D:\Computer Science\Component 3\Prototype>java Tester [INFO] <QUESTION_PANEL_LIST> Running addQuestionPanel [INFO] <QUESTION_PANEL_LIST> Running writeDatabase</pre>
8	Proto	The list of question panels is able to be written to file	<pre>D:\Computer Science\Component 3\Prototype>java Tester [INFO] <QUESTION_PANEL_LIST> Running addQuestionPanel [INFO] <QUESTION_PANEL_LIST> Running writeDatabase</pre> 		
9	Proto	The list of question panels is able to be read from file. The id of the question panel in the list is successfully printed after being loaded.	<pre>D:\Computer Science\Component 3\Prototype>java Tester [INFO] <QUESTION_PANEL_LIST> Running loadDatabase Q0000001</pre>		
10	Proto	Multiple Symbol not found errors due to misspellings	<pre>Tester.java:25: error: cannot find symbol QuestionPanel qP = new QuestionPanel.QuestoinPanelBuilder("TEST0 1").add(new JButton("Test")).build(); ^ symbol: class JButton location: class Tester Tester.java:25: error: cannot find symbol QuestionPanel qP = new QuestionPanel.QuestoinPanelBuilder("TEST0 1").add(new JButton("Test")).build(); ^ symbol: class QuestoinPanelBuilder location: class QuestionPanel Note: Some messages have been simplified; recompile with -Xdiags:verbose to get full output 4 errors</pre>	Fix the spelling errors	<pre>D:\Computer Science\Component 3\Prototype>javac *.java D:\Computer Science\Component 3\Prototype></pre> Compiles without errors

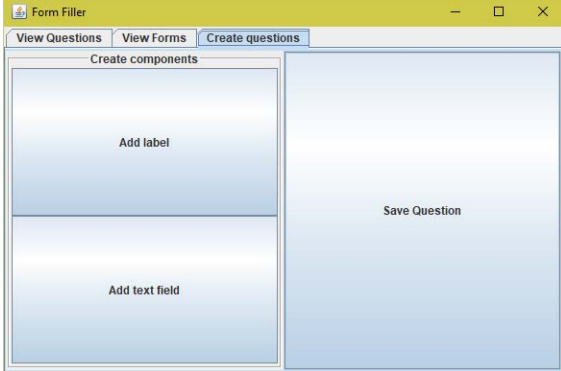
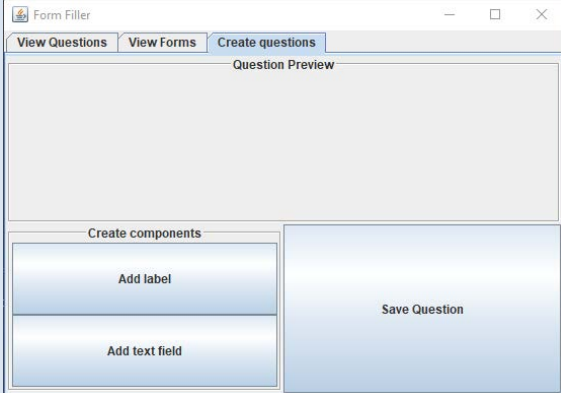
11	Proto	Runtime error when building a new JPanel because of an illegal cast	<pre> D:\Computer Science\Component 3\Prototype>java Tester [INFO] <QUESTION_PANEL_BUILDER> Running add [INFO] <QUESTION_PANEL> Running build [INFO] <UTILITIES> Running trimArray Exception in thread "main" java.lang.ClassCastException: [Ljava.lang.Object; cannot be cast to javax.swing.JComponent; at QuestionPanel\$QuestionPanelBuilder.build(QuestionPanel.java:64) at Tester.runTest(Tester.java:25) at Tester.main(Tester.java:8) </pre>	Change the type to JComponent rather an object in the trim array method so no cast is needed	<pre> D:\Computer Science\Component 3\Prototype>java Tester [INFO] <QUESTION_PANEL_BUILDER> Running add [INFO] <QUESTION_PANEL_BUILDER> Running build [INFO] <QUESTION_PANEL_BUILDER> Running trimArray [INFO] <QUESTION_PANEL> Running setup TEST01 </pre> 
12	Proto	Simple question panel successfully displayed in a JFrame	<pre> D:\Computer Science\Component 3\Prototype>java Tester [INFO] <QUESTION_PANEL_BUILDER> Running add [INFO] <QUESTION_PANEL_BUILDER> Running build [INFO] <QUESTION_PANEL_BUILDER> Running trimArray [INFO] <QUESTION_PANEL> Running setup TEST01 </pre> 		
13	Proto	Null pointer exception when trying to add loaded question panel to the window because the load database method was never called.	<pre> D:\Computer Science\Component 3\Prototype>java Tester Exception in thread "main" java.lang.NullPointerException at java.awt.Container.addImpl(Unknown Source) at java.awt.Container.add(Unknown Source) at javax.swing.JFrame.addImpl(Unknown Source) at java.awt.Container.add(Unknown Source) at Tester.runTest(Tester.java:37) at Tester.main(Tester.java:9) </pre>	Add a call to the loadDatabase() method before trying to add the question panel to the frame	<pre> D:\Computer Science\Component 3\Prototype>java Tester [INFO] <QUESTION_PANEL_LIST> Running loadDatabase </pre> 
14	Proto	The question panel is successfully loaded from file and displayed	<pre> D:\Computer Science\Component 3\Prototype>java Tester [INFO] <QUESTION_PANEL_LIST> Running loadDatabase </pre> 		

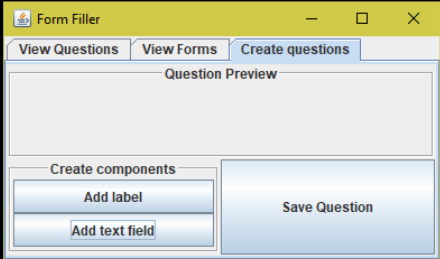
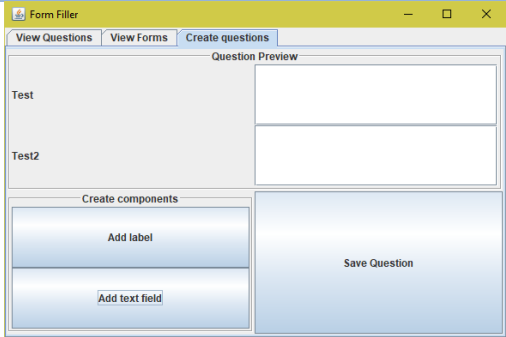
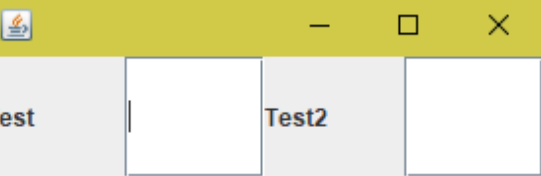
20	Proto	Blank create questions window created			
21	Proto	Error compiling create questions window because of incorrect reference to QuestionPanelBuilder		Change QuestionPanelBuilder to QuestionPanel.QuestionPanelBuilder	
22	Proto	An example question panel opens when the question window is opened to visually show what the question will look like but it's too small		Set the size before making it visible so it's not tiny	
23	Proto	There is a save question button			

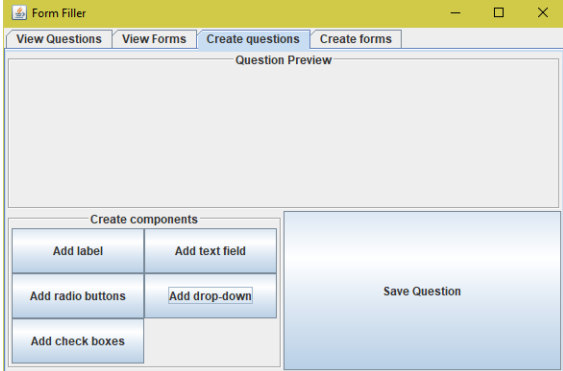
24	Proto	The user is prompted for the type and difficulty of the question after they press save	<pre> D:\Computer Science\Component 3\Prototype>java Tester [INFO] <QUESTION_LIST> Running loadDatabase [INFO] <QUESTION_LIST> 2 Questions loaded from file [INFO] <QUESTION_PANEL_LIST> Running loadDatabase [INFO] <QUESTION_PANEL_LIST> 2 QuestionPanels loaded from file [INFO] <QUESTION_CREATION_WINDOW> Running prepareGUI [INFO] <QUESTION_CREATION_WINDOW> addLabelButton pressed [INFO] <QUESTION_CREATION_WINDOW> Running addLabel [INFO] <QUESTION_PANEL_BUILDER> Running add [INFO] <QUESTION_CREATION_WINDOW> addTextFieldButton pressed [INFO] <QUESTION_CREATION_WINDOW> Running addTextField [INFO] <QUESTION_PANEL_BUILDER> Running add [INFO] <QUESTION_CREATION_WINDOW> saveQuestionButton pressed [INFO] <QUESTION_CREATION_WINDOW> Running getFinalDetails </pre>  		
25	Proto	The add question method runs after the user has entered the type and difficulty of the question	<pre> D:\Computer Science\Component 3\Prototype>java Tester [INFO] <QUESTION_LIST> Running loadDatabase [INFO] <QUESTION_LIST> 2 Questions loaded from file [INFO] <QUESTION_PANEL_LIST> Running loadDatabase [INFO] <QUESTION_PANEL_LIST> 2 QuestionPanels loaded from file [INFO] <QUESTION_CREATION_WINDOW> Running prepareGUI [INFO] <QUESTION_CREATION_WINDOW> addLabelButton pressed [INFO] <QUESTION_CREATION_WINDOW> Running addLabel [INFO] <QUESTION_PANEL_BUILDER> Running add [INFO] <QUESTION_CREATION_WINDOW> addTextFieldButton pressed [INFO] <QUESTION_CREATION_WINDOW> Running addTextField [INFO] <QUESTION_PANEL_BUILDER> Running add [INFO] <QUESTION_CREATION_WINDOW> saveQuestionButton pressed [INFO] <QUESTION_CREATION_WINDOW> Running getFinalDetails [INFO] <QUESTION_PANEL_BUILDER> Running build [INFO] <QUESTION_PANEL_BUILDER> Running trimArray [INFO] <QUESTION_PANEL> Running setup [INFO] <QUESTION_PANEL_LIST> Running addQuestionPanel [INFO] <QUESTION_LIST> Successfully added question and questionPanel </pre>		
26	Proto	Save dialog after a question has been added.			
27	Proto	Question is saved after pressed the yes button			

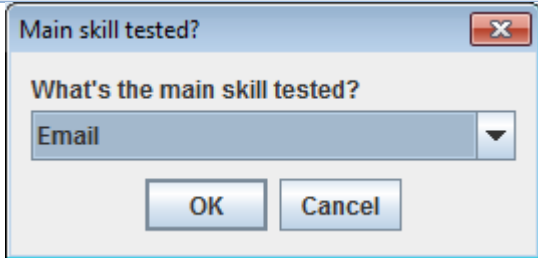
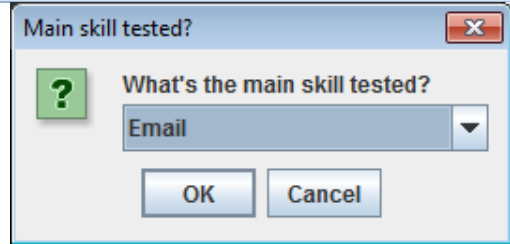
					
28	Proto	Null pointer exception when trying to attempt the question that was just saved because the question panel wasn't saved.		Call the save method on the question panel list as well as the question list	
29		Layout of the visual representation is incorrect there are too few rows.	<pre>J:\Computer Science\Component 3\Prototype>java Tester [INFO] <QUESTION_CREATION_WINDOW> Running prepareGUI [INFO] <QUESTION_CREATION_WINDOW> addLabelButton pressed [INFO] <QUESTION_CREATION_WINDOW> Running addLabel [INFO] <QUESTION_PANEL_BUILDER> Running add [INFO] <QUESTION_CREATION_WINDOW> addTextFieldButton pressed [INFO] <QUESTION_PANEL_BUILDER> Running add [INFO] <QUESTION_CREATION_WINDOW> addLabelButton pressed [INFO] <QUESTION_CREATION_WINDOW> Running addLabel [INFO] <QUESTION_PANEL_BUILDER> Running add [INFO] <QUESTION_CREATION_WINDOW> addTextFieldButton pressed [INFO] <QUESTION_PANEL_BUILDER> Running add</pre> 	Change the arguments to the gridlayout to allow the panel to have infinite rows by changing the first argument to a 0.	<pre>J:\Computer Science\Component 3\Prototype>java Tester [INFO] <QUESTION_CREATION_WINDOW> Running prepareGUI [INFO] <QUESTION_CREATION_WINDOW> addLabelButton pressed [INFO] <QUESTION_CREATION_WINDOW> Running addLabel [INFO] <QUESTION_PANEL_BUILDER> Running add [INFO] <QUESTION_CREATION_WINDOW> addTextFieldButton pressed [INFO] <QUESTION_PANEL_BUILDER> Running add [INFO] <QUESTION_CREATION_WINDOW> addLabelButton pressed [INFO] <QUESTION_CREATION_WINDOW> Running addLabel [INFO] <QUESTION_PANEL_BUILDER> Running add [INFO] <QUESTION_CREATION_WINDOW> addTextFieldButton pressed [INFO] <QUESTION_PANEL_BUILDER> Running add</pre> 
30	Proto	Error compiling QuestionList because java.util wasn't imported and so random wasn't imported	<pre>J:\Computer Science\Component 3\Prototype>javac *.java QuestionList.java:204: error: cannot find symbol Random r = new Random(); ^</pre>	Import java.util.*	<pre>J:\Computer Science\Component 3\Prototype>javac *.java J:\Computer Science\Component 3\Prototype></pre> Successful compile
31	Proto	A question is able to be got by id	<pre>J:\Computer Science\Component 3\Prototype>java Tester [INFO] <QUESTION_LIST> Running loadDatabase [INFO] <QUESTION_LIST> 2 Questions loaded from file [INFO] <QUESTION_PANEL_LIST> Running loadDatabase [INFO] <QUESTION_PANEL_LIST> 2 QuestionPanels loaded from file [INFO] <QUESTION_LIST> Running getQuestionById TEST02,3>Password entry</pre>		

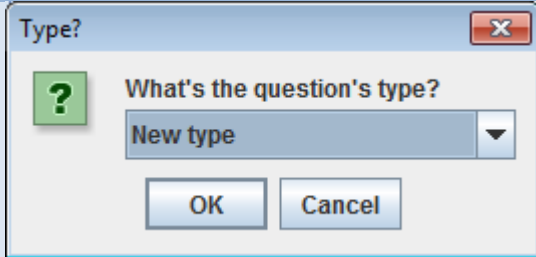
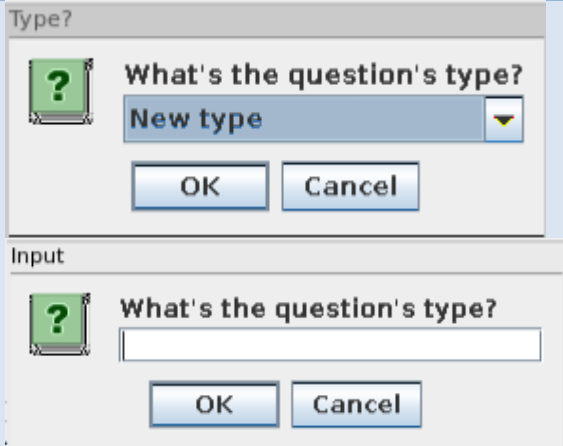
32	Proto	A unique 8 digit id is generated.	<pre> D:\Computer Science\Component 3\Prototype>java Tester [INFO] <QUESTION_LIST> Running loadDatabase [INFO] <QUESTION_LIST> 2 Questions loaded from file [INFO] <QUESTION_PANEL_LIST> Running loadDatabase [INFO] <QUESTION_PANEL_LIST> 2 QuestionPanels loaded from file [INFO] <QUESTION_LIST> Running generateId [INFO] <QUESTION_LIST> Running getQuestionById Q11578249 </pre>		
33	Proto	Null pointer exception when attempting to view a created question because the question and question panel have mismatched ids	<pre> [INFO] <QUESTION_PANEL_LIST> Running addQuestionPanel [INFO] <QUESTION_LIST> Successfully added question and questionPanel [INFO] <MAIN_MENU> openQuestionDisplayWindowButton pressed [INFO] <GUI> Running openQuestionDisplayWindow [INFO] <QUESTION_DISPLAY_WINDOW> Running prepareGUI [INFO] <QUESTION_DISPLAY_WINDOW> Running populateTable [INFO] <QUESTION_DISPLAY_WINDOW> attemptButton pressed [INFO] <QUESTION_DISPLAY_WINDOW> Running openQuestionInWindow [INFO] <QUESTION_LIST> Running getPanelById [INFO] <QUESTION_PANEL_LIST> Running getPanelById Exception in thread "AWT-EventQueue-0" java.lang.NullPointerException </pre>	Call the <code>getFreeID()</code> method when the question creation window opens and pass the id to both the question and question panel	<pre> D:\Computer Science\Component 3\Prototype>java Main [INFO] <GUI> Running setup [INFO] <QUESTION_LIST> Running loadDatabase [INFO] <QUESTION_LIST> 2 Questions loaded from file [INFO] <QUESTION_PANEL_LIST> Running loadDatabase [INFO] <QUESTION_PANEL_LIST> 2 QuestionPanels loaded from file [INFO] <FORM_LIST> Running loadDatabase [INFO] <FORM_LIST> 2 Forms loaded from file [INFO] <MAIN_MENU> Running prepareGUI [INFO] <MAIN_MENU> openQuestionCreationWindowButton pressed [INFO] <GUI> Running openQuestionCreationWindow [INFO] <QUESTION_LIST> Running generateId [INFO] <QUESTION_LIST> Running getQuestionById [INFO] <QUESTION_CREATION_WINDOW> Running prepareGUI [INFO] <QUESTION_CREATION_WINDOW> addLabelButton pressed [INFO] <QUESTION_CREATION_WINDOW> Running addLabel [INFO] <QUESTION_PANEL_BUILDER> Running add [INFO] <QUESTION_CREATION_WINDOW> addTextFieldButton pressed [INFO] <QUESTION_CREATION_WINDOW> Running addTextField [INFO] <QUESTION_PANEL_BUILDER> Running add [INFO] <QUESTION_CREATION_WINDOW> saveQuestionButton pressed [INFO] <QUESTION_CREATION_WINDOW> Running getFinalDetails [INFO] <QUESTION_PANEL_BUILDER> Running build [INFO] <QUESTION_PANEL_BUILDER> Running trimArray [INFO] <QUESTION_PANEL> Running setup [INFO] <QUESTION_PANEL_LIST> Running addQuestionPanel [INFO] <QUESTION_LIST> Successfully added question and questionPanel [INFO] <MAIN_MENU> openQuestionDisplayWindowButton pressed [INFO] <GUI> Running openQuestionDisplayWindow [INFO] <QUESTION_DISPLAY_WINDOW> Running prepareGUI [INFO] <QUESTION_DISPLAY_WINDOW> Running populateTable [INFO] <QUESTION_DISPLAY_WINDOW> attemptButton pressed [INFO] <QUESTION_DISPLAY_WINDOW> Running openQuestionInWindow [INFO] <QUESTION_LIST> Running getPanelById [INFO] <QUESTION_PANEL_LIST> Running getPanelById </pre> 
34	Proto	Two newly created questions can be opened showing that the unique id system is working			

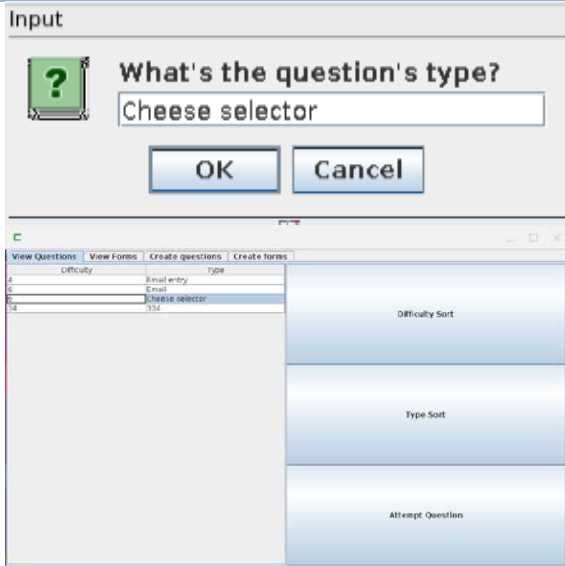
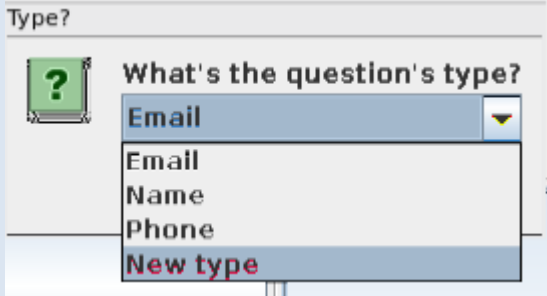
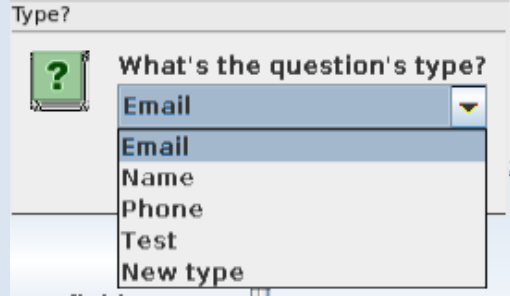
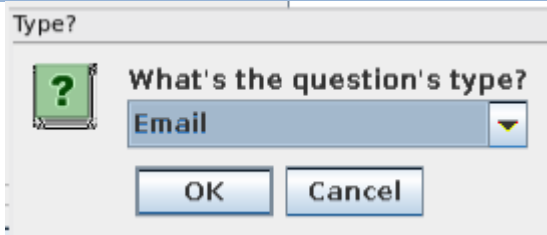
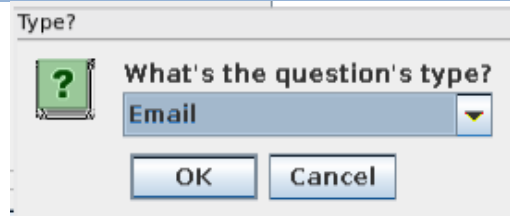
35	Proto	Error compiling the question creation panel because the TitledBorder class cannot be found. Also loweredetached can't be found	<pre> D:\Computer Science\Component 3\Prototype>javac *.java QuestionCreationPanel.java:72: error: cannot find symbol TitledBorder border = BorderFactory.createTitledBorder(^ symbol: class TitledBorder location: class QuestionCreationPanel QuestionCreationPanel.java:72: error: cannot find symbol TitledBorder border = BorderFactory.createTitledBorder(^ symbol: variable loweredetached location: class QuestionCreationPanel QuestionCreationPanel.java:74: error: cannot find symbol border.setTitleJustification(TitledBorder.CENTER); ^ symbol: variable TitledBorder location: class QuestionCreationPanel 3 errors </pre>	Import <code>javax.swing.border.*</code> and declare a border called <code>lowerdetached</code> at the top of the class	<pre> D:\Computer Science\Component 3\Prototype>javac *.java D:\Computer Science\Component 3\Prototype> </pre> Successful compile.
36	Proto	The create component buttons have a titled border			
37	Proto	Clearer and more consolidated question creation section			

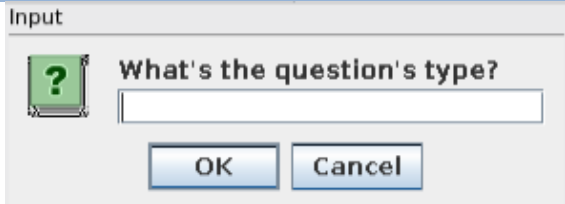
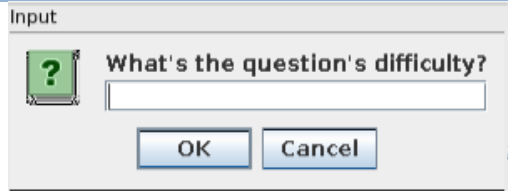
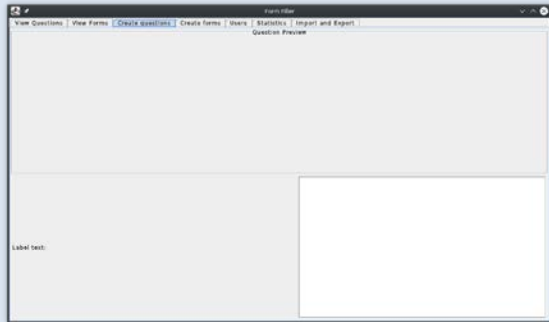
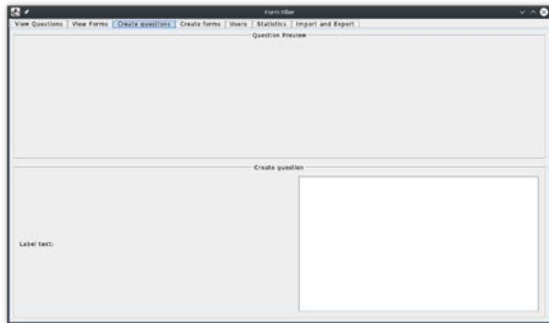
38	Proto	Question preview not updating as the components are added.	<pre> J:\Computer Science\Component 3\Prototype>java Main [INFO] <GUI> Running setup [INFO] <QUESTION_LIST> Running loadDatabase [INFO] <QUESTION_LIST> 2 Questions loaded from file [INFO] <QUESTION_PANEL_LIST> Running loadDatabase [INFO] <QUESTION_PANEL_LIST> 2 QuestionPanels loaded from file [INFO] <FORM_LIST> Running loadDatabase [INFO] <FORM_LIST> 2 Forms loaded from file [INFO] <QUESTION_DISPLAY_PANEL> Running prepareGUI [INFO] <QUESTION_DISPLAY_PANEL> Running populateTable [INFO] <FORM_DISPLAY_PANEL> Running prepareGUI [INFO] <FORM_DISPLAY_PANEL> Running populateTable [INFO] <QUESTION_LIST> Running generateId [INFO] <QUESTION_LIST> Running getQuestionById [INFO] <QUESTION_CREATION_PANEL> Running prepareGUI [INFO] <QUESTION_CREATION_PANEL> addLabelButton pressed [INFO] <QUESTION_CREATION_PANEL> Running addLabel [INFO] <QUESTION_PANEL_BUILDER> Running add [INFO] <QUESTION_CREATION_PANEL> addTextFieldButton pressed [INFO] <QUESTION_CREATION_PANEL> Running addTextField [INFO] <QUESTION_PANEL_BUILDER> Running add </pre> 	Remove the accidental second call to the <code>prepareVisualPanel()</code> method	
39	Proto	Created questions with multiple lines have the wrong layout		Set rows to 0 QuestionPanel as it was only 1	

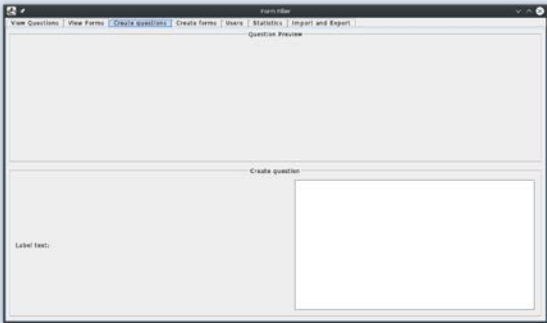
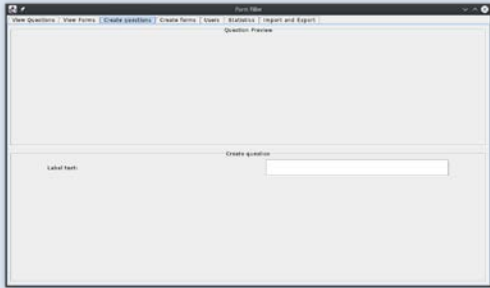
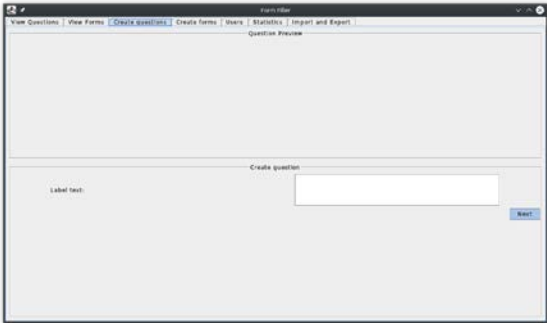
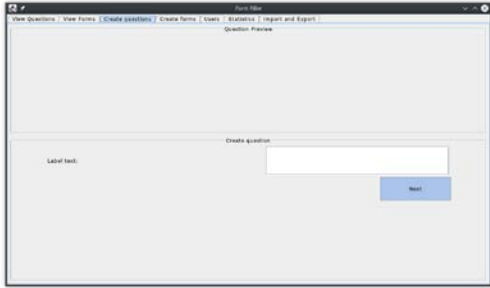
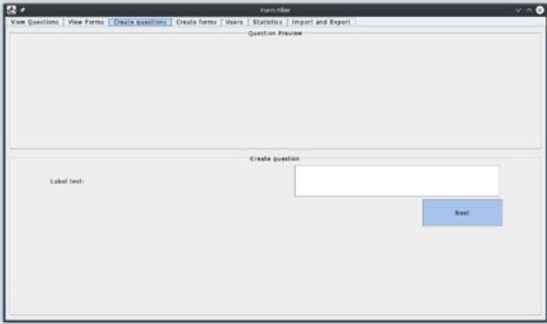
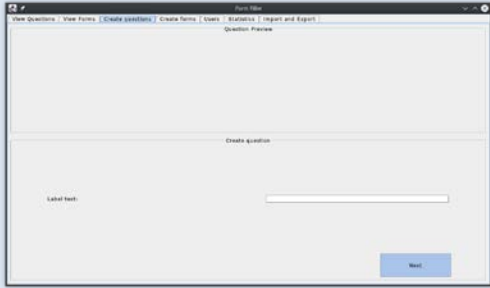
40	Proto	Buttons for various other components in the question creation panel			
41	Proto	The array of question Types is loaded from file at startup	<pre>[INFO] <FORM_LIST> 7 Forms loaded from file [INFO] <FORM_LIST> Running loadSkills Email,Name,Phone [INFO] <QUESTION_DISPLAY_PANEL> R...</pre>		

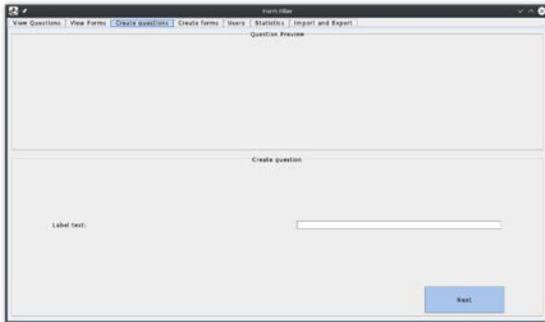
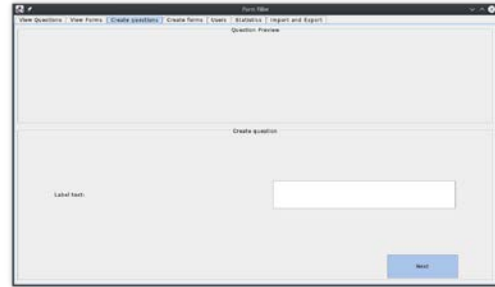
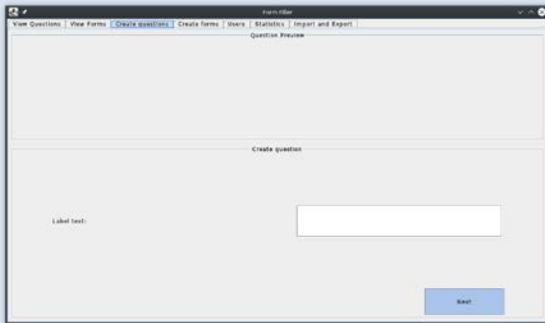
42	Proto	Multiple symbol not found errors when compiling QuestionCreationPanel due to not importing java.io.*;	<pre> E:\Computer Science\Component 3\Prototype>javac -cp .;LGoodDateP *.java QuestionCreationPanel.java:332: error: incompatible types: String erted to Component ^ "What's the main skill tested?", ^ QuestionCreationPanel.java:344: error: cannot find symbol BufferedReader br = new BufferedReader(n illsFileName)); // Open the database file name ^ symbol: class BufferedReader location: class QuestionCreationPanel QuestionCreationPanel.java:344: error: cannot find symbol BufferedReader br = new BufferedReader(n illsFileName)); // Open the database file name ^ symbol: class BufferedReader location: class QuestionCreationPanel QuestionCreationPanel.java:344: error: cannot find symbol BufferedReader br = new BufferedReader(n illsFileName)); // Open the database file name ^ symbol: class FileReader location: class QuestionCreationPanel QuestionCreationPanel.java:364: error: cannot find symbol FileWriter fw = new FileWriter(skillsFil re a new file writer ^ symbol: class FileWriter location: class QuestionCreationPanel QuestionCreationPanel.java:364: error: cannot find symbol FileWriter fw = new FileWriter(skillsFil re a new file writer ^ symbol: class FileWriter location: class QuestionCreationPanel Tester.java:163: error: cannot find symbol new SkillSelectionFrame(forms.getSkills()); ^ symbol: class SkillSelectionFrame location: class Tester Tester.java:163: error: cannot find symbol new SkillSelectionFrame(forms.getSkills()); ^ symbol: method getSkills() location: variable forms of type FormList Note: Some messages have been simplified; recompile with -Xdiags full output 8 errors </pre>	add "import java.io.*;" to the top of the code.	<pre> E:\Computer Science\Component 3\Prototype>javac -cp .;LGoodDateP *.java QuestionCreationPanel.java:333: error: incompatible types: String erted to Component ^ "What's the main skill tested?", ^ Note: Some messages have been simplified; recompile with -Xdiags full output 1 error </pre> The errors because java.io.* wasn't imported no longer occur.
43	Proto	Incompatible types error because the JOptionPane.showInputDialog arguments weren't correctly given.	<pre> E:\Computer Science\Component 3\Prototype>javac -cp .;LGoodDateP *.java QuestionCreationPanel.java:333: error: incompatible types: String erted to Component ^ "What's the main skill tested?", ^ Note: Some messages have been simplified; recompile with -Xdiags full output 1 error </pre>	Supply the method with the correct arguments.	<pre> E:\Computer Science\Component 3\Prototype>javac -cp .;LGoodDateP *.java </pre> Successful compile
44	Proto	There is a drop down that allows the admin to select the main skill that the question tests. However there is no question mark icon.		Change the type of message from PLAIN_MESSAGE to QUESTION_MESSAGE	

45	Proto	The type entry box has been replaced with the main skills tested box as they served the same purpose. There is also now a new type option so that new types can be specified.			
46	Proto	Error compiling QuestionList because of incorrect syntax when declaring a for loop.	<pre> QuestionList.java:315: error: illegal start of expression for (int i = 0; <= types.length; i++) ^ 1 error </pre>	Add an i before the <=	<pre> (GUI)harry@localhost:/media/removable/HARRY_BEST/Computer Science/Component 3/Prototy pes\$ sh compile0.sh (GUI)harry@localhost:/media/removable/HARRY_BEST/Computer Science/Component 3/Prototy pes\$ </pre> Successful compile.
47	Proto	When new type is selected the user is prompted to enter the question's type into a text field.			

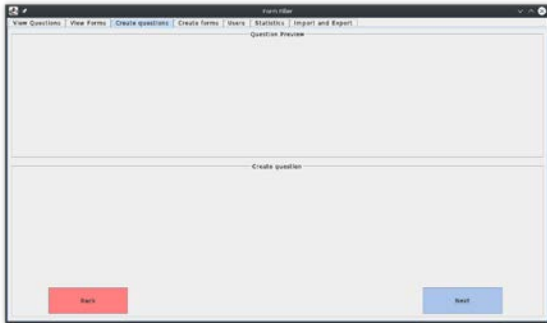
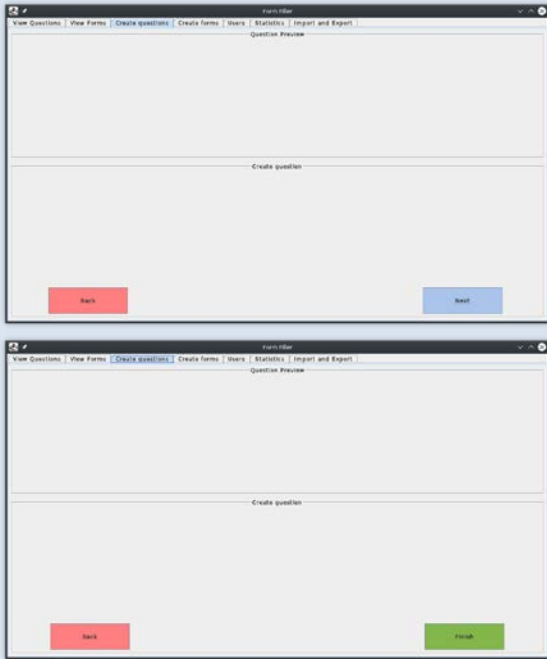
48	Proto	The question is stored with the new type entered.			
49	Proto	The newly added type "cheese" isn't shown in the type selector after a restart.		Recompile all of the files because the code for adding it after a restart was already present in the source file.	 A newly added type 'Test' is present after a restart
50	Proto	The admin is prompted to enter a type even if they don't select new type.		There was a semicolon at the end of the line with the if statement on. Removing it fixed the bug.	

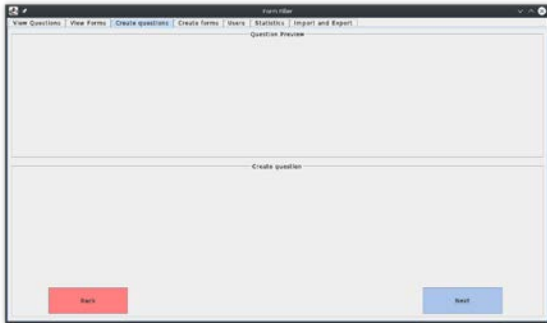
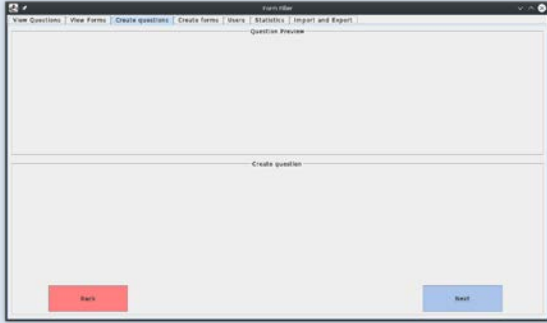
					
51	Soft Dev	The create questions panel has a basic form of the first screen from the redesign implemented.			
52	Soft Dev	There is now a border around the lower half of the screen.			

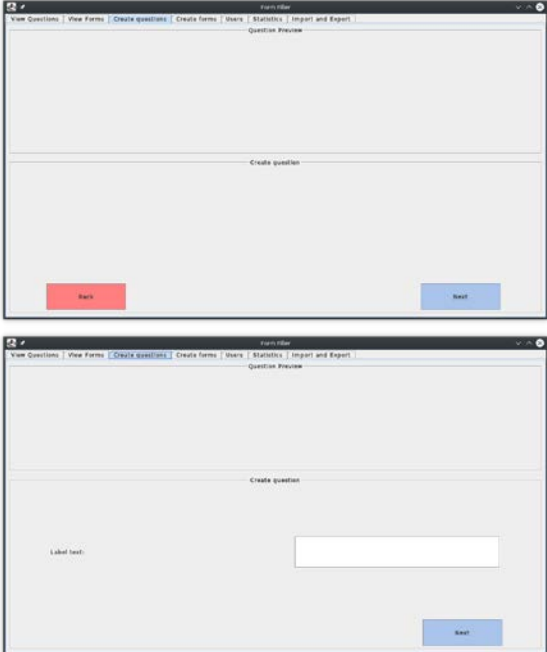
53	Soft Dev	The text field is too large.		This is because although the panel has a preferred size, it's contained within a GridBagLayout. GridBagLayouts ignore the preferred size. Changing the panel that contains the panel with the text field to use a box layout will fix this.	
54	Soft Dev	There is a next button however it is too small.		Set the maximum and preferred size of the button to be larger	
55	Soft Dev	The label entry should be centered and the next button should be at the bottom		Put vertical glue between the top and the label entry and the label entry and the next button. This 'glue' will expand to fill the empty space and will push the label entry to the	

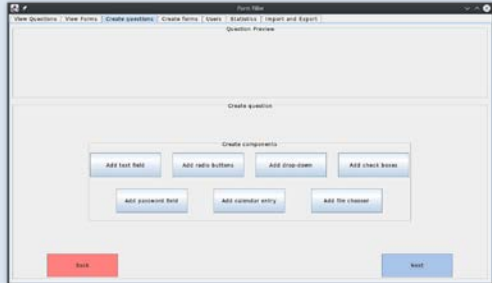
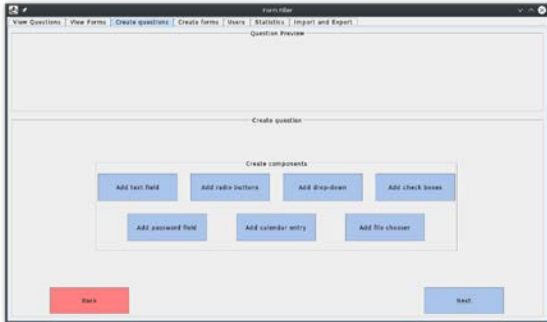
				middle and the next button to the bottom	
56	Soft Dev	The label entry is too short.		Set the preferred height to be taller.	
57	Soft Dev	Presses to the next button are detected	 <pre> [INFO] <FORM_SELECTION_PANEL> Running prepareGUI [INFO] <FORM_SELECTION_PANEL> Running populateTable [INFO] <QUESTION_CREATION_PANEL> nextButton pressed </pre>		

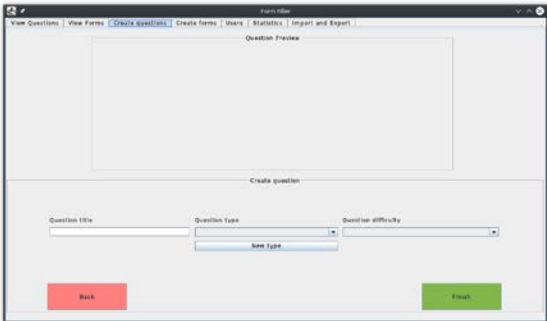
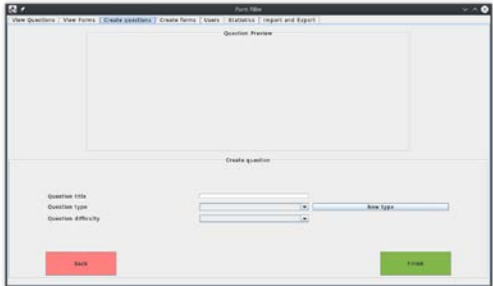
58	Soft Dev	Error when compiling QuestionCreationPanel because of missed semicolon		Add the semicolon	
59	Soft Dev	Null pointer exception when pressing the next button		This is because the array that the function was trying to access had not been initialised. Initialising the array fixed this.	
60	Soft Dev	When the next button is pressed the addLabel panel is replaced with the addComponents panel and the back button appears			

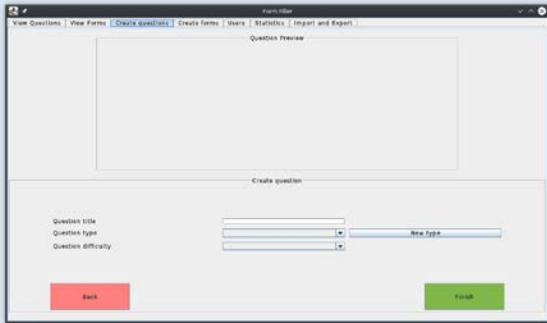
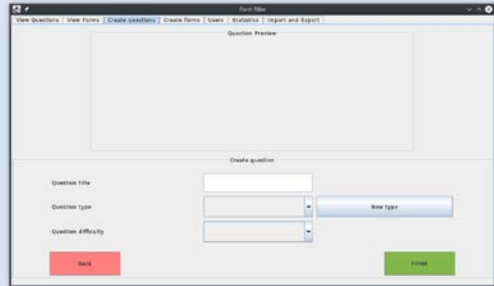
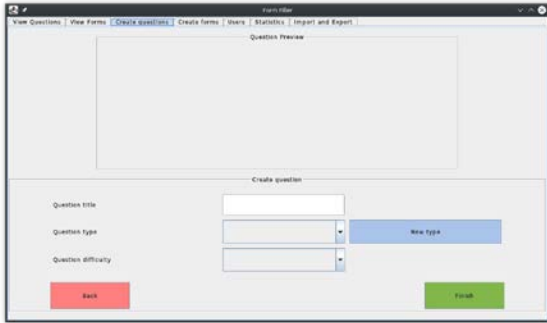
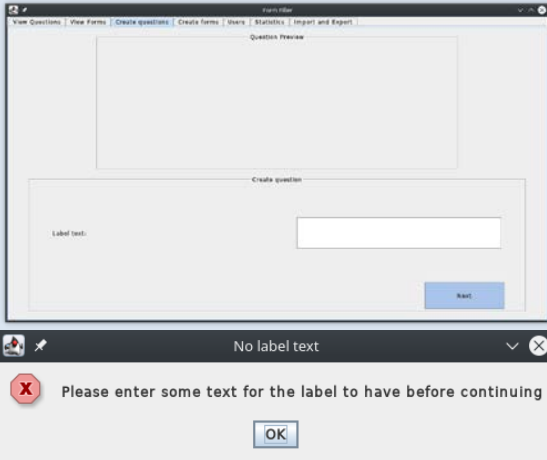
			 The addComponents panel is blank as nothing has been added to it yet		
61	Soft Dev	When the next button is pressed on the addComponents panel is pressed, the next button is replaced by the finish button and the addComponents panel is replaced by the addFinalDetails panel.	 The panel has been replaced. The addFinalDetails panel is also blank.		

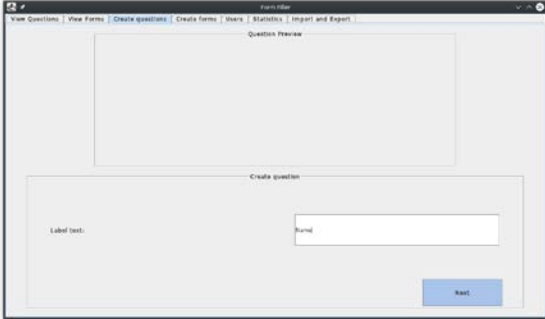
62	Soft Dev	Presses to the back button are detected	 <pre>[INFO] <QUESTION_CREATION_PANEL> nextButton pressed [INFO] <QUESTION_CREATION_PANEL> backButton pressed</pre>		
63	Soft Dev	When the back button is pressed on the addFinalDetails screen the finish button is replaced by the next button and the addFinalDetails screen is replaced by the addComponents screen.	 		

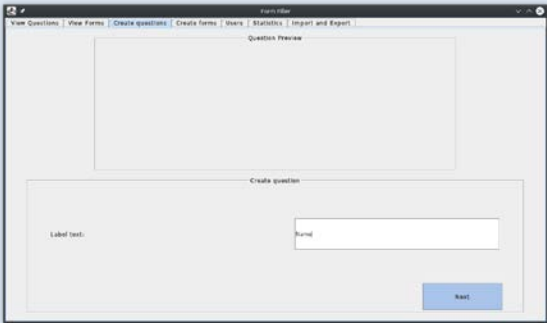
64	Soft Dev	When the back button is pressed on the addComponents screen the back button is removed and the addComponents screen is replaced by the addLabel screen.			
65	Soft Dev	Null pointer exception when the prepareGUI method is run on the questionCreationPanel	<pre>[INFO] <QUESTION_CREATION_PANEL> Running prepareGUI Exception in thread "main" java.lang.NullPointerException at java.awt.Container.addImpl(Container.java:1095) at java.awt.Container.add(Container.java:419) at QuestionCreationPanel.prepareAddComponentPanel(QuestionCreationPanel.java:188) at QuestionCreationPanel.prepareCreateQuestionPanel(QuestionCreationPanel.java:135) at QuestionCreationPanel.prepareMainPanel(QuestionCreationPanel.java:228) at QuestionCreationPanel.prepareGUI(QuestionCreationPanel.java:75) at QuestionCreationPanel.<init>(QuestionCreationPanel.java:67) at GUI.prepareGUI(GUI.java:82) at GUI.setup(GUI.java:46) at GUI.<init>(GUI.java:23) at Main.main(Main.java:5)</pre>	This is because the component creation buttons JPanel hadn't been initialised before adding it to the addComponentPanel. Calling prepareComponentCreationButtons() before adding them fixed this	<pre>[INFO] <QUESTION_CREATION_PANEL> Running prepareGUI Exception in thread "main" java.awt.AWTError: BoxLayout can't be shared at javax.swing.BoxLayout.checkContainer(BoxLayout.java:465) at javax.swing.BoxLayout.uninstall(BoxLayout.java:248) at javax.swing.BoxLayout.addLayoutComponent(BoxLayout.java:282) at java.awt.Container.addImpl(Container.java:1136) at java.awt.Container.add(Container.java:419) at QuestionCreationPanel.prepareComponentCreationButtons(QuestionCreationPanel.java:281) at QuestionCreationPanel.prepareAddComponentPanel(QuestionCreationPanel.java:185) at QuestionCreationPanel.prepareCreateQuestionPanel(QuestionCreationPanel.java:137) at QuestionCreationPanel.prepareMainPanel(QuestionCreationPanel.java:238) at QuestionCreationPanel.prepareGUI(QuestionCreationPanel.java:75) at QuestionCreationPanel.<init>(QuestionCreationPanel.java:67) at GUI.prepareGUI(GUI.java:82) at GUI.setup(GUI.java:46) at GUI.<init>(GUI.java:23) at Main.main(Main.java:5)</pre> Different error

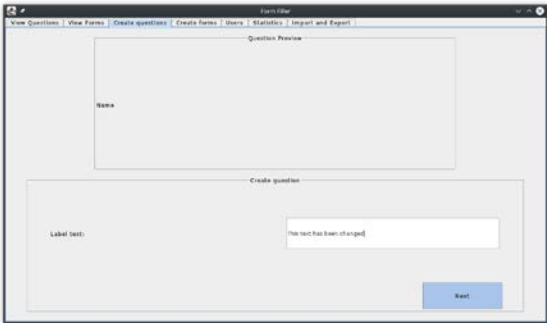
66	Soft Dev	BoxLayout can't be shared error.	<pre> [INFO] <QUESTION_CREATION_PANEL> Running prepareGUI Exception in thread "main" java.awt.AWTError: BoxLayout can't be shared at javax.swing.BoxLayout.checkContainer(BoxLayout.java:465) at javax.swing.BoxLayout.invalidateLayout(BoxLayout.java:249) at javax.swing.BoxLayout.addLayoutComponent(BoxLayout.java:282) at java.awt.Container.addImpl(Container.java:1130) at java.awt.Container.add(Container.java:419) at QuestionCreationPanel.prepareComponentCreationButtons(QuestionCreationPanel.java:281) at QuestionCreationPanel.prepareAddComponentPanel(QuestionCreationPanel.java:195) at QuestionCreationPanel.prepareCreateQuestionPanel(QuestionCreationPanel.java:137) at QuestionCreationPanel.prepareMainPanel(QuestionCreationPanel.java:238) at QuestionCreationPanel.prepareGUI(QuestionCreationPanel.java:75) at QuestionCreationPanel.<init>(QuestionCreationPanel.java:67) at GUI.prepareGUI(GUI.java:82) at GUI.setup(GUI.java:46) at GUI.<init>(GUI.java:23) at Main.main(Main.java:5) </pre>	This happened because when creating a BorderLayout for row2 I accidentally put row1 in the constructor of the BorderLayout. Changing this to row2 should fix this.	<pre> [INFO] <QUESTION_CREATION_PANEL> Running prepareGUI [INFO] <FORM_LIST> Running getFreeID [INFO] <FORM_LIST> Running generateId [INFO] <FORM_LIST> Running getFormByID </pre> PrepareGUI runs without errors
67	Soft Dev	The layout of the addComponentsPanel is incorrect.		Remove the rightmost horizontal glue of the first row. Add horizontal glue to the left of the second row. Decrease the width and height of the addComponentsPanel.	 The layout is fixed
68	Soft Dev	The add component buttons have the correct background colour			

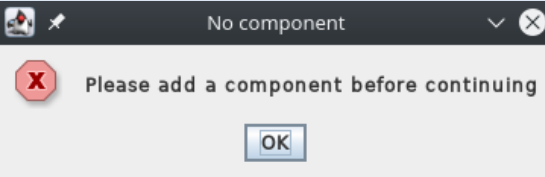
69	Soft Dev	The question preview is smaller when the addComponents panel is visible compared to when the add label panel is visible. The question preview panel should remain the same size.		Setting a preferred size for the Create question panel should fix this.	 It no longer changes size
70	Soft Dev	The add final details panel is incorrectly formatted. It needs to be transposed.		Swap all of the changes to gridx with the ones from gridy to transpose	 Correct order

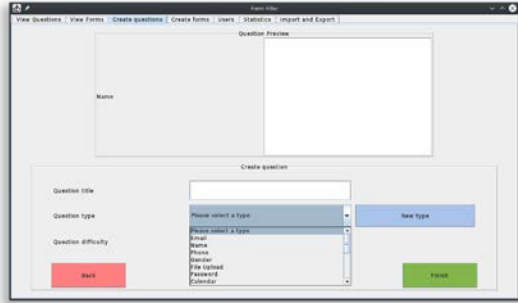
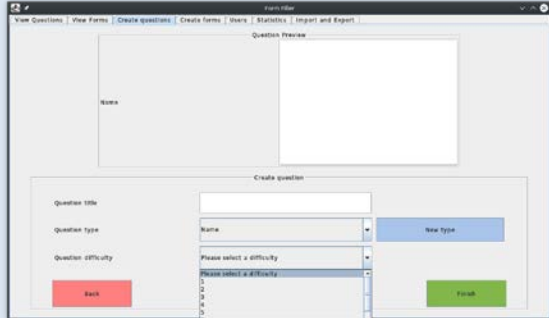
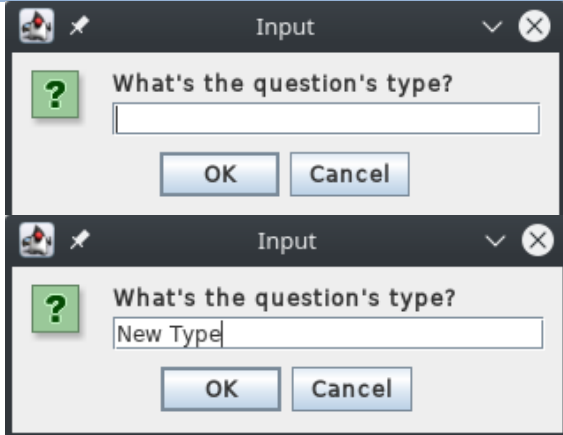
71	Soft Dev	The add final details entries are too short (they are squashed) as the height of the panel they are in is too small.		Increase for the height of the panel.	
72	Soft Dev	The new type button has the correct background			
73	Soft Dev	When the next button is pressed and there is no text in the label text field an error message appears and doesn't allow the user to move to the next step			

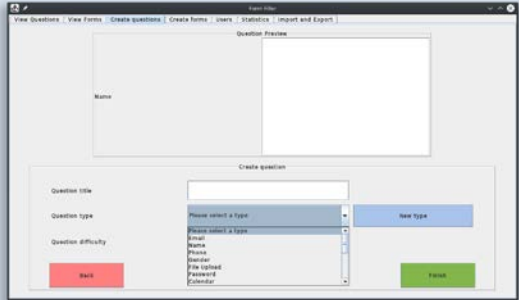
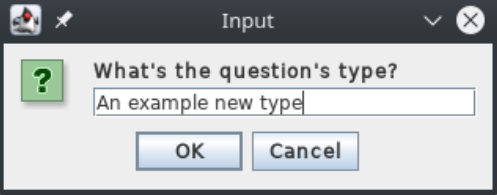
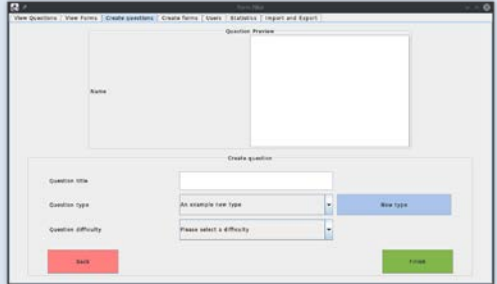
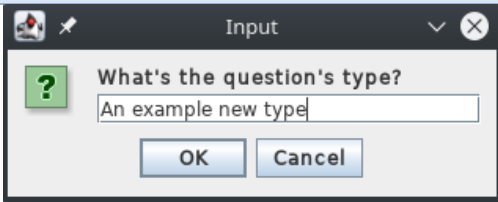
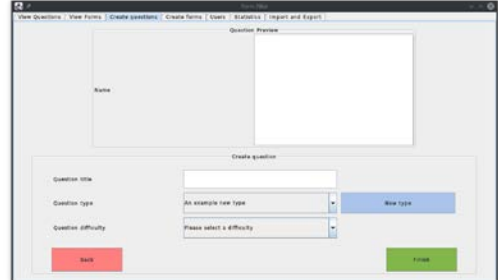
74	Soft Dev	When the next button is pressed and there is text in the label text field a label with that text is added to the question and the addComponents Panel is shown	 		
75	Soft Dev	If back then forwards is pressed there is no change to the label text (duplicate labels aren't added)			

			 		
76	Soft Dev	If back then forwards is pressed and the label text in the text field is changed, the label is updated with the new text.			

			 		
77	Soft Dev	All types of components can be added. (This is the same code from the prototype) So no further tests			

78	Soft Dev	If another add component button is pressed when there already is a component added, the already added component gets overwritten. In this case add checkboxes was pressed when there was already a text field.	 		
80	Soft Dev	If next is pressed when no component has been added an error message appears and the user is not allowed to continue.	 		

81	Soft Dev	The question type drop-down is populated with the question types in the system. There is also a “please select a type” option			
82	Soft Dev	The question difficulty drop down is populated with the difficulties 1-10 and a “Please select a difficulty” option.			
83	Soft Dev	When the new type button is pressed a dialog appears asking for the new type.			

84	Soft Dev	When the new type is added the combo box doesn't update	 The new type should be selected	Add a call to <code>removeAllItems()</code> on the combobox then add each new type by using <code>addItem()</code>	  The new type is automatically added and selected
85	Soft Dev	When the user specifies a new type the type is automatically added to the combobox and selected.	 		

86

Soft Dev

If all of the details are entered and the user presses finish the question is added to the database and they are prompted if they want to save the question to file.

QuizMaker

View Questions | View Forms | Create questions | Create forms | Users | Statistics | Import and Export

Question Preview

Middle Name

Create question

Question title: Middle Name entry

Question type: Name

Question difficulty: 5

Back Finish

Save?

Question added! Would you like to save?

Yes No

QuizMaker

View Questions | View Forms | Create questions | Create forms | Users | Statistics | Import and Export

Title	Difficulty	Type
PLACHOUDEN	1	Service
PLACHOUDEN	1	Name
PLACHOUDEN	1	Service
PLACHOUDEN	2	Phone
PLACHOUDEN	3	Birth day
PLACHOUDEN	3	Photo upload
PLACHOUDEN	3	Video upload
Middle Name entry	5	Name

Sort and Filter

Difficulty Sort: 1 2 3 4 5 6 7 8 9 10

Type Sort

Apply difficulty filter

Type filter

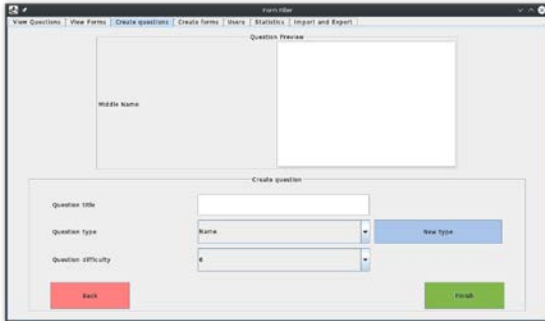
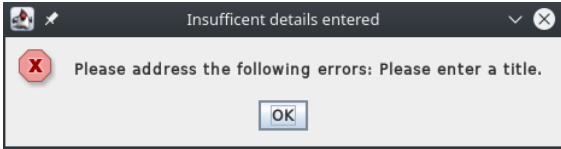
Reset type filter

Reset sorts and filters

Create question

Attempt question

Question can be seen in the view questions panel

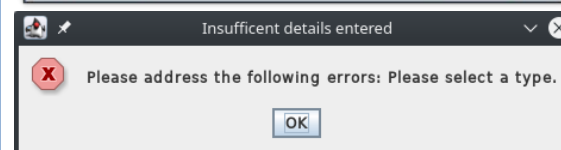
			 The question can also be opened to be attempted		
87	Soft Dev	If any of the final details fields are not filled in then an error message is produced.	 Title field not filled in.  Type not filled in		

Question title: Middle Name

Question type: Please select a type

Question difficulty: Please select a difficulty

Buttons: Back, Finish



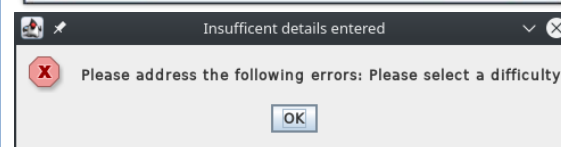
Difficulty not filled in

Question title: Middle Name

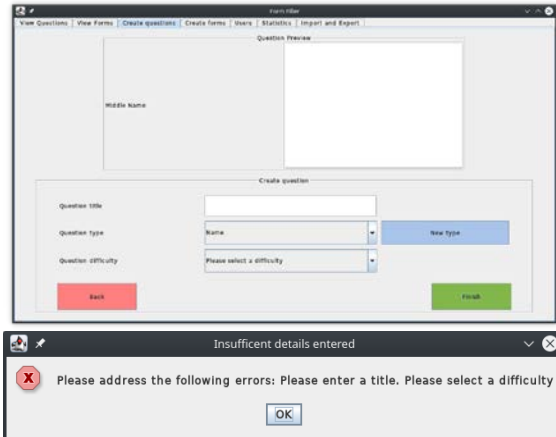
Question type: Name

Question difficulty: Please select a difficulty

Buttons: Back, Finish



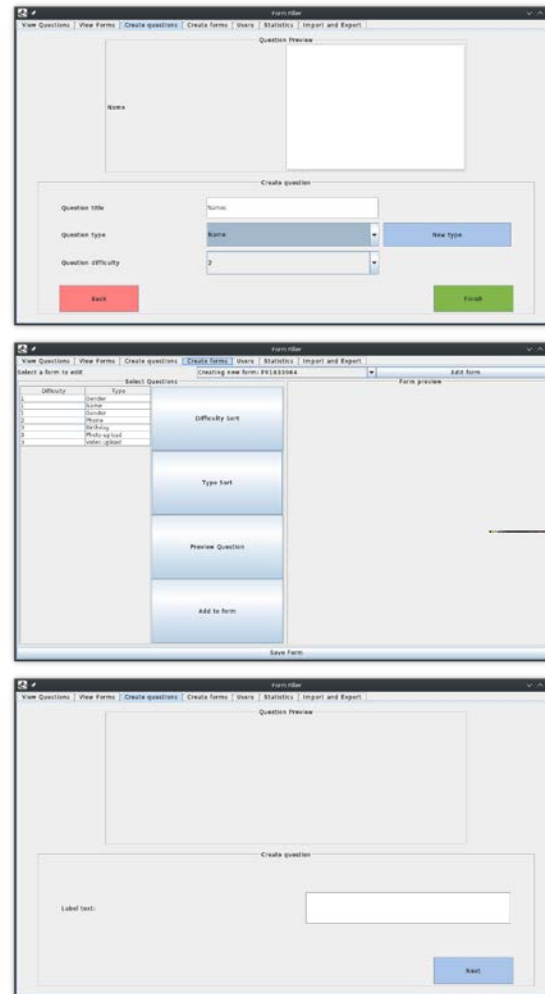
Any combination of these errors works.

			 An example combination of the title and the difficulty not being filled in.		
88	Soft Dev	When enter is pressed in the label text field it's the same as pressing the next button			

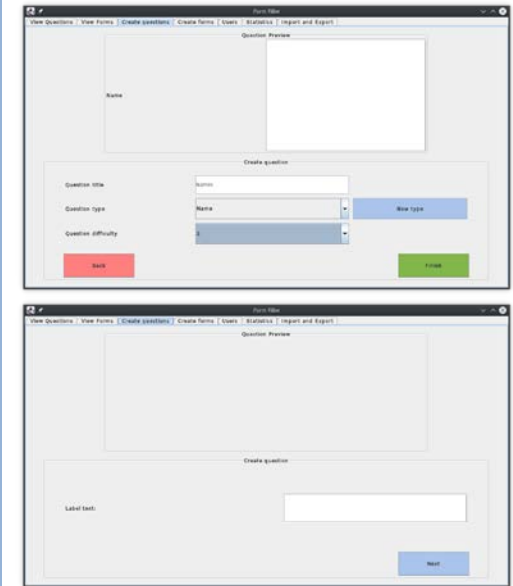
88

Soft Dev

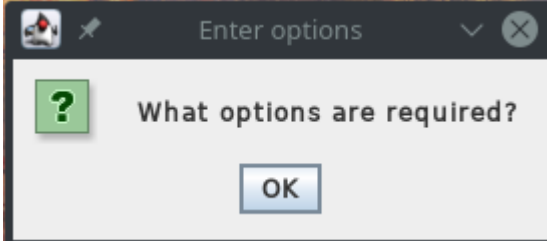
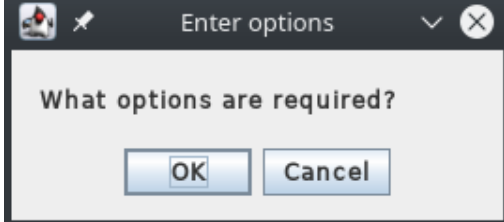
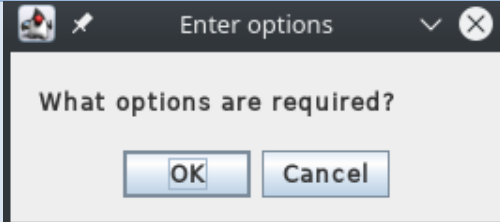
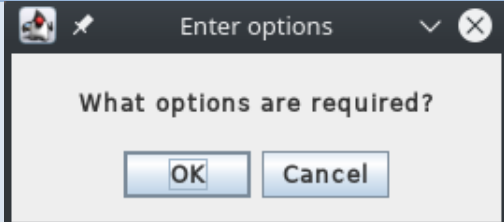
When the question is added to the system the create question tab resets. However after the reset the selected tab becomes the form creation tab rather than the create questions tab.

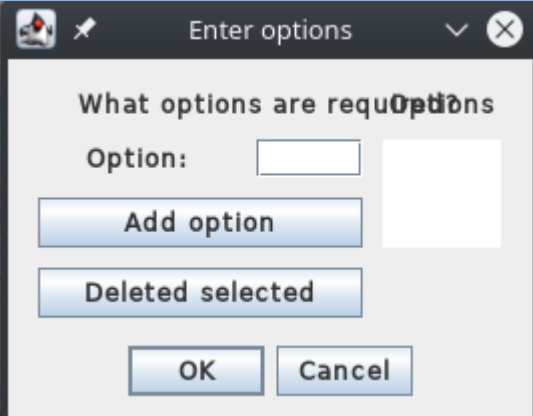
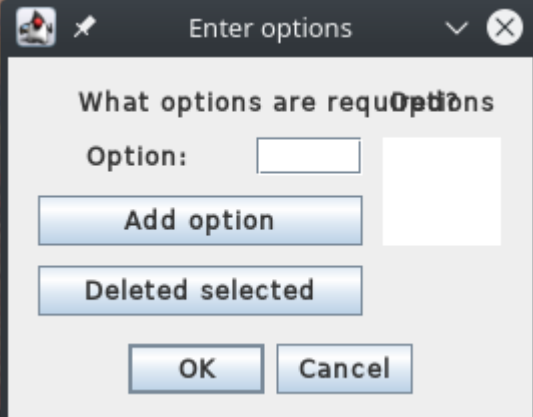
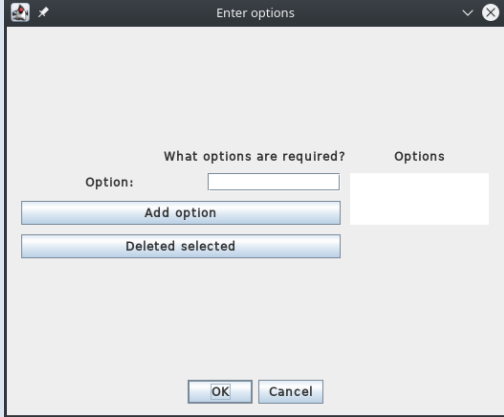


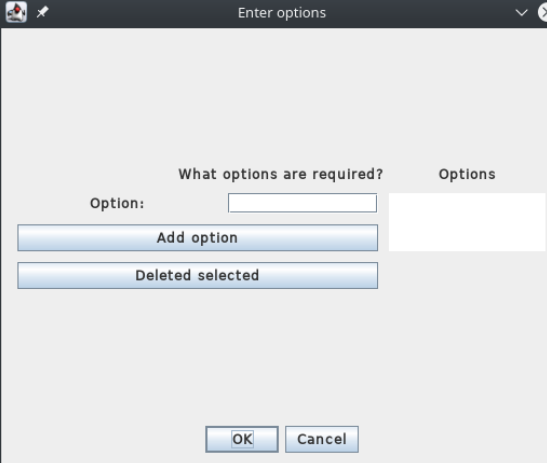
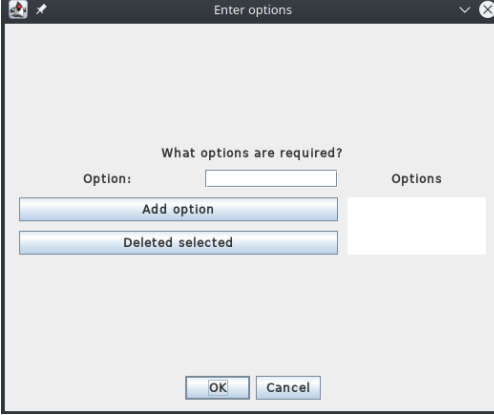
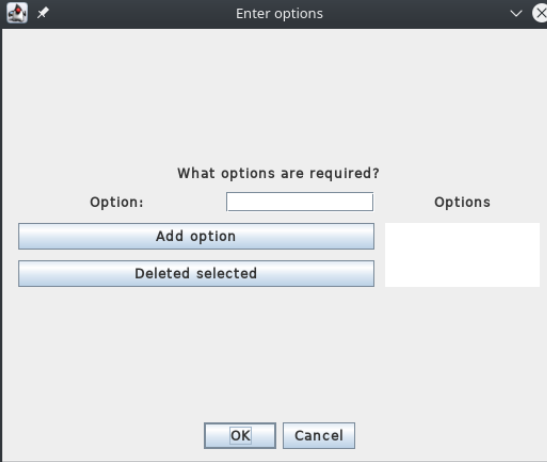
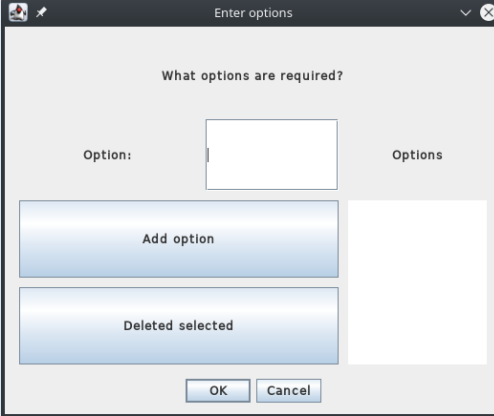
Set the component of the question creation tab to be the new QuestionCreationP anel rather than adding a new tab at teh same index. Use `tabs.setComponentAt()`

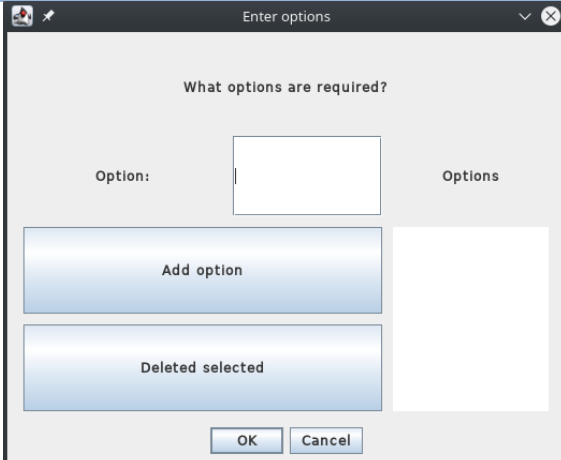
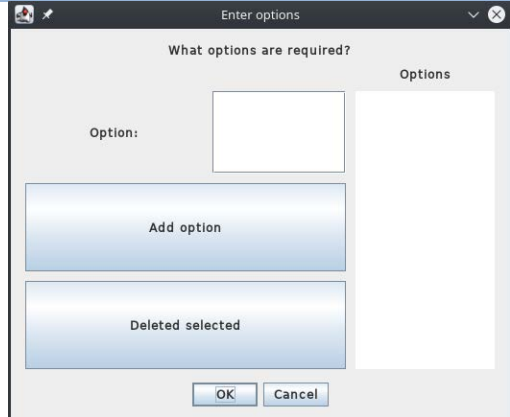
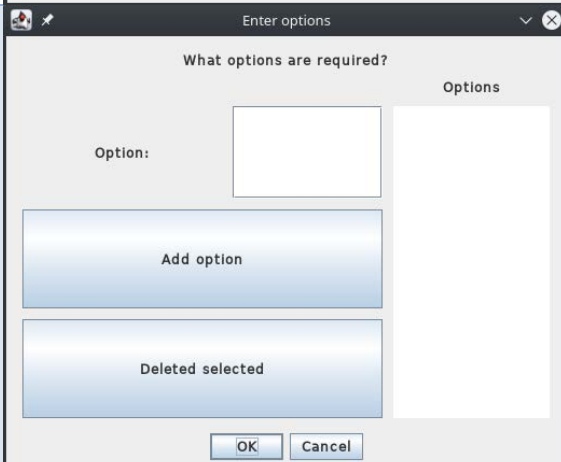
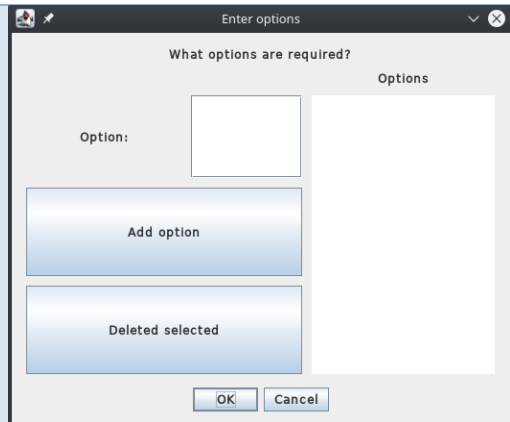


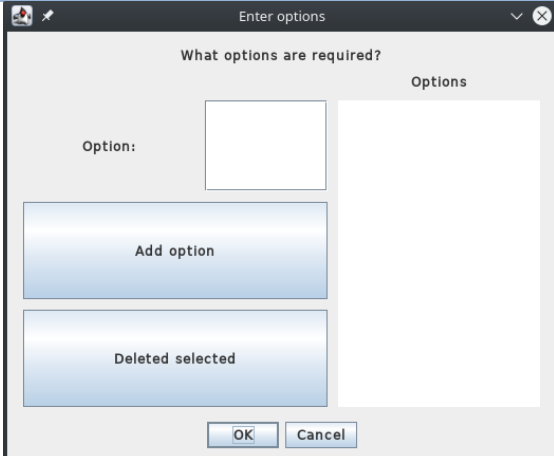
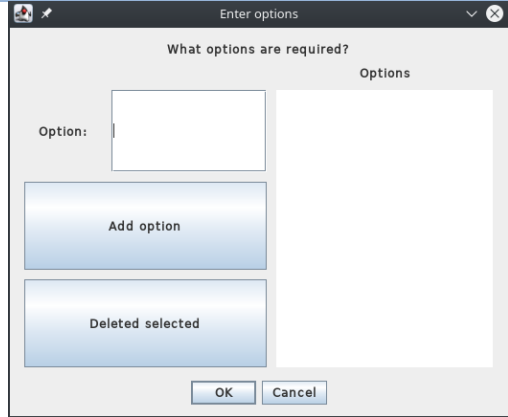
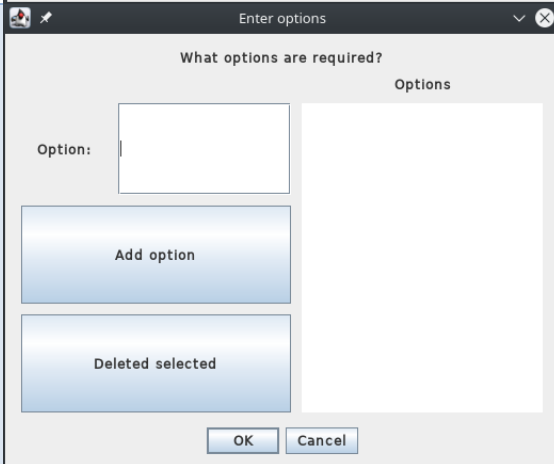
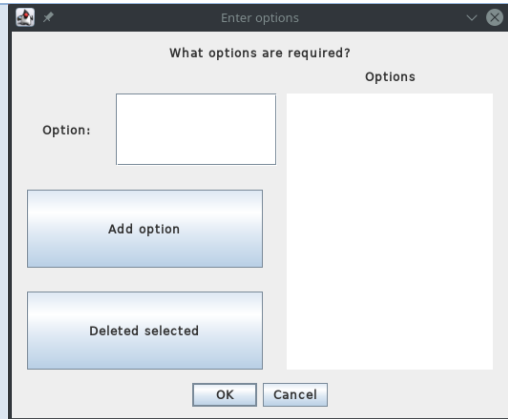
Panel is reset without changing which tab the user is on.

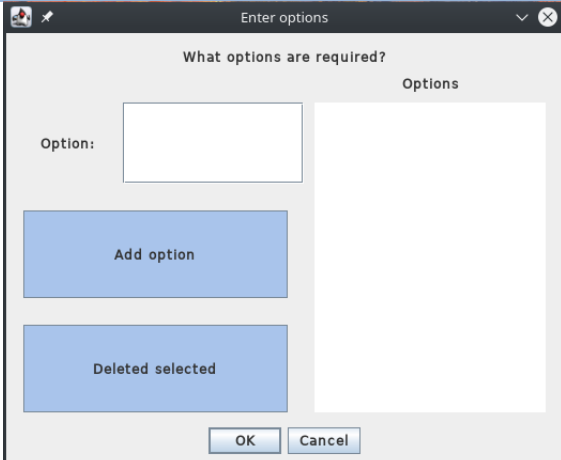
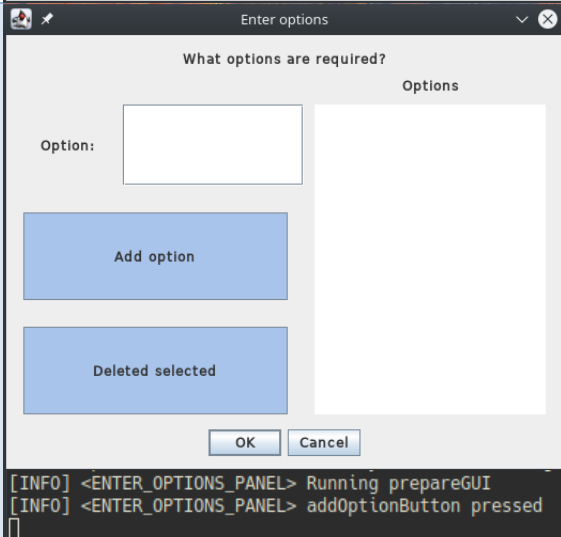
89	Soft Dev	The question preview panel is better sized to match the design.			
90	Soft Dev	There is an enter options dialog with a label. But there is no cancel button as there is a question mark icon that needs to be removed.		Add an additional argument of <code>JOptionPane.OK_CANCEL_OPTION</code> to the pane to add the cancel button. Adding an additional argument of <code>JOptionPane.PLAIN_MESSAGE</code> as the look and feel will get rid of the icon.	 There is a cancel option and no icon.
91	Soft Dev	The “what options are required” label isn’t centered.		Add an additional argument of <code>SwingConstants.CENTER</code> to the <code>JLabel</code> constructor this specifies that the text should be centered.	 The label text is centered now.

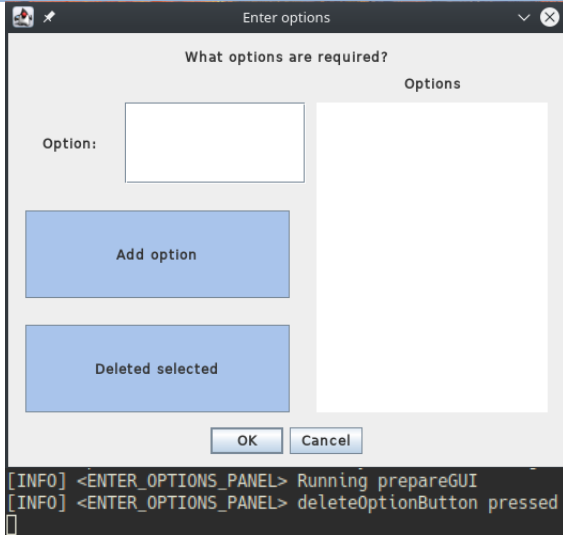
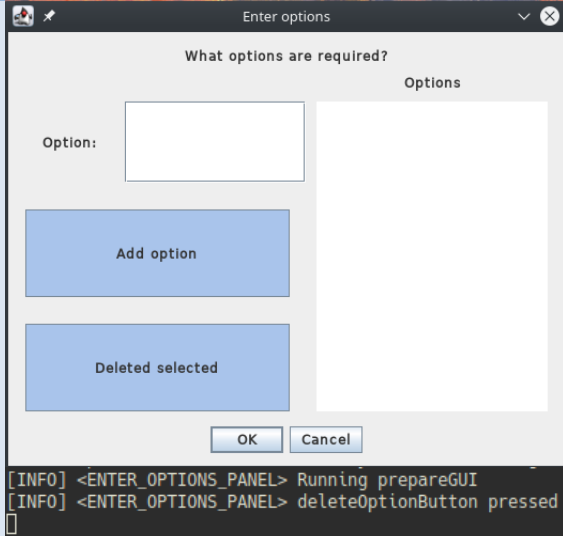
92	Soft Dev	The other components are somewhat correctly arranged. The window is too small however and the options Jlist isn't large enough and the header label isn't correctly placed.			
93	Soft Dev	The Pane is too small		Use .setPreferredSize() to set a larger size	 The pane is larger

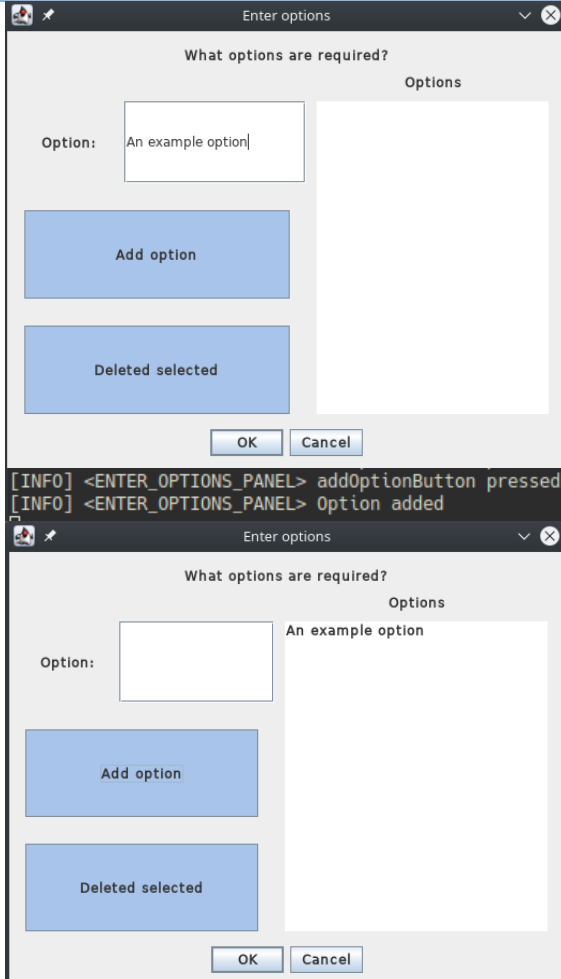
94	Soft Dev	The options header and the options JList are incorrectly placed		Change the grid y of the header from 0 to 1. Change the grid x of them both from 3 to 2. This will shift them down and across to the left one.	
95	Soft Dev	The panel doesn't take up all the space in the window vertically.		Set the weighty of the constraints to 1 so that the components take up all the extra space vertically.	

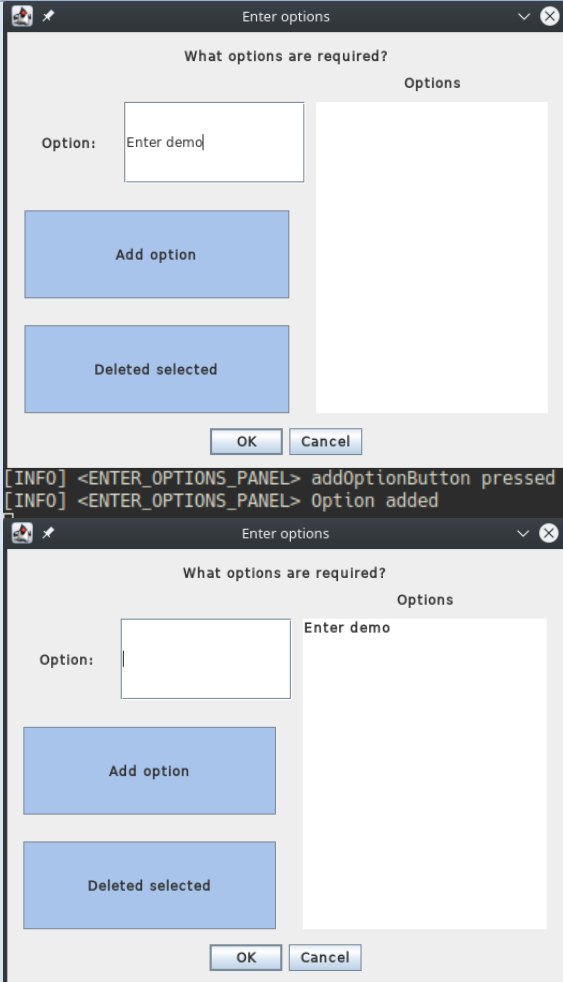
96	Soft Dev	The header labels take up the same vertical space as the rest of the components this wastes a lot of space.		Set the weight of the rows with the labels in to 0 so that they do not take up as much space.	
97	Soft Dev	The JList doesn't take up enough space		Increase the weight of it's column	

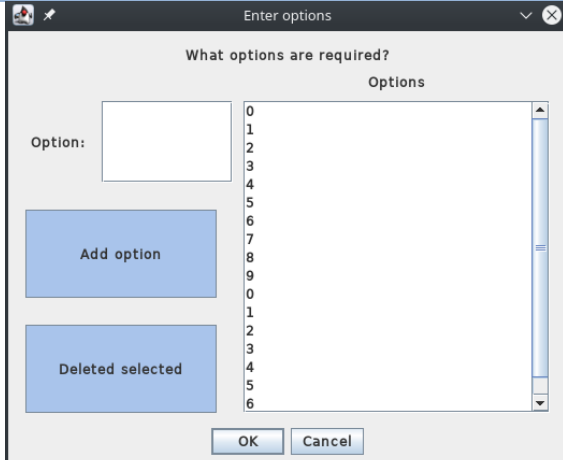
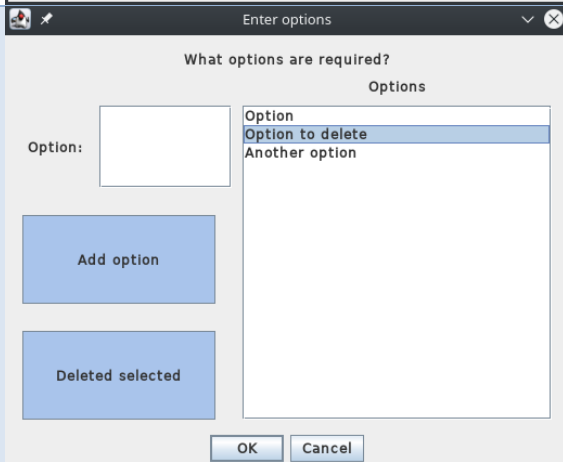
98	Soft Dev	The option text field isn't long enough		Reduce the weight of the column of that contains the option label so that it take up less space and makes the text field larger	
99	Soft Dev	The add option and delete selected option are too large		Add top and bottom padding to them using an insets object.	

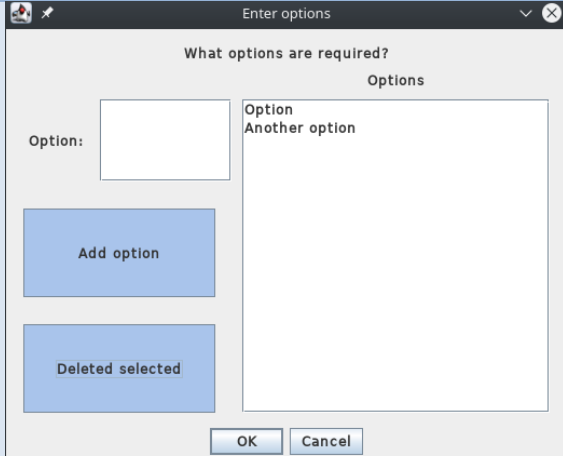
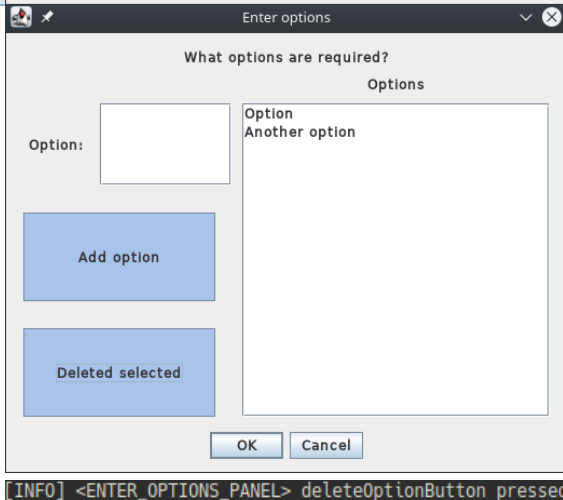
100	Soft Dev	The buttons are of the correct colour			
101	Soft Dev	Presses to the add option button are detected.			

102	Soft Dev	Presses to the deleteSelected button are detected			
103	Soft Dev	When the add option button is pressed and the option text field is empty no option is added.			

104	Soft Dev	When the add option button is pressed and the option text field has text in it the option is added and the text field is cleared.			
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105	Soft Dev	When enter is pressed inside the text field it's the same as pressing the addOptionButton	 The image displays two sequential screenshots of a software dialog box titled "Enter options". Top Screenshot: The dialog box has a title bar "Enter options" with a close button. The main content area is titled "What options are required?". It contains a text input field labeled "Option:" with the text "Enter demo" entered. Below the input field are two buttons: "Add option" and "Deleted selected". To the right of the input field is a large empty rectangular area labeled "Options". At the bottom are "OK" and "Cancel" buttons. Log Messages: Between the two screenshots, the following log messages are visible: [INFO] <ENTER_OPTIONS_PANEL> addOptionButton pressed [INFO] <ENTER_OPTIONS_PANEL> Option added Bottom Screenshot: The dialog box is in the same state as the top one, but the "Option:" text field is now empty, and the "Options" list area now contains the text "Enter demo".
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106	Soft Dev	When too many options to display are added to the list a scroll bar appears which allows the list of the scrolled. The shift in layout is a result of adding the scrollbar (even when the scrollbar isn't visible the panel has this new layout)			
107	Soft Dev	When an option is selected and the delete selected button is pressed, the option is deleted.	 <pre data-bbox="712 1133 1272 1187">[INFO] <ENTER_OPTIONS_PANEL> deleteOptionButton pressed [INFO] <ENTER_OPTIONS_PANEL> Option deleted</pre>		

					
108	Soft Dev	When the delete selected button is pressed and no option is selected nothing happens.			

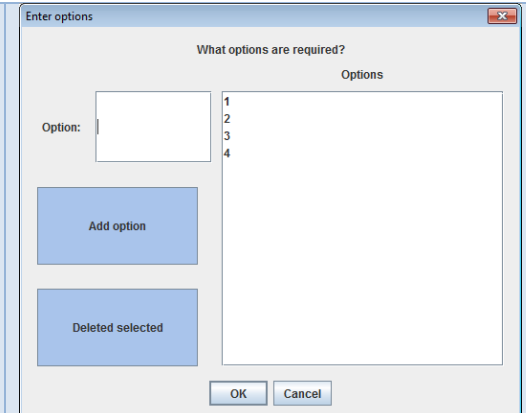
109

Soft Dev

When the `getOptions` method of the `OptionPane` is called there is a `ClassCastException`.

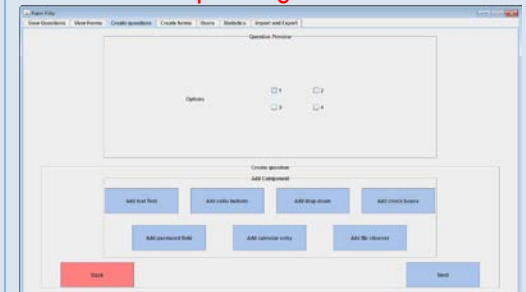
```
Exception in thread "AWT-EventQueue-0" java.lang.ClassCastException: [Ljava.lang.Object;
cannot be cast to [Ljava.lang.String;
at EnterOptionsPanel.getOptions(EnterOptionsPanel.java:94)
at QuestionCreationPanel.addCheckboxes(QuestionCreationPanel.java:640)
at QuestionCreationPanel.actionPerformed(QuestionCreationPanel.java:404)
at javax.swing.AbstractButton.fireActionPerformed(Unknown Source)
at javax.swing.DefaultButtonModel.fireActionPerformed(Unknown Source)
at javax.swing.DefaultButtonModel.setPressed(Unknown Source)
at javax.swing.plaf.basic.BasicButtonListener.mouseReleased(Unknown Source)
at java.awt.Component.processMouseEvent(Unknown Source)
at javax.swing.JComponent.processMouseEvent(Unknown Source)
at java.awt.Component.processEvent(Unknown Source)
at java.awt.Container.processEvent(Unknown Source)
at java.awt.Component.dispatchEventImpl(Unknown Source)
at java.awt.Container.dispatchEventImpl(Unknown Source)
at java.awt.Component.dispatchEvent(Unknown Source)
at java.awt.LightweightDispatcher.retargetMouseEvent(Unknown Source)
at java.awt.LightweightDispatcher.processMouseEvent(Unknown Source)
at java.awt.LightweightDispatcher.dispatchEvent(Unknown Source)
at java.awt.Container.dispatchEventImpl(Unknown Source)
at java.awt.Window.dispatchEventImpl(Unknown Source)
at java.awt.Component.dispatchEvent(Unknown Source)
at java.awt.EventQueue.dispatchEventImpl(Unknown Source)
at java.awt.EventQueue.access$500(Unknown Source)
at java.awt.EventQueue$3.run(Unknown Source)
at java.awt.EventQueue$3.run(Unknown Source)
at java.security.AccessController.doPrivileged(Native Method)
at java.security.ProtectionDomain$JavaSecurityAccessImpl.doIntersectionPrivilege(Unknown Source)
at java.security.ProtectionDomain$JavaSecurityAccessImpl.doIntersectionPrivilege(Unknown Source)
at java.awt.EventQueue$4.run(Unknown Source)
at java.awt.EventQueue$4.run(Unknown Source)
at java.security.AccessController.doPrivileged(Native Method)
at java.security.ProtectionDomain$JavaSecurityAccessImpl.doIntersectionPrivilege(Unknown Source)
at java.awt.EventQueue.dispatchEvent(Unknown Source)
at java.awt.EventDispatchThread.pumpOneEventForFilters(Unknown Source)
at java.awt.EventDispatchThread.pumpEventsForFilter(Unknown Source)
at java.awt.EventDispatchThread.pumpEventsForHierarchy(Unknown Source)
at java.awt.EventDispatchThread.pumpEvents(Unknown Source)
at java.awt.EventDispatchThread.pumpEvents(Unknown Source)
at java.awt.EventDispatchThread.run(Unknown Source)
```

The `toArray()` method of the default list method returns an array of objects however the `questionCreationPanel` expects a `String` array to be returned. Converting each element to a `String` and storing them into a `String` array and returning the `String` array instead should fix this.

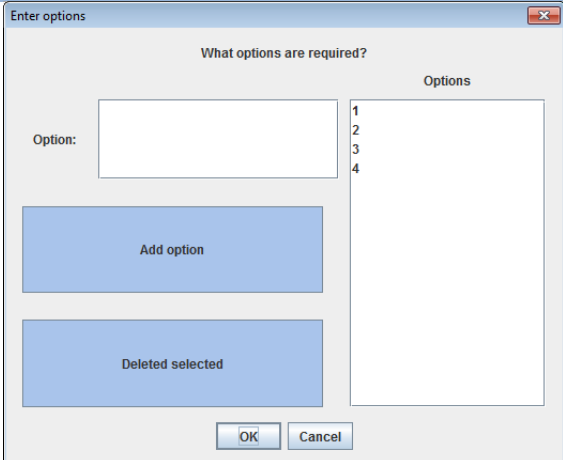
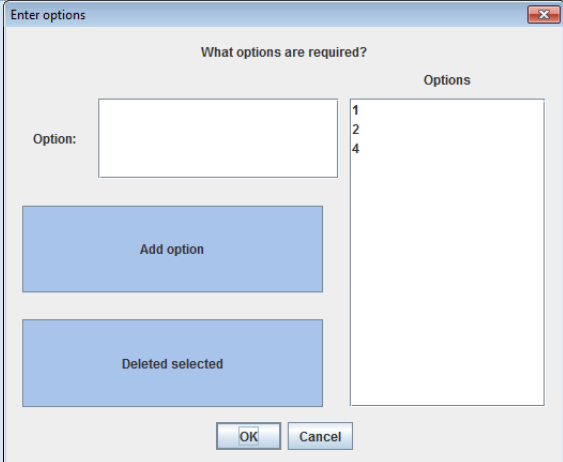


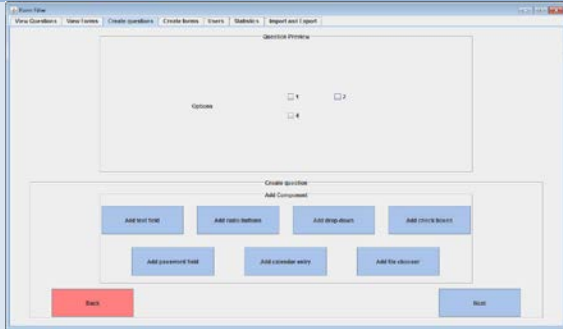
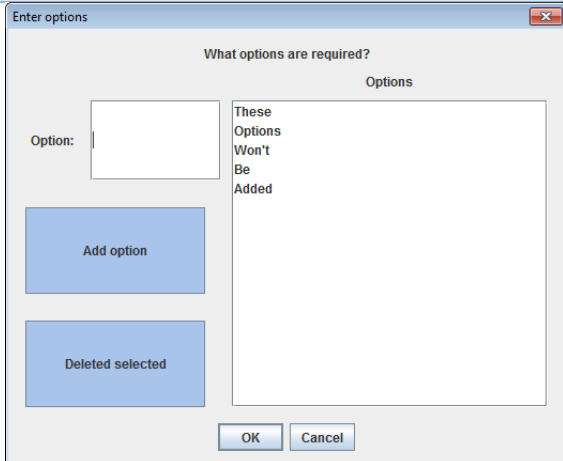
```
[INFO] <QUESTION_CREATION_PANEL> addCheckboxesButton pressed
[INFO] <QUESTION_CREATION_PANEL> Running addCheckboxes
[INFO] <ENTER_OPTIONS_PANEL> addOptionButton pressed
[INFO] <ENTER_OPTIONS_PANEL> Option added
[INFO] <ENTER_OPTIONS_PANEL> addOptionButton pressed
[INFO] <ENTER_OPTIONS_PANEL> Option added
[INFO] <ENTER_OPTIONS_PANEL> addOptionButton pressed
[INFO] <ENTER_OPTIONS_PANEL> Option added
[INFO] <ENTER_OPTIONS_PANEL> addOptionButton pressed
[INFO] <ENTER_OPTIONS_PANEL> Option added
[INFO] <CHECKBOX_PANEL_BUILDER> Running add
[INFO] <CHECKBOX_PANEL_BUILDER> Running add
[INFO] <CHECKBOX_PANEL_BUILDER> Running add
[INFO] <CHECKBOX_PANEL_BUILDER> Running add
[INFO] <CHECKBOX_PANEL_BUILDER> Running build
[INFO] <CHECKBOX_PANEL_BUILDER> Running trimArray
[INFO] <CHECKBOX_PANEL_BUILDER> Running preparePanel
```

No error when pressing ok



The options are shown in the checkboxes (in this case a checkbox was being added)

110	Soft Dev	When the add checkboxes button (or any add component button that uses the optionEntryPanel) is pressed the same option panel is opened so that the options can be edited.			
111	Soft Dev	If the options are changed in the option entry panel the component that is added to the question is updated to reflect these new options.			

			 In this case number 3 was removed.		
112	Soft Dev	If options are entered but cancel is pressed, no component is added to the question.	  No options added		

113

Soft Dev

If not options are entered and the ok button is pressed the component is still added but with no options. This means that the user is allowed to continue to the next screen even though there is no visible component.

Enter options

What options are required?

Options

Option:

Windows Snip

Add option

Deleted selected

OK Cancel

Question Preview

Big example

Create question

Add Component

Add text text Add radio buttons Add drag/drop Add choice buttons

Add paragraph text Add column entry Add file upload

Back Next

Ok is pressed and no visible component is added to the question.

Question Preview

Big example

Create question

Question title

Question type

Question difficulty

Back Next

Add a check to ensure that at least 1 option has been added before adding the component to the question.

Enter options

What options are required?

Options

Option:

Windows Snip

Add option

Deleted selected

OK Cancel

Question Preview

Big example

Create question

Add Component

Add text text Add radio buttons Add drag/drop Add choice buttons

Add paragraph text Add column entry Add file upload

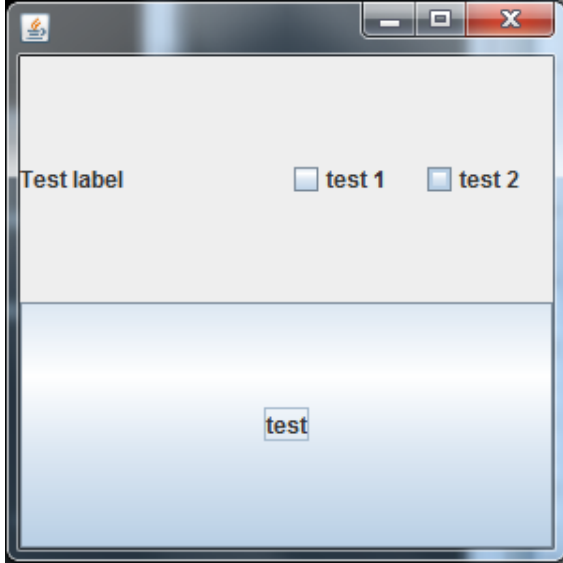
Back Next

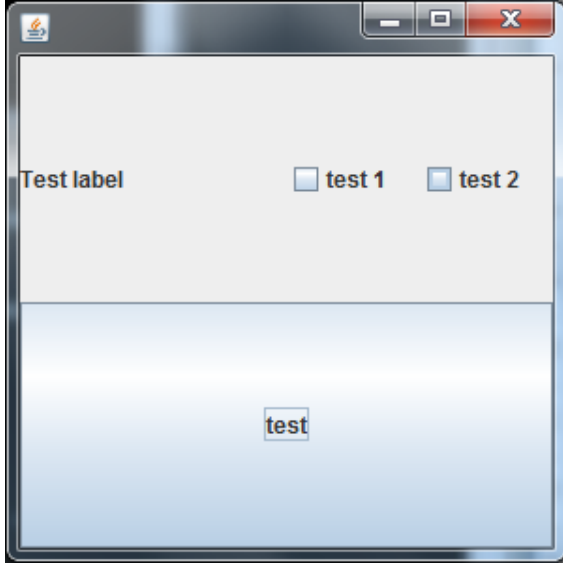
No component

Please add a component before continuing

OK

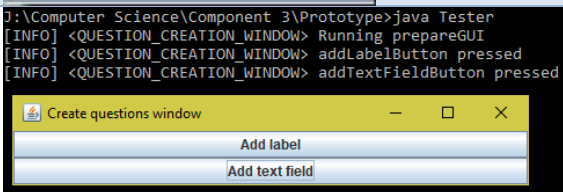
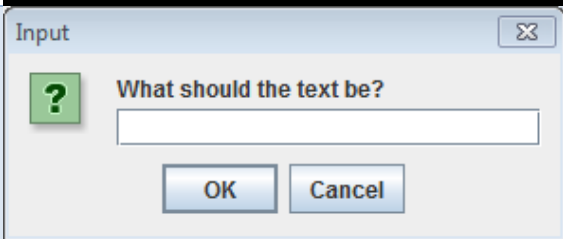
When no options are entered the user isn't allowed to continue.

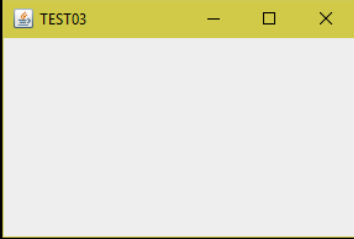
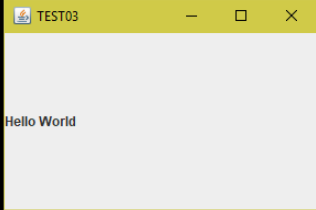
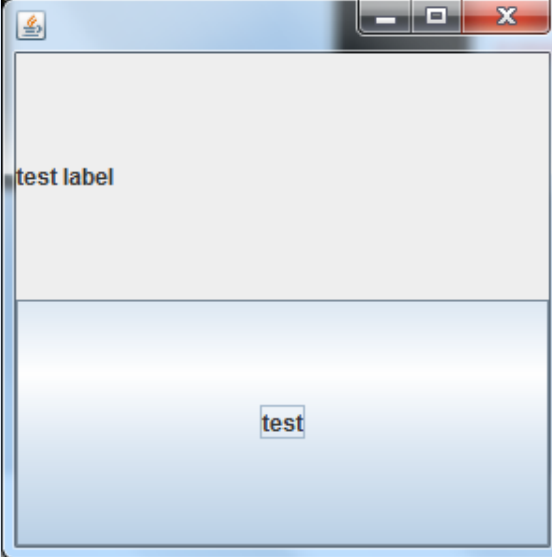
			The next screen can be accessed even though there is not an entry added the question.		
114	Soft Dev	When the toString method a questionPanel is called it's id and a list of all of the components in it are output.	 <pre>Q12345678\$label:Test label checkboxes:test 1:false,test 2:false</pre>		
115	Soft Dev	ArrayOutOfBoundsException when loading a QuestionPanel from string	<pre>Exception in thread "main" java.lang.ArrayIndexOutOfBoundsException: 1 at QuestionPanel.<init>(QuestionPanel.java:33) at Tester.testComponentSaving(Tester.java:73) at Tester.main(Tester.java:33)</pre>	This is because the string was being split at the "\$" symbol. However, this is a special symbol therefore it needs to be escaped before using it to split. Splitting at "\\ \$"	<pre>\$./runTW.sh test 1:false,test 2:false test 1:false test 2:false [INFO] <CHECKBOX_PANEL_BUILDER> Running preparePanel [INFO] <QUESTION_PANEL> Running setup</pre> No errors when running

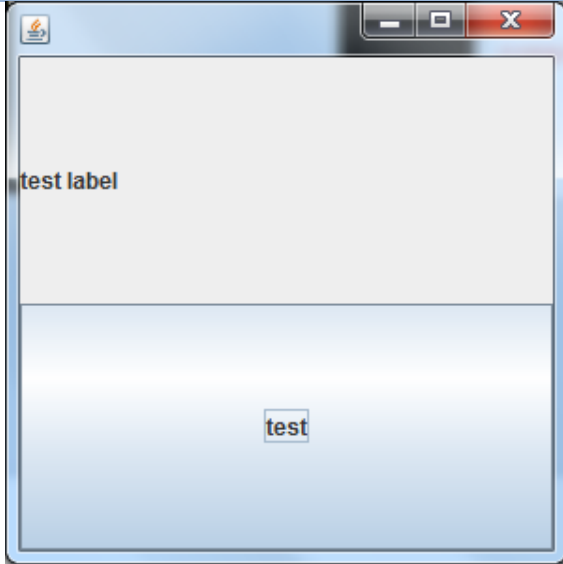
				will work as it means that the "\$" is escaped correctly.	
116	Soft Dev	A questionPanel can be perfectly recreated from it's output string	 This panel was recreated from the string output above.		
117	Soft Dev	Error loading the question panel database from file after implementing new loading logic which loads the database in order of question panel id	<pre>[INFO] <QUESTION_PANEL_LIST> Running addQuestionPanel [ERROR] <QUESTION_PANEL_LIST> Error loading database java.lang.ArrayIndexOutOfBoundsException: -1 java.lang.ArrayIndexOutOfBoundsException: -1 at QuestionPanelList.addQuestionPanel(QuestionPanelList.java:111) at QuestionPanelList.loadDatabase(QuestionPanelList.java:78) at QuestionPanelList.<init>(QuestionPanelList.java:13) at QuestionList.<init>(QuestionList.java:23) at GUI.setup(GUI.java:39) at GUI.<init>(GUI.java:38) at Main.main(Main.java:5)</pre>	This is because the logic that inserted the question panel before another one in the array inserted at the index-1 this resulted in an array out of	<pre>[INFO] <QUESTION_PANEL> Running setup [INFO] <QUESTION_PANEL_LIST> Running addQuestionPanel [INFO] <RADIOBUTTON_PANEL> Running preparePanel [INFO] <QUESTION_PANEL> Running setup [INFO] <QUESTION_PANEL_LIST> Running addQuestionPanel [INFO] <QUESTION_PANEL> Running setup [INFO] <QUESTION_PANEL_LIST> Running addQuestionPanel [INFO] <RADIOBUTTON_PANEL> Running preparePanel [INFO] <QUESTION_PANEL> Running setup [INFO] <QUESTION_PANEL_LIST> Running addQuestionPanel [INFO] <QUESTION_PANEL_LIST> 4 QuestionPanels loaded from file</pre> The question panel database is successfully loaded from file.

				bounds exception when the index to insert before was index 0. The question panel should simply be inserted at index rather than index-1. This will fix the exception and the algorithm will function as intended.	
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2: Ability to add labels

Test	Stage	Result Problem	Evidence	Action and Justification (If Needed)	Evidence of Action (If Needed)
1	Proto	The create questions window has a button to add a label to the question being created.			
2	Proto	Presses of the add label button are detected	<pre>D:\Computer Science\Component 3\Prototype>java Tester [INFO] <QUESTION_CREATION_WINDOW> Running prepareGUI [INFO] <QUESTION_CREATION_WINDOW> addLabelButton pressed [INFO] <QUESTION_CREATION_WINDOW> addTextFieldButton pressed</pre> 		
3	Proto	A prompt appears asking the user what the text of the label should be			

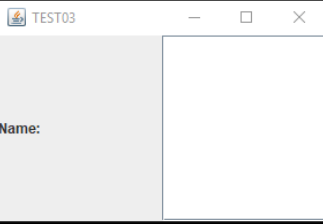
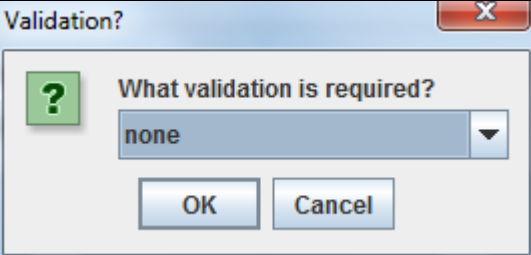
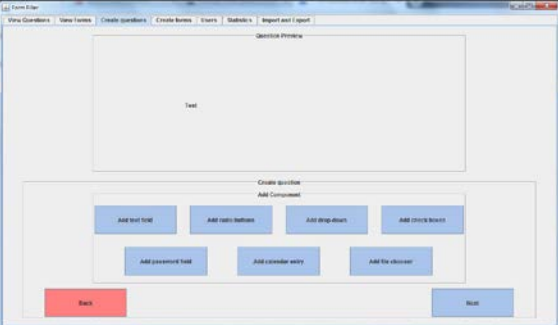
4	Proto	Added label doesn't appear until the window is resized	<pre>J:\Computer Science\Component 3\Prototype>java Tester [INFO] <QUESTION_CREATION_WINDOW> Running prepareGUI [INFO] <QUESTION_CREATION_WINDOW> addLabelButton pressed [INFO] <QUESTION_CREATION_WINDOW> Running addLabel [INFO] <QUESTION_PANEL_BUILDER> Running add</pre> 	Call the revalidate method of the panel after adding the component	<pre>J:\Computer Science\Component 3\Prototype>java Tester [INFO] <QUESTION_CREATION_WINDOW> Running prepareGUI [INFO] <QUESTION_CREATION_WINDOW> addLabelButton pressed [INFO] <QUESTION_CREATION_WINDOW> Running addLabel [INFO] <QUESTION_PANEL_BUILDER> Running add</pre> 
5	Soft Dev	When the toString method is called on a JSaveableLabel a string detailing the text in it is output	 <pre>label:test label</pre>		

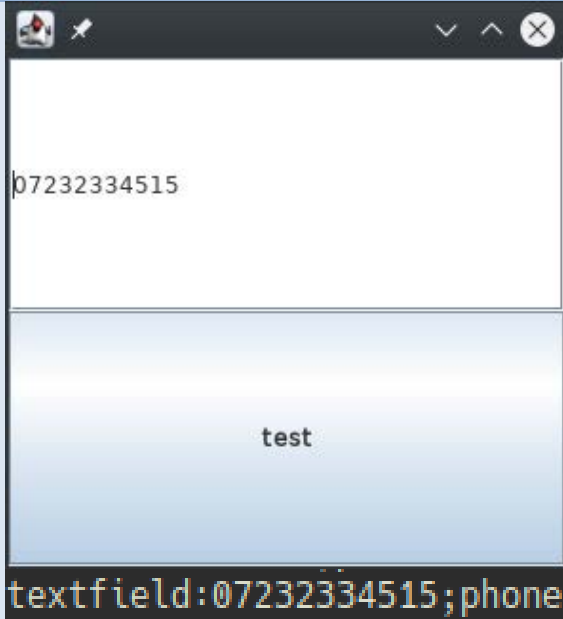
6	Soft Dev	When the output string from the toString method of a JSaveableLabel is used to create a new JSaveableLabel, the label is recreated exactly from the string.			
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This label was recreated from the string output above.

3: Ability to add text fields

Test	Stage	Result Problem	Evidence	Action and Justification (If Needed)	Evidence of Action (If Needed)
1	Proto	The create questions window has a button to add a text field to the question being created.			
2	Proto	Presses of the add text field button are detected	<pre> D:\Computer Science\Component 3\Prototype>java Tester [INFO] <QUESTION_CREATION_WINDOW> Running prepareGUI [INFO] <QUESTION_CREATION_WINDOW> addLabelButton pressed [INFO] <QUESTION_CREATION_WINDOW> addTextFieldButton pressed </pre> 		

3	Proto	Text field can be added	<pre> D:\Computer Science\Component 3\Prototype>java Tester [INFO] <QUESTION_CREATION_WINDOW> Running prepareGUI [INFO] <QUESTION_CREATION_WINDOW> addLabelButton pressed [INFO] <QUESTION_CREATION_WINDOW> Running addLabel [INFO] <QUESTION_PANEL_BUILDER> Running add [INFO] <QUESTION_CREATION_WINDOW> addTextFieldButton pressed [INFO] <QUESTION_PANEL_BUILDER> Running add </pre> 		
4	Soft Dev	If cancel is pressed on the text field validation dialog, no text field is added to the question.	 		

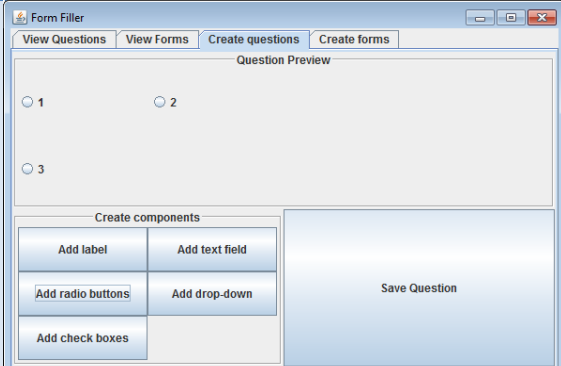
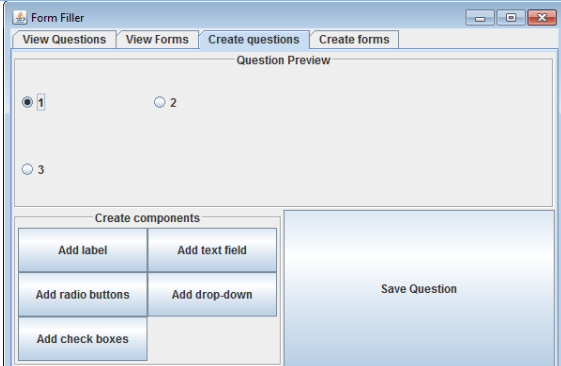
5	Soft Dev	When the toString method is called on a JValidatedTextField a string detailing its type (e.g. phone number), and the text entered is output			
6	Soft Dev	When the output string from the toString method of a JValidatedTextField is used to create a new JValidatedTextField, the textfield is recreated exactly from the string.			

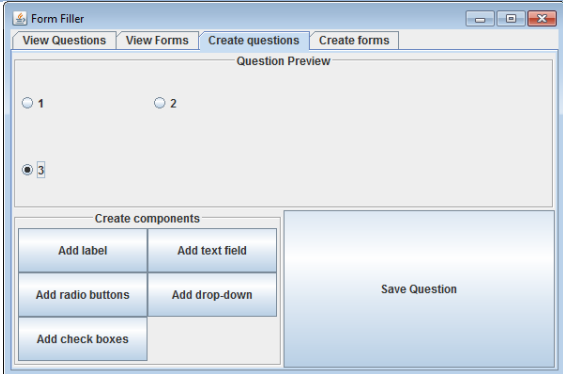
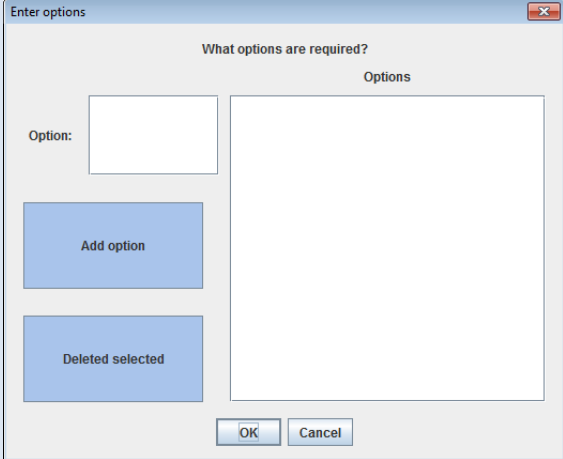
			This text field has been recreated from the string output above. Phone number: Please enter a valid phone number When the <code>getEditText()</code> method is called on the textfield the phone number error text is output. This proves that the text field is of type <code>phonenum</code>		
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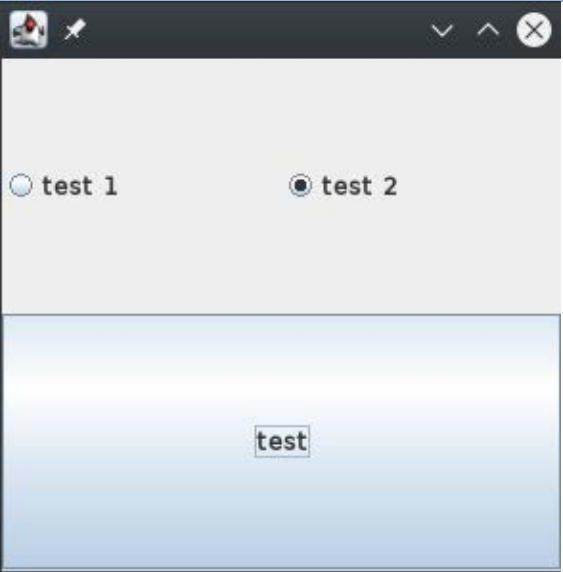
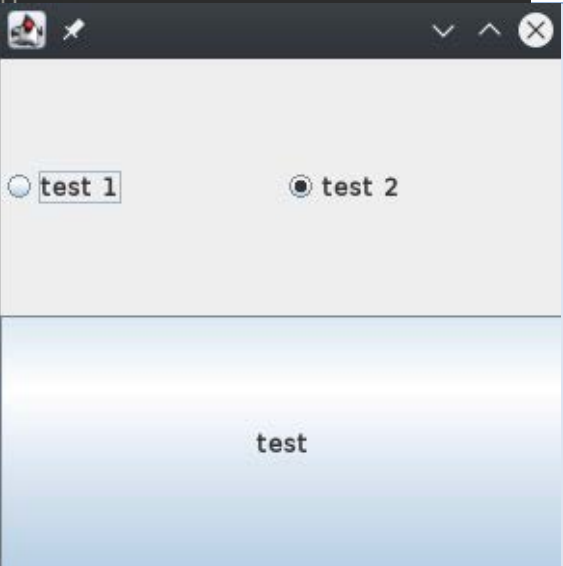
4: Ability to specify the type of data that should be entered into the text field

Test	Stage	Result	Problem	Evidence	Action and Justification (If Needed)	Evidence of Action (If Needed)
1	Proto	Question display window fails to compile	<pre> J:\Computer Science\Component 3\Prototype>javac Main.java .\QuestionDisplayWindow.java:14: error: cannot find symbol private DefaultTableModel questionTableModel = new DefaultTableMode ^ symbol: class DefaultTableModel location: class QuestionDisplayWindow .\QuestionDisplayWindow.java:14: error: cannot find symbol private DefaultTableModel questionTableModel = new DefaultTableMode ^ symbol: class DefaultTableModel location: class QuestionDisplayWindow 2 errors </pre>	<pre> J:\Computer Science\Component 3\Prototype>javac Main.java </pre>	Import javax.swing.table.*	<pre> J:\Computer Science\Component 3\Prototype>javac Main.java </pre>

5: Ability to add radio buttons

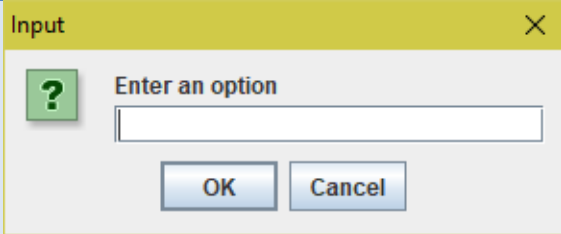
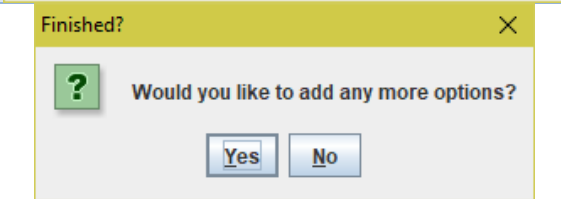
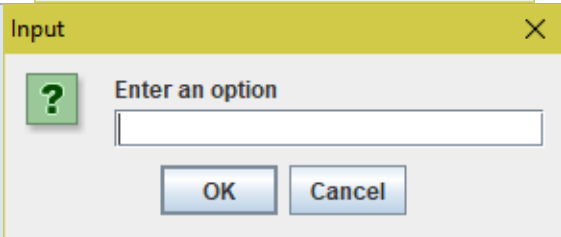
Test	Stage	Result Problem	Evidence	Action and Justification (If Needed)	Evidence of Action (If Needed)
1	Proto	Radio buttons can be added to the question. This code is reused from the add checkbox code.			
2	Proto	Only 1 radio button at a time can be selected			

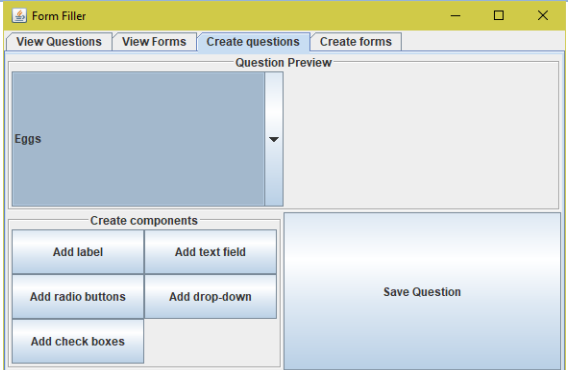
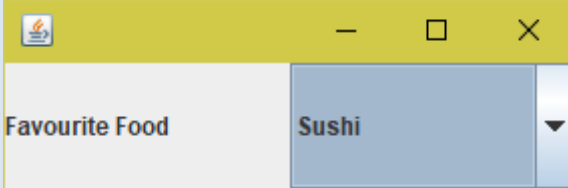
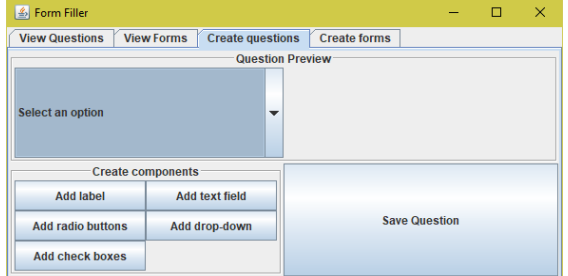
					
3	Soft Dev	When the add radio buttons button is pressed the new option entry dialog is opened			

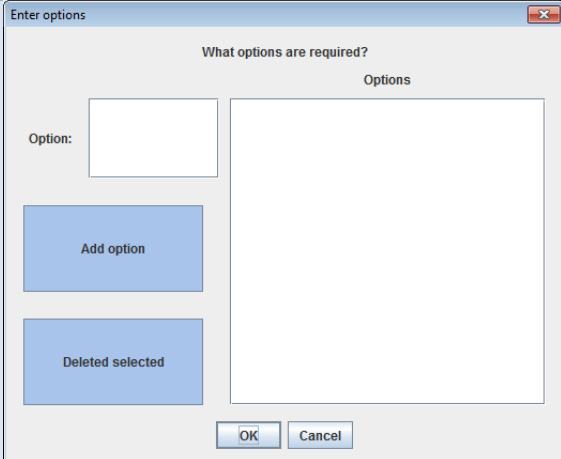
4	Soft Dev	When the toString method is called on the RadioButtonPanel a string detailing the options and which ones are selected is output.	 <pre>radiobuttons:test 1,false.test 2,true</pre>		
5	Soft Dev	When the output string from the toString method of a radio button panel is used to create a new RadioButtonPanel, the panel is recreated exactly from the string.			

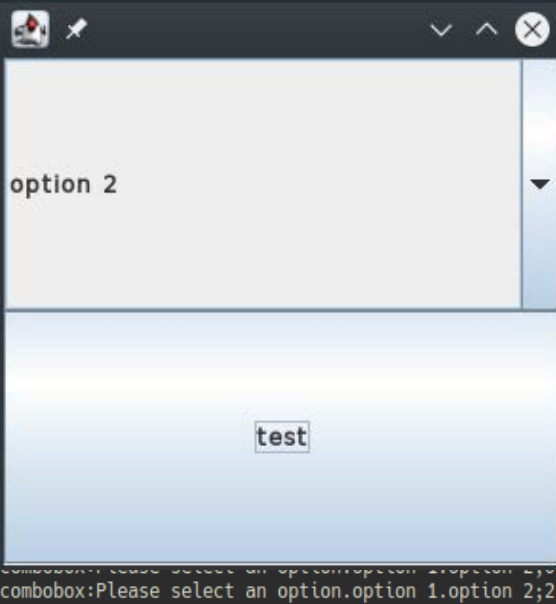
			This panel was created using the string output above.web		
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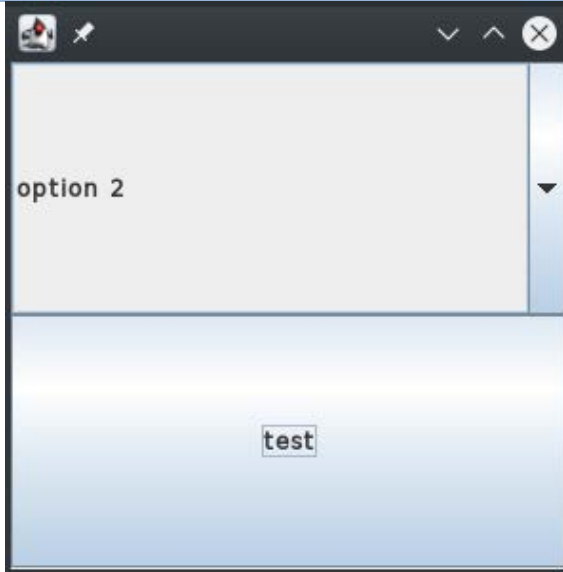
6: Ability to add drop-downs

Test	Stage	Result Problem	Evidence	Action and Justification (If Needed)	Evidence of Action (If Needed)
1	Proto	The user is prompted to enter an option for the combobox when the add dropdown button is pressed			
2	Proto	The user is asked if they want to enter any more options after entering an option			
3	Proto	If the user pressed yes to the any more options button another prompt to add an option appears			

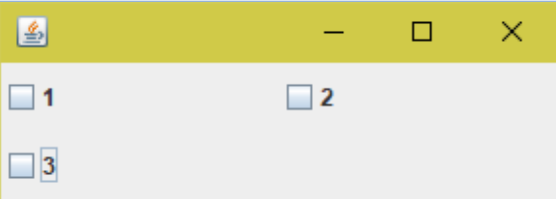
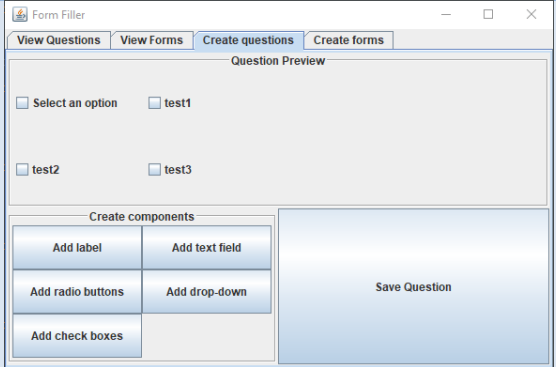
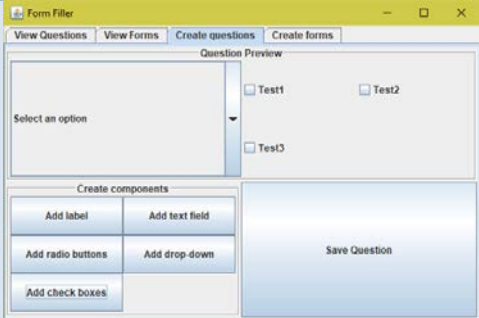
4	Proto	When the user pressed no the combobox is added to the question			
5	Proto	A saved question with a combobox is successfully loaded and displayed			
6	Proto	Select an option is automatically added as the first option for a combo box.			

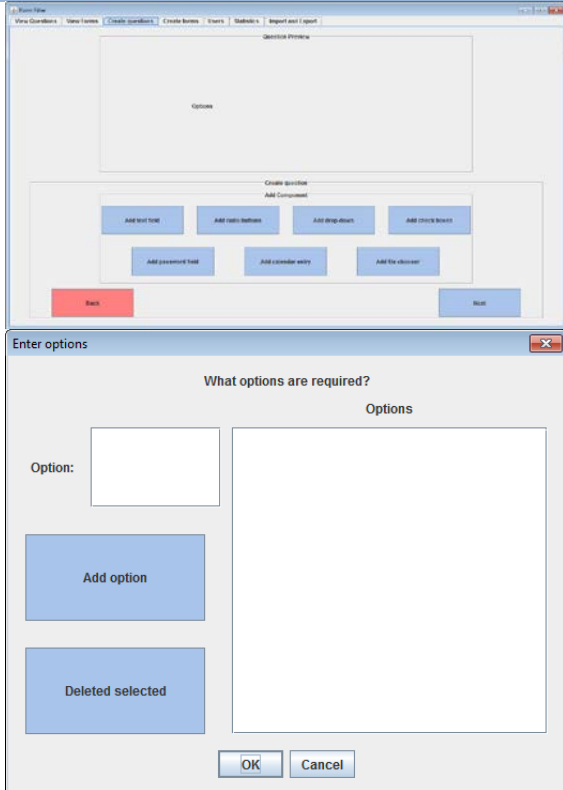
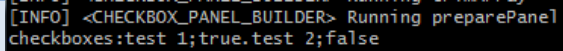
7	Soft Dev	If the add drop-down button is pressed the new option entry opens.			
8	Soft Dev	There is an error compiling JValidatedComboBox because super() isn't the first call in the constructor.	<pre>./components/JValidatedComboBox.java:16: error: call to super must be first statement in constructor super(options); ^ ./components/JValidatedComboBox.java:25: error: incompatible types: String cannot be converted to String[] String[] options = componentData[0]; ^</pre>	Super should be called first in the constructor to add the options to the combo box. However, this isn't possible as the options are only retrieved later in the constructor. To fix this the addItem() method should be used to add each item in turn.	<pre>harry@harry-pc-linux:/media/harry/HARRY BEST/Computer Science/Component 3/SOFTDEV/Project/Form-Filler\$./compile.sh Picked up _JAVA_OPTIONS: -Dswt.useSystemAAFontSettings=on -Dswing.aatext=true Note: RegisterJava uses or overrides a deprecated API. Note: Recompile with -Xlint:deprecation for details. Picked up _JAVA_OPTIONS: -Dswt.useSystemAAFontSettings=on -Dswing.aatext=true harry@harry-pc-linux:/media/harry/HARRY BEST/Computer Science/Component 3/SOFTDEV/Project/Form-Filler\$</pre> Successful compile

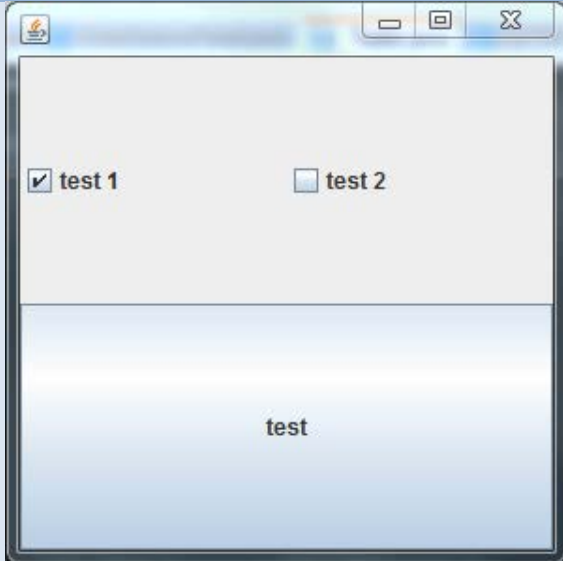
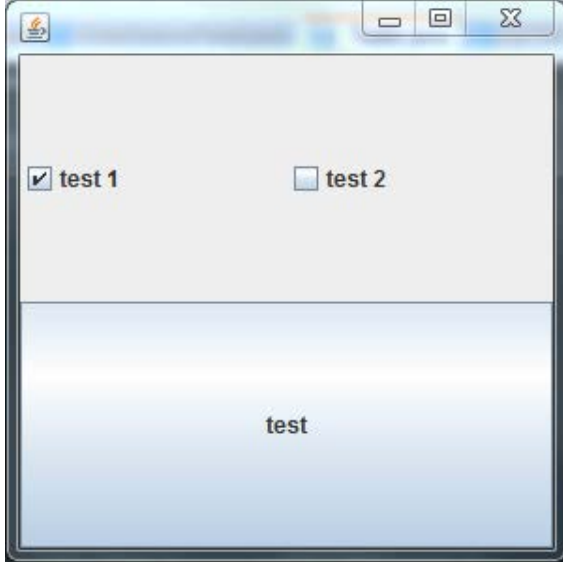
9	Soft Dev	When the toString method is called on JValidatedComboBox a string detailing the options and which ones are selected is output.	 The options are output and the index of the selected option is output at the end of the string.		
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10	Soft Dev	When the output string from the toString method of a JValidatedCombobox is used to create a new JValidatedCombobox, the combobox is recreated exactly from the string.	 This combobox was recreated from the string output above		
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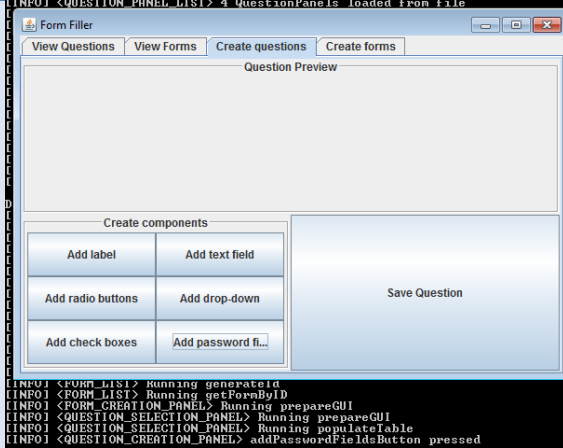
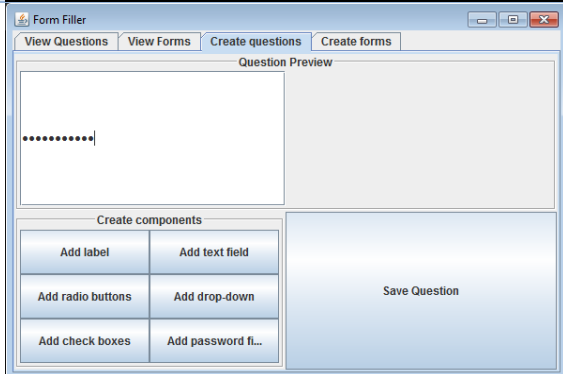
7: Ability to add checkboxes

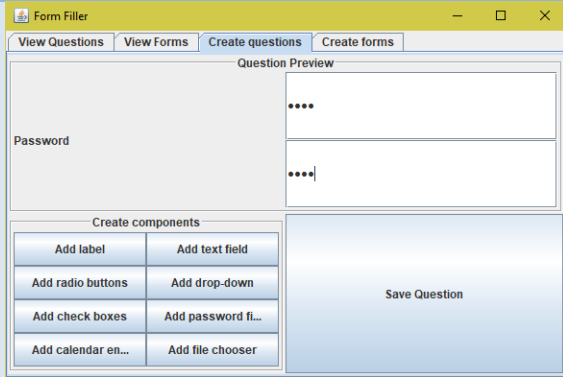
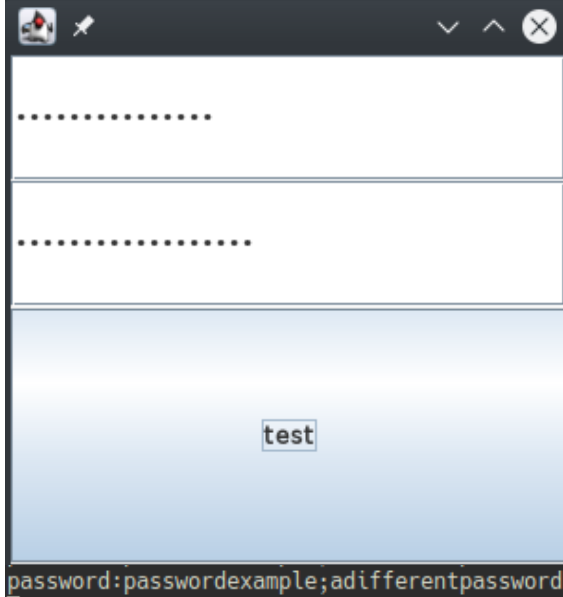
Test	Stage	Result Problem	Evidence	Action and Justification (If Needed)	Evidence of Action (If Needed)
1	Proto	Failure to compile CheckBoxPanel as no return type was specified for the preparePanel() method	<pre>J:\Computer Science\Component 3\Prototype>javac *.java CheckBoxPanel.java:15: error: invalid method declaration; return type required private preparePanel() ^ 1 error</pre>	Add a void return type	 Successful compile
2	Proto	CheckBoxPanel successfully created and added to a window. This is from the tester class and is just calling the methods.			
3	Proto	Checkboxes can be added to the question. The method of adding using calls the same get options method that adding a combobox does therefore I've not shown the adding options windows prompts. Because the same method has been used there is a select an option box when there shouldn't.		Move the code that adds the "select an option" option to the add combobox method and remove it from the getOptions() method.	 No more select an option check box but there is still a select an option for the combobox

4	Soft Dev	When the add checkboxes option is pressed the new entry options dialog appears			
5	Soft Dev	When the toString method is called on the checkboxPanel a string detailing the options and which ones are selected is output.			

					
6	Soft Dev	When a checkbox panel is created using the string output from the toString method it is recreated exactly.	 This checkbox panel has been recreated using the string output from the first one.		

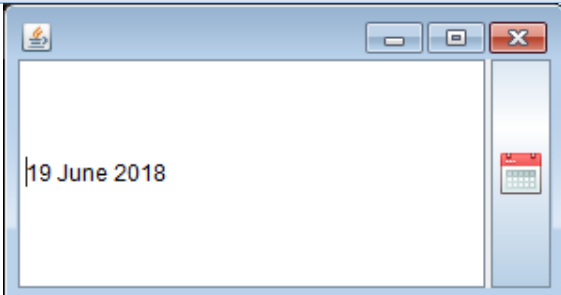
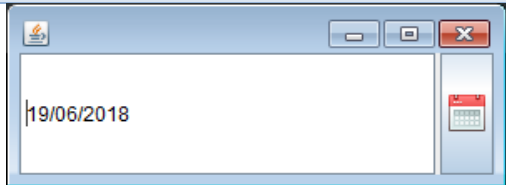
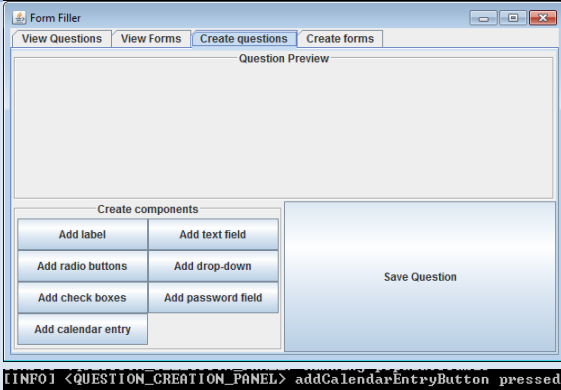
8: Ability to add password fields

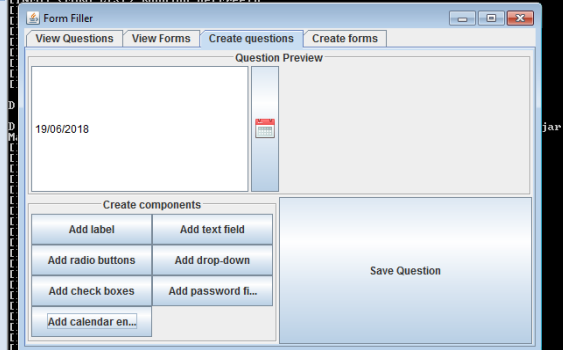
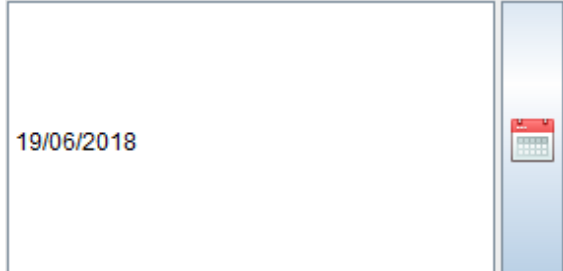
Test	Stage	Result Problem	Evidence	Action and Justification (If Needed)	Evidence of Action (If Needed)
1	Proto	There is an add passwords field button and presses of it are detected	 <pre> INFO: <FORM_LIST> Running generateId INFO: <FORM_LIST> Running getFormById INFO: <FORM_CREATION_PANEL> Running prepareGUI INFO: <QUESTION_SELECTION_PANEL> Running prepareGUI INFO: <QUESTION_SELECTION_PANEL> Running populateTable INFO: <QUESTION_CREATION_PANEL> addPasswordFieldButton pressed </pre>		
2	Proto	Password fields can be added.			

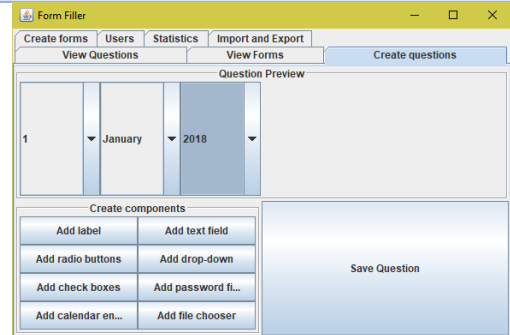
3	Proto	Two password fields are added.			
4	Soft Dev	When the toString method is called on a JValidatedPasswordField a string detailing the passwords entered is output.			

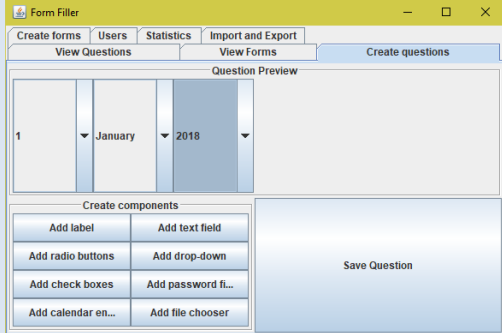
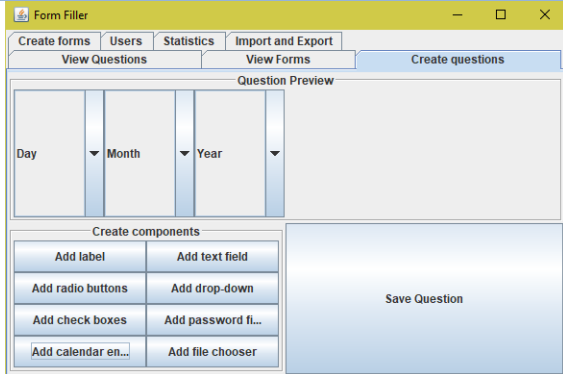
5	Soft Dev	When the output string from the toString method of a JValidatedPasswordField is used to create a new JValidatedPasswordField, the passwordfield is recreated exactly from the string.	 This password field was recreated from the string output above. <pre>password:passwordexample;adifferentpassword</pre> This is the string that this password field outputs. This is to show the the passwords entered are the same as in the password fields above.		
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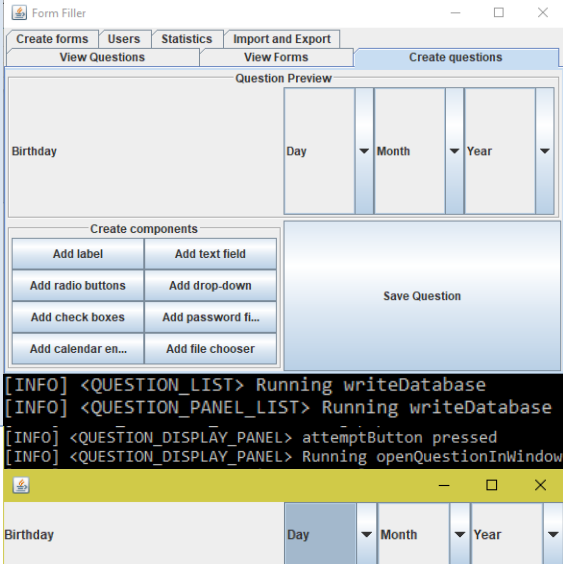
9: Ability to add calendar entry

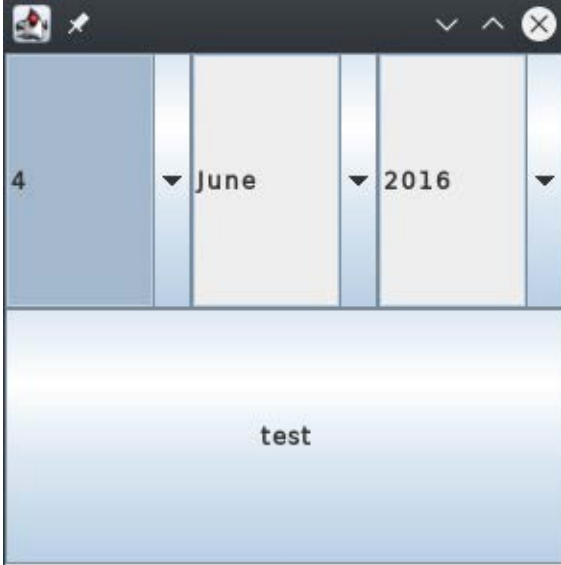
Test	Stage	Result Problem	Evidence	Action and Justification (If Needed)	Evidence of Action (If Needed)
1	Proto	Failure to compile the calendar panel because java.net.URL wasn't imported and so the calendar icon for the open calendar button couldn't be imported	<pre> D:\Computer Science\Component 3\Prototype>javac Main.java .\QuestionDisplayWindow.java:14: error: cannot find symbol private DefaultTableModel questionTableModel = new DefaultTableMode 1(questionData, tableHeaders); ^ symbol: class DefaultTableModel location: class QuestionDisplayWindow .\QuestionDisplayWindow.java:14: error: cannot find symbol private DefaultTableModel questionTableModel = new DefaultTableMode 1(questionData, tableHeaders); ^ symbol: class DefaultTableModel location: class QuestionDisplayWindow 2 errors </pre>	Import java.net.URL	<pre> D:\Computer Science\Component 3\Prototype>javac -cp .;lib\GoodDatePicker-10.3.1.jar *.java D:\Computer Science\Component 3\Prototype>cd components </pre> Successful compile
2	Proto	Date is the wrong format. Should be DD-MM-YYYY		Configure the calendar entry before it's added to the window by using the DatePickerSettings object	
3	Proto	There is an add calendar entry button and presses are detected			

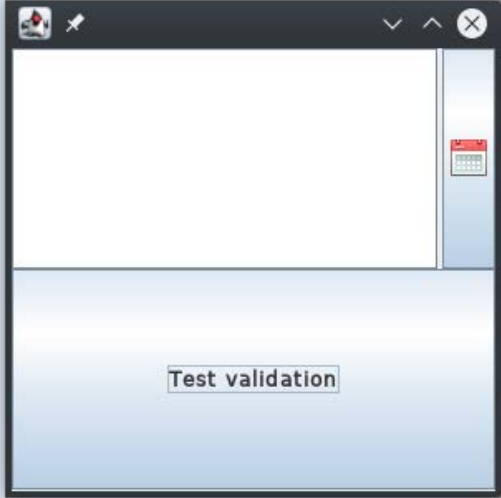
4	Proto	Failure to compile QuestionCreationPanel after adding the code that adds a calendar entry because the relevant classes and packages hadn't been imported.	<pre> D:\Computer Science\Component 3\Prototype>javac -cp .;LGoodDatePicker-10.3.1.jar *.java QuestionCreationPanel.java:273: error: cannot find symbol DatePickerSettings dateSettings = new DatePickerSettings(); ^ symbol: class DatePickerSettings location: class QuestionCreationPanel QuestionCreationPanel.java:273: error: cannot find symbol DatePickerSettings dateSettings = new DatePickerSettings(); ^ symbol: class DatePickerSettings location: class QuestionCreationPanel QuestionCreationPanel.java:276: error: cannot find symbol DatePicker datePicker = new DatePicker(dateSettings); ^ symbol: class DatePicker location: class QuestionCreationPanel QuestionCreationPanel.java:276: error: cannot find symbol DatePicker datePicker = new DatePicker(dateSettings); ^ symbol: class DatePicker location: class QuestionCreationPanel QuestionCreationPanel.java:279: error: cannot find symbol URL dateImageUrl = QuestionCreationPanel.class.getResource("data pickerbutton1.png"); ^ symbol: class URL location: class QuestionCreationPanel 5 errors </pre>	Import all the required packages	<pre> D:\Computer Science\Component 3\Prototype>javac -cp .;LGoodDatePicker-10.3.1.jar *.java </pre>	Successful compile
5	Proto	When the add calendar entry button is pressed a calendar entry is added to the question	 <pre> [INFO] <QUESTION_CREATION_PANEL> addCalendarEntryButton pressed [INFO] <QUESTION_CREATION_PANEL> Running addCalendarEntry [INFO] <QUESTION_PANEL_BUILDER> Running add </pre>			
6	Proto	A date is already set when the calendar entry is created.		Remove the line that sets the date today. (datePicker.setDateToToday();)		The text field is empty when the calendar entry is added.
7	Proto	Error saving question with a calendar entry panel to file because the convert class of the calendar entry isn't	<pre> [INFO] <QUESTION_LIST> Running writeDatabase [INFO] <QUESTION_PANEL_LIST> Running writeDatabase [ERROR] <QUESTION_PANEL_LIST> Writing database java.io.NotSerializableException: com.github.lgooddatepicker.internaltools.Convert </pre>			

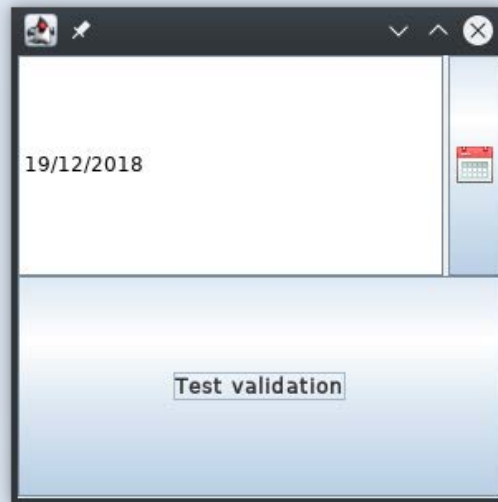
8	Proto	serialisable Error compiling JValidatedDatePicker after implementing serializeable because java.io.Serializable wasn't imported	<pre>D:\Computer Science\Component 3\Prototype\components>javac -cp .;JGoodDatePicker-10.3.1.jar *.java JValidatedDatePicker.java:9: error: cannot find symbol public class JValidatedDatePicker extends DatePicker implements JValidatedComponent, Serializable ^ symbol: class Serializable 1 error</pre>	Import java.io.*;	<pre>D:\Computer Science\Component 3\Prototype\components>javac -cp .;JGoodDatePicker-10.3.1.jar *.java</pre> Successful compile
9	Proto	Error loading previously saved datepicker from file as I had made changes to the class file and the classes no longer matched.	<pre>D:\Computer Science\Component 3\Prototype>java -cp .;JGoodDatePicker-10.3.1.jar Main [INFO] <GUI> Running setup [INFO] <QUESTION_LIST> Running loadDatabase [INFO] <QUESTION_LIST> 11 Questions loaded from file [INFO] <QUESTION_PANEL_LIST> Running loadDatabase [ERROR] <QUESTION_PANEL_LIST> Error loading database java.io.InvalidClassException: components.JValidatedDatePicker; local class incompatible: stream classdesc serialVersionUID = 3123422476854404716, local class serialVersionUID = -5936361165231247652</pre>	Delete the database and remake the question.	<pre>D:\Computer Science\Component 3\Prototype>java -cp .;JGoodDatePicker-10.3.1.jar Main [INFO] <GUI> Running setup [INFO] <QUESTION_LIST> Running loadDatabase [INFO] <QUESTION_LIST> 1 Questions loaded from file [INFO] <QUESTION_PANEL_LIST> Running loadDatabase [INFO] <QUESTION_PANEL_LIST> 1 QuestionPanels loaded from file</pre> No errors when loading question panels
10	Proto	Array index out of bounds exception when adding a date picker to a question.	<pre>[INFO] <QUESTION_CREATION_PANEL> Running addCalendarEntry Exception in thread "AWT-EventQueue-0" java.lang.ArrayIndexOutOfBoundsException: 100 at components.JValidatedDatePicker.setupDatePicker(JValidatedDatePicker.java:42) at components.JValidatedDatePicker.<init>(JValidatedDatePicker.java:30) at QuestionCreationPanel.actionPerformed(QuestionCreationPanel.java:289) at javax.swing.AbstractButton.fireActionPerformed(Unknown Source) at javax.swing.AbstractButton\$Handler.actionPerformed(Unknown Source) at javax.swing.DefaultButtonModel.fireActionPerformed(Unknown Source) at javax.swing.DefaultButtonModel.setPressed(Unknown Source) at javax.swing.plaf.basic.BasicButtonListener.mouseReleased(Unknown Source) at java.awt.Component.processMouseEvent(Unknown Source) at javax.swing.JComponent.processMouseEvent(Unknown Source) at java.awt.Component.processEvent(Unknown Source) at java.awt.Container.processEvent(Unknown Source) at java.awt.Component.dispatchEventImpl(Unknown Source) at java.awt.Container.dispatchEventImpl(Unknown Source) at java.awt.Component.dispatchEvent(Unknown Source) at java.awt.LightweightDispatcher.retargetMouseEvent(Unknown Source) at java.awt.LightweightDispatcher.processMouseEvent(Unknown Source) at java.awt.LightweightDispatcher.dispatchEvent(Unknown Source) at java.awt.Container.dispatchEventImpl(Unknown Source) at java.awt.Window.dispatchEventImpl(Unknown Source) at java.awt.Component.dispatchEvent(Unknown Source) at java.awt.EventQueue.dispatchEvent(Unknown Source) at java.awt.EventQueue.access\$500(Unknown Source) at java.awt.EventQueue\$3.run(Unknown Source) at java.awt.EventQueue\$3.run(Unknown Source) at java.security.AccessController.doPrivileged(Native Method) at java.security.ProtectionDomain\$JavaSecurityAccessImpl.doIntersectionPrivilege(Unknown Source) at java.security.ProtectionDomain\$JavaSecurityAccessImpl.doIntersectionPrivilege(Unknown Source) at java.awt.EventQueue\$4.run(Unknown Source) at java.awt.EventQueue\$4.run(Unknown Source) at java.security.AccessController.doPrivileged(Native Method) at java.security.ProtectionDomain\$JavaSecurityAccessImpl.doIntersectionPrivilege(Unknown Source) at java.awt.EventQueue.dispatchEvent(Unknown Source) at java.awt.EventDispatchThread.pumpOneEventForFilters(Unknown Source) at java.awt.EventDispatchThread.pumpEventsForFilter(Unknown Source) at java.awt.EventDispatchThread.pumpEventsForHierarchy(Unknown Source) at java.awt.EventDispatchThread.pumpEvents(Unknown Source) at java.awt.EventDispatchThread.pumpEvents(Unknown Source) at java.awt.EventDispatchThread.run(Unknown Source)</pre>	The array to store the years is filled backwards (from the end to the start) however although the for loop starts at 100 and the end condition is when i is less than zero the final statement is i++ rather than i-- therefore as the array is only of size 100 i quickly becomes larger than 100 and the exception occurs. Changing i++ to i-- will fix this. Also the loop should start at 99 rather than 100	 A drop down datepicker can be added to the question

				because if the array is of size 100 it goes up to index 99 not index 100	
11	Proto	A drop down date picker can be added to the question	 The screenshot shows the 'Form Filler' application window. The 'Create questions' tab is active. In the 'Question Preview' section, a date picker is displayed with '1' in the first dropdown, 'January' in the second, and '2018' in the third. Below the preview, the 'Create components' section includes buttons for 'Add label', 'Add text field', 'Add radio buttons', 'Add drop-down', 'Add check boxes', 'Add password fi...', 'Add calendar en...', and 'Add file chooser'. A 'Save Question' button is also present.		
12	Proto	The dropdowns start at day, month, year rather than starting with a date filled in. This means the user has to fill in the question and it won't pass the validation check from the start	 This screenshot shows the 'Form Filler' application window with the 'Create questions' tab active. The 'Question Preview' section displays a date picker where the first dropdown is labeled 'Day', the second is 'Month', and the third is 'Year'. The 'Create components' section and 'Save Question' button are also visible, identical to the previous screenshot.		

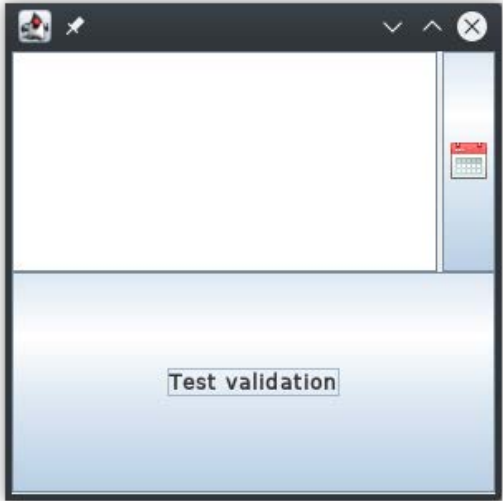
13	Proto	The date picker is successfully saved to file without any errors			
14	Proto	The saved date picker is successfully loaded from file.			
15	Soft Dev	When the toString method is called on a JValidatedDatepicker a string detailing the date selected is output.			

			datepicker:4.6.3		
16	Soft Dev	When the output string from the toString method of a JValidatedDatepicker is used to create a new JValidatedDatepicker, the datepicker is recreated exactly from the string.	 This datepicker was recreated from the output option string from above.		

17	Soft Dev	The date picker with calendar entry is used. When the toString method is called it outputs a string containing the date which is selected.	 No date entered datepicker:-1 -1 shows that no date was selected		
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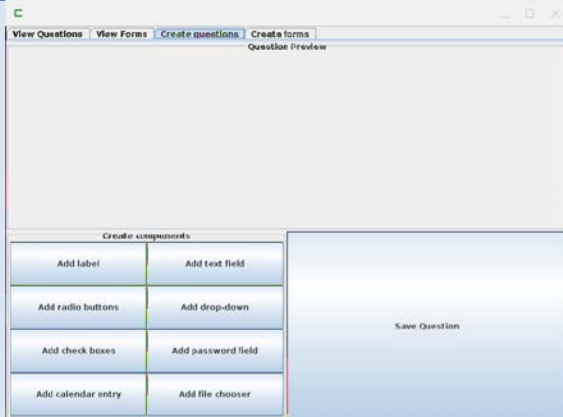
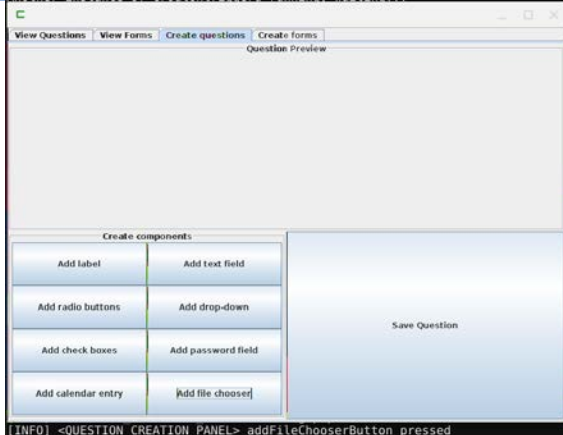


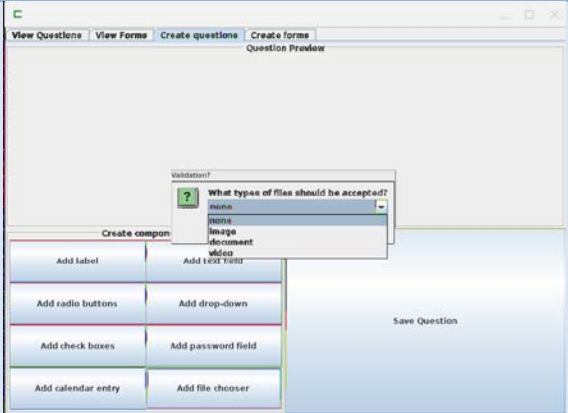
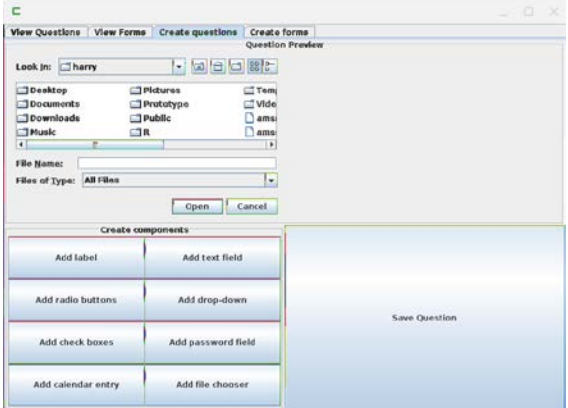
Date entered
datepicker:2018-12-19
The selected date was output

18	Soft Dev	The date picker can be recreated from its output string.	 This date picker was recreated from the string output by the empty datepicker 		
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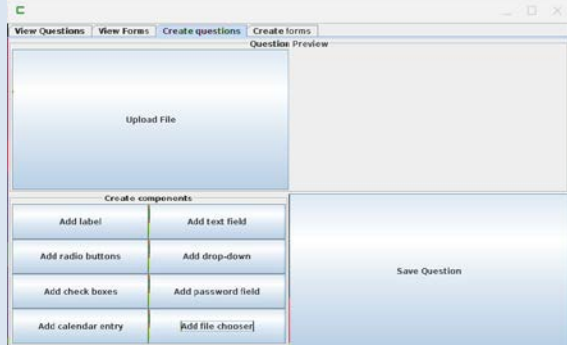
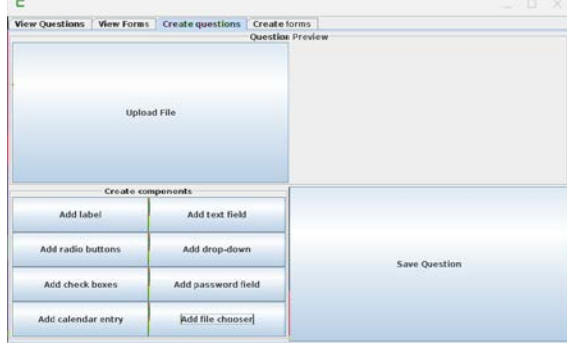
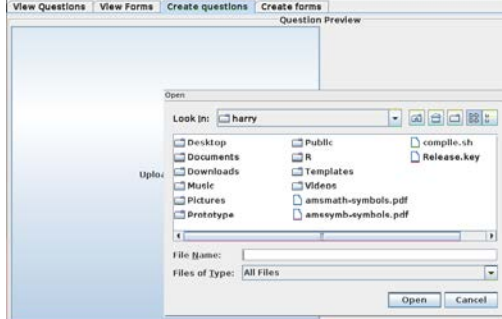
			This date picker was recreated from the string output by the filled in datepicker		
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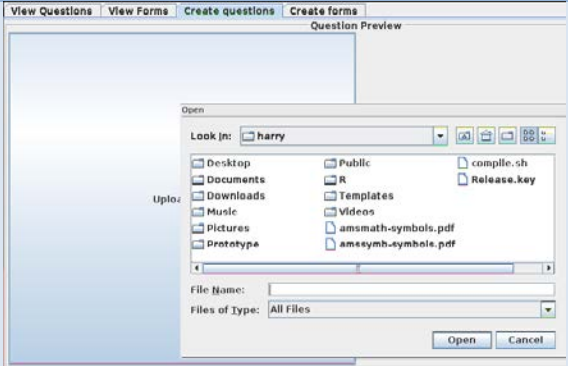
10: Ability to add a document upload field

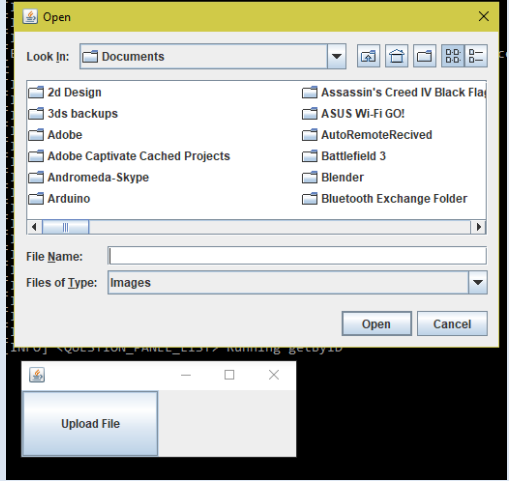
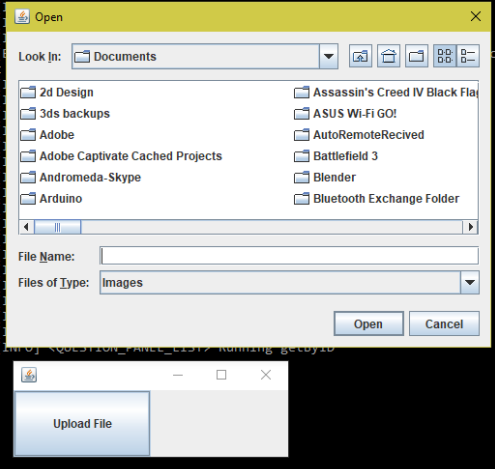
Test	Stage	Result Problem	Evidence	Action and Justification (If Needed)	Evidence of Action (If Needed)
1	Proto	There is a button in the create questions panel to add a file chooser			
2	Proto	Presses to the add file chooser button are detected			

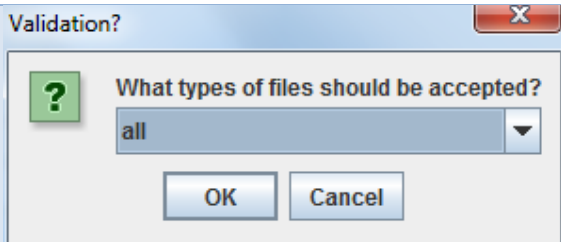
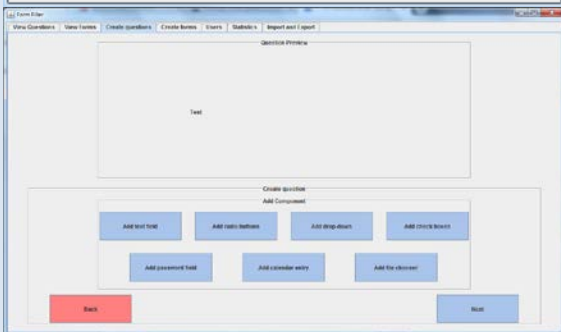
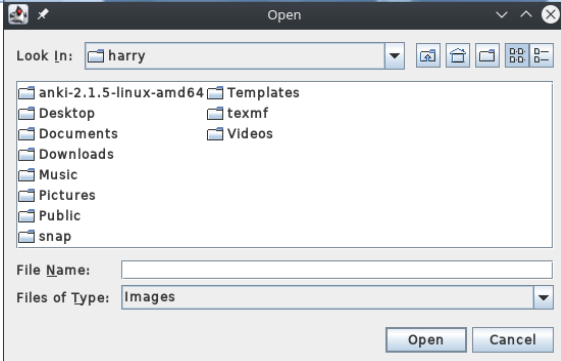
3	Proto	When the add file chooser button is pressed a dialog appears allowing the user to select what types of files should be allowed to be uploaded			
4	Proto	A file chooser is added to the form. However rather than a button to open the dialog the actual file chooser window has been added the question.		Add a JButton that when pressed calls the showOpenDialog method of the file chooser to open the file chooser in a new window.	

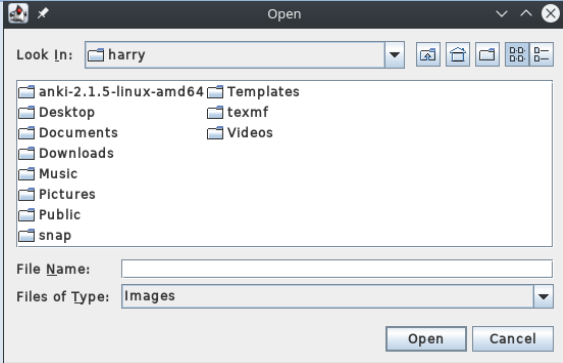
5	Proto	Multiple errors when compiling JValidatedFileChooser	<pre> (GUI)harry@localhost:~/Prototypes\$./compile0.sh ./components/JValidatedFileChooser.java:8: error: cannot find symbol public class JValidatedFileChooser extends JPanel implements JValidatedComponent, ActionListener ^ symbol: class ActionListener ./components/JValidatedFileChooser.java:29: error: cannot find symbol public void actionPerformed(ActionEvent evt) ^ symbol: class ActionEvent location: class JValidatedFileChooser ./components/JValidatedFileChooser.java:33: error: method showOpenDialog in class JFileChooser cannot be applied to given types; fileChooser.showOpenDialog(); ^ required: Component found: no arguments reason: actual and formal argument lists differ in length 3 errors JValidatedFileChooser.java:8: error: cannot find symbol public class JValidatedFileChooser extends JPanel implements JValidatedComponent, ActionListener ^ symbol: class ActionListener JValidatedFileChooser.java:29: error: cannot find symbol public void actionPerformed(ActionEvent evt) ^ symbol: class ActionEvent location: class JValidatedFileChooser JValidatedFileChooser.java:33: error: method showOpenDialog in class JFileChooser cannot be applied to given types; fileChooser.showOpenDialog(); ^ required: Component found: no arguments reason: actual and formal argument lists differ in length 3 errors </pre>	Resolve each error	
6	Proto	Symbol not found error when compiling JValidatedFileChooser.	<pre> (GUI)harry@localhost:~/Prototypes\$./compile0.sh ./components/JValidatedFileChooser.java:8: error: cannot find symbol public class JValidatedFileChooser extends JPanel implements JValidatedComponent, ActionListener ^ symbol: class ActionListener ./components/JValidatedFileChooser.java:29: error: cannot find symbol public void actionPerformed(ActionEvent evt) ^ symbol: class ActionEvent location: class JValidatedFileChooser </pre>	ActionListener and ActionEvent etc. could not be found because java.awt.event.* wasn't imported so java.awt.event.* should be imported	<pre> (GUI)harry@localhost:~/Prototypes\$./compile0.sh ./components/JValidatedFileChooser.java:22: error: cannot find symbol if (evt.getSource() == openButton) ^ symbol: variable getSource location: variable evt of type ActionEvent ./components/JValidatedFileChooser.java:34: error: method showOpenDialog in class JFileChooser cannot be applied to given types; fileChooser.showOpenDialog(); ^ required: Component found: no arguments reason: actual and formal argument lists differ in length 2 errors JValidatedFileChooser.java:32: error: cannot find symbol if (evt.getSource() == openButton) ^ symbol: variable getSource location: variable evt of type ActionEvent JValidatedFileChooser.java:34: error: method showOpenDialog in class JFileChooser cannot be applied to given types; fileChooser.showOpenDialog(); ^ required: Component found: no arguments reason: actual and formal argument lists differ in length 2 errors (GUI)harry@localhost:~/Prototypes\$ </pre>
7	Proto	Symbol not found error for evt.getSource	<pre> (GUI)harry@localhost:~/Prototypes\$./compile0.sh ./components/JValidatedFileChooser.java:32: error: cannot find symbol if (evt.getSource() == openButton) ^ symbol: variable getSource location: variable evt of type ActionEvent </pre>	getSource is a method and so should be called by getSource(). Without the brackets it would be an attribute hence the symbol not found error because ActionEvents don't have a getSource	<pre> (GUI)harry@localhost:~/Prototypes\$./compile0.sh ./components/JValidatedFileChooser.java:34: error: method showOpenDialog in class JFileChooser cannot be applied to given types; fileChooser.showOpenDialog(); ^ required: Component found: no arguments reason: actual and formal argument lists differ in length 1 error JValidatedFileChooser.java:34: error: method showOpenDialog in class JFileChooser cannot be applied to given types; fileChooser.showOpenDialog(); ^ required: Component found: no arguments reason: actual and formal argument lists differ in length 1 error (GUI)harry@localhost:~/Prototypes\$ </pre>

8	Proto	Error compiling because showOpenDialog had no arguments passed to it.	<pre>(GUI)harry@localhost:~/Prototype\$./compile0.sh ./components/JValidatedFileChooser.java:34: error: method showOpenDialog in class JFileChooser cannot be applied to given types; fileChooser.showOpenDialog(); ^ required: Component found: no arguments reason: actual and formal argument lists differ in length 1 error JValidatedFileChooser.java:34: error: method showOpenDialog in class JFileChooser cannot be applied to given types; fileChooser.showOpenDialog(); ^ required: Component found: no arguments reason: actual and formal argument lists differ in length 1 error (GUI)harry@localhost:~/Prototype\$</pre>	attribute. Pass 'this' to it to pass the JValidatedFileChooser JPanel to it.	<pre>(GUI)harry@localhost:~/Prototype\$./compile0.sh (GUI)harry@localhost:~/Prototype\$</pre> Succesful compile.
9	Proto	When the add file chooser button is pressed a button that is labeled upload file is added to the question.			
10	Proto	When the upload file button is pressed nothing happens.		This was because the upload file button didn't have an actionlistener added to it therefore it wasn't actually creating actionEvents when it was pressed.	 A file chooser is opened when the button is pressed

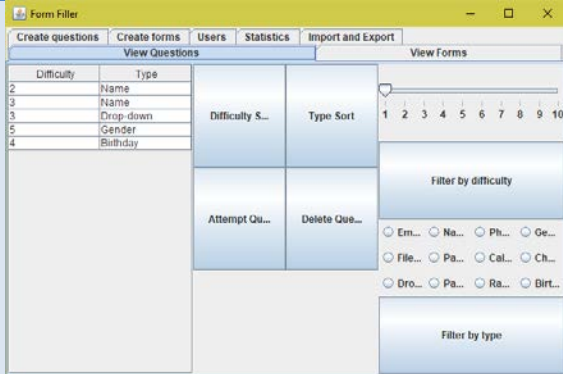
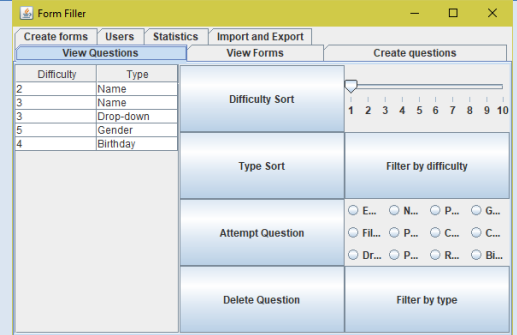
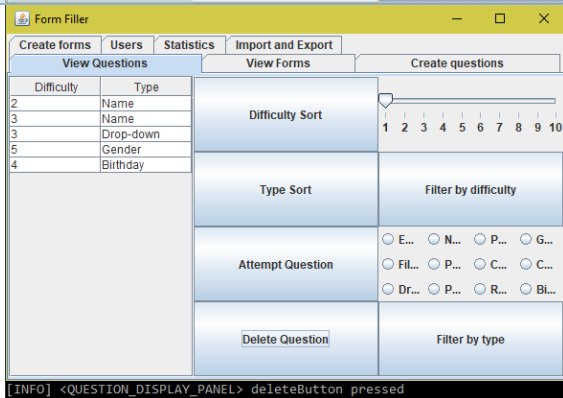
11	Proto	A file chooser is opened when the upload file button is pressed			
12	Proto	Null pointer exception when attempting to attempt a form that has a file chooser in it.		This is because the FileNameExtension Filter is not serialisable so there is an error when loading the questionPanels from file and they aren't loaded. Marking the FileChooser field as transient will stop it being serialised and so will avoid this error. However this means that the FileChooser field will have to be reinitialized when the question is loaded from file.	<pre>[INFO] <QUESTION_PANEL_LIST> Running loadDatabase [INFO] <QUESTION_PANEL_LIST> 1 QuestionPanels loaded from file</pre>

13	Proto	Null pointer exception when pressing the upload file button.	<pre> [INFO] <QUESTION_PANEL_LIST> Running getbyID Exception in thread "AWT-EventQueue-8" java.lang.NullPointerException at components.JValidatedFileChooser.setFileExtensionFilter(JValidatedFileChooser.java:48) at components.JValidatedFileChooser.actionPerformed(JValidatedFileChooser.java:87) at javax.swing.AbstractButton.fireActionPerformed(Unknown Source) at javax.swing.AbstractButton\$Handler.actionPerformed(Unknown Source) at javax.swing.DefaultButtonModel.fireActionPerformed(Unknown Source) at javax.swing.DefaultButtonModel.setPressed(Unknown Source) at javax.swing.plaf.basic.BasicButtonListener.mouseReleased(Unknown Source) at java.awt.Component.processMouseEvent(Unknown Source) at javax.swing.JComponent.processMouseEvent(Unknown Source) at java.awt.Component.processEvent(Unknown Source) at java.awt.Container.processEvent(Unknown Source) at java.awt.Component.dispatchEventImpl(Unknown Source) at java.awt.Container.dispatchEventImpl(Unknown Source) at java.awt.Component.dispatchEvent(Unknown Source) at java.awt.LightweightDispatcher.retargetMouseEvent(Unknown Source) at java.awt.LightweightDispatcher.processMouseEvent(Unknown Source) at java.awt.LightweightDispatcher.dispatchEvent(Unknown Source) at java.awt.Container.dispatchEventImpl(Unknown Source) at java.awt.Window.dispatchEventImpl(Unknown Source) at java.awt.Component.dispatchEvent(Unknown Source) at java.awt.EventQueue.dispatchEventImpl(Unknown Source) at java.awt.EventQueue.access\$500(Unknown Source) at java.awt.EventQueue\$3.run(Unknown Source) at java.awt.EventQueue\$3.run(Unknown Source) at java.security.AccessController.doPrivileged(Native Method) at java.security.ProtectionDomain\$JavaSecurityAccessImpl.doIntersectionPrivilege(Unknown Source) at java.security.ProtectionDomain\$JavaSecurityAccessImpl.doIntersectionPrivilege(Unknown Source) at java.awt.EventQueue\$4.run(Unknown Source) at java.awt.EventQueue\$4.run(Unknown Source) at java.security.AccessController.doPrivileged(Native Method) at java.security.ProtectionDomain\$JavaSecurityAccessImpl.doIntersectionPrivilege(Unknown Source) at java.awt.EventQueue.dispatchEvent(Unknown Source) at java.awt.EventDispatchThread.pumpOneEventForFilters(Unknown Source) at java.awt.EventDispatchThread.pumpEventsForFilter(Unknown Source) at java.awt.EventDispatchThread.pumpEventsForHierarchy(Unknown Source) at java.awt.EventDispatchThread.pumpEvents(Unknown Source) at java.awt.EventDispatchThread.pumpEvents(Unknown Source) at java.awt.EventDispatchThread.run(Unknown Source) </pre>	This is because the FileChooser hadn't been reinitialised. Reinitialising the FileChooser should fix this.	 File chooser opens.
14	Proto	File chooser successfully loaded from file with the correct FileNameExtensionFilter also.			

15	Soft Dev	When cancel is pressed on the validation dialog no file chooser is add to the question.	 		
16	Soft Dev	When the toString method is called on a JValidatedFileChooser a string detailing the types of files allowed is output.	 <pre>filechooser:image</pre>		

17	Soft Dev	When the output string from the toString method of a JValidatedFileChooser is used to create a new JValidatedFileChooser, the filechooser is recreated exactly from the string.	 This filechooser was created from the string output above. As you can see the recreated file chooser only accepts images.		
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11: Ability to delete Questions

Test	Stage	Result Problem	Evidence	Action and Justification (If Needed)	Evidence of Action (If Needed)
1	Proto	There is a delete button in the view questions panel. However the layout of the buttons is wrong.		Make the button panel grid layout 4 rows and 1 column so that all of the buttons are in the same column.	 The buttons are in the correct layout
2	Proto	Presses to the delete button are detected			
3	Proto	When the delete button is pressed the selected question is deleted.			

Form Filler

Create forms Users Statistics Import and Export

View Questions View Forms Create questions

Difficulty	Type
4	Birthday
3	Drop-down
5	Gender
2	Name
3	Name
4	Name

Difficulty Sort 1 2 3 4 5 6 7 8 9 10

Type Sort Filter by difficulty

Attempt Question

Filter by type

Delete Question

```
[INFO] <QUESTION_DISPLAY_PANEL> deleteButton pressed
[INFO] <QUESTION_LIST> Running removeQuestion
[INFO] <QUESTION_LIST> Running writeDatabase
[INFO] <QUESTION_PANEL_LIST> Running writeDatabase
[INFO] <QUESTION_PANEL_LIST> Running removeQuestionPanel
[INFO] <QUESTION_PANEL_LIST> Running writeDatabase
```

Form Filler

Create forms Users Statistics Import and Export

View Questions View Forms Create questions

Difficulty	Type
4	Birthday
3	Drop-down
5	Gender
2	Name
3	Name

Difficulty Sort 1 2 3 4 5 6 7 8 9 10

Type Sort Filter by difficulty

Attempt Question

Filter by type

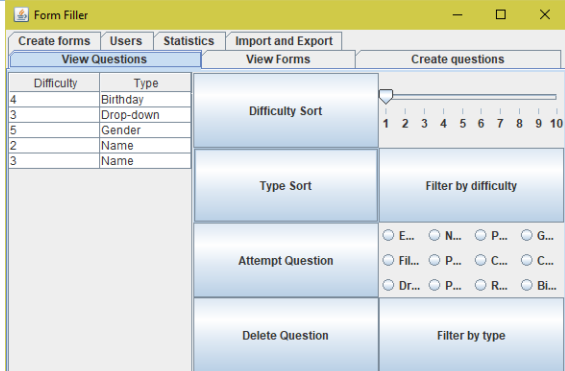
Delete Question

The question of type name with difficulty 4 is no longer present

4

Proto

The question doesn't appear again after restarting the system - the deletion is permanent.



The screenshot shows the 'Form Filler' application window. The 'View Questions' tab is active, displaying a table with the following data:

Difficulty	Type
4	Birthday
3	Drop-down
5	Gender
2	Name
3	Name

Below the table, there are several interactive elements:

- Difficulty Sort:** A dropdown menu showing a difficulty level of 1.
- Type Sort:** A button.
- Attempt Question:** A button.
- Delete Question:** A button.
- Filter by difficulty:** A section with radio buttons for E..., N..., P..., G..., Fil..., P..., C..., C..., Dr..., P..., R..., and Bl...
- Filter by type:** A button.

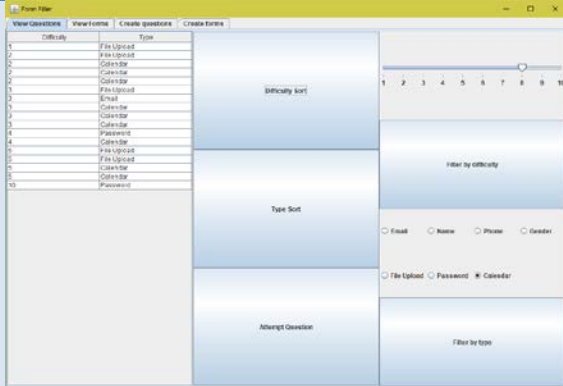
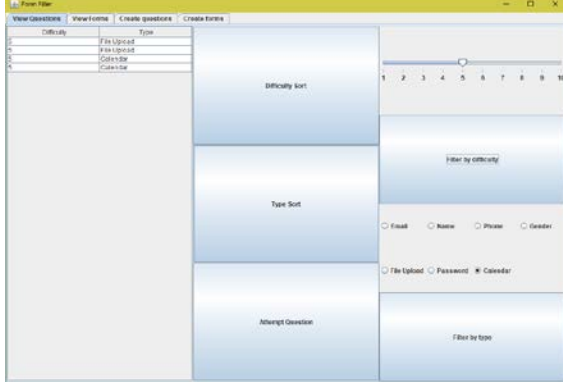
12: Ability to sort questions by difficulty

Test	Stage	Result Problem	Evidence	Action and Justification (If Needed)	Evidence of Action (If Needed)
1	Proto	Questions sorted in descending order rather than ascending order of difficulty	<pre> D:\Computer Science\Component 3\Prototype>java Question Creating a question list [INFO] <QUESTION_LIST> Running loadDatabase [INFO] <QUESTION_LIST> Running printQuestions 3,Address 4,Name 6,Credit Card 2,Email address 4,Password 9,Date 10,Passport numbers 10,Photo Upload [INFO] <QUESTION_LIST> Running sortDifficulty [INFO] <QUESTION_LIST> Running printQuestions 10,Passport numbers 10,Photo Upload 9,Date 6,Credit Card 4,Name 4,Password 3,Address 2,Email address </pre>	Change the < to > in the bubble sort	<pre> D:\Computer Science\Component 3\Prototype>java Question Creating a question list [INFO] <QUESTION_LIST> Running loadDatabase [INFO] <QUESTION_LIST> Running printQuestions 3,Address 4,Name 6,Credit Card 2,Email address 4,Password 9,Date 10,Passport numbers 10,Photo Upload [INFO] <QUESTION_LIST> Running sortDifficulty [INFO] <QUESTION_LIST> Running printQuestions 2,Email address 3,Address 4,Name 4,Password 6,Credit Card 9,Date 10,Passport numbers 10,Photo Upload </pre>
2	Proto	Questions can be sorted in difficulty order	<pre> D:\Computer Science\Component 3\Prototype>java Question Creating a question list [INFO] <QUESTION_LIST> Running loadDatabase [INFO] <QUESTION_LIST> Running printQuestions 3,Address 4,Name 6,Credit Card 2,Email address 4,Password 9,Date 10,Passport numbers 10,Photo Upload [INFO] <QUESTION_LIST> Running sortDifficulty [INFO] <QUESTION_LIST> Running printQuestions 2,Email address 3,Address 4,Name 4,Password 6,Credit Card 9,Date 10,Passport numbers 10,Photo Upload </pre>		

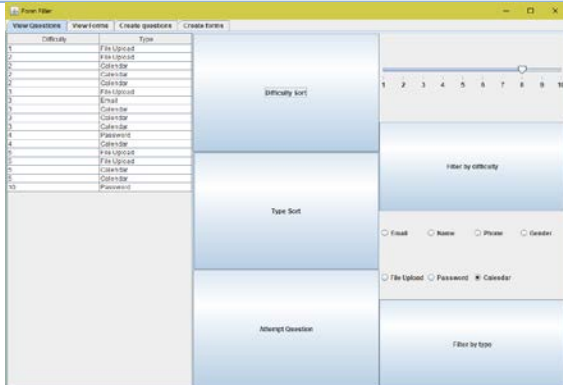
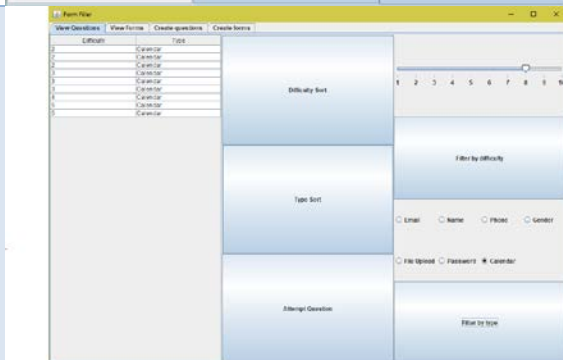
13: Ability to sort questions by type

Test	Stage	Result	Problem	Evidence	Action and Justification (If Needed)	Evidence of Action (If Needed)
1	Proto	Questions can be sorted by type alphabetically		<pre> D:\Computer Science\Component 3\Prototype>java Question Creating a question list [INFO] <QUESTION_LIST> Running loadDatabase [INFO] <QUESTION_LIST> Running printQuestions 3,Address 4,Name 6,Credit Card 2,Email address 4>Password 9,Date 10,Passport numbers 10,Photo Upload 4,Address 3,Email address [INFO] <QUESTION_LIST> Running sortType [INFO] <QUESTION_LIST> Running printQuestions 3,Address 4,Address 6,Credit Card 9,Date 2,Email address 3,Email address 4,Name 4>Password 10,Passport numbers 10,Photo Upload </pre>		

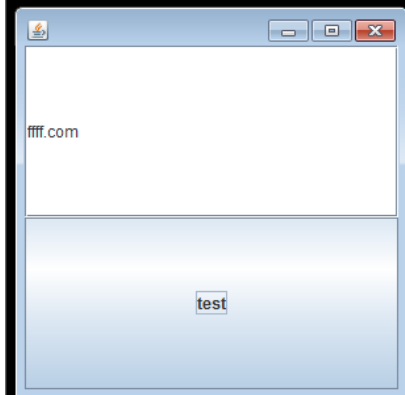
14: Ability to filter questions by difficulty

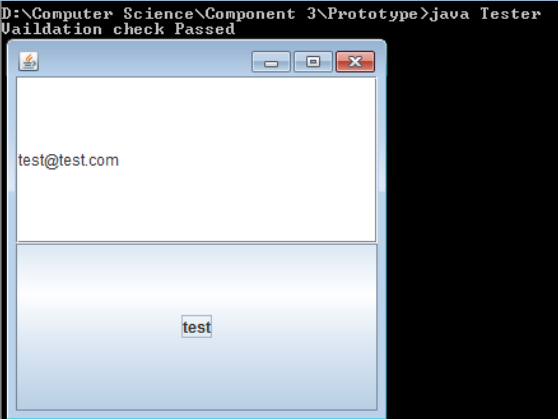
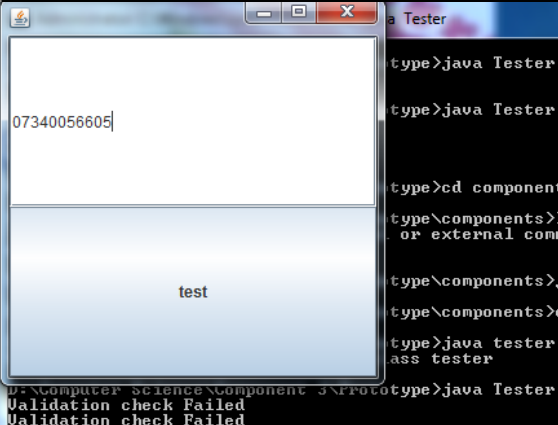
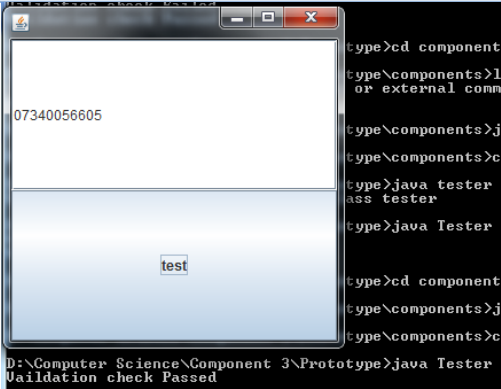
Test	Stage	Result Problem	Evidence	Action and Justification (If Needed)	Evidence of Action (If Needed)
1	Proto	There is a slider that is labelled 1 to 10 and a filter by difficulty button	 The screenshot shows a window titled 'Filter' with a tabbed interface. The 'View Questions' tab is active. On the right, there is a 'Difficulty' slider ranging from 1 to 10, currently set to 5. Below the slider is a button labeled 'Filter by difficulty'. Other buttons include 'Filter by type' and 'Attempt Question'.		
2	Proto	When the slider is set to specific difficulty and the filter by difficulty button is pressed only questions of that difficulty are displayed	 The screenshot shows the same 'Filter' window. The 'Filter by difficulty' button has been pressed, and the list of questions on the left is now filtered to show only questions of the selected difficulty (5). The slider remains at 5.		

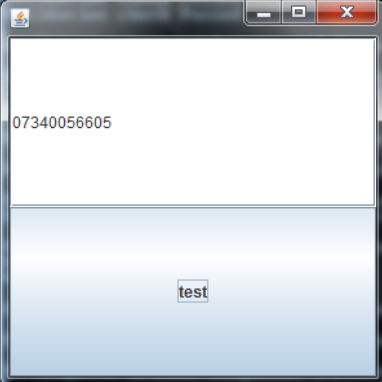
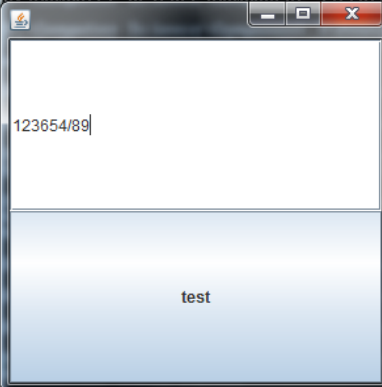
15: Ability to filter Questions by type

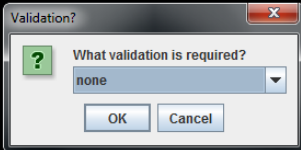
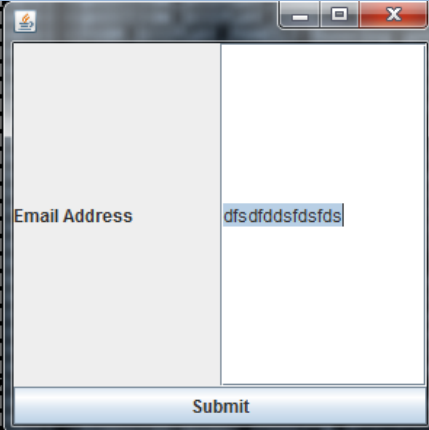
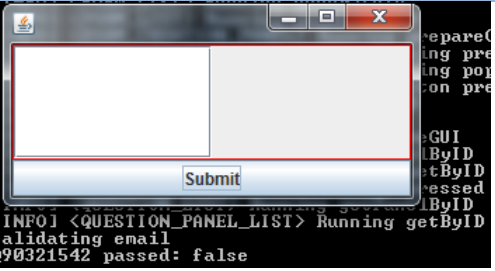
Test	Stage	Result Problem	Evidence	Action and Justification (If Needed)	Evidence of Action (If Needed)
1	Proto	Question display window fails to compile because the symbol JButtonGroup can't be found.	<pre>J:\Computer Science\Component 3\Prototype>compile QuestionDisplayPanel.java:29: error: cannot find symbol private JButtonGroup typeRadioButtonGroup = new JButtonGroup(); ^ symbol: class JButtonGroup location: class QuestionDisplayPanel</pre>	Change JButtonGroup to ButtonGroup	<pre>J:\Computer Science\Component 3\Prototype>compile</pre> Successful compile
2	Proto	There is a radio button for every question type.			
3	Proto	When a radio button is pressed and the type filter button is pressed only questions of that type are displayed.			

16: Email addresses, dates, and phone numbers should have format checks

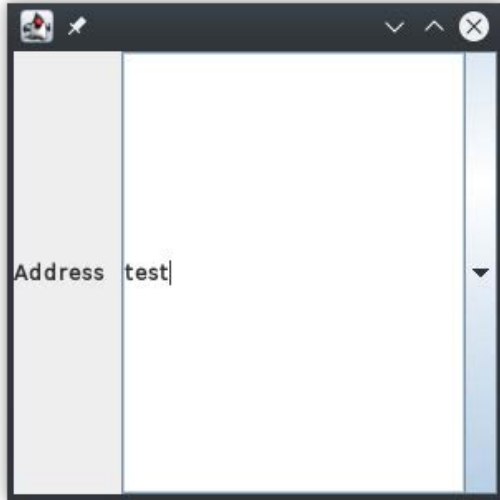
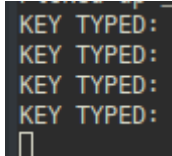
Test	Stage	Result Problem	Evidence	Action and Justification (If Needed)	Evidence of Action (If Needed)
1	Proto	JValidatedTextBox fails to compile because the variable Pattern couldn't be found. This is because the regex class wasn't imported	D:\Computer Science\Component 3\Prototype\components>javac *.java JValidatedTextField.java:13: error: cannot find symbol return Pattern.matches("<?>[a-z0-9!#\$%&'*/+=~`{ }~-]+(?:\\.\\w+ z0-9!#%\$&'*/+=~`{ }~ ^\\.?)~\\.?\$)"; ^ symbol: variable Pattern location: class JValidatedTextField 1 error	Import <code>java.util.regex.*;</code>	D:\Computer Science\Component 3\Prototype\components>javac *.java D:\Computer Science\Component 3\Prototype\components> Successful compile
2	Proto	A non-valid email address fails the check	D:\Computer Science\Component 3\Prototype>java Tester Validation check Failed 		

3	Proto	A valid email address passes the check			
4	Proto	Phone number validation fails even though it should pass. This is because I used a regular expression to validate it that only works for US phone numbers		Replace the regex with one that works for UK phone numbers	 Phone number passes now

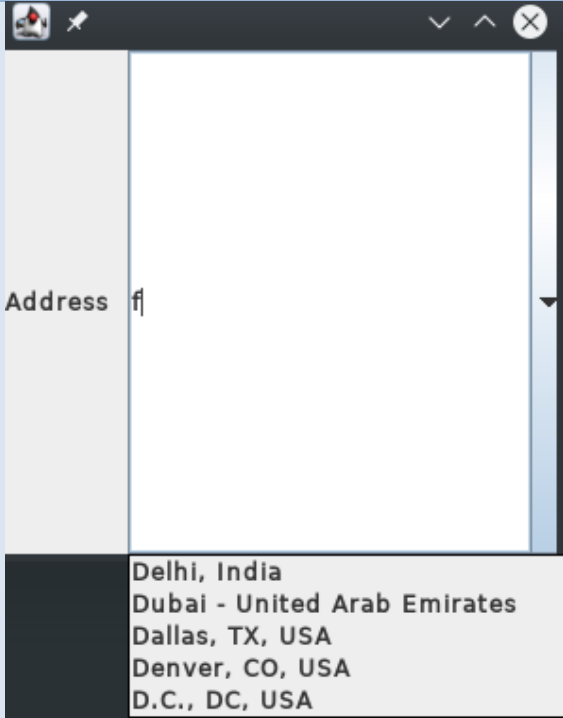
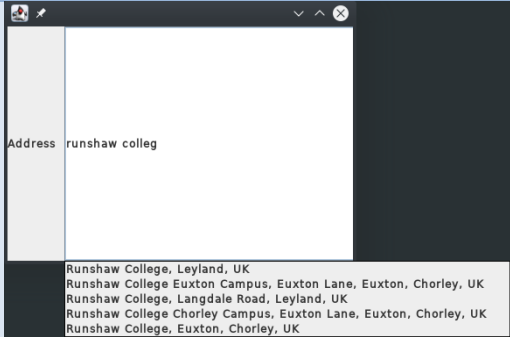
5	Proto	A valid phone number passes the check	 <pre> type>cd component type\components>ls or external comm type\components>j type\components>c type>java tester ass tester type>java Tester type>cd component type\components>j type\components>c D:\Computer Science\Component 3\Prototype>java Tester Validation check Passed </pre>		
6	Proto	An invalid phone number fails the check	 <pre> type>cd component type\components>ls or external com type\components> type\components> type>java tester ass tester type>java Tester type>cd component type\components> type\components> type>java Tester D:\Computer Science\Component 3\Prototype>java Tester Validation check Failed </pre>		

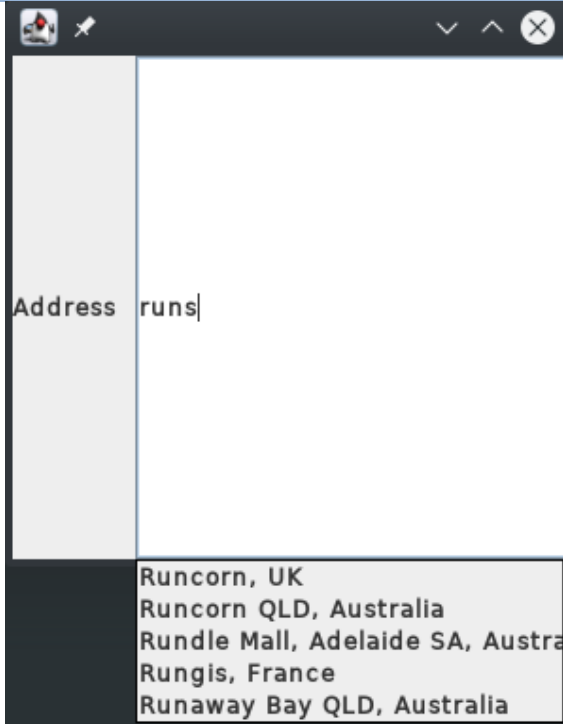
7	Proto	Prompt asking user to specify validation for a text field appears after they add a text field.	<pre> D:\Computer Science\Component 3\Prototype>java Main [INFO] <GUI> Running setup [INFO] <QUESTION_LIST> Running loadDatabase [INFO] <QUESTION_LIST> 2 Questions loaded from file [INFO] <QUESTION_PANEL_LIST> Running loadDatabase [INFO] <QUESTION_PANEL_LIST> 2 QuestionPanels loaded from file [INFO] <FORM_LIST> Running loadDatabase [INFO] <FORM_LIST> 1 Forms loaded from file [INFO] <QUESTION_DISPLAY_PANEL> Running prepareGUI [INFO] <QUESTION_DISPLAY_PANEL> Running populateTable [INFO] <FORM_DISPLAY_PANEL> Running prepareGUI [INFO] <FORM_DISPLAY_PANEL> Running populateTable [INFO] <QUESTION_LIST> Running generateId [INFO] <QUESTION_LIST> Running getQuestionById [INFO] <QUESTION_CREATION_PANEL> Running prepareGUI [INFO] <FORM_LIST> Running getFreeId [INFO] <FORM_LIST> Running generateId [INFO] <FORM_LIST> Running getFormById [INFO] <FORM_CREATION_PANEL> Running prepareGUI [INFO] <QUESTION_SELECTION_PANEL> Running prepareGUI [INFO] <QUESTION_SELECTION_PANEL> Running populateTable [INFO] <QUESTION_CREATION_PANEL> addTextFieldButton pressed [INFO] <QUESTION_CREATION_PANEL> Running addTextField </pre> 		
8	Proto	When added to a form the email address passes the check even though it should fail.	 <pre> [INFO] <QUESTION_PANEL_LIST> Running getFormById Q71496887 passed: true </pre>	Recompile all files	 Validation occurs now <pre> [INFO] <QUESTION_PANEL_LIST> Running getFormById Validating email Q90321542 passed: false </pre>

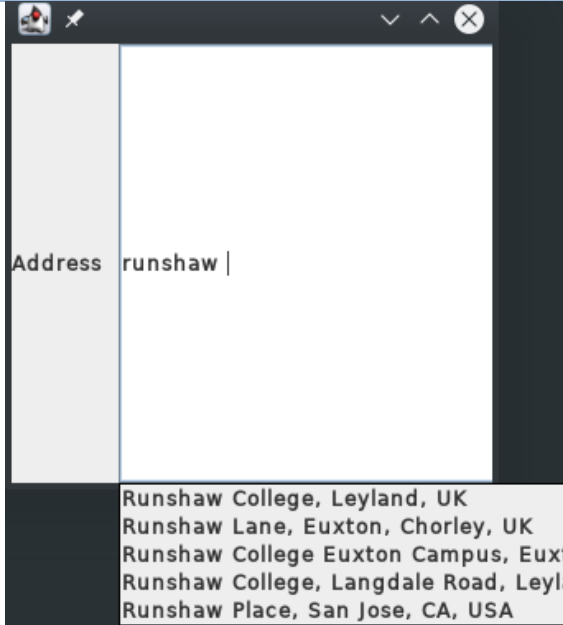
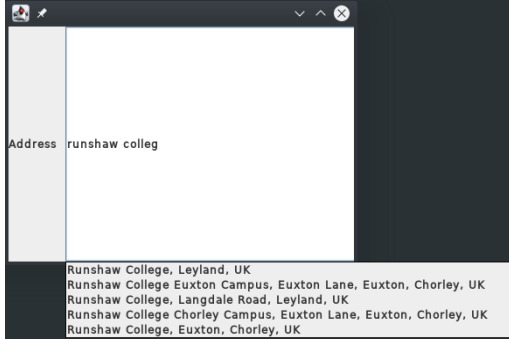
17: Addresses should be looked up via the google maps API to autofill information

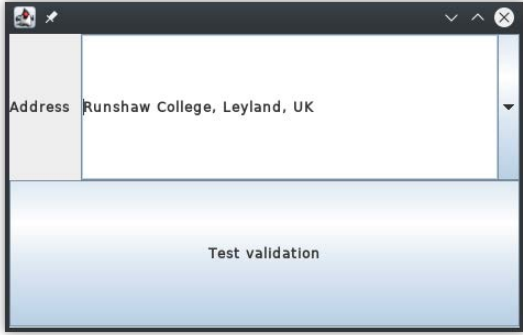
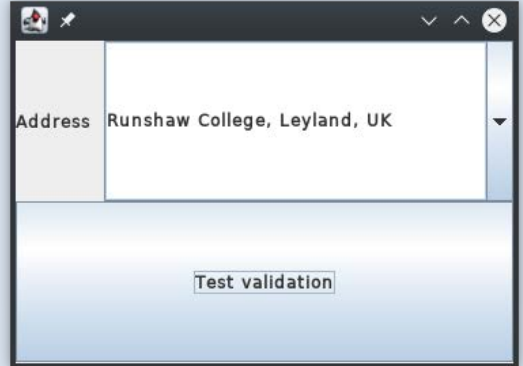
Test	Stage	Result Problem	Evidence	Action and Justification (If Needed)	Evidence of Action (If Needed)
1	Soft Dev	There is a basic address entry question component. The component has a dropdown that can be typed into an a label which asks for the address.			
2	Soft Dev	Whenever a key is typed, the key typed event is called / handled.	 		

3	Soft Dev	Class not found error when running test address entry. This is because the google maps library has various dependencies that were not included in the jar.	<pre> - ./runTest.sh Picked up _JAVA_OPTIONS: -Dawt.useSystemAAFontSettings=on -Dswing.aatext=true Exception in thread "main" java.lang.NoClassDefFoundError: okhttp3/Authenticator at com.google.maps.GeoApiContext\$Builder.<init>(GeoApiContext.java:318) at components.JValidatedLocationEntry.setup(JValidatedLocationEntry.java:26) at components.JValidatedLocationEntry.<init>(JValidatedLocationEntry.java:20) at Tester.testLocationEntry(Tester.java:45) at Tester.main(Tester.java:40) Caused by: java.lang.ClassNotFoundException: okhttp3.Authenticator at java.net.URLClassLoader.findClass(URLClassLoader.java:382) at java.lang.ClassLoader.loadClass(ClassLoader.java:424) at sun.misc.Launcher\$AppClassLoader.loadClass(Launcher.java:349) at java.lang.ClassLoader.loadClass(ClassLoader.java:357) ... 5 more c </pre>	To fix this the dependencies need to be installed. Because I have the source code of the library I will build a new jar with all of the dependencies included.	<pre> - ./runTest.sh Picked up _JAVA_OPTIONS: -Dawt.useSystemAAFontSettings=on -Dswing.aatext=true SLF4J: Failed to load class "org.slf4j.impl.StaticLoggerBinder". SLF4J: Defaulting to no-operation (NOP) logger implementation SLF4J: See http://www.slf4j.org/codes.html#StaticLoggerBinder for further details. </pre> The error no longer occurs
4	Soft Dev	A test api call works. In this test the string "Runshaw PR25 3DQ" was used.	<pre> [INFO] <JVALIDATED_LOCATION_ENTRY> Getting results [INFO] <JVALIDATED_LOCATION_ENTRY> Recieved results Runshaw College, Langdale Road, Leyland PR25 3DQ, UK </pre> The place "runshaw college" is returned.		

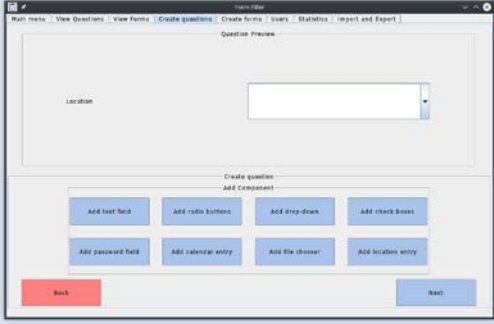
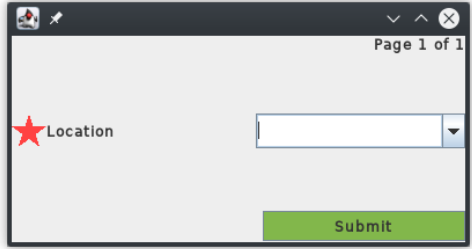
5	Soft Dev	It's only possible to type one letter in at a time into the combobox and any further letters entered just overwrite the original letter.		To fix this the text of the combobox should be set to be what it was before making the google maps query.	 It's possible to type longer words
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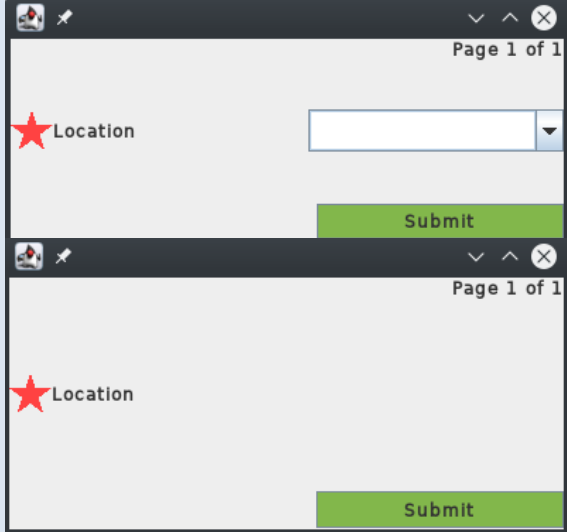
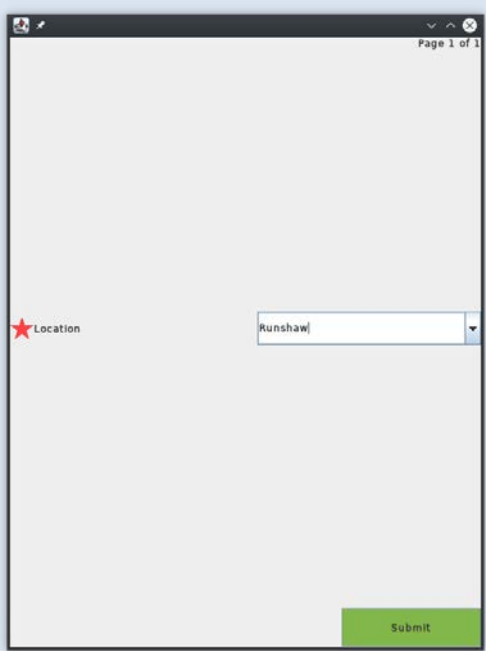
6	Soft Dev	As you type into the address box the results list automatically appears and is populated with results from the google maps autocomplete api.			
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			  In this example runshaw college was used <pre>locationentry:Runshaw College, Leyland, UK</pre>		
7	Soft Dev	When the toString method is called a string containing the address that was entered is output.			

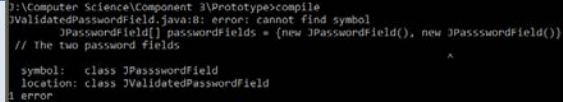
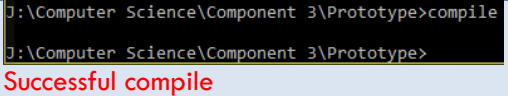
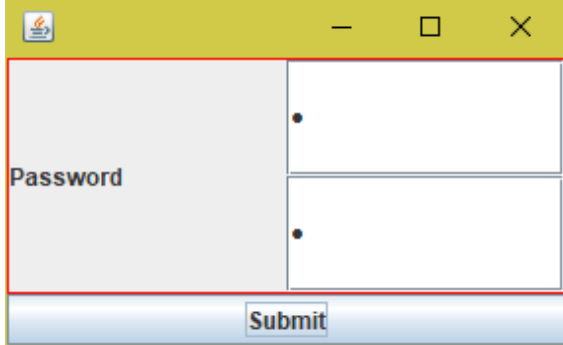
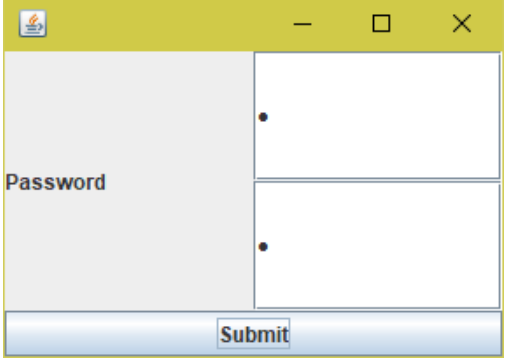
8	Soft Dev	The locationEntry can be recreated from the output string	 This location entry was recreated from the string above		
9	Soft Dev	When an address (or any text this is just a presence check) is entered the validation check passes	 <pre>locationentry:Runshaw College, Leyland, UK true</pre> true is output showing that the check passed		

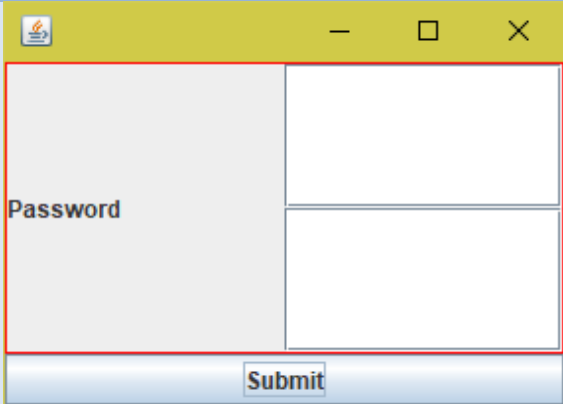
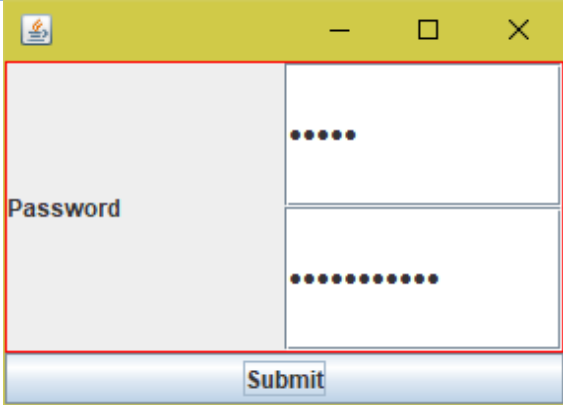
10	Soft Dev	When no address is entered the validation check fails	 <pre>locationentry: false</pre> false is output showing that the check failed		
11	Soft Dev	There is an add location entry button on the create questions screen.			
12	Soft Dev	Presses to the add location entry button are detected	<pre>[INFO] <QUESTION_CREATION_PANEL> addLocationEntryButton pressed</pre>		

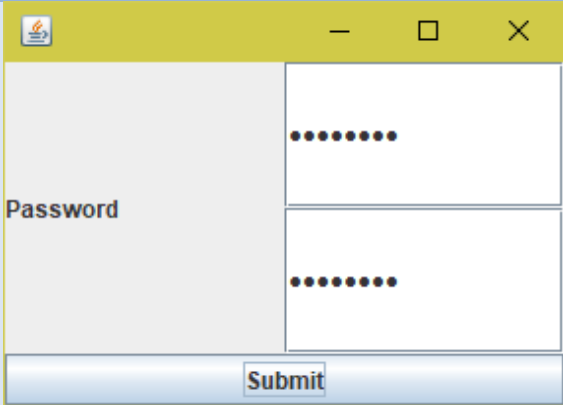
13	Soft Dev	A location entry is successfully added to the question however it's too small		This is because the combobox isn't in a gridlayout this means that it won't automatically fill all available space. Adding the combo box to a gridlayout should fix this.	
14	Soft Dev	ArrayIndexOutOfBoundsException when loading a address entry that has no text added.	<pre>[INFO] <QUESTION_PANEL_LIST> Running getByID Exception in thread "AWT-EventQueue-1" java.lang.ArrayIndexOutOfBoundsException: 1 at components.JValidatedLocationEntry.<init>(JValidatedLocationEntry.java:39) at QuestionPanel.importComponent(QuestionPanel.java:81) at QuestionPanel.<init>(QuestionPanel.java:42) at QuestionPanel.clone(QuestionPanel.java:104) at QuestionPanelList.getByID(QuestionPanelList.java:146) at QuestionList.getPanelByID(QuestionList.java:86) at GUI.buildFormComponentArray(GUI.java:185) at GUI.openForm(GUI.java:159) at QuestionDisplayPanel.openQuestionInWindow(QuestionDisplayPanel.java:398) at QuestionDisplayPanel.actionPerformed(QuestionDisplayPanel.java:464) at javax.swing.AbstractButton.fireActionPerformed(AbstractButton.java:2022) at javax.swing.AbstractButton\$Handler.actionPerformed(AbstractButton.java:2348) at javax.swing.DefaultButtonModel.fireActionPerformed(DefaultButtonModel.java:402) at javax.swing.DefaultButtonModel.setPressed(DefaultButtonModel.java:259) at javax.swing.plaf.basic.BasicButtonListener.mouseReleased(BasicButtonListener.java:259)</pre>	This is because when the location entry is loaded the text at index 1 of the saveString is set to be the text of the combobox. However if the entry was empty when it was saved, there is no text at that index hence the exception. Adding a length check to ensure that there is data at that index when loading will fix this.	 The location entry is successfully loaded from file.

15	Soft Dev	When the location entry is attempted the entry box moves further right every character entered and end up off screen.		This is because the window is too small to display the drop down list. Making the window wider should fix this.	 The combobox is still visible.
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18: Passwords should have double entry verification

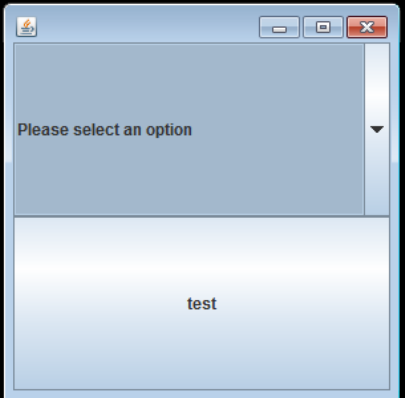
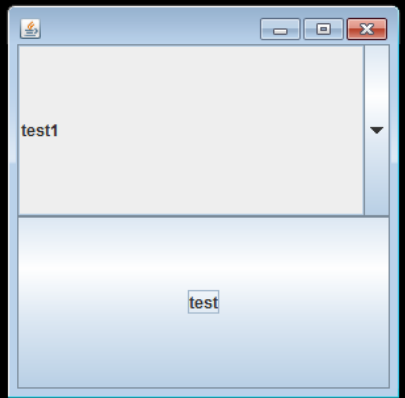
Test	Stage	Result	Problem	Evidence	Action and Justification (If Needed)	Evidence of Action (If Needed)
1	Proto	JValidatedPassword field fails to compile due to a symbol error.			The symbol error was due to mistyping JPasswordField as JPassswordField. Fixing this typo should resolve this error.	
2	Proto	The validation check fails even if both fields are the same.			This is because the validation check if passwordField1.equals(passwordField2) which checks to see if the password fields are the same object. It really should be passwordField1.getPassword().equals(passwordField2.getPassword()) which will check if the passwords entered in them are the same	 The check passes if both fields are the same

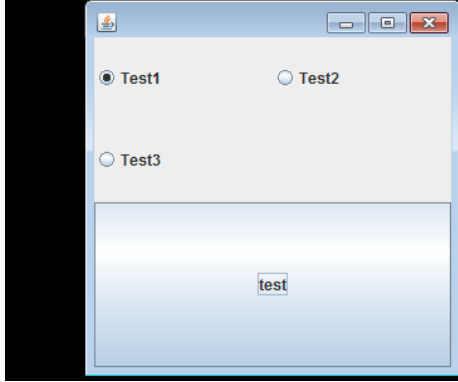
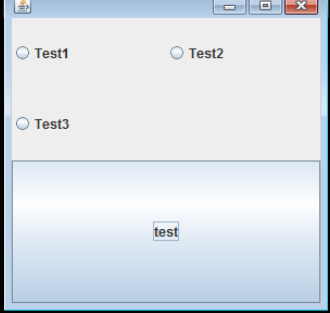
3	Proto	If both password fields are empty the validation check fails			
4	Proto	If the passwords in both fields is different the validation check fails			

5	Proto	If the passwords in both fields are the same the validation check passes			
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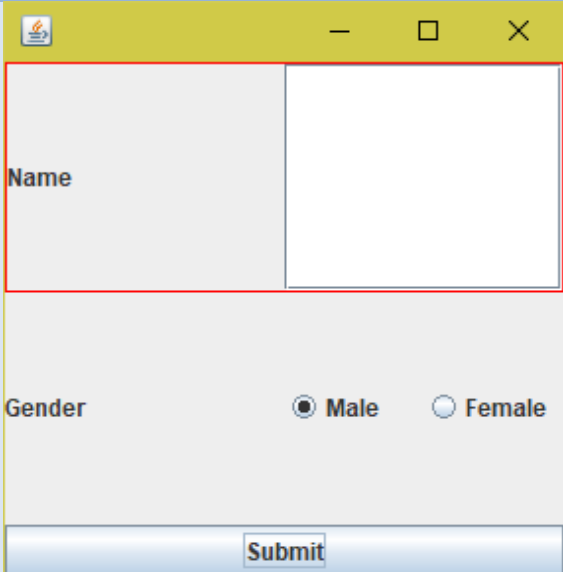
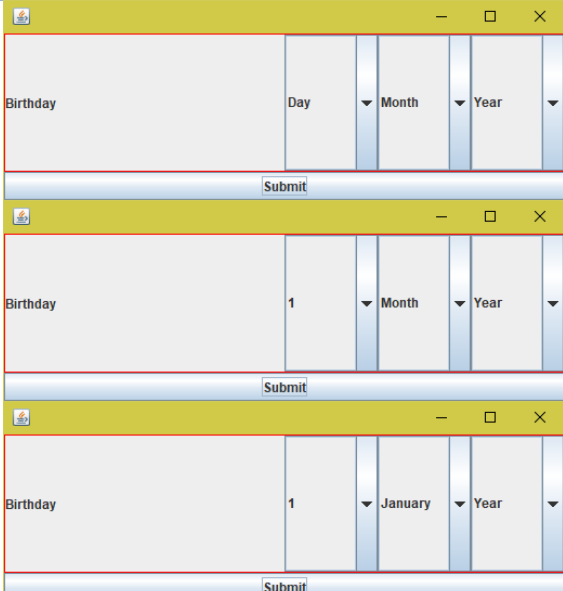
19: All fields should have presence check

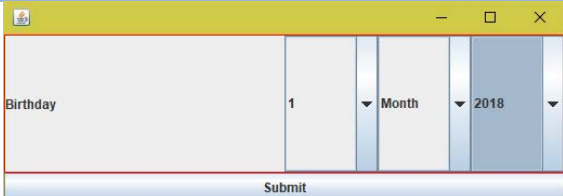
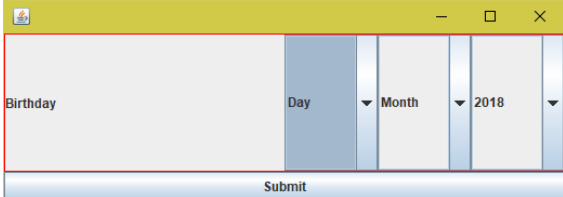
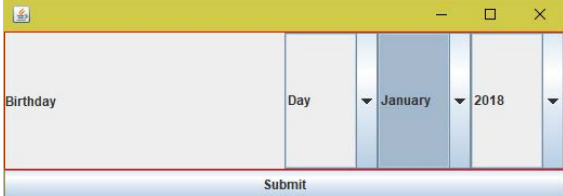
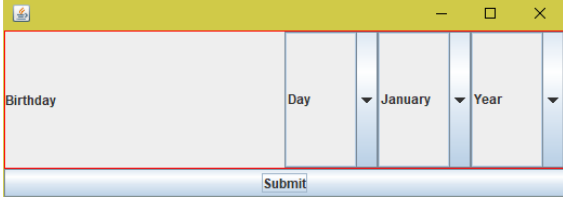
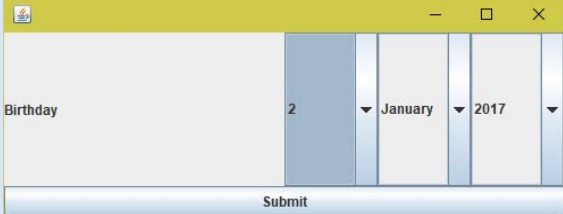
Test	Stage	Result Problem	Evidence	Action and Justification (If Needed)	Evidence of Action (If Needed)
1	Proto	JVaildatedComboBox is unable to be used because I didn't implement the constructor therefore it's not able to be used.	D:\Computer Science\Component 3\Prototype>javac *.java test1.java:14: error: constructor JValidatedComboBox in class JValidatedComboBox cannot be applied to given types; box = new JValidatedComboBox(new String[] {"Please select an option", "test1", "test2"}); ^ required: no arguments found: String[] reason: actual and formal argument lists differ in length	Add a constructor that takes a string array and passes it to the JComboBox constructor	D:\Computer Science\Component 3\Prototype>javac *.java D:\Computer Science\Component 3\Prototype> Successful compile
2	Proto	Warning message when compiling JValidatedComboBox because of extending JComboBox rather than JComboBox<String>	D:\Computer Science\Component 3\Prototype>javac -Xlint:unchecked components/*.java components\JValidatedComboBox.java:10: warning: [unchecked] unchecked call to JComboBox(E) as a member of the raw type JComboBox super((String[]) options); ^ where E is a type-variable: E extends Object declared in class JComboBox 1 warning	Extend JComboBox<String> rather than just JComboBox	D:\Computer Science\Component 3\Prototype>javac -Xlint:unchecked components/*.java D:\Computer Science\Component 3\Prototype> Successful compile with no warnings
3	Proto	Error Compiling JValidatedComboBox because getItem() actually returns the selected item rather than the index of the selected item	D:\Computer Science\Component 3\Prototype>javac components/*.java components\JValidatedComboBox.java:15: error: incompatible types: Object and int if (getItem() == 0) // If an item other than the choose ^ an option has not been selected 1 error	Use getItemIndex() rather than getItem()	D:\Computer Science\Component 3\Prototype>javac components/*.java D:\Computer Science\Component 3\Prototype> Successful compile

4	Proto	Presence check fails when no option is selected	<pre>D:\Computer Science\Component 3\Prototype>java Tester Validation check failed</pre> 		
5	Proto	Presence check passes when an option is selected	<pre>D:\Computer Science\Component 3\Prototype>java Tester Validation check passed</pre> 		
6	Proto	Error compiling QuestionCreationPanel after changing the JComboBox to JValidatedComboBox because I didn't remove <String>.	<pre>D:\Computer Science\Component 3\Prototype>java Tester Validation check passed D:\Computer Science\Component 3\Prototype>javac *.java QuestionCreationPanel.java:213: error: type JValidatedComboBox does not take parameters addComponent(new JValidatedComboBox<String>(adjustedOptions)); ^ 1 error</pre>	Remove <String>	<pre>D:\Computer Science\Component 3\Prototype>javac *.java D:\Computer Science\Component 3\Prototype> Successful compile</pre>
7		Error compiling RadioButtonPanel after	<pre>D:\Computer Science\Component 3\Prototype>javac components/*.java components\RadioButtonPanel.java:4: error: cannot find symbol public class RadioButtonPanel extends JPanel implements JValidatedComponent // R ^ better class for managing check boxes symbol: class JValidatedComponent 1 error</pre>	Add package components to the	<pre>D:\Computer Science\Component 3\Prototype>javac components/*.java components\RadioButtonPanel.java:4: error: cannot find symbol public class RadioButtonPanel extends JPanel implements JValidatedComponent // R ^ better class for managing check boxes symbol: class JValidatedComponent 1 error</pre>

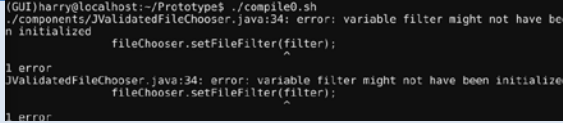
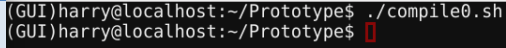
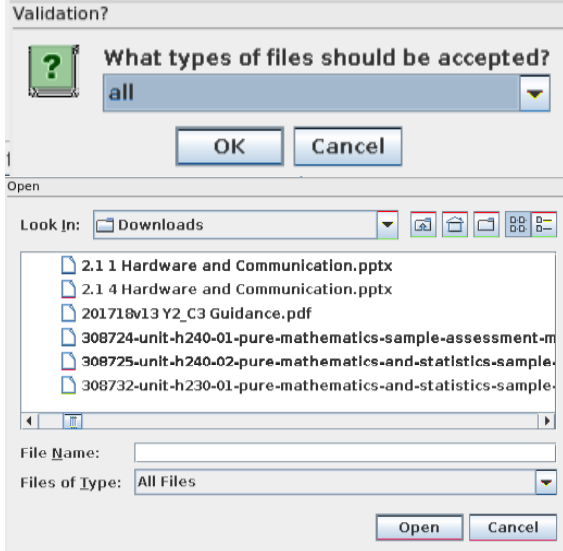
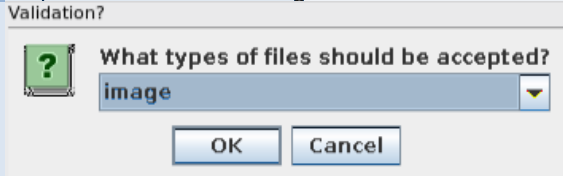
	Proto	moving to the components directory and using trying to use JValidatedComponent as package components wasn't included at the top of the source file		top of the source file	Different error
8	Proto	Error when compiling RadioButtonPanel because of a spelling error	<pre> D:\Computer Science\Component 3\Prototype>javac components/*.java components\RadioButtonPanel.java:41: error: cannot find symbol public class RadioButtonPanel extends JPanel implements JValidatedComponent // R better class for managing check boxes ^ symbol: class JValidatedComponent 1 error </pre>	Correct vaildatedAnswer to validatedAnswer	<pre> D:\Computer Science\Component 3\Prototype>javac components/*.java components\RadioButtonPanel.java:41: error: missing return statement 2 1 error </pre> Different error
9	Proto	Error compiling RadioButtonPanel because of having no return statement in the validateAnswer method	<pre> D:\Computer Science\Component 3\Prototype>javac components/*.java components\RadioButtonPanel.java:41: error: missing return statement 2 1 error </pre>	Add a return statement	<pre> D:\Computer Science\Component 3\Prototype>javac components/*.java D:\Computer Science\Component 3\Prototype> </pre> Successful Compile
10	Proto	Validation check fails when a button is selected and passes when no button is selected. This is because it returns true (passed) if getSelection() == null therefore it returns true when no button is pressed and so wrongly passes the test.	<pre> D:\Computer Science\Component 3\Prototype>java Tester [INFO] <RADIOBUTTON_PANEL_BUILDER> Running add [INFO] <RADIOBUTTON_PANEL_BUILDER> Running add [INFO] <RADIOBUTTON_PANEL_BUILDER> Running add [INFO] <RADIOBUTTON_PANEL_BUILDER> Running add [INFO] <RADIOBUTTON_PANEL_BUILDER> Running build [INFO] <RADIOBUTTON_PANEL_BUILDER> Running trimArray [INFO] <RADIOBUTTON_PANEL_BUILDER> Running preparePanel Validation check Failed </pre> 	Change the condition for it passing from it passing from getSelection() == null to getSelection() != null	<pre> D:\Computer Science\Component 3\Prototype>java Tester [INFO] <RADIOBUTTON_PANEL_BUILDER> Running add [INFO] <RADIOBUTTON_PANEL_BUILDER> Running add [INFO] <RADIOBUTTON_PANEL_BUILDER> Running add [INFO] <RADIOBUTTON_PANEL_BUILDER> Running add [INFO] <RADIOBUTTON_PANEL_BUILDER> Running build [INFO] <RADIOBUTTON_PANEL_BUILDER> Running trimArray [INFO] <RADIOBUTTON_PANEL_BUILDER> Running preparePanel Validation check Failed </pre>  Check fails when no button is pressed. This means that the checks and fails are the correct way around now.
11		QuestionPanel fails to compile because	<pre> D:\Computer Science\Component 3\Prototype>javac *.java QuestionPanel.java:49: error: cannot find symbol validateAnswer() ^ symbol: method validateAnswer() location: variable c of type JComponent 1 error </pre>	Cast the component to a	Successful compile

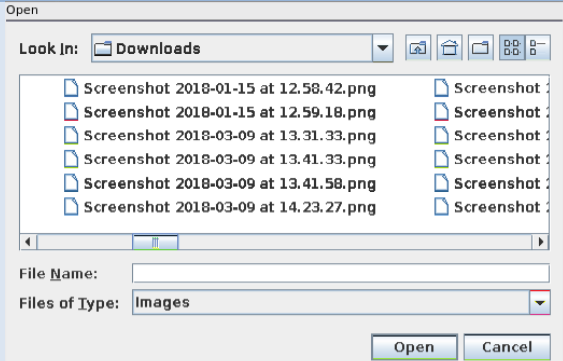
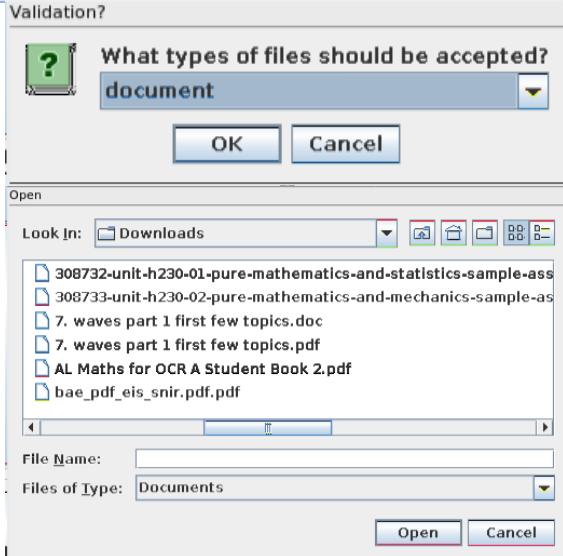
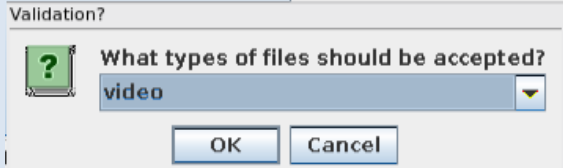
	Proto	although the code checked that the JComponent was of type JValidatedComponent it didn't actually cast it to JValidated component so the validateAnswer() method wasn't available.		JValidatedComponent	
12	Proto	Error loading QuestionPanels from file as I changed the QuestionPanel class and added a new method therefore the stored panels are invalid	<pre> D:\Computer Science\Component 3\Prototype>java Main [INFO] <GUI> Running setup [INFO] <QUESTION_LIST> Running loadDatabase [INFO] <QUESTION_LIST> 4 Questions loaded from file [INFO] <QUESTION_PANEL_LIST> Running loadDatabase [INFO] <QUESTION_PANEL_LIST> Error loading database java.io.InvalidClassException: on: QuestionPanel; local class incompatible: stream classdesc serialVersionUID = 3322550085877381098, local class serialVersionUID = 7525872287083739345 [INFO] <QUESTION_PANEL_LIST> 0 QuestionPanels loaded from file [INFO] <FORM_LIST> Running loadDatabase [INFO] <FORM_LIST> 4 Forms loaded from file [INFO] <QUESTION_DISPLAY_PANEL> Running prepareGUI [INFO] <QUESTION_DISPLAY_PANEL> Running populateTable [INFO] <FORM_DISPLAY_PANEL> Running prepareGUI [INFO] <FORM_DISPLAY_PANEL> Running populateTable [INFO] <QUESTION_LIST> Running generateId [INFO] <QUESTION_LIST> Running getQuestionById [INFO] <QUESTION_CREATION_PANEL> Running prepareGUI [INFO] <FORM_LIST> Running getFreeID [INFO] <FORM_LIST> Running generateId [INFO] <FORM_LIST> Running getFormById [INFO] <FORM_CREATION_PANEL> Running prepareGUI [INFO] <QUESTION_SELECTION_PANEL> Running prepareGUI [INFO] <QUESTION_SELECTION_PANEL> Running populateTable </pre>	Delete all the stored questions and remake them	<pre> D:\Computer Science\Component 3\Prototype>java Main [INFO] <GUI> Running setup [INFO] <QUESTION_LIST> Running loadDatabase [INFO] <QUESTION_LIST> 2 Questions loaded from file [INFO] <QUESTION_PANEL_LIST> Running loadDatabase [INFO] <QUESTION_PANEL_LIST> 2 QuestionPanels loaded from file [INFO] <FORM_LIST> Running loadDatabase [INFO] <FORM_LIST> 1 Forms loaded from file [INFO] <QUESTION_DISPLAY_PANEL> Running prepareGUI [INFO] <QUESTION_DISPLAY_PANEL> Running populateTable [INFO] <FORM_DISPLAY_PANEL> Running prepareGUI [INFO] <FORM_DISPLAY_PANEL> Running populateTable [INFO] <QUESTION_LIST> Running generateId [INFO] <QUESTION_LIST> Running getQuestionById [INFO] <QUESTION_CREATION_PANEL> Running prepareGUI [INFO] <FORM_LIST> Running getFreeID [INFO] <FORM_LIST> Running generateId [INFO] <FORM_LIST> Running getFormById [INFO] <FORM_CREATION_PANEL> Running prepareGUI [INFO] <QUESTION_SELECTION_PANEL> Running prepareGUI [INFO] <QUESTION_SELECTION_PANEL> Running populateTable </pre> The new questions that I have created have been successfully loaded from file

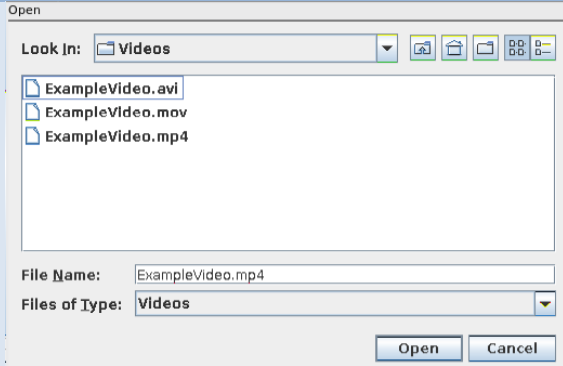
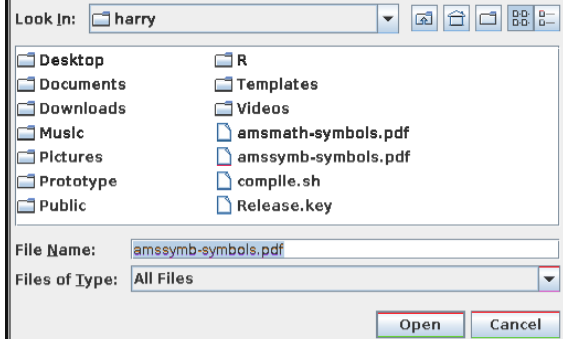
13	Proto	An empty text field fails the validation check.			
14	Proto	The datepicker fails the validation check if any or all of the fields have not been filled in.			

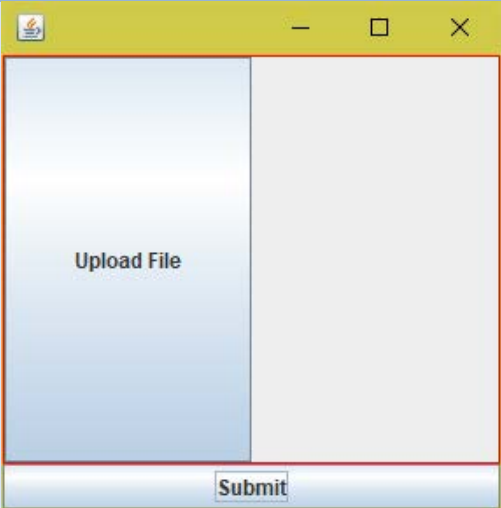
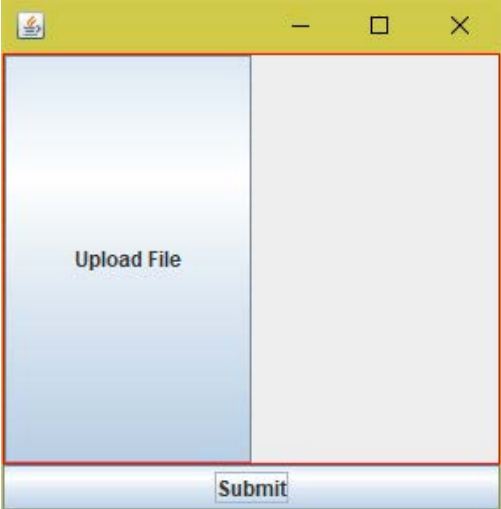
			   		
15	Proto	The datepicker passes the validation check if all of the fields have been filled in.			

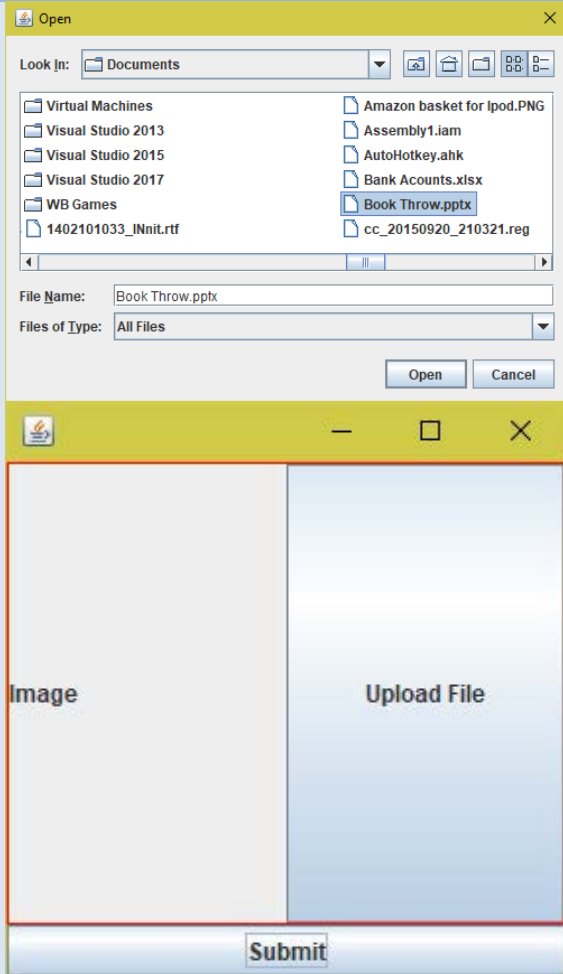
20: Document upload fields should only allow certain types of files to be uploaded

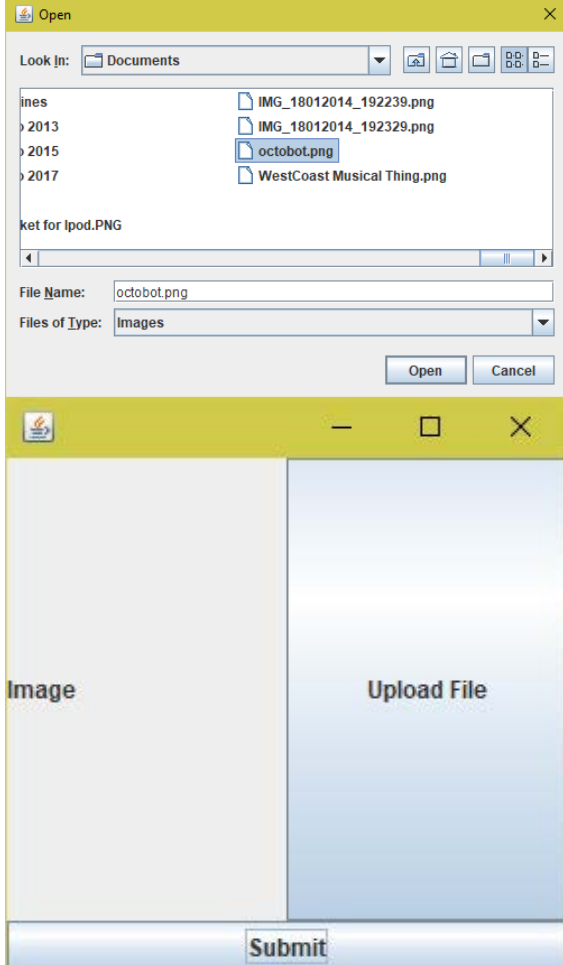
Test	Stage	Result Problem	Evidence	Action and Justification (If Needed)	Evidence of Action (If Needed)
1	Proto	Error when compiling JValidatedFileChooser		Rather than declaring filter as null store a new FileExtensionFilter in it so that it won't equal null.	
2	Proto	When the all filter is applied all files are displayed.			
3	Proto	When the image filter is applied only image files are displayed.			

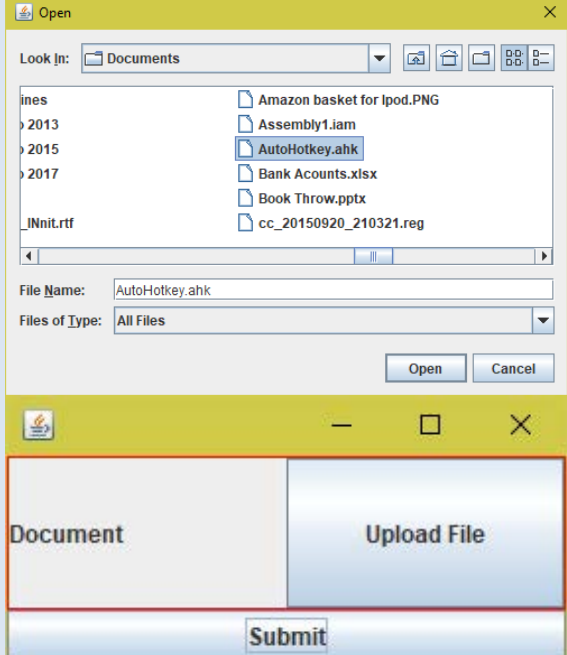
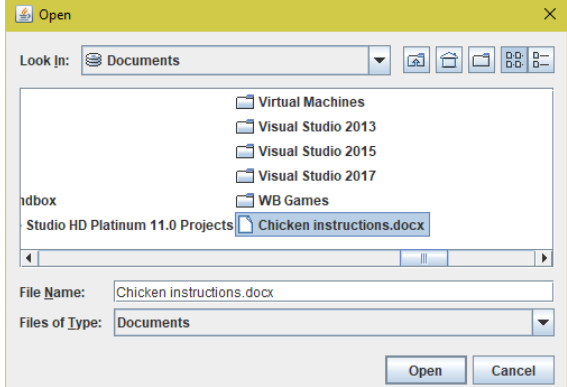
					
4	Proto	When the document filter is applied only document files are displayed.			
5	Proto	When the video filter is applied only video files are displayed.			

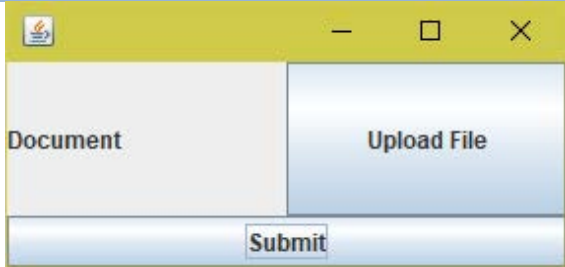
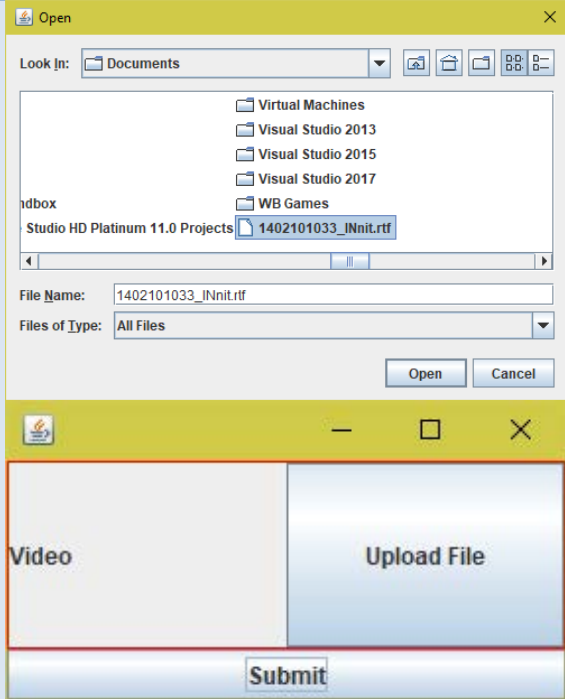
					
6	Proto	When a file is selected the extension is outputted.	 <pre data-bbox="712 912 1272 963">[INFO] <QUESTION_PANEL_BUILDER> Running add pdf</pre>		

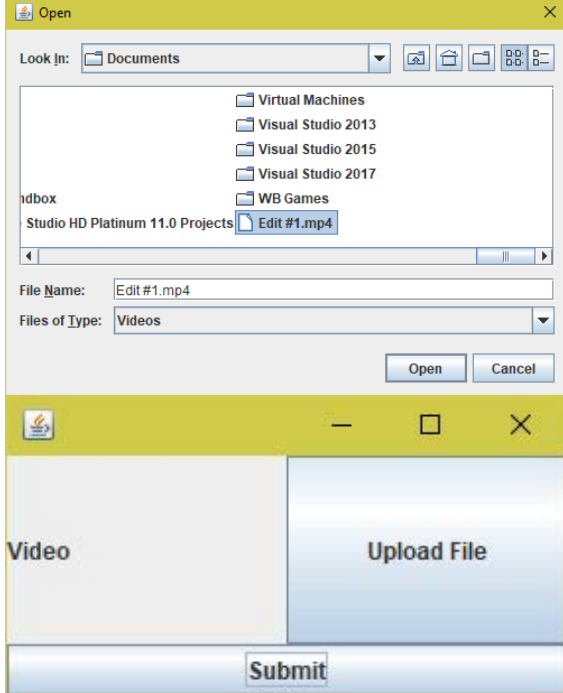
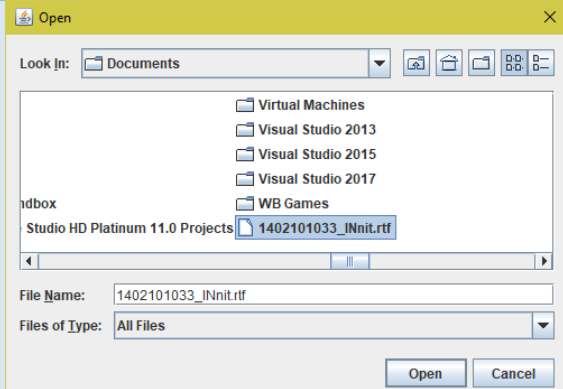
7	Proto	Null pointer Exception when pressing the submit form button when the upload file button hasn't been pressed yet.	<pre> [INFO] <QUESTION_PANEL_LIST> Running getByID Exception in thread "Awt-EventQueue-0" java.lang.NullPointerException at components.JValidatedFileChooser.validateAnswer(JValidatedFileChooser.java:57)) at QuestionPanel.validateAnswers(QuestionPanel.java:56) at FormDisplay.validateQuestions(FormDisplay.java:49) at FormDisplay.actionPerformed(FormDisplay.java:50) at javax.swing.AbstractButton.fireActionPerformed(Unknown Source) at javax.swing.DefaultButtonModel.fireActionPerformed(Unknown Source) at javax.swing.DefaultButtonModel.setPressed(Unknown Source) at javax.swing.plaf.basic.BasicButtonListener.mouseReleased(Unknown Source) at java.awt.Component.processMouseEvent(Unknown Source) at javax.swing.JComponent.processMouseEvent(Unknown Source) at java.awt.Component.processEvent(Unknown Source) at java.awt.Container.processEvent(Unknown Source) at java.awt.Component.dispatchEventImpl(Unknown Source) at java.awt.Container.dispatchEventImpl(Unknown Source) at java.awt.Component.dispatchEvent(Unknown Source) at java.awt.LightweightDispatcher.retargetMouseEvent(Unknown Source) at java.awt.LightweightDispatcher.processMouseEvent(Unknown Source) at java.awt.LightweightDispatcher.dispatchEvent(Unknown Source) at java.awt.Container.dispatchEventImpl(Unknown Source) at java.awt.Window.dispatchEventImpl(Unknown Source) at java.awt.Component.dispatchEvent(Unknown Source) at java.awt.EventQueue.dispatchEventImpl(Unknown Source) at java.awt.EventQueue.access\$500(Unknown Source) at java.awt.EventQueue\$3.run(Unknown Source) at java.awt.EventQueue\$3.run(Unknown Source) at java.security.AccessController.doPrivileged(Native Method) at java.security.ProtectionDomain\$JavaSecurityAccessImpl.doIntersectionPrivilege (Unknown Source) at java.security.ProtectionDomain\$JavaSecurityAccessImpl.doIntersectionPrivilege (Unknown Source) at java.awt.EventQueue\$4.run(Unknown Source) at java.awt.EventQueue\$4.run(Unknown Source) at java.security.AccessController.doPrivileged(Native Method) at java.security.ProtectionDomain\$JavaSecurityAccessImpl.doIntersectionPrivilege (Unknown Source) at java.awt.EventQueue.dispatchEvent(Unknown Source) at java.awt.EventDispatchThread.pumpOneEventForFilters(Unknown Source) at java.awt.EventDispatchThread.pumpEventsForFilter(Unknown Source) at java.awt.EventDispatchThread.pumpEventsForHierarchy(Unknown Source) at java.awt.EventDispatchThread.pumpEvents(Unknown Source) at java.awt.EventDispatchThread.pumpEvents(Unknown Source) at java.awt.EventDispatchThread.run(Unknown Source) </pre>	When the button is pressed a FileChooser object is created. Before the button is pressed there is no FileChooser object hence when the validation check is performed there is a nullpointer exception. To resolve make the check fail if the fileChooser variable's value is null.	 There is no null pointer exception and the validation check fails.
8	Proto	NullPointerException when the submit form button is pressed when the file chooser has been opened and then closed without selecting a file.	<pre> [INFO] <QUESTION_PANEL_LIST> Running getByID Exception in thread "Awt-EventQueue-0" java.lang.NullPointerException at components.JValidatedFileChooser.validateAnswer(JValidatedFileChooser.java:61)) at QuestionPanel.validateAnswers(QuestionPanel.java:56) at FormDisplay.validateQuestions(FormDisplay.java:49) at FormDisplay.actionPerformed(FormDisplay.java:50) at javax.swing.AbstractButton.fireActionPerformed(Unknown Source) at javax.swing.DefaultButtonModel.fireActionPerformed(Unknown Source) at javax.swing.DefaultButtonModel.setPressed(Unknown Source) at javax.swing.plaf.basic.BasicButtonListener.mouseReleased(Unknown Source) at java.awt.Component.processMouseEvent(Unknown Source) at javax.swing.JComponent.processMouseEvent(Unknown Source) at java.awt.Component.processEvent(Unknown Source) at java.awt.Container.processEvent(Unknown Source) at java.awt.Component.dispatchEventImpl(Unknown Source) at java.awt.Container.dispatchEventImpl(Unknown Source) at java.awt.Component.dispatchEvent(Unknown Source) at java.awt.LightweightDispatcher.retargetMouseEvent(Unknown Source) at java.awt.LightweightDispatcher.processMouseEvent(Unknown Source) at java.awt.LightweightDispatcher.dispatchEvent(Unknown Source) at java.awt.Container.dispatchEventImpl(Unknown Source) at java.awt.Window.dispatchEventImpl(Unknown Source) at java.awt.Component.dispatchEvent(Unknown Source) at java.awt.EventQueue.dispatchEventImpl(Unknown Source) at java.awt.EventQueue.access\$500(Unknown Source) at java.awt.EventQueue\$3.run(Unknown Source) at java.awt.EventQueue\$3.run(Unknown Source) at java.security.AccessController.doPrivileged(Native Method) at java.security.ProtectionDomain\$JavaSecurityAccessImpl.doIntersectionPrivilege (Unknown Source) at java.security.ProtectionDomain\$JavaSecurityAccessImpl.doIntersectionPrivilege (Unknown Source) at java.awt.EventQueue\$4.run(Unknown Source) at java.awt.EventQueue\$4.run(Unknown Source) at java.security.AccessController.doPrivileged(Native Method) at java.security.ProtectionDomain\$JavaSecurityAccessImpl.doIntersectionPrivilege (Unknown Source) at java.awt.EventQueue.dispatchEvent(Unknown Source) at java.awt.EventDispatchThread.pumpOneEventForFilters(Unknown Source) at java.awt.EventDispatchThread.pumpEventsForFilter(Unknown Source) at java.awt.EventDispatchThread.pumpEventsForHierarchy(Unknown Source) at java.awt.EventDispatchThread.pumpEvents(Unknown Source) at java.awt.EventDispatchThread.pumpEvents(Unknown Source) at java.awt.EventDispatchThread.run(Unknown Source) </pre>	This nullPointerException occurs because no file is selected. To resolve make the validation check fail when there is no file selected.	 The validation check fails when no file has been selected.

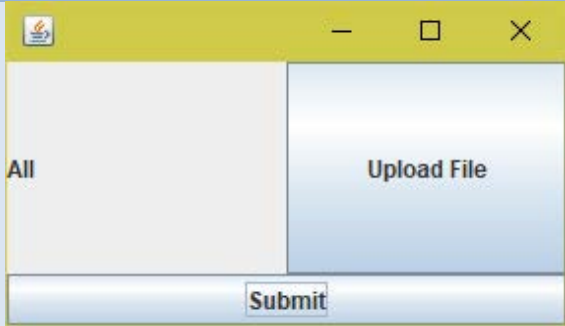
9	Proto	When a file other than an image is selected using the image file chooser the validation check fails.	
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10	Proto	When an image file is selected using an image file chooser the validation check passes.			
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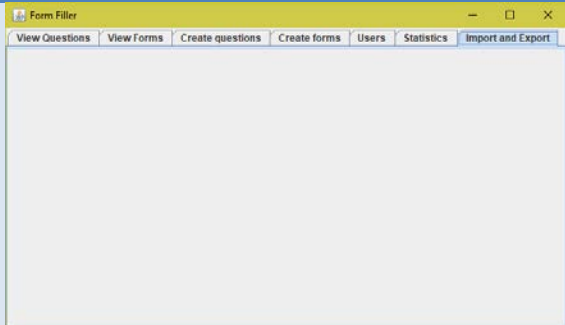
11	Proto	When a file other than a document file is selected with a document file chooser the validation check fails			
12	Proto	When a document file is selected using a document file chooser the validation check passes.			

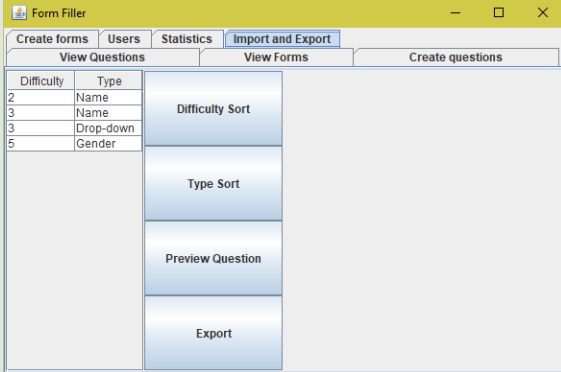
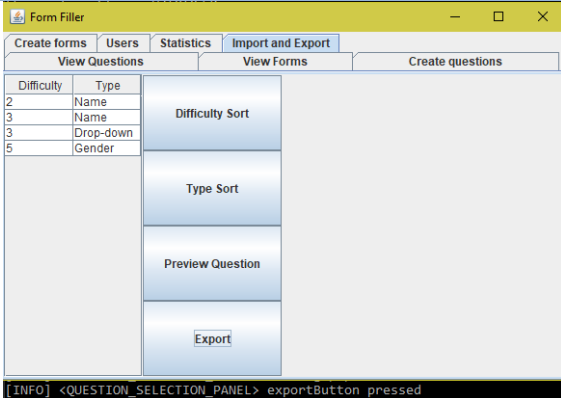
					
13	Proto	When a file other than a video file is selected with a video file chooser the validation check fails			

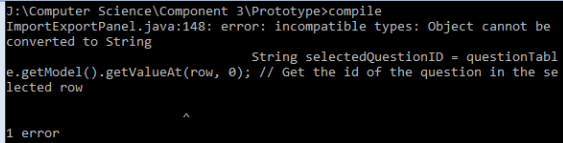
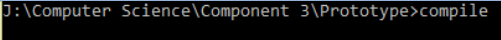
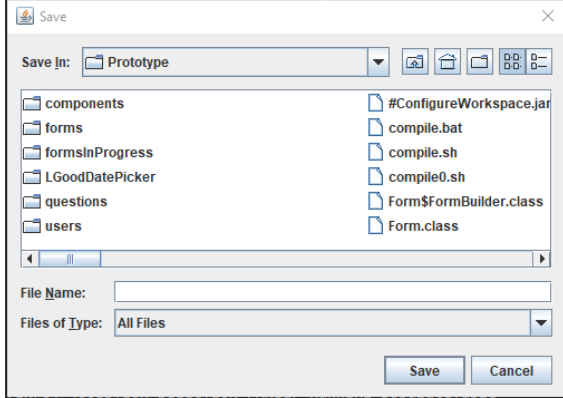
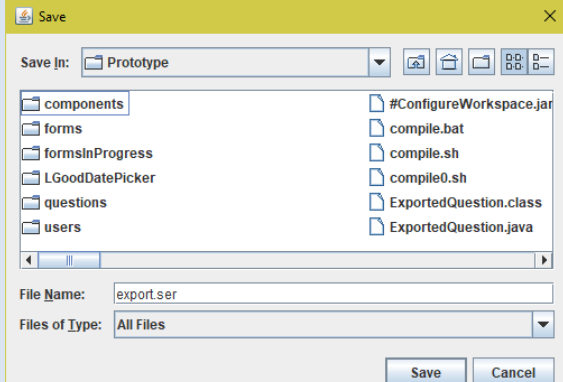
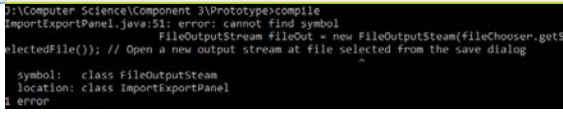
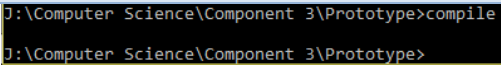
14	Proto	When a video file is selected using a video file chooser the validation check passes.	 The screenshot shows a standard Windows 'Open' file dialog. The 'Look In' dropdown is set to 'Documents'. The file list contains several folders like 'Virtual Machines', 'Visual Studio 2013', etc., and a file named 'Edit #1.mp4' which is highlighted. Below the list, the 'File Name' field contains 'Edit #1.mp4' and the 'Files of Type' dropdown is set to 'Videos'. 'Open' and 'Cancel' buttons are at the bottom right of the dialog.		
15	Proto	When any type of file is selected using a any file chooser the validation check passes.	 The screenshot shows a standard Windows 'Open' file dialog. The 'Look In' dropdown is set to 'Documents'. The file list contains the same folders as in the previous screenshot, and a file named '1402101033_INnit.rtf' which is highlighted. Below the list, the 'File Name' field contains '1402101033_INnit.rtf' and the 'Files of Type' dropdown is set to 'All Files'. 'Open' and 'Cancel' buttons are at the bottom right of the dialog.		

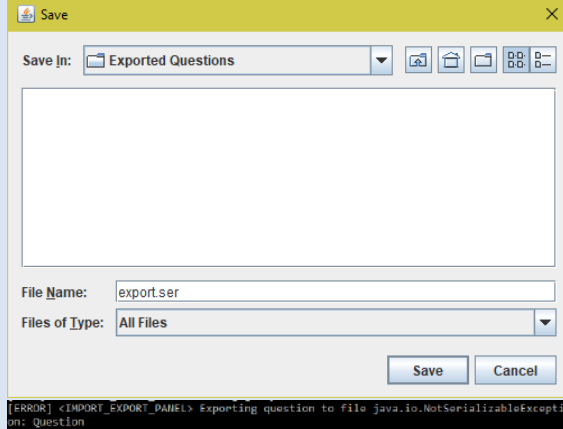
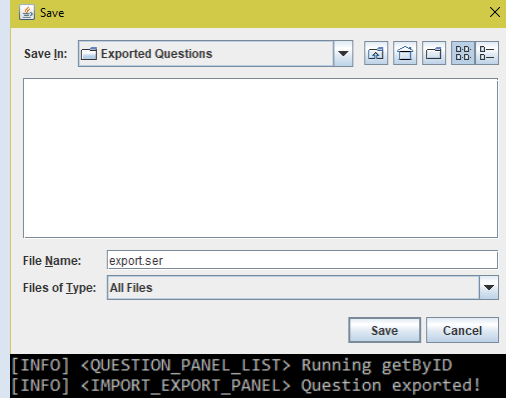
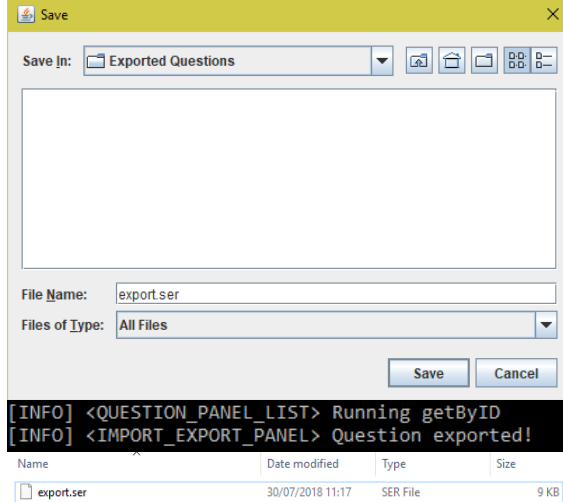
					
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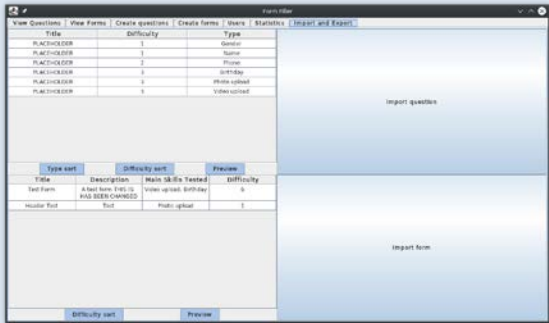
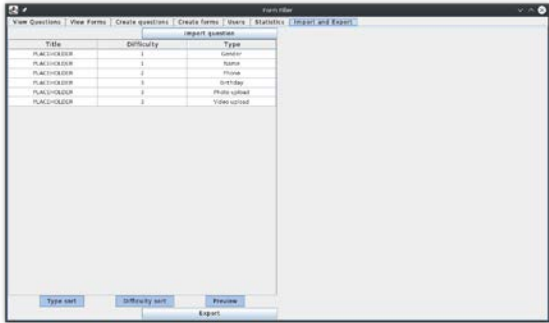
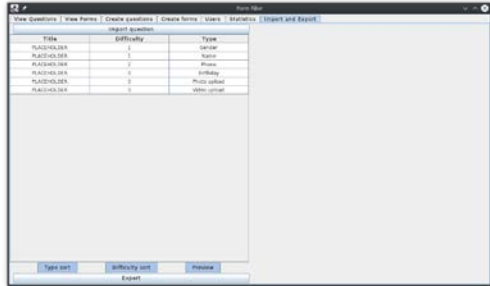
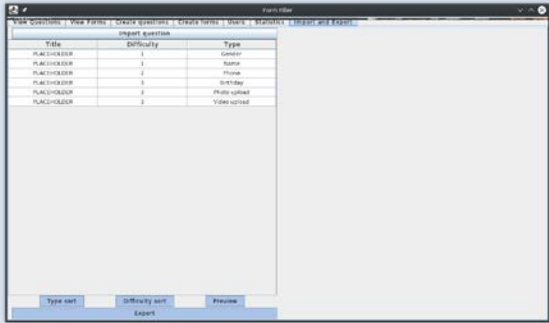
21: Ability to export a Question to file

Test	Stage	Result Problem	Evidence	Action and Justification (If Needed)	Evidence of Action (If Needed)
1	Proto	There is an import and export tab.			

2	Proto	Multiple symbol not found errors when compiling ImportExportPanel.java .	<pre> D:\Computer Science\Component 3\Prototype>compile ImportExportPanel.java:147: error: reached end of file while parsing } ^ 1 error D:\Computer Science\Component 3\Prototype>compile ImportExportPanel.java:38: error: cannot find symbol private DefaultTableModel questionTableModel = new DefaultTableModel(questionData, tableHeaders); ^ symbol: class DefaultTableModel location: class ImportExportPanel.QuestionSelectionPanel ImportExportPanel.java:38: error: cannot find symbol private DefaultTableModel questionTableModel = new DefaultTableModel(questionData, tableHeaders); ^ symbol: class DefaultTableModel location: class ImportExportPanel.QuestionSelectionPanel ImportExportPanel.java:83: error: cannot find symbol TableColumnModel tcm = questionTable.getColumnModel(); ^ symbol: class TableColumnModel location: class ImportExportPanel.QuestionSelectionPanel 3 errors </pre>	These errors are due to javax.swing.table.*; not being imported imported it will fix these errors	<pre> D:\Computer Science\Component 3\Prototype>compile D:\Computer Science\Component 3\Prototype> </pre> Successful compile
3	Proto	The same question selector that is used in the FormCreationPanel is used here to allow the users to selected questions to be exported.			
4	Proto	Presses to the export button are detected			

5	Proto	Incompatible types error when compiling ImportExportPanel.java		add ' + "" ' to the end of the line to convert the object to a String so that it can be stored in the variable	 Successful compile
6	Proto	When the export button is pressed a save file dialog is opened.			
7	Proto	The file chooser starts with a default filename with the correct (.ser) extension.			
8	Proto	Symbol not found error when compiling ImportExportPanel.java because of a typo.		Correct FileOutputStream to FileOutputSteam	 Successful compile

9	Proto	Error exporting the question due to a NotSerializableException.		This is due to the objects of type Question not being serializable. Making the Question class implement Serializable will fix this.	 No errors when exporting the question								
10	Proto	Question and QuestionPanel are successfully exported to file	 <table><thead><tr><th>Name</th><th>Date modified</th><th>Type</th><th>Size</th></tr></thead><tbody><tr><td>export.ser</td><td>30/07/2018 11:17</td><td>SER File</td><td>9 KB</td></tr></tbody></table>	Name	Date modified	Type	Size	export.ser	30/07/2018 11:17	SER File	9 KB		
Name	Date modified	Type	Size										
export.ser	30/07/2018 11:17	SER File	9 KB										

11	Soft Dev	The import export panel uses the new FormSelectionPanel and the new QuestionSelectionPanel			
12	Soft Dev	The import and export buttons are too short.		This is because their minimum width wasn't set so they defaulted to the width that they are. Setting a very wide minimum width will ensure that they fill the panel.	
13	Soft Dev	The export button is the correct colour.			

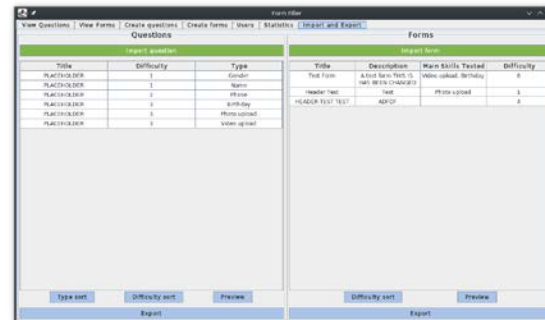
14

Soft Dev

When the import export tab is selected the question table refreshes.



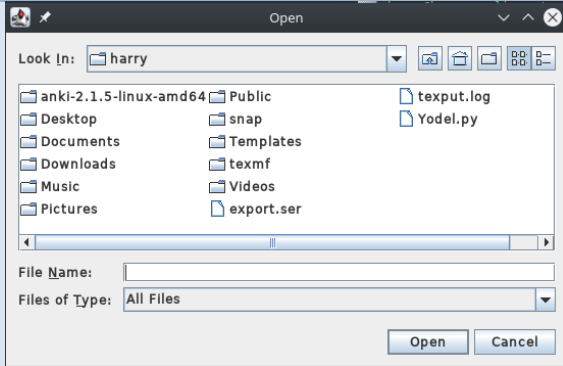
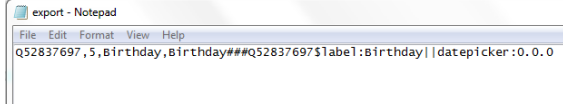
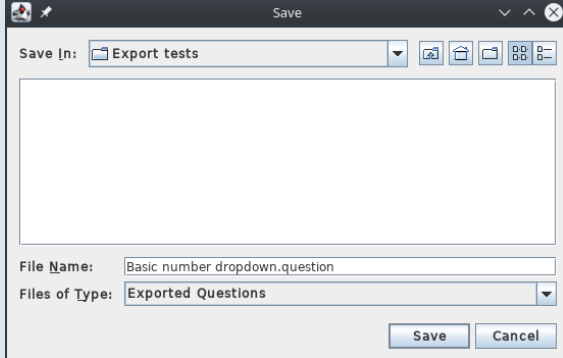
Tab not selected

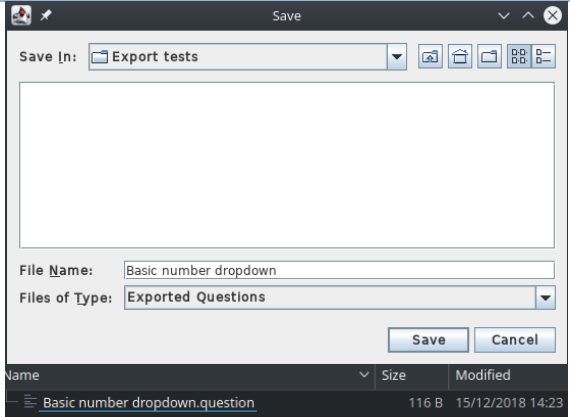


Tab selected

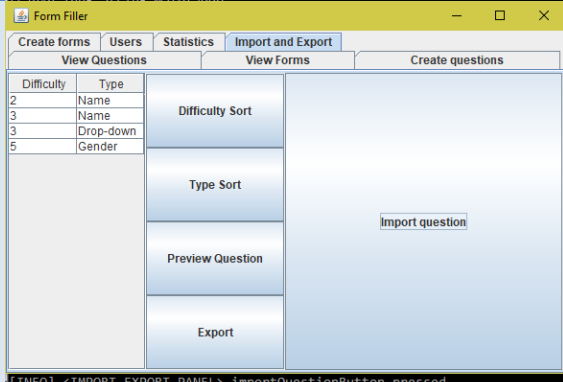
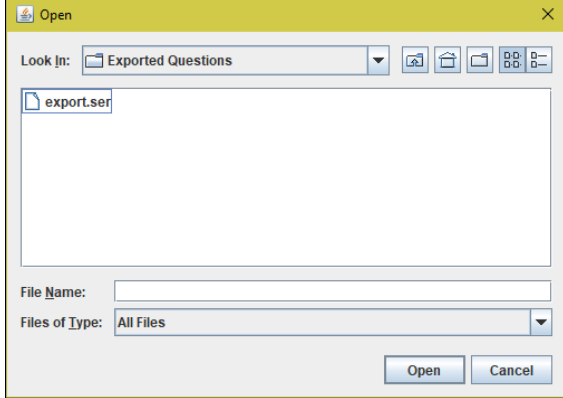
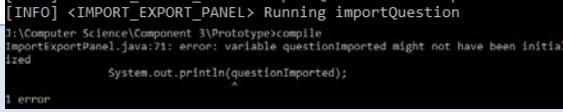
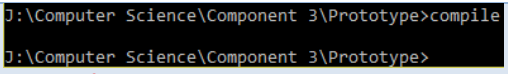
[INFO] <SELECT_QUESTIONS_PANEL> Running populateTable
[INFO] <SELECT_FORMS_PANEL> Running populateTable

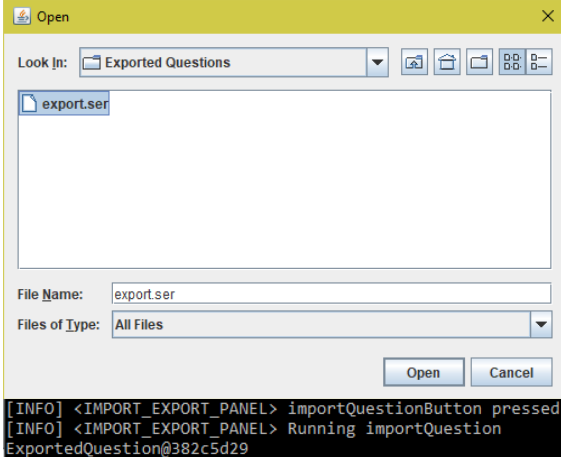
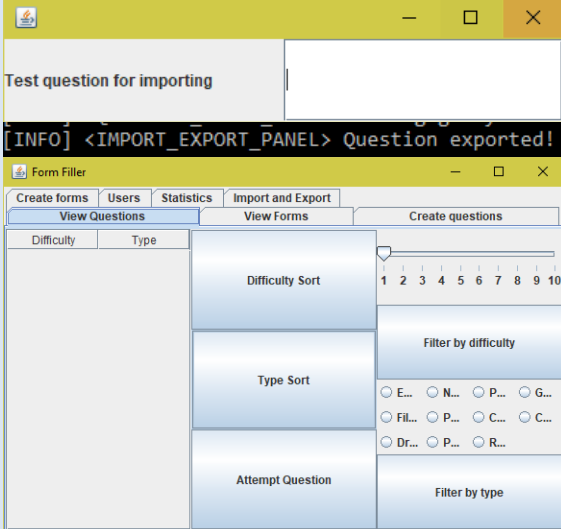
Table refreshed

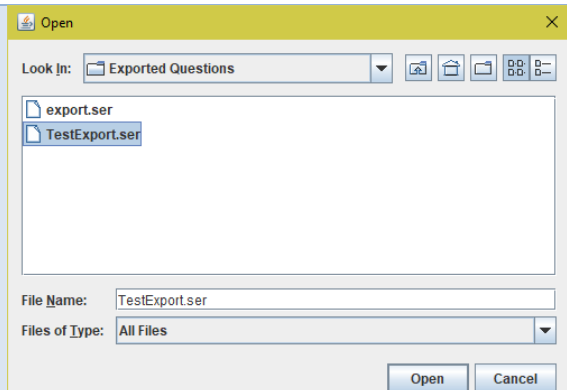
15	Soft Dev	If the user presses cancel when exporting a question the export process is stopped.			
16	Soft Dev	When a question is exported it's exported to a text file rather than a serialised object.			
17	Soft Dev	Rather than a question being exported with the file name "export" the file name is automatically set to the title of the question			

18	Soft Dev	The .question extension is automatically added by the system.	 Here you can see the question exported.	
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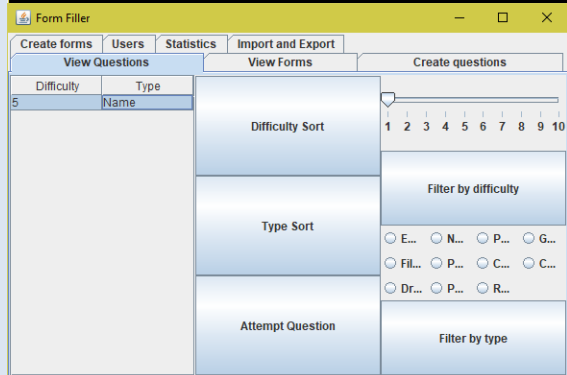
22: Ability to import a Question from file

Test	Stage	Result Problem	Evidence	Action and Justification (If Needed)	Evidence of Action (If Needed)
1	Proto	There is an import question button and presses to it are detected			
2	Proto	An open file dialog opens when the import question button is pressed			
3	Proto	Variable might not have been initialized error when compiling ImportExportPanel.java .		Store null in the questionImported variable when creating it rather than leaving it	

4	Proto	ExportedQuestion objects can be successfully imported.		empty.	
5	Proto	A question is successfully exported and then imported	 QuestionDB empty		



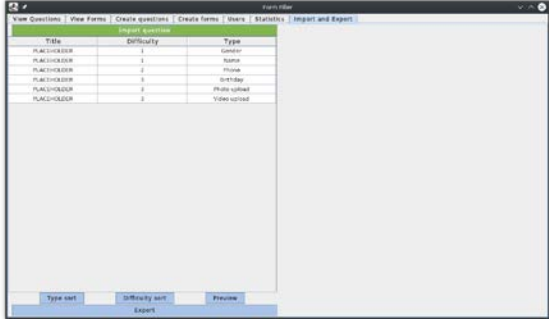
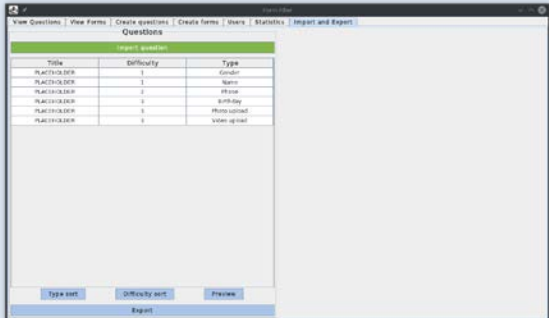
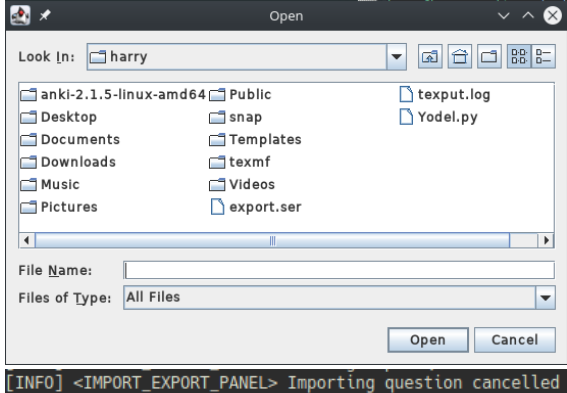
[INFO] <IMPORT_EXPORT_PANEL> Running importQuestion
[INFO] <QUESTION_PANEL_LIST> Running addQuestionPanel
[INFO] <QUESTION_LIST> Successfully added question and questionPanel

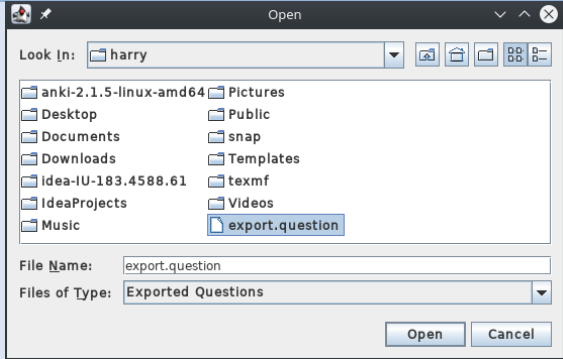


Question now present in view window



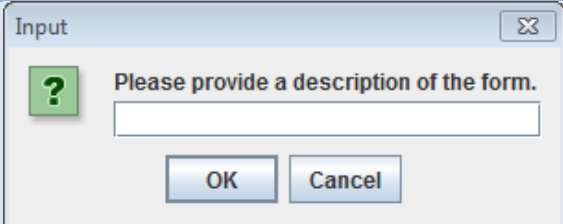
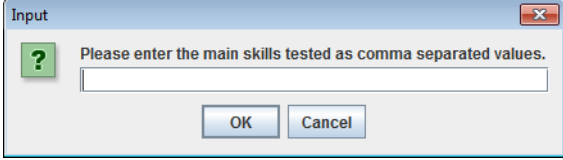
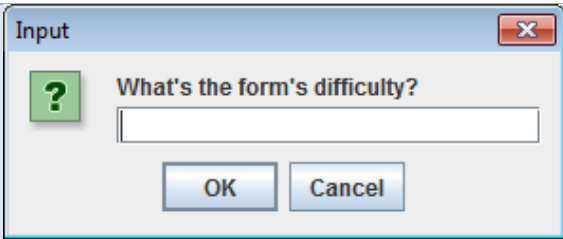
Question successfully imported

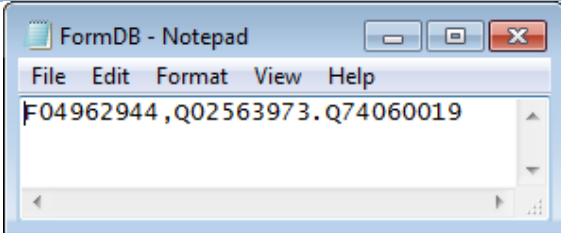
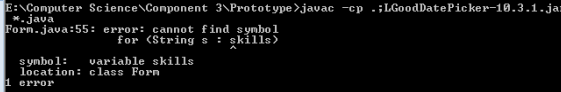
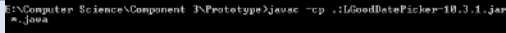
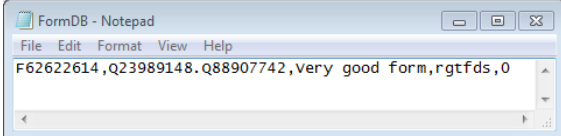
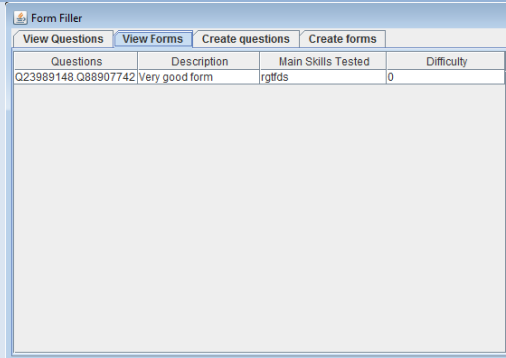
6	Soft Dev	The import question button is the correct colour with the correct colour text.			
7	Soft Dev	There is a border titled questions that surrounds the questions section.			
8	Soft Dev	If the user presses cancel when importing a question the import process is stopped.			

9	Soft Dev	The file chooser that allows the user to select an exported question to import only shows files of type .question. These are exported questions.			
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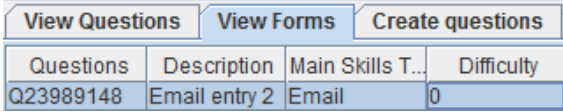
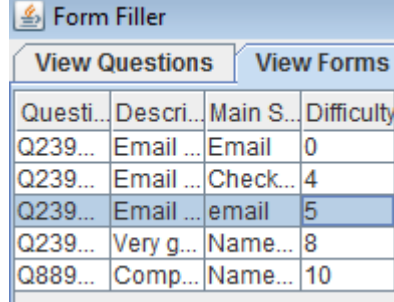
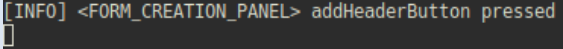
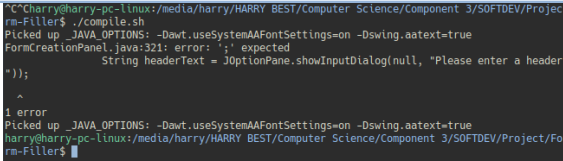
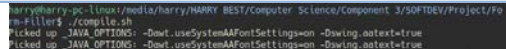
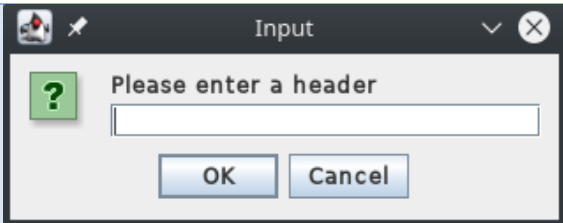
25: Ability to create forms

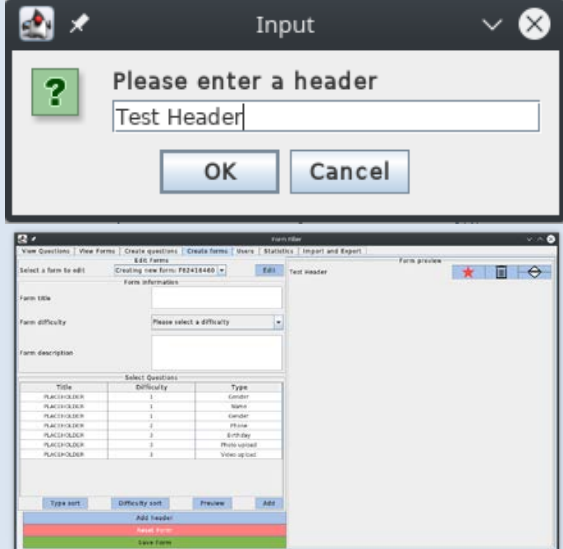
Test	Stage	Result Problem	Evidence	Action and Justification (If Needed)	Evidence of Action (If Needed)
1	Proto	Form class to compile because tempId wasn't defined before using	<pre>D:\Computer Science\Component 3\Prototype> D:\Computer Science\Component 3\Prototype>javac *.java Form.java:21: error: cannot find symbol id = tempId; ^ symbol: variable tempId location: class FormBuilder 1 error D:\Computer Science\Component 3\Prototype>javac *.java D:\Computer Science\Component 3\Prototype></pre>	Define tempId	<pre>D:\Computer Science\Component 3\Prototype>javac *.java Successful compile</pre>
2	Proto	A form is successfully created with ID TESTF01 and questions TEST01 and TEST02	<pre>D:\Computer Science\Component 3\Prototype>java Tester [INFO] <FORM_BUILDER> Running add [INFO] <FORM_BUILDER> Running add [INFO] <FORM_BUILDER> Running build [INFO] <FORM_BUILDER> Running trimArray TESTF01 [TEST01, TEST02]</pre>		
3	Proto	Semantic error when compiling due to attempting to remove a character from a string using the - operator.	<pre>D:\Computer Science\Component 3\Prototype>javac Tester.java .\Form.java:27: error: bad operand types for binary operator '-' arrayAsString -= "."; // Get rid of trailing . ^ first type: String second type: String 1 error</pre>	Use String.substring() to remove the last character instead.	<pre>D:\Computer Science\Component 3\Prototype>javac Tester.java .\Form.java:35: error: cannot find symbol return id + "." + arrayAsString; ^ symbol: variable arrayAsString location: class Form .\Form.java:40: error: cannot find symbol return toString.split(","); ^ symbol: variable toString location: class Form 2 errors</pre>
4	Proto	Multiple cannot find symbol errors	<pre>D:\Computer Science\Component 3\Prototype>javac Tester.java .\Form.java:35: error: cannot find symbol return id + "." + arrayAsString; ^ symbol: variable arrayAsString location: class Form .\Form.java:40: error: cannot find symbol return toString.split(","); ^ symbol: variable toString location: class Form 2 errors</pre>	Add a () on the end of the calls to the methods	<pre>D:\Computer Science\Component 3\Prototype>javac Tester.java D:\Computer Science\Component 3\Prototype> Successful compile</pre>
5	Proto	Form toString and toStringArray methods work.	<pre>D:\Computer Science\Component 3\Prototype>java Tester [INFO] <FORM_BUILDER> Running add [INFO] <FORM_BUILDER> Running add [INFO] <FORM_BUILDER> Running build [INFO] <FORM_BUILDER> Running trimArray TESTF01,TEST01.TEST02 [TESTF01, TEST01.TEST02]</pre>		

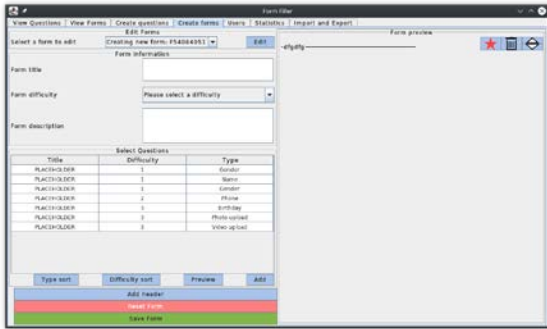
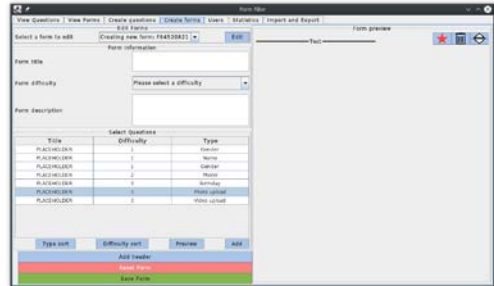
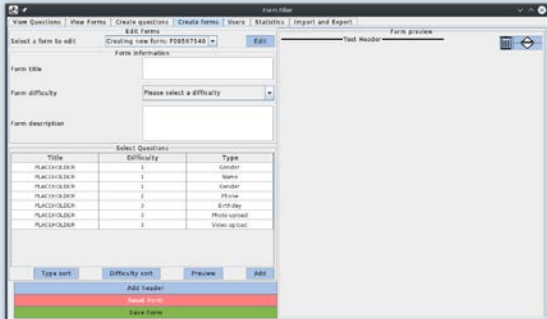
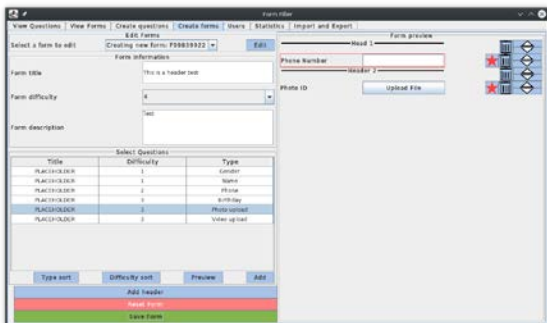
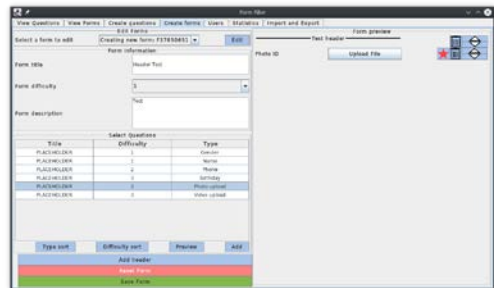
15	Proto	Error compiling form.java because the setFinalDetails() method doesn't return anything.	E:\Computer Science\Component 3\Prototype>javac -cp .;LGoodDatePi *.java Form.java:86: error: missing return statement ? ^ 1 error	Add a "return this" to the end of the setFinalDetails method	E:\Computer Science\Component 3\Prototype>javac -cp .;LGoodDatePi *.java FormCreationPanel.java:145: error: no suitable method found for s ts> String[] mainSkillsTested = JOptionPane.showInput enter the main skills tested as comma separated values. ").split(ain skills tested and split at the commas ^ method String.split(String,int) is not applicable (actual and formal argument lists differ in length) method String.split(String) is not applicable (actual and formal argument lists differ in length) 1 error
16	Proto	Error compiling form creation panel because a delimiter wasn't specified for the split() method on the main skills tested option pane	E:\Computer Science\Component 3\Prototype>javac -cp .;LGoodDatePi *.java FormCreationPanel.java:145: error: no suitable method found for s ts> String[] mainSkillsTested = JOptionPane.showInput enter the main skills tested as comma separated values. ").split(ain skills tested and split at the commas ^ method String.split(String,int) is not applicable (actual and formal argument lists differ in length) method String.split(String) is not applicable (actual and formal argument lists differ in length) 1 error	Pass "," to the split method to split at each comma	E:\Computer Science\Component 3\Prototype>javac -cp .;LGoodDatePi *.java FormCreationPanel.java:145: error: no suitable method found for s ts> String[] mainSkillsTested = JOptionPane.showInput enter the main skills tested as comma separated values. ").split(ain skills tested and split at the commas ^ method String.split(String,int) is not applicable (actual and formal argument lists differ in length) method String.split(String) is not applicable (actual and formal argument lists differ in length) 1 error
17	Proto	The user is prompted to enter a description of the form after pressing save form			
18	Proto	The user is prompted to enter the main skills tested as comma separated values after entering a description of the form			
19	Proto	The user is prompted to enter the form's difficulty after entering the main skills tested			

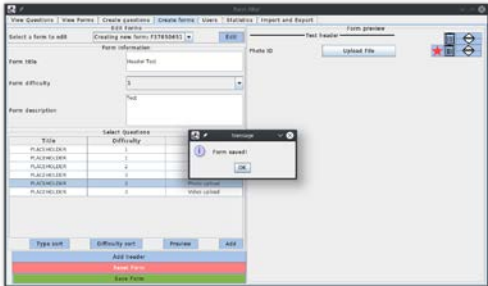
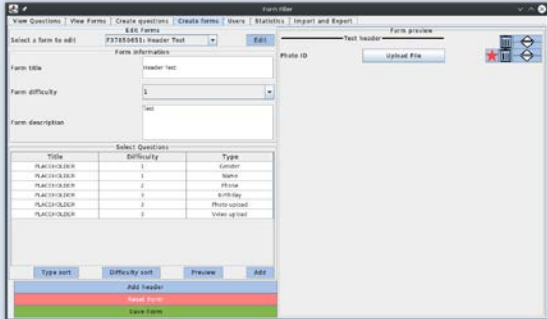
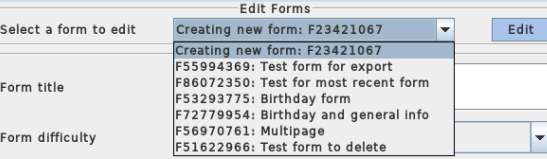
20	Proto	The additional form attributes aren't saved to the file because they aren't returned when the toString method is called			
21	Proto	Symbol not found exception when compiling Form.java as the array is called mainSkillsTested, not skills.		Replace skills with mainSkillsTested	
22	Proto	The additional form attributes are saved to the file.			
23	Proto	Null pointer exception when the mainSkillsTestedToString method is called for a loaded form because the new attributes weren't loaded correctly from the file.		Assign the loaded data from file to the correct variables in the form object. I also added the additional columns to the table to ensure that the table wasn't the issue.	

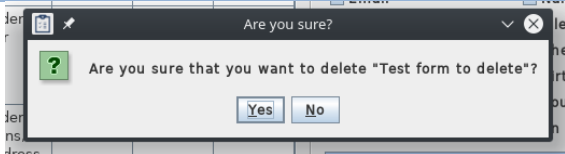
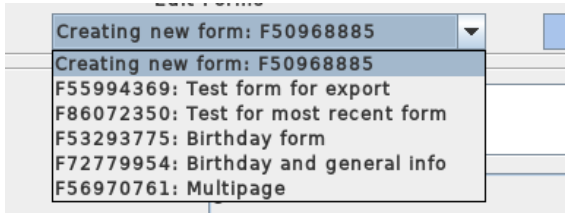
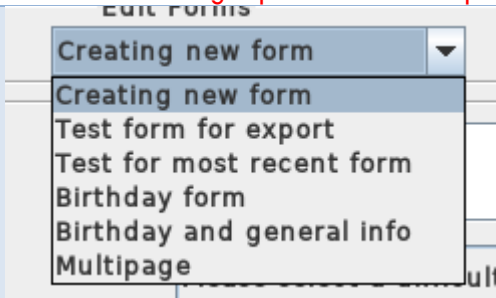
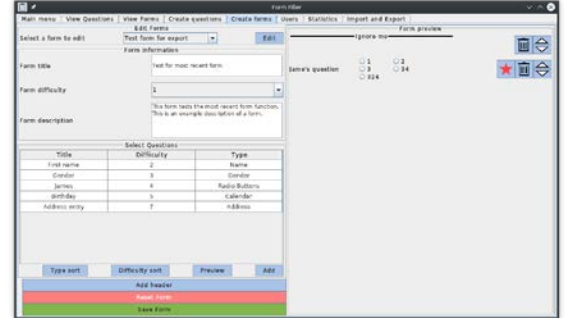
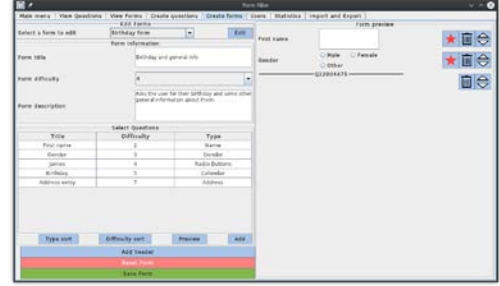
Form is displayed and loaded correctly

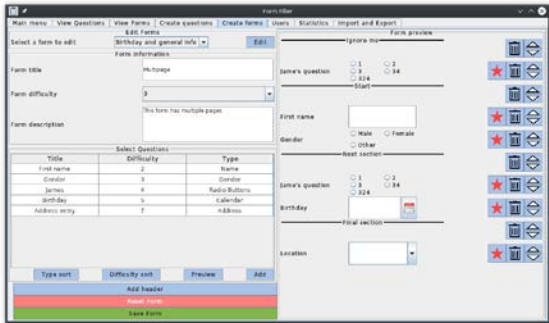
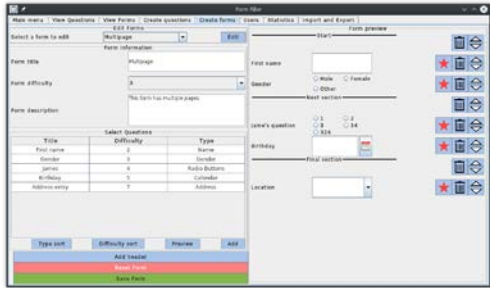
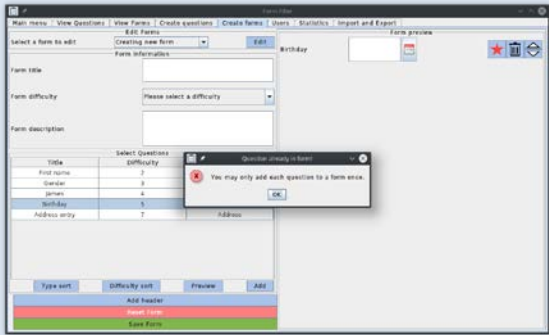
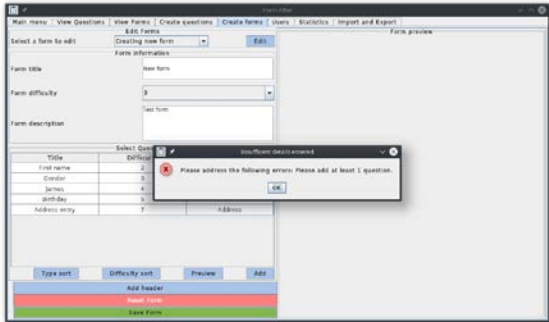
24	Proto	Each form added is being saved with a difficulty of 0 regardless of what is entered in the enter the difficulty box.		This happened because the difficulty attribute of the FormBuilder wasn't assigned to the form in the constructor so the default value of 0 was kept. Assigning the value of the difficulty attribute of the form builder to the form in the constructor will fix this.	 The highlighted form has been added with a correct non zero difficulty
25	Soft Dev	Presses to the add header button are detected			
26	Soft Dev	Error when compiling form creation panel.		Although the error says that a semicolon is missing, this isn't the issue. The line of code just has one too many close brackets. Removing the bracket will fix this error.	 Successful compile
27	Soft Dev	When the add header button is pressed a dialog appears asking the user for a header.			

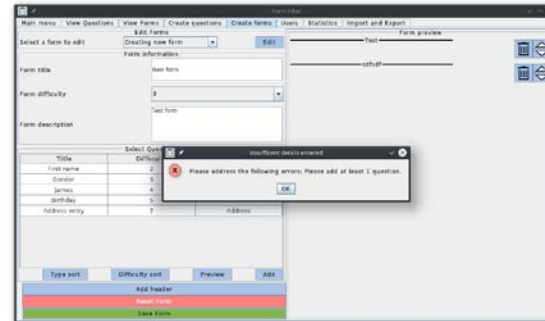
28	Soft Dev	Symbol not found error when compiling FormCreationPanel after changing the name of a variable.	<pre> harry@harry-pc:~/linux/media/harry/HARRY BEST/Computer Science/Component 3/50FTDEV/Project/Form-Filler\$./compile.sh Picked up _JAVA_OPTIONS: -Dwt.useSystemAAFontSettings=on -Dswing.aatext=true FormCreationPanel.java:460: error: cannot find symbol JPanel questionPanel = questions.getPanelByID(questionID); ^ symbol: variable questionID location: class FormCreationPanel FormCreationPanel.java:474: error: cannot find symbol actionButtonPanel.setName(questionID); // Allows the question that the butto ^ ns belong to to be identified. symbol: variable questionID location: class FormCreationPanel FormCreationPanel.java:515: error: cannot find symbol upAndDownPanel.setName(questionID); // Allows the question that the buttons ^ belong to to be identified. symbol: variable questionID location: class FormCreationPanel FormCreationPanel.java:530: error: cannot find symbol formBeingCreated = formBeingCreated.add(questionID, true); // Add the questi ^ on to the form symbol: variable questionID location: class FormCreationPanel 4 errors Picked up _JAVA_OPTIONS: -Dwt.useSystemAAFontSettings=on -Dswing.aatext=true </pre>	This error has occurred because there are a few references to the variable that I haven't changed the name of. Changing those to the new name should fix this.	<pre> "C:\Cherry\harry-pc\linux\media\harry\HARRY BEST\Computer Science\Component 3\50FTDEV\Project\rm-Filler\$./compile.sh Picked up _JAVA_OPTIONS: -Dwt.useSystemAAFontSettings=on -Dswing.aatext=true Picked up _JAVA_OPTIONS: -Dwt.useSystemAAFontSettings=on -Dswing.aatext=true Successful compile </pre>
29	Soft Dev	When the header text is entered and ok is pressed a label with the correct header text is added to the form.			

30	Soft Dev	When a header is added to the form the text isn't centered.		This is because the title justification of the border wasn't set so it defaulted to aligning to the left. Setting the title justification of the border using	
31	Soft Dev	Headers don't have the required button. This is intentional because a header isn't an answerable question.			
32	Soft Dev	There is a null pointer exception when saving a form that has headers in it.		This happens because when the build method of the FormBuilder class runs it populates an array with all of the type so of the questions that are in the form. The issue is that when it tries to get the type of a header element as the header is not a question there is a	

			<pre> [INFO] <FORM_BUILDER> Running trimRequiredQuestionsArray Exception in thread "AWT-EventQueue-1" java.lang.NullPointerException at Form\$FormBuilder.build(Form.java:273) at FormCreationPanel.saveForm(FormCreationPanel.java:288) at FormCreationPanel.actionPerformed(FormCreationPanel.java:336) at javax.swing.AbstractButton.fireActionPerformed(AbstractButton.java:2022) at javax.swing.AbstractButton\$Handler.actionPerformed(AbstractButton.java:2348) at javax.swing.DefaultButtonModel.fireActionPerformed(DefaultButtonModel.java:482) at javax.swing.DefaultButtonModel.setPressed(DefaultButtonModel.java:259) at javax.swing.plaf.basic.BasicButtonListener.mouseReleased(BasicButtonListener.java:252) at java.awt.Component.processMouseEvent(Component.java:6539) at javax.swing.JComponent.processMouseEvent(JComponent.java:3324) at java.awt.Component.processEvent(Component.java:6304) at java.awt.Container.dispatchEventImpl(Container.java:2239) at java.awt.Container.dispatchEventImpl(Container.java:4889) at java.awt.Container.dispatchEventImpl(Container.java:2297) at java.awt.Component.dispatchEvent(Component.java:4711) at java.awt.LightweightDispatcher.retargetMouseEvent(Container.java:4904) at java.awt.LightweightDispatcher.processMouseEvent(Container.java:4535) at java.awt.LightweightDispatcher.dispatchEvent(Container.java:4476) at java.awt.Container.dispatchEventImpl(Container.java:2283) at java.awt.Window.dispatchEventImpl(Window.java:2746) at java.awt.Component.dispatchEvent(Component.java:4711) at java.awt.EventQueue.dispatchEventImpl(EventQueue.java:760) at java.awt.EventQueue.access\$509(EventQueue.java:97) at java.awt.EventQueue\$3.run(EventQueue.java:709) at java.awt.EventQueue\$3.run(EventQueue.java:703) at java.security.AccessController.doPrivileged(Native Method) at java.security.ProtectionDomain\$JavaSecurityAccessImpl.doIntersectionPrivilege(ProtectionDomain.java:74) at java.security.ProtectionDomain\$JavaSecurityAccessImpl.doIntersectionPrivilege(ProtectionDomain.java:84) at java.awt.EventQueue\$4.run(EventQueue.java:733) at java.awt.EventQueue\$4.run(EventQueue.java:731) at java.security.AccessController.doPrivileged(Native Method) at java.security.ProtectionDomain\$JavaSecurityAccessImpl.doIntersectionPrivilege(ProtectionDomain.java:74) at java.awt.EventQueue.dispatchEvent(EventQueue.java:730) at org.gnome.Accessibility.AtkWrapper\$6.dispatchEvent(AtkWrapper.java:715) at java.awt.EventDispatchThread.pumpOneEventForFilters(EventDispatchThread.java:205) at java.awt.EventDispatchThread.pumpEventsForFilter(EventDispatchThread.java:116) at java.awt.EventDispatchThread.pumpEventsForHierarchy(EventDispatchThread.java:105) at java.awt.EventDispatchThread.pumpEvents(EventDispatchThread.java:101) at java.awt.EventDispatchThread.pumpEvents(EventDispatchThread.java:93) at java.awt.EventDispatchThread.run(EventDispatchThread.java:82) </pre>	null pointer exception. Adding a check to determine whether a form component is a header or a question before getting its type should fix this.	 Form can be successfully saved
33	Soft Dev	A form with headers in it can correctly be loaded to edit	 This form has been loaded to edit		
34	Soft Dev	When the create form tab is clicked the edit form drop down is refreshed.	 The test form to delete is in the drop down		

			 The form is deleted  The form is no longer present in the drop down.		
35	Soft Dev	The drop down menu no longer includes the form ids as these are not relevant to the user.			
36	Soft Dev	When editing a form, if another form is selected to be edited the form preview isn't cleared and the new form is added at the bottom of the old one.	 A form is being edited.	This is because although the form preview panel was being cleared, the array that stored all of the question panels wasn't being emptied between editing forms this meant that when the question panels were added to	 A form is being edited

			 Another form has been selected to be edited. As you can see the first form hasn't been removed from the preview.	the preview all of the old ones were also being added. Emptying the question panel array between editing forms will fix this.	 Another form is selected to be edited and the first form has been removed
37	Soft Dev	When the user tries to add a question to a form more than once the system stops them and displays an error.	 In this case the calendar question was tried to be added twice.		
38	Soft Dev	The user is required to add at least 1 question to a form before they are allowed to save.			

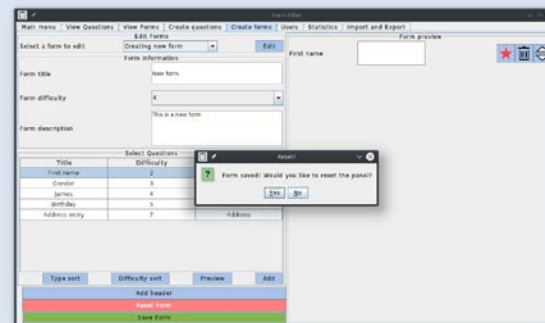
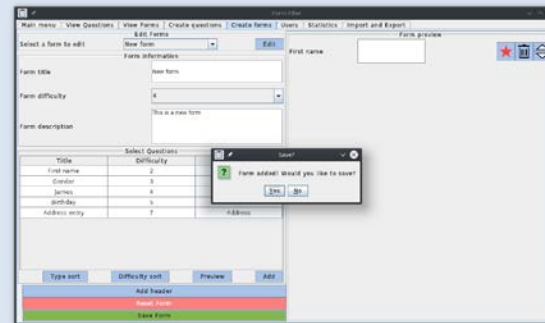


A form with headers

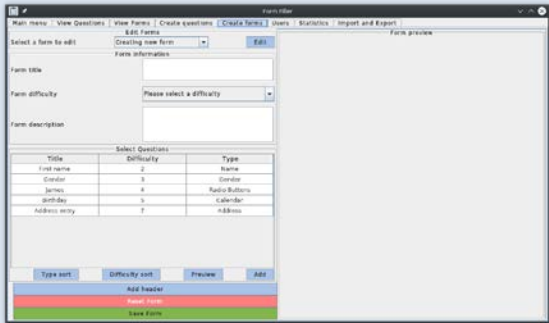
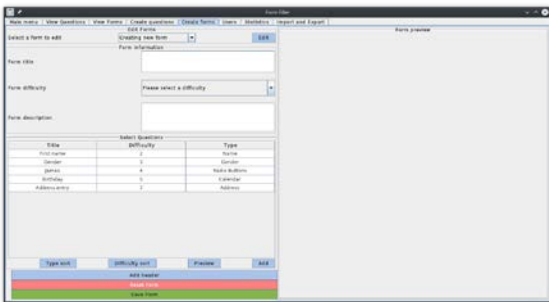
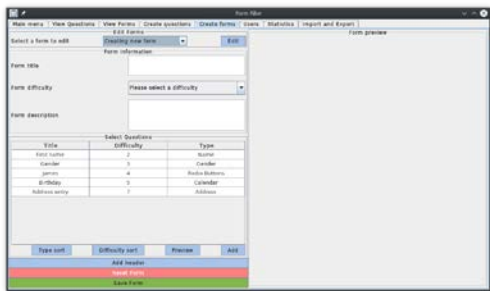
39

Soft Dev

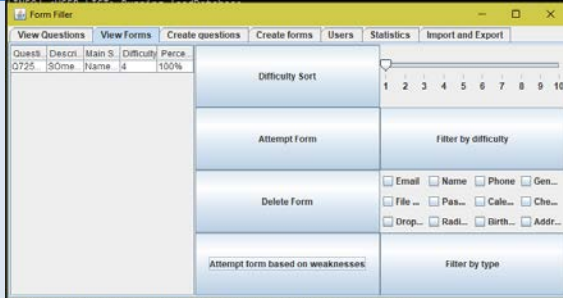
When a form is added the user is asked if they want to reset the panel.

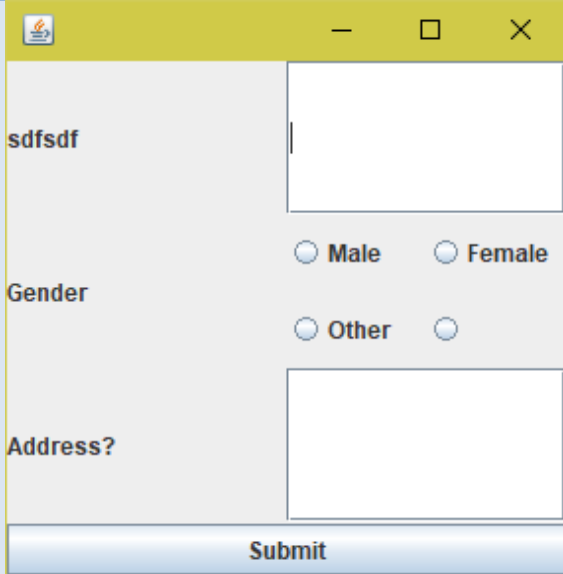


The user selected yes and the panel is reset

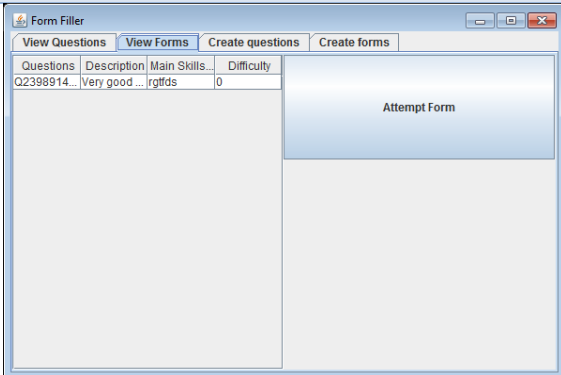
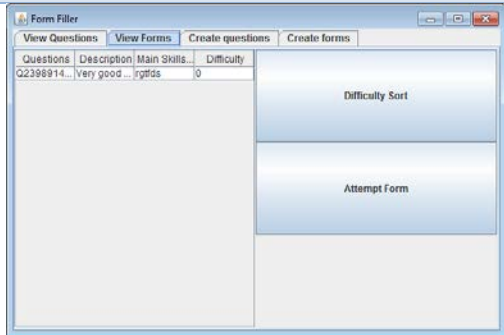
					
40	Soft Dev	The width of the save form etc. buttons in the bottom left is incorrect. They should be longer.		Adding the buttons to their own panel with a grid layout then adding the panel to the screen should fix this. This is because the gridlayout will ensure that the buttons entirely fill the panel because a gridlayout ignores the preferred, maximum, and minimum sizes of components.	 The buttons fill the width fully.

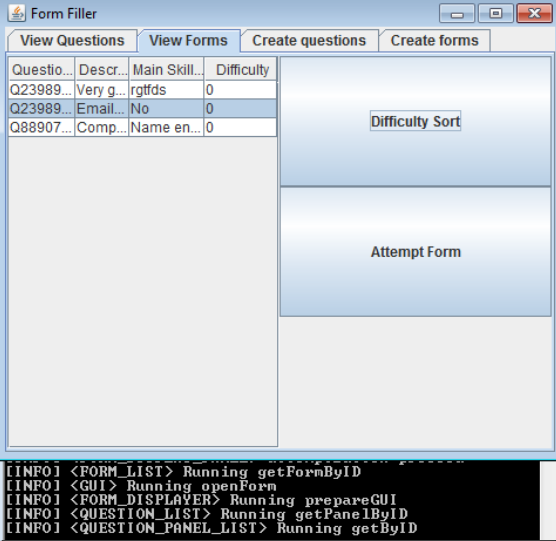
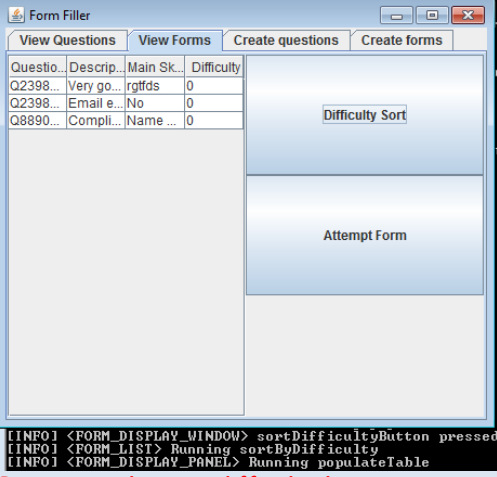
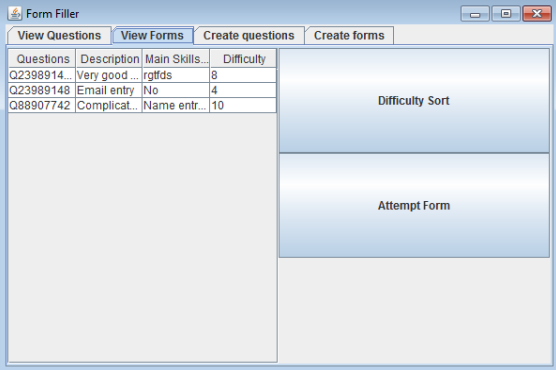
26: Ability to create forms from user weaknesses

Test	Stage	Result Problem	Evidence	Action and Justification (If Needed)	Evidence of Action (If Needed)
1	Proto	There is a create form from user weaknesses button in the view forms panel and presses to it are detected.			
2	Proto	Symbol not found error when compiling GUI.java because FormBuilder couldn't be found.	<pre> [INFO <FORM_DISPLAY_PANEL> attemptUserWeaknessesFormButton pressed D:\Computer Science\Component 3\Prototype>compile GUI.java:132: error: cannot find symbol FormBuilder questionForm = new Form.FormBuilder("USER_WEAKNESSES", questions ^ symbol: class FormBuilder location: class GUI 1 error </pre>	Change FormBuilder to Form.FormBuilder as FormBuilder is a subclass of Form.	<pre> D:\Computer Science\Component 3\Prototype>compile D:\Computer Science\Component 3\Prototype> </pre> Successful compile
3	Proto	When the attempt form based on user weaknesses button is pressed the user's question stat array is sorted descending based on the average number of attempts taken to correct an error for each question	<pre> [INFO <FORM_DISPLAY_PANEL> attemptUserWeaknessesFormButton pressed 000916722 1 Q07726485 2 Q87781316 4 Q72529010 1 Q50319011 5 Q35477436 1 Q46927134 2 Q51308304 1 Q81533244 2 AFTER SORT Q50319011 5 Q87781316 4 Q67726485 2 Q46927134 2 Q81533244 2 Q00916722 1 Q72529010 1 Q35477436 1 Q51308304 1 </pre> This output shows the question id for each question and the average number of attempts taken to correct		

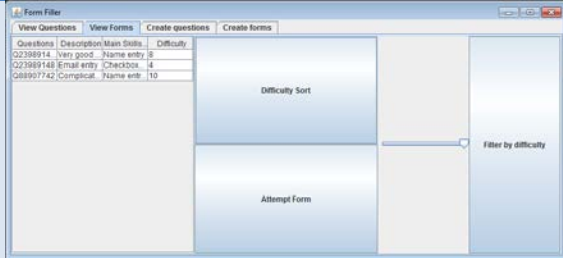
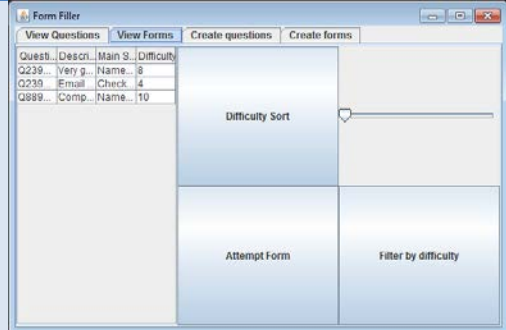
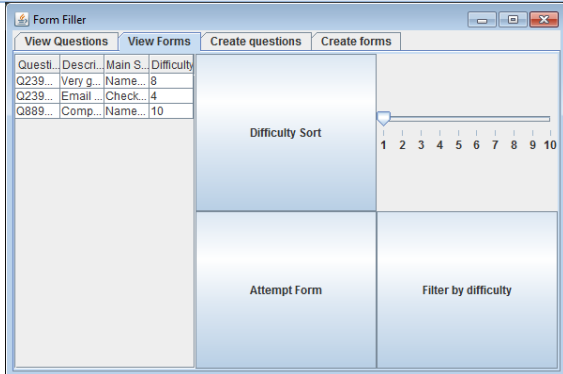
4	Proto	The top 3 valid questions are taken from the array to be attempted. Valid means that there is a question associated with the questionStat object as if a question is deleted there can be orphaned question stat objects.	<pre>AFTER SORT Q50319011 5 Q87781316 4 Q67726485 2 Q46927134 2 Q81533244 2 Q00916722 1 Q72529010 1 Q35477436 1 Q51308304 1 Q50319011 Q46927134 Q81533244</pre> The bottom three questions are the ones that have been picked. As you can see Q87787316 and Q67726485 haven't been picked as the questions that they correspond to no longer exist.		
5	Proto	When the attempt form based on user weaknesses button is pressed a form containing the 3 questions, that the user struggles to correct the most amount of times, is opened for the user to attempt.			

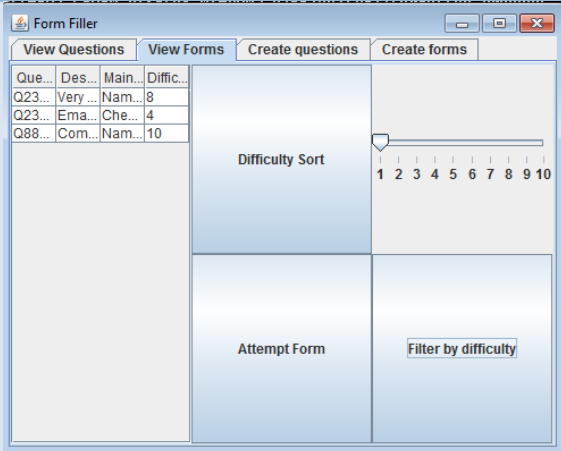
27: Ability to sort forms by difficulty

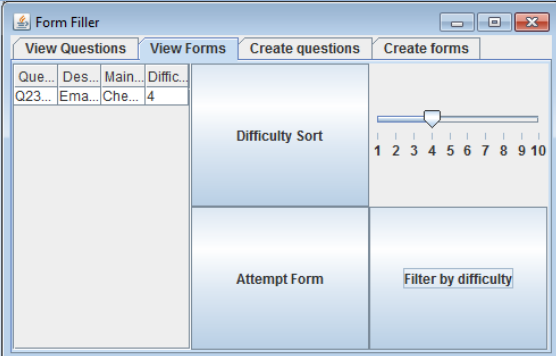
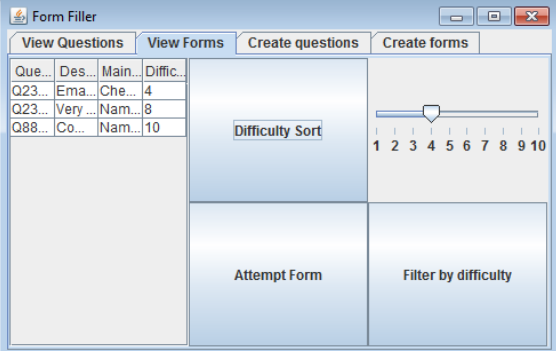
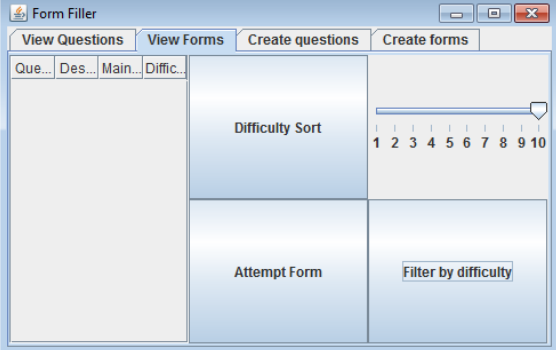
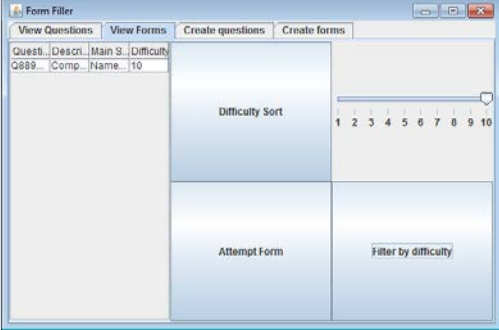
Test	Stage	Result Problem	Evidence	Action and Justification (If Needed)	Evidence of Action (If Needed)
1	Proto	FormList fails to compile after adding the difficulty sort because the form has no method getDifficulty()	<pre> D:\Computer Science\Component 3\Prototype>javac -cp .;LGoodDatePicker-10.3.1.jar *.java FormList.java:79: error: cannot find symbol if (FormArray[i].getDifficulty() > formArray[i+1].getDifficulty()) // Check to see if it's greater than the one after ^ symbol: method getDifficulty() location: class Form FormList.java:79: error: cannot find symbol if (FormArray[i].getDifficulty() > formArray[i+1].getDifficulty()) // Check to see if it's greater than the one after ^ symbol: method getDifficulty() location: class Form 2 errors D:\Computer Science\Component 3\Prototype>cd components </pre>	Add a getDifficulty() method to the form class that returns the difficulty of the form.	<pre> D:\Computer Science\Component 3\Prototype>javac -cp .;LGoodDatePicker-10.3.1.jar *.java D:\Computer Science\Component 3\Prototype>cd components </pre> Successful compile
2	Proto	The is no sort difficulty button because it wasn't added to the button panel		Add the button to the button panel	

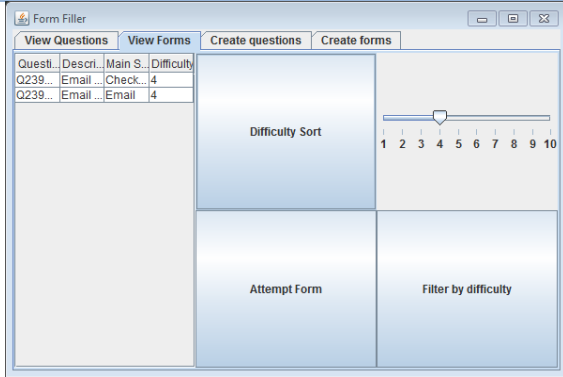
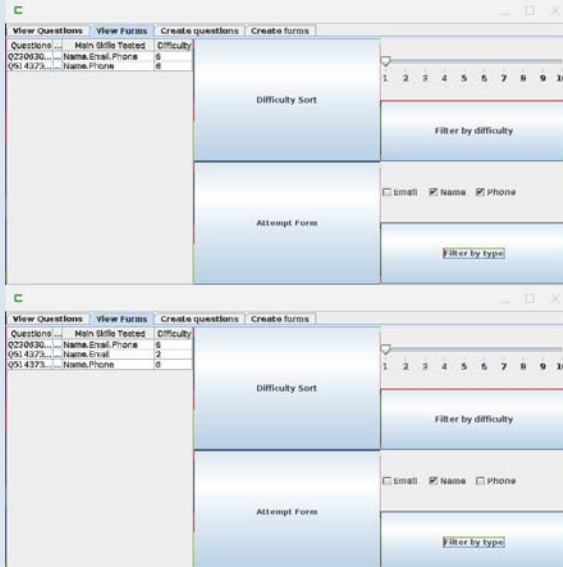
3	Proto	Presses to the sort difficulty button aren't detected. This is because the sort difficulty button hadn't had an action listener added to it.		Set the ViewFormsPanel as the sortDifficulty button's action listener	 Presses to the sort difficulty button are detected
4	Proto	The forms are sorted into descending difficulty after the difficulty sort button is pressed			

28: Ability to filter forms by difficulty

Test	Stage	Result Problem	Evidence	Action and Justification (If Needed)	Evidence of Action (If Needed)
1	Proto	There is a slider and a filter difficulty button however they have the wrong layout the button should be below the slider		Change the grid layout from having 2 columns to only having 1 column to put the button and the slider in the same column.	 Correct layout
2	Proto	The difficulty slider has ticks from 1 to 10			

3	Proto	Presses to the difficulty filter button are detected			
4	Proto	Symbol error when compiling FormList due to not putting () after a method call.	<pre> [INFO] <FORM_DISPLAY_WINDOW> difficultyFilterButton pressed E:\Computer Science\Component 3\Prototype>javac -cp .;LGoodDa *.java FormList.java:102: error: cannot find symbol if <formArray[i].getDifficulty == dif of the correct difficulty ^ symbol: variable getDifficulty location: class Form </pre>	Change getDifficulty to getDifficulty()	<pre> E:\Computer Science\Component 3\Prototype>javac -cp .;LGoodDa *.java FormList.java:113: error: incompatible types: Form cannot be trimmedResults = results[i]; ^ 1 error </pre>
5	Proto	Type error when trimming the array that contains the results of the filter. This is because an index wasn't specified to store each item.	<pre> E:\Computer Science\Component 3\Prototype>javac -cp .;LGoodDa *.java FormList.java:113: error: incompatible types: Form cannot be trimmedResults = results[i]; ^ 1 error </pre>	Store results[i] at index i in the trimmed results array by changed trimmedResults to trimmedResults[i]	<pre> E:\Computer Science\Component 3\Prototype>javac -cp .;LGoodDa *.java </pre> Successful compile

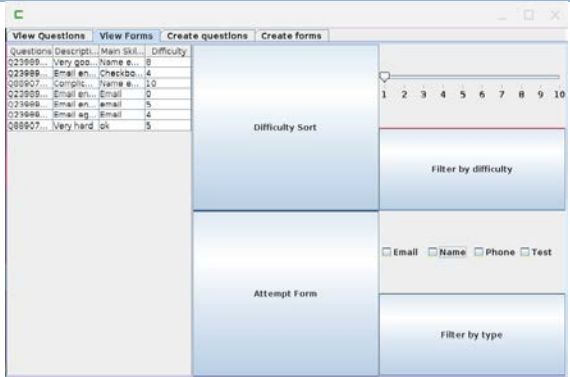
6	Proto	When 4 is selected on the difficulty slider and the filter by difficulty button is pressed only the forms with difficulty 4 are displayed. The filter is working as intended. This also works for the other difficulties on the slider	 <pre> [INFO] <FORM_DISPLAY_WINDOW> difficultyFilterButton pressed [INFO] <FORM_LIST> Running filterByDifficulty [INFO] <FORM_DISPLAY_PANEL> Running populateTable </pre>		
7	Proto	When the slider is set to 10 and the filter is applied the form with difficulty 10 isn't displayed even though it should be.	 	The code to iterate through the forms was copied from the difficulty sort method and so stopped iterating one before the last form as is needed for a bubble sort. The stop point of the for loop should be changed from 'nextFormLocation - 1' to 'nextFormLocation' to rectify this	 The form with difficulty 10 is displayed.

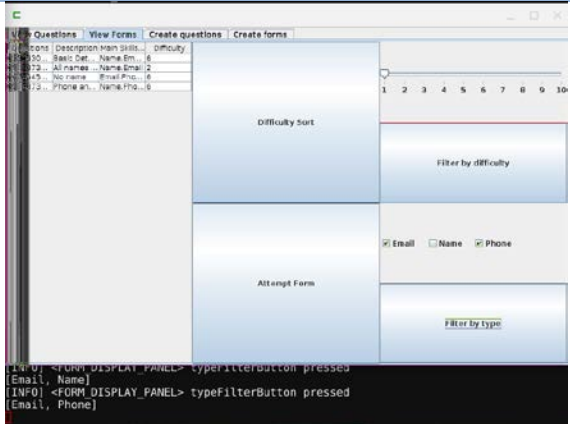
8	Proto	All forms that match the filter are displayed.			
9	Proto	The type filter works correctly. Only the forms that fully match the filter criteria are displayed.			

			View Questions View Forms Create questions Create forms Questions... Main Skills Tested Difficulty Q270936... Name Email Phone 6 Difficulty Sort 1 2 3 4 5 6 7 8 9 10 Filter by difficulty Attempt Form ☒ Email ☒ Name ☒ Phone Filter by type		
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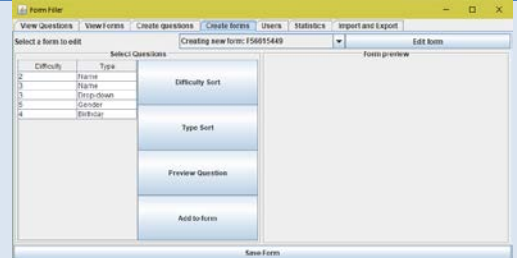
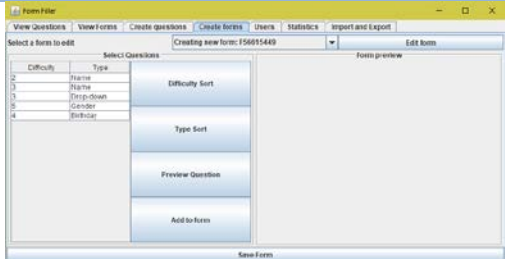
29: Ability to filter forms by types of questions tested

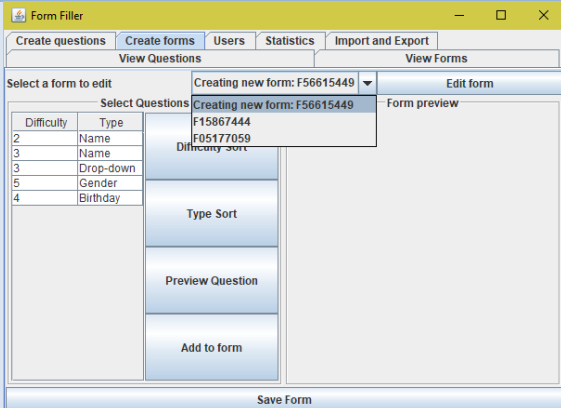
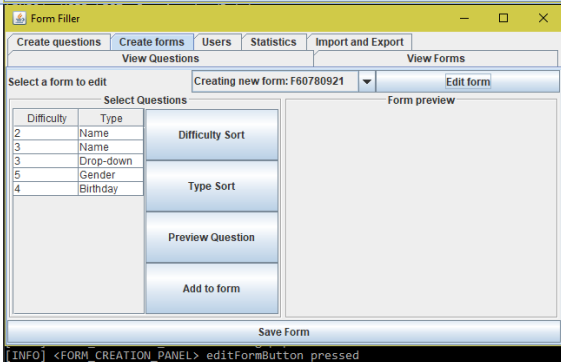
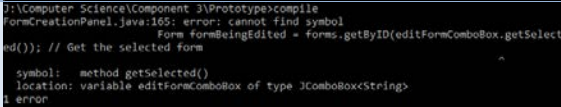
Test	Stage	Result Problem	Evidence	Action and Justification (If Needed)	Evidence of Action (If Needed)
1	Proto	Symbol not found error because there is no instance of QuestionList in the formDisplayPanel	<pre>(GUI)harry@localhost:~/Prototypes\$./compile0.sh FormDisplayPanel.java:101: error: cannot find symbol String questionTypes = questions.getTypes(); // Get the types ^ symbol: variable questions location: class FormDisplayPanel</pre>	Create a new variable of type QuestionList and pass the questions instance from the GUI class to the FormDisplayPanel in the FormDisplayPanel's constructor	<pre>(GUI)harry@localhost:~/Prototypes\$./compile0.sh FormDisplayPanel.java:103: error: incompatible types: String[] cannot be converted to String String questionTypes = questions.getTypes(); // Get the types ^ FormDisplayPanel.java:105: error: cannot find symbol typeCheckBoxes = new JCheckBox[questionTypes.length]; // Create an array large enough to store the check boxes ^ symbol: variable length location: variable questionTypes of type String FormDisplayPanel.java:109: error: array required, but String found typeCheckBoxes[i] = new JCheckBox(questionTypes[i]); // Create a checkbox with the correct name ^ FormDisplayPanel.java:143: error: method filterByType in class FormList cannot be applied to given types; populateTable(forms, filterByType()); ^ required: String[] found: no arguments reason: actual and formal argument lists differ in length 4 errors (GUI)harry@localhost:~/Prototypes\$</pre> Error no longer present in javac output
2	Proto	Type error due to attempting to store a String array in a string	<pre>(GUI)harry@localhost:~/Prototypes\$./compile0.sh FormDisplayPanel.java:103: error: incompatible types: String[] cannot be converted to String String questionTypes = questions.getTypes(); // Get the types ^</pre>	Change the type of the variable questionTypes from a String to a String[]	<pre>(GUI)harry@localhost:~/Prototypes\$./compile0.sh FormDisplayPanel.java:143: error: method filterByType in class FormList cannot be applied to given types; populateTable(forms, filterByType()); ^ required: String[] found: no arguments reason: actual and formal argument lists differ in length 1 error (GUI)harry@localhost:~/Prototypes\$</pre> Error no longer present in javac output
3	Proto	Error when calling filterByType because an array of types to filter by wasn't passed to the method.	<pre>(GUI)harry@localhost:~/Prototypes\$./compile0.sh FormDisplayPanel.java:143: error: method filterByType in class FormList cannot be applied to given types; populateTable(forms, filterByType()); ^ required: String[] found: no arguments reason: actual and formal argument lists differ in length 1 error (GUI)harry@localhost:~/Prototypes\$</pre>	Comment out the line for now as currently it's unnecessary.	<pre>(GUI)harry@localhost:~/Prototypes\$./compile0.sh (GUI)harry@localhost:~/Prototypes\$</pre> Successful compile

4	Proto	There are checkboxes for each type of question above a filter by type button.			
5	Proto	Presses to the filter by type button are detected.	[INFO] <FORM DISPLAY PANEL> typeFilterButton pressed		
6	Proto	Type error because of storing a string in a variable of type string array.	<pre>(GUI)harry@localhost:~/Prototype\$./compile0.sh FormDisplayPanel.java:141: error: incompatible types: String cannot be converted to String[] typesSelected = cB.getText(); ^</pre>	Store the string at index nextLocation in the array. And increment nextLocation after adding the string to the array	<pre>(GUI)harry@localhost:~/Prototype\$./compile0.sh FormDisplayPanel.java:168: error: 'void' type not allowed here populateTable(forms.filterByType(getTypesSelected())); ^ error (GUI)harry@localhost:~/Prototype\$</pre>
7	Proto	Type error when compiling FormDisplayPanel.	<pre>(GUI)harry@localhost:~/Prototype\$./compile0.sh FormDisplayPanel.java:168: error: 'void' type not allowed here populateTable(forms.filterByType(getTypesSelected())); ^ error</pre>	The method filterByType doesn't return anything. Changing void to Form[] and making the method return a Form array should fix this.	<pre>(GUI)harry@localhost:~/Prototype\$./compile0.sh (GUI)harry@localhost:~/Prototype\$</pre> Successful compile

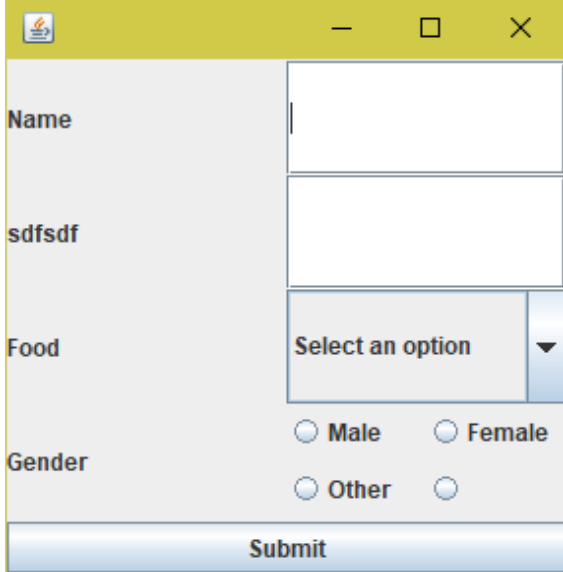
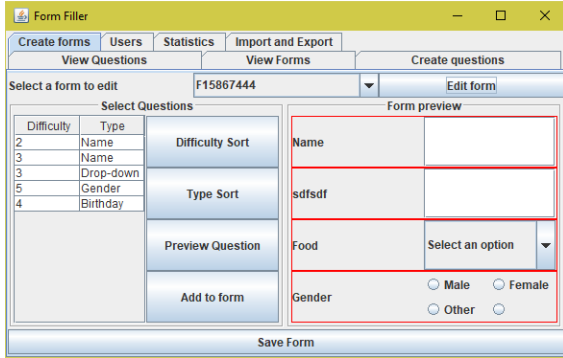
8	Proto	When the filter by types button is pressed the types that have their checkboxes checked are outputted.	 The screenshot shows a software interface with a table of items. The table has columns for 'Phone', 'Description', 'Main', 'Email', 'Name', and 'Difficulty'. Below the table, there are two sections: 'Difficulty sort' and 'Attempt Form'. To the right of the 'Attempt Form' section, there are three checkboxes: 'Email' (checked), 'Name' (unchecked), and 'Phone' (checked). Below these checkboxes is a button labeled 'Filter by type'. The console log at the bottom shows the following output: <pre> [INFO] <FORM_DISPLAY_PANEL> typeFilterButton pressed [Email, Name] [INFO] <FORM_DISPLAY_PANEL> typeFilterButton pressed [Email, Phone] </pre>		
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30: Ability to edit forms

Test	Stage	Result Problem	Evidence	Action and Justification (If Needed)	Evidence of Action (If Needed)
1	Proto	Null pointer exception when running prepareEditPanel() in FormCreationPanel.java.	<pre>[INFO] <FORM_CREATION_PANEL> Running prepareGUI Exception in thread "main" java.lang.NullPointerException at FormCreationPanel.prepareEditPanel(FormCreationPanel.java:66) at FormCreationPanel.prepareMainPanel(FormCreationPanel.java:89) at FormCreationPanel.prepareGUI(FormCreationPanel.java:51) at FormCreationPanel.<init>(FormCreationPanel.java:43) at GUI.prepareGUI(GUI.java:80) at GUI.setup(GUI.java:43) at GUI.<init>(GUI.java:23) at Main.main(Main.java:5)</pre>	The method iterates over the form array from FormList however that array has null items in it and the method doesn't check for null items and assumes all the indexes have proper data in them hence the null pointer exception. Getting a trimmed version of the array from FormList rather than the full array will fix this	 The form creation panel opens successfully
2	Proto	There is an area at the top of the form creation panel that allows the user to select a past form to edit			

3	Proto	All of the forms in the database are shown in the dropdown. (In this case only 2)			
4	Proto	Presses to the edit form button are detected.			
5	Proto	Symbol error when compiling FormCreationPanel.java because the method getSelected() cannot be found.		Change getSelected() to getSelectedItem() because that's what the method is actually called	<pre>J:\Computer Science\Component 3\Prototype>compile J:\Computer Science\Component 3\Prototype></pre> Successful compile

6	Proto	When the edit button is pressed the form is loaded into the edit panel.	Form Filler View Questions View Forms Create questions Create forms Users Statistics Import and Export Select a form to edit F15867444 Edit form Select Questions <table><tr><th>Difficulty</th><th>Type</th></tr><tr><td>2</td><td>Name</td></tr><tr><td>3</td><td>Name</td></tr><tr><td>3</td><td>Drop-down</td></tr><tr><td>5</td><td>Gender</td></tr><tr><td>4</td><td>Birthday</td></tr></table> Difficulty Sort Type Sort Preview Question Add to form Form preview Name sdfsdf Food Select an option ☐ Male ☐ Female ☐ Other ☐ Save Form <pre>[INFO] <FORM_CREATION_PANEL> editFormButton pressed [INFO] <FORM_LIST> Running getFormByID [INFO] <QUESTION_LIST> Running getPanelByID [INFO] <QUESTION_PANEL_LIST> Running getByName [INFO] <FORM_BUILDER> Running add [INFO] <QUESTION_LIST> Running getPanelByID [INFO] <QUESTION_PANEL_LIST> Running getByName [INFO] <FORM_BUILDER> Running add [INFO] <QUESTION_LIST> Running getPanelByID [INFO] <QUESTION_PANEL_LIST> Running getByName [INFO] <FORM_BUILDER> Running add [INFO] <QUESTION_LIST> Running getPanelByID [INFO] <QUESTION_PANEL_LIST> Running getByName [INFO] <FORM_BUILDER> Running add [INFO] <QUESTION_LIST> Running getPanelByID [INFO] <QUESTION_PANEL_LIST> Running getByName [INFO] <FORM_BUILDER> Running add</pre>	Difficulty	Type	2	Name	3	Name	3	Drop-down	5	Gender	4	Birthday	
Difficulty	Type															
2	Name															
3	Name															
3	Drop-down															
5	Gender															
4	Birthday															
7	Proto	A new question can be added to the form	Form Filler View Questions View Forms Create questions Create forms Users Statistics Import and Export Select a form to edit F15867444 Edit form Select Questions <table><tr><th>Difficulty</th><th>Type</th></tr><tr><td>2</td><td>Name</td></tr><tr><td>3</td><td>Name</td></tr><tr><td>3</td><td>Drop-down</td></tr><tr><td>5</td><td>Gender</td></tr><tr><td>4</td><td>Birthday</td></tr></table> Difficulty Sort Type Sort Preview Question Add to form Form preview Name sdfsdf Food Select an option ☐ Male ☐ Female ☐ Other ☐ Day Month Year Save Form	Difficulty	Type	2	Name	3	Name	3	Drop-down	5	Gender	4	Birthday	
Difficulty	Type															
2	Name															
3	Name															
3	Drop-down															
5	Gender															
4	Birthday															

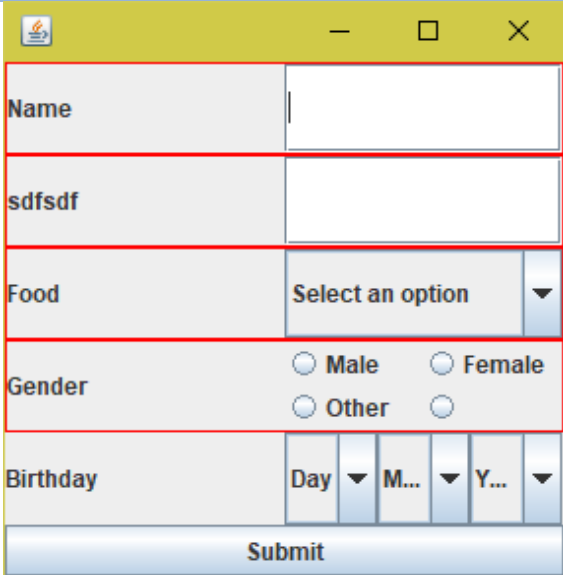
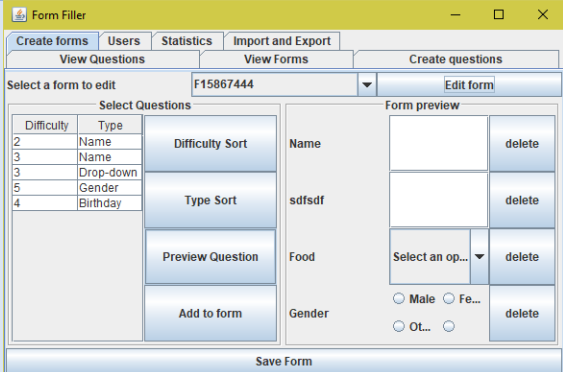
			<pre>[INFO] <QUESTION_SELECTION_PANEL> addButton pressed [INFO] <QUESTION_LIST> Running getPanelByID [INFO] <QUESTION_PANEL_LIST> Running getByID [INFO] <FORM_BUILDER> Running add</pre>		
8	Proto	When a form is edited the new version can be saved and it overwrites the old one.	 Form before editing  Form loaded in editor		

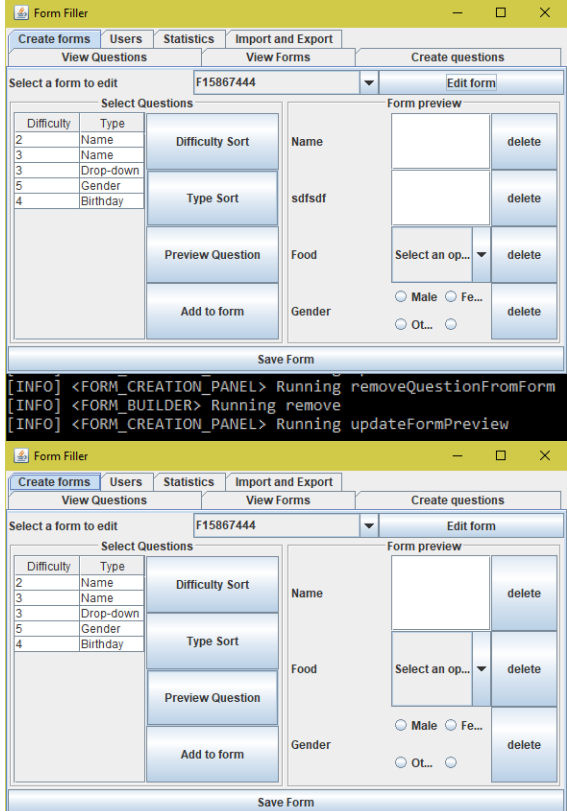
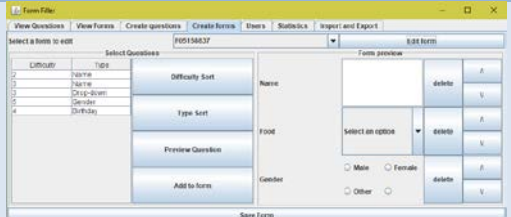
Difficulty	Type
2	Name
3	Name
3	Drop-down
5	Gender
4	Birthday

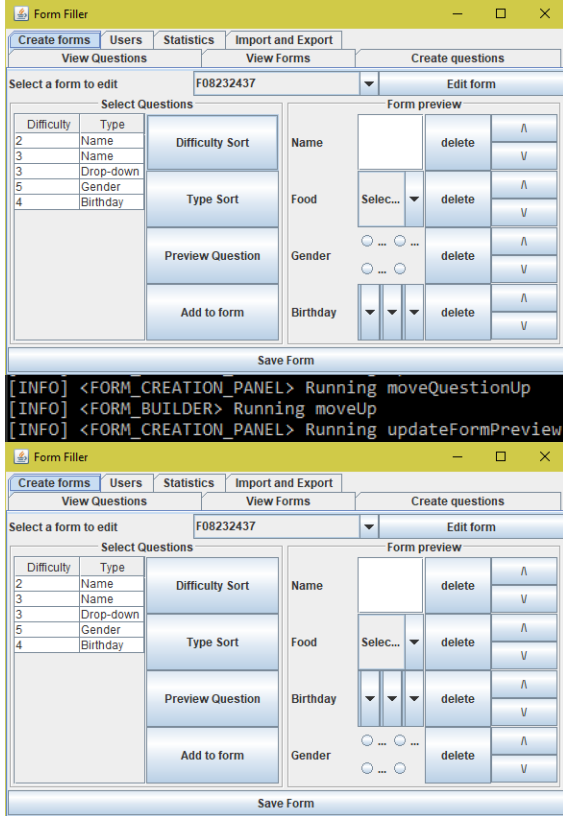
Added new question

Que...	Des...	Main...	Diffic...	Perc...
Q72...	Upd...	Na...	4	0%

After saving form only the edited form exists - no duplicate

			 Edited form		
9	Proto	There is a delete button next to every question in the form			

10	Proto	Questions can be deleted from the form by pressing their delete buttons			
11	Proto	There are up and down arrows for each question however the delete buttons have disappeared.		Add the delete button and the JPanel containing the up and down arrows to the same parent JPanel and then add that JPanel to the questionPreview. This avoids them	 Both the delete button and the up and down arrows are visible

				both (the delete button and up and down arrows) occupying the same space in the border layout - which is what has caused the delete buttons to disappear.	
12	Proto	When the up button is pressed the question is moved up the form.			

13

Proto

When the down button is pressed the question is moved down the form.

The screenshots show the 'Form Filler' application interface. The top screenshot shows the 'Form preview' with questions Name, Food, Birthday, and Gender. The bottom screenshot shows the same form after the 'moveDown' action, where the questions have been shifted down.

```

[INFO] <FORM_CREATION_PANEL> Running moveQuestionDown
[INFO] <FORM_BUILDER> Running moveDown
[INFO] <FORM_CREATION_PANEL> Running updateFormPreview
  
```

14

Proto

When the down arrow is pressed on a question at the bottom the question disappears and there is a null pointer exception

The screenshot shows the 'Form Filler' application interface. The 'Form preview' shows questions Name, Food, Birthday, and Gender. The question at the bottom (Gender) is highlighted, indicating it is the one being interacted with.

This is because the check to see if there is a question below the question wanting to move down is based on `nextQuestionLocation`. However it really should be based on `nextQuestionLocation`

The screenshot shows the 'Form Filler' application interface. The 'Form preview' shows questions Name, Food, Birthday, and Gender. The question at the bottom (Gender) is highlighted, indicating it is the one being interacted with.

Form Filler

Create forms

Users

Statistics

Import and Export

View Questions

View Forms

Create questions

Select a form to edit

F65621127

Edit form

Select Questions

Form preview

Difficulty

Type

2

Name

3

Drop-down

5

Gender

4

Birthday

Difficulty Sort

Type Sort

Preview Question

Add to form

Food

Selec...

delete

Λ

Birthday

Λ

delete

V

Gender

Λ

delete

V

Save Form

```

Exception in thread "AWT-EventQueue-0" java.lang.NullPointerException
    at java.awt.Container.addImpl(Unknown Source)
    at java.awt.Container.add(Unknown Source)
    at FormCreationPanel.updateFormPreview(FormCreationPanel.java:326)
    at FormCreationPanel.moveQuestionDown(FormCreationPanel.java:315)
    at FormCreationPanel.actionPerformed(FormCreationPanel.java:197)
    at javax.swing.AbstractButton.fireActionPerformed(Unknown Source)
    at javax.swing.AbstractButtonHandler.actionPerformed(Unknown Source)
    at javax.swing.DefaultButtonModel.fireActionPerformed(Unknown Source)
    at javax.swing.DefaultButtonModel.setPressed(Unknown Source)
    at javax.swing.plaf.basic.BasicButtonListener.mouseReleased(Unknown Source)
    at java.awt.Component.processMouseEvent(Unknown Source)
    at javax.swing.JComponent.processMouseEvent(Unknown Source)
    at java.awt.Component.processEvent(Unknown Source)
    at java.awt.Container.processEvent(Unknown Source)
    at java.awt.Component.dispatchEventImpl(Unknown Source)
    at java.awt.Container.dispatchEventImpl(Unknown Source)
    at java.awt.Component.dispatchEvent(Unknown Source)
    at java.awt.LightweightDispatcher.retargetMouseEvent(Unknown Source)
    at java.awt.LightweightDispatcher.processMouseEvent(Unknown Source)
    at java.awt.LightweightDispatcher.dispatchEvent(Unknown Source)
    at java.awt.Container.dispatchEventImpl(Unknown Source)
    at java.awt.Window.dispatchEventImpl(Unknown Source)
    at java.awt.Component.dispatchEvent(Unknown Source)
    at java.awt.EventQueue.dispatchEventImpl(Unknown Source)
    at java.awt.EventQueue.access$500(Unknown Source)
    at java.awt.EventQueue$3.run(Unknown Source)
    at java.awt.EventQueue$3.run(Unknown Source)
    at java.security.AccessController.doPrivileged(Native Method)
    at java.security.ProtectionDomain$JavaSecurityAccessImpl.doIntersectionPrivilege
    at java.security.ProtectionDomain$JavaSecurityAccessImpl.doIntersectionPrivilege
    at java.awt.EventQueue$4.run(Unknown Source)
    at java.awt.EventQueue$4.run(Unknown Source)
    at java.security.AccessController.doPrivileged(Native Method)
    at java.security.ProtectionDomain$JavaSecurityAccessImpl.doIntersectionPrivilege
    at java.awt.EventQueue.dispatchEvent(Unknown Source)
    at java.awt.EventDispatchThread.pumpOneEventForFilters(Unknown Source)
    at java.awt.EventDispatchThread.pumpEventsForFilter(Unknown Source)
    at java.awt.EventDispatchThread.pumpEventsForHierarchy(Unknown Source)
    at java.awt.EventDispatchThread.pumpEvents(Unknown Source)
    at java.awt.EventDispatchThread.pumpEvents(Unknown Source)
    at java.awt.EventDispatchThread.run(Unknown Source)

```

ion -1 as there isn't a question at nextQuestionLocation (this means the check returns that there is a question below it when there actually isn't). Changing this should fix it.

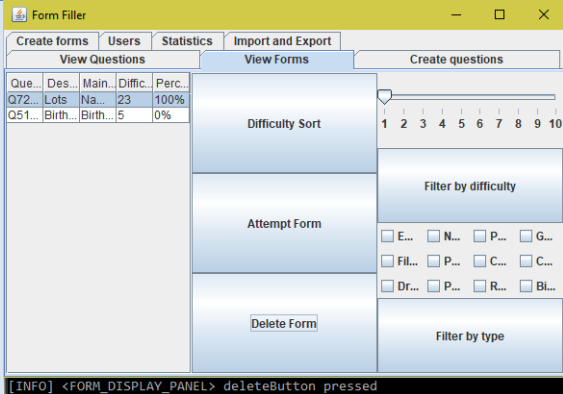
```

[INFO] <FORM_BUILDER> Running moveDown
[INFO] <FORM_CREATION_PANEL> Running updateFormPreview
[INFO] <FORM_CREATION_PANEL> Running moveQuestionDown
[INFO] <FORM_BUILDER> Running moveDown
[INFO] <FORM_CREATION_PANEL> Running updateFormPreview
[INFO] <FORM_CREATION_PANEL> Running moveQuestionDown
[INFO] <FORM_BUILDER> Running moveDown
[INFO] <FORM_CREATION_PANEL> Running updateFormPreview
[INFO] <FORM_CREATION_PANEL> Running moveQuestionDown
[INFO] <FORM_BUILDER> Running moveDown
[INFO] <FORM_CREATION_PANEL> Running updateFormPreview

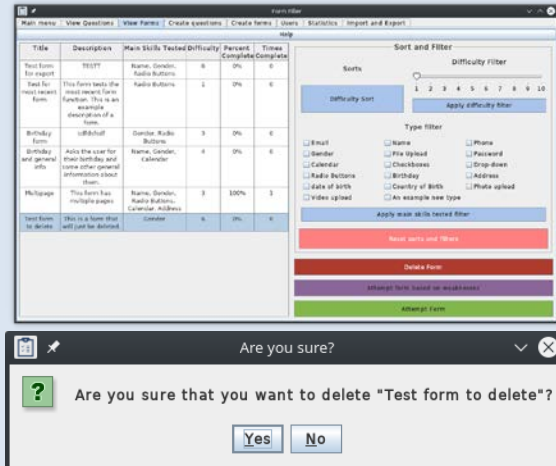
```

No more errors when pressing move down on the bottom question of a form.

31: Ability to delete forms

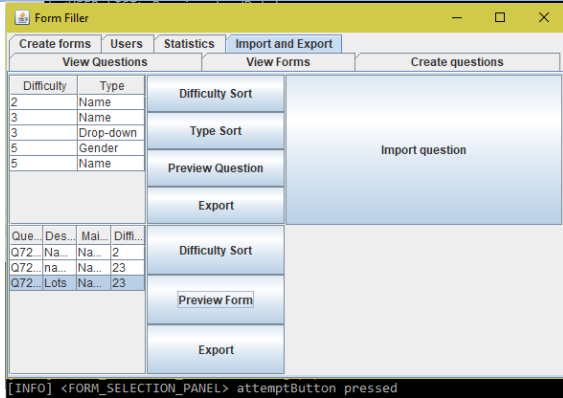
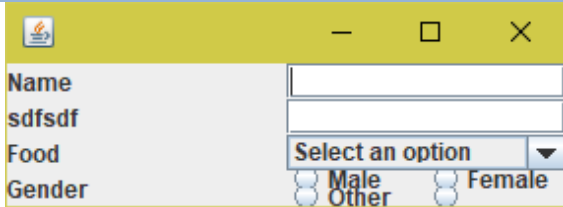
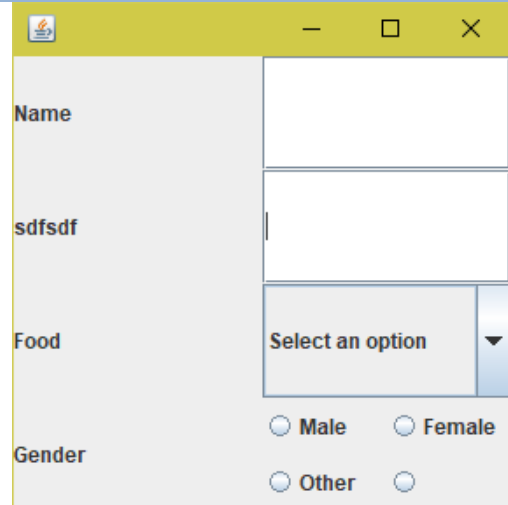
Test	Stage	Result	Problem	Evidence	Action and Justification (If Needed)	Evidence of Action (If Needed)
1	Proto		There is a delete form button in the view forms panel and presses to it are detected.	 The screenshot shows the 'Form Filler' application. The 'View Forms' panel is active, displaying a table of forms with columns: Que..., Des., Main, Diffic., and Perc... The table contains two rows: 'Q72... Lots Na... 23 100%' and 'Q51... Birth... 5 0%'. Below the table is a 'Delete Form' button. The 'Filter by difficulty' and 'Filter by type' sections are also visible.		
2	Proto		Null pointer exception when running deleteForm	<pre>Exception in thread "AWT-EventQueue-0" java.lang.NullPointerException at FormList.removeForm(FormList.java:204) at FormDisplayPanel.actionPerformed(FormDisplayPanel.java:212) at javax.swing.AbstractButton.fireActionPerformed(Unknown Source) at javax.swing.DefaultButtonModel.fireActionPerformed(Unknown Source) at javax.swing.DefaultButtonModel.setPressed(Unknown Source) at javax.swing.plaf.basic.BasicButtonListener.mouseReleased(Unknown Source) at java.awt.Component.processMouseEvent(Unknown Source) at javax.swing.JComponent.processMouseEvent(Unknown Source) at java.awt.Component.processEvent(Unknown Source) at java.awt.Container.processEvent(Unknown Source) at java.awt.Component.dispatchEventImpl(Unknown Source) at java.awt.Container.dispatchEventImpl(Unknown Source) at java.awt.Component.dispatchEvent(Unknown Source) at java.awt.LightweightDispatcher.retargetMouseEvent(Unknown Source) at java.awt.LightweightDispatcher.dispatchEvent(Unknown Source) at java.awt.Container.dispatchEventImpl(Unknown Source) at java.awt.Window.dispatchEventImpl(Unknown Source) at java.awt.Component.dispatchEvent(Unknown Source) at java.awt.EventQueue.dispatchEventImpl(Unknown Source) at java.awt.EventQueue.access\$509(Unknown Source) at java.awt.EventQueue\$3.run(Unknown Source) at java.awt.EventQueue\$3.run(Unknown Source) at java.security.AccessController.doPrivileged(Native Method) at java.security.ProtectionDomain\$JavaSecurityAccessImpl.doIntersectionPrivilege(Unknown Source) at java.security.ProtectionDomain\$JavaSecurityAccessImpl.doIntersectionPrivilege(Unknown Source) at java.awt.EventQueue\$4.run(Unknown Source) at java.awt.EventQueue\$4.run(Unknown Source) at java.security.AccessController.doPrivileged(Native Method) at java.security.ProtectionDomain\$JavaSecurityAccessImpl.doIntersectionPrivilege(Unknown Source) at java.awt.EventQueue.dispatchEvent(Unknown Source) at java.awt.EventDispatchThread.pumpOneEventForFilters(Unknown Source) at java.awt.EventDispatchThread.pumpEventsForFilter(Unknown Source) at java.awt.EventDispatchThread.pumpEventsForHierarchy(Unknown Source) at java.awt.EventDispatchThread.pumpEvents(Unknown Source) at java.awt.EventDispatchThread.pumpEvents(Unknown Source) at java.awt.EventDispatchThread.run(Unknown Source)</pre>	This is caused by copying the null elements from the old array to the new array as each element is checked to see if it's the form to be deleted. When this check is done on a null element a null pointer exception is thrown. Only copying elements up to nextFormLocation will ensure that no null elements are	No null pointer exception

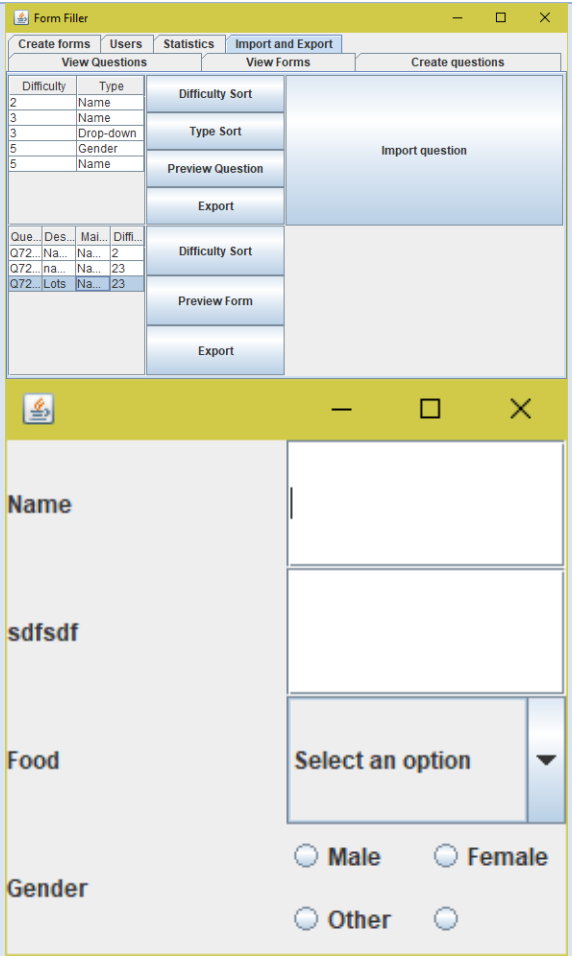
				copied and so will fix this error.	
3	Proto	A form is successfully deleted from the array.	<pre>[INFO] <FORM_DISPLAY_PANEL> deleteButton pressed [F15867444,Q72529010.Q50319011.Q35477436.Q46027134,Lots.Name.Name.Drop-down.Gender,23, Q51327059,Q51308304,Birthday.Birthday,5,null,null,null,null,null,null,null,null,n null,null,null,null,null,null,null,null,null,null,null,null,null,null,n null,null,null,null,null,null,null,null,null,null,null,null,null,null,n null,null,null,null,null,null,null,null,null,null,null,null,null,null,n null,null,null,null,null,null,null,null,null,null,null,null,null,null,n null,null,null,null,null,null,null,null,null,null,null,null,null,null,n null,null,null,null,null,null,null,null,null,null,null,null,null,null,n null,null,null,null,null,null,null,null,null,null,null,null,null,null,n null,null,null,null,null,null,null,null,null,null,null,null,null,null,n null,null,null,null,null,null,null,null,null,null,null,null,null,null,n null,null,null,null,null,null,null,null,null,null,null,null,null,null,n null,null,null,null,null,null,null,null,null,null,null,null,null,null,n null,null,null,null,null,null,null,null,null,null,null,null,null,null,n null,null,null,null,null,null,null,null,null,null,null,null,null,null,n null,null,null,null,null,null,null,null,null,null,null,null,null,null,n null,null,null,null,null,null,null,null,null,null,null,null,null,null,n null,null,null,null,null,null,null,null,null,null,null,null,null,null,n null,null,null,null,null,null,null,null,null,null,null,null,null,null,n null,null,null,null,null,null,null,null,null,null,null,null,null,null,n null,null,null,null,null,null,null,null,null,null,null,null,null,null,n null,null,null,null,null,null,null,null,null,null,null,null,null,null,n null,null,null,null,null,null,null,null,null,null,null,null,null,null,n null,null,null,null,null,null,null,null,null,null,null,null,null,null,n null,null,null,null,null,null,null,null,null,null,null,null,null,null,n null,null,null,null,null,null,null,null,null,null,null,null,null,null,n null,null,null,null,null,null,null,null,null,null,null,null,null,null,n null,null,null,null,null,null,null,null,null,null,null,null,null,null,n null,null,null,null,null,null,null,null,null,null,null,null,null,null,n null,null,null,null,null,null,null,null,null,null,null,null,null,null,n null,null,null,null,null,null,null,null,null,null,null,null,null,null,n null,null,null,null,null,null,null,null,null,null,null,null,null,null,n null,null,null,null,null,null,null,null,null,null,null,null,null,null,n null,null,null,null,null,null,null,null,null,null,null,null,null,null,n null,null,null,null,null,null,null,null,null,null,null,null,null,null,n null,null,null,null,null,null,null,null,null,null,null,null,null,null,n null,null,null,null,null,null,null,null,null,null,null,null,null,null,n null,null,null,null,null,null,null,null,null,null,null,null,null,null,n null,null,null,null,null,null,null,null,null,null,null,null,null,null,n null,null,null,null,null,null,null,null,null,null,null,null,null,null,n null,null,null,null,null,null,null,null,null,null,null,null,null,null,n null,null,null,null,null,null,null,null,null,null,null,null,null,null,n null,null,null,null,null,null,null,null,null,null,null,null,null,null,n null,null,null,null,null,null,null,null,null,null,null,null,null,null,n null,null,null,null,null,null,null,null,null,null,null,null,null,null,n null,null,null,null,null,null,null,null,null,null,null,null,null,null,n null,null,null,null,null,null,null,null,null,null,null,null,null,null,n null,null,null,null,null,null,null,null,null,null,null,null,null,null,n null,null,null,null,null,null,null,null,null,null,null,null,null,null,n null,null,null,null,null,null,null,null,null,null,null,null,null,null,n null,null,null,null,null,null,null,null,null,null,null,null,null,null,n null,null,null,null,null,null,null,null,null,null,null,null,null,null,n null,null,null,null,null,null,null,null,null,null,null,null,null,null,n null,null,null,null,null,null,null,null,null,null,null,null,null,null,n null,null,null,null,null,null,null,null,null,null,null,null,null,null,n null,null,null,null,null,null,null,null,null,null,null,null,null,null,n null,null,null,null,null,null,null,null,null,null,null,null,null,null,n null,null,null,null,null,null,null,null,null,null,null,null,null,null,n null,null,null,null,null,null,null,null,null,null,null,null,null,null,n null,null,null,null,null,null,null,null,null,null,null,null,null,null,n null,null,null,null,null,null,null,null,null,null,null,null,null,null,n null,null,null,null,null,null,null,null,null,null,null,null,null,null,n null,null,null,null,null,null,null,null,null,null,null,null,null,null,n null,null,null,null,null,null,null,null,null,null,null,null,null,null,n null,null,null,null,null,null,null,null,null,null,null,null,null,null,n null,null,null,null,null,null,null,null,null,null,null,null,null,null,n null,null,null,null,null,null,null,null,null,null,null,null,null,null,n null,null,null,null,null,null,null,null,null,null,null,null,null,null,n null,null,null,null,null,null,null,null,null,null,null,null,null,null,n null,null,null,null,null,null,null,null,null,null,null,null,null,null,n null,null,null,null,null,null,null,null,null,null,null,null,null,null,n null,null,null,null,null,null,null,null,null,null,null,null,null,null,n null,null,null,null,null,null,null,null,null,null,null,null,null,null,n null,null,null,null,null,null,null,null,null,null,null,null,null,null,n null,null,null,null,null,null,null,null,null,null,null,null,null,null,n null,null,null,null,null,null,null,null,null,null,null,null,null,null,n null,null,null,null,null,null,null,null,null,null,null,null,null,null,n null,null,null,null,null,null,null,null,null,null,null,null,null,null,n null,null,null,null,null,null,null,null,null,null,null,null,null,null,n null,null,null,null,null,null,null,null,null,null,null,null,null,null,n null,null,null,null,null,null,null,null,null,null,null,null,null,null,n null,null,null,null,null,null,null,null,null,null,null,null,null,null,n null,null,null,null,null,null,null,null,null,null,null,null,null,null,n null,null,null,null,null,null,null,null,null,null,null,null,null,null,n null,null,null,null,null,null,null,null,null,null,null,null,null,null,n null,null,null,null,null,null,null,null,null,null,null,null,null,null,n null,null,null,null,null,null,null,null,null,null,null,null,null,null,n null,null,null,null,null,null,null,null,null,null,null,null,null,null,n null,null,null,null,null,null,null,null,null,null,null,null,null,null,n null,null,null,null,null,null,null,null,null,null,null,null,null,null,n null,null,null,null,null,null,null,null,null,null,null,null,null,null,n null,null,null,null,null,null,null,null,null,null,null,null,null,null,n null,null,null,null,null,null,null,null,null,null,null,null,null,null,n null,null,null,null,null,null,null,null,null,null,null,null,null,null,n null,null,null,null,null,null,null,null,null,null,null,null,null,null,n null,null,null,null,null,null,null,null,null,null,null,null,null,null,n null,null,null,null,null,null,null,null,null,null,null,null,null,null,n null,null,null,null,null,null,null,null,null,null,null,null,null,null,n null,null,null,null,null,null,null,null,null,null,null,null,null,null,n null,null,null,null,null,null,null,null,null,null,null,null,null,null,n null,null,null,null,null,null,null,null,null,null,null,null,null,null,n null,null,null,null,null,null,null,null,null,null,null,null,null,null,n null,null,null,null,null,null,null,null,null,null,null,null,null,null,n null,null,null,null,null,null,null,null,null,null,null,null,null,null,n null,null,null,null,null,null,null,null,null,null,null,null,null,null,n null,null,null,null,null,null,null,null,null,null,null,null,null,null,n null,null,null,null,null,null,null,null,null,null,null,null,null,null,n null,null,null,null,null,null,null,null,null,null,null,null,null,null,n null,null,null,null,null,null,null,null,null,null,null,null,null,null,n null,null,null,null,null,null,null,null,null,null,null,null,null,null,n null,null,null,null,null,null,null,null,null,null,null,null,null,null,n null,null,null,null,null,null,null,null,null,null,null,null,null,null,n null,null,null,null,null,null,null,null,null,null,null,null,null,null,n null,null,null,null,null,null,null,null,null,null,null,null,null,null,n null,null,null,null,null,null,null,null,null,null,null,null,null,null,n null,null,null,null,null,null,null,null,null,null,null,null,null,null,n null,null,null,null,null,null,null,null,null,null,null,null,null,null,n null,null,null,null,null,null,null,null,null,null,null,null,null,null,n null,null,null,null,null,null,null,null,null,null,null,null,null,null,n null,null,null,null,null,null,null,null,null,null,null,null,null,null,n null,null,null,null,null,null,null,null,null,null,null,null,null,null,n null,null,null,null,null,null,null,null,null,null,null,null,null,null,n null,null,null,null,null,null,null,null,null,null,null,null,null,null,n null,null,null,null,null,null,null,null,null,null,null,null,null,null,n null,null,null,null,null,null,null,null,null,null,null,null,null,null,n null,null,null,null,null,null,null,null,null,null,null,null,null,null,n null,null,null,null,null,null,null,null,null,null,null,null,null,null,n null,null,null,null,null,null,null,null,null,null,null,null,null,null,n null,null,null,null,null,null,null,null,null,null,null,null,null,null,n null,null,null,null,null,null,null,null,null,null,null,null,null,null,n null,null,null,null,null,null,null,null,null,null,null,null,null,null,n null,null,null,null,null,null,null,null,null,null,null,null,null,null,n null,null,null,null,null,null,null,null,null,null,null,null,null,null,n null,null,null,null,null,null,null,null,null,null,null,null,null,null,n null,null,null,null,null,null,null,null,null,null,null,null,null,null,n null,null,null,null,null,null,null,null,null,null,null,null,null,null,n null,null,null,null,null,null,null,null,null,null,null,null,null,null,n null,null,null,null,null,null,null,null,null,null,null,null,null,null,n null,null,null,null,null,null,null,null,null,null,null,null,null,null,n null,null,null,null,null,null,null,null,null,null,null,null,null,null,n null,null,null,null,null,null,null,null,null,null,null,null,null,null,n null,null,null,null,null,null,null,null,null,null,null,null,null,null,n null,null,null,null,null,null,null,null,null,null,null,null,null,null,n null,null,null,null,null,null,null,null,null,null,null,null,null,null,n null,null,null,null,null,null,null,null,null,null,null,null,null,null,n null,null,null,null,null,null,null,null,null,null,null,null,null,null,n null,null,null,null,null,null,null,null,null,null,null,null,null,null,n null,null,null,null,null,null,null,null,null,null,null,null,null,null,n null,null,null,null,null,null,null,null,null,null,null,null,null,null,n null,null,null,null,null,null,null,null,null,null,null,null,null,null,n null,null,null,null,null,null,null,null,null,null,null,null,null,null,n null,null,null,null,null,null,null,null,null,null,null,null,null,null,n null,null,null,null,null,null,null,null,null,null,null,null,null,null,n null,null,null,null,null,null,null,null,null,null,null,null,null,null,n null,null,null,null,null,null,null,null,null,null,null,null,null,null,n null,null,null,null,null,null,null,null,null,null,null,null,null,null,n null,null,null,null,null,null,null,null,null,null,null,null,null,null,n null,null,null,null,null,null,null,null,null,null,null,null,null,null,n null,null,null,null,null,null,null,null,null,null,null,null,null,null,n null,null,null,null,null,null,null,null,null,null,null,null,null,null,n null,null,null,null,null,null,null,null,null,null,null,null,null,null,n null,null,null,null,null,null,null,null,null,null,null,null,null,null,n null,null,null,null,null,null,null,null,null,null,null,null,null,null,n null,null,null,null,null,null,null,null,null,null,null,null,null,null,n null,null,null,null,null,null,null,null,null,null,null,null,null,null,n null,null,null,null,null,null,null,null,null,null,null,null,null,null,n null,null,null,null,null,null,null,null,null,null,null,null,null,null,n null,null,null,null,null</pre>		

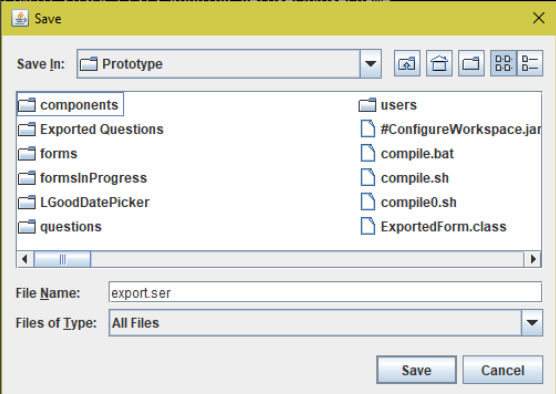
5	Soft Dev	There is a confirmation dialog when the delete form button is pressed.	 The screenshot shows the 'FormEditor' application window. The main window has a menu bar with 'Main menu', 'View Questions', 'View Forms', 'Create questions', 'Create forms', 'Users', 'Statistics', 'Import and Export'. Below the menu is a table with columns: Title, Description, Main Skills Tested, Difficulty, Percent Complete, and Times Completed. The table lists several forms, including 'Test form for report', 'Test for test lesson form', 'Birthday form', 'Birthday and general info', 'Multipage', and 'Test form to delete'. A 'Sort and Filter' panel is on the right, showing a 'Difficulty filter' slider and a 'Type filter' with checkboxes for various form types. A confirmation dialog is open in the foreground, asking 'Are you sure?' and 'Are you sure that you want to delete "Test form to delete"?'. The dialog has 'Yes' and 'No' buttons.		
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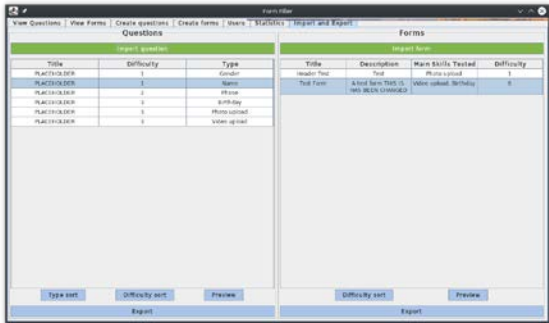
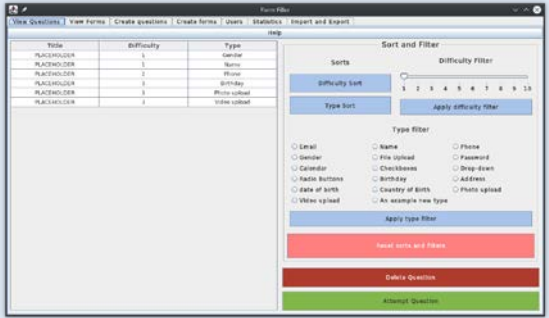
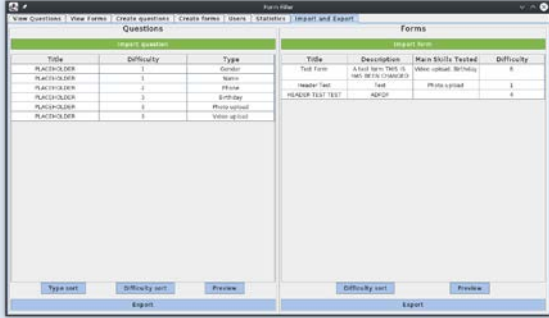
32: Ability to export forms to file

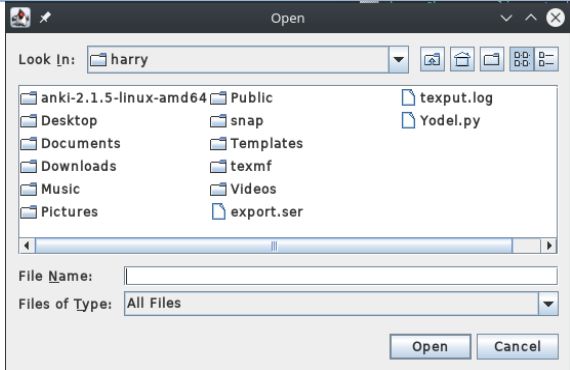
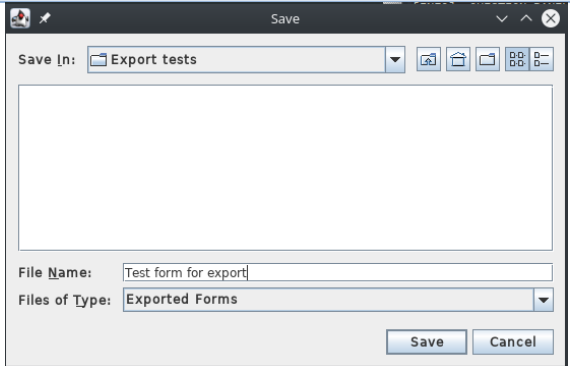
Test	Stage	Result Problem	Evidence	Action and Justification (If Needed)	Evidence of Action (If Needed)
1	Proto	There is a panel that allows the user to select forms	 The screenshot shows the 'FormEditor' application window. The main window has a menu bar with 'Main menu', 'View Questions', 'View Forms', 'Create questions', 'Create forms', 'Users', 'Statistics', 'Import and Export'. Below the menu is a table with columns: Title, Description, Main Skills Tested, Difficulty, Percent Complete, and Times Completed. The table lists several forms, including 'Test form for report', 'Test for test lesson form', 'Birthday form', 'Birthday and general info', 'Multipage', and 'Test form to delete'. A 'Sort and Filter' panel is on the right, showing a 'Difficulty filter' slider and a 'Type filter' with checkboxes for various form types. A panel titled 'Import question' is open in the foreground, showing a list of forms to select from. The panel has buttons for 'Import question', 'Export', 'Preview question', 'Export', 'Import question', 'Export', 'Preview question', and 'Export'.		

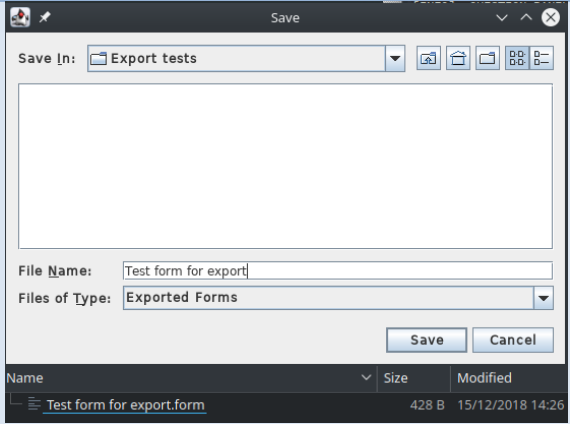
2	Proto	Presses to the preview form button are detected	 (The preview form button is called attemptButton in the code)		
3	Proto	Symbol error compiling ImportExportPanel.java .	<pre> D:\Computer Science\Component 3\Prototype>compile ImportExportPanel.java:196: error: cannot find symbol for (String question : selectedForm.getQuestions()) // For each ^ symbol: method getQuestions() location: variable selectedForm of type Form 1 error D:\Computer Science\Component 3\Prototype> </pre>	Change getQuestions() to getQuestionIDs() because the method is called getQuestionIDs() not getQuestions	<pre> D:\Computer Science\Component 3\Prototype>compile D:\Computer Science\Component 3\Prototype> </pre> Successful compile
4	Proto	When the preview button is pressed the form is displayed however the window is too small and so the form is displayed incorrectly		Increase the height of the window from 100px to 300px	

5	Proto	When the preview form button is pressed the selected form is previewed.	 The screenshot shows the 'Form Filler' application window. The top menu bar includes 'Create forms', 'Users', 'Statistics', and 'Import and Export'. Below the menu, there are tabs for 'View Questions', 'View Forms', and 'Create questions'. The 'View Questions' tab is active, displaying a list of questions with columns for 'Difficulty' and 'Type'. The 'View Forms' tab is also visible, showing buttons for 'Difficulty Sort', 'Type Sort', 'Preview Question', 'Export', 'Import question', 'Difficulty Sort', 'Preview Form', and 'Export'. The 'Create questions' tab is also visible, showing a form with fields for 'Name', 'sdfsdf', 'Food', and 'Gender'. The 'Form Filler' window is titled 'Form Filler' and has a yellow header bar.		The form is displayed properly
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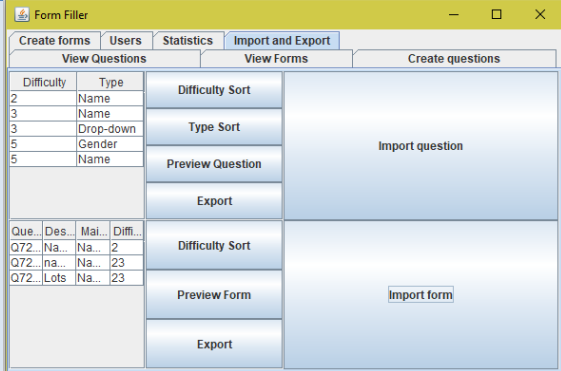
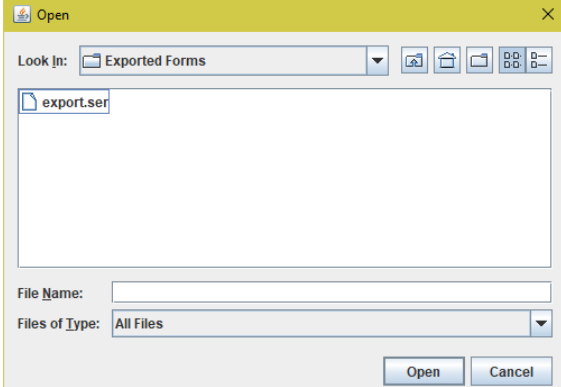
			<pre> [INFO] <FORM_SELECTION_PANEL> attemptButton pressed [INFO] <FORM_SELECTION_PANEL> Running openFormInWindow [INFO] <FORM_LIST> Running getFormByID [INFO] <QUESTION_LIST> Running getPanelByID [INFO] <QUESTION_PANEL_LIST> Running getByID [INFO] <QUESTION_LIST> Running getPanelByID [INFO] <QUESTION_PANEL_LIST> Running getByID [INFO] <QUESTION_LIST> Running getPanelByID [INFO] <QUESTION_PANEL_LIST> Running getByID [INFO] <QUESTION_LIST> Running getPanelByID [INFO] <QUESTION_PANEL_LIST> Running getByID [INFO] <QUESTION_LIST> Running getPanelByID [INFO] <QUESTION_PANEL_LIST> Running getByID </pre>		
6	Proto	When the export form button is pressed a file chooser is opened to allow the user to select where to save the form	 <pre> [INFO] <FORM_SELECTION_PANEL> exportButton pressed [INFO] <IMPORT_EXPORT_PANEL> Running exportForm </pre>		
7	Proto	When the save button is pressed the form is successfully exported	<pre> [INFO] <FORM_SELECTION_PANEL> exportButton pressed [INFO] <IMPORT_EXPORT_PANEL> Running exportForm [INFO] <FORM_LIST> Running getFormByID [INFO] <QUESTION_LIST> Running getQuestionByID [INFO] <QUESTION_LIST> Running getPanelByID [INFO] <QUESTION_PANEL_LIST> Running getByID [INFO] <QUESTION_LIST> Running getQuestionByID [INFO] <QUESTION_LIST> Running getPanelByID [INFO] <QUESTION_PANEL_LIST> Running getByID [INFO] <QUESTION_LIST> Running getQuestionByID [INFO] <QUESTION_LIST> Running getPanelByID [INFO] <QUESTION_PANEL_LIST> Running getByID [INFO] <QUESTION_LIST> Running getQuestionByID [INFO] <QUESTION_LIST> Running getPanelByID [INFO] <QUESTION_PANEL_LIST> Running getByID [INFO] <QUESTION_LIST> Running getQuestionByID [INFO] <QUESTION_LIST> Running getPanelByID [INFO] <QUESTION_PANEL_LIST> Running getByID [INFO] <IMPORT_EXPORT_PANEL> Form exported! </pre> export.ser 30/07/2018 13:55 SER File 17 KB		

8	Soft Dev	The forms section of the import export panel has the correct design.			
9	Soft Dev	When the import export tab is selected the form table refreshes.	 Tab not selected  Tab selected <pre>[INFO] <SELECT_QUESTIONS_PANEL> Running populateTable [INFO] <SELECT_FORMS_PANEL> Running populateTable</pre> Table refreshed		

10	Soft Dev	If the user presses cancel when exporting a form the export process is stopped.	 [INFO] <IMPORT_EXPORT_PANEL> Exporting form cancelled		
11	Soft Dev	When a form is exported it's now exported as a text string to file.	<pre> P35946309.s header:077076293_Q36849304_q20678054_TESTIT_Raw:Sender:Radio Buttons,6,Test form for export,false,true,true:truesha </pre> This is a small section of the string the full string is too long to display.		
12	Soft Dev	Rather than a form being exported with the file name "export" the file name is automatically set to the title of the form			

13	Soft Dev	The .form extension is automatically added by the system.	 Here you can see the form exported.		
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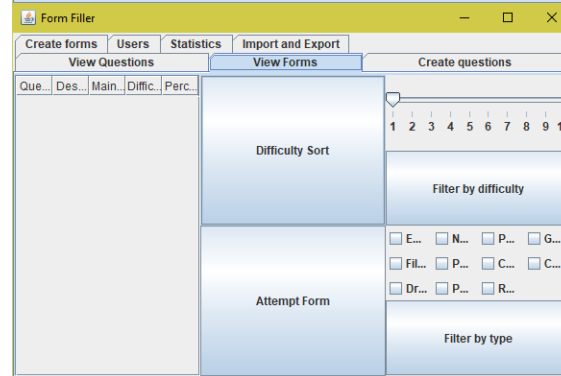
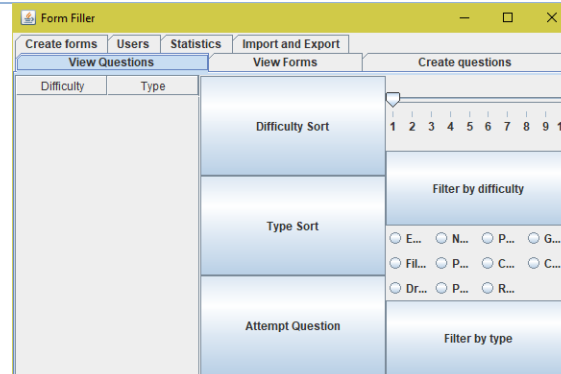
33: Ability to import forms from file

Test	Stage	Result Problem	Evidence	Action and Justification (If Needed)	Evidence of Action (If Needed)
1	Proto	There is an import form button and presses to it are detected			
2	Proto	When the import form button is pressed a file chooser opens, allowing the user to select a form to import.			
3	Proto	An exported ExportedForm object is successfully loaded from file	<pre>[INFO] <IMPORT_EXPORT_PANEL> importFormButton pressed [INFO] <IMPORT_EXPORT_PANEL> Running importForm [INFO] <IMPORT_EXPORT_PANEL> importFormButton pressed [INFO] <IMPORT_EXPORT_PANEL> Running importForm ExportedForm@61687235</pre>		

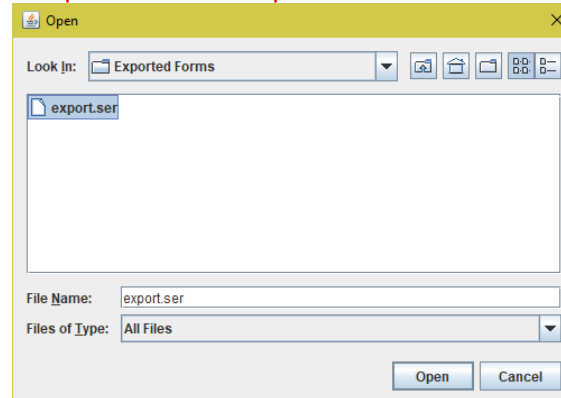
4

Proto

An exported form is imported along with all the necessary questions.



No questions or forms present in the databases



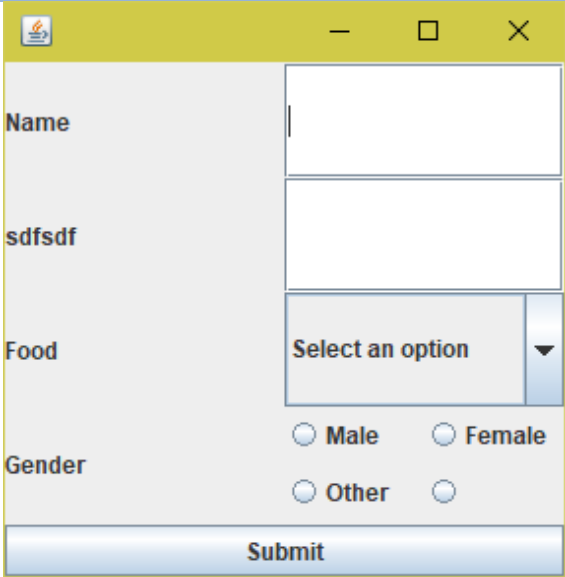
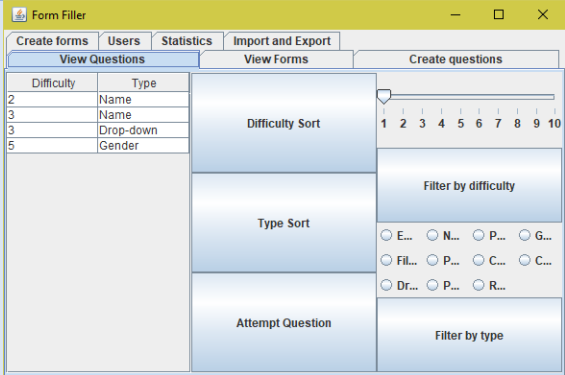
```
[INFO] <IMPORT_EXPORT_PANEL> importFormButton pressed
[INFO] <IMPORT_EXPORT_PANEL> Running importForm
[INFO] <QUESTION_LIST> Running getQuestionByID
[INFO] <QUESTION_PANEL_LIST> Running addQuestionPanel
[INFO] <QUESTION_LIST> Successfully added question and questionPanel
[INFO] <QUESTION_LIST> Running getQuestionByID
[INFO] <QUESTION_PANEL_LIST> Running addQuestionPanel
[INFO] <QUESTION_LIST> Successfully added question and questionPanel
[INFO] <QUESTION_LIST> Running getQuestionByID
[INFO] <QUESTION_PANEL_LIST> Running addQuestionPanel
[INFO] <QUESTION_LIST> Successfully added question and questionPanel
[INFO] <QUESTION_LIST> Running getQuestionByID
[INFO] <QUESTION_PANEL_LIST> Running addQuestionPanel
[INFO] <QUESTION_LIST> Successfully added question and questionPanel
[INFO] <QUESTION_LIST> Running writeDatabase
[INFO] <QUESTION_PANEL_LIST> Running writeDatabase
[INFO] <FORM_LIST> Running writeDatabase
[INFO] <IMPORT_EXPORT_PANEL> Form imported!
```

Form imported

The top screenshot shows the 'Form Filler' application with the 'View Questions' tab selected. It displays a table with columns 'Difficulty' and 'Type'. The table contains four rows of data. To the right of the table are three filter buttons: 'Difficulty Sort', 'Type Sort', and 'Attempt Question'. Further right are two filter sections: 'Filter by difficulty' with a range selector from 1 to 10, and 'Filter by type' with radio buttons for various question types.

The bottom screenshot shows the same application with the 'View Forms' tab selected. It displays a table with columns 'Que...', 'Des...', 'Main...', 'Diffic...', and 'Perc...'. The first row of data is visible. To the right of the table are three filter buttons: 'Difficulty Sort', 'Type Sort', and 'Attempt Form'. Further right are two filter sections: 'Filter by difficulty' with a range selector from 1 to 10, and 'Filter by type' with checkboxes for various question types.

The questions and form are now present in the databases

					
5	Proto	Questions that are already in the questions database aren't imported when importing the form to avoid duplicates.	The form can be attempted  All questions are present		

No form present

```
[INFO] <IMPORT_EXPORT_PANEL> importFormButton pressed
[INFO] <IMPORT_EXPORT_PANEL> Running importForm
[INFO] <QUESTION_LIST> Running getQuestionByID
[INFO] <QUESTION_LIST> Running getQuestionByID
[INFO] <QUESTION_LIST> Running getQuestionByID
[INFO] <QUESTION_LIST> Running getQuestionByID
[INFO] <QUESTION_LIST> Running writeDatabase
[INFO] <QUESTION_PANEL_LIST> Running writeDatabase
[INFO] <FORM_LIST> Running writeDatabase
[INFO] <IMPORT_EXPORT_PANEL> Form imported!
```

Form imported

Que...	Des...	Main...	Diffic...	Perc...
Q72...	Lots	Na...	23	0%

Form present in form database

Questions aren't duplicated

6

Soft Dev

When a form is imported that has the same id of an existing form (is the same form) the form in the database is overwritten with the imported one.

The form highlighted is edited. So that the text says new version.

[INFO] <IMPORT_EXPORT_PANEL> Form exported!

It's then exported.

The screenshot shows the 'Form Editor' application window. It features a table with the following data:

Title	Description	Main Skills Tested	Difficulty	Percent Complete/Complete	Times
Test Form	A test form THIS IS THE NEW VERSION	Video upload, Birthday	5	0%	PLACEHOLD
Another test	test	Photo upload	3	0%	PLACEHOLD
Test Form	test	Photo upload	3	0%	PLACEHOLD

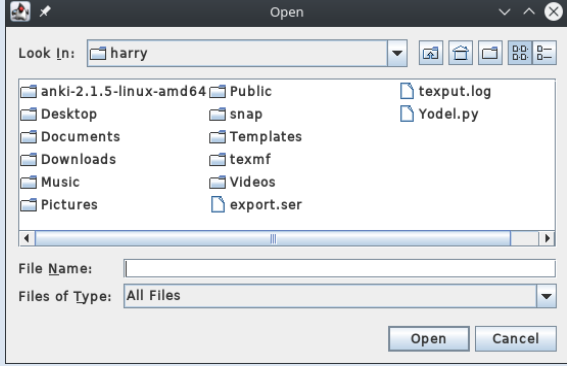
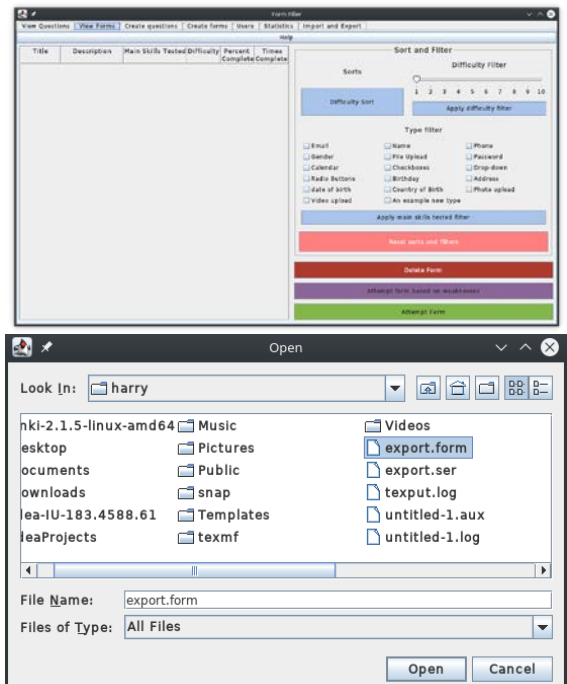
Below the table is a large empty area. To the right is a 'Sort and Filter' panel with a 'Difficulty filter' slider (1-10), a 'Type filter' section with checkboxes for Email, Gender, Calendar, Radio Buttons, Date of Birth, Video upload, Name, Pix Upload, Checkboxes, Birthday, Country of Birth, and Photo upload, and buttons for 'Apply main skills tested filter', 'Reset sorts and filters', 'Delete Form', 'Import/Export Form based on accessibility', and 'Import Form'.

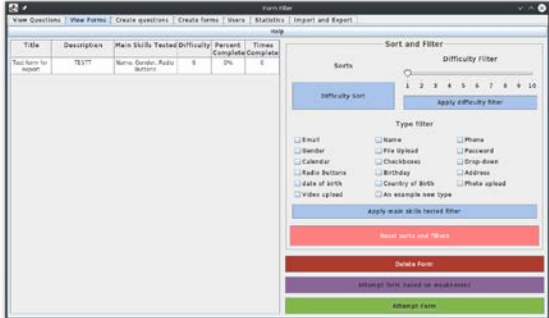
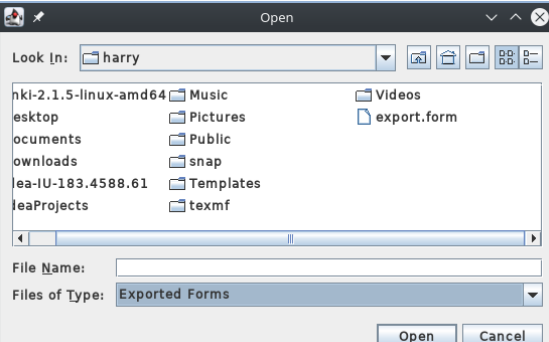
A different version of the system with the old version of the form. The exported form is imported.

[INFO] <IMPORT_EXPORT_PANEL> Form imported!

This screenshot is identical to the one above, showing the 'Form Editor' application with the same table of test forms and the 'Sort and Filter' panel.

The old version was overwritten.

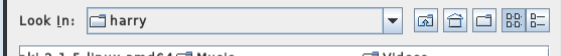
7	Soft Dev	If the user presses cancel when importing a form the import process is stopped.	 [INFO] <IMPORT_EXPORT_PANEL> Importing form cancelled		
8	Soft Dev	An exported form can be successfully imported.			

		<pre>[INFO] <QUESTION_PANEL> Running setup F55904360,a_header.Q77876291.Q36849364.Q08878854.TESTT.Name.Gender.Radio Buttons,6.Test form for export,false,true true,true,V007876291,2.Name,First name#Q77876291[labelFirst name][textfield:none---Q36849364,3.Gender,Ge ndere#Q36849364[labelGender][radioButtons:Male:False, Female:False, Other:False---Q0878854,4.Radio Buttons,Ja me#Q0878854[label:Jame's question][radioButtons:1:false,2:false,3:false,34:false,324:false [INFO] <FORM_LIST> Running getFormById [INFO] <QUESTION_LIST> Running writeToDatabase [INFO] <QUESTION_PANEL_LIST> Running writeToDatabase [INFO] <FORM_LIST> Running writeToDatabase [INFO] <IMPORT_EXPORT_PANEL> Form imported!</pre> The form is successfully imported and can be seen in the view forms table 	
9	Soft Dev	The filechooser that is used to select a form to import only shows files of type exportedForm (these files have extension.form) 	

The form is successfully imported and can be seen in the view forms table

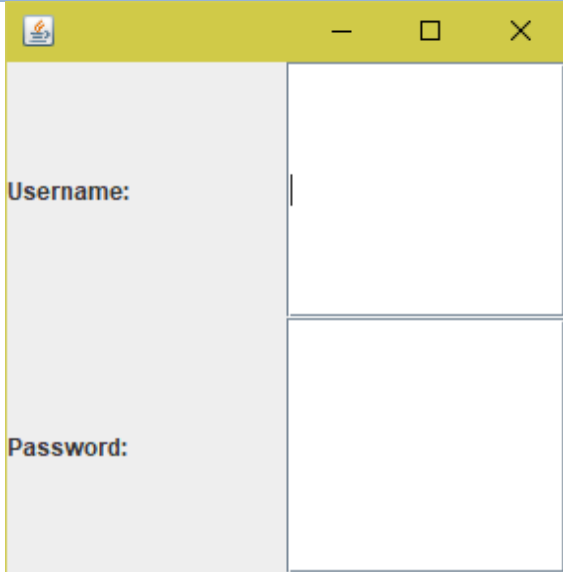
9 Soft Dev

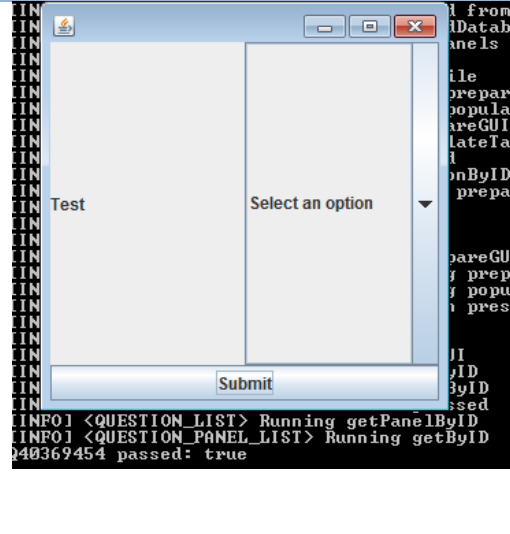
The filechooser that is used to select a form to import only shows files of type `exportedForm` (these files have extension `.form`)

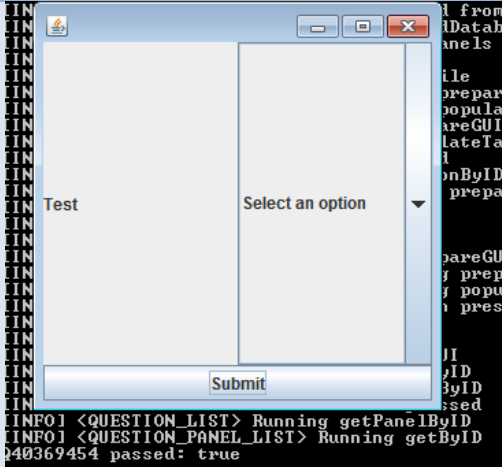
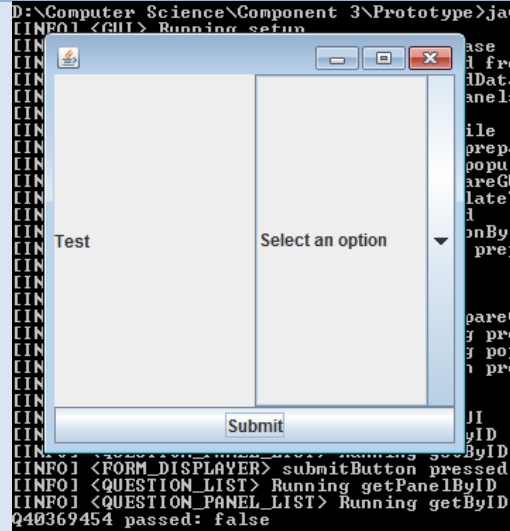
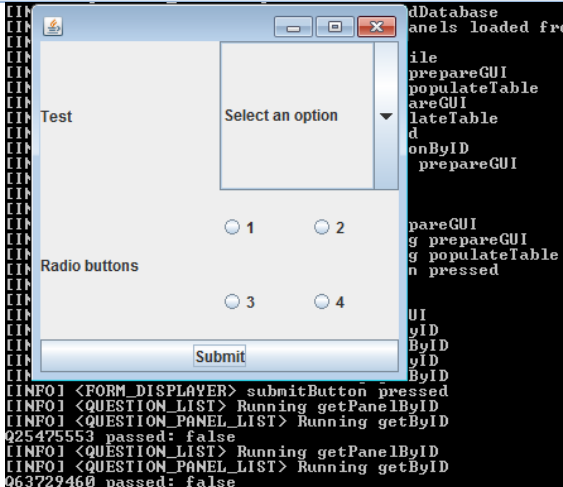


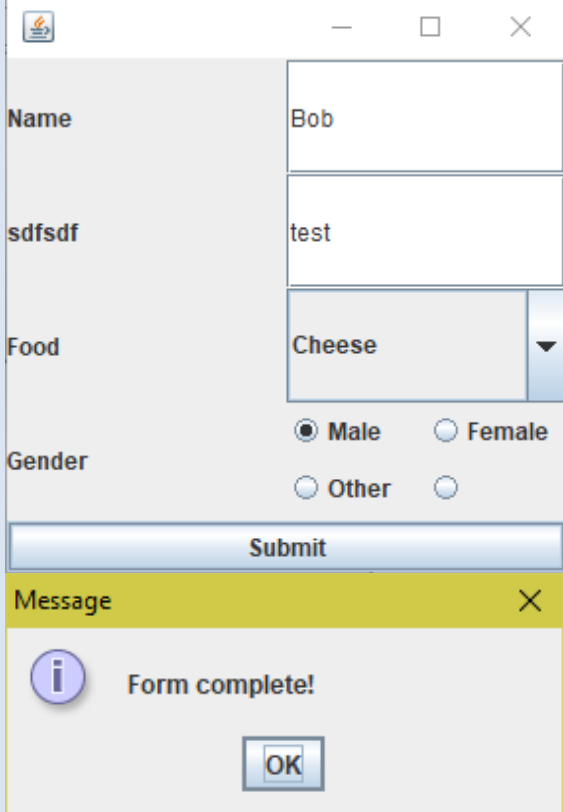
34: Ability to attempt a form

Test	Stage	Result Problem	Evidence	Action and Justification (If Needed)	Evidence of Action (If Needed)
1	Proto	Error compiling the method that opens the JFrame that opens the form because Swing wasn't imported	<pre> D:\Computer Science\Component 3\Prototype>javac *.java GUI.java:34: error: cannot find symbol JFrame formWindow = new JFrame(); // Create a new window ^ symbol: class JFrame location: class GUI GUI.java:34: error: cannot find symbol JFrame formWindow = new JFrame(); // Create a new window ^ symbol: class JFrame location: class GUI GUI.java:36: error: cannot find symbol formWindow.setLayout(new GridLayout(0,1)); // Set a grid layout ^ symbol: class GridLayout location: class GUI GUI.java:39: error: cannot find symbol for (Question q : f.getQuestions()) // For each question in the form ^ symbol: method getQuestions() location: variable f of type Form GUI.java:44: error: cannot find symbol test.setVisible(true); ^ symbol: variable test location: class GUI 5 errors </pre>	Import swing	<pre> D:\Computer Science\Component 3\Prototype>javac *.java GUI.java:42: error: cannot find symbol for (Question q : f.getQuestions()) // For each question in the form ^ symbol: method getQuestions() location: variable f of type Form 1 error </pre> Different compile error now
2	Proto	Error compiling the method that opens the JFrame that opens the form because I hadn't actually programmed the getQuestions() method	<pre> D:\Computer Science\Component 3\Prototype>javac *.java GUI.java:42: error: cannot find symbol for (Question q : f.getQuestions()) // For each question in the form ^ symbol: method getQuestions() location: variable f of type Form 1 error </pre>	Program the get questions method	<pre> D:\Computer Science\Component 3\Prototype>javac *.java GUI.java:42: error: incompatible types: String cannot be converted to Question for (Question q : f.getQuestions()) // For each question in the form ^ 1 error </pre> Method is found.
3	Proto	Method to open the form in a window expects an array of questions but the getQuestions() method actually returns an array of strings question ids.	<pre> D:\Computer Science\Component 3\Prototype>javac *.java GUI.java:42: error: incompatible types: String cannot be converted to Question for (Question q : f.getQuestions()) // For each question in the form ^ 1 error </pre>	Change the variable to a string and use the getPanelByID() method instead	<pre> D:\Computer Science\Component 3\Prototype>javac *.java D:\Computer Science\Component 3\Prototype> </pre> Successful compile

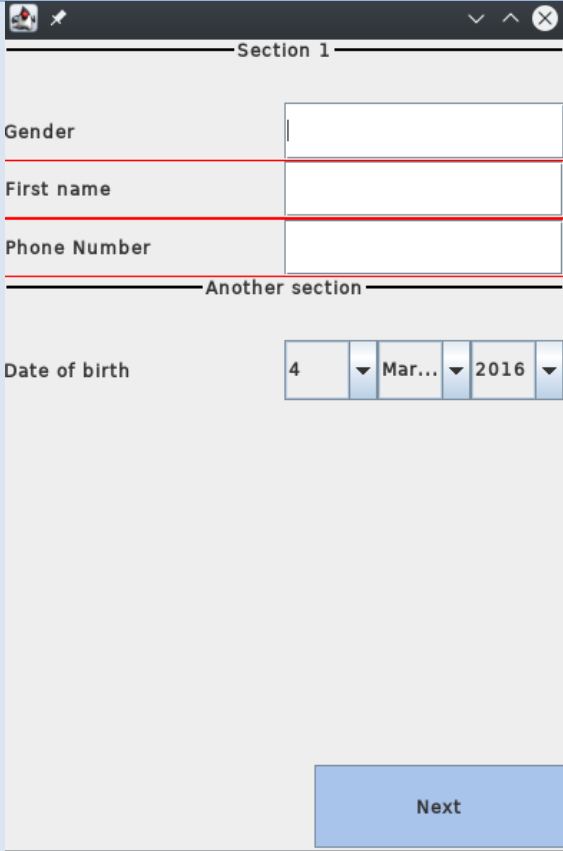
4	Proto	Form is displayed correctly. I.e. the questions that make up the form are displayed as there is no submit button etc. yet.	 <pre> D:\Computer Science\Component 3\Prototype>java Tester [INFO] <FORM_BUILDER> Running add [INFO] <FORM_BUILDER> Running add [INFO] <FORM_BUILDER> Running build [INFO] <FORM_BUILDER> Running trimArray TESTF01 [TEST01, TEST02] [INFO] <GUI> Running setup [INFO] <QUESTION_LIST> Running loadDatabase [INFO] <QUESTION_PANEL_LIST> Running loadDatabase [INFO] <QUESTION_PANEL_LIST> 2 QuestionPanels loaded from file [INFO] <MAIN_MENU> Running prepareGUI [INFO] <GUI> Running openForm [INFO] <QUESTION_LIST> Running getPanelByID [INFO] <QUESTION_PANEL_LIST> Running getByID [INFO] <QUESTION_LIST> Running getPanelByID [INFO] <QUESTION_PANEL_LIST> Running getByID </pre>		
5	Proto	Error compiling FormDisplayer as questions symbol wasn't able to be found	<pre> D:\Computer Science\Component 3\Prototype>javac *.java FormDisplayer.java:55: error: cannot find symbol formPanel.add<questions.getPanelByID<questionID>>; // Ge ^ symbol: variable questions location: class FormDisplayer error: </pre>	Add a questions variable and store the QuestionList instance from the GUI class into it in the FormDisplayer constructor	<pre> D:\Computer Science\Component 3\Prototype>javac *.java D:\Computer Science\Component 3\Prototype> </pre> Successful compile

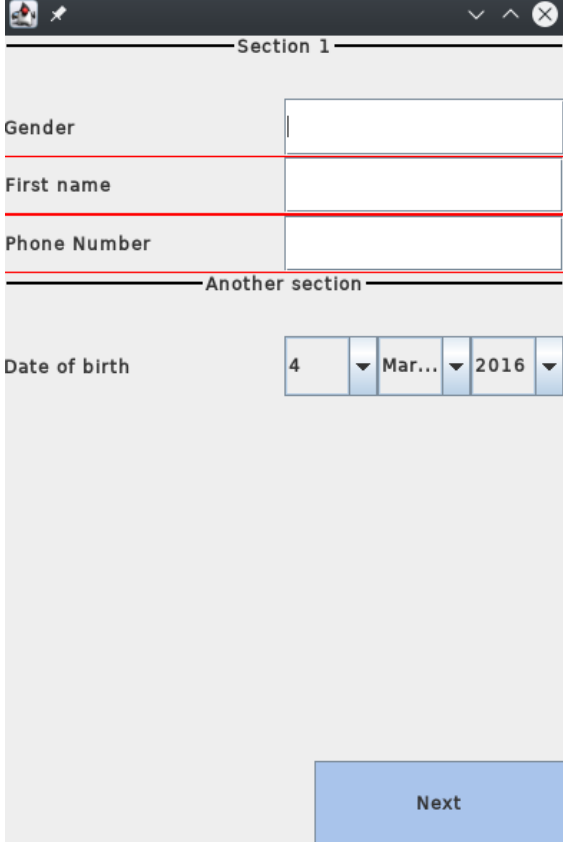
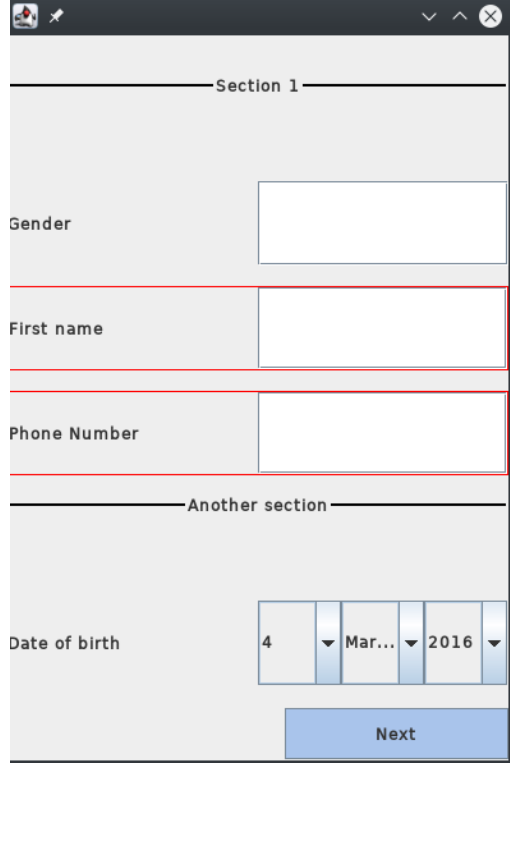
6	Proto	Error compiling Form because of a type error. The array of questions is actually an array of questionIDs hence a String array rather than a Question array. There is also a similar error in FormDisplayer	<pre> D:\Computer Science\Component 3\Prototype>javac *.java Form.java:60: error: incompatible types: String[] cannot be converted to Question[] return questions; ^ D:\Computer Science\Component 3\Prototype>javac *.java FormDisplayer.java:47: error: incompatible types: String cannot be converted to Question for (Question q : f.getQuestionIDs()) // For each question in the form ^ 2 errors </pre>	Change the return type to String[]. Change Question in FormDisplayer to String	<pre> D:\Computer Science\Component 3\Prototype>javac *.java FormDisplayer.java:49: error: cannot find symbol questions.getQuestionPanelID(q).validateAnswers(); ^ symbol: method getQuestionPanelID(String) location: variable questions of type QuestionList FormDisplayer.java:59: error: cannot find symbol validateQuestions(); ^ symbol: method validateQuestions() location: class FormDisplayer 2 errors </pre> Different errors are produced now
7	Proto	Symbol not found errors due to incorrectly typing the method names	<pre> D:\Computer Science\Component 3\Prototype>javac *.java FormDisplayer.java:49: error: cannot find symbol questions.getQuestionPanelID(q).validateAnswers(); ^ symbol: method getQuestionPanelID(String) location: variable questions of type QuestionList FormDisplayer.java:59: error: cannot find symbol validateQuestions(); ^ symbol: method validateQuestions() location: class FormDisplayer 2 errors </pre>	Change the method calls to the correct method names	<pre> D:\Computer Science\Component 3\Prototype>javac *.java D:\Computer Science\Component 3\Prototype> </pre> Successful compile
8	Proto	When the submit button is pressed the form isn't validated. There is no output to do with validation in the console. This is because there were actually no lines that outputted to the console in the code.	 <pre> [INFO] <FORM_DISPLAYER> submit button pressed [INFO] <QUESTION_LIST> Running getPanelByID [INFO] <QUESTION_PANEL_LIST> Running getByID </pre>	Add a line to print the output to the console	 <pre> [INFO] <QUESTION_LIST> Running getPanelByID [INFO] <QUESTION_PANEL_LIST> Running getByID 240369454 passed: true </pre>

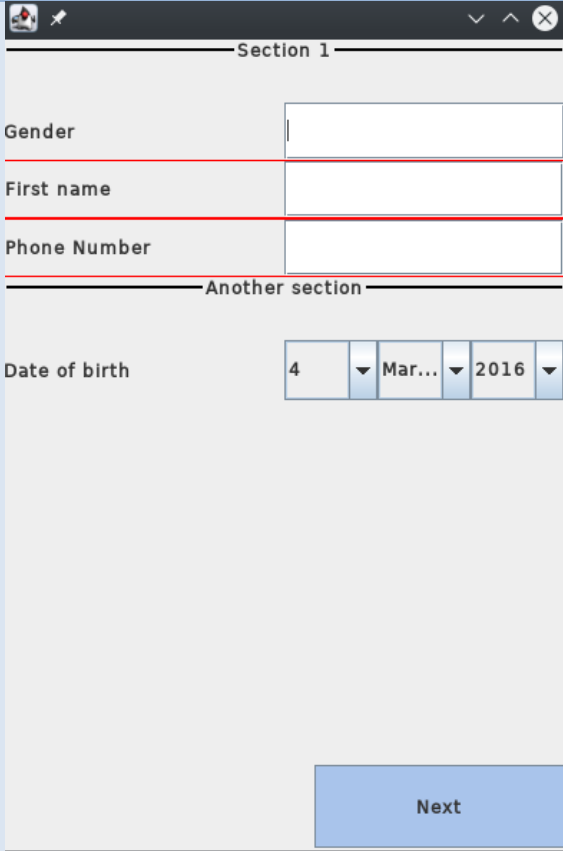
9	Proto	The validation check is passed when it should actually have failed.	 <pre> [INFO] <QUESTION_LIST> Running getPanelById [INFO] <QUESTION_PANEL_LIST> Running getByID 940369454 passed: true </pre>	Change the fail case to when option selected is "select an option" rather than otherwise.	 <pre> D:\Computer Science\Component 3\Prototype>jav [INFO] <GUI> Running setup [INFO] <FORM_DISPLAYER> submitButton pressed [INFO] <QUESTION_LIST> Running getPanelById [INFO] <QUESTION_PANEL_LIST> Running getByID 940369454 passed: false </pre>
10	Proto	All questions in a form are validated when the submit button is pressed.	 <pre> Database panels loaded from file prepareGUI populateTable areGUI lateTable d onById prepareGUI prepareGUI g prepareGUI g populateTable n pressed UI yID ByID yID ByID [INFO] <FORM_DISPLAYER> submitButton pressed [INFO] <QUESTION_LIST> Running getPanelById [INFO] <QUESTION_PANEL_LIST> Running getByID 925475553 passed: false [INFO] <QUESTION_LIST> Running getPanelById [INFO] <QUESTION_PANEL_LIST> Running getByID 963729460 passed: false </pre>		

11	Proto	When a form is filled in successfully and the submit button is pressed (i.e. the form has been completed) a dialog box appears telling the user that they have successfully completed the form.	 The screenshot shows a web form with the following fields: 'Name' (text input with 'Bob'), 'sdfsdf' (text input with 'test'), 'Food' (dropdown menu with 'Cheese'), and 'Gender' (radio buttons for 'Male', 'Female', and 'Other', with 'Male' selected). A 'Submit' button is at the bottom of the form. Below the form, a yellow message dialog box is displayed with the title 'Message' and the text 'Form complete!'. The dialog box has an 'OK' button.		
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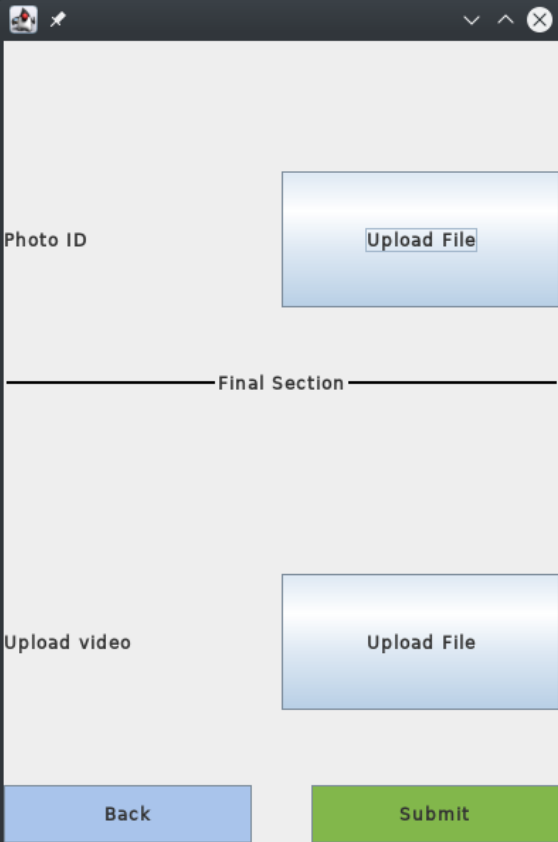
12	Soft Dev	If there is a header in the form it is displayed in the form previewer.			
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13	Soft Dev	There is a next page button.			
----	----------	------------------------------	---	--	--

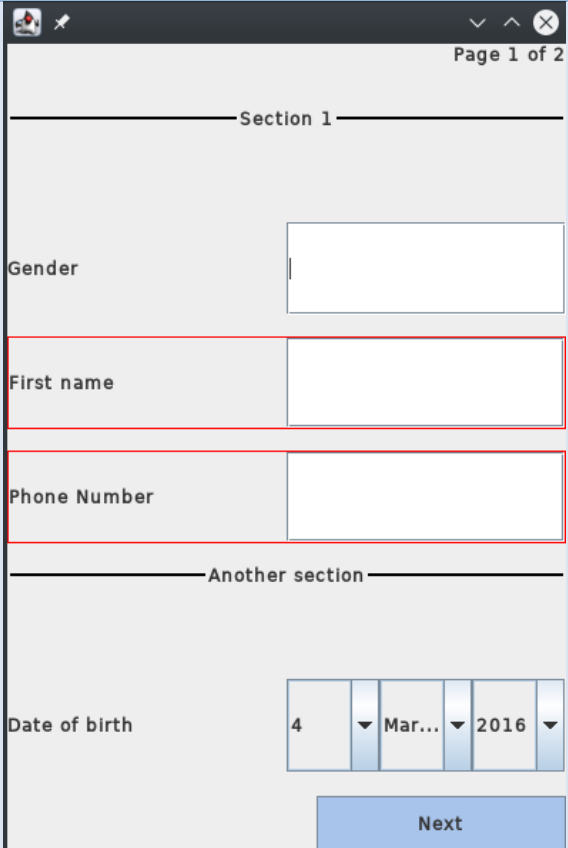
14	Soft Dev	The button panel at the bottom takes up too much space. The questions in the form should be spread across the entire window.	 The screenshot shows a window titled 'Section 1' with three input fields: 'Gender', 'First name', and 'Phone Number'. Below these is a section titled 'Another section' containing a 'Date of birth' field with a date picker showing '4 Mar... 2016'. A blue 'Next' button is positioned at the bottom right of the window.	This is because a BorderLayout was used and BorderLayouts don't respect the preferred sizes of the components. Using a Box layout fixed this.	 This screenshot shows the same form window after a Box layout was used. The components are more compactly arranged, and the 'Next' button is more visible at the bottom right.
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15	Soft Dev	When the next button is pressed the form goes to the next page	 A screenshot of a web browser window displaying a form titled "Section 1". The form has a light gray background and is divided into two sections by a horizontal line. The first section, "Section 1", contains three input fields: "Gender", "First name", and "Phone Number". The second section, "Another section", contains a "Date of birth" field with a date picker showing "4", "Mar...", and "2016". A blue "Next" button is located at the bottom right of the form. The browser window has a standard title bar with minimize, maximize, and close buttons.
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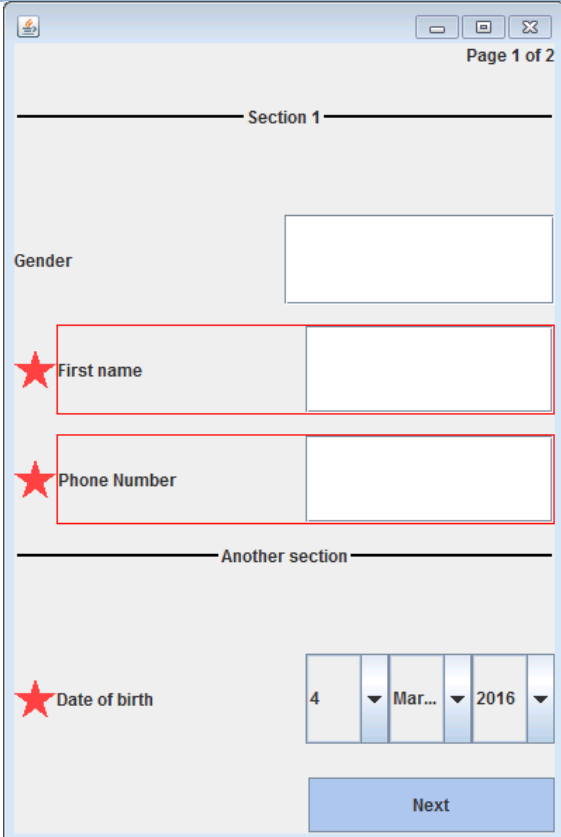
The screenshot shows a web application with a dark header bar containing a logo, a cursor icon, and navigation arrows. The main content area is divided into two sections by a horizontal line. The top section, labeled 'Photo ID', contains a light blue button labeled 'Upload File'. The bottom section, labeled 'Final Section', contains a light blue button labeled 'Upload File', a light blue button labeled 'Back', and a green button labeled 'Submit'.

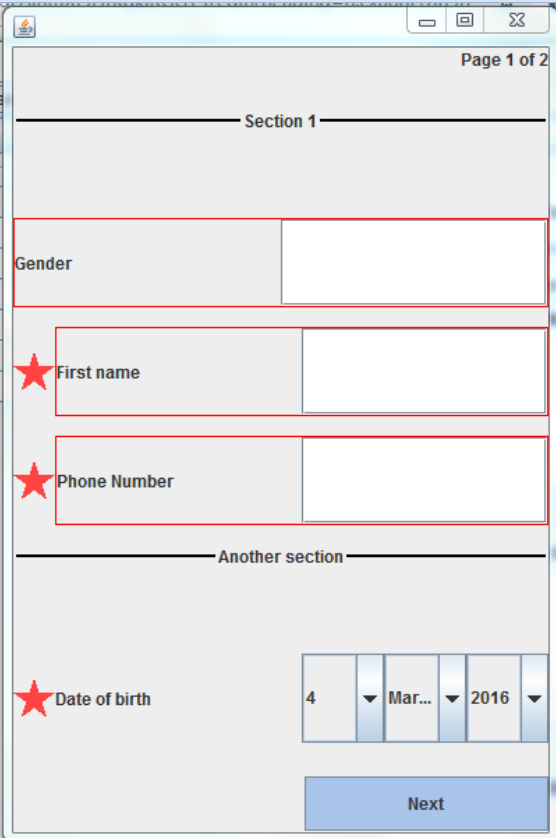
16	Soft Dev	When the back button is pressed, the form goes to the previous page.			
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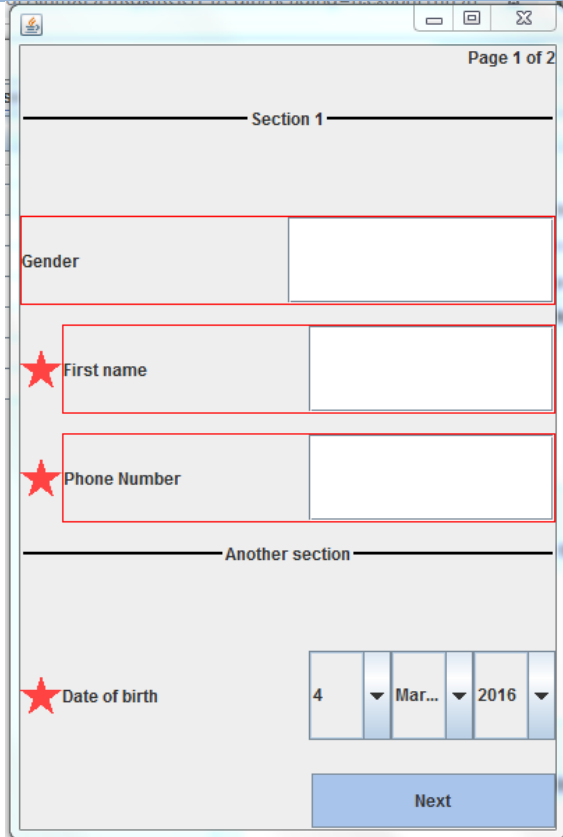
			Section 1 Gender First name Phone Number Another section Date of birth 4 ▼ Mar... ▼ 2016 ▼ Next		
--	--	--	---	--	--

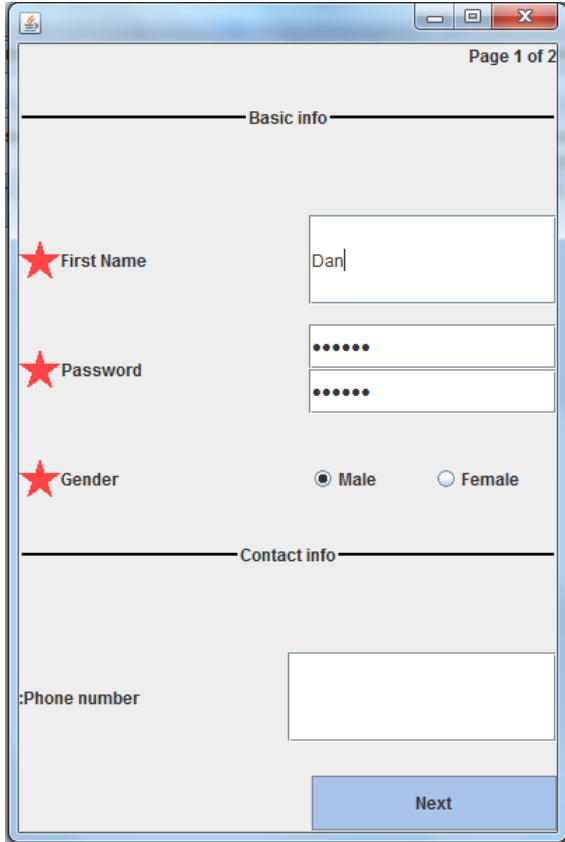
17	Soft Dev	There is a page indicator in the top right that displays the current page.			
----	----------	--	---	--	--

			Page 2 of 2 Photo ID Upload File Final Section Upload video Upload File Back Submit		
--	--	--	---	--	--

18	Soft Dev	Questions that are required have a red star next to them.			
19	Soft Dev	Error compiling FormDisplayer.java because of an compatible type error.	<pre> hes171624308L424-STUDPC-353 MING64 /D/Computer Science/Component 3/SOFTDEV/Proj ect/Form-Filler (attempt-form-redesign) \$./compilew.sh FormDisplayer.java:336: error: incompatible types: Component cannot be converted to JComponent for (JComponent c : pages[0].getComponents()) // Get the compone nts in the page and iterate over them ^ Note: Register.java uses or overrides a deprecated API. Note: Recompile with -Xlint:deprecation for details. 1 error </pre>	JPanel.getCompon ents() returns an array of Components not an array of JComponents therefore c should be of type Component. Changing c to be of type component	<pre> hes171624308L424-STUDPC-353 MING64 /D/Computer Science/Component 3/SOFTDEV/Proj ect/Form-Filler (attempt-form-redesign) \$./compilew.sh Note: Register.java uses or overrides a deprecated API. Note: Recompile with -Xlint:deprecation for details. </pre> Successful compile

20	Soft Dev	If the next button is pressed and all of the questions aren't correctly filled in the the user isn't allowed to move onto the next page of the form.	 Next button is pressed	should fix this.	
----	----------	---	---	------------------	--

					
21	Soft Dev	Failure to compile JValidatedDatePicker.java because pass wasn't declared correctly.	<pre> \$./compilew.sh .\components\JValidatedDatePicker.java:68: error: cannot find symbol pass = false; ^ symbol: variable pass location: class JValidatedDatePicker .\components\JValidatedDatePicker.java:72: error: cannot find symbol pass = true; ^ symbol: variable pass location: class JValidatedDatePicker .\components\JValidatedDatePicker.java:75: error: cannot find symbol return pass; ^ symbol: variable pass location: class JValidatedDatePicker </pre>	Pass wasn't declared as "boolean pass" it was declared as just "pass" therefore this symbol not found error was caused. Correctly declaring pass will fix this	<pre> \$./compilew.sh Note: Register.java uses or overrides a deprecated API. Note: Recompile with -Xlint:deprecation for details. </pre> Successful compile

22	Soft Dev	If a question isn't filled in and it's not required then it automatically passes the validation check. (The other question types aren't included because their validation checks are the same as their validation checks (i.e. there isn't an incorrect way to fill them in))	Phone number  Next is pressed		
----	----------	---	--	--	--

Page 2 of 2

Birthday

Birthday

Day	▼	Month	▼	Year	▼
-----	---	-------	---	------	---

Back Submit

The next screen is progressed to.

Date picker

Page 2 of 2

Contact info

Phone number

07233334605

Birthday

Day ▼ Month ▼ Year ▼

Back Submit

Message

i Form complete!

OK

Password field

Page 2 of 2

Contact info

:Phone number

07233334605

Birthday

3 ▼ Janu... ▼ 2016 ▼

Back Submit

File upload

Page 2 of 2

Contact info

Phone number

Birthday

Photo

Photo ID

Message

 Form complete!

23

Soft Dev

If a not required question is filled in incorrectly then it will fail it's validation check.

(The other question types aren't included because their validation checks are the same as their validation checks (i.e. there isn't an incorrect way to fill them in))

Page 1 of 2

Basic info

★ First Name Dan

★ Password

★ Gender ☒ Male ☐ Female

Contact info

Phone number dsfdsf

Next

Phone number has failed it's check

Date Picker

Page 2 of 2

Contact info

:Phone number 07233334605

Birthday 3 ▼ Month ▼ Year ▼

Back Submit

Date picker failed its validation check

Password field

Page 1 of 2

Basic info

First Name

Password

Gender ☒ Male ☐ Female

Birthday

File chooser

Page 2 of 2

Contact info

Phone number

07232332305

Birthday

3

▼

Febr...

▼

1951

▼

Photo

Photo ID

Upload File

Back

Submit

Fails validation check after incorrect type of file uploaded

24

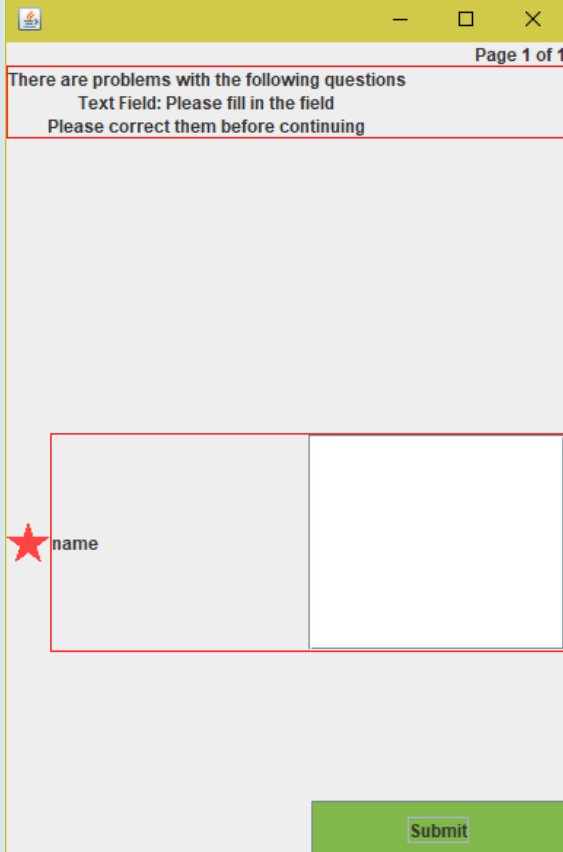
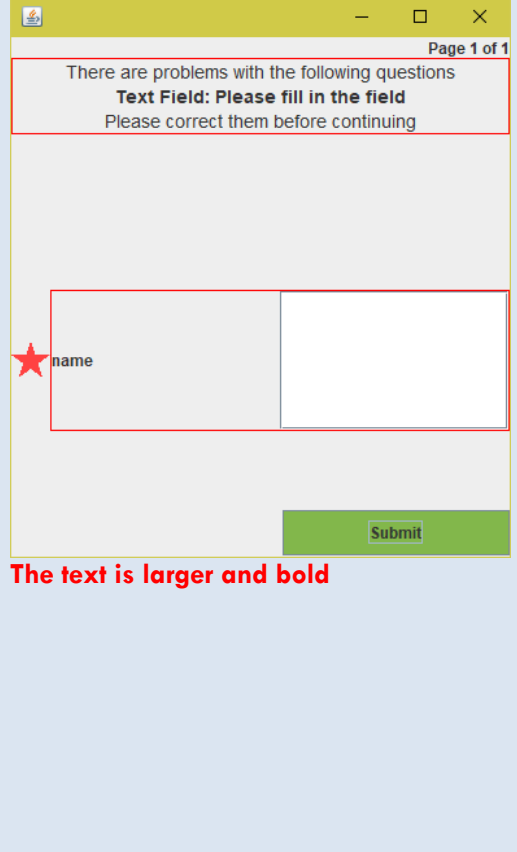
Soft Dev

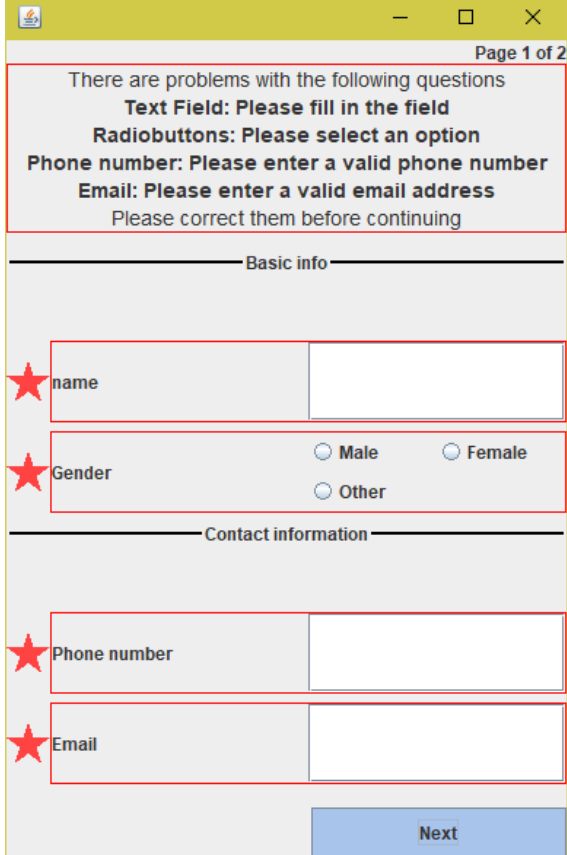
Error when compiling
FormDisplay.java
due to “\” being an
illegal escape
character.

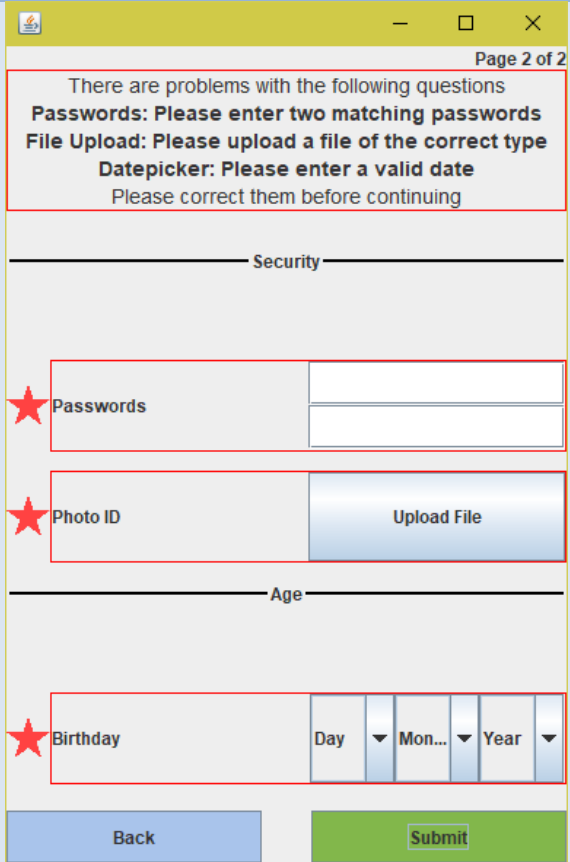
```
C:\Users\Amit> cd C:/Computer Science/Component-1/SOP/TDEx/Project/Face-Pill  
er (attestof-fur-radegid)  
C /comp-test.h  
forOndisplay.java:486: error: illegal escape character  
    before continuing dbvs</div><br/>  
                                     A  
forOndisplay.java:486: error: illegal escape character  
    before continuing dbvs</div><br/>  
                                     A  
errorMessages = "Please correct  
  
the errors
```

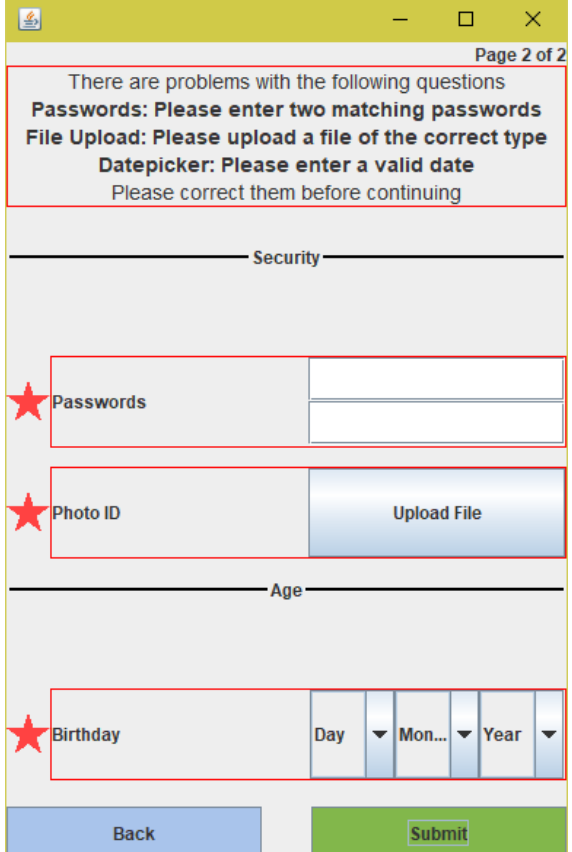
The “\”s should be replaced with “/” as using a forward slash is the correct syntax for html in JLabels

```
harry@harry-PC: ~/Documents/3/Computer Science/Component 3/SOFTDEV/Project/Form-Filler (attempt-form-redesign)
$ ./compile.sh
Note: Register.java uses or overrides a deprecated API.
Note: Recompile with -Xlint:deprecation for details.
```

25	Soft Dev	There is a text box that appears detailing the errors that the user has made in the pages of the form. However the text is too small and is incorrectly centered		The centering can be fixed by making the JLabel center the text within in (currently the text is only centered relative to itself). The text can be made larger by using <code>.setFont(currentFont.deriveFont(Font.PLAIN, 14));</code> on the JLabel where currentFont is the current font that the JLabel is using. This alters the font so that it's larger	 The text is larger and bold
----	----------	---	---	---	---

26	Soft Dev	The correct error messages are displayed for text fields, radio buttons, phone number, and email entry.	 Page 1 of 2 There are problems with the following questions Text Field: Please fill in the field Radiobuttons: Please select an option Phone number: Please enter a valid phone number Email: Please enter a valid email address Please correct them before continuing Basic info ★ name ★ Gender ☐ Male ☐ Female ☐ Other Contact information ★ Phone number ★ Email Next		
----	----------	---	---	--	--

27	Soft Dev	Password entry, file upload, and datepickers show the correct error messages	 The screenshot shows a web form titled "Page 2 of 2" with a yellow header bar. A red-bordered box at the top contains the following error messages: - There are problems with the following questions Passwords: Please enter two matching passwords File Upload: Please upload a file of the correct type Datepicker: Please enter a valid date - Please correct them before continuing The form is divided into sections by horizontal lines: Security: Contains a "Passwords" field with two input boxes, marked with a red star. Photo ID: Contains a "Photo ID" field with an "Upload File" button, marked with a red star. Age: Contains a "Birthday" field with "Day", "Mon...", and "Year" dropdown menus, marked with a red star. At the bottom, there are two buttons: "Back" (blue) and "Submit" (green).
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28	Soft	When an error is corrected it's removed from the list.	 Page 2 of 2 There are problems with the following questions Passwords: Please enter two matching passwords File Upload: Please upload a file of the correct type Datepicker: Please enter a valid date Please correct them before continuing Security ★ Passwords ★ Photo ID Upload File Age ★ Birthday Day Mon... Year Back Submit		
----	------	--	--	--	--

Page 2 of 2

There are problems with the following questions
File Upload: Please upload a file of the correct type
Datepicker: Please enter a valid date
Please correct them before continuing

Security

★ Passwords

.....
.....

★ Photo ID

Upload File

Age

★ Birthday

Day ▼ Mon... ▼ Year ▼

Back Submit

Only password is shown here but it works for all types as the logic is the same.

29

Soft Dev

The attempt form window has the correct icon and its title is the title of the form.

Multipage
Page 1 of 2

Start

★ First name

★ Gender

☐ Male ☐ Female
☐ Other

Next section

★ Jame's question

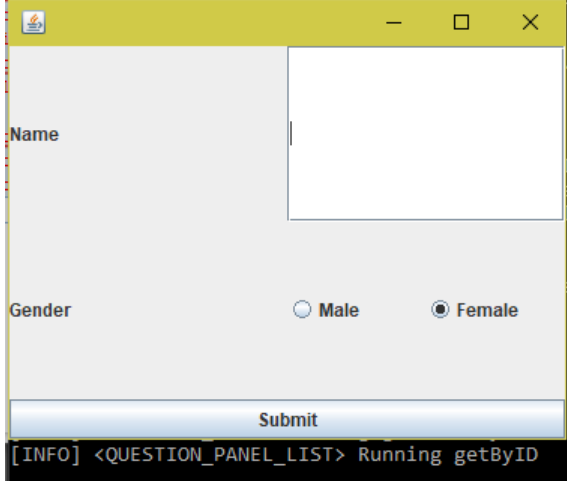
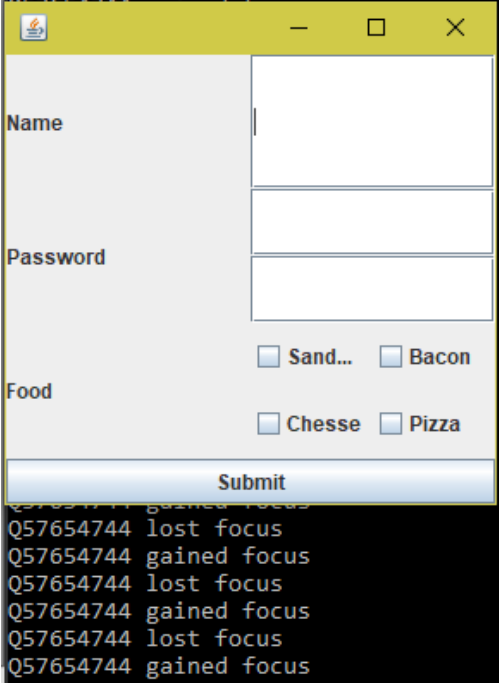
☐ 1 ☐ 2
☐ 3 ☐ 34
☐ 324

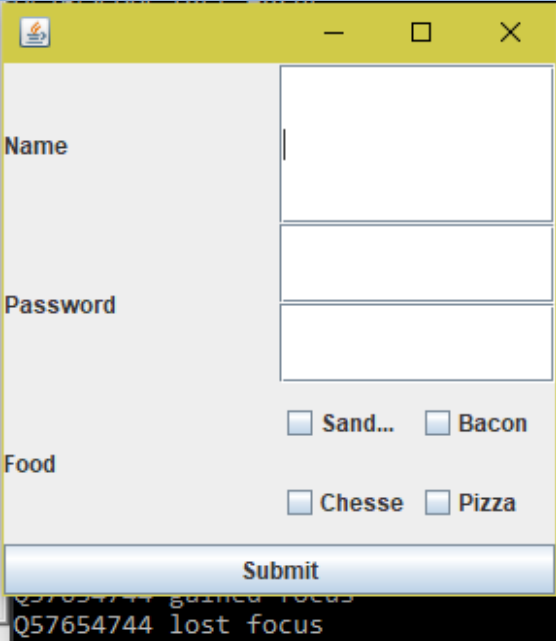
★ Birthday

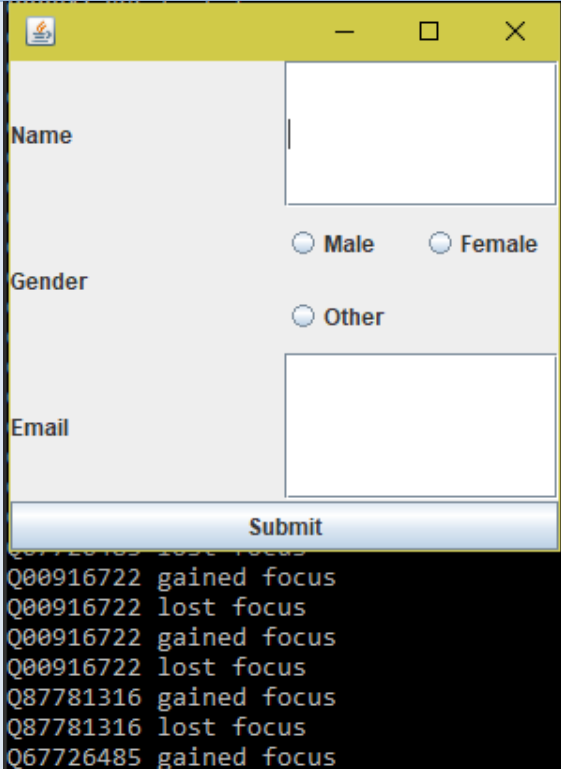
Next

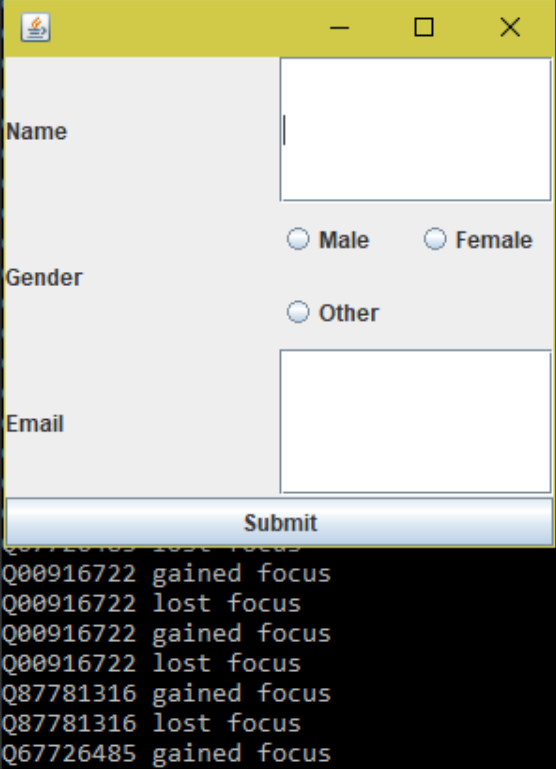
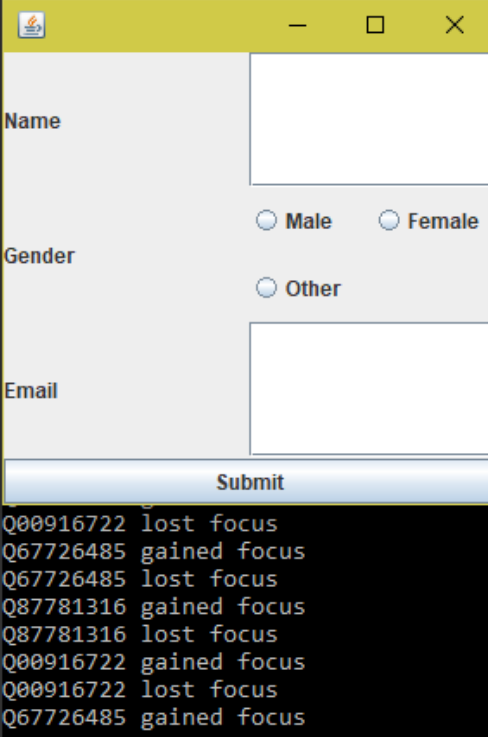
36: The time taken to complete each question should be stored

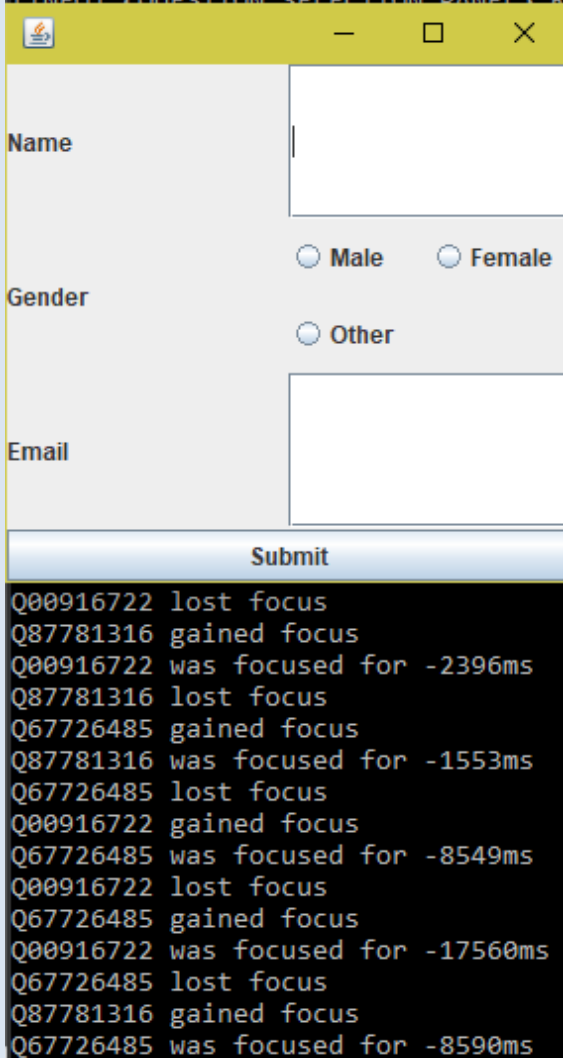
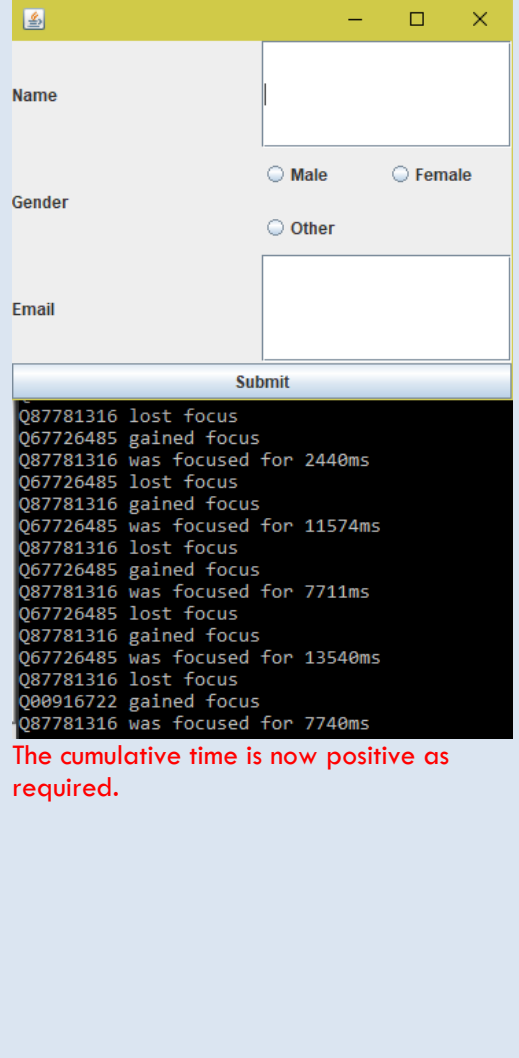
Test	Stage	Result	Problem	Evidence	Action and Justification (If Needed)	Evidence of Action (If Needed)
1	Proto	Symbol not found error when compiling form displayer. The method getQuestionID() couldn't be found.	<pre>J:\Computer Science\Component 3\Prototype>compile FormDisplayer.java:113: error: cannot find symbol System.out.println(evt.getSource().getQuestionID() + " gained focus"); ^ symbol: method getQuestionID() location: class Object FormDisplayer.java:118: error: cannot find symbol System.out.println(evt.getSource().getQuestionID() + " lost focus"); ^ symbol: method getQuestionID() location: class Object 2 errors</pre>	This is because FocusEvent.getSou rce() returns an Object of type Object. Checking that the source of the event is an instance of QuestionPanel and then casting to QuestionPanel will allow the QuestionPanel methods to be accessed.	<pre>J:\Computer Science\Component 3\Prototype>compile J:\Computer Science\Component 3\Prototype></pre> Successful compile	

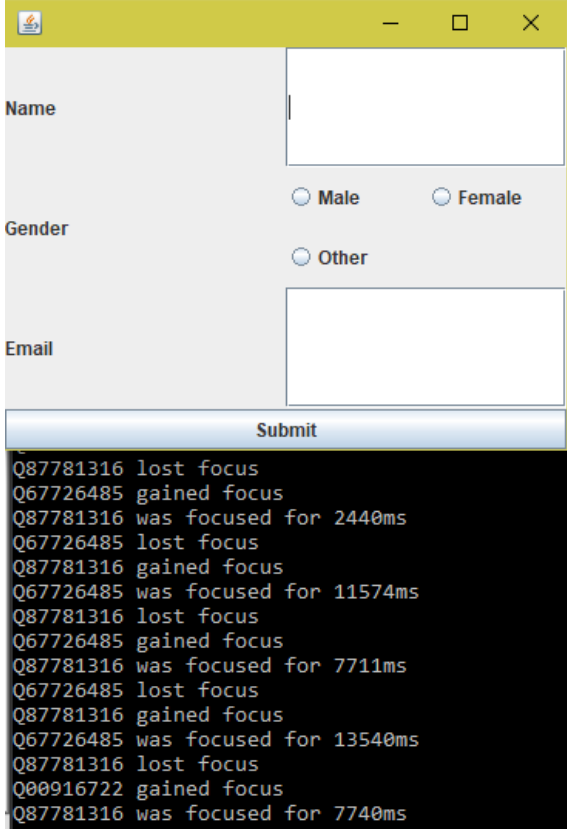
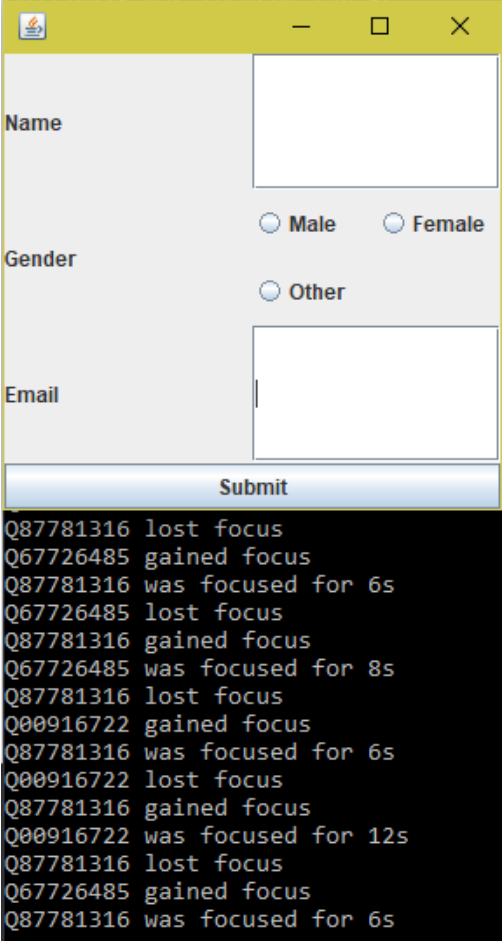
2	Proto	No focus gained or lost events are being triggered when clicking on the different questions of the form.		The questionPanel is based on a JPanel and JPanels don't support detecting focus.  Changes to which question is focused are detected
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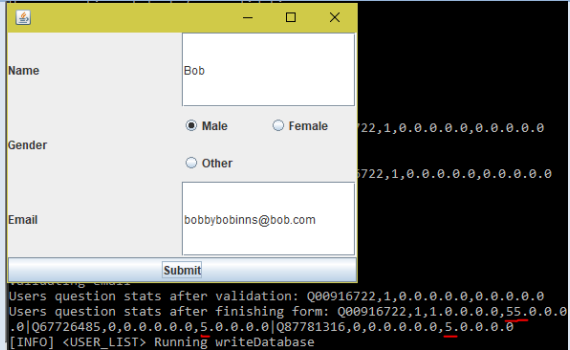
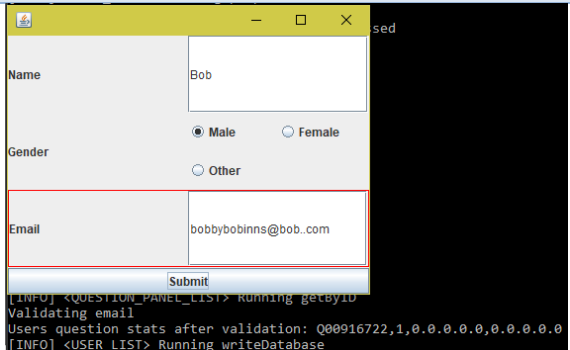
3	Proto	The mouse focus events are only created when the mouse is over a JLabel and not for the other components of a question.	 Q57654744 is said to have lost focus however the cursor is in its text field	Add the mousetlistener to every component that is part of each question rather than just the QuestionPanel	
4	Proto	Error loading the QuestionPanel database after adding the mousetlistener logic because the serialised QuestionPanel class is different to the updated/alterd one with the new logic	<pre>[ERROR] <QUESTION_PANEL_LIST> error loading database java.io.InvalidClassException: QuestionPanel; local class incompatible: stream classdesc serialVersionUID = 6084869774848849290, local class serialVersio nUID = -427236675837759023 [INFO] <QUESTION_PANEL_LIST> 0 QuestionPanels loaded from file</pre>	Delete the panels and remake them	<pre>[INFO] <QUESTION_PANEL_LIST> Running loadDatabase [INFO] <QUESTION_PANEL_LIST> 4 QuestionPanels loaded from file</pre> Newly created QuestionPanels successfully loaded from file

5	Proto	Focus to all the components of each question are correctly detected.	 The screenshot shows a web form with the following elements: Name: A text input field. Gender: Radio buttons for Male, Female, and Other. Email: A text input field. Submit: A button at the bottom of the form. Below the form, a terminal window displays the following focus events: <pre> Q00916722 gained focus Q00916722 lost focus Q00916722 gained focus Q00916722 lost focus Q87781316 gained focus Q87781316 lost focus Q67726485 gained focus </pre>	
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6	Proto	When the mouse moves from one component of a question to another component of the same question a new focus gained and focus lost event is fired even though it's the same question	 You can see it here with Q00916722	When handling the focus lost and gained events check if it's moving between the same question	 Focus lost and gained events are only acted on if the focus changes to a different question.
---	-------	--	--	--	---

7	Proto	The cumulative number of milliseconds that each question has been focused for is outputted everytime focus changes however the time is negative.	 <pre> Q00916722 lost focus Q87781316 gained focus Q00916722 was focused for -2396ms Q87781316 lost focus Q67726485 gained focus Q87781316 was focused for -1553ms Q67726485 lost focus Q00916722 gained focus Q67726485 was focused for -8549ms Q00916722 lost focus Q67726485 gained focus Q00916722 was focused for -17560ms Q67726485 lost focus Q87781316 gained focus Q67726485 was focused for -8590ms </pre>	Rather than doing <code>startTime - currentTime</code> change it to <code>currentTime - startTime</code> to ensure that the result is positive.	 <pre> Q87781316 lost focus Q67726485 gained focus Q87781316 was focused for 2440ms Q67726485 lost focus Q87781316 gained focus Q67726485 was focused for 11574ms Q87781316 lost focus Q67726485 gained focus Q87781316 was focused for 7711ms Q67726485 lost focus Q87781316 gained focus Q67726485 was focused for 13540ms Q87781316 lost focus Q00916722 gained focus Q87781316 was focused for 7740ms </pre> The cumulative time is now positive as required.
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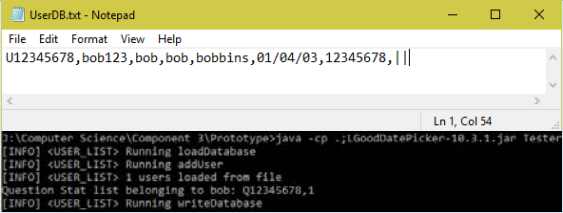
8	Proto	The time is reported in milliseconds but this level of precision isn't necessary. It would be more convenient if it was reported in seconds.	 <pre> Q87781316 lost focus Q67726485 gained focus Q87781316 was focused for 2440ms Q67726485 lost focus Q87781316 gained focus Q67726485 was focused for 11574ms Q87781316 lost focus Q67726485 gained focus Q87781316 was focused for 7711ms Q67726485 lost focus Q87781316 gained focus Q67726485 was focused for 13540ms Q87781316 lost focus Q00916722 gained focus Q87781316 was focused for 7740ms </pre>	Rather than dividing by 1 000 000 when converting the time from nanoseconds divide by 1 000 000 000 to convert it to seconds.	 <pre> Q87781316 lost focus Q67726485 gained focus Q87781316 was focused for 6s Q67726485 lost focus Q87781316 gained focus Q67726485 was focused for 8s Q87781316 lost focus Q00916722 gained focus Q87781316 was focused for 6s Q00916722 lost focus Q87781316 gained focus Q00916722 was focused for 12s Q87781316 lost focus Q67726485 gained focus Q87781316 was focused for 6s </pre> The time is now reported in seconds.
---	-------	---	---	--	---

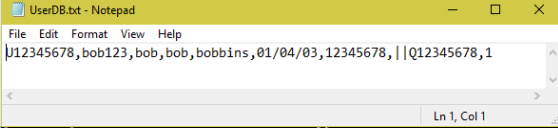
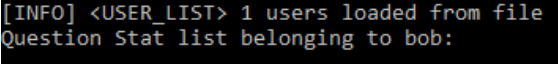
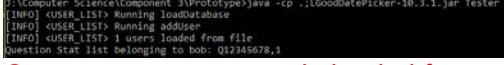
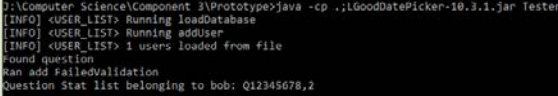
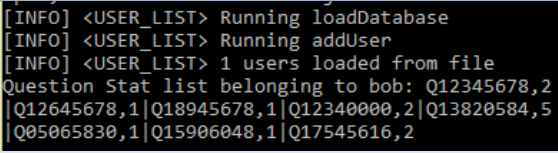
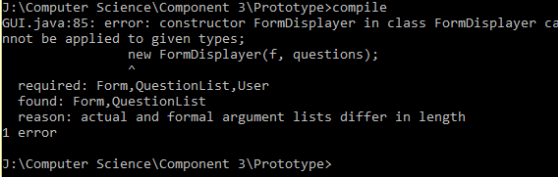
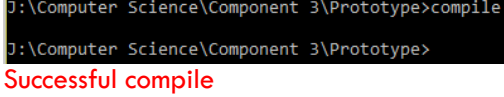
9	Proto	When the user successfully completes a form the time that it took them to complete each question is added to the timeTakenToComplete array in each relevant QuestionStat object.	 The times are underlined here in red.		
10	Proto	The times are only stored when the user successfully completes the form. This is as intended.	 As you can see there are no times stored.		
11	Proto	The times taken to complete each question are successfully written to file	<pre> 12345678,bob123,bob,bob,bobbins,01/04/03,12345678, Q00916722,2,1.1.0.0.0,168.55.0.0.0 Q67726485,1,0.0.0.0.0,6.5.0.0.0 Q87781316,1,0.0.0.0.0,6.5.0.0.0 </pre> This is all one line in the database it's just split up for better readability. The time taken to complete is underlined in red		
12	Proto	The times taken to complete each question are successfully loaded from file	<pre> Users question stats before validation: Q00916722,1,1.1.0.0.0,168.55.0.0.0 Q67726485,0,0.0.0.0.0,6.5.0.0.0 Q87781316,0,0.0.0.0.0,6.5.0.0.0 </pre>		

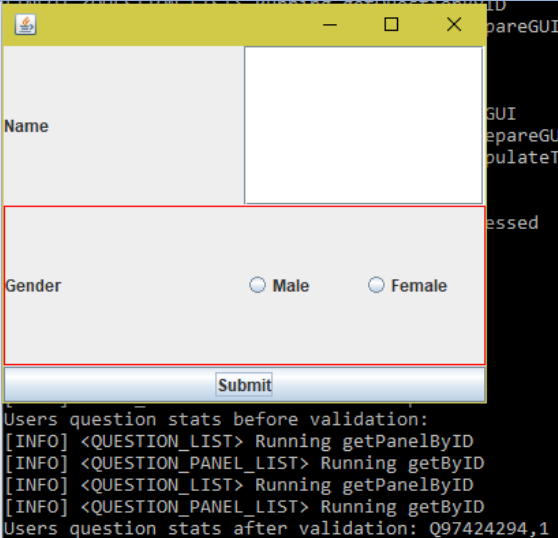
13	Proto	The time for the last question in the form isn't stored if the submit button is immediately pressed after filling it in.	Users question stats .0.0.0,0.0.168.55.0	This is because when the submit button is pressed the questionFocusChange() method isn't called to log to the time. Calling questionFocusChange() when the submit button is pressed will fix this.	Users question stats .1.0.0,58.0.0.0.168 The time is stored now.
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37: The number of times that they've failed the validation check for a question should be stored

Test	Stage	Result Problem	Evidence	Action and Justification (If Needed)	Evidence of Action (If Needed)
1	Proto	A Question stat object is created and then a failed validation check is added and then this change is outputted. The change from a 0 to a 1 shows that the question validation check has been failed once.	D:\Computer Science\Component 3\Prototype>java -cp .;LGoodDatePicker-10.3.1.jar Tester 000000001,0 Added a failed check 000000001,1		
2		The question stat object can be added	D:\Computer Science\Component 3\Prototype>java -cp .;LGoodDatePicker-10.3.1.jar Tester 000000001,0 Added a failed check 000000001,1 Added Question stat to list 000000001,1		

	Proto	to a list of question stat objects.			
3	Proto	Error loading the user database from file when there are no saved question stat objects.	<pre>[INFO] <USER_LIST> Running loadDatabase [ERROR] <USER_LIST> Error loading database java.lang.ArrayIndexOutOfBoundsException: 1 Exception in thread "main" java.lang.ArrayIndexOutOfBoundsException: 0 at GUI.getUser(GUI.java:47) at GUI.setup(GUI.java:32) at GUI.<init>(GUI.java:18) at Main.main(Main.java:5)</pre>	Add a check in the loadDatabase method of UserList to check whether there are any saved question stat objects and if there aren't create a new empty questionStatList rather than attempting to load. Also as is the delimiter between the userData and questionStatList data when calling .split the needs to be escaped so that the line is split properly because is a special character.	<pre>[INFO] <USER_LIST> Running addUser [INFO] <USER_LIST> 1 users loaded from file</pre> User successfully loaded from file
4	Proto	When calling addFailedValidation with a question id that wasn't previously in the questionStatList a questionStat object is automatically created for that id.	<pre>D:\Computer Science\Component 3\Prototype>java -cp .;lGoodDatePicker-10.3.1.jar Tester Question Stat list: Added a failed check for a question that didn't previously exist Question Stat list: Q12345678,1</pre>		
5	Proto	When a the writeDatabase() method of UserList is called the questionStatList for the user is also written to file on the	 <pre>D:\Computer Science\Component 3\Prototype>java -cp .;lGoodDatePicker-10.3.1.jar Tester [INFO] <USER_LIST> Running loadDatabase [INFO] <USER_LIST> Running addUser [INFO] <USER_LIST> 1 users loaded from file Question Stat list belonging to bob: Q12345678,1 [INFO] <USER_LIST> Running writeDatabase</pre>		

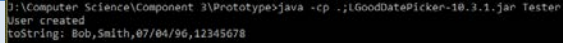
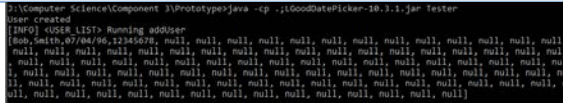
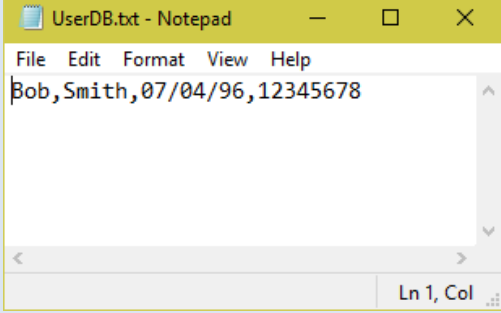
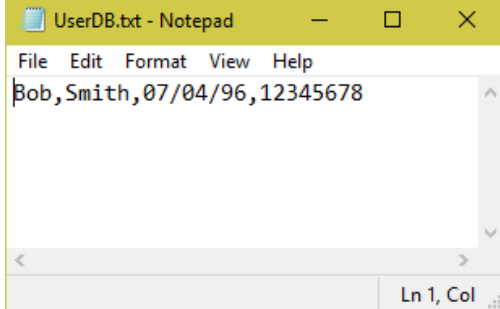
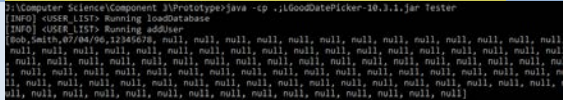
		same line			
6	Proto	The question stat list is not correctly loaded from file.		This is because the QuestionStatList constructor that takes the data loaded from the database hadn't actually been implemented. Implementing this constructor should fix this issue	
7	Proto	When calling addFailedValidation with an already present questionStat object's id the failed counter is incremented			
8	Proto	QuestionStatLists with multiple QuestionStat objects are successfully written to file			
9	Proto	QuestionStatLists with multiple QuestionStat objects are successfully loaded from file			
10	Proto	Error compiling GUI after adding the logic that tracks the number of validation checks failed to form displayer because		Add the current user variable to the end of the form displayer call in GUI.	

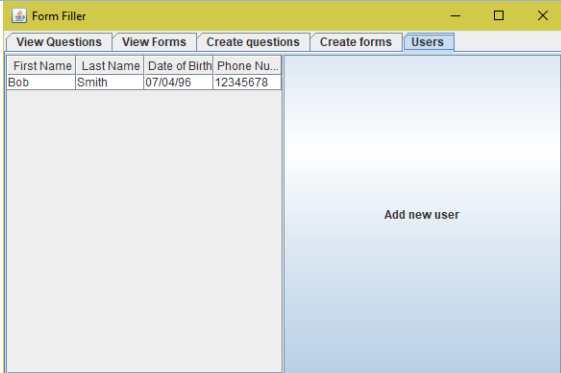
		when formDisplayer was called it wasn't passed an instance of user			
11	Proto	When a user attempts a form and gets a question wrong this is tracked and added to the questionStat. As you can see here the validation check for the second question fails and is added to the users question stats			
12	Proto	FormDisplayer fails to compile due to a symbol not found error after adding code to save the users data after pressing submit form.	<pre> J:\Computer Science\Component 3\Prototype>compile FormDisplayer.java:62: error: cannot find symbol users.writeDatabase(); ^ symbol: variable users location: class FormDisplayer 1 error </pre>	This because there is no UserList instance. To resolve create a variable called users and store the UserList instance from the GUI class in it. The UserList instance from the GUI class should be passed to the FormDisplayer's constructor.	<pre> J:\Computer Science\Component 3\Prototype>compile J:\Computer Science\Component 3\Prototype> Successful compile </pre>
13	Proto	After submit form is pressed and the user's stats are updated and they are saved to file	<pre> Users question stats after validation: Q57654744,1 Q97424294,1 [INFO] <USER_LIST> Running writeDatabase UB5381714,eFenty12,uQ7HfVv6h5,e1iza,Fenty,13/04/1978,2134951916, Q57654744,1 Q97424294,1 </pre>		

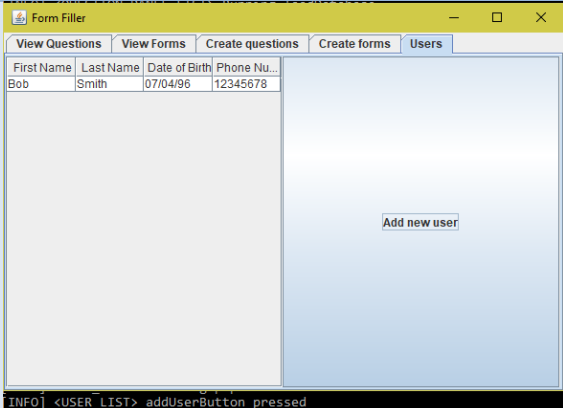
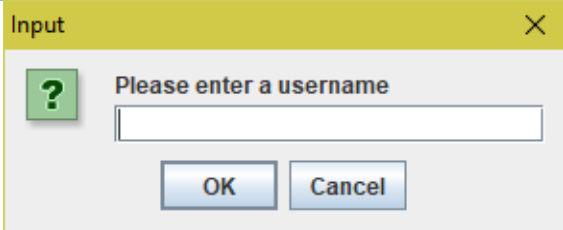
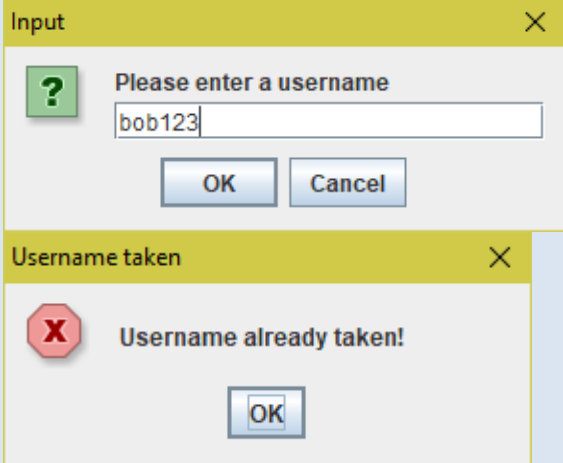
14	Proto	Multiple errors due to incompatible types when compiling QuestionStat.java	<pre> J:\Computer Science\Component 3\Prototype>compile QuestionStat.java:18: error: incompatible types: String[] cannot be converted to int[] numberOfAttemptsNeededToCorrect = loadedData[2].split("\\."); ^ QuestionStat.java:38: error: incompatible types: int cannot be converted to String String output = numberOfAttemptsNeededToCorrect[0]; ^ 2 errors </pre>		
15	Proto	Incompatible type error when compiling QuestionStat.java because String.split() returns a string array and the array is being attempted to be stored in an int array.	<pre> J:\Computer Science\Component 3\Prototype>compile QuestionStat.java:18: error: incompatible types: String[] cannot be converted to int[] numberOfAttemptsNeededToCorrect = loadedData[2].split("\\."); ^ </pre>	When splitting the loadedData string loop through it and convert each item into an int and copy across to a new array rather than attempting to store it in the int array immediately after splitting.	<pre> J:\Computer Science\Component 3\Prototype>compile QuestionStat.java:38: error: incompatible types: int cannot be converted to String String output = numberOfAttemptsNeededToCorrect[0]; ^ 1 error J:\Computer Science\Component 3\Prototype> </pre> Error no longer present in javac output
16	Proto	Incompatible type error when compiling QuestionStat.java because numberOfAttemptsNeededToCorrect[0] is an int but it's being stored in a string.	<pre> J:\Computer Science\Component 3\Prototype>compile QuestionStat.java:38: error: incompatible types: int cannot be converted to String String output = numberOfAttemptsNeededToCorrect[0]; ^ 1 error J:\Computer Science\Component 3\Prototype> </pre>	Concatenate an empty string to it to convert it to a string. i.e. + ""	<pre> J:\Computer Science\Component 3\Prototype>compile J:\Computer Science\Component 3\Prototype> </pre> Successful compile
17	Proto	The number of attempts to correct a question is successfully stored to file	Q57654744,11,0.0.0.0.0 Q97424294,11,0.0.0.0.0		
18	Proto	After manually altering the data in the database file to be something other than the default, the attempts of attempts to correct a question is successfully loaded	Q57654744,11,1.2.3.4.5 Q9742 In the database <pre>Q57654744,11,1.2.3.4.5 </pre> Output from the system		

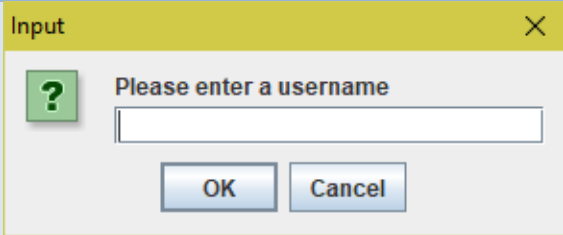
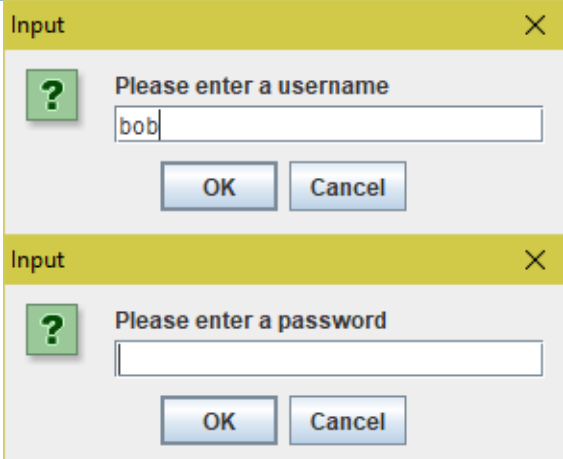
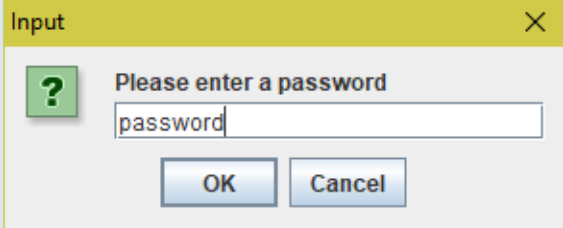
				getQuestionStatById()	
22	Proto	When the user gets all the form correct the number of times that it took them to correct each question gets added to the numberOfAttemptsNeededToCorrect array for the relevant QuestionStat	<pre>[INFO] <USER_LIST> Running writeDatabase Users question stats after finishing form: Q57654744,4,4.0.0.0.0 Q97424294,3,3.0.0.0.0</pre> The number of attempts needed to correct is underlined.		
23	Proto	The number of timesNeededToCorrect array isn't loaded from file.	<pre>Users question stats before validation: Q57654744,4,0.0.0.0.0 Q97424294,3,0.0.0.0.0</pre>	This is because the user database was never written to file after updating these arrays. Adding a call to users.writeDatabase() will fix this	<pre>Users question stats before validation: Q57654744,5,5.0.0.0.0 Q97424294,4,4.0.0.0.0</pre> The array's content is retained between system restarts.
24	Proto	After completing another form the array is updated with the number of attempts it took them to correct each question	<pre>Users question stats after finishing form: Q57654744,7,2.5.0.0.0 Q97424294,5,1.4.0.0.0</pre> The new data is underlined.		

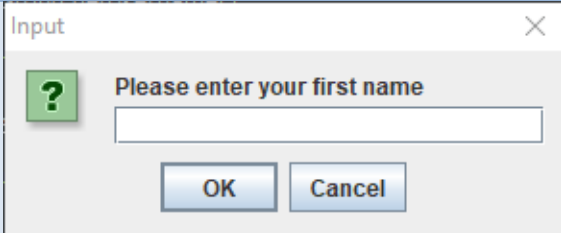
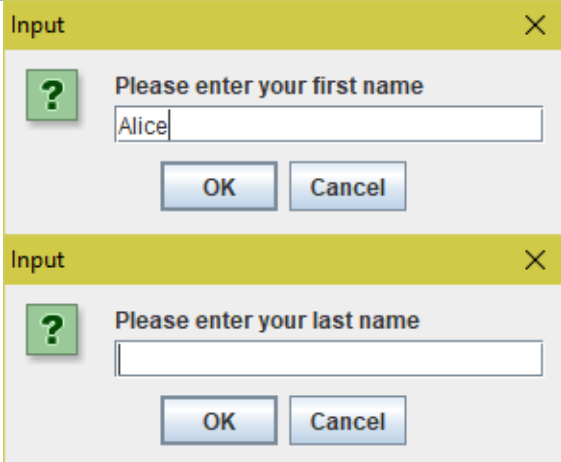
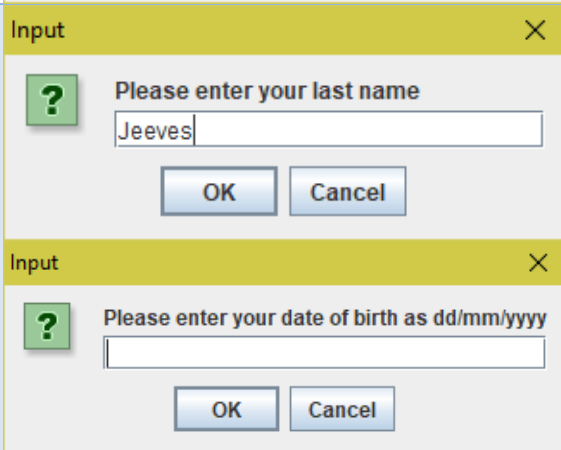
38: Ability to add users

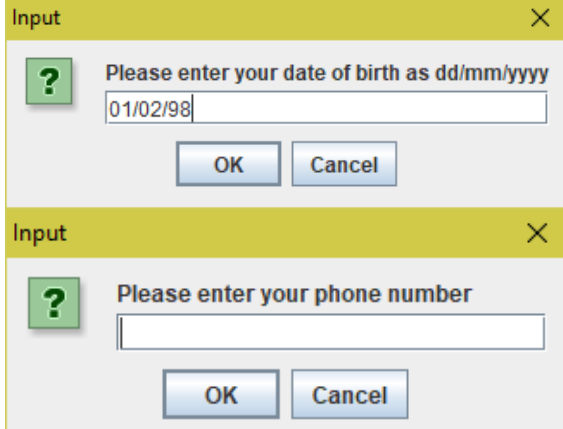
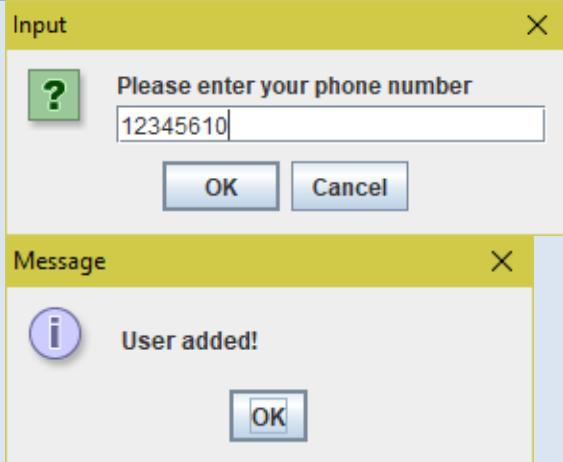
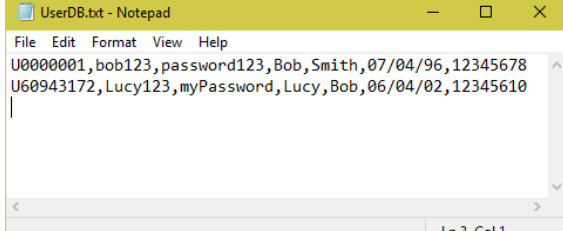
Test	Stage	Result Problem	Evidence	Action and Justification (If Needed)	Evidence of Action (If Needed)
1	Proto	A user can be created.			
2	Proto	User can be added to UserList			
3	Proto	Error writing the User database to file because the users directory hadn't been created.		Create the users directory	 User list written to file
4	Proto	The user list can be written to file			
5	Proto	The user list can be loaded from file			

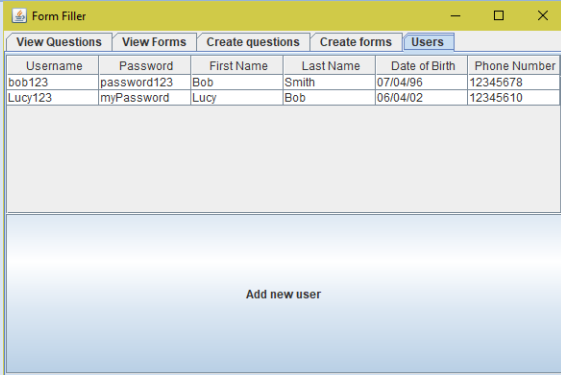
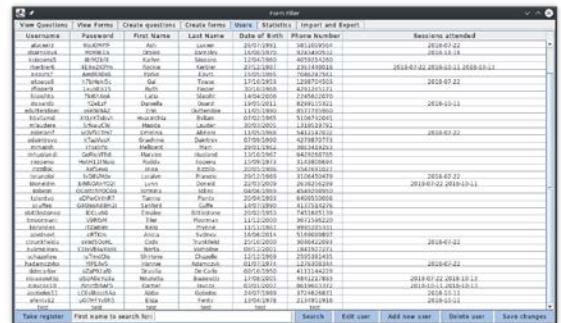
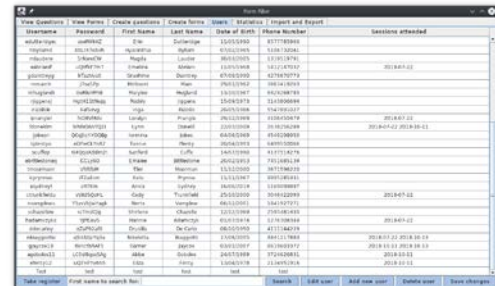
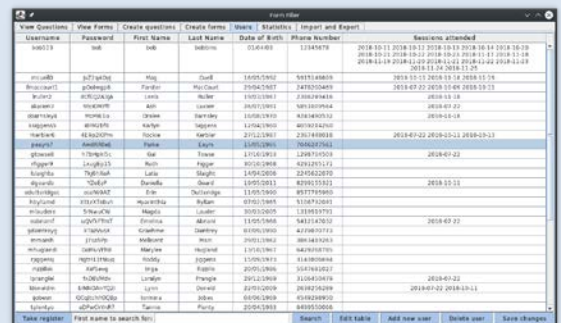
6	Proto	Error compiling user list because Random couldn't be found.	<pre> D:\Computer Science\Component 3\Prototype>compile UserList.java:41: error: cannot find symbol Random r = new Random(); // Declare a random number generator ^ symbol: class Random location: class UserList UserList.java:41: error: cannot find symbol Random r = new Random(); // Declare a random number generator ^ symbol: class Random location: class UserList UserList.java:54: error: cannot find symbol Random r = new Random(); ^ symbol: class Random location: class UserList UserList.java:54: error: cannot find symbol Random r = new Random(); ^ symbol: class Random location: class UserList 4 errors </pre>	Import java.util.*; as Random is part of the util package.	<pre> D:\Computer Science\Component 3\Prototype>compile </pre> Successful compile
7	Proto	Unique ids can be generated for each user	<pre> D:\Computer Science\Component 3\Prototype>java -cp .;iGoodDatePicker-10.3.1.jar Tester [INFO] <USER LIST> Running loadDatabase [INFO] <USER LIST> Running addUser [INFO] <USER LIST> 1 users loaded from file [INFO] <USER LIST> Running getFreeID [INFO] <USER LIST> Running generateId [INFO] <USER LIST> Running getUserById Unique id: U79456362 </pre>		
8	Proto	There is a add new user button	 The screenshot shows a window titled 'Form Filler' with several tabs: 'View Questions', 'View Forms', 'Create questions', 'Create forms', and 'Users'. The 'Users' tab is active, displaying a table with columns: 'First Name', 'Last Name', 'Date of Birth', and 'Phone Nu...'. The table contains one row with the data: 'Bob', 'Smith', '07/04/96', and '12345678'. Below the table, there is a large blue button labeled 'Add new user'.		

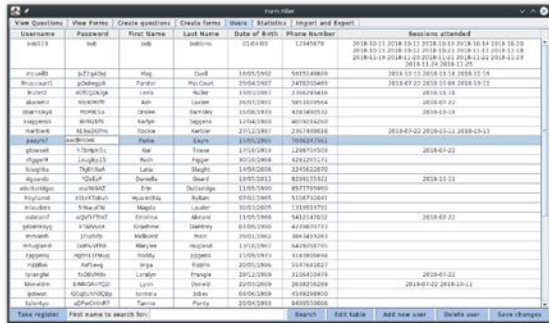
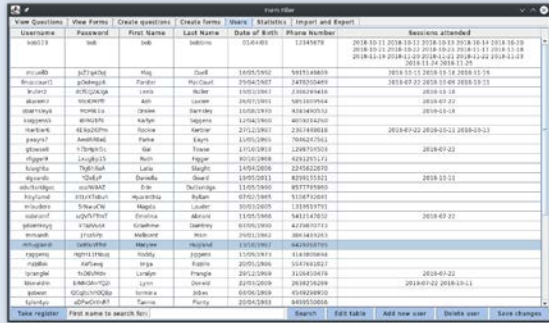
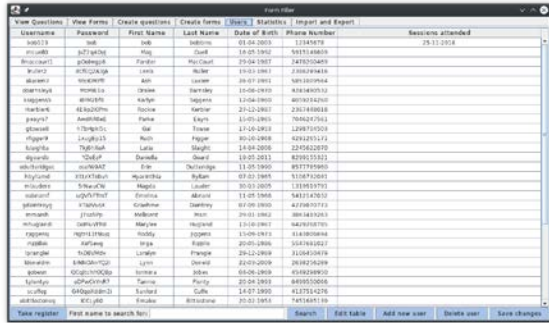
9	Proto	Presses to the add user button are detected			
10	Proto	When the add user button is pressed a dialog opens asking for the username of the new user			
11	Proto	If an already used username is entered an error message appears and the user has to enter a new username.			

					
12	Proto	When a unique username is entered the user is then prompted to enter a password for the new user			
13	Proto	After entering a password the user is prompted to enter their first name			

					
14	Proto	After entering their first name the user is prompted to enter their last name			
15	Proto	After entering their last name the user is prompted to enter their date of birth			

15	Proto	After entering their date of birth the user is prompted to enter their phone number			
16	Proto	After entering their phone number the user is added to the database			
17	Proto	The new user is saved in the user database file			

18	Proto	The new user can be seen in the user table			
19	Soft Dev	When a new user is added, the table refreshes however the rows do not resize.		Add a call to the <code>resizeRows()</code> method in the <code>populateTable()</code> method this means that whenever the table is populated the rows will also resize.	
20	Soft Dev	The table is non editable by default.		The table cannot be edited	The rows resize and are the correct height.

21	Soft Dev	When the edit table button has been pressed the table becomes editable.		
22	Soft Dev	When the save changes button is pressed the table is no longer editable.		
23	Soft Dev	The sessions attended is now logged as dd-mm-yyyy rather than yyyy-mm-dd		

24	Soft Dev	Multiple errors when compiling UserPanel.java	<pre> ^ C:\Harry\harry-pc-linux\media\harry\HARRY BEST\Computer Science\Component 3\SOFTDEV\Project rm-Filler\$./compile.sh Picked up _JAVA_OPTIONS: -Dawt.useSystemAAFontSettings=on -Dswing.aatext=true UserPanel.java:216: error: cannot find symbol return date.length() == 10 && splitDate.length == 3 && splitDate[0].length == ^ symbol: variable length location: class String UserPanel.java:217: error: cannot find symbol && splitDate[1].length == 2 && splitDate[2].length == 2 && splitDate[1] <= 1 ^ 2 && splitDate[2] <= 31; symbol: variable length location: class String UserPanel.java:217: error: cannot find symbol && splitDate[1].length == 2 && splitDate[2].length == 2 && splitDate[1] <= 1 ^ 2 && splitDate[2] <= 31; symbol: variable length location: class String UserPanel.java:217: error: bad operand types for binary operator '<=' && splitDate[1].length == 2 && splitDate[2].length == 2 && splitDate[1] <= 1 ^ first type: String second type: int UserPanel.java:217: error: bad operand types for binary operator '<=' && splitDate[1].length == 2 && splitDate[2].length == 2 && splitDate[1] <= 1 ^ first type: String second type: int 5 errors Picked up _JAVA_OPTIONS: -Dawt.useSystemAAFontSettings=on -Dswing.aatext=true UserPanel.java:216: error: cannot find symbol return date.length() == 10 && splitDate.length == 3 && splitDate[0].length == ^ symbol: variable length location: class String UserPanel.java:217: error: cannot find symbol && splitDate[1].length == 2 && splitDate[2].length == 2 && splitDate[1] <= 1 ^ 2 && splitDate[2] <= 31; symbol: variable length location: class String UserPanel.java:217: error: cannot find symbol && splitDate[1].length == 2 && splitDate[2].length == 2 && splitDate[1] <= 1 ^ 2 && splitDate[2] <= 31; symbol: variable length location: class String </pre>		
25	Soft Dev	Symbol not found error	<pre> ^ Picked up _JAVA_OPTIONS: -Dawt.useSystemAAFontSettings=on -Dswing.aatext=true UserPanel.java:216: error: cannot find symbol return date.length() == 10 && splitDate.length == 3 && splitDate[0].length == ^ symbol: variable length location: class String UserPanel.java:217: error: cannot find symbol && splitDate[1].length == 2 && splitDate[2].length == 2 && splitDate[1] <= 1 ^ 2 && splitDate[2] <= 31; symbol: variable length location: class String UserPanel.java:217: error: cannot find symbol && splitDate[1].length == 2 && splitDate[2].length == 2 && splitDate[1] <= 1 ^ 2 && splitDate[2] <= 31; symbol: variable length location: class String </pre>	Use .length() rather than .length as length is a method for string not an attribute.	<pre> Picked up _JAVA_OPTIONS: -Dawt.useSystemAAFontSettings=on -Dswing.aatext=true UserPanel.java:217: error: bad operand types for binary operator '<=' && splitDate[1].length() == 2 && splitDate[2].length() == 2 && splitDate[1] ^ <= 12 && splitDate[2] <= 31; ^ first type: String second type: int UserPanel.java:217: error: bad operand types for binary operator '<=' && splitDate[1].length() == 2 && splitDate[2].length() == 2 && splitDate[1] ^ <= 12 && splitDate[2] <= 31; ^ first type: String second type: int 2 errors Picked up _JAVA_OPTIONS: -Dawt.useSystemAAFontSettings=on -Dswing.aatext=true </pre>
26	Soft Dev	“bad operand types for binary operator ‘<=’” error when compiling userPanel.java	<pre> Picked up _JAVA_OPTIONS: -Dawt.useSystemAAFontSettings=on -Dswing.aatext=true UserPanel.java:217: error: bad operand types for binary operator '<=' && splitDate[1].length() == 2 && splitDate[2].length() == 2 && splitDate[1] ^ <= 12 && splitDate[2] <= 31; ^ first type: String second type: int UserPanel.java:217: error: bad operand types for binary operator '<=' && splitDate[1].length() == 2 && splitDate[2].length() == 2 && splitDate[1] ^ <= 12 && splitDate[2] <= 31; ^ first type: String second type: int 2 errors Picked up _JAVA_OPTIONS: -Dawt.useSystemAAFontSettings=on -Dswing.aatext=true </pre>	This is because the statement is comparing a string to an int using <=. The string should be converted to an int before the comparison using Integer.parseInt()	<pre> harry@harry-pc-linux\media\harry\HARRY BEST\Computer Science\Component 3\SOFTDEV\Project\Fo rm-Filler\$./compile.sh Picked up _JAVA_OPTIONS: -Dawt.useSystemAAFontSettings=on -Dswing.aatext=true Picked up _JAVA_OPTIONS: -Dawt.useSystemAAFontSettings=on -Dswing.aatext=true harry@harry-pc-linux\media\harry\HARRY BEST\Computer Science\Component 3\SOFTDEV\Project\Fo rm-Filler\$ </pre>

Soft Dev

Please enter a username

Form Filter

View Questions

View Items

Create questions

Create Items

Statistics

Import and Export

Username	Password	First Name	Last Name	Date of Birth	Phone Number	Session attended
hammad1	123456789	hammad	hammad	07-02-1985	01994233445	
hammad2	987654321	hammad	hammad	08-03-2004	01144444444	
hammad3	123456789	hammad	hammad	11-05-1998	0342474332	
hammad4	987654321	hammad	hammad	07-08-1980	04278710773	
hammad5	123456789	hammad	hammad	08-01-1982	03005431663	
hammad6	987654321	hammad	hammad	13-09-1967	03420506101	
hammad7	123456789	hammad	hammad	05-09-1973	01743690154	
hammad8	987654321	hammad	hammad	26-08-1985	05547810277	
hammad9	123456789	hammad	hammad	29-12-2000	03304545792	
hammad10	987654321	hammad	hammad	20-03-2004	0330920208	
hammad11	123456789	hammad	hammad	04-06-1984	05040600048	
hammad12	987654321	hammad	hammad	20-04-1983	04080500103	
hammad13	123456789	hammad	hammad	04-01-1990	04317414276	
hammad14	987654321	hammad	hammad	20-12-1983	03474607100	
hammad15	123456789	hammad	hammad	11-12-2000	03471886239	
hammad16	987654321	hammad	hammad	11-11-1987	03090200120	
hammad17	123456789	hammad	hammad	06-02-2014	03160600187	
hammad18	987654321	hammad	hammad	20-10-2000	03040422063	
hammad19	123456789	hammad	hammad	06-02-2014	03160600179	
hammad20	987654321	hammad	hammad	02-12-1989	03090346406	
hammad21	123456789	hammad	hammad	01-07-1974	03275003144	
hammad22	987654321	hammad	hammad	06-10-1985	03171144208	
hammad23	123456789	hammad	hammad	07-08-2005	03040277883	
hammad24	987654321	hammad	hammad	03-01-2007	03019600072	
hammad25	123456789	hammad	hammad	24-07-1980	03750400018	
hammad26	987654321	hammad	hammad	03-04-1979	03234400000	

Filter options

View name as

Search

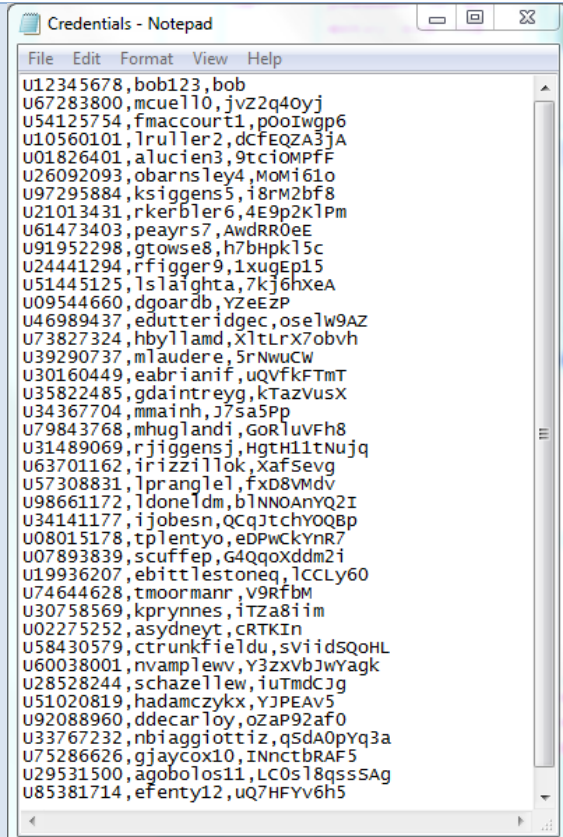
Exit login

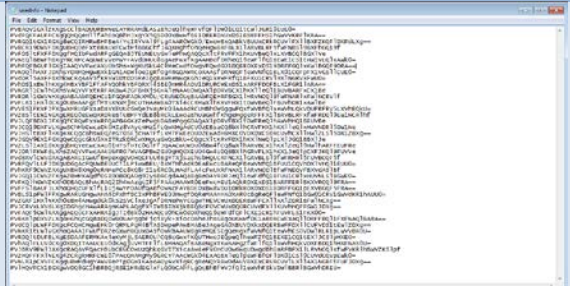
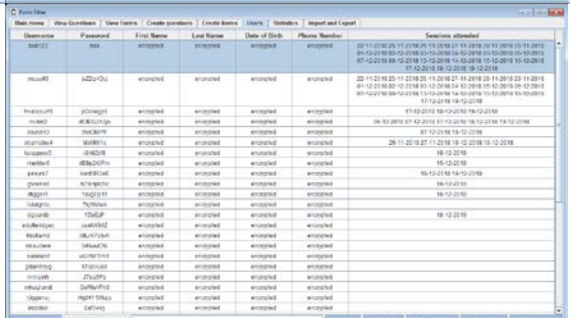
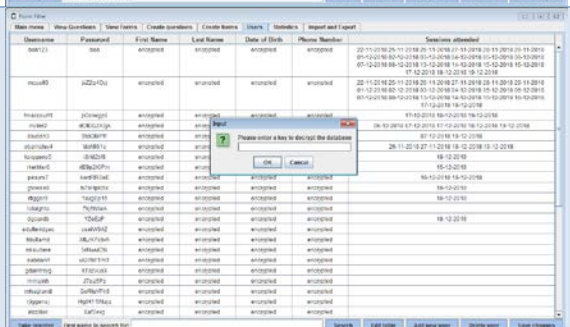
Add new user

delete user

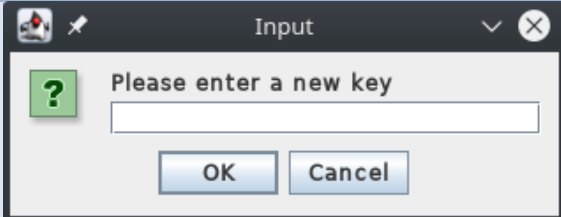
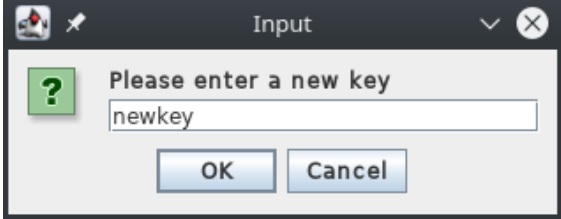
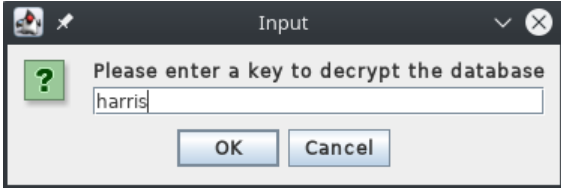
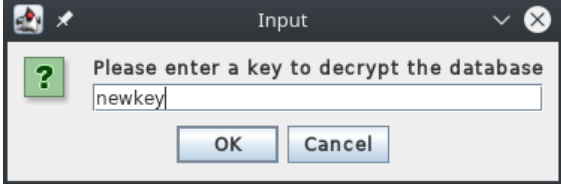
View reports

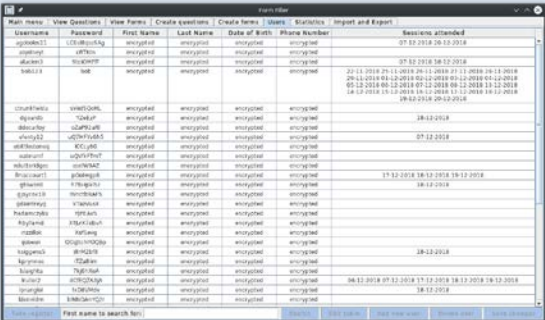
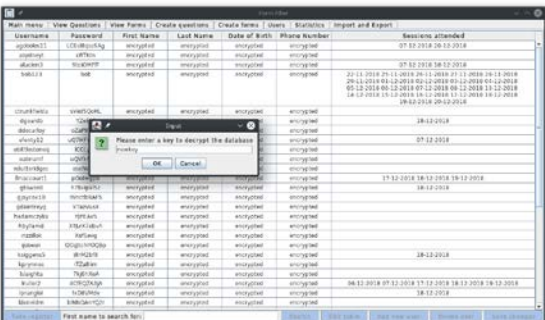
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28	Soft Dev	The credentials of the users are stored in a separate file			
29	Soft Dev	The question stat data is stored in a separate file			

30	Soft Dev	The sensitive user information is stored in an encrypted file.			
31	Soft Dev	When the user table is viewed the sensitive data is marked as encrypted. All other non-sensitive information is automatically loaded.W			
32	Soft Dev	When the user panel is selected a dialog appears asking the user to enter a key to decrypt the database.			

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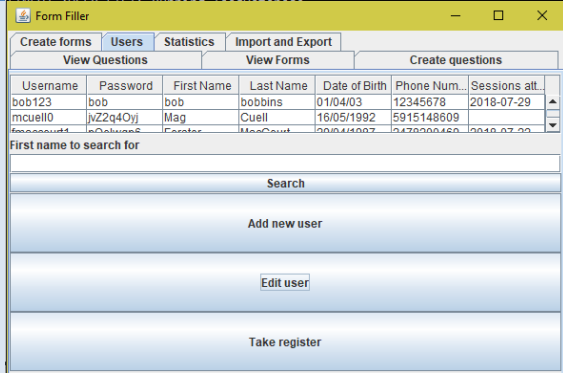
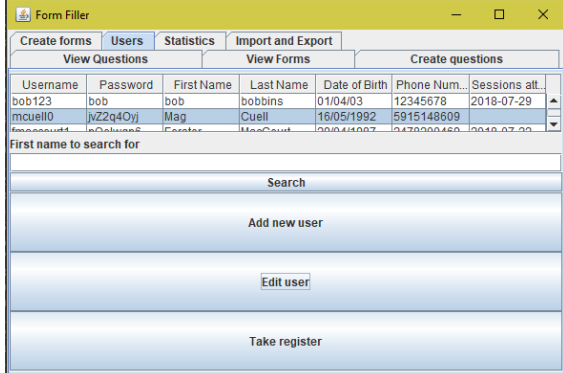
36	Soft Dev	If the user presses yes they are given a text field to enter the new key into			
37	Soft Dev	When the user enters a new key, the database is encrypted with that key.	  The old key is entered  Error decrypting The new key is entered  The database is successfully decrypted		

				
38	Soft Dev	If the database isn't decrypted all of the buttons are disabled		
39	Soft Dev	When the database is decrypted successfully the buttons are enabled.		

[illegible]

The buttons are now enabled

39: Ability to edit users

Test	Stage	Result Problem	Evidence	Action and Justification (If Needed)	Evidence of Action (If Needed)
1	Proto	There is an edit user button and presses to it are detected	 <pre>[INFO] <USER_LIST> editUserButton pressed</pre>		
2	Proto	When the edit button is pressed the username of the user whose row is selected is outputted.	 <pre>[INFO] <USER_LIST> editUserButton pressed [INFO] <USER_LIST> Running getUserById mcuello</pre>		

3	Proto	When the edit user button is pressed the same dialog boxes that are used to create a user appear however they are populated with the user's data allowing it to be changed.	Input ×  Please enter a username bob123 OK Cancel Input ×  Please enter a password bob OK Cancel Input ×  Please enter your first name bob OK Cancel Input ×  Please enter your last name bobbins OK Cancel Input ×  Please enter your date of birth as dd/mm/yyyy 01/04/03 OK Cancel		
---	-------	--	---	--	--

			Input ? Please enter your phone number 12345678 OK Cancel																																					
4	Proto	A sessions attended dialog box also appears allowing the user to alter the sessions attended data.	Input ? Please enter the sessions attended as dd-mm-yyyy separated by commas 2018-07-29 OK Cancel																																					
5	Proto	When the user has finished altering the information the user's details are changed.	<table><tr><th>Username</th><th>Password</th><th>First Name</th><th>Last Name</th><th>Date of Birth</th><th>Phone Num...</th><th>Sessions att...</th></tr><tr><td>bob123</td><td>bob</td><td>bob</td><td>bobbins</td><td>01/04/03</td><td>12345678</td><td>2018-07-29</td></tr><tr><td>mcuell0</td><td>jvZ2q4Oyj</td><td>Mag</td><td>Cuell</td><td>16/05/1992</td><td>5915148609</td><td></td></tr><tr><td>fmaccourt1</td><td>pOolwpp6</td><td>Forster</td><td>MacCourt</td><td>29/04/1987</td><td>2478200469</td><td>2018-07-22</td></tr><tr><td>lruller2</td><td>dCREQZA3A</td><td>Leela</td><td>Ruller</td><td>19/03/1967</td><td>2386249416</td><td>2018-07-29</td></tr></table> Input ? Please enter a username mcuell0EDIT OK Cancel Input ? Please enter a password jvZ2q4OyjEDIT OK Cancel Input ? Please enter your first name MagEDIT OK Cancel	Username	Password	First Name	Last Name	Date of Birth	Phone Num...	Sessions att...	bob123	bob	bob	bobbins	01/04/03	12345678	2018-07-29	mcuell0	jvZ2q4Oyj	Mag	Cuell	16/05/1992	5915148609		fmaccourt1	pOolwpp6	Forster	MacCourt	29/04/1987	2478200469	2018-07-22	lruller2	dCREQZA3A	Leela	Ruller	19/03/1967	2386249416	2018-07-29		
Username	Password	First Name	Last Name	Date of Birth	Phone Num...	Sessions att...																																		
bob123	bob	bob	bobbins	01/04/03	12345678	2018-07-29																																		
mcuell0	jvZ2q4Oyj	Mag	Cuell	16/05/1992	5915148609																																			
fmaccourt1	pOolwpp6	Forster	MacCourt	29/04/1987	2478200469	2018-07-22																																		
lruller2	dCREQZA3A	Leela	Ruller	19/03/1967	2386249416	2018-07-29																																		

Input

?

Please enter your last name

CueIIEDIT

OK

Cancel

Input

?

Please enter your date of birth as dd/mm/yyyy

16/05/2001

OK

Cancel

Input

?

Please enter your phone number

1234567899

OK

Cancel

Input

?

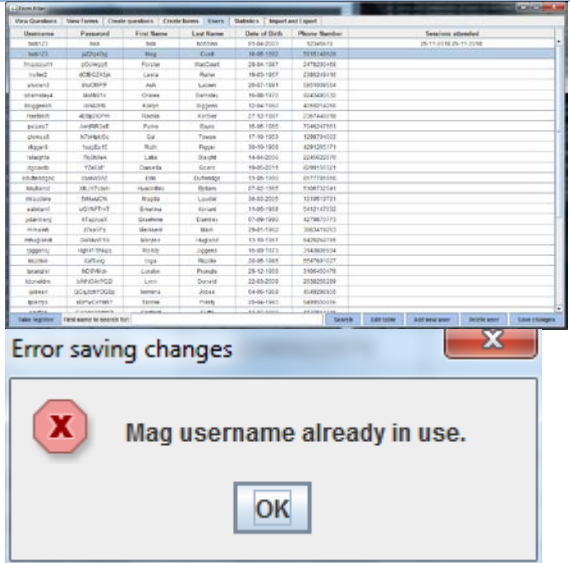
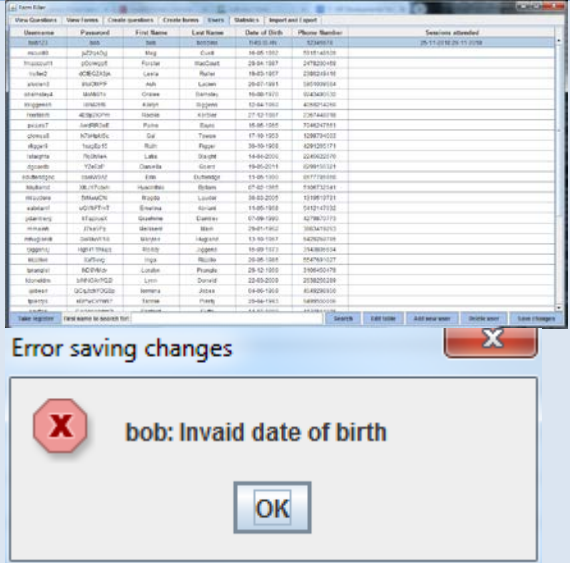
Please enter the sessions attended as dd-mm-yyyy separated by commas

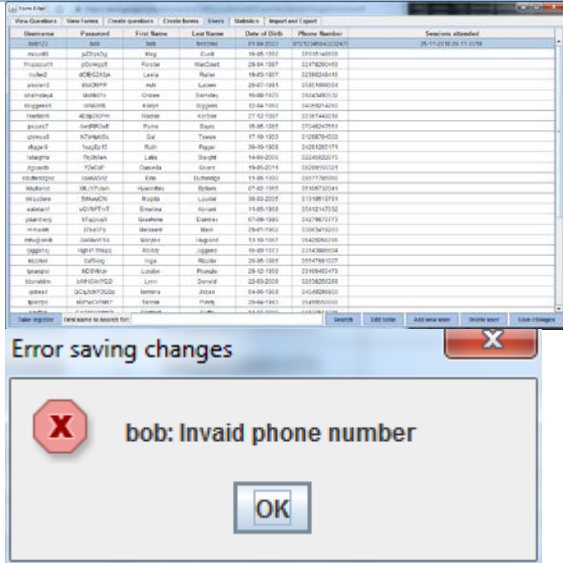
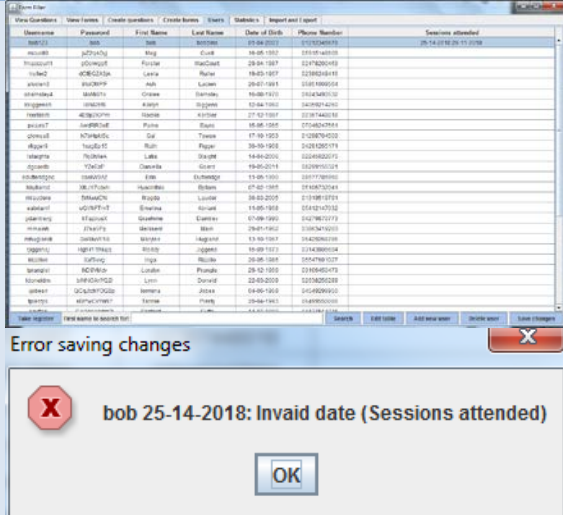
01-02-04

OK

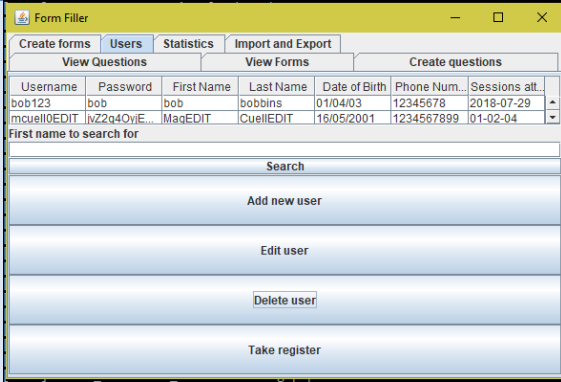
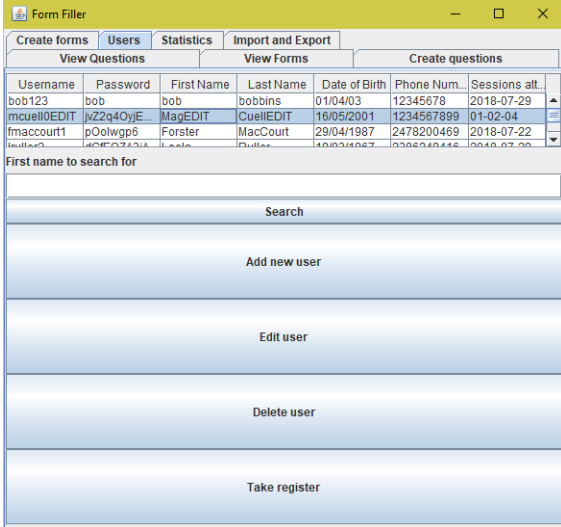
Cancel

Username	Password	First Name	Last Name	Date of Birth	Phone Number	Sessions attended
bob123	bob	bob	bobbins	01/04/03	12345678	2018-07-29
macb00101	macb00101	Macb001	CueIIEDIT	16/05/2001	1234567899	01-02-04
maccourt1	macb00101	MacCourt	MacCourt	29/04/1982	2475200469	2018-07-22

6	Soft Dev	When a users username is changed to one that is already in use then the system tells the user this and doesn't save the changes.	
7	Soft Dev	When the user's date of birth is changed to an invalid date the system tells the user this and doesn't save the changes	

8	Soft Dev	When the user's phone number is changed to an invalid number the system tells the user this and doesn't save the changes	 The screenshot shows a web application interface for managing users. At the top, there's a navigation bar with links like 'View Users', 'Create questions', 'Create forms', 'Events', 'Statistics', and 'Import and Export'. Below this is a table listing users with columns: Username, Password, First Name, Last Name, Date of Birth, Phone Number, and Sessions attended. The user 'bob' is highlighted, showing an invalid phone number. An error dialog box is open, displaying a red 'X' icon and the message 'bob: Invalid phone number' with an 'OK' button.
9	Soft Dev	When the user's sessions attended string has an invalid date the system tells the user this and doesn't save the changes.	 This screenshot shows the same application interface as the previous one. In the user list, the 'Sessions attended' field for user 'bob' contains an invalid date. An error dialog box is displayed, showing a red 'X' icon and the message 'bob 25-14-2018: Invalid date (Sessions attended)' with an 'OK' button.

40: Ability to delete users

Test	Stage	Result Problem	Evidence	Action and Justification (If Needed)	Evidence of Action (If Needed)
1	Proto	There is a delete user button and presses to it are detected	 The screenshot shows the 'Form Filler' application with the 'Users' tab selected. The 'Delete user' button is highlighted. The user list includes bob123, mcuelloEDIT, and CuellEDIT. The 'Delete user' button is highlighted.		
2	Proto	When the delete user button is pressed the selected user is deleted.	 The screenshot shows the 'Form Filler' application with the 'Users' tab selected. The 'Delete user' button is highlighted. The user list includes bob123, mcuelloEDIT, and CuellEDIT. The 'Delete user' button is highlighted.		

```
[INFO] <USER_LIST> deleteUserButton pressed
[INFO] <USER_LIST> Running removeUser
[INFO] <USER_LIST> Running writeDatabase
```

Username	Password	First Name	Last Name	Date of Birth	Phone Num...	Sessions att...
bob123	bob	bob	bobbins	01/04/03	12345678	2018-07-29
fmaccourt1	pOolwgp6	Forster	MacCourt	29/04/1987	2478200469	2018-07-22
lruller2	dCfEQZA3JA	Leela	Ruller	19/03/1967	2386249416	2018-07-29

First name to search for

Search

Add new user

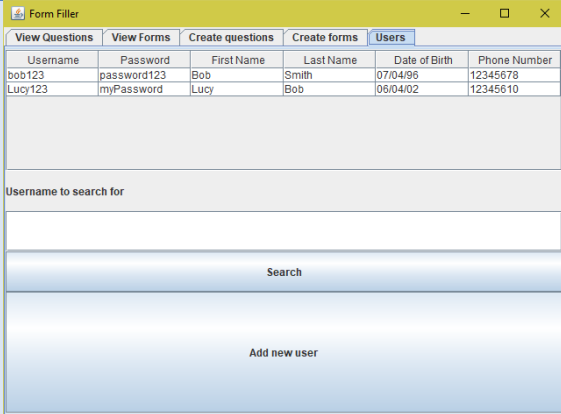
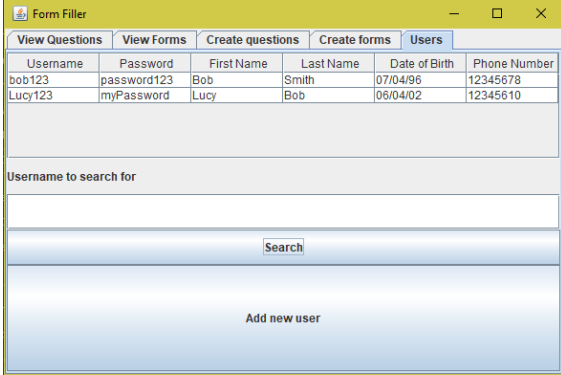
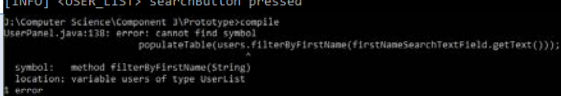
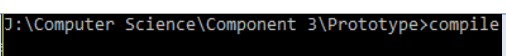
Edit user

Delete user

Take register

The user mcuello has been deleted.

41: Ability to search through users

Test	Stage	Result Problem	Evidence	Action and Justification (If Needed)	Evidence of Action (If Needed)
1	Proto	There is a text field and search button to search by username			
2	Proto	Presses to the search button are detected			
3	Proto	Symbol not found error when compiling UserPanel due to a typo.		change filterByFirstname to filterByFirstName in UserList.java	

4

Proto

When a first name is searched for all the users with that first name are displayed.

The image displays two screenshots of a web application titled "Form Filler". Both screenshots show the "Users" tab, which contains a table of user information and a search interface.

Top Screenshot:

Username	Password	First Name	Last Name	Date of Birth	Phone Number
bob123	password123	Bob	Smith	07/04/96	12345678

First name to search for

bob

Search

Add new user

Bottom Screenshot:

Username	Password	First Name	Last Name	Date of Birth	Phone Number
Lucy123	myPassword	Lucy	Bob	06/04/02	12345610
Lucy1234	password	Lucy	Bobbins	06/04/04	12345611

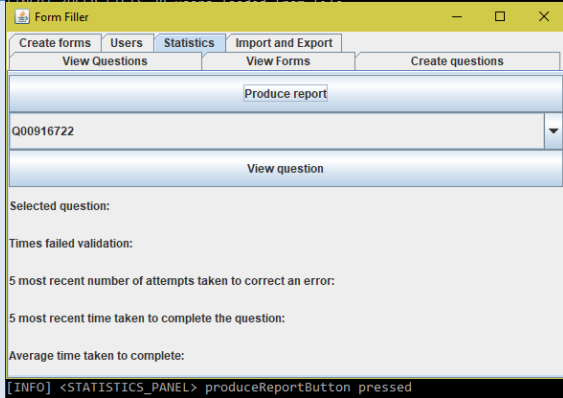
First name to search for

Lucy

Search

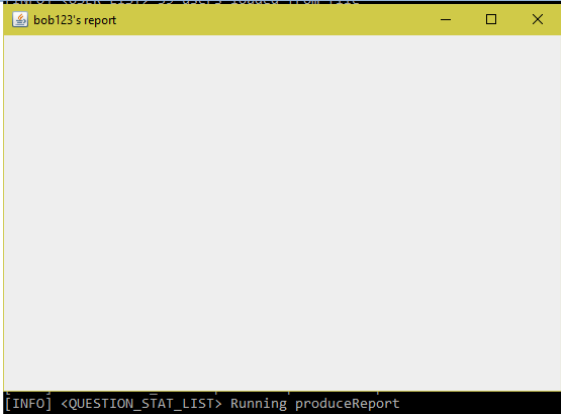
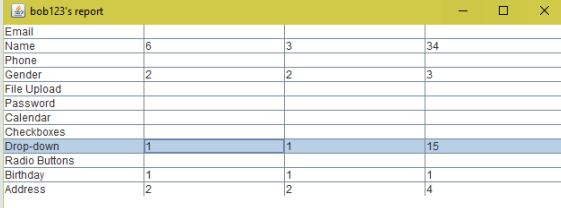
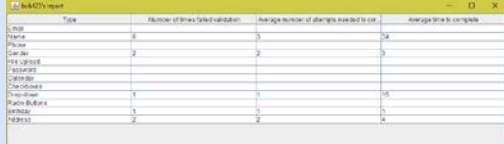
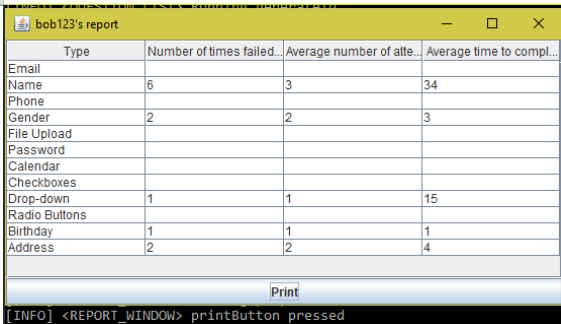
Add new user

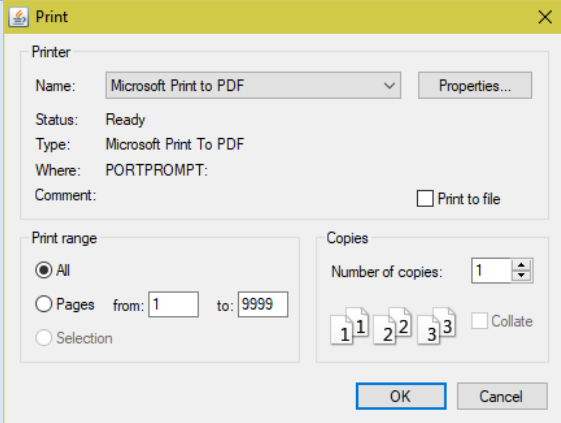
42: Ability to produce a report on the user's weak areas and progress

Test	Stage	Result Problem	Evidence	Action and Justification (If Needed)	Evidence of Action (If Needed)
1	Proto	There is a produce report button in the statistics panel and presses to it are detected			
2	Proto	Error compiling QuestionStatList.java due to symbol not found errors.	<pre> J:\Computer Science\Component 3\Prototype>compile QuestionStatList.java:56: error: cannot find symbol Question currentQuestion = question.getQuestion ^ symbol: variable question location: class QuestionStatList QuestionStatList.java:66: error: cannot find symbol System.out.println(currentType + " Times failed validation: " + totalTimesFailedValidation); ^ symbol: variable totalTimesFailedValidation location: class QuestionStatList 2 errors </pre>	Replace question with questions as question was a typo. Move the definition of totalTimeFailedValidation out of the for loop so that it can be used later on in the method.	<pre> J:\Computer Science\Component 3\Prototype>compile QuestionStatList.java:69: error: missing return statement } ^ 1 error </pre> Different error
3	Proto	Error compiling QuestionStatList.java because of a missing return statement.	<pre> J:\Computer Science\Component 3\Prototype>compile QuestionStatList.java:69: error: missing return statement } ^ 1 error </pre>	Add "return data" to the end of the method as data is the object that should be returned	<pre> J:\Computer Science\Component 3\Prototype>compile J:\Computer Science\Component 3\Prototype> </pre> Successful compile

4	Proto	Null pointer exception when running QuestionStatList.produceReport()	<pre> [INFO] <STATISTICS_PANEL> produceReportButton pressed [INFO] <QUESTION_LIST> Running getQuestionByID Exception in thread "AWT-EventQueue-0" java.lang.NullPointerException at QuestionStatList.produceReport(QuestionStatList.java:59) at StatisticsPanel.actionPerformed(StatisticsPanel.java:143) at javax.swing.AbstractButton.fireActionPerformed(Unknown Source) at javax.swing.AbstractButton\$Handler.actionPerformed(Unknown Source) at javax.swing.DefaultButtonModel.fireActionPerformed(Unknown Source) at javax.swing.DefaultButtonModel.setPressed(Unknown Source) at javax.swing.plaf.basic.BasicButtonListener.mouseReleased(Unknown Source) at java.awt.Component.processMouseEvent(Unknown Source) at javax.swing.JComponent.processMouseEvent(Unknown Source) at java.awt.Component.processEvent(Unknown Source) at java.awt.Container.processEvent(Unknown Source) at java.awt.Component.dispatchEventImpl(Unknown Source) at java.awt.Container.dispatchEventImpl(Unknown Source) at java.awt.Component.dispatchEvent(Unknown Source) at java.awt.LightweightDispatcher.retargetMouseEvent(Unknown Source) at java.awt.LightweightDispatcher.processMouseEvent(Unknown Source) at java.awt.LightweightDispatcher.dispatchEvent(Unknown Source) at java.awt.Container.dispatchEventImpl(Unknown Source) at java.awt.Window.dispatchEventImpl(Unknown Source) at java.awt.Component.dispatchEvent(Unknown Source) at java.awt.EventQueue.dispatchEventImpl(Unknown Source) at java.awt.EventQueue.access\$500(Unknown Source) at java.awt.EventQueue\$3.run(Unknown Source) at java.awt.EventQueue\$3.run(Unknown Source) at java.security.AccessController.doPrivileged(Native Method) at java.security.ProtectionDomain\$JavaSecurityAccessImpl.doIntersectionPrivilege(Unknown Source) at java.security.ProtectionDomain\$JavaSecurityAccessImpl.doIntersectionPrivilege(Unknown Source) at java.awt.EventQueue\$4.run(Unknown Source) at java.awt.EventQueue\$4.run(Unknown Source) at java.security.AccessController.doPrivileged(Native Method) at java.security.ProtectionDomain\$JavaSecurityAccessImpl.doIntersectionPrivilege(Unknown Source) at java.awt.EventQueue.dispatchEvent(Unknown Source) at java.awt.EventDispatchThread.pumpOneEventForFilters(Unknown Source) at java.awt.EventDispatchThread.pumpEventsForFilter(Unknown Source) at java.awt.EventDispatchThread.pumpEventsForHierarchy(Unknown Source) at java.awt.EventDispatchThread.pumpEvents(Unknown Source) at java.awt.EventDispatchThread.pumpEvents(Unknown Source) at java.awt.EventDispatchThread.run(Unknown Source) </pre>	Sometimes a questionStat can exist that corresponds to no question because the question that it corresponds to was deleted. Adding a check to ensure that the returned question (when looking up the question stat) isn't null should fix this error as questionStats without questions would be ignored.	<pre> [INFO] <QUESTION_STAT_LIST> Running produceReport Email Times failed validation: 0 Name Times failed validation: 1 Phone Times failed validation: 0 Gender Times failed validation: 0 File Upload Times failed validation: 0 Password Times failed validation: 0 Calendar Times failed validation: 0 Checkboxes Times failed validation: 0 Drop-down Times failed validation: 1 Radio Buttons Times failed validation: 0 Birthday Times failed validation: 1 Address Times failed validation: 0 </pre> No null pointer exception when running produce report.
5	Proto	When the produce report is pressed, the total number of times that the user has failed the validation check for each type of question is output.	<pre> [INFO] <STATISTICS_PANEL> produceReportButton pressed [INFO] <QUESTION_STAT_LIST> Running produceReport Email Times failed validation: 0 Name Times failed validation: 1 Phone Times failed validation: 0 Gender Times failed validation: 0 File Upload Times failed validation: 0 Password Times failed validation: 0 Calendar Times failed validation: 0 Checkboxes Times failed validation: 0 Drop-down Times failed validation: 1 Radio Buttons Times failed validation: 0 Birthday Times failed validation: 1 Address Times failed validation: 0 </pre>		

6	Proto	Division by zero error when calculating the average number of attempts taken to correct an error for question types that haven't yet been attempted by the user.	<pre> exception in thread "AWT-EventQueue-0" java.lang.ArithmeticException: / by zero at QuestionStatList.produceReport(QuestionStatList.java:78) at StatisticsPanel.actionPerformed(StatisticsPanel.java:143) at javax.swing.AbstractButton.fireActionPerformed(Unknown Source) at javax.swing.AbstractButton\$Handler.actionPerformed(Unknown Source) at javax.swing.DefaultButtonModel.fireActionPerformed(Unknown Source) at javax.swing.DefaultButtonModel.setPressed(Unknown Source) at javax.swing.plaf.basic.BasicButtonListener.mouseReleased(Unknown Source) at java.awt.Component.processMouseEvent(Unknown Source) at javax.swing.JComponent.processMouseEvent(Unknown Source) at java.awt.Container.processEvent(Unknown Source) at java.awt.Container.dispatchEventImpl(Unknown Source) at java.awt.Container.dispatchEventImpl(Unknown Source) at java.awt.Container.dispatchEventImpl(Unknown Source) at java.awt.LightweightDispatcher.retargetMouseEvent(Unknown Source) at java.awt.LightweightDispatcher.processMouseEvent(Unknown Source) at java.awt.LightweightDispatcher.dispatchEvent(Unknown Source) at java.awt.Container.dispatchEventImpl(Unknown Source) at java.awt.Window.dispatchEventImpl(Unknown Source) at java.awt.Component.dispatchEvent(Unknown Source) at java.awt.EventQueue.dispatchEventImpl(Unknown Source) at java.awt.EventQueue.access\$500(Unknown Source) at java.awt.EventQueue\$3.run(Unknown Source) at java.awt.EventQueue\$3.run(Unknown Source) at java.security.AccessController.doPrivileged(Native Method) at java.security.ProtectionDomain\$JavaSecurityAccessImpl.doIntersectionPrivilege(Unknown Source) at java.security.ProtectionDomain\$JavaSecurityAccessImpl.doIntersectionPrivilege(Unknown Source) at java.awt.EventQueue\$4.run(Unknown Source) at java.awt.EventQueue\$4.run(Unknown Source) at java.security.AccessController.doPrivileged(Native Method) at java.security.ProtectionDomain\$JavaSecurityAccessImpl.doIntersectionPrivilege(Unknown Source) at java.awt.EventQueue.dispatchEvent(Unknown Source) at java.awt.EventDispatchThread.pumpOneEventForFilters(Unknown Source) at java.awt.EventDispatchThread.pumpEventsForFilter(Unknown Source) at java.awt.EventDispatchThread.pumpEventsForHierarchy(Unknown Source) at java.awt.EventDispatchThread.pumpEvents(Unknown Source) at java.awt.EventDispatchThread.run(Unknown Source) </pre>	Because the question type hasn't yet been attempted by the user when dividing by the total number of questions a division by zero occurs because the total number of questions of that type is 0. Adding a check to skip these calculations for question types that haven't been attempted yet will fix this error.	<pre> [INFO] <QUESTION_STAT_LIST> Running produceReport Name Times failed validation: 1 Average # attempts to correct: 0 Gender Times failed validation: 0 Average # attempts to correct: 0 Drop-down Times failed validation: 1 Average # attempts to correct: 1 Birthday Times failed validation: 1 Average # attempts to correct: 1 Address Times failed validation: 0 Average # attempts to correct: 0 </pre> No division by zero error
7	Proto	The average number of attempts that it takes the user to correct an error for each question type is output for the question types that they've attempted.	<pre> [INFO] <QUESTION_STAT_LIST> Running produceReport Name Times failed validation: 1 Average # attempts to correct: 0 Gender Times failed validation: 0 Average # attempts to correct: 0 Drop-down Times failed validation: 1 Average # attempts to correct: 1 Birthday Times failed validation: 1 Average # attempts to correct: 1 Address Times failed validation: 0 Average # attempts to correct: 0 </pre>		
8	Proto	The average time taken to complete each question type is outputted.	<pre> Name Times failed validation: 82 Average # attempts to correct: 3 Average time: 34 Gender Times failed validation: 50 Average # attempts to correct: 2 Average time: 3 Drop-down Times failed validation: 23 Average # attempts to correct: 1 Average time: 15 Birthday Times failed validation: 34 Average # attempts to correct: 1 Average time: 1 Address Times failed validation: 59 Average # attempts to correct: 2 Average time: 4 </pre>		
9	Proto	Error compiling QuestionStatList.java due to an incompatible types error.	<pre> J:\Computer Science\Component 3\Prototype>compile QuestionStatList.java:91: error: incompatible types: int cannot be converted to String rowData[1] = totalNumberOfAttemptsNeededToCorrect; ^ QuestionStatList.java:92: error: incompatible types: int cannot be converted to String rowData[2] = averageNumberOfAttemptsToCorrect; ^ QuestionStatList.java:93: error: incompatible types: long cannot be converted to String rowData[3] = averageTimeTakenToComplete; ^ 3 errors </pre>	The data array that the averages etc. are being stored in is a String array. Converting the	<pre> J:\Computer Science\Component 3\Prototype>compile J:\Computer Science\Component 3\Prototype> </pre> Successful compile

				values to strings by concatenating an empty string (+ "") will fix this.	
10	Proto	When the produce report button is pressed an empty window opens to display the report.			
11	Proto	The table is present however the headers are not.		Add the table to a JScrollPane before adding it to the window	 Table displayed correctly with headers.
12	Proto	There is a print button and presses to it are detected.			

13	Proto	When the print button is pressed a print dialog opens		
----	-------	---	--	--

14

Proto

When the report is printed information is cut off because the document is portrait

Type	Number of times failed validation	Average number of attempts needed
Email		
Name	6	3
Phone		
Gender	2	2
File Upload		
Password		
Calendar		
Checkboxes		
Drop-down	1	1
Radio Buttons		
Birthday	1	1
Address	2	2

Make the printer print landscape by setting up a page format object and calling .setOrientation(PageFormat.LANDSCAPE) on it and then passing it to the print job

Type	Number of times failed validation	Average number of attempts needed
Email		
Name	6	3
Phone		
Gender	2	2
File Upload		
Password		
Calendar		
Checkboxes		
Drop-down	1	1
Radio Buttons		
Birthday	1	1
Address	2	2

The printout is landscape now

15

Proto

Most the data from the table is cropped off.

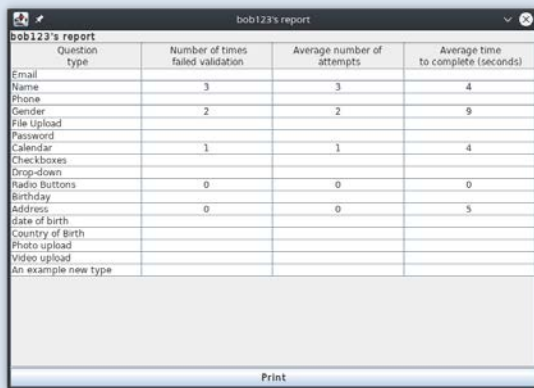
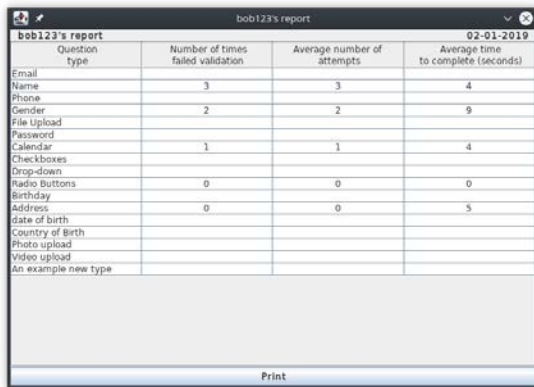
Type	Number of times failed validation	Average number of attempts needed
Email		
Name	6	3
Phone		
Gender	2	2
File Upload		
Password		
Calendar		
Checkboxes		
Drop-down	1	1
Radio Buttons		
Birthday	1	1
Address	2	2

Scale the JFrame that is being printed to a smaller size so that it all fits in the print area.

Type	Number of times failed validation	Average number of attempts needed	Average time to complete
Email			
Name	6	3	34
Phone			
Gender	2	2	3
File Upload			
Password			
Calendar			
Checkboxes			
Drop-down	1	1	13
Radio Buttons			
Birthday	1	1	1
Address	2	2	4

The entire table fits on the page

16	Proto	The column headings are cut off	<table><tr><th>Type</th><th>Number of times failed validation</th><th>Average number of attempts needed to correct</th><th>Average time to complete</th></tr><tr><td>Email</td><td></td><td></td><td></td></tr><tr><td>Name</td><td>5</td><td>3</td><td>34</td></tr><tr><td>Phone</td><td></td><td></td><td></td></tr><tr><td>Gender</td><td>2</td><td>2</td><td>3</td></tr><tr><td>File Upload</td><td></td><td></td><td></td></tr><tr><td>Password</td><td></td><td></td><td></td></tr><tr><td>Calendar</td><td></td><td></td><td></td></tr><tr><td>Checkboxes</td><td></td><td></td><td></td></tr><tr><td>Drop-Down</td><td>1</td><td>1</td><td>15</td></tr><tr><td>Radio Buttons</td><td></td><td></td><td></td></tr><tr><td>Birthday</td><td>1</td><td>1</td><td>1</td></tr><tr><td>Address</td><td>2</td><td>2</td><td>4</td></tr></table>	Type	Number of times failed validation	Average number of attempts needed to correct	Average time to complete	Email				Name	5	3	34	Phone				Gender	2	2	3	File Upload				Password				Calendar				Checkboxes				Drop-Down	1	1	15	Radio Buttons				Birthday	1	1	1	Address	2	2	4	Format them with html such as "<html><center>Number of times failed validation<html>" so that they wrap multiple lines and take up less horizontal space.	<table><tr><th>Question type</th><th>Number of times failed validation</th><th>Average number of attempts needed to correct</th><th>Average time to complete</th></tr><tr><td>Email</td><td></td><td></td><td></td></tr><tr><td>Name</td><td>5</td><td>3</td><td>34</td></tr><tr><td>Phone</td><td></td><td></td><td></td></tr><tr><td>Gender</td><td>2</td><td>2</td><td>3</td></tr><tr><td>File Upload</td><td></td><td></td><td></td></tr><tr><td>Password</td><td></td><td></td><td></td></tr><tr><td>Calendar</td><td></td><td></td><td></td></tr><tr><td>Checkboxes</td><td></td><td></td><td></td></tr><tr><td>Drop-down</td><td>1</td><td>1</td><td>15</td></tr><tr><td>Radio Buttons</td><td></td><td></td><td></td></tr><tr><td>Birthday</td><td>1</td><td>1</td><td>1</td></tr><tr><td>Address</td><td>2</td><td>2</td><td>4</td></tr></table>	Question type	Number of times failed validation	Average number of attempts needed to correct	Average time to complete	Email				Name	5	3	34	Phone				Gender	2	2	3	File Upload				Password				Calendar				Checkboxes				Drop-down	1	1	15	Radio Buttons				Birthday	1	1	1	Address	2	2	4
Type	Number of times failed validation	Average number of attempts needed to correct	Average time to complete																																																																																																										
Email																																																																																																													
Name	5	3	34																																																																																																										
Phone																																																																																																													
Gender	2	2	3																																																																																																										
File Upload																																																																																																													
Password																																																																																																													
Calendar																																																																																																													
Checkboxes																																																																																																													
Drop-Down	1	1	15																																																																																																										
Radio Buttons																																																																																																													
Birthday	1	1	1																																																																																																										
Address	2	2	4																																																																																																										
Question type	Number of times failed validation	Average number of attempts needed to correct	Average time to complete																																																																																																										
Email																																																																																																													
Name	5	3	34																																																																																																										
Phone																																																																																																													
Gender	2	2	3																																																																																																										
File Upload																																																																																																													
Password																																																																																																													
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19	Soft Dev	The numbers in the columns in the report are centered.			
20	Soft Dev	The date is displayed at the top of the report			

21

Soft Dev

The report is printed in portrait however the image is stretched strangely and is difficult to read.

bob123's report		02-01-2019	
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Phone			
Gender	2	2	9
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Checkboxes			
Drop-down			
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Address	0	0	5
date of birth			
Country of Birth			
Photo upload			
Video upload			
An example new type			

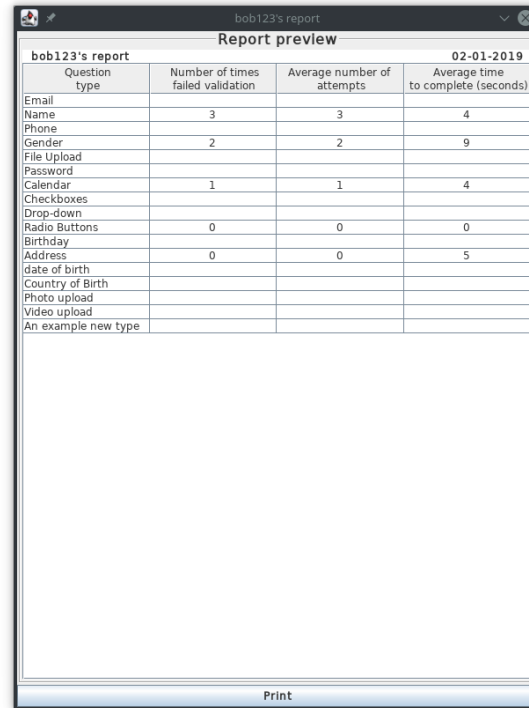
This is because the height and width of the report are scaled by different amounts. Scaling them both by the same amount should fix this.

bob123's report		02-01-2019	
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Gender	2	2	9
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Address	0	0	5
date of birth			
Country of Birth			
Photo upload			
Video upload			
An example new type			

22

Soft Dev

The report preview window is sized more like an a4 piece of paper. This means that it more closely matches the design from the design document.



Question type	Number of times failed validation	Average number of attempts	Average time to complete (seconds)
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Name	3	3	4
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Gender	2	2	9
File Upload			
Password			
Calendar	1	1	4
Checkboxes			
Drop-down			
Radio Buttons	0	0	0
Birthday			
Address	0	0	5
date of birth			
Country of Birth			
Photo upload			
Video upload			
An example new type			

Print

23

Soft Dev

The print report button is the correct colour.

Report preview

bob123's report

Question type	Number of times failed validation	Average number of attempts	Average time to complete (seconds)
Email			
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Phone			
Gender	2	2	9
File Upload			
Password			
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Checkboxes			
Drop-down			
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date of birth			
Country of Birth			
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Video upload			
An example new type			

Print

24

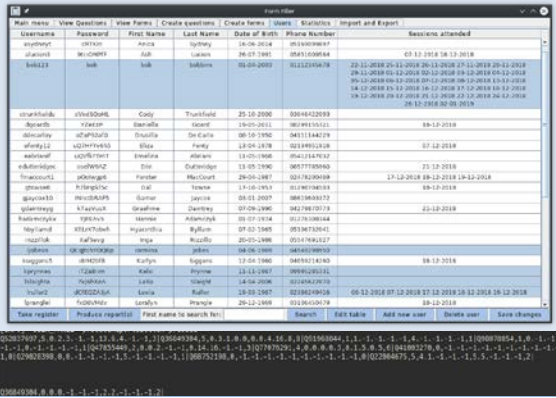
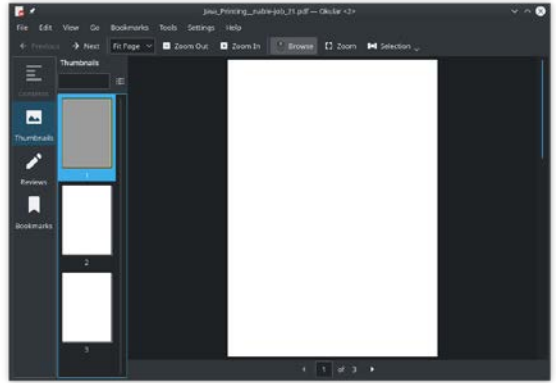
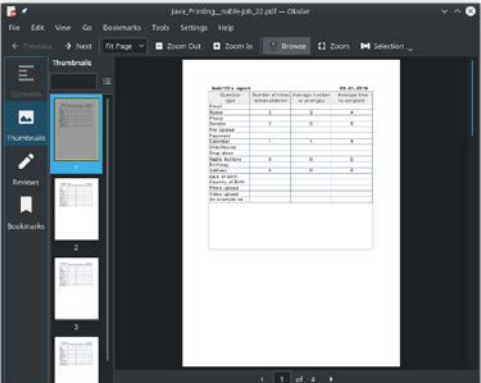
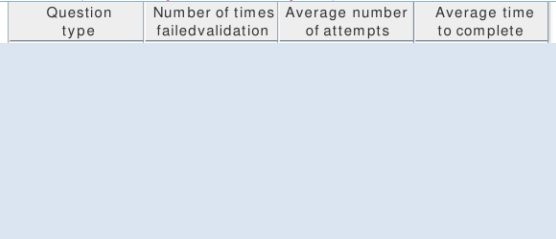
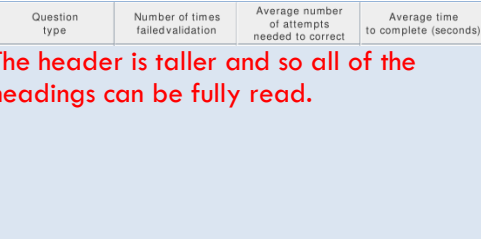
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There is a produce multiple reports button and presses to it are detected.

Admin Editor

Menu: View Questions | View Forms | Create questions | Create forms | Users | Statistics | Import and Export

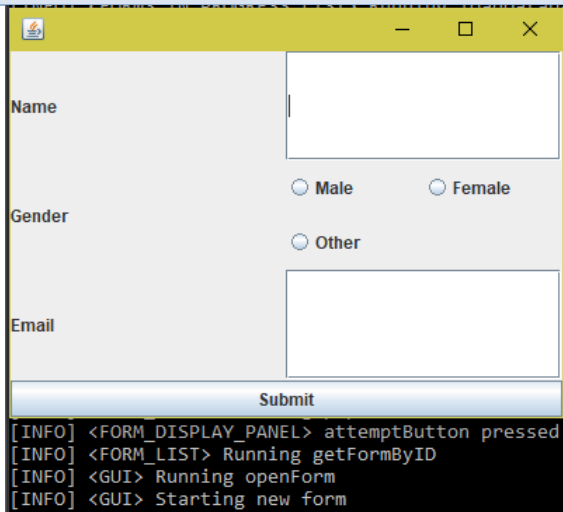
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30-04-2023 01-05-2023 03-05-2023 05-05-2023 07-05-2023 09-05-2023 11-05-2023 13-05-2023 15-05-2023 17-05-2023 19-05-2023 21-05-2023 23-05-2023 25-05-2023 27-05-2023 29-05-2023 31-05-2023 02-06-2023 04-06-2023 06-06-2023 08-06-2023 10-06-2023 12-06-2023 14-06-2023 16-06-2023 18-06-2023 20-06-2023 22-06-2023 24-06-2023 26-06-2023 28-06-2023 30-06-2023 01-07-2023 03-07-2023 05-07-2023 07-07-2023 09-07-2023 11-07-2023 13-07-2023 15-07-2023 17-07-2023 19-07-2023 21-07-2023 23-07-2023 25-07-2023 27-07-2023 29-07-2023 31-07-2023 02-08-2023 04-08-2023 06-08-2023 08-08-2023 10-08-2023 12-08-2023 14-08-2023 16-08-2023 18-08-2023 20-08-2023 22-08-2023 24-08-2023 26-08-2023 28-08-2023 30-08-2023 01-09-2023 03-09-2023 05-09-2023 07-09-2023 09-09-2023 11-09-2023 13-09-2023 15-09-2023 17-09-2023 19-09-2023 21-09-2023 23-09-2023 25-09-2023 27-09-2023 29-09-2023 01-10-2023 03-10-2023 05-10-2023 07-10-2023 09-10-2023 11-10-2023 13-10-2023 15-10-2023 17-10-2023 19-10-2023 21-10-2023 23-10-2023 25-10-2023 27-10-2023 29-10-2023 31-10-2023 02-11-2023 04-11-2023 06-11-2023 08-11-2023 10-11-2023 12-11-2023 14-11-2023 16-11-2023 18-11-2023 20-11-2023 22-11-2023 24-11-2023 26-11-2023 28-11-2023 30-11-2023 01-12-2023 03-12-2023 05-12-2023 07-12-2023 09-12-2023 11-12-2023 13-12-2023 15-12-2023 17-12-2023 19-12-2023 21-12-2023 23-12-2023 25-12-2023 27-12-2023 29-12-2023 31-12-2023 02-01-2024 04-01-2024 06-01-2024 08-01-2024 10-01-2024 12-01-2024 14-01-2024 16-01-2024 18-01-2024 20-01-2024 22-01-2024 24-01-2024 26-01-2024 28-01-2024 30-01-2024 02-02-2024 04-02-2024 06-02-2024 08-02-2024 10-02-2024 12-02-2024 14-02-2024 16-02-2024 18-02-2024 20-02-2024 22-02-2024 24-02-2024 26-02-2024 28-02-2024 01-03-2024 03-03-2024 05-03-2024 07-03-2024 09-03-2024 11-03-2024 13-03-2024 15-03-2024 17-03-2024 19-03-2024 21-03-2024 23-03-2024 25-03-2024 27-03-2024 29-03-2024 31-03-2024 02-04-2024 04-04-2024 06-04-2024 08-04-2024 10-04-2024 12-04-2024 14-04-2024 16-04-2024 18-04-2024 20-04-2024 22-04-2024 24-04-2024 26-04-2024 28-04-2024 30-04-2024 01-05-2024 03-05-2024 05-05-2024 07-05-2024 09-05-2024 11-05-2024 13-05-2024 15-05-2024 17-05-2024 19-05-2024 21-05-2024 23-05-2024 25-05-2024 27-05-2024 29-05-2024 31-05-2024 02-06-2024 04-06-2024

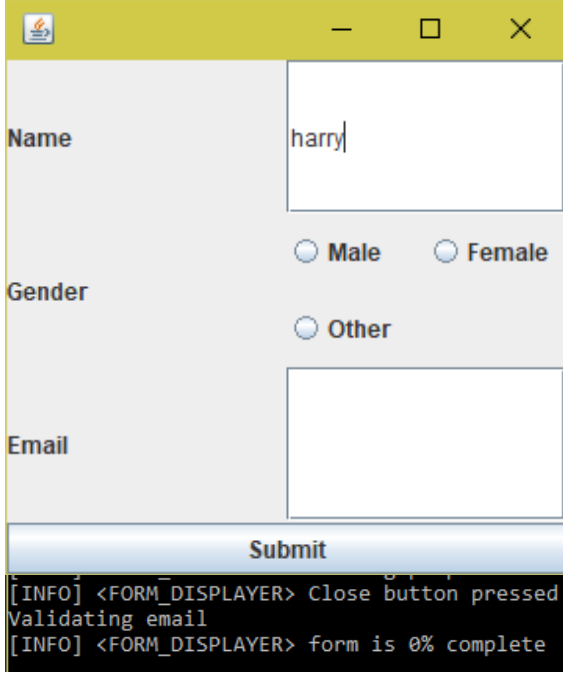
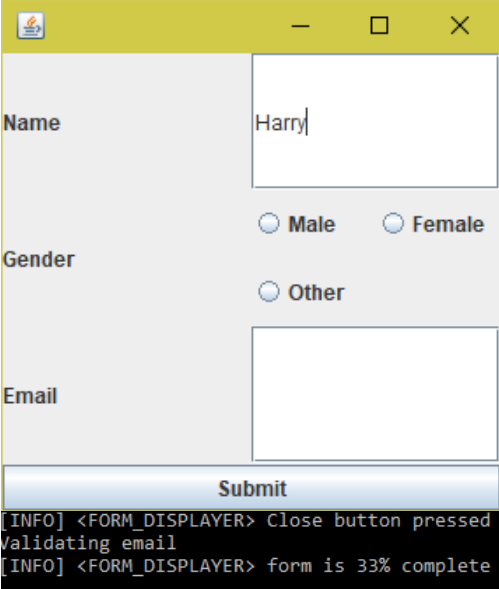
25	Soft Dev	The strings of the question stat lists of the users selected are printed to console when the produce report button is pressed.			
26	Soft Dev	When printing multiple reports the reports are all blank	 As you can see in this screenshot the reports are blank. (I have printed to pdf)	This because the JPanels that are being printed haven't been added to a JFrame therefore they are not drawn. Adding them to an invisible JFrame before printing should fix this.	 The reports are printed.
27	Soft Dev	Some of the header text is cut off.		This is because the header is not tall enough to display all of the text. Setting the size of the header text to be larger should fix this.	

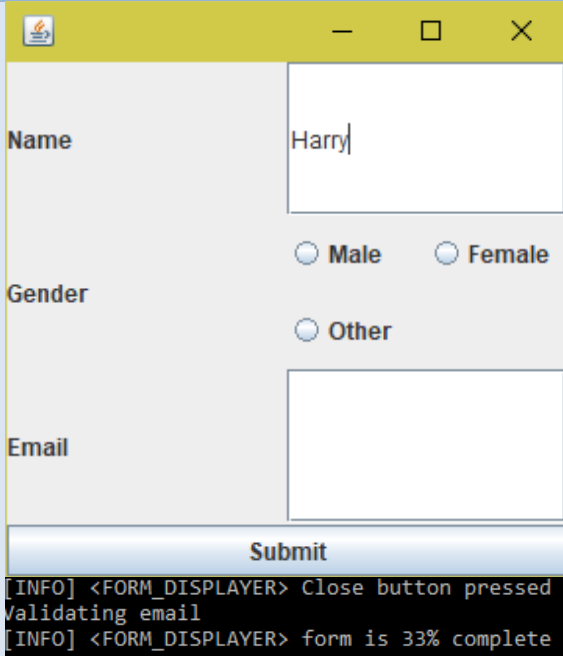
43: How much of each form a user has attempted should be tracked.

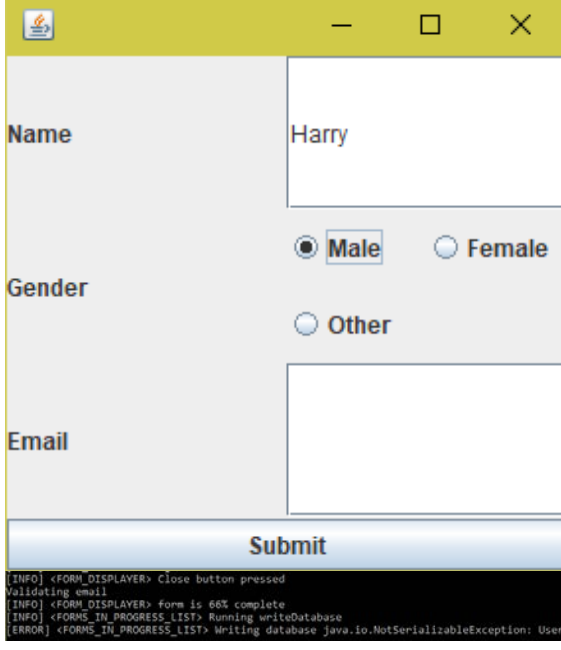
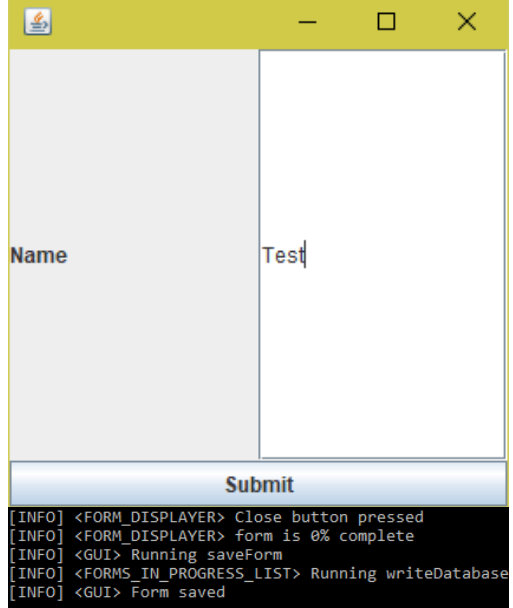
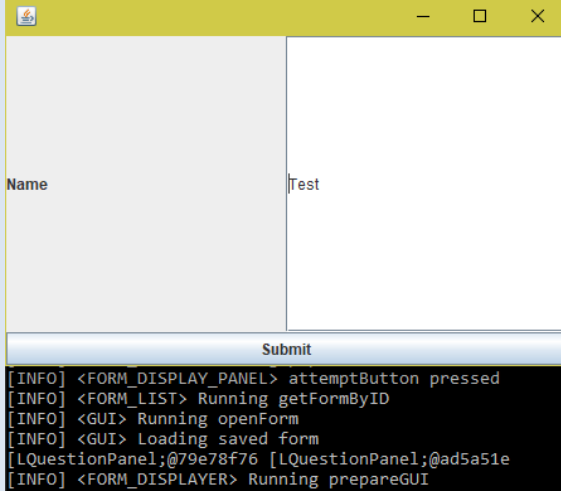
Test	Stage	Result Problem	Evidence	Action and Justification (If Needed)	Evidence of Action (If Needed)
1	Proto	FormInProgress.java fails to compile because the Symbol Serializable cannot be found		Import java.io.*; as Serializable is part of the java.io package	
2	Proto	A FormInProgress object can be added to the FormsInProgress list.	The added from object is underlined in red		
3	Proto	Error writing the FormsInProgress array to file.	The correct directories in are created (underlined in red).	The correct directories are created however FormsInProgressDB .ser is created as as directory when it should be a file. Stopping this directory from being created should fix the error.	Database successfully serialized The correct directories are created and nothing more
4	Proto	The formsInProgress array can be written to file			

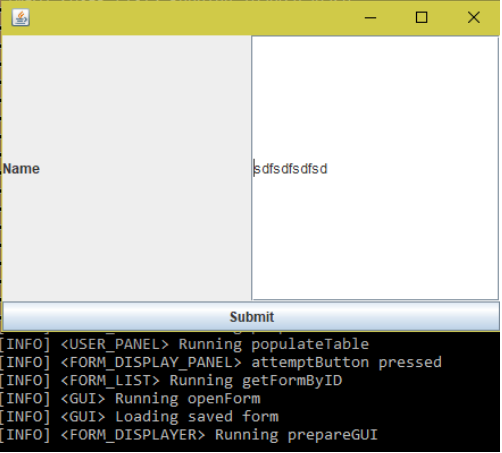
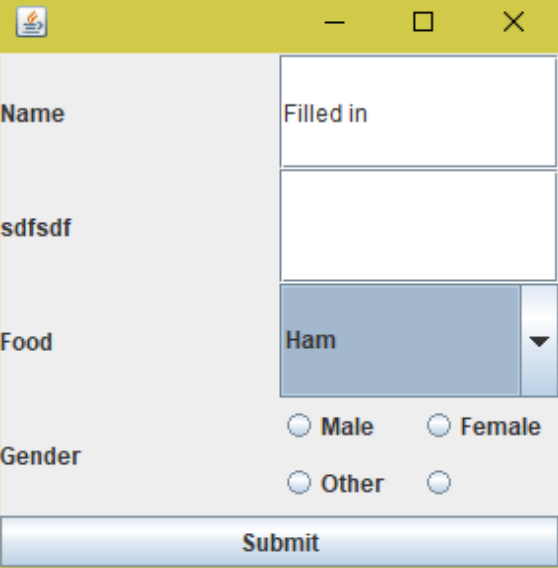
10	Proto	Multiple incompatible type errors when compiling GUI.java	<pre> D:\Computer Science\Component 3\Prototype\compile GUI.java:36: error: incompatible types: User cannot be converted to String formsInProgress = new FormsInProgressList(currentUser); ^ GUI.java:104: error: incompatible types: String[] cannot be converted to String formQuestionPanels[i] = questions.getPanelByID(questionIDs); ^ // Add the relevant question panel to the formQuestionPanels array GUI.java:108: error: incompatible types: Component[] cannot be converted to QuestionPanel[] new FormDisplay(f, formQuestionPanels, currentUser, users); ^ </pre>	Resolve each error.	
11	Proto	Type error when initialising a new FormsInProgressList	<pre> GUI.java:36: error: incompatible types: User cannot be converted to String formsInProgress = new FormsInProgressList(currentUser); ^ </pre>	Call .getUsername() on user to get the username and pass that to the constructor because the username of the username should be passed to the FormsInProgressList constructor not the user object	<pre> GUI.java:104: error: incompatible types: String[] cannot be converted to String formQuestionPanels[i] = questions.getPanelByID(questionIDs); ^ // Add the relevant question panel to the formQuestionPanels array GUI.java:108: error: incompatible types: Component[] cannot be converted to QuestionPanel[] new FormDisplay(f, formQuestionPanels, currentUser, users); ^ </pre> Error no longer present in javac output
12	Proto	Type error when calling getPanelByID	<pre> GUI.java:104: error: incompatible types: String[] cannot be converted to String formQuestionPanels[i] = questions.getPanelByID(questionIDs); ^ // Add the relevant question panel to the formQuestionPanels array </pre>	add [i] to the end of questionIDs to get the String in the relevant place in the array rather than the String array	<pre> GUI.java:108: error: incompatible types: Component[] cannot be converted to QuestionPanel[] new FormDisplay(f, formQuestionPanels, currentUser, users); ^ </pre> Error no longer present in javac output
13	Proto	Type error when initialising a new FormDisplay	<pre> GUI.java:108: error: incompatible types: Component[] cannot be converted to QuestionPanel[] new FormDisplay(f, formQuestionPanels, currentUser, users); ^ </pre>	Change the type of formQuestionPanels from Component[] to QuestionPanel[] because it's storing questionPanels and the constructor expects a QuestionPanel	<pre> FormsInProgressList.java:35: error: cannot find symbol if (form.getID().equals(searchFormID)) // If the correct form has ^ been found symbol: variable searchFormID location: class FormsInProgressList FormDisplay.java:82: error: incompatible types: QuestionPanel cannot be converted to String String questionPanel = questionPanels[i]; ^ FormDisplay.java:84: error: cannot find symbol boolean passed = questionPanel.validateAnswers(); // Validates the an ^ swer symbol: method validateAnswers() location: variable questionPanel of type String FormDisplay.java:85: error: cannot find symbol currentUser.getQuestionStats().getQuestionStatByID(questionPa ^ nel.getID()).addFailedValidation(); symbol: method getQuestionID() location: variable questionPanel of type String 4 errors </pre> No more errors due to GUI.java when compiling

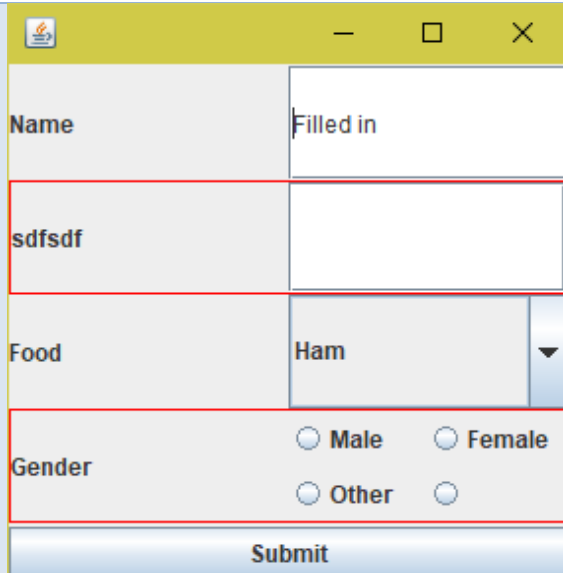
14	Proto	Symbol not found error when compiling FormsInProgressList.java	<pre>FormsInProgressList.java:35: error: cannot find symbol if (form.getFormID().equals(searchFormID)) // If the correct form has ^ been found symbol: variable searchFormID location: class FormsInProgressList</pre>	array. Change searchFormID to searchID as that's the identifier of the variable that stores the id to be searched for	<pre>FormDisplay.java:82: error: incompatible types: QuestionPanel cannot be converted to String String questionPanel = questionPanel[i]; ^ FormDisplay.java:84: error: cannot find symbol boolean passed = questionPanel.validateAnswers(); // Validates the an ^ user symbol: method validateAnswers() location: variable questionPanel of type String FormDisplay.java:89: error: cannot find symbol currentUser.getQuestionStats().getQuestionStatByID(questionPa ^ nel.getQuestionID()).addFailedValidation(); symbol: method getQuestionID() location: variable questionPanel of type String 3 errors</pre> No more errors due to FormsInProgressList.java when compiling
15	Proto	Multiple type error when compiling FormDisplay.java due to a variable that stores a QuestionPanel being of type String rather than QuestionPanel	<pre>FormDisplay.java:82: error: incompatible types: QuestionPanel cannot be converted to String String questionPanel = questionPanel[i]; ^ FormDisplay.java:84: error: cannot find symbol boolean passed = questionPanel.validateAnswers(); // Validates the an ^ user symbol: method validateAnswers() location: variable questionPanel of type String FormDisplay.java:89: error: cannot find symbol currentUser.getQuestionStats().getQuestionStatByID(questionPa ^ nel.getQuestionID()).addFailedValidation(); symbol: method getQuestionID() location: variable questionPanel of type String 3 errors</pre>	Change the type of the variable from String to QuestionPanel because the variable should be of type QuestionPanel because a QuestionPanel is to be stored in it	<pre>J:\Computer Science\Component 3\Prototype>compile J:\Computer Science\Component 3\Prototype></pre> Successful compile
16	Proto	When a form is attempted and a copy of it isn't stored in the user's FormsInProgressList then the form is opened with all the fields blank.			

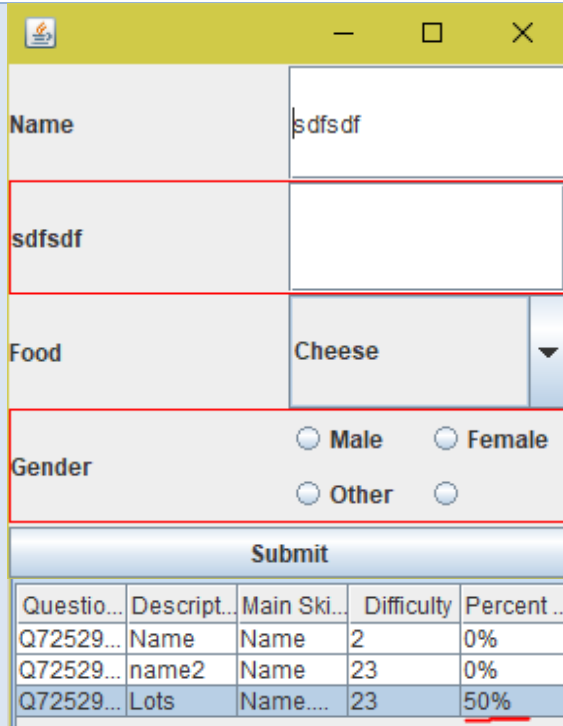
17	Proto	When the close button of the form is pressed the button press is detected and close button pressed is outputted just before the form closes.	<pre>[INFO] <FORM_DISPLAYER> Running prepareGUI [INFO] <FORM_DISPLAYER> Close button pressed</pre>		
18	Proto	Even though 1 question is filled out when the close button is pressed the form is 0% complete is outputted rather than 33% complete.	 <pre>[INFO] <FORM_DISPLAYER> Close button pressed Validating email [INFO] <FORM_DISPLAYER> form is 0% complete</pre>	Change the order of operations when calculating the percentage. NumberCorrectTotalNumberOfQuestions returns a decimal and even though it is being multiplied by 100 (later on) it's being rounded down to 0 because it's less than zero and being stored in an int. Multipling NumberCorrect by 100 and then doing the division will fix this issue.	 <pre>[INFO] <FORM_DISPLAYER> Close button pressed Validating email [INFO] <FORM_DISPLAYER> form is 33% complete</pre> The correct percentage complete is displayed

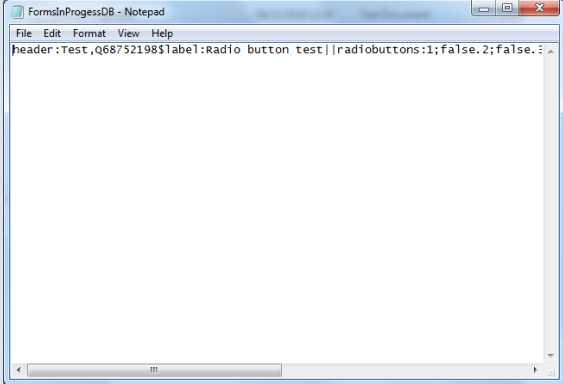
19	Proto	When the close button on a form is pressed (the close window button) the percentage that the form is complete is outputted. This is calculated as the percentage of questions that passed their validation checks	 <pre>[INFO] <FORM_DISPLAYER> Close button pressed Validating email [INFO] <FORM_DISPLAYER> form is 33% complete</pre>		
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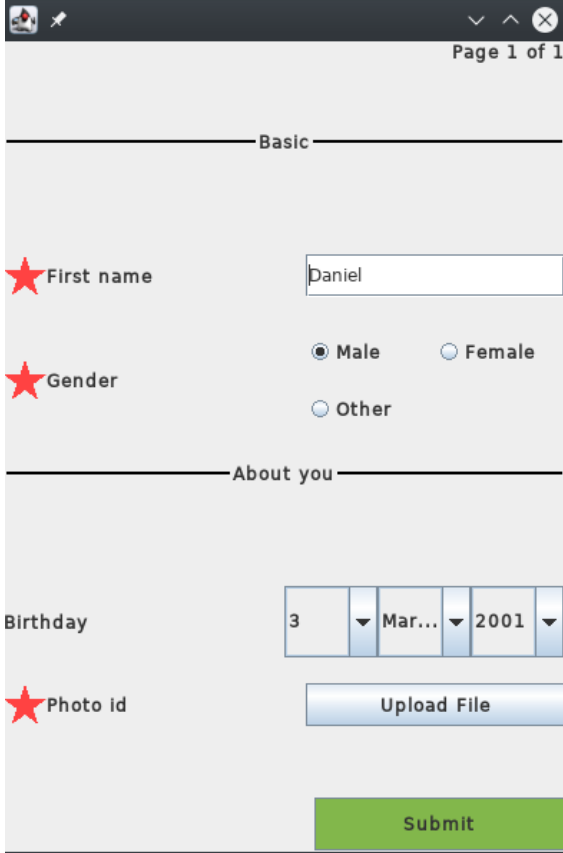
20	Proto	Not serialisable exception when writing the FormsInProgressList to file	 <pre> [INFO] <FORM_DISPLAYER> Close button pressed Validating email [INFO] <FORM_DISPLAYER> form is 66% complete [INFO] <FORMS_IN_PROGRESS_LIST> Running writeDatabase [ERROR] <FORMS_IN_PROGRESS_LIST> Writing database java.io.NotSerializableException: User </pre>	This is because there are references to the QuestionPanels in the FormDisplay class therefore when the QuestionPanels are serialised the FormDisplay is also serialised. Removing the QuestionPanels from the FormDisplay using this.removeAll() should fix this.	 <pre> [INFO] <FORM_DISPLAYER> Close button pressed [INFO] <FORM_DISPLAYER> form is 0% complete [INFO] <GUI> Running saveForm [INFO] <FORMS_IN_PROGRESS_LIST> Running writeDatabase [INFO] <GUI> Form saved </pre> Form in progress object successfully saved to file
21	Proto	Form successfully loaded from file and progress has been saved.	 <pre> [INFO] <FORM_DISPLAY_PANEL> attemptButton pressed [INFO] <FORM_LIST> Running getFormByID [INFO] <GUI> Running openForm [INFO] <GUI> Loading saved form [LQuestionPanel;@79e78f76 [LQuestionPanel;@ad5a51e [INFO] <FORM_DISPLAYER> Running prepareGUI </pre>		

22	Proto	When the mouse enters the saved from large amounts of null point exceptions are created.	<pre> Exception in thread "Awt-EventQueue-0" java.lang.NullPointerException at formDisplay.mouseEntered(formDisplay.java:170) at java.awt.AWTEventMulticaster.mouseEntered(Unknown Source) at java.awt.Component.processMouseEvent(Unknown Source) at javax.swing.JComponent.processMouseEvent(Unknown Source) at java.awt.Component.processEvent(Unknown Source) at java.awt.Container.processEvent(Unknown Source) at java.awt.Container.dispatchEventImpl(Unknown Source) at java.awt.Container.dispatchEvent(Unknown Source) at java.awt.LightweightDispatcher.retargetMouseEvent(Unknown Source) at java.awt.LightweightDispatcher.trackMouseEventExit(Unknown Source) at java.awt.LightweightDispatcher.processMouseEvent(Unknown Source) at java.awt.LightweightDispatcher.dispatchEvent(Unknown Source) at java.awt.Container.dispatchEventImpl(Unknown Source) at java.awt.Window.dispatchEventImpl(Unknown Source) at java.awt.Component.dispatchEvent(Unknown Source) at java.awt.EventQueue.dispatchEventImpl(Unknown Source) at java.awt.EventQueue.access\$508(Unknown Source) at java.awt.EventQueue\$3.run(Unknown Source) at java.awt.EventQueue\$3.run(Unknown Source) at java.security.AccessController.doPrivileged(Native Method) at java.security.ProtectionDomain\$JavaSecurityAccessImpl.doIntersectionPrivilege(Unknown Source) at java.security.ProtectionDomain\$JavaSecurityAccessImpl.doIntersectionPrivilege(Unknown Source) at java.awt.EventQueue\$4.run(Unknown Source) at java.awt.EventQueue\$4.run(Unknown Source) at java.security.AccessController.doPrivileged(Native Method) at java.security.ProtectionDomain\$JavaSecurityAccessImpl.doIntersectionPrivilege(Unknown Source) at java.awt.EventQueue.dispatchEvent(Unknown Source) at java.awt.EventDispatchThread.pumpOneEventForFilters(Unknown Source) at java.awt.EventDispatchThread.pumpEventsForFilter(Unknown Source) at java.awt.EventDispatchThread.pumpEventsForHierarchy(Unknown Source) at java.awt.EventDispatchThread.pumpEvents(Unknown Source) at java.awt.EventDispatchThread.pumpEvents(Unknown Source) at java.awt.EventDispatchThread.run(Unknown Source) </pre>	This could be because there were mouse listeners added to the questions before saving and after loading the object that they point to doesn't exist any more. Removing the mouseListener from each QuestionPanel before saving should fix this	 No more null pointer exceptions when the mouse in in the form.
23	Proto	The progress of more complicated forms is successfully saved.	 After program restart		

																									
24	Proto	Error compiling FormDisplayPanel.java because the variable i has already been defined previously in the method	<pre>D:\Computer Science\Component 3\Prototype>compile FormDisplayPanel.java:133: error: variable i is already defined in method populateTable(Form []) for (int i = 0; i < formData.length; i++) ^ 1 error</pre>	Change the i to a j so that it no longer conflicts with the outer loop.	<pre>D:\Computer Science\Component 3\Prototype>compile D:\Computer Science\Component 3\Prototype> Successful compile</pre>																				
25	Proto	Although there is now a percent complete column the table in the FormDisplayPanel is incorrectly populated	<table><tr><th>Questions</th><th>Description</th><th>Main Skills Te...</th><th>Difficulty</th><th>Percent Com...</th></tr><tr><td></td><td></td><td></td><td></td><td>0%</td></tr><tr><td>Q72529010</td><td></td><td></td><td></td><td>0%</td></tr><tr><td></td><td>Lots</td><td></td><td></td><td>0%</td></tr></table>	Questions	Description	Main Skills Te...	Difficulty	Percent Com...					0%	Q72529010				0%		Lots			0%	Although i was change to j in the loop constructor I didn't actually change it from i to j inside the loop. Replacing all the is with js should fix this	
Questions	Description	Main Skills Te...	Difficulty	Percent Com...																					
				0%																					
Q72529010				0%																					
	Lots			0%																					
26	Proto	The table is correctly populated	<table><tr><th>Questions</th><th>Descripti...</th><th>Main Skil...</th><th>Difficulty</th><th>Percent ...</th></tr><tr><td>Q72529...</td><td>Name</td><td>Name</td><td>2</td><td>0%</td></tr><tr><td>Q72529...</td><td>name2</td><td>Name</td><td>23</td><td>0%</td></tr><tr><td>Q72529...</td><td>Lots</td><td>Name.N...</td><td>23</td><td>0%</td></tr></table>	Questions	Descripti...	Main Skil...	Difficulty	Percent ...	Q72529...	Name	Name	2	0%	Q72529...	name2	Name	23	0%	Q72529...	Lots	Name.N...	23	0%		
Questions	Descripti...	Main Skil...	Difficulty	Percent ...																					
Q72529...	Name	Name	2	0%																					
Q72529...	name2	Name	23	0%																					
Q72529...	Lots	Name.N...	23	0%																					

27	Proto	When a partially completed form is loaded from file the percentage complete is correct			
28	Soft Dev	There is a null pointer exception when the saveForm method is run	<pre>[INFO] <GUI> Running saveForm Exception in thread "AWT-EventQueue-0" java.lang.NullPointerException at GUI.saveForm(GUI.java:171) at FormDisplayer.windowClosing(FormDisplayer.java:606) at java.awt.Window.processWindowEvent(Unknown Source) at javax.swing.JFrame.processWindowEvent(Unknown Source) at java.awt.Window.dispatchEvent(Unknown Source) at java.awt.Component.dispatchEventImpl(Unknown Source) at java.awt.Container.dispatchEventImpl(Unknown Source) at java.awt.Window.dispatchEventImpl(Unknown Source) at java.awt.Component.dispatchEvent(Unknown Source) at java.awt.EventQueue.dispatchEventImpl(Unknown Source) at java.awt.EventQueue.access\$500(Unknown Source) at java.awt.EventQueue\$3.run(Unknown Source) at java.awt.EventQueue\$3.run(Unknown Source)</pre>	This this because when the form has't been attempted previously it's not in the FormInProgress database. When a form is saved it's retrieved from the form in progress database, updated, and then saved. Because	<pre>[INFO] <FORMS_IN_PROGRESS_LIST> Running writeDatabase [INFO] <GUI> Form saved</pre> No error when saving the form.

				the form wasn't in the database when it was retrieved there was a null pointer exception. Making sure that the form is added to the database before saving (if it's not present will fix this)	
29	Soft Dev	When a form is closed it's formInProgress object is saved to file.	<pre> [INFO] <FORM_DISPLAYER> Close button pressed [INFO] <FORM_DISPLAYER> Form is 100% complete [INFO] <GUI> Running saveForm F22688749,Test.Q68752198,dsfdsfdsf,File Upload,2,dgdsfds,false,true 100 [INFO] <FORMS_IN_PROGRESS_LIST> Running writeDatabase [INFO] <FORM_IN_PROGRESS> Running toString OS: header:Test,Q68752198\$label:Radio button test radiobuttons:1;false.2;false.3;false.3;false.4;false.44;false.43;true.3;false.3;false.42;false.34;false.3;false.3;false.4;false.44;false.43;true.3;false.3;false.42;false.34;false.3;false.4;false.44;false.43;true.3;false.3;false.42;false.34;false [INFO] <GUI> Form saved </pre> 		

30	Soft Dev	When a form is attempted that has been filled in before the users past answers are loaded	 This is a form that the user attempted and then closed the system. As you can see their answers have been saved and loaded from file.		
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31

Soft Dev

When a form is completed the times completed count in the view forms panel is incremented

The screenshot shows a web application titled 'Form Builder'. It has a navigation bar with links: 'View Questions', 'View Forms', 'Create questions', 'Create forms', 'Users', 'Statistics', 'Logout and Logout'. Below the navigation bar is a table with the following columns: 'Title', 'Description', 'Main Skills Tested', 'Difficulty', 'Percent', 'Times Completed', and 'Complete'. The table contains one row with the following data: 'Back info', 'Add for team info', 'Name, Gender, birthday, no signat', '3', '75%', and '0'. To the right of the table is a 'Sort and Filter' panel. It has a 'Sorts' section with a 'Difficulty sort' button. Below that is a 'Type filter' section with checkboxes for 'Email', 'Name', 'Phone', 'Gender', 'File Upload', 'Password', 'Calendar', 'Checkbox', 'Drop down', 'Radio Buttons', 'Birthday', 'Address', 'Date of birth', 'Country of Birth', 'Photo upload', 'Video upload', and 'No example new type'. There are buttons for 'Apply main skills tested filter', 'Reset sorts and filters', 'Delete Form', 'Strategic form based on workload', and 'Adapted Form'.

The form has been completed 0 times

Page 1 of 1

Basic

★ First name

★ Gender ☒ Male ☐ Female

Message

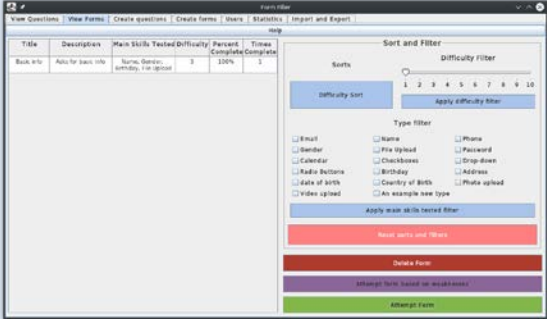
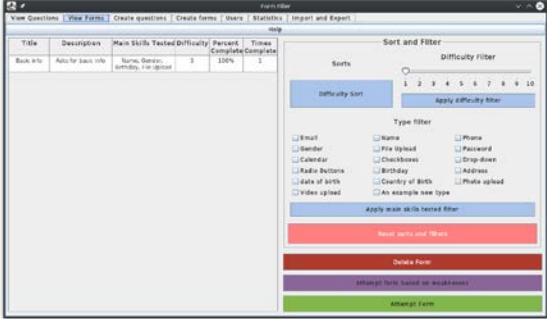
Form complete!

OK

Birthday

★ Photo id

The form is attempted and completed

			 The times completed counter has been incremented		
32	Soft Dev	When a form that is 100% complete is attempted again the form is reset.	 <pre>[INFO] <GUI> Running openForm [INFO] <GUI> Resetting form</pre>		

Page 1 of 1

Basic

★ First name

★ Gender ☐ Male ☐ Female
☐ Other

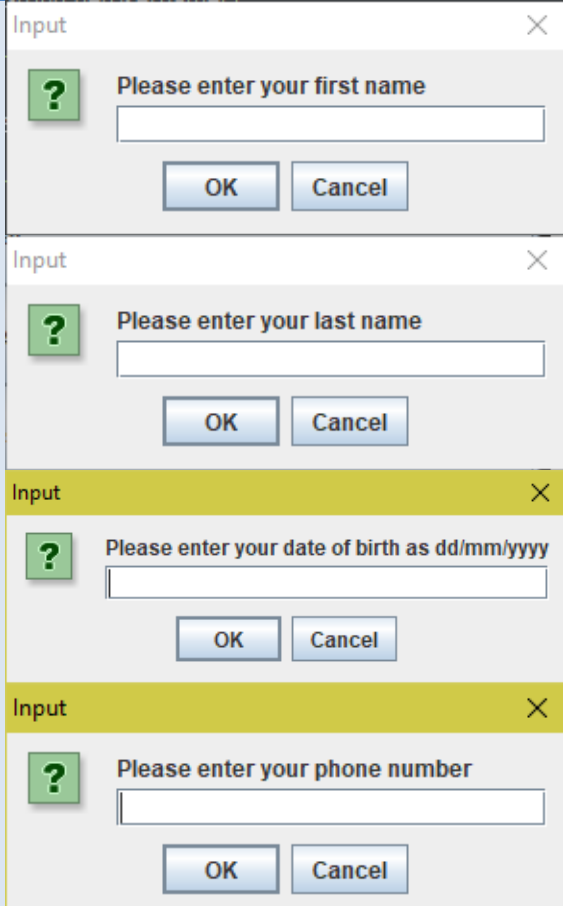
About you

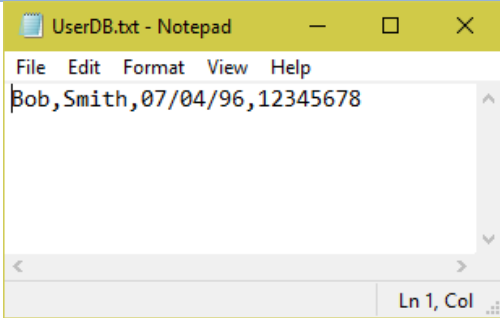
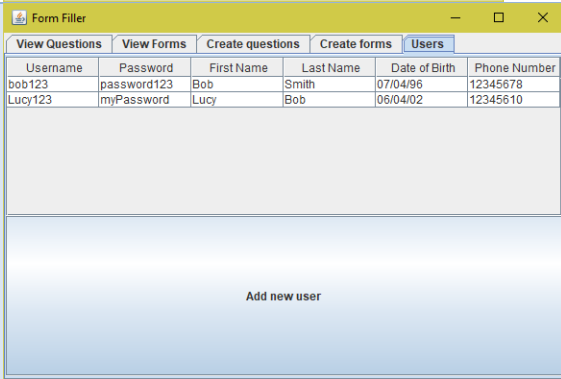
Birthday Day Mo... Year

★ Photo id

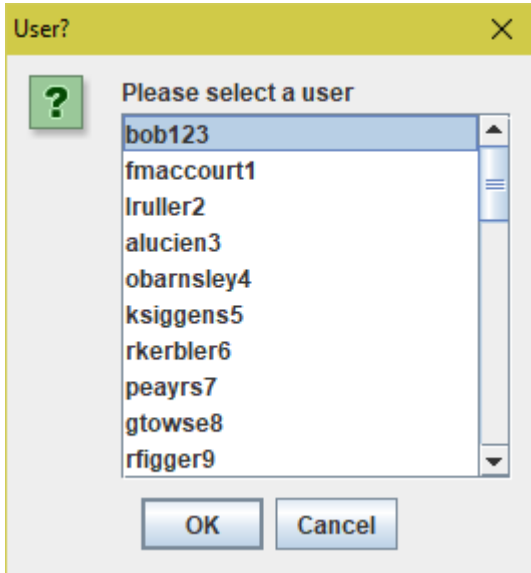
The form has been reset.

44: Firstname, lastname, Date of birth and phone number should be stored for each user

Test	Stage	Result Problem	Evidence	Action and Justification (If Needed)	Evidence of Action (If Needed)
1	Proto	When creating a user, the admin is prompted for their: First name, Last name, date of birth, and phonenumber	 The evidence shows four sequential input dialog boxes. Each dialog has a title bar with 'Input' and a close button. The first dialog asks 'Please enter your first name'. The second asks 'Please enter your last name'. The third asks 'Please enter your date of birth as dd/mm/yyyy'. The fourth asks 'Please enter your phone number'. Each dialog contains a text input field and 'OK' and 'Cancel' buttons.		

2	Proto	This data is saved in the user database file.																					
3	Proto	The data is displayed in the view users table.	 <table border="1"> <thead> <tr> <th>Username</th><th>Password</th><th>First Name</th><th>Last Name</th><th>Date of Birth</th><th>Phone Number</th></tr> </thead> <tbody> <tr> <td>bob123</td><td>password123</td><td>Bob</td><td>Smith</td><td>07/04/96</td><td>12345678</td></tr> <tr> <td>Lucy123</td><td>myPassword</td><td>Lucy</td><td>Bob</td><td>06/04/02</td><td>12345610</td></tr> </tbody> </table> Add new user	Username	Password	First Name	Last Name	Date of Birth	Phone Number	bob123	password123	Bob	Smith	07/04/96	12345678	Lucy123	myPassword	Lucy	Bob	06/04/02	12345610		
Username	Password	First Name	Last Name	Date of Birth	Phone Number																		
bob123	password123	Bob	Smith	07/04/96	12345678																		
Lucy123	myPassword	Lucy	Bob	06/04/02	12345610																		

45: The attendance for each user should be automatically tracked

Test	Stage	Result Problem	Evidence	Action and Justification (If Needed)	Evidence of Action (If Needed)
1	Proto	Error compiling User.java due to a symbol not found error.	<pre>D:\Computer Science\Component 3\Prototype>compile User.java:36: error: cannot find symbol String today = LocalDate.now().toString(); // Get today's date ^ symbol: variable LocalDate location: class User 1 error</pre>	Import java.time.*; as LocalDate is part of the java.time package	<pre>D:\Computer Science\Component 3\Prototype>compile D:\Computer Science\Component 3\Prototype></pre>
2	Proto	When a user logs in today's date is added to their sessions present at array - they are marked present	bob,bob,bobbins,01/04/03,12345678,2018-07-29 Before logging in only 290718 is in the database  Selecting bob		

			Form Filler Create forms Users Statistics Import and Export View Questions View Forms Create questions <table><tr><th>Difficulty</th><th>Type</th></tr><tr><td>4</td><td>Birthday</td></tr><tr><td>3</td><td>Drop-down</td></tr><tr><td>5</td><td>Gender</td></tr><tr><td>2</td><td>Name</td></tr><tr><td>3</td><td>Name</td></tr><tr><td>5</td><td>Address</td></tr></table> Difficulty Sort Type Sort Attempt Question Delete Question Filter by difficulty Filter by type Logged in .bob123,bob,bob,bobbins,01/04/03,12345678,2018-07-29.2018-08-01 Today's date (010818) is now present in the array.	Difficulty	Type	4	Birthday	3	Drop-down	5	Gender	2	Name	3	Name	5	Address		
Difficulty	Type																		
4	Birthday																		
3	Drop-down																		
5	Gender																		
2	Name																		
3	Name																		
5	Address																		
3	Proto	If the user logs again on the same day then the date isn't added to the array again to avoid duplicates.	User? ? Please select a user bob123 fmaccourt1 lruller2 alucien3 obarnsley4 ksiggens5 rkerbler6 peayrs7 gtowse8 rfigger9 OK Cancel Selecting bob																

Selecting bob

Form Filler

Create forms

Users

Statistics

Import and Export

View Questions

View Forms

Create questions

Difficulty	Type
4	Birthday
3	Drop-down
5	Gender
2	Name
3	Name
5	Address

Difficulty Sort

Type Sort

Attempt Question

Delete Question

1 2 3 4 5 6 7 8 9 10

Filter by difficulty

☐ E... ☐ N... ☐ P... ☐ G...
☐ Fil... ☐ P... ☐ C... ☐ C...
☐ Dr... ☐ P... ☐ R... ☐ Bl...
☐ A...

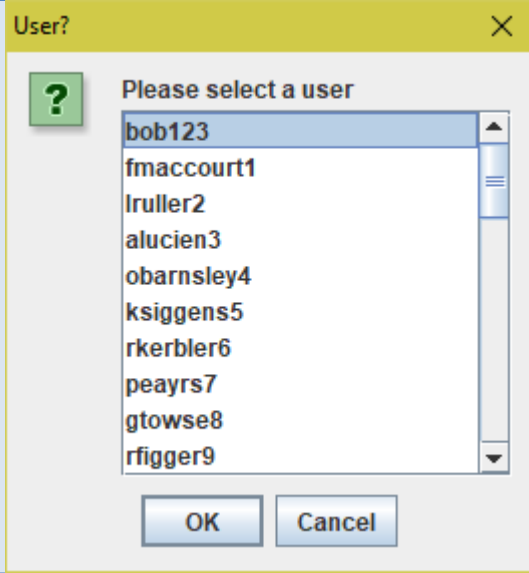
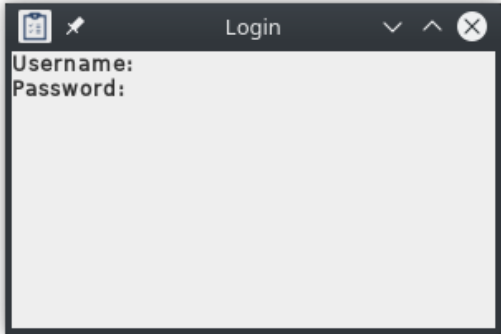
Filter by type

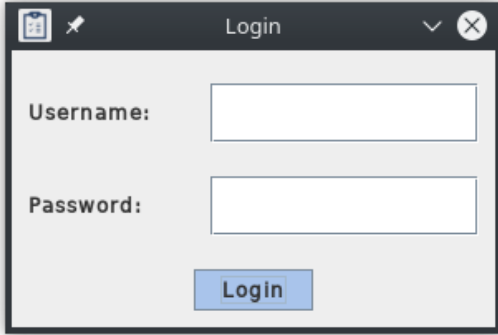
Logged in

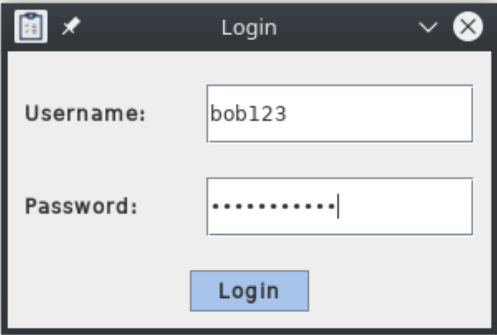
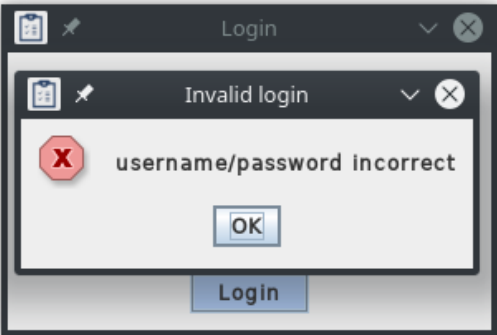
bob123,bob,bob,bobbins,01/04/03,12345678,2018-07-29.2018-08-01

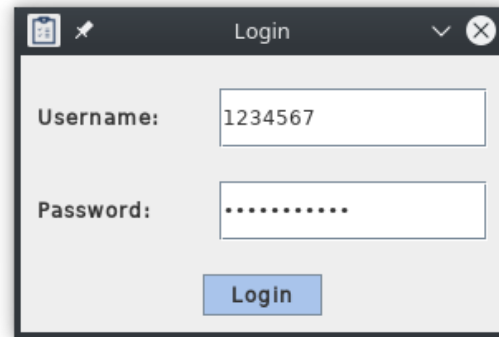
Today's date is only present in the array once

48: Login page which allows users or admins to login

Test	Stage	Result Problem	Evidence	Action and Justification (If Needed)	Evidence of Action (If Needed)
1	Proto	The user can select an account to login to when starting the program. Full login functionality and proper credential validation has not been added for ease of testing			
2	Soft Dev	There is a window with username and password labels.			

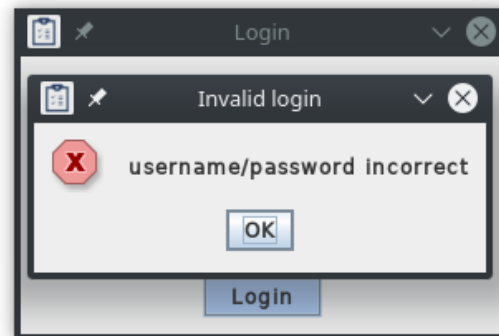
3	Soft Dev	There is a username field, password field, and login button.			
4	Soft Dev	The layout of the components more closely matches the layout in the design.			
5	Soft Dev	Presses to the login button are detected. Pressing enter in the password field also acts as pressing the login button	<pre>[INFO] <LOGIN_FRAME> Running prepareGUI [INFO] <LOGIN_FRAME> loginButton pressed</pre>		

6	Soft Dev	When an incorreft password or username is entered an error message is displayed	 Bob's password is bob so this password is clearly incorrect. 		
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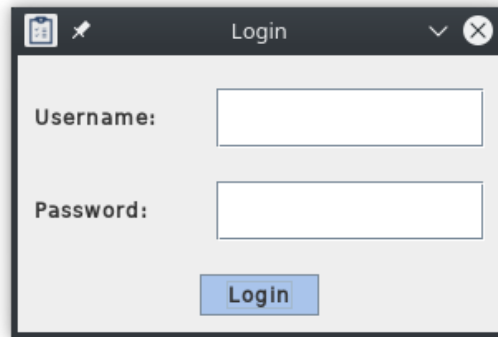


A screenshot of a 'Login' dialog box. It has a title bar with a minimize icon, a maximize icon, and a close icon. The dialog contains two input fields: 'Username:' with the text '1234567' and 'Password:' with masked characters '.....'. Below the fields is a blue 'Login' button.

1234567 isn't a valid username

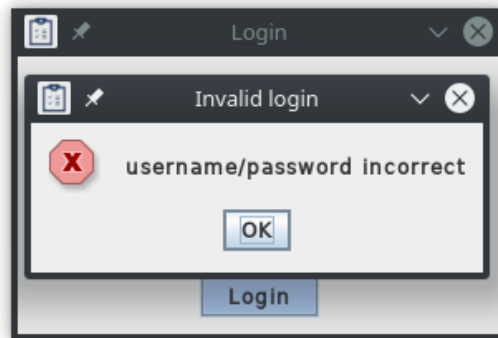


A screenshot of an 'Invalid login' error dialog box. It has a title bar with a minimize icon, a maximize icon, and a close icon. The dialog contains a red octagonal icon with a white 'X' and the text 'username/password incorrect'. Below the text is a blue 'OK' button. The dialog is shown on top of the 'Login' dialog box, which is partially visible behind it.



A screenshot of a 'Login' dialog box. It has a title bar with a minimize icon, a maximize icon, and a close icon. The dialog contains two text input fields: 'Username:' and 'Password:'. Both fields are empty. Below the fields is a 'Login' button.

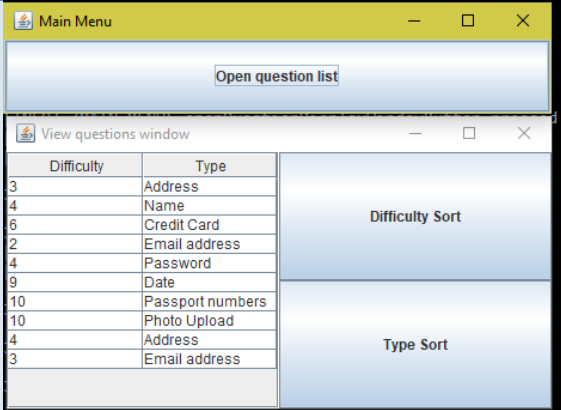
Both fields are empty

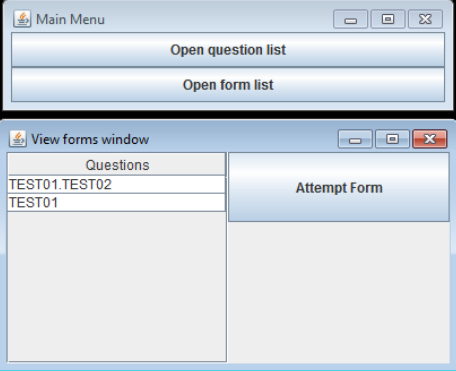
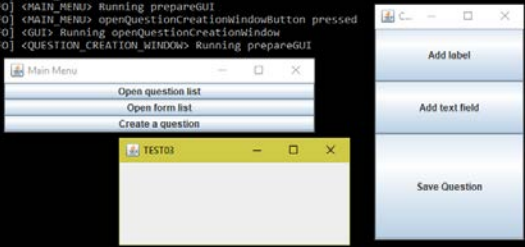


A screenshot of an 'Invalid login' error dialog box. It has a title bar with a minimize icon, a maximize icon, and a close icon. The dialog contains a red octagonal icon with a white 'X' and the text 'username/password incorrect'. Below the text is an 'OK' button. The dialog is shown on top of the 'Login' dialog box, which is partially visible behind it.

7	Soft Dev	When a correct login is entered, the system starts with that user logged in.			
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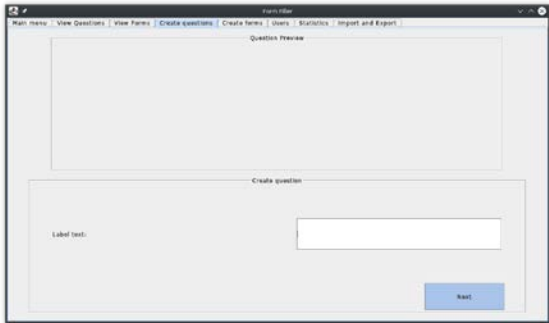
49: There should be a main menu panel

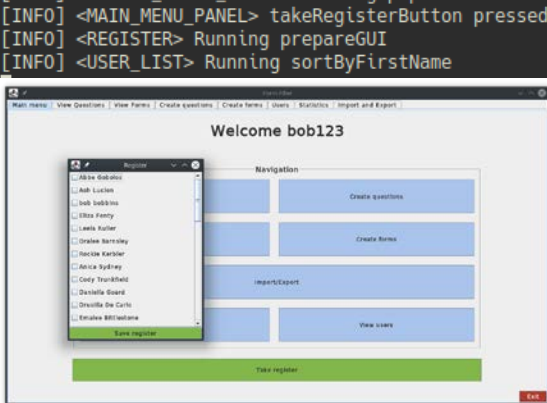
Test	Stage	Result Problem	Evidence	Action and Justification (If Needed)	Evidence of Action (If Needed)
1	Proto	There is a main menu window with a button that opens a question display window	 <pre> D:\Computer Science\Component 3\Prototype>java Main [INFO] <GUI> Running setup [INFO] <QUESTION_LIST> Running loadDatabase [INFO] <MAIN_MENU> Running prepareGUI [INFO] <MAIN_MENU> openQuestionDisplayWindowButton pressed [INFO] <GUI> Running openQuestionDisplayWindow [INFO] <QUESTION_DISPLAY_WINDOW> Running prepareGUI [INFO] <QUESTION_DISPLAY_WINDOW> Running populateTable </pre>		
2	Proto	Error compiling the mainMenu because not the correct arguments weren't passed to the FormDisplayWindow constructor	<pre> D:\Computer Science\Component 3\Prototype>javac *.java GUI.java:41: error: constructor FormDisplayWindow in class FormDisplayWindow not be applied to given types: new FormDisplayWindow(); ^ required: FormList,GUI found: no arguments reason: actual and formal argument lists differ in length 1 error </pre>	Add the correct arguments	<pre> D:\Computer Science\Component 3\Prototype>javac *.java D:\Computer Science\Component 3\Prototype> </pre> Successful compile

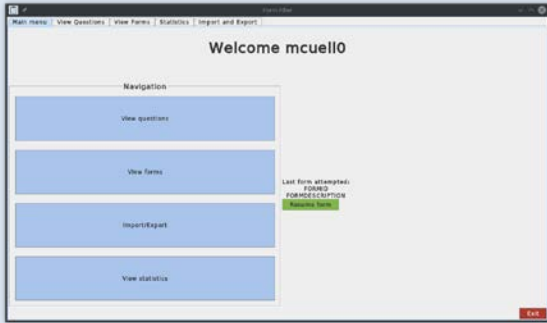
3	Proto	There is a button to open the form list which opens the view forms window	<pre> D:\Computer Science\Component 3\Prototype>java Main [INFO] <GUI> Running setup [INFO] <QUESTION_LIST> Running loadDatabase [INFO] <QUESTION_LIST> 2 Questions loaded from file [INFO] <QUESTION_PANEL_LIST> Running loadDatabase [INFO] <QUESTION_PANEL_LIST> 2 QuestionPanels loaded from file [INFO] <FORM_LIST> Running loadDatabase [INFO] <FORM_LIST> 2 Forms loaded from file [INFO] <MAIN_MENU> Running prepareGUI [INFO] <MAIN_MENU> openFormDisplayWindowButton pressed [INFO] <GUI> Running openFormDisplayWindow [INFO] <FORM_DISPLAY_WINDOW> Running prepareGUI [INFO] <FORM_DISPLAY_WINDOW> Running populateTable </pre> 		
4	Proto	There is a button which opens the create a question window	<pre> D:\Computer Science\Component 3\Prototype>java Main [INFO] <GUI> Running setup [INFO] <QUESTION_LIST> Running loadDatabase [INFO] <QUESTION_LIST> 2 Questions loaded from file [INFO] <QUESTION_PANEL_LIST> Running loadDatabase [INFO] <QUESTION_PANEL_LIST> 2 QuestionPanels loaded from file [INFO] <FORM_LIST> Running loadDatabase [INFO] <FORM_LIST> 2 Forms loaded from file [INFO] <MAIN_MENU> Running prepareGUI [INFO] <MAIN_MENU> openQuestionCreationWindowButton pressed [INFO] <GUI> Running openQuestionCreationWindow [INFO] <QUESTION_CREATION_WINDOW> Running prepareGUI </pre> 		

5	Soft Dev	There is a panel in the main menu tab of the system which displays the current user's username			
6	Soft Dev	The username text is larger and centered at the top			
7	Soft Dev	The text is even larger and is centered properly as specified in the design.			

8	Soft Dev	There are buttons for all elements of the system. The buttons are in the layout specified in the design.			
9	Soft Dev	There is a “navigation” title and border around the buttons			
10	Soft Dev	All of the navigation buttons are functional	 The create questions button was pressed		

					
11	Soft Dev	There is an exit button in the bottom right corner.			
12	Soft Dev	When the exit button is pressed the program closes.	 <pre data-bbox="712 1206 1276 1246"> [INFO] <MAIN_MENU_PANEL> - exit button pressed [INFO] <MAIN_MENU_PANEL> - Terminating program harry@harry-pc-Linux:~/media/harry/HARRY_BEST/Computer Science/Component 3/SoftDev/Project/Form-Filler\$ </pre> The program has been closed		

13	Soft Dev	There is a take register button			
14	Soft Dev	When the take register button is pressed a register opens	<pre>[INFO] <MAIN_MENU_PANEL> takeRegisterButton pressed [INFO] <REGISTER> Running prepareGUI [INFO] <USER_LIST> Running sortByFirstName</pre> 		
15	Soft Dev	When a regular user logs in they only have navigation buttons for the panels that they should have access to.			

16	Soft Dev	When a regular user logs in they no longer have access to the admin tabs at the top.			
17	Soft Dev	There is a continue form button and a placeholder label.			
18	Soft Dev	NullPointerException when getting the ID of the most recent form a user has attempted when they haven't attempted a form yet	<pre>[INFO] <FORM_DISPLAY_PANEL> Running populateTable Exception in thread "main" java.lang.NullPointerException at FormsInProgressList.getByID(FormsInProgressList.java:44) at FormsInProgressList.isFormPresent(FormsInProgressList.java:26) at FormDisplayPanel.populateTable(FormDisplayPanel.java:443) at FormDisplayPanel.prepareTable(FormDisplayPanel.java:215) at FormDisplayPanel.prepareGUI(FormDisplayPanel.java:180) at FormDisplayPanel.<init>(FormDisplayPanel.java:72) at GUI.prepareGUI(GUI.java:107) at GUI.setup(GUI.java:55) at GUI.<init>(GUI.java:30) at Main.main(Main.java:5)</pre>	Checking whether they have attempted a form yet before getting the id will fix this. This should be done by checking that the most recent form attempted isn't null.	<pre>[INFO] <FORM_DISPLAY_PANEL> Running prepareGUI [INFO] <FORM_DISPLAY_PANEL> Running populateTable [INFO] <STATISTICS_PANEL> Running prepareGUI</pre> No more null pointer exception.

19	Soft Dev	If no form has been attempted yet, the text of the label tells the user this.			
20	Soft Dev	When the resume form button is pressed the most recently attempted form is opened for the user.	 <pre data-bbox="712 869 1276 997">[INFO] <MAIN_MENU_PANEL> resumeFormButton pressed [INFO] <FORM_LIST> Running getFormByID [INFO] <GUI> Running openForm [INFO] <GUI> Loading saved form [INFO] <FORM_DISPLAYER> Running getQuestionPanels</pre>		

Page 1 of 1

Ignore me

★ Jame's question

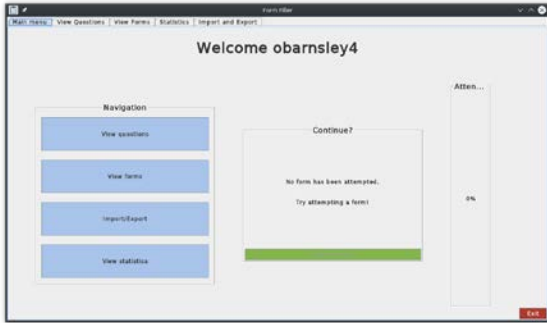
☐ 1 ☐ 2

☐ 3 ☐ 34

☐ 324

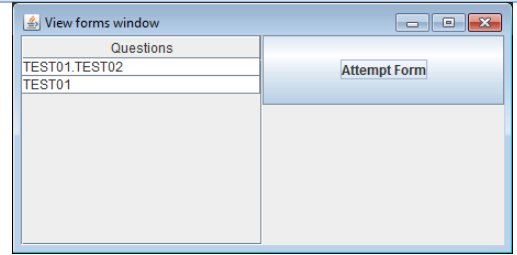
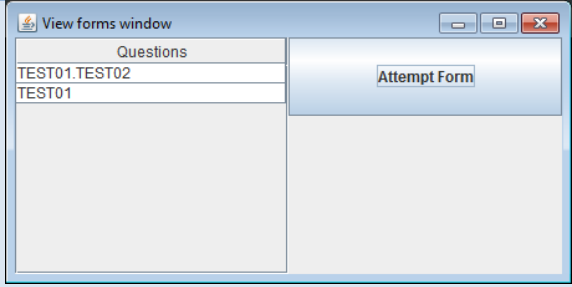
Submit

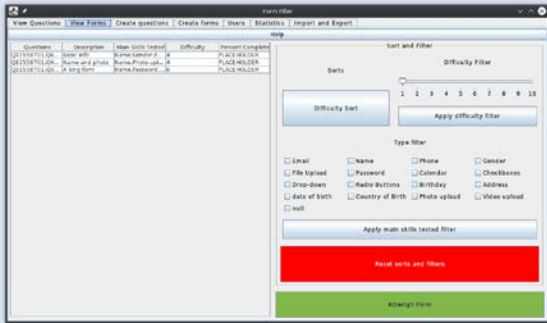
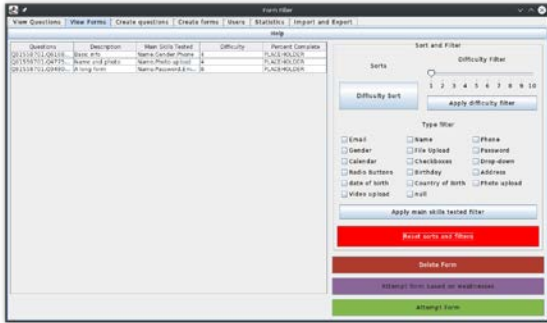
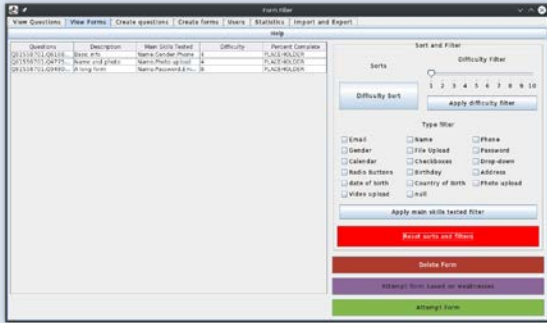
The form is opened

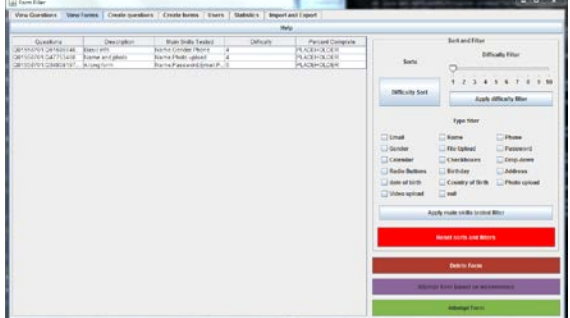
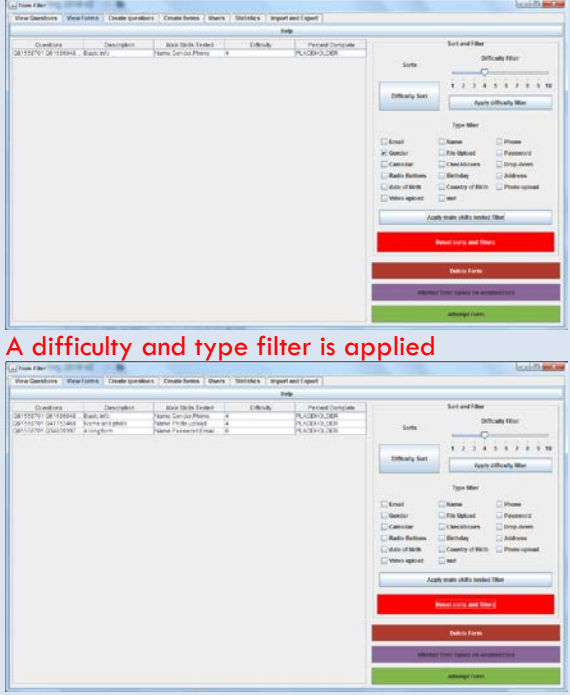
21	Soft Dev	The layout of the resume form panel better matches the one in the design			
22	Soft Dev	There is an attendance panel however it is not wide enough.		This is because the panel doesn't have a minimum size. Setting a minimum size should fix this.	
23	Soft Dev	The font of the percentage is larger so that it can be more easily read.			
24	Soft Dev	Error when compiling MainMenuPanel.java	<pre>- ./compile.sh Picked up _JAVA_OPTIONS: -Dawt.useSystemAAFontSettings=on -Dswing.aatext=true MainMenuPanel.java:129: error: cannot find symbol Date fiveWeeksAgo = new Date(System.currentTimeMillis() - (30 * DAY_IN_MS)); // Calculate ^ the date five weeks ago symbol: class Date location: class MainMenuPanel MainMenuPanel.java:129: error: cannot find symbol Date fiveWeeksAgo = new Date(System.currentTimeMillis() - (30 * DAY_IN_MS)); // Calculate ^ the date five weeks ago</pre>	This is because the symbol Date cannot be found. To fix this the time and date	<pre>harry-pc-linux: /media/harry/HARRY BEST/Computer Science/Component 3/Software/Project/Form-Filler [main- harry@] ~ - ./compile.sh Picked up _JAVA_OPTIONS: -Dawt.useSystemAAFontSettings=on -Dswing.aatext=true Note: Some input files use or override a deprecated API. Note: Recompile with -Xlint:deprecation for details. Picked up _JAVA_OPTIONS: -Dawt.useSystemAAFontSettings=on -Dswing.aatext=true</pre>

				packages should be imported.	
25	Soft Dev	The users percentage attendance is now correctly displayed.			

50: There should be a panel which will display the forms to the user and allow them to search through them

Test	Stage	Result Problem	Evidence	Action and Justification (If Needed)	Evidence of Action (If Needed)
1	Proto	Form display window fails to compile due to a typo	<pre> D:\Computer Science\Component 3\Prototype>javac *.java FormDisplayWindow.java:103: error: cannot find symbol Form selectedForm = forms.getFormByID(getformTable.getMo del().getValueAt(row, 0).toString()); // Get the form id and get the form ^ symbol: variable getformTable location: class FormDisplayWindow 1 error </pre>	Remove the first get from getformTable	<pre> D:\Computer Science\Component 3\Prototype>javac *.java D:\Computer Science\Component 3\Prototype> </pre> Successful compile
2	Proto	Null pointer exception when trying to open the form display window because the formList had never been initialized	<pre> D:\Computer Science\Component 3\Prototype>java Main [INFO] <GUI> Running setup [INFO] <QUESTION_LIST> Running loadDatabase [INFO] <QUESTION_PANEL_LIST> Running loadDatabase [INFO] <QUESTION_PANEL_LIST> 2 QuestionPanels loaded from file [INFO] <MAIN_MENU> Running prepareGUI [INFO] <FORM_DISPLAY_WINDOW> Running prepareGUI Exception in thread "main" java.lang.NullPointerException at FormDisplayWindow.prepareGUI(FormDisplayWindow.java:60) at FormDisplayWindow.<init>(FormDisplayWindow.java:28) at GUI.setup(GUI.java:26) at GUI.<init>(GUI.java:14) at Main.main(Main.java:5) </pre>	Initialize the formList before passing it to the FormDisplayWindow	 Window opens
3	Proto	FormDisplayWindow successfully opens and displays the forms. At this stage it just displays the list of questions that make up the form			
			The rest of the testing for this panel is under the relevant sort and filter objectives in the forms section.		

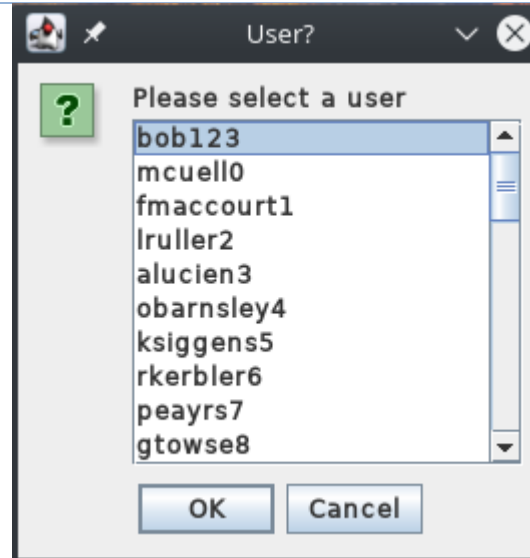
4	Soft Dev	The layout from the QuestionDisplayPanel has been ported to the view form panel. This was done without errors as the panels are very similar.			
5	Soft Dev	There is an attempt form from user weaknesses button.			
6	Soft Dev	The table doesn't fill the entire height of the window.		Increase the gridHeight of the table to 4	

7	Soft Dev	The presses to the reset sorts and filter button are not detected.	 <pre data-bbox="712 523 1279 592">[INFO] <QUESTION_SELECTION_PANEL> Running populateTable [INFO] <FORM_SELECTION_PANEL> Running prepareGUI [INFO] <FORM_SELECTION_PANEL> Running populateTable</pre>	This is because the reset sorts and filters method isn't called when the button is pressed as the action performed method doesn't have an entry for the reset sorts and filters button.	<pre data-bbox="1545 199 2056 244">[INFO] <FORM_DISPLAY_PANEL> resetButton pressed [INFO] <FORM_DISPLAY_PANEL> Running populateTable</pre>
8	Soft Dev	When the reset sorts and filters button is pressed the sorts and filters are reset.	 A difficulty and type filter is applied The reset button was pressed and the sorts and filters were reset.		

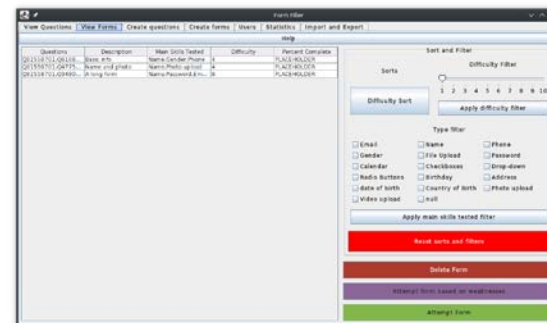
9

Soft Dev

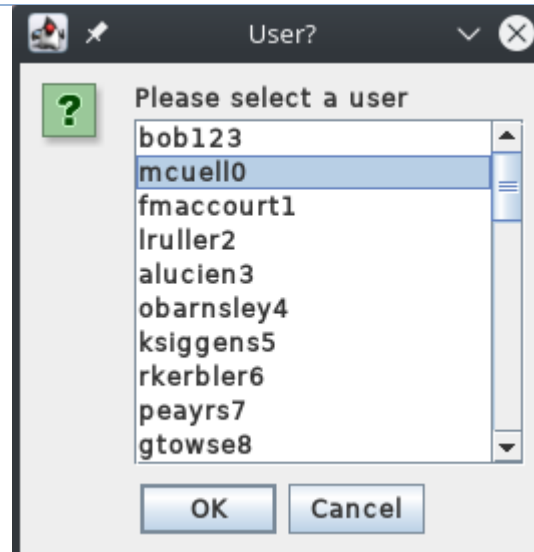
The delete form button is only present when an admin user is logged in



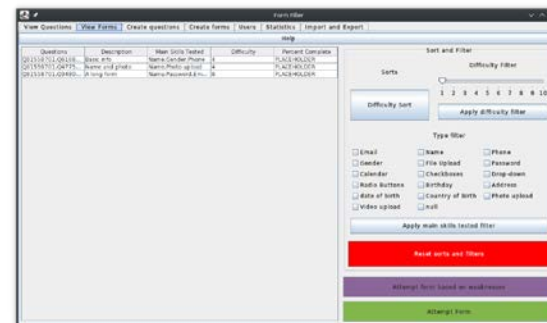
Bob is selected to be logged in (bob is an admin)



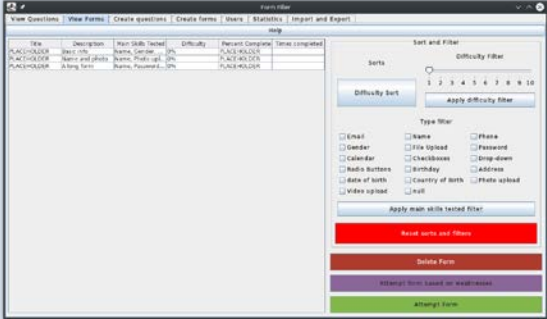
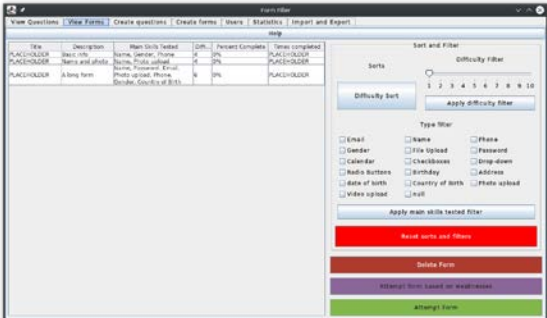
The delete form button is present



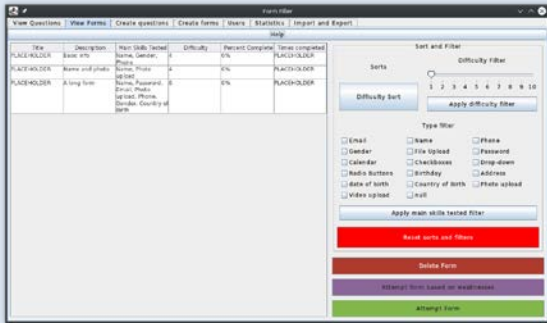
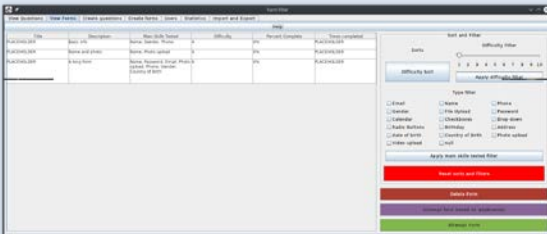
mcuell0 is selected to be logged in (mcuell0 is not an admin)



There is no delete form button

10	Soft Dev	The column headings are correct but the data is in the incorrect columns. The data is in the correct columns up until the difficulty column. There is no difficulty.		Add difficulty data to the table. This will fix this issue and will shift the data back into the correct columns	
11	Soft Dev	The main skills tested column automatically wraps the text so that it's all visible.			

12	Soft Dev	When the main skills tested column is clicked there is a null pointer exception.	<pre> [INFO] <FORM_SELECTION_PANEL> Running populateTable Exception in thread "AWT-EventQueue-1" java.lang.NullPointerException at FormDisplayPanel\$WordWrapCellRenderer.getCellRendererComponent(FormDisplayPanel.java:485) at javax.swing.JTable\$AccessibleJTable.getAccessibleChild(JTable.java:7437) at javax.swing.JTable\$AccessibleJTable.getAccessibleAt(JTable.java:7424) at javax.swing.JTable\$AccessibleJTable.valueChanged(JTable.java:6939) at javax.swing.DefaultListSelectionModel.fireValueChanged(DefaultListSelectionModel.java:184) at javax.swing.DefaultListSelectionModel.fireValueChanged(DefaultListSelectionModel.java:164) at javax.swing.DefaultListSelectionModel.fireValueChanged(DefaultListSelectionModel.java:211) at javax.swing.DefaultListSelectionModel.changeSelection(DefaultListSelectionModel.java:405) at javax.swing.DefaultListSelectionModel.changeSelection(DefaultListSelectionModel.java:415) at javax.swing.DefaultListSelectionModel.setSelectionInterval(DefaultListSelectionModel.java:459) at javax.swing.JTable.changeSelectionModel(JTable.java:2392) at javax.swing.JTable.changeSelection(JTable.java:2461) at javax.swing.plaf.basic.BasicTableUI\$Handler.adjustSelection(BasicTableUI.java:1115) at javax.swing.plaf.basic.BasicTableUI\$Handler.mousePressed(BasicTableUI.java:1038) at java.awt.AWTEventMulticaster.mousePressed(AWTEventMulticaster.java:280) at java.awt.Component.processMouseEvent(Component.java:6536) at javax.swing.JComponent.processMouseEvent(JComponent.java:3324) at java.awt.Component.processEvent(Component.java:6304) at java.awt.Container.processEvent(Container.java:2239) at java.awt.Component.dispatchEventImpl(Component.java:4889) at java.awt.Container.dispatchEventImpl(Container.java:2297) at java.awt.Component.dispatchEvent(Component.java:4711) at java.awt.LightweightDispatcher.retargetMouseEvent(Container.java:4904) at java.awt.LightweightDispatcher.processMouseEvent(Container.java:4532) at java.awt.LightweightDispatcher.dispatchEvent(Container.java:4476) at java.awt.Container.dispatchEventImpl(Container.java:2283) at java.awt.Window.dispatchEventImpl(Window.java:2746) at java.awt.Component.dispatchEvent(Component.java:4711) at java.awt.EventQueue.dispatchEventImpl(EventQueue.java:760) at java.awt.EventQueue.access\$500(EventQueue.java:97) at java.awt.EventQueue\$3.run(EventQueue.java:709) at java.awt.EventQueue\$3.run(EventQueue.java:703) at java.security.AccessController.doPrivileged(Native Method) at java.security.ProtectionDomain\$JavaSecurityAccessImpl.doIntersectionPrivilege(ProtectionDomain.java:74) at java.security.ProtectionDomain\$JavaSecurityAccessImpl.doIntersectionPrivilege(ProtectionDomain.java:84) at java.awt.EventQueue\$4.run(EventQueue.java:733) at java.awt.EventQueue\$4.run(EventQueue.java:731) at java.security.AccessController.doPrivileged(Native Method) at java.security.ProtectionDomain\$JavaSecurityAccessImpl.doIntersectionPrivilege(ProtectionDomain.java:74) at java.awt.EventQueue.dispatchEvent(EventQueue.java:730) at org.GNOME.Accessibility.AtkWrapper\$6.dispatchEvent(AtkWrapper.java:715) at java.awt.EventDispatchThread.pumpOneEventForFilters(EventDispatchThread.java:205) at java.awt.EventDispatchThread.pumpEventsForFilter(EventDispatchThread.java:116) at java.awt.EventDispatchThread.pumpEventsForHierarchy(EventDispatchThread.java:105) at java.awt.EventDispatchThread.pumpEvents(EventDispatchThread.java:101) </pre>	Adding a check to test if a null value is given to the table resize method should fix this.	<pre> [INFO] <USER_PANEL> Running prepareGUI [INFO] <USER_PANEL> Running populateTable [INFO] <STATISTICS_PANEL> Running prepareGUI [INFO] <QUESTION_SELECTION_PANEL> Running prepareGUI [INFO] <QUESTION_SELECTION_PANEL> Running populateTable [INFO] <FORM_SELECTION_PANEL> Running prepareGUI [INFO] <FORM_SELECTION_PANEL> Running populateTable </pre> The null pointer exception happens no more
----	----------	--	--	---	--

14	Soft Dev	When a table is made wider the rows do not shrink in height. They only grow in height when the table is made narrower	 Normal size window.  Window made larger Heights of the rows do not shrink	This is because the row resizing logic only makes the row larger if it's not tall enough. It doesn't handle making the row smaller. Implementing a TableColumnModelListener to listen for when the columns change size and then, when a column changes size, iterating through each row and setting the row height to the largest required height of all the components in the row would fix this.	<pre> harry@harry-pc-linux:/media/harry/HARRY_BEST/Computer_Science/Component_3/SoftDev/Project/Form- r\$./compile.sh Picked up _JAVA_OPTIONS: -Dawt.useSystemAAFontSettings=on -Dswing.aatext=true FormDisplayPanel.java:13: error: FormDisplayPanel is not abstract and does not override abstract method columnSelectionChanged(ListSelectionEvent) in TableColumnModelListener public class FormDisplayPanel extends JPanel implements ActionListener, TableColumnModelListene r ^ 1 error Picked up _JAVA_OPTIONS: -Dawt.useSystemAAFontSettings=on -Dswing.aatext=true </pre> Error when compiling after implementing the TableColumnModelListener
15	Soft Dev	Error when compiling FormDisplayPanel after implementing TableColumnModelListener because the columnSelectionChanged Method wasn't implemented.	<pre> harry@harry-pc-linux:/media/harry/HARRY_BEST/Computer_Science/Component_3/SoftDev/Project/Form- r\$./compile.sh Picked up _JAVA_OPTIONS: -Dawt.useSystemAAFontSettings=on -Dswing.aatext=true FormDisplayPanel.java:13: error: FormDisplayPanel is not abstract and does not override abstract method columnSelectionChanged(ListSelectionEvent) in TableColumnModelListener public class FormDisplayPanel extends JPanel implements ActionListener, TableColumnModelListene r ^ 1 error Picked up _JAVA_OPTIONS: -Dawt.useSystemAAFontSettings=on -Dswing.aatext=true </pre>	Implement all of the required methods of the TableColumnModelListener interface.	<pre> harry@harry-pc-linux:/media/harry/HARRY_BEST/Computer_Science/Component_3/SoftDev/Project/Form- r\$./compile.sh Picked up _JAVA_OPTIONS: -Dawt.useSystemAAFontSettings=on -Dswing.aatext=true Picked up _JAVA_OPTIONS: -Dawt.useSystemAAFontSettings=on -Dswing.aatext=true </pre> Successful compile

16

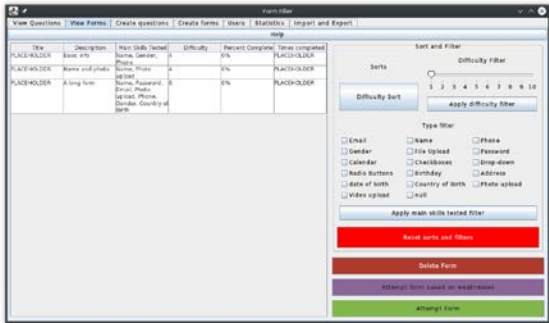
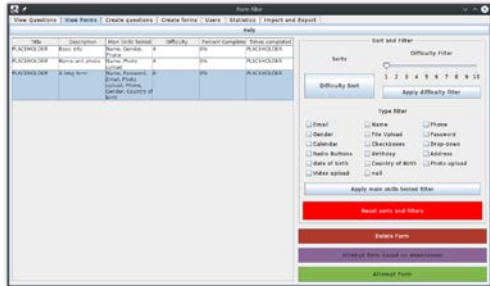
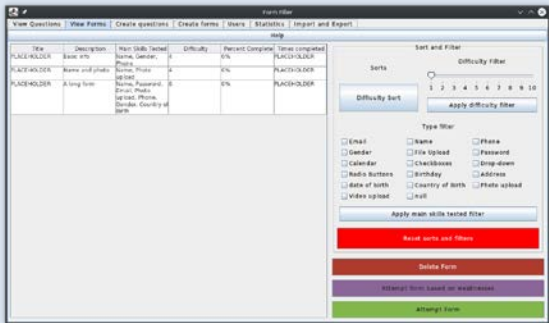
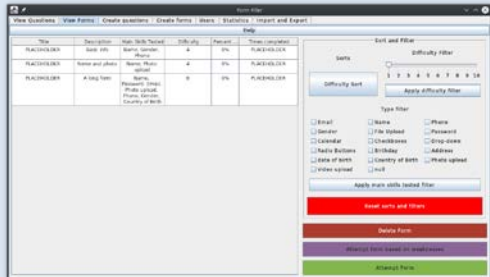
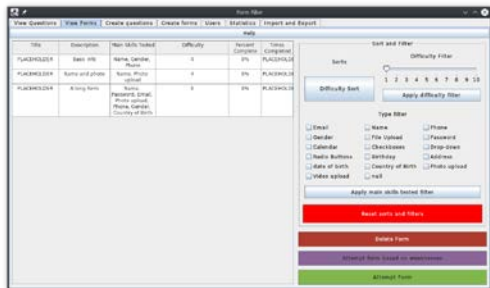
Soft Dev

The table correctly
resizes when it changes
size to ensure that all
of the text is visible.

Regular size

Window made wider

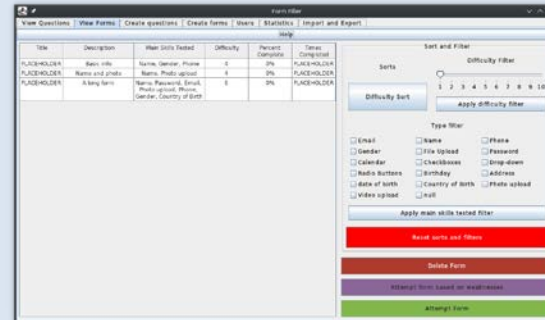
Window made narrower

17	Soft Dev	When a column in the table is clicked it doesn't become highlighted		Change the cell renderer so that if the cell is selected it changes the background to the highlight colour.	
18	Soft Dev	The text in the table cells isn't centered.		This is because a JTextArea was used to hold the text. JTextArea doesn't support centering the text. Using a JTextPane will fix this.	
19	Soft Dev	The table header text is center aligned but you can't see it all because the headers are not tall enough.		Make the headers automatically resize like the cells do so that the text can be seen	

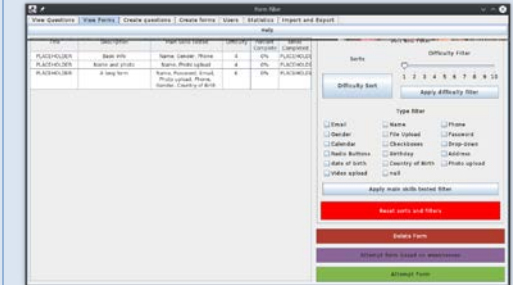
20

Soft Dev

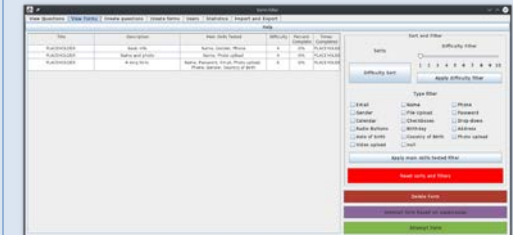
The widths of the difficulty, times completed, and percentage complete columns are small than the rest. However they still resize when the table resizes. I would rather them be even smaller and a fixed size.



Rather than using set preferred size, use setMaxSize() and setMinSize() on the columns. Setting the max and min size to the same value should make the columns a fixed width



The columns are now a better size. They do not change size as the table is resized.

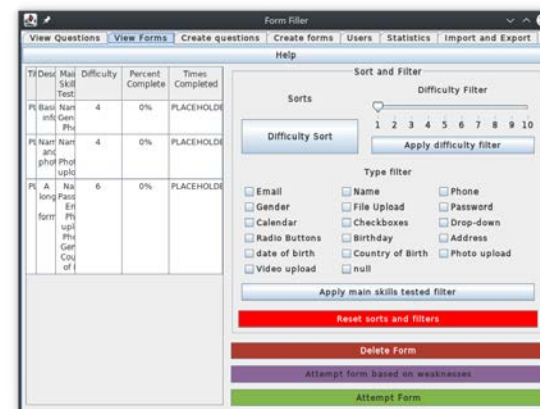


The table is wider in this screenshot and the columns haven't changed width.

21

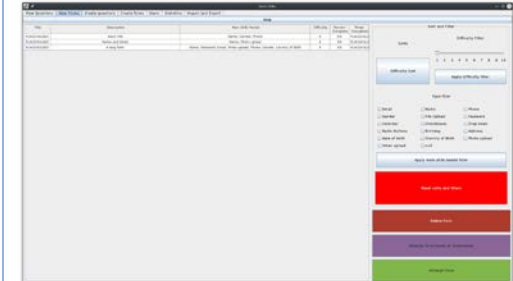
Soft Dev

Setting the min width means that when the window is small the table doesn't resize correctly and the difficulty etc. columns take up too much space.

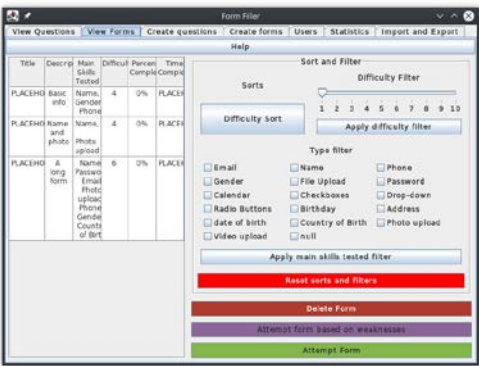
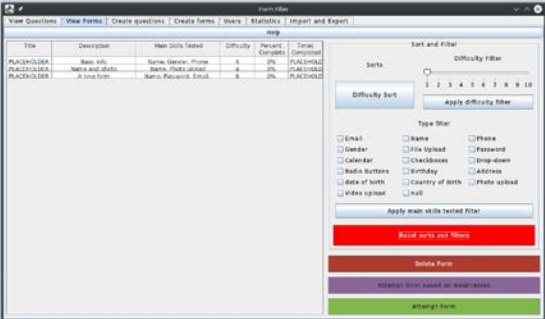
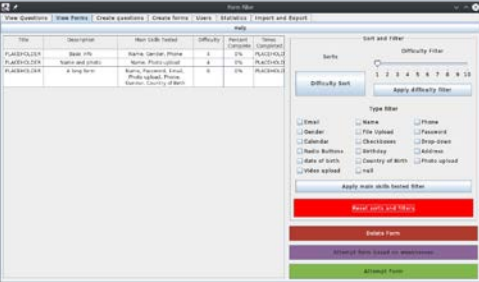
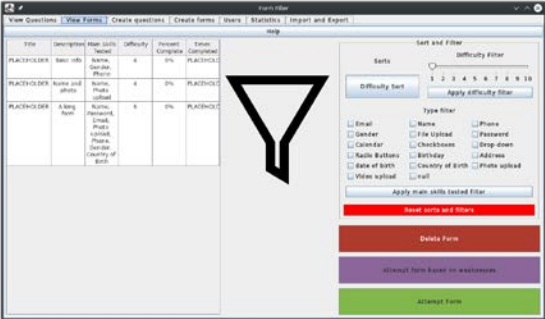
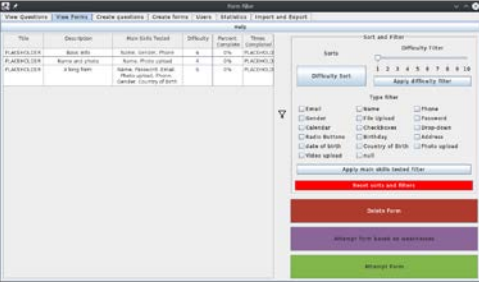


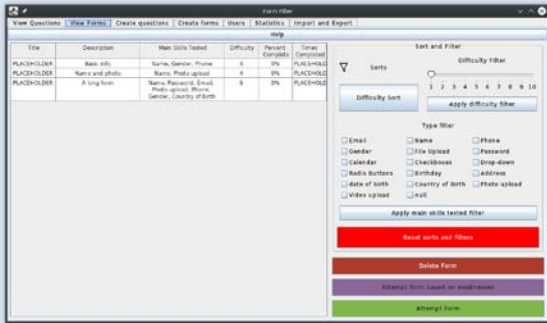
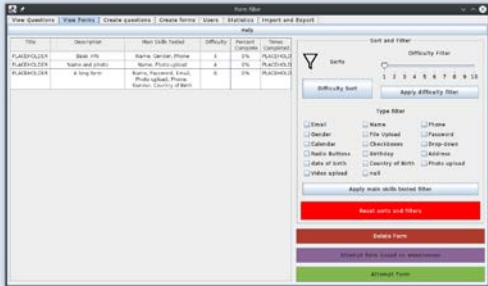
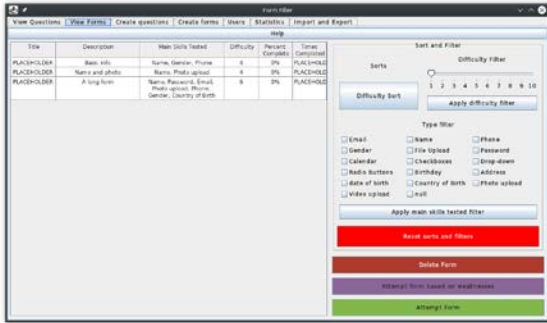
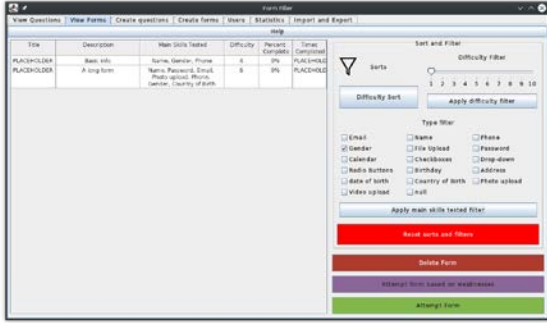
Window is small

Just setting the max width, rather than the min width, will mean that the columns won't take up a lot of space when the window is large and will get smaller when the window is small.



The columns don't take up a large amount of space when the table is large.

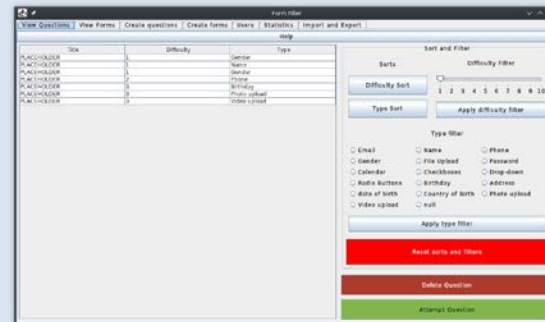
					 The columns become smaller when the window is small
22	Soft Dev	When the reset sorts button is pressed the rows default to their smallest height and this means that the text isn't visible.		Add a call to <code>resizeRows()</code> (the method that I wrote to make the rows the correct height) when the table is populated.	 Rows are still the correct size after resetting the sorts and filters.
23	Soft Dev	There is a filter icon.		The filter icon is too large. To fix this resize using <code>getImage().getScaledInstance(20, 20, Image.SCALE_DEFAULT)</code>	 The icon is smaller now

24	Soft Dev	The icon is in the correct place however it is too small.		Change the size from 20x20 to 40x40	
25	Soft Dev	The label is invisible by default and appears when a filter is applied	 No filter is applied  A filter is applied and so the icon is visible.		

26

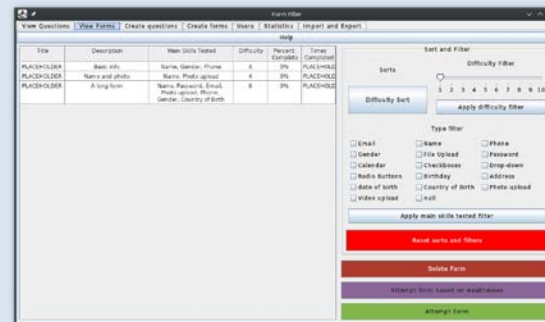
Soft Dev

When the view forms
tab is clicked the table
is refreshed



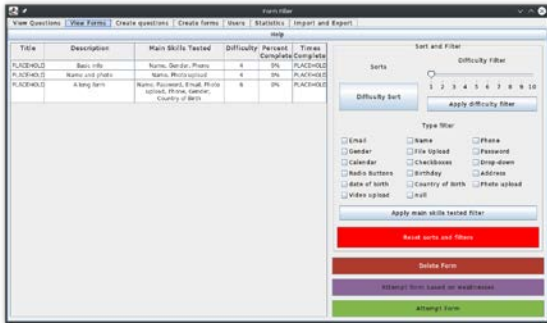
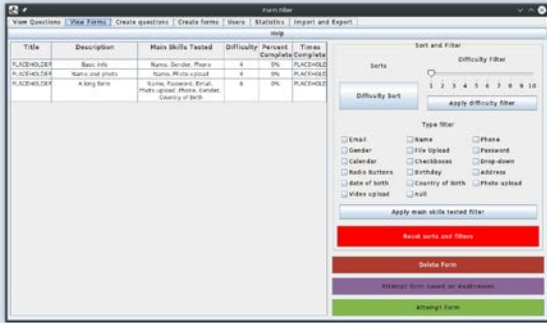
```
[INFO] <QUESTION_SELECTION_PANEL> Running prepareGUI
[INFO] <QUESTION_SELECTION_PANEL> Running populateTable
[INFO] <FORM_SELECTION_PANEL> Running prepareGUI
[INFO] <FORM_SELECTION_PANEL> Running populateTable
```

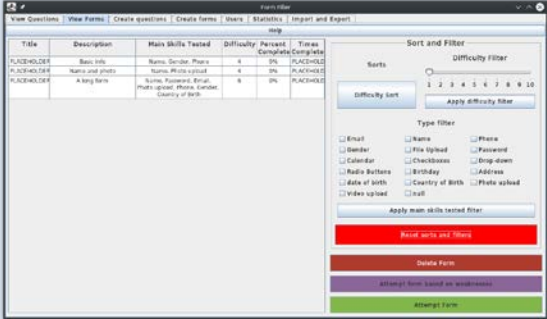
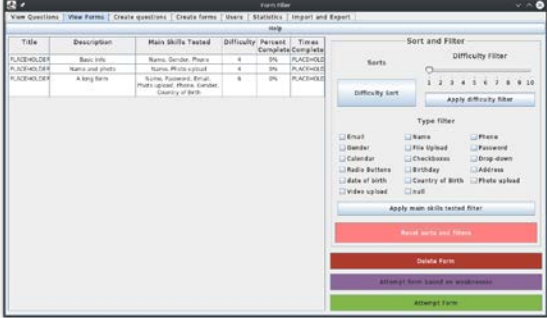
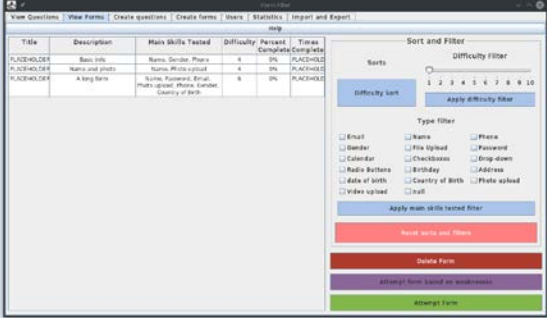
The view questions tab is open and the form panel hasn't been refreshed.



```
[INFO] <QUESTION_SELECTION_PANEL> Running populateTable
[INFO] <FORM_SELECTION_PANEL> Running prepareGUI
[INFO] <FORM_SELECTION_PANEL> Running populateTable
[INFO] <FORM_DISPLAY_PANEL> Running populateTable
```

The view forms panel is selected and the most recent function that was run is now the populate table function of form display panel showing that the table was refreshed when the tab was selected.

27	Soft Dev	The table headers have larger and bold text.			
28	Soft Dev	The table text is larger			
29	Soft Dev	Error when compiling FormDisplayPanel because TitledBorder has no method getFont or setFont	<pre>C:\Harry\harry-pc-linux\media\harry\HARRY BEST\Computer Science\component 3\SOFTDEV\Project\F Pillers\$./compile.sh Picked up _JAVA_OPTIONS: -Dawt.useSystemAAFontSettings=on -Dswing.aatext=true FormDisplayPanel.java:183: error: method getFont in class TitledBorder cannot be applied to giv en types: Font currentFont = border.getFont(); ^ required: Component found: no arguments reason: actual and formal argument lists differ in length FormDisplayPanel.java:184: error: cannot find symbol border.setFont(currentFont.deriveFont(Font.BOLD, 14)); // Make the font larger ^ symbol: method setFont(Font) location: variable border of type TitledBorder 2 errors Picked up _JAVA_OPTIONS: -Dawt.useSystemAAFontSettings=on -Dswing.aatext=true</pre>	Change getFont and setFont to getTitleFont and setTitleFont as these are the methods that deal with the fonts of the border	<pre>C:\Harry\harry-pc-linux\media\harry\HARRY BEST\Computer Science\component 3\SOFTDEV\Project\F Pillers\$./compile.sh Picked up _JAVA_OPTIONS: -Dawt.useSystemAAFontSettings=on -Dswing.aatext=true Picked up _JAVA_OPTIONS: -Dawt.useSystemAAFontSettings=on -Dswing.aatext=true harry\harry-pc-linux\media\harry\HARRY BEST\Computer Science\component 3\SOFTDEV\Project\F Pillers\$</pre>

30	Soft Dev	The label text has been made larger.			
31	Soft Dev	The colour of the reset button has been changed to better match the colour of the one in the design.			
32	Soft Dev	The sort and filter buttons have had their colours updated to match those in the design.			

33

Soft Dev

When a form is completed the percentage complete in the table isn't updated as soon as the form is closed.

Title	Description	Main Skills Tested	Difficulty	Percent Complete	Status
Basic info	Basic info	File upload	2	0%	PLACED IN
Basic info	Basic info	File upload	2	0%	PLACED IN

The form "Basic info" is 0% complete.
The form is attempted and filled in.

To fix this there should be a call to `FormDisplayPanel.refreshTable()` to refresh the table when the form is closed.

The form "Basic info" is 0% complete.
The form is attempted and filled in.

The form is closed

Page 1 of 1

Basic

★ First name

★ Gender ☐ Male ☐ Female
☐ Other

About you

Birthday

★ Photo id

Submit

The form is closed

Sort and Filter

Title	Description	Skills Tested	Difficulty	Percent Complete	Status
Skills	Advanced	File Upload	2	95%	PLACED
Skills	Basic	Form, Layout, Buttons, etc.	1	100%	PLACED

Sort and Filter

Difficulty Filter

1 2 3 4 5 6 7 8 9 10

Apply difficulty filter

Type Filter

☐ Detail ☐ Basic ☐ Photo

☐ Gender ☐ File Upload ☐ Placement

☐ Calendar ☐ Checkboxes ☐ Drop down

☐ Radio Buttons ☐ Birthday ☐ Address

☐ Date of Birth ☐ Country of Birth ☐ Photo upload

☐ Video upload ☐ Do not use any tags

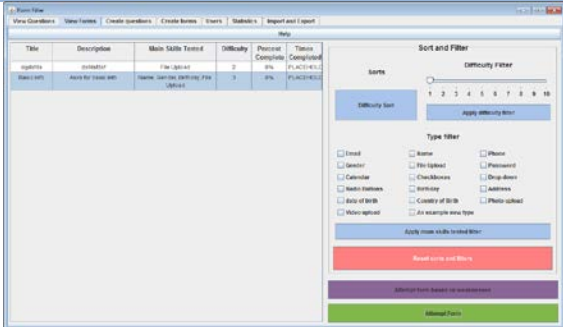
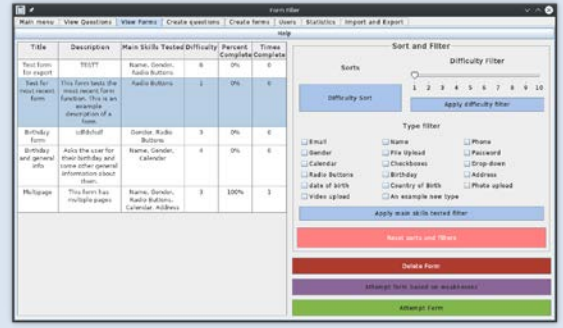
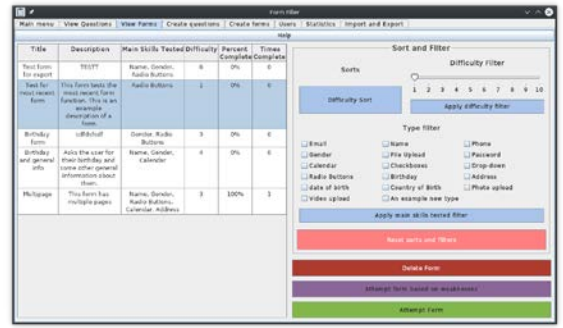
Apply type filter

Reset sort and filter

Reset sort and filter

Advanced Search

The percentage complete is updated.

			 The percentage complete isn't updated.		
34	Soft Dev	Only one row at a time is allowed to be selected.			
35	Soft Dev	The table is non editable			

36

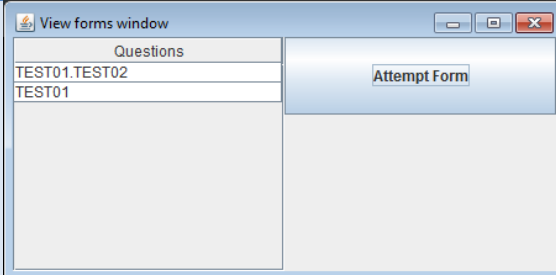
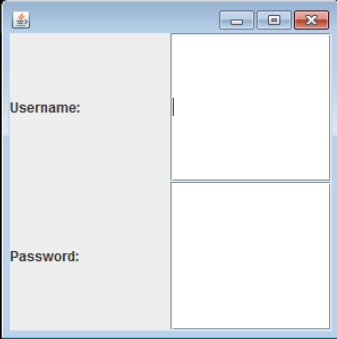
Soft Dev

Double clicking on a row opens the form to be attempted.

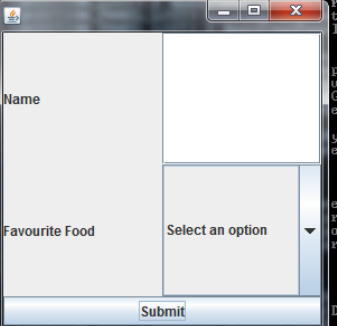
The screenshot displays a software development tool interface. On the left, a table lists various test forms. On the right, a form view is shown, which is a detailed representation of one of the forms listed in the table. The table has columns for Title, Description, Main Skills Tested, Difficulty, Percent Complete, and Times Completed. The form view on the right includes fields for First name, Gender (Male, Female, Other), a Next section button, a Name's question with radio button options (1, 2, 3, 24), and a Birthday field with a calendar icon. A 'Next' button is at the bottom right of the form view.

Title	Description	Main Skills Tested	Difficulty	Percent Complete	Times Completed
Test form for report	Test form for report	Name, Gender, Radio Buttons	6	0%	0
Test form for report	This form tests the most recent form function. This is an example description of a form.	Radio Buttons	3	0%	0
Birthday form	Test form for birthday	Gender, Radio Buttons	3	0%	0
Birthday and general info	Tests the user for birthday and some other general information about them.	Name, Gender, Calendar	4	0%	0
Multiple	This form has multiple pages.	Name, Gender, Radio Buttons, Calendar, Address	5	0%	1

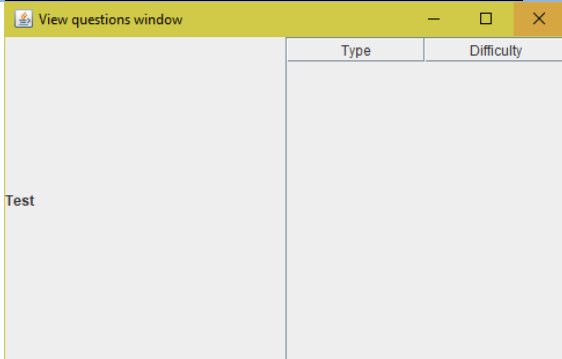
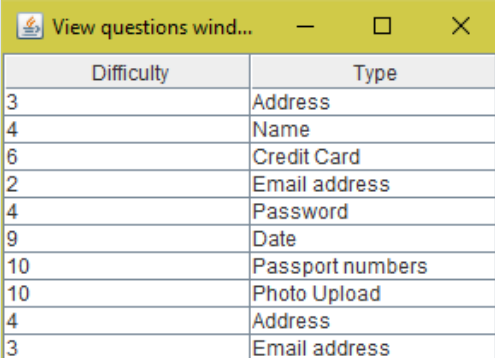
51: Allow user to select a form to be attempted

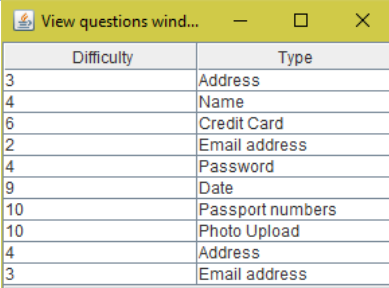
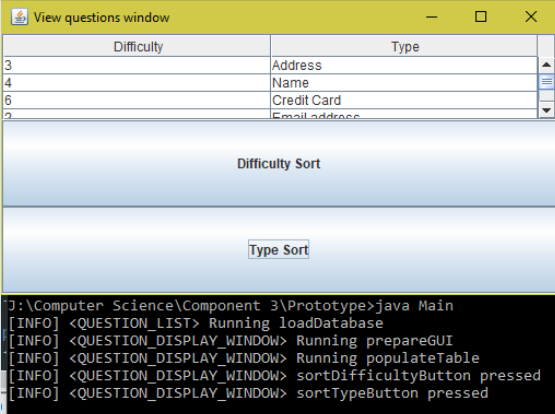
Test	Stage	Result Problem	Evidence	Action and Justification (If Needed)	Evidence of Action (If Needed)
1	Proto	There is an attempt form button			
2	Proto	Form is opened when selected and the attempt form button is pressed.	<pre> D:\Computer Science\Component 3\Prototype>java Main [INFO] <GUI> Running setup [INFO] <QUESTION_LIST> Running loadDatabase [INFO] <QUESTION_PANEL_LIST> Running loadDatabase [INFO] <QUESTION_PANEL_LIST> 2 QuestionPanels loaded from file [INFO] <FORM_LIST> Running loadDatabase [INFO] <FORM_LIST> Running addForm [INFO] <FORM_LIST> Running addForm [INFO] <MAIN_MENU> Running prepareGUI [INFO] <FORM_DISPLAY_WINDOW> Running prepareGUI [INFO] <FORM_DISPLAY_WINDOW> Running populateTable [INFO] <FORM_DISPLAY_WINDOW> attemptButton pressed [INFO] <FORM_LIST> Running getFormById [INFO] <GUI> Running openForm [INFO] <QUESTION_LIST> Running getPanelById [INFO] <QUESTION_PANEL_LIST> Running getPanelById [INFO] <QUESTION_LIST> Running getPanelById [INFO] <QUESTION_PANEL_LIST> Running getPanelById </pre> 		

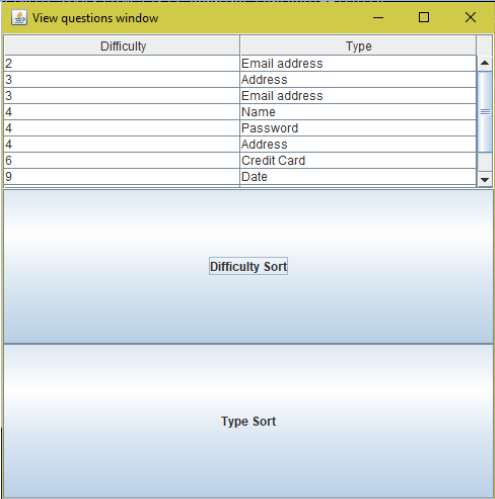
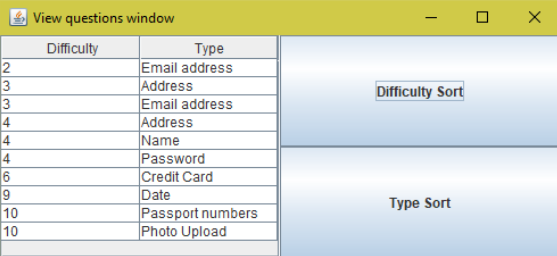
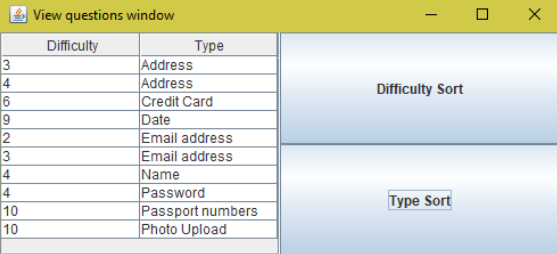
52: Allow form to be submitted

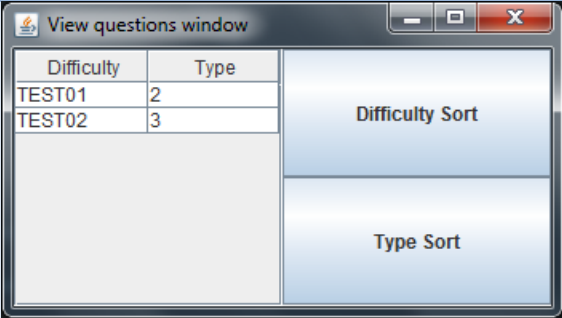
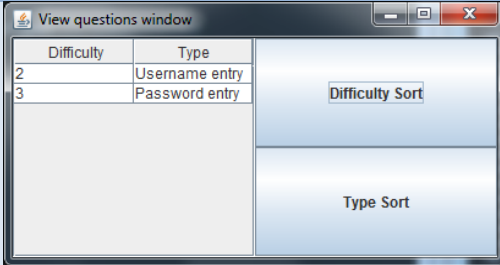
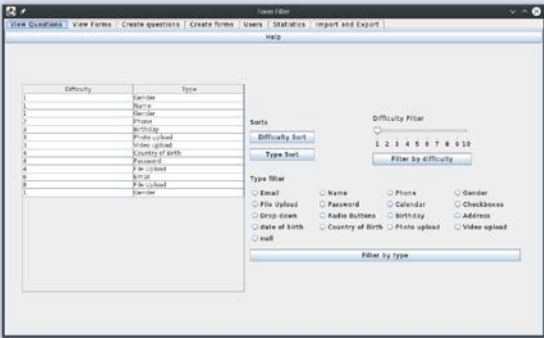
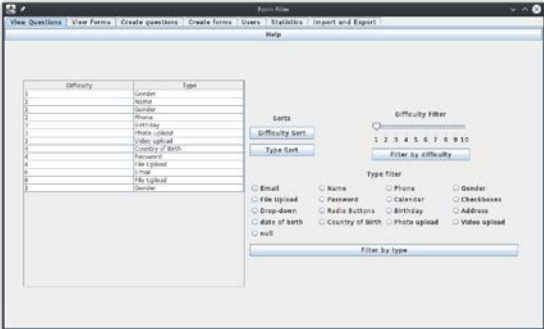
Test	Stage	Result Problem	Evidence	Action and Justification (If Needed)	Evidence of Action (If Needed)
1	Proto	FormDisplayer fails to compile because I type JButton rather than JButton	<pre>D:\Computer Science\Component 3\Prototype>javac *.java FormDisplayer.java:12: error: cannot find symbol private JButton submitButton = new JButton("Submit"); ^ symbol: class JButton location: class FormDisplayer</pre>	Change JButton to JButton	<pre>D:\Computer Science\Component 3\Prototype>javac *.java FormDisplayer.java:35: error: cannot find symbol formPanel.add(questions.getPanelByID(questionID)); // Ge ^ symbol: variable questions location: class FormDisplayer 1 error</pre> Error no longer present
2	Proto	There is a submit button at the bottom of the form and presses to it are detected	<pre>D:\Computer Science\Component 3\Prototype>java Main [INFO] <GUI> Running setup [INFO] <QUESTION_LIST> Running loadDatabase [INFO] ...om file [INFO] ...atabase [INFO] ...s loaded from file [INFO] ... [INFO] ...pareGUI [INFO] ...ulateTable [INFO] ...GUI [INFO] ...eTable [INFO] ...yID [INFO] ...epareGUI [INFO] ... [INFO] ...eGUI [INFO] ...epareGUI [INFO] ...opulateTable [INFO] ...ressed [INFO] ... [INFO] <QUESTION_PANEL_LIST> Running getByID [INFO] <FORM_DISPLAYER> submitButton pressed</pre> 		
			The rest of the testing relevant to this objective can be found at "Ability to attempt a form"		

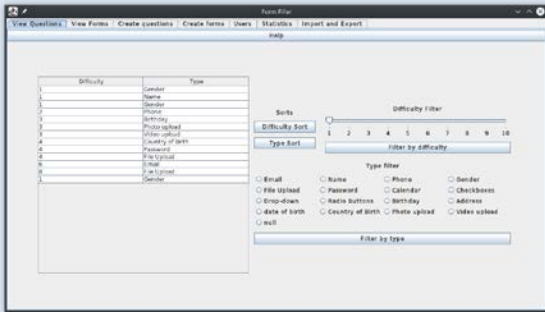
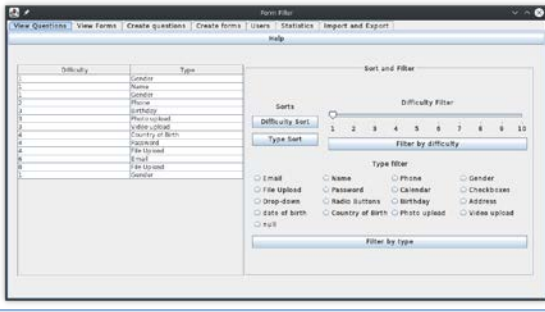
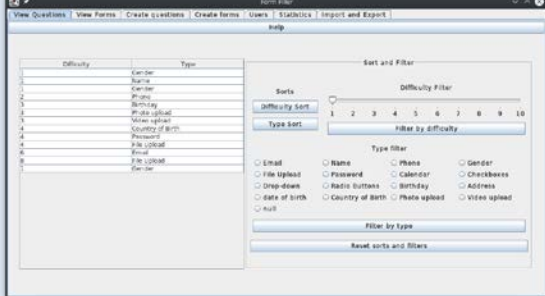
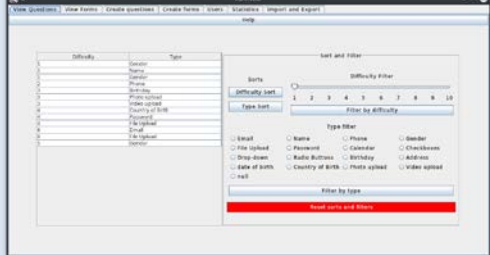
53: There should be a panel which will display the questions to the user and allow them to search through them

Test	Stage	Result	Problem	Evidence	Action and Justification (If Needed)	Evidence of Action (If Needed)																						
1		Proto	Question display window fails to compile	<pre>D:\Computer Science\Component 3\Prototype>javac Main.java .\QuestionDisplayWindow.java:14: error: cannot find symbol private DefaultTableModel questionTableModel = new DefaultTableMode ^ symbol: class DefaultTableModel location: class QuestionDisplayWindow .\QuestionDisplayWindow.java:14: error: cannot find symbol private DefaultTableModel questionTableModel = new DefaultTableMode ^ symbol: class DefaultTableModel location: class QuestionDisplayWindow 2 errors</pre>	Import javax.swing.table.*	<pre>D:\Computer Science\Component 3\Prototype>javac Main.java</pre>																						
2		Proto	Empty table is displayed with correct headers																									
3		Proto	Question Table won't populate	<pre>D:\Computer Science\Component 3\Prototype>javac QuestionDisplayWindow.java QuestionDisplayWindow.java:48: error: method populateTable in class QuestionDisplayWindow cannot be applied to given types; populateTable(); ^ required: Question[] found: no arguments reason: actual and formal argument lists differ in length 1 error</pre>	Pass the populate table function some data	 <table><thead><tr><th>Difficulty</th><th>Type</th></tr></thead><tbody><tr><td>3</td><td>Address</td></tr><tr><td>4</td><td>Name</td></tr><tr><td>6</td><td>Credit Card</td></tr><tr><td>2</td><td>Email address</td></tr><tr><td>4</td><td>Password</td></tr><tr><td>9</td><td>Date</td></tr><tr><td>10</td><td>Passport numbers</td></tr><tr><td>10</td><td>Photo Upload</td></tr><tr><td>4</td><td>Address</td></tr><tr><td>3</td><td>Email address</td></tr></tbody></table>	Difficulty	Type	3	Address	4	Name	6	Credit Card	2	Email address	4	Password	9	Date	10	Passport numbers	10	Photo Upload	4	Address	3	Email address
Difficulty	Type																											
3	Address																											
4	Name																											
6	Credit Card																											
2	Email address																											
4	Password																											
9	Date																											
10	Passport numbers																											
10	Photo Upload																											
4	Address																											
3	Email address																											

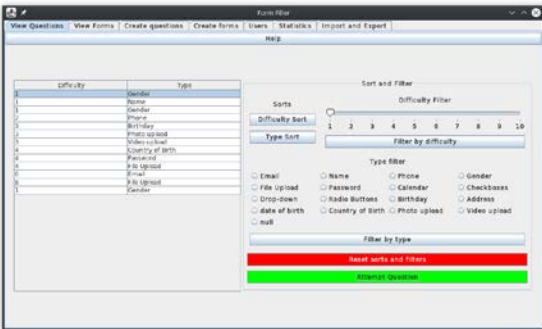
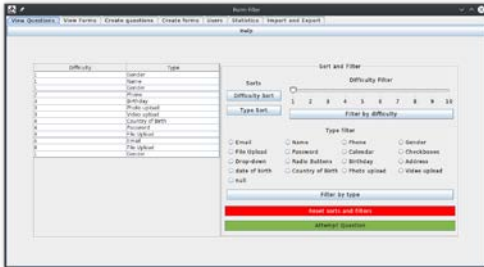
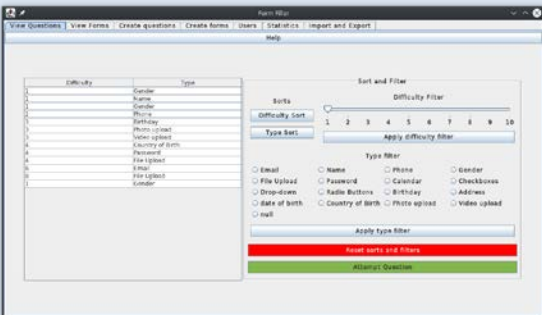
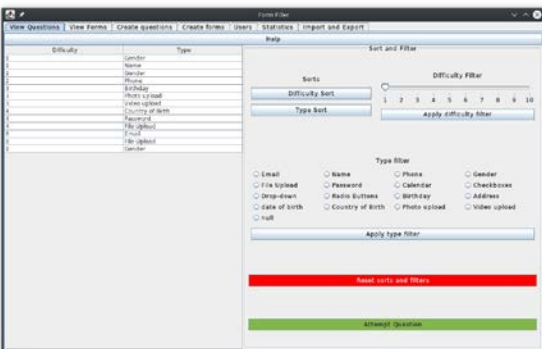
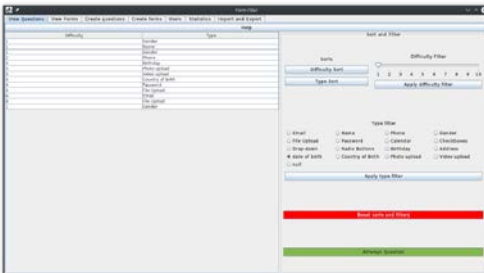
4		Proto	Question table displays questions			
5		Proto	Presses of the difficulty and type sort buttons are detected			

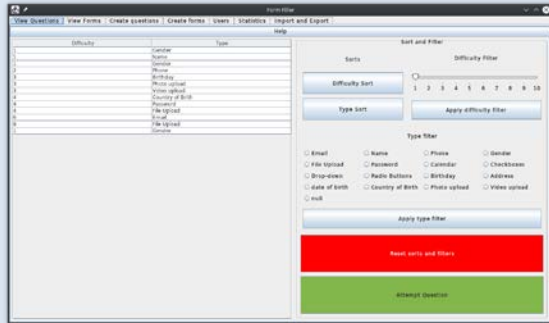
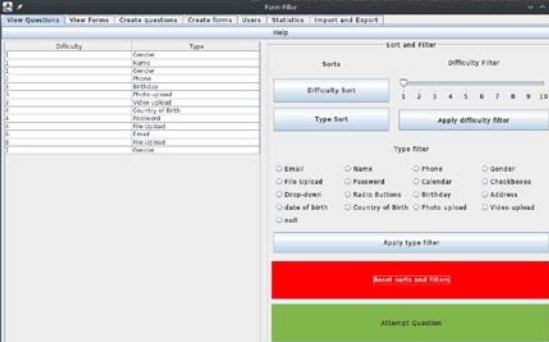
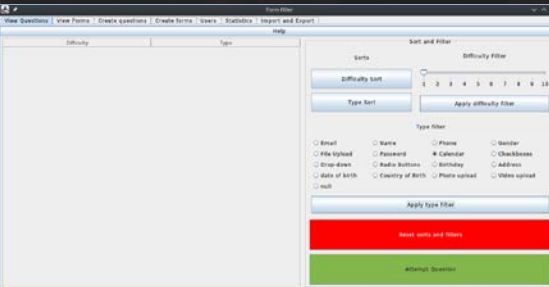
6		Proto	Table doesn't update as a result of the sort as no data was passed to the populate table method	<pre> C:\Computer Science\Component 3\Prototype>java Main QuestionDisplayWindow.java:78: error: method populateTable in class QuestionDisplayWindow cannot be applied to given types; populateTable(); ^ required: Question[] found: no arguments reason: actual and formal argument lists differ in length 1 error </pre>	Pass the array from the question list to the method	 <pre> J:\Computer Science\Component 3\Prototype>java Main [INFO] <QUESTION_LIST> Running loadDatabase [INFO] <QUESTION_DISPLAY_WINDOW> Running prepareGUI [INFO] <QUESTION_DISPLAY_WINDOW> Running populateTable [INFO] <QUESTION_DISPLAY_WINDOW> sortByDifficultyButton pressed [INFO] <QUESTION_LIST> Running sortByDifficulty [INFO] <QUESTION_DISPLAY_WINDOW> Running populateTable </pre>
7		Proto	The table is able to be sorted by difficulty when the difficulty sort button is pressed			
8		Proto	The table is able to be sorted by type when the type sort button is pressed			

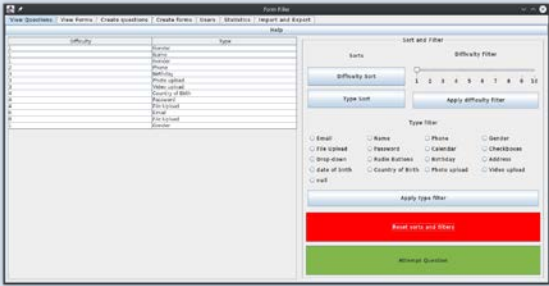
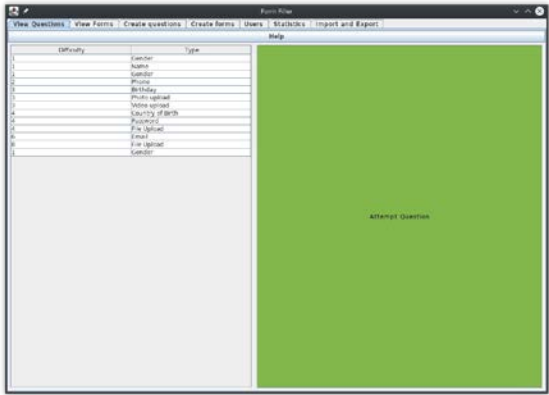
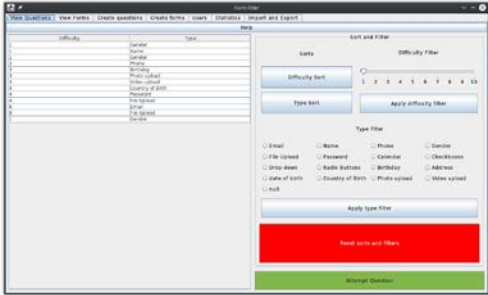
9		Proto	After adding an ID to each question the data is displayed in the incorrect column		Add a method to question that returns the data without the ID	
				The rest of the testing relevant to this objective and be found in the sort and filter objectives in the question section.		
10		Soft Dev	The panel has a layout that is approaching the one that is in the design document.		Remove the padding around the sides. And make further improvements	
11		Soft Dev	Labels above the sorts and filters are now centered			

12		Soft Dev	Difficulty filter options now span 2 rows. This was achieved by giving the difficulty filter panel constraints a weightx of 1.			
13		Soft Dev	There is a border that is titled "Sort and filter" above the sort and filter panel.			
14		Soft Dev	There is now a reset sorts and filters button.		It isn't red with white text as specified in the design. Set the background of the button to red and the foreground to white.	

It's now red with white text

15		Soft Dev	There is a green attempt question button		The green is too bright and harsh. Change from Color.GREEN to Color(130,183,75) as this is a more pleasant green.	
16		Soft Dev	The sort and filter buttons now have the correct text			
17		Soft Dev	All of the components now fully fill the screen when the screen is resized.		It would be better if the table grew larger faster than the sorts panel. To make this happen I will give it a higher weight than the sort and filter panel	

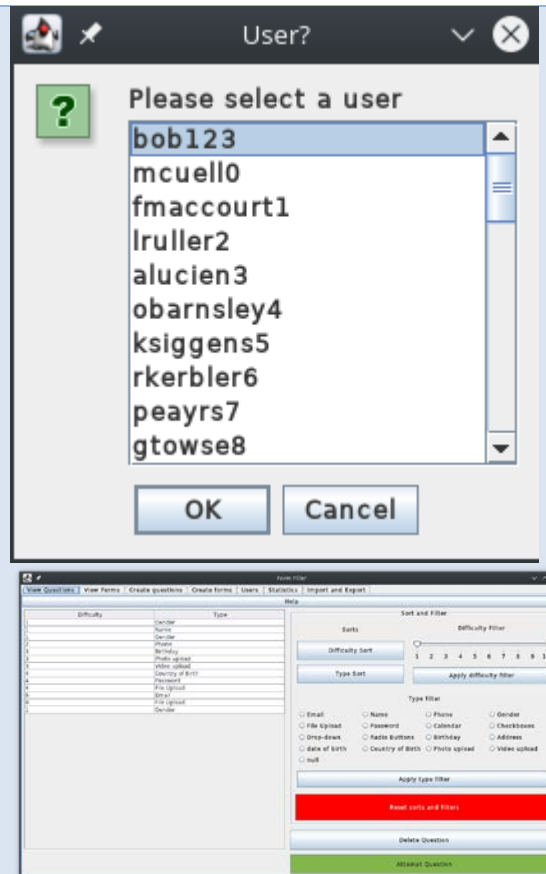
18		Soft Dev	The buttons components within the sort and filter panel space up the extra space vertically as well as horizontally when the window is resized.		
19		Soft Dev	Presses to the reset sorts and filters button are detected	 <pre>[INFO] <QUESTION_SELECTION_PANEL> Running prepareGUI [INFO] <QUESTION_SELECTION_PANEL> Running populateTable [INFO] <FORM_SELECTION_PANEL> Running prepareGUI [INFO] <FORM_SELECTION_PANEL> Running populateTable [INFO] <QUESTION_DISPLAY_PANEL> resetButton pressed</pre>	
20		Soft Dev	Pressing the reset sorts and filters button resets the sorts and filters and deselects any selected radio button.	 <pre>[INFO] <QUESTION_DISPLAY_PANEL> resetButton pressed [INFO] <QUESTION_DISPLAY_PANEL> Running populateTable</pre>	

						
21		Soft Dev	Attempt button fills the entire right of the screen. This happened when I moved it out of the sorts and filters panel.		Change the height of the button from 2 rows to just one row	

22

Soft Dev

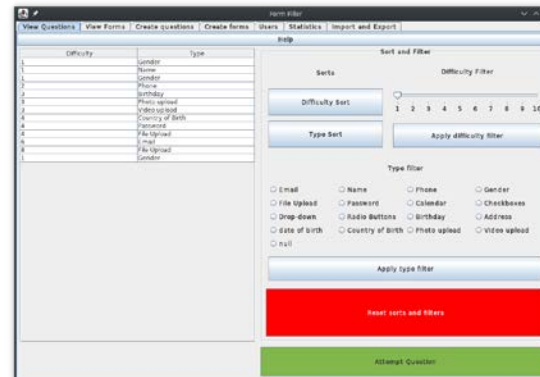
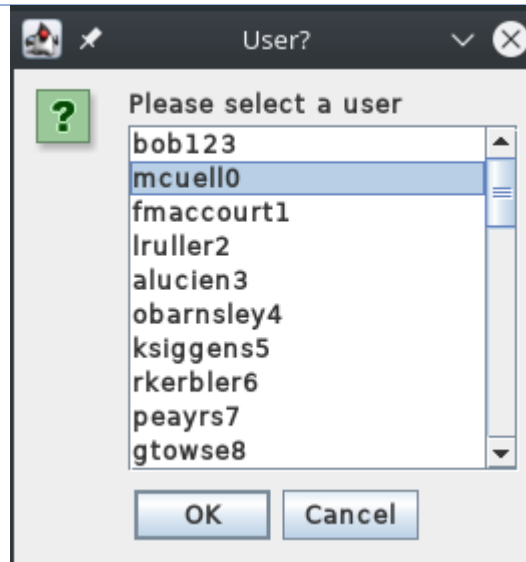
There is a delete question button when an admin user is logged in. (Bob is an admin user)

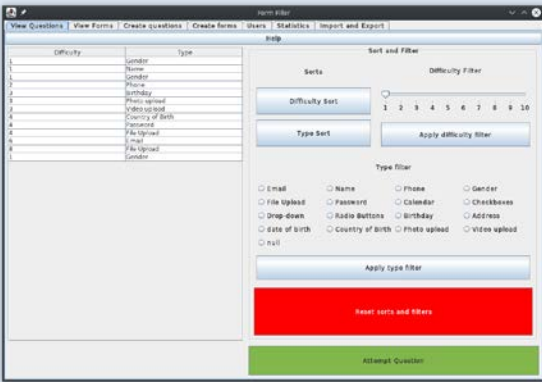
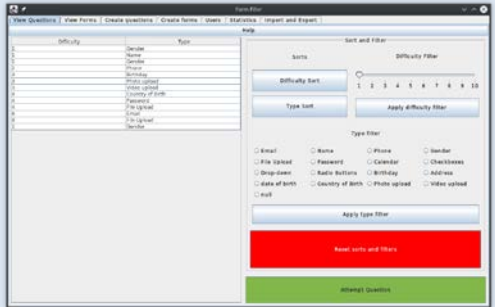
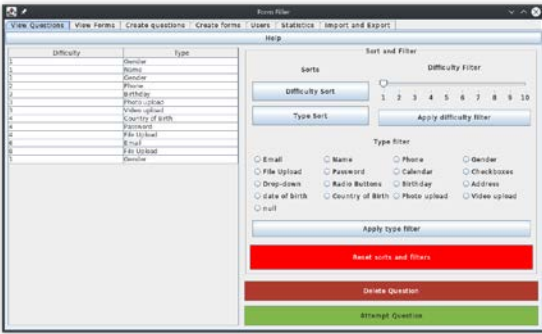


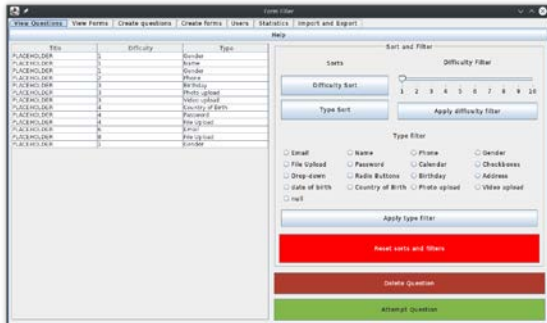
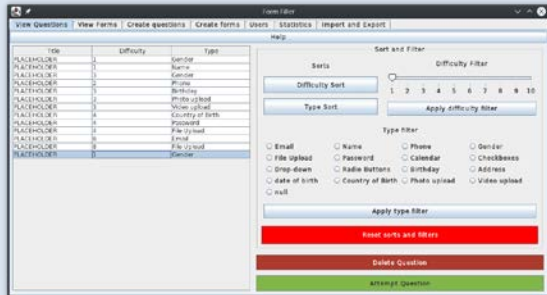
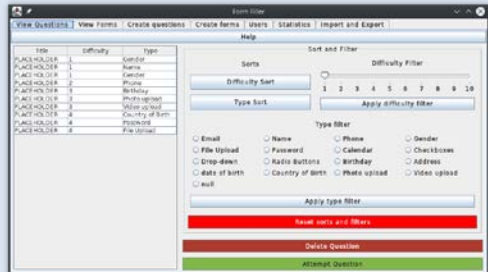
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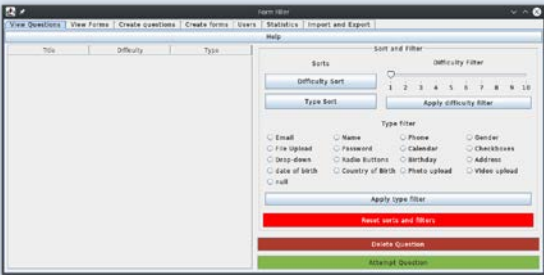
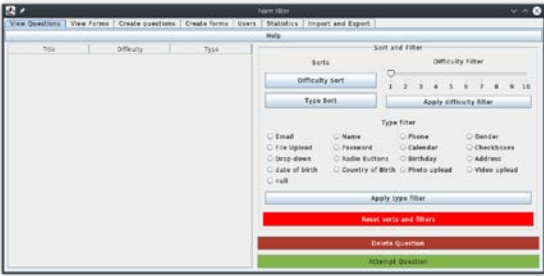
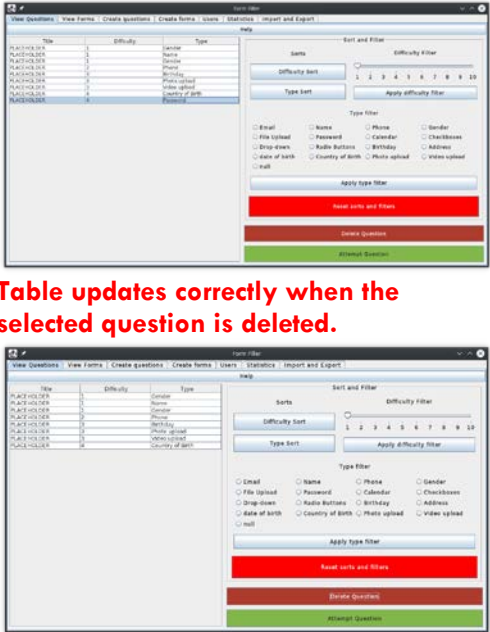
Soft Dev

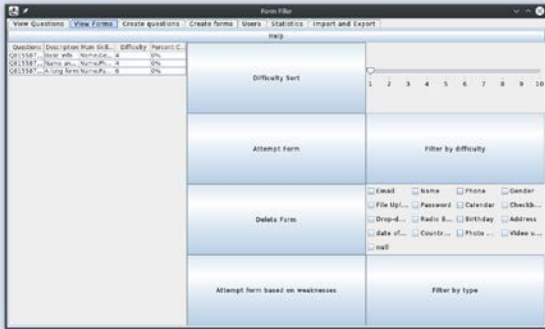
There is no delete question button when a non-admin user is logged in. (mcuell0 is not an admin)

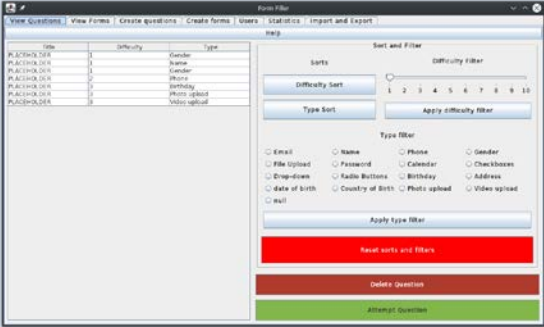


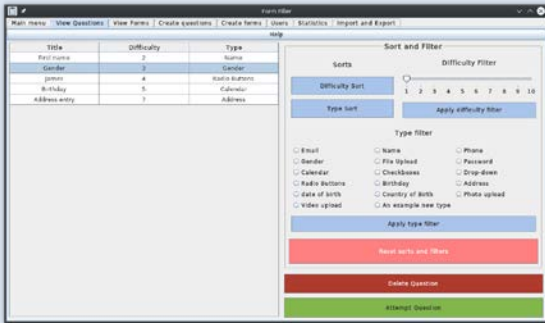
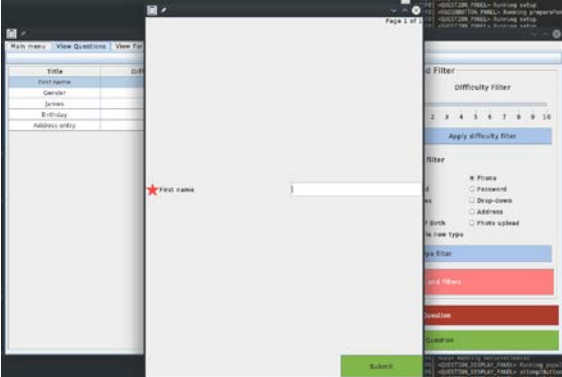
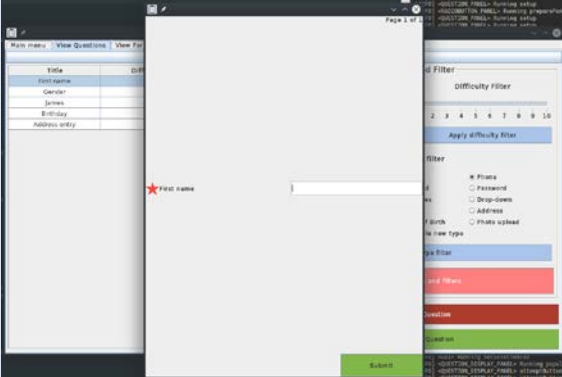
24		Soft Dev	The questions table doesn't span all 3 rows.		Change the <code>gridBagLayout</code> constraint's <code>gridHeight</code> attribute to 3 for the table.	
25		Soft Dev	Delete question button matches the style from the design			
26		Soft Dev	Error compiling view questions window because of a type error.	<pre>harry@harry-pc-linux:~/Documents/SoftDev/form-filler\$./compile.sh Picked up _JAVA_OPTIONS: -Dawt.useSystemAAFontSettings=on -Dswing.aatext=true QuestionDisplayPanel.java:275: error: incompatible types: int cannot be converted to String String[] question = {currentQuestion.getID(), currentQuestion.ge ^ (tTitle(), currentQuestion.getDifficulty(), currentQuestion.getType())}; // Convert the question t o a String array 1 error Picked up _JAVA_OPTIONS: -Dawt.useSystemAAFontSettings=on -Dswing.aatext=true</pre>	add + "" on the question's difficulty to convert it to a string so it can be displayed in the table	<pre>harry@harry-pc-linux:~/Documents/SoftDev/form-filler\$./compile.sh Picked up _JAVA_OPTIONS: -Dawt.useSystemAAFontSettings=on -Dswing.aatext=true Picked up _JAVA_OPTIONS: -Dawt.useSystemAAFontSettings=on -Dswing.aatext=true harry@harry-pc-linux:~/Documents/SoftDev/form-filler\$</pre> Successful compile

27		Soft Dev	The title of each question is displayed in a new title column of the window. Each question currently has the title PLACEHOLDER.			
28		Soft Dev	Error when compiling question display panel because QuestionList.sortByType () doesn't return an array.	<pre>harry@harry-pc-linux:~/Documents/SoftDev/form-filler\$./compile.sh Picked up JAVA_OPTIONS: -Dawt.useSystemAAFontSettings=on -Dswing.aatext=true QuestionDisplayPanel.java:259: error: incompatible types: void cannot be converted to Question[] questionData = questions.sortByType(); ^ QuestionDisplayPanel.java:263: error: incompatible types: void cannot be converted to Question[] questionData = questions.sortByDifficulty(); ^ 2 errors Picked up JAVA_OPTIONS: -Dawt.useSystemAAFontSettings=on -Dswing.aatext=true</pre>	Call .getArray() to return the array. The array is required to populate the table after a difficulty sort	<pre>harry@harry-pc-linux:~/Documents/SoftDev/form-filler\$./compile.sh Picked up JAVA_OPTIONS: -Dawt.useSystemAAFontSettings=on -Dswing.aatext=true Picked up JAVA_OPTIONS: -Dawt.useSystemAAFontSettings=on -Dswing.aatext=true harry@harry-pc-linux:~/Documents/SoftDev/form-filler\$</pre> Successful compile
29		Soft Dev	Type filter applied when the difficulty filter button is pressed. And vice versa.	<pre>[INFO] <QUESTION_DISPLAY_PANEL> difficultyFilterButton pressed [INFO] <QUESTION_LIST> Running filterByType [INFO] <QUESTION_DISPLAY_PANEL> Running populateTable [INFO] <QUESTION_DISPLAY_PANEL> typeFilterButton pressed [INFO] <QUESTION_LIST> Running filterByDifficulty [INFO] <QUESTION_DISPLAY_PANEL> Running populateTable</pre>	Swap the method calls over	<pre>[INFO] <QUESTION_DISPLAY_PANEL> typeFilterButton pressed [INFO] <QUESTION_LIST> Running filterByType [INFO] <QUESTION_DISPLAY_PANEL> Running populateTable [INFO] <QUESTION_DISPLAY_PANEL> difficultyFilterButton pressed [INFO] <QUESTION_LIST> Running filterByDifficulty [INFO] <QUESTION_DISPLAY_PANEL> Running populateTable</pre>
30		Soft Dev	Deleting a question breaks the database.		This is because the two string method of the question outputs the attributes in a different order than the question list reads them in. Making the orders the same should fix this.	 The database order is maintained. However the table doesn't automatically refresh correctly when a question is deleted.

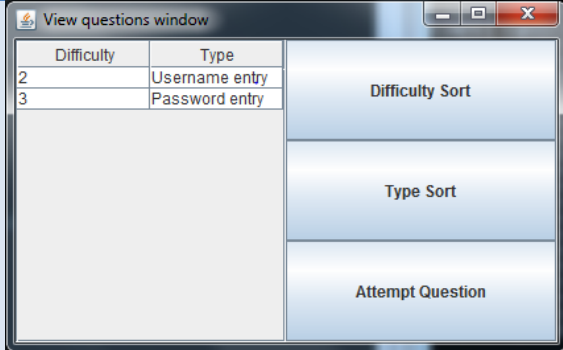
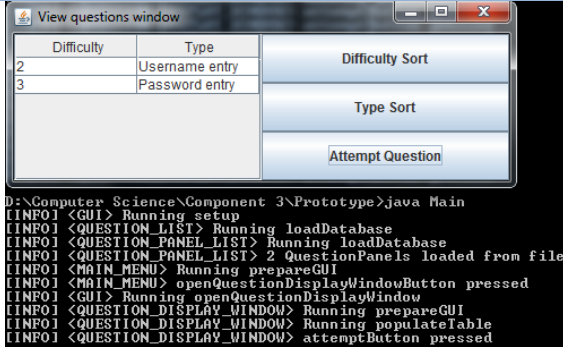
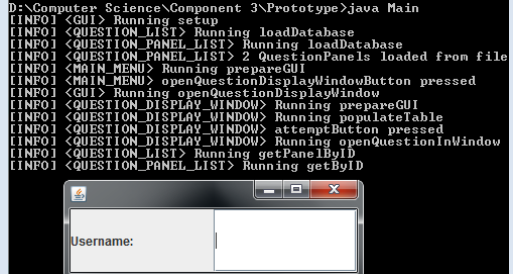
						
31		Soft Dev	Table doesn't automatically refresh correctly when a question is deleted.	 Table goes blank.	This is an issue with the refresh table method. When no filters or sorts are applied it defaults to an empty table rather than a table that contains all of the questions. Making it default to a table with all of the questions will fix this.	 Table updates correctly when the selected question is deleted.

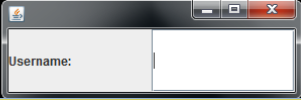
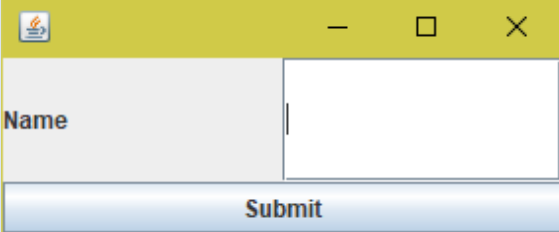
32	Soft Dev	Array index out of bounds exception when pressing the delete question button. The button still works correctly however. This is because two events are fired when the delete question button is pressed.	<pre>[INFO] <QUESTION_DISPLAY_PANEL> deleteButton pressed Exception in thread "AWT-EventQueue-1" java.lang.ArrayIndexOutOfBoundsException: -1 at java.util.Vector.elementAt(Vector.java:737) at java.util.Vector.elementAt(Vector.java:486) at javax.swing.table.DefaultTableModel.getValueAt(DefaultTableModel.java:648) at QuestionDisplayPanel.actionPerformed(QuestionDisplayPanel.java:425) at javax.swing.AbstractButton.fireActionPerformed(AbstractButton.java:2022) at javax.swing.AbstractButtonHandler.actionPerformed(AbstractButton.java:2348) at javax.swing.DefaultButtonModel.fireActionPerformed(DefaultButtonModel.java:402) at javax.swing.DefaultButtonModel.setPressed(DefaultButtonModel.java:259) at javax.swing.plaf.basic.BasicButtonListener.mouseReleased(BasicButtonListener.java:252) at java.awt.Component.processMouseEvent(Component.java:6539) at javax.swing.JComponent.processMouseEvent(JComponent.java:3324) at java.awt.Component.processEvent(Component.java:6384) at java.awt.Container.processEvent(Container.java:2239) at java.awt.Component.dispatchEventImpl(Component.java:4889) at java.awt.Container.dispatchEventImpl(Container.java:2297) at java.awt.Component.dispatchEvent(Component.java:4711) at java.awt.LightweightDispatcher.retargetMouseEvent(Container.java:4904) at java.awt.LightweightDispatcher.processMouseEvent(Container.java:4535) at java.awt.LightweightDispatcher.dispatchEvent(Container.java:4476) at java.awt.Container.dispatchEventImpl(Container.java:2263) at java.awt.Window.dispatchEventImpl(Window.java:2746) at java.awt.Component.dispatchEvent(Component.java:4711) at java.awt.EventQueue.dispatchEventImpl(EventQueue.java:768) at java.awt.EventQueue.access\$568(EventQueue.java:97) at java.awt.EventQueue\$3.run(EventQueue.java:789) at java.awt.EventQueue\$3.run(EventQueue.java:783) at java.security.AccessController.doPrivileged(Native Method) at java.security.ProtectionDomain\$JavaSecurityAccessImpl.doIntersectionPrivilege(ProtectionDomain.java:74) at java.security.ProtectionDomain\$JavaSecurityAccessImpl.doIntersectionPrivilege(ProtectionDomain.java:84) at java.awt.EventQueue\$4.run(EventQueue.java:733) at java.awt.EventQueue\$4.run(EventQueue.java:721) at java.security.AccessController.doPrivileged(Native Method) at java.security.ProtectionDomain\$JavaSecurityAccessImpl.doIntersectionPrivilege(ProtectionDomain.java:74) at java.awt.EventQueue.dispatchEvent(EventQueue.java:738) at org.springframework.aop.framework.AopWrapper\$6.dispatchEvent(AopWrapper.java:715) at java.awt.EventDispatchThread.pumpOneEventForFilters(EventDispatchThread.java:205) at java.awt.EventDispatchThread.pumpEventsForFilter(EventDispatchThread.java:116) at java.awt.EventDispatchThread.pumpEventsForHierarchy(EventDispatchThread.java:105) at java.awt.EventDispatchThread.pumpEvents(EventDispatchThread.java:101) at java.awt.EventDispatchThread.pumpEvents(EventDispatchThread.java:93) at java.awt.EventDispatchThread.run(EventDispatchThread.java:82) [INFO] <QUESTION_DISPLAY_PANEL> Running populateTable [INFO] <QUESTION_DISPLAY_PANEL> deleteButton pressed [INFO] <QUESTION_DISPLAY_PANEL> deleteButton pressed</pre>	The delete question button has had the action listener added to it twice. Removing one will fix this.	<pre>[INFO] <QUESTION_DISPLAY_PANEL> deleteButton pressed [INFO] <QUESTION_LIST> Running removeQuestion [INFO] <QUESTION_LIST> Running writeDatabase [INFO] <QUESTION_PANEL_LIST> Running writeDatabase [INFO] <QUESTION_PANEL_LIST> Running removeQuestionPanel [INFO] <QUESTION_PANEL_LIST> Running writeDatabase [INFO] <QUESTION_DISPLAY_PANEL> Running populateTable</pre> Exception no longer occurs when running delete question
33	Soft Dev	Incompatible types error when compiling GUI.java	<pre>harry@harry-pc-linux:~/Documents/SoftDev/form-filler\$./compile.sh Picked up _JAVA_OPTIONS: -Dawt.useSystemAAFontSettings=on -Dswing.aatext=true GUI.java:155: error: incompatible types: Object cannot be converted to JTabbedPane JTabbedPane sourcePane = changeEvent.getSource(); ^ 1 error Picked up _JAVA_OPTIONS: -Dawt.useSystemAAFontSettings=on -Dswing.aatext=true</pre>	Cast the object to JTabbedPane by putting (JTabbedPane) before it	<pre>harry@harry-pc-linux:~/Documents/SoftDev/form-filler\$./compile.sh Picked up _JAVA_OPTIONS: -Dawt.useSystemAAFontSettings=on -Dswing.aatext=true Picked up _JAVA_OPTIONS: -Dawt.useSystemAAFontSettings=on -Dswing.aatext=true harry@harry-pc-linux:~/Documents/SoftDev/form-filler\$</pre> Successful compile
34	Soft Dev	When the question display panel gains focus, the table refreshes.	 View forms panel.		

			 Console before switching to the view questions panel <pre>[INFO] <QUESTION_SELECTION_PANEL> Running prepareGUI [INFO] <QUESTION_SELECTION_PANEL> Running populateTable [INFO] <FORM_SELECTION_PANEL> Running prepareGUI [INFO] <FORM_SELECTION_PANEL> Running populateTable</pre> Console after <pre>[INFO] <QUESTION_SELECTION_PANEL> Running populateTable [INFO] <FORM_SELECTION_PANEL> Running prepareGUI [INFO] <FORM_SELECTION_PANEL> Running populateTable [INFO] <QUESTION_DISPLAY_PANEL> Running populateTable</pre> As you can see populate table has been called and so the table refreshes	
35	Soft Dev	The colours and fonts have been updated to better match those from the design. This was ported from the view forms redesign and little change to the code was needed.		

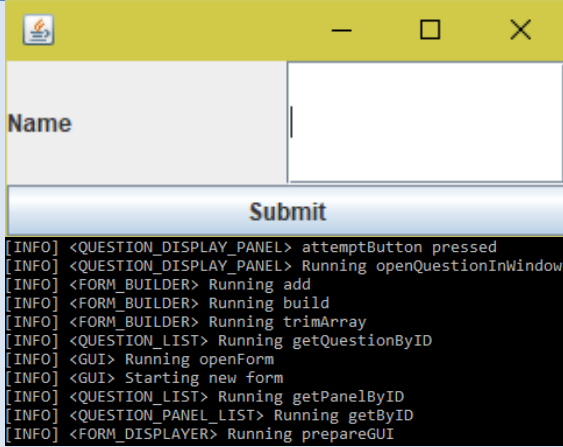
36		Soft Dev	Only 1 row of the table can be selected at a time.		
37		Soft Dev	The table isn't able to be edited.		
38		Soft Dev	Double clicking a question causes it to be opened to be attempted.		

54: Allow user to select a question to be attempted

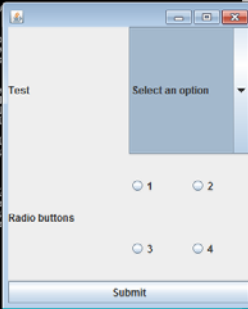
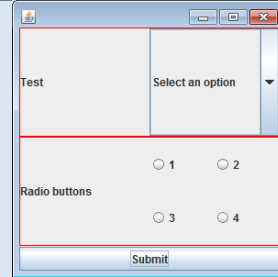
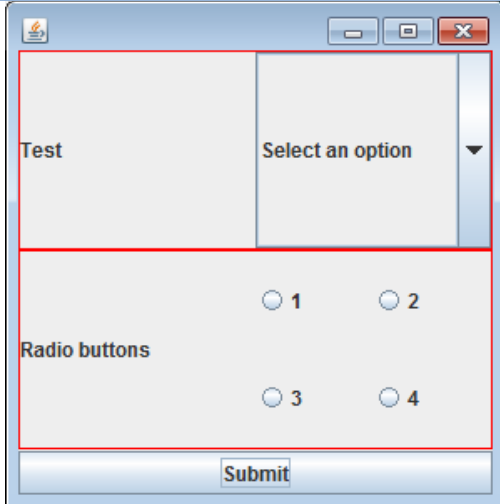
Test	Stage	Result Problem	Evidence	Action and Justification (If Needed)	Evidence of Action (If Needed)
1	Proto	Attempt question button displayed in view questions window			
2	Proto	Event is detected when the attempt question button is pressed	 <pre> D:\Computer Science\Component 3\Prototype>java Main [INFO] <GUI> Running setup [INFO] <QUESTION_LIST> Running loadDatabase [INFO] <QUESTION_PANEL_LIST> Running loadDatabase [INFO] <QUESTION_PANEL_LIST> 2 QuestionPanels loaded from file [INFO] <MAIN_MENU> Running prepareGUI [INFO] <MAIN_MENU> openQuestionDisplayWindowButton pressed [INFO] <GUI> Running openQuestionDisplayWindow [INFO] <QUESTION_DISPLAY_WINDOW> Running prepareGUI [INFO] <QUESTION_DISPLAY_WINDOW> Running populateTable [INFO] <QUESTION_DISPLAY_WINDOW> attemptButton pressed [INFO] <QUESTION_DISPLAY_WINDOW> attemptButton pressed </pre>		
3	Proto	Question is opened in a window too small	 <pre> D:\Computer Science\Component 3\Prototype>java Main [INFO] <GUI> Running setup [INFO] <QUESTION_LIST> Running loadDatabase [INFO] <QUESTION_PANEL_LIST> Running loadDatabase [INFO] <QUESTION_PANEL_LIST> 2 QuestionPanels loaded from file [INFO] <MAIN_MENU> Running prepareGUI [INFO] <MAIN_MENU> openQuestionDisplayWindowButton pressed [INFO] <GUI> Running openQuestionDisplayWindow [INFO] <QUESTION_DISPLAY_WINDOW> Running prepareGUI [INFO] <QUESTION_DISPLAY_WINDOW> Running populateTable [INFO] <QUESTION_DISPLAY_WINDOW> attemptButton pressed [INFO] <QUESTION_DISPLAY_WINDOW> Running openQuestionInWindow [INFO] <QUESTION_LIST> Running getPanelById [INFO] <QUESTION_PANEL_LIST> Running getPanelById </pre>	Set the size of the opened window	

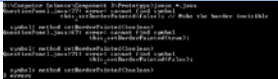
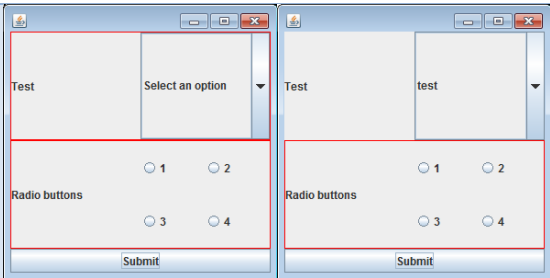
4	Proto	Question can be opened in it's own window	<pre> D:\Computer Science\Component 3\Prototype>java Main [INFO] <GUI> Running setup [INFO] <QUESTION_LIST> Running loadDatabase [INFO] <QUESTION_PANEL_LIST> Running loadDatabase [INFO] <QUESTION_PANEL_LIST> 2 QuestionPanels loaded from file [INFO] <MAIN_MENU> Running prepareGUI [INFO] <MAIN_MENU> openQuestionDisplayWindowButton pressed [INFO] <GUI> Running openQuestionDisplayWindow [INFO] <QUESTION_DISPLAY_WINDOW> Running prepareGUI [INFO] <QUESTION_DISPLAY_WINDOW> Running populateTable [INFO] <QUESTION_DISPLAY_WINDOW> attemptButton pressed [INFO] <QUESTION_DISPLAY_WINDOW> Running openQuestionInWindow [INFO] <QUESTION_LIST> Running getPanelByID [INFO] <QUESTION_PANEL_LIST> Running getByID </pre> 		
5	Proto	The question is added to a form with only it and then the form is opened to be attempted. This gives the question a submit button.	 <pre> [INFO] <QUESTION_DISPLAY_PANEL> attemptButton pressed [INFO] <QUESTION_DISPLAY_PANEL> Running openQuestionInWindow [INFO] <FORM_BUILDER> Running add [INFO] <FORM_BUILDER> Running build [INFO] <FORM_BUILDER> Running trimArray [INFO] <QUESTION_LIST> Running getQuestionByID [INFO] <GUI> Running openForm [INFO] <GUI> Starting new form [INFO] <QUESTION_LIST> Running getPanelByID [INFO] <QUESTION_PANEL_LIST> Running getByID [INFO] <FORM_DISPLAYER> Running prepareGUI </pre>		

55: Allow question to be submitted

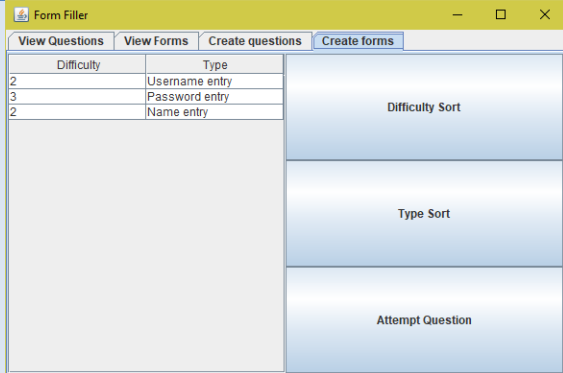
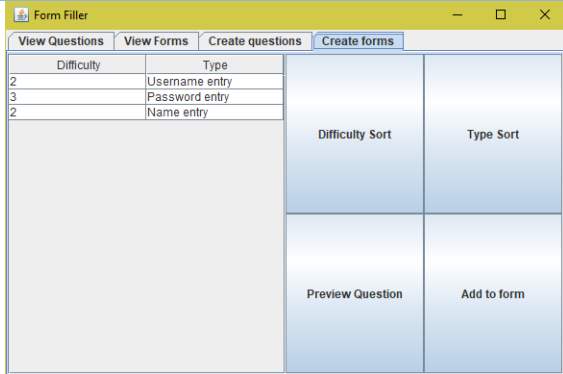
Test	Stage	Result Problem	Evidence	Action and Justification (If Needed)	Evidence of Action (If Needed)
1	Proto	When a question is opened to be attempted the same submit logic as the forms is used because it is just a form with one question. Therefore all features such as validation, saving progress, and statistic tracking work the same as with the forms. For testing see the form objectives.			

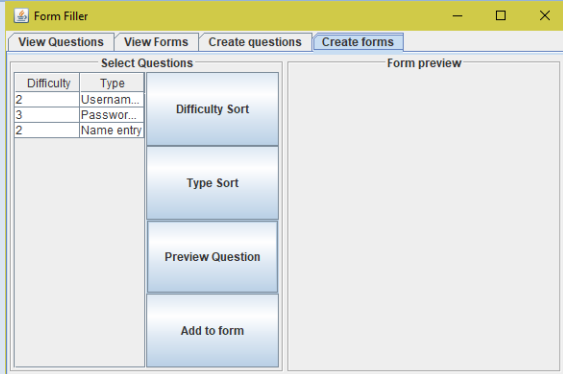
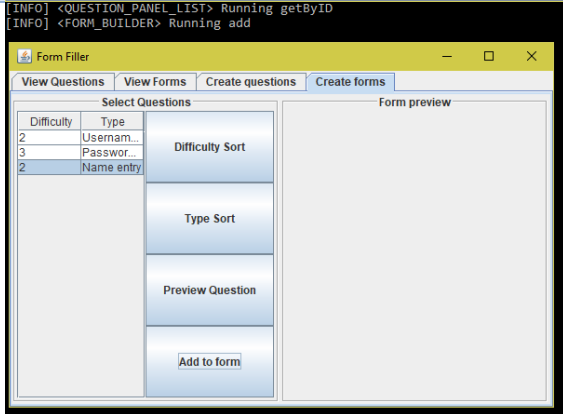
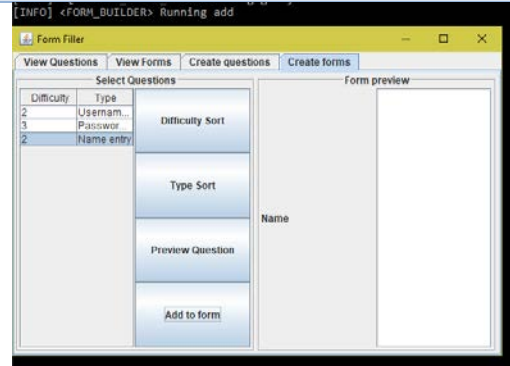
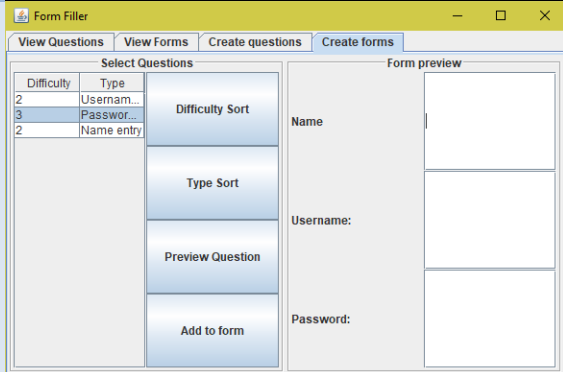
56: Questions that have failed their validation check should be highlighted

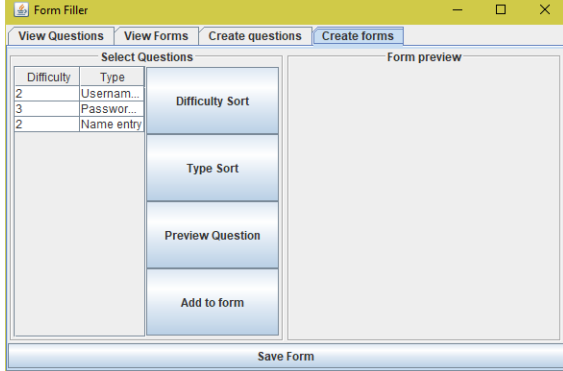
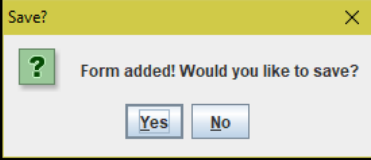
Test	Stage	Result	Problem	Evidence	Action and Justification (If Needed)	Evidence of Action (If Needed)
1	Proto	Red border doesn't display when the submit button is pressed. This is because the validate answers method wasn't actually called.	<pre> D:\Computer Science\Component 3\Prototype\Java [INFO] <GUI> Running setup [INFO] <QUESTION_LIST> Running loadDatabase [INFO] <QUESTION_LIST> 2 Questions loaded from file [INFO] <QUESTION_PANEL_LIST> Running loadDatabase [INFO] <QUESTION_PANEL_LIST> 2 QuestionPanes loaded [INFO] <FORM_LIST> Running loadDatabase [INFO] <FORM_LIST> 2 Forms loaded from file [INFO] <QUESTION_DISPLAY_PANEL> Running prepareGUI [INFO] <QUESTION_DISPLAY_PANEL> Running populate [INFO] <QUESTION_LIST> Running generateId [INFO] <QUESTION_LIST> Running getQuestionById [INFO] <QUESTION_CREATION_PANEL> Running prepareGUI [INFO] <QUESTION_CREATION_PANEL> Running populate [INFO] <FORM_LIST> Running generateId [INFO] <FORM_LIST> Running getFormById [INFO] <QUESTION_SELECTION_PANEL> Running prepareGUI [INFO] <QUESTION_SELECTION_PANEL> Running populate [INFO] <QUESTION_DISPLAY_PANEL> Running prepareGUI [INFO] <QUESTION_DISPLAY_PANEL> Running populate [INFO] <FORM_LIST> Running getFormById [INFO] <FORM_DISPLAY> Running prepareGUI [INFO] <QUESTION_LIST> Running getPanelById [INFO] <QUESTION_PANEL_LIST> Running getPanelById [INFO] <QUESTION_PANEL_LIST> Running getPanelById [INFO] <QUESTION_PANEL_LIST> Running getPanelById [INFO] <FORM_DISPLAY> submit button pressed </pre>		Add a call to the validate answer method	
2	Proto	Red border displayed when validation check failed.				
3	Proto	Multiple errors when compiling question panel because of an accidentally deleted }	<pre> D:\Computer Science\Component 3\Prototype\Java > javac QuestionPanel.java:37: error: illegal start of expression public String getQuestionID() ^ QuestionPanel.java:37: error: ';' expected public String getQuestionID() ^ QuestionPanel.java:44: error: illegal start of expression public void actionPerformed(ActionEvent evt) ^ QuestionPanel.java:44: error: ';' expected public void actionPerformed(ActionEvent evt) ^ QuestionPanel.java:44: error: illegal start of expression public void actionPerformed(ActionEvent evt) ^ QuestionPanel.java:44: error: ';' expected public void actionPerformed(ActionEvent evt) ^ QuestionPanel.java:44: error: illegal start of expression public void actionPerformed(ActionEvent evt) ^ QuestionPanel.java:44: error: ';' expected public void actionPerformed(ActionEvent evt) ^ </pre>	Add the } back in	<pre> D:\Computer Science\Component 3\Prototype\Java > javac QuestionPanel.java:37: error: illegal start of expression public String getQuestionID() ^ QuestionPanel.java:37: error: ';' expected public String getQuestionID() ^ QuestionPanel.java:44: error: illegal start of expression public void actionPerformed(ActionEvent evt) ^ QuestionPanel.java:44: error: ';' expected public void actionPerformed(ActionEvent evt) ^ QuestionPanel.java:44: error: illegal start of expression public void actionPerformed(ActionEvent evt) ^ QuestionPanel.java:44: error: ';' expected public void actionPerformed(ActionEvent evt) ^ QuestionPanel.java:44: error: illegal start of expression public void actionPerformed(ActionEvent evt) ^ QuestionPanel.java:44: error: ';' expected public void actionPerformed(ActionEvent evt) ^ QuestionPanel.java:44: error: illegal start of expression public void actionPerformed(ActionEvent evt) ^ QuestionPanel.java:44: error: ';' expected public void actionPerformed(ActionEvent evt) ^ </pre>	Different errors now

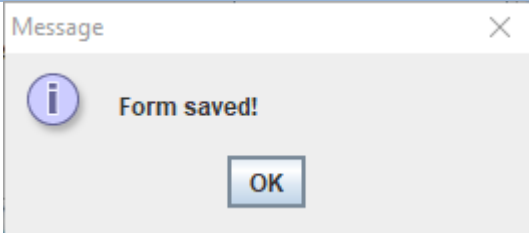
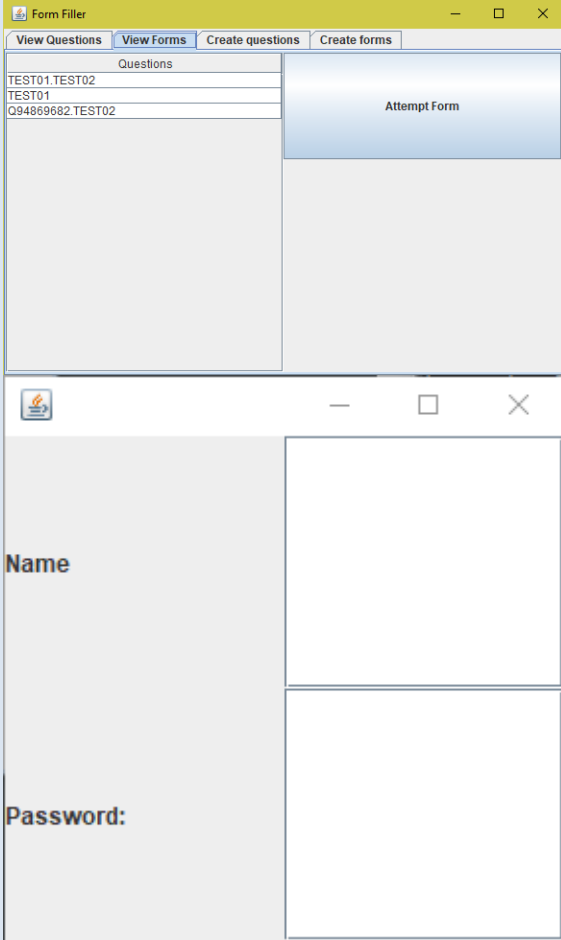
4	Proto	method setBorderPainted() cannot be found		Use setBorder rather than setBorderPainted	
5	Proto	The border is red when the question has been answered wrong and then disappears is invisible when the question is corrected			

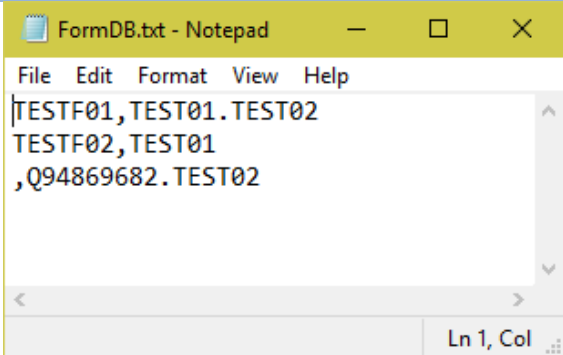
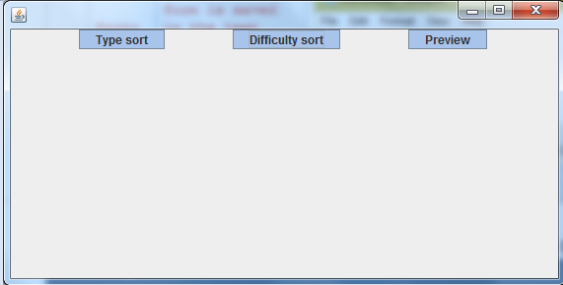
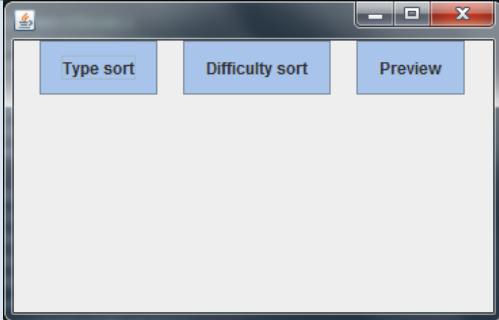
57: The user should be able to create forms by selecting questions from a list

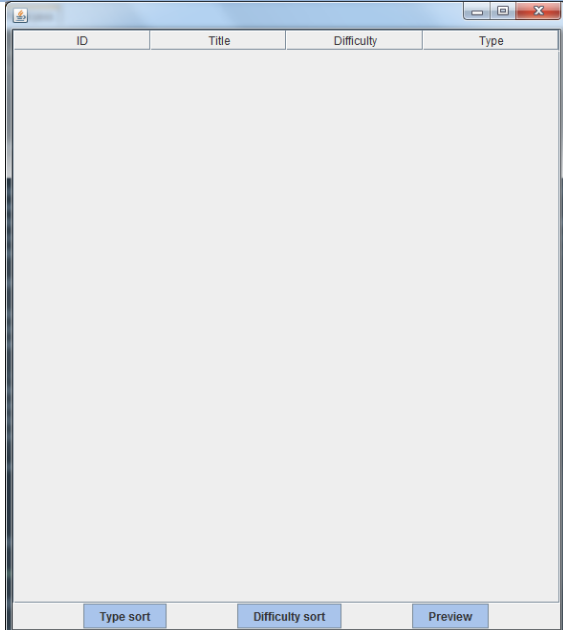
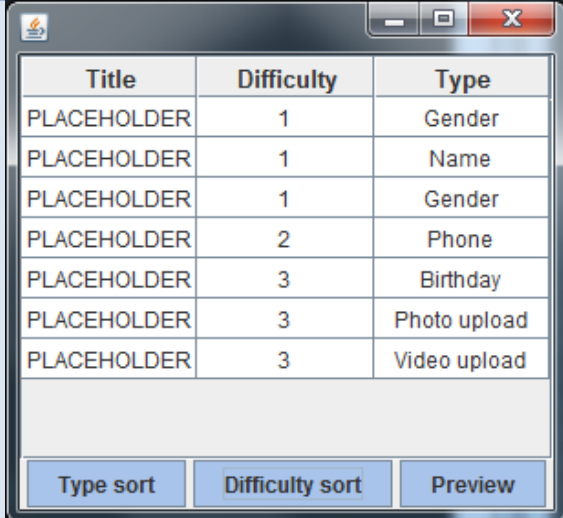
Test	Stage	Result Problem	Evidence	Action and Justification (If Needed)	Evidence of Action (If Needed)
1	Proto	There is a create forms panel which displays a view questions panel. Because I've reused this panel all of the buttons work correctly.			
2	Proto	There is an add to form button and the attempt question button has been renamed to "Preview Question"			

3	Proto	There is a form preview area.			
4	Proto	Form preview panel only updates when the window resizes		Call revalidate() after adding the question	
5	Proto	Form preview shows questions that have been added to the form			

6	Proto	There is a save from button and presses to it are detected.	 <pre>[INFO] <FORM_CREATION_PANEL> saveFormButton pressed</pre>		
7	Proto	Error compiling as a formbuilder rather than a form was passed to the addForm method.	<pre>D:\Computer Science\Component 3\Prototype>javac *.java FormCreationPanel.java:115: error: incompatible types: FormBuilder cannot be converted to Form forms.addForm(formBeingCreated); // Add the form to ^ Note: Some messages have been simplified; recompile with -Xdiags:verbose to get full output 1 error</pre>	Call the .build() method on the form builder when passing to convert it to a form	<pre>D:\Computer Science\Component 3\Prototype>javac *.java D:\Computer Science\Component 3\Prototype></pre> Successful compile
8	Proto	Prompt to write database to file appears after pressing the save form button and a confirmation window appears after pressing yes to confirm that it's been saved.	<pre>[INFO] <QUESTION_SELECTION_PANEL> addButton pressed [INFO] <QUESTION_LIST> Running getPanelByID [INFO] <QUESTION_PANEL_LIST> Running getByID [INFO] <FORM_BUILDER> Running add [INFO] <QUESTION_SELECTION_PANEL> addButton pressed [INFO] <QUESTION_LIST> Running getPanelByID [INFO] <QUESTION_PANEL_LIST> Running getByID [INFO] <FORM_BUILDER> Running add [INFO] <FORM_CREATION_PANEL> saveFormButton pressed [INFO] <FORM_BUILDER> Running build [INFO] <FORM_BUILDER> Running trimArray</pre> 		

					
9	Proto	Newly created form appears in the view forms window and can be opened.			

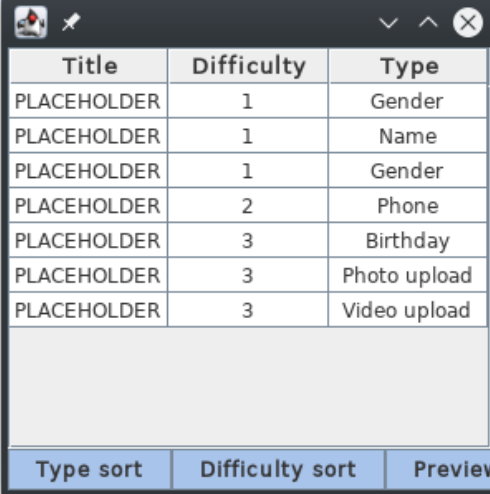
10	Proto	Newly created form is saved in the text file			
11	Soft Dev	This panel will be used to selected to the question from the table. There are type sort difficulty sort and preview buttons. However they are of the incorrect width		Change the maximum size of the buttons to 80,40 rather than 50,20	

12	Soft Dev	There is an empty table. With the correct headers however the id column needs to be hidden.		Set the column to be hidden in the column model.	
13	Soft Dev	The table is populated with the questions info and the theme is the same as the other tables in the system.			

14

Soft Dev

The type sort button works



Title	Difficulty	Type
PLACEHOLDER	1	Gender
PLACEHOLDER	1	Name
PLACEHOLDER	1	Gender
PLACEHOLDER	2	Phone
PLACEHOLDER	3	Birthday
PLACEHOLDER	3	Photo upload
PLACEHOLDER	3	Video upload

Type sort Difficulty sort Preview

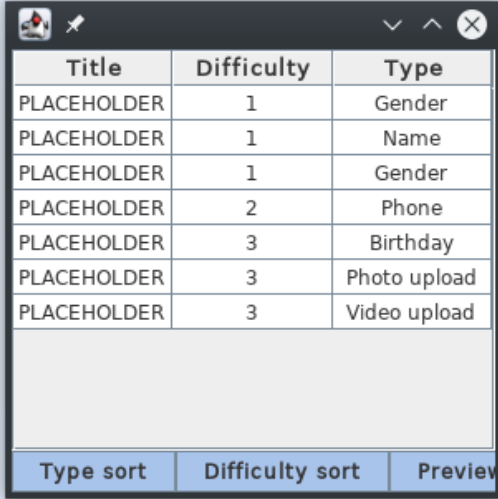
The questions are not in type order

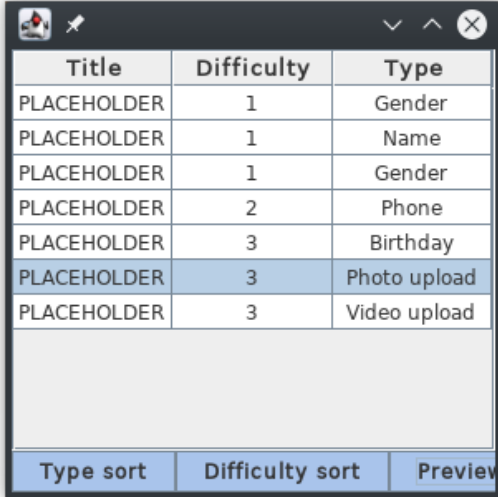
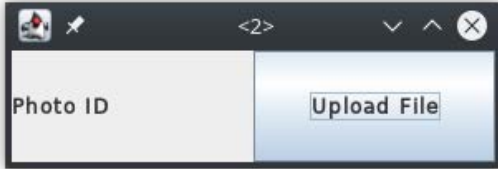
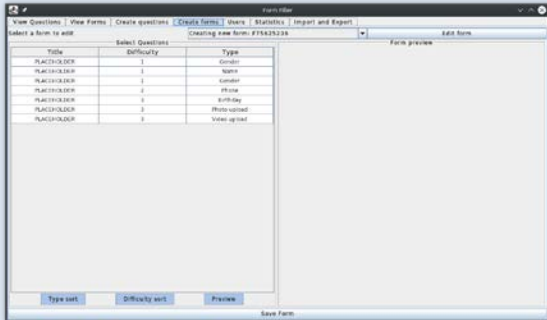


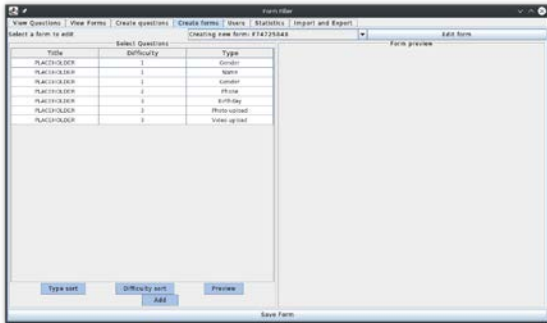
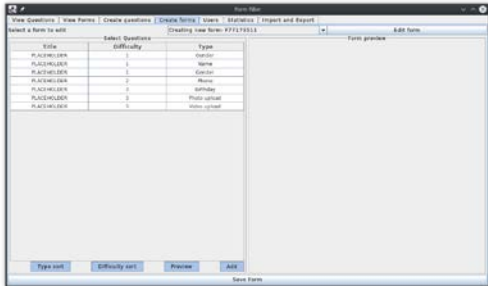
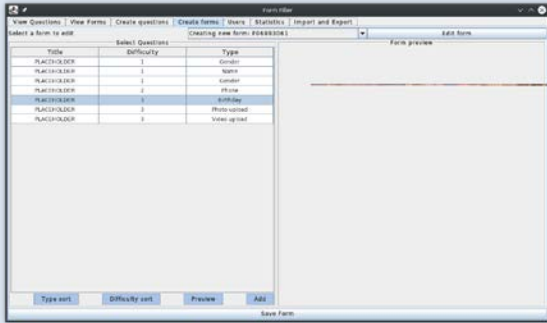
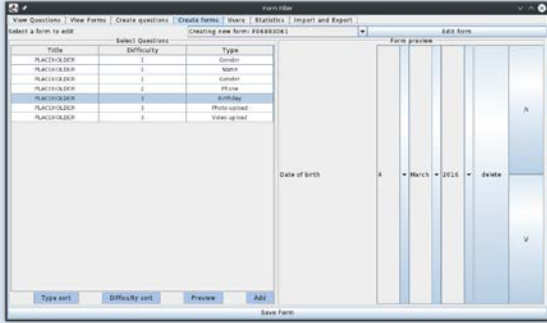
Title	Difficulty	Type
PLACEHOLDER	3	Birthday
PLACEHOLDER	1	Gender
PLACEHOLDER	1	Gender
PLACEHOLDER	1	Name
PLACEHOLDER	2	Phone
PLACEHOLDER	3	Photo upload
PLACEHOLDER	3	Video upload

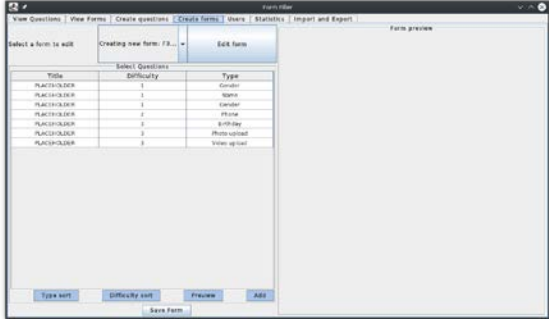
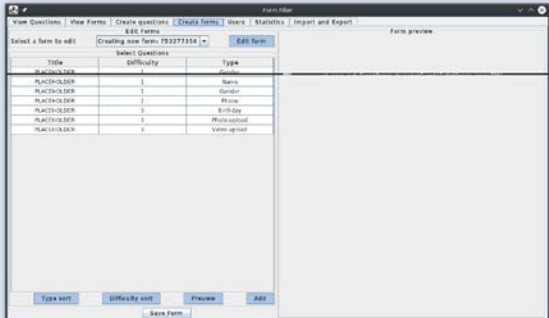
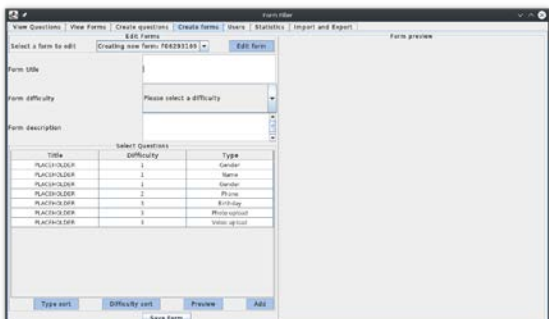
Type sort Difficulty sort Preview

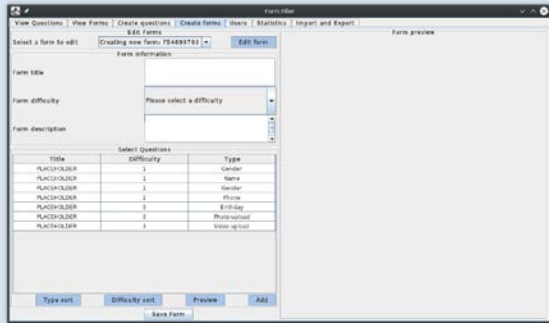
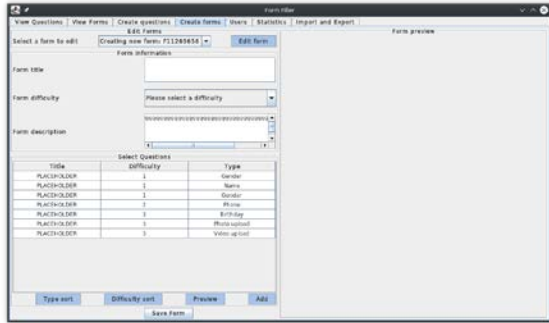
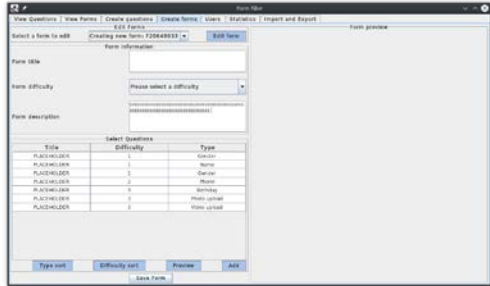
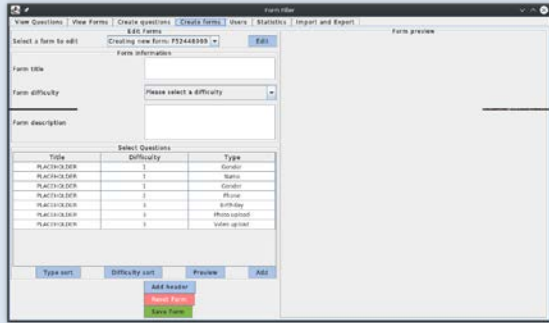
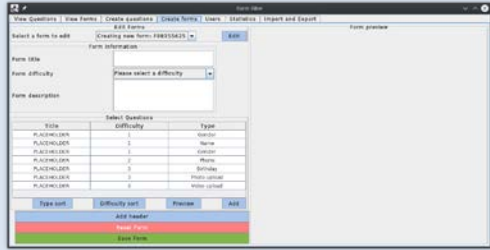
The questions are in type order after pressing

			the type sort button		
15	Soft Dev	The difficulty sort button works	the type sort button  The questions are not in difficulty order 		

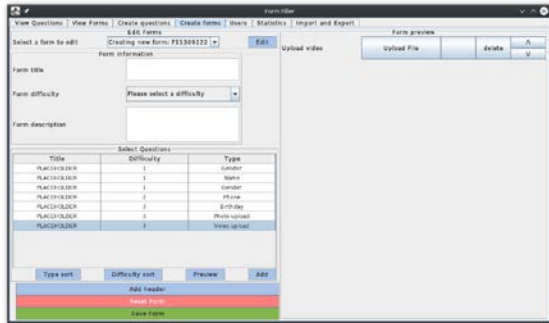
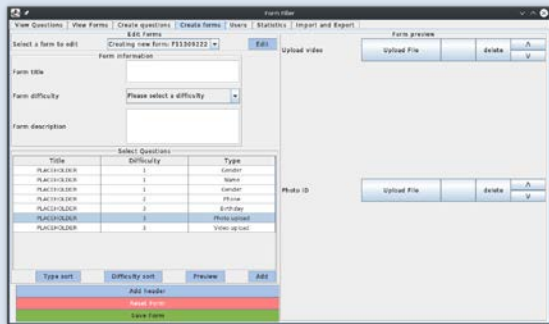
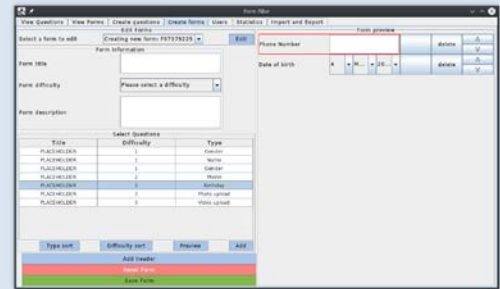
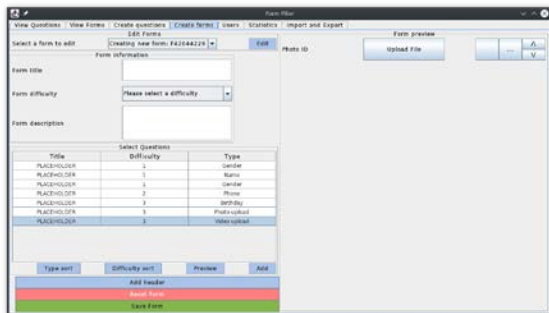
			The questions are in difficulty order after pressing the difficulty sort button		
16	Soft Dev	When the user selects a question and presses the preview button the question is opened in a window for them to preview.	 		
17	Soft Dev	The new question selection panel has been added to the create forms panel			

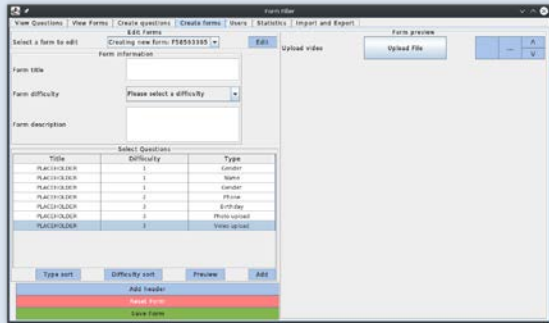
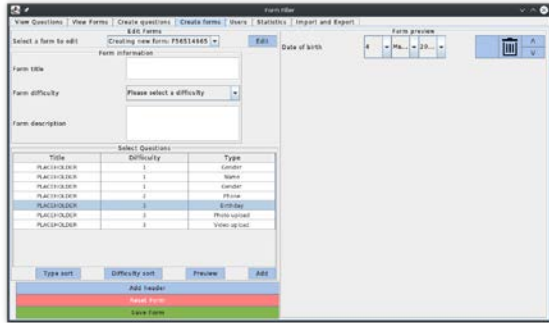
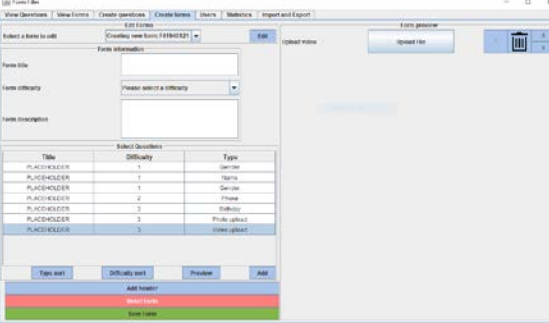
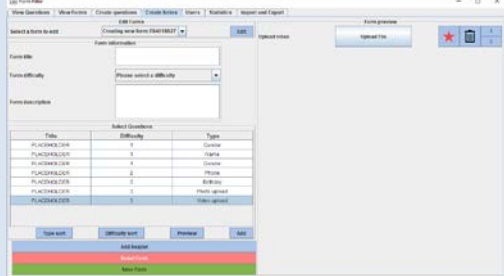
18	Soft Dev	There is an add button however it's in the wrong place.		Add it to the buttonPanel with the other buttons so that it's in the same row as them.  It's in the correct location now.
19	Soft Dev	If a question is selected and the add button is pressed, the question is added to the form.	 	

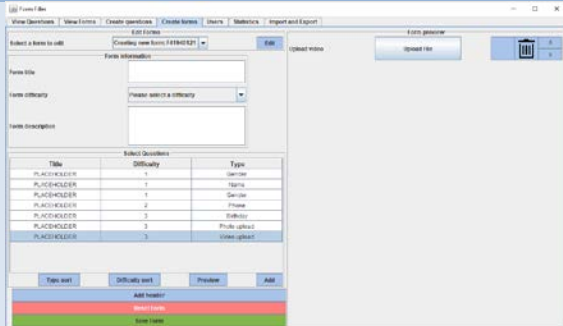
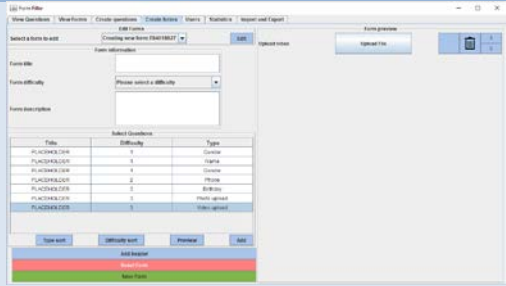
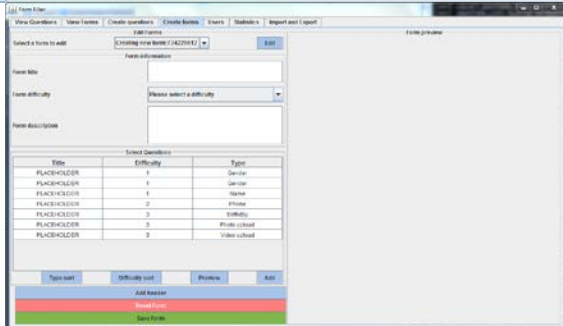
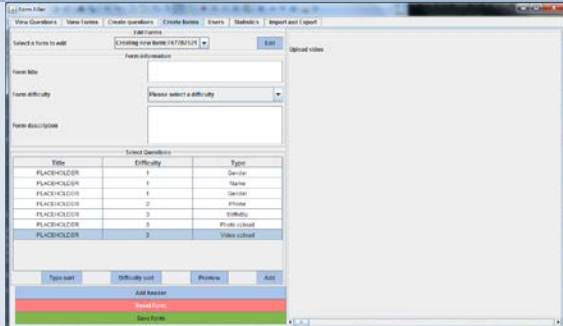
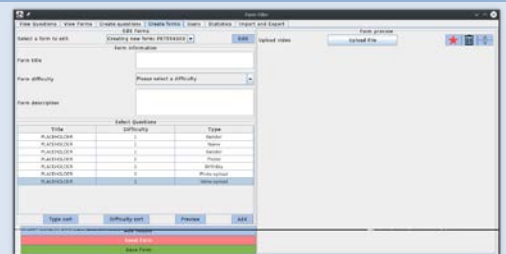
20	Soft Dev	The layout of the panel is more like the one in the design.			
21	Soft Dev	The edit forms area of the screen has the correct layout.			
22	Soft Dev	There is a basic implementation of the form information section			

23	Soft Dev	The form information panel has a border.			
24	Soft Dev	When a large amount of text is added to the description field it scrolls horizontally, this is undesired. It should wrap around.		Set the lineWrap of the text area to true by using the .setLineWrap() method	
25	Soft Dev	The buttons at the bottom aren't the full width even though that have a wide preferred size.		Set the minimum and maximum size of the buttons to be the preferred size.	

The buttons are wide enough

26	Soft Dev	When a question is added to the form it takes up a more reasonable amount of space as it has a maximum height. There is also a new blank button to be used for marking the question as required or not.			
27	Soft Dev	When multiple questions are added to the form there is a large gap between them.		Rather than adding vertical glue between each component only add it at the bottom. Vertical glue is used to fill the blank space therefore by removing it the blank space will disappear.	
28	Soft Dev	There is a gap between the question preview and the buttons			

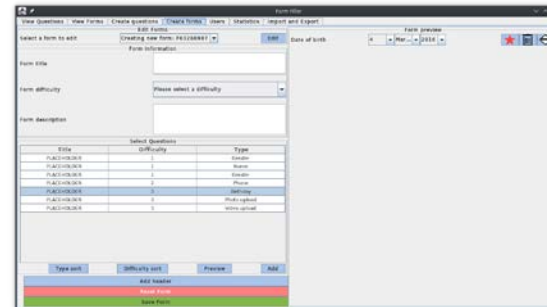
29	Soft Dev	The option buttons to the right of the question are of the correct colours.			
30	Soft Dev	The delete button has a rubbish bin icon.			
31	Soft Dev	The required button has a red asterisk in it when it's pressed. If it's pressed again, the asterisk disappears		Use an image of an asterisk rather than text. This will ensure that the asterisk is larger and is the same size regardless of font.	

					
32	Soft Dev	When any of the form information fields are not filled in an error message is produced when the save form button is pressed detailing which fields aren't filled in.	 Insufficient details entered Please address the following errors: Please enter a title. Please enter a description. Please select a difficulty OK		
33	Soft Dev	After adding the form preview scroller the preview scrolls horizontally this is not desired it should only scroll vertically.		Disable the horizontal scrollbar of the JScrollPane so that it never appears and set a preferred width for the question row so that it doesn't become wider than the form preview panel.	

34

Soft Dev

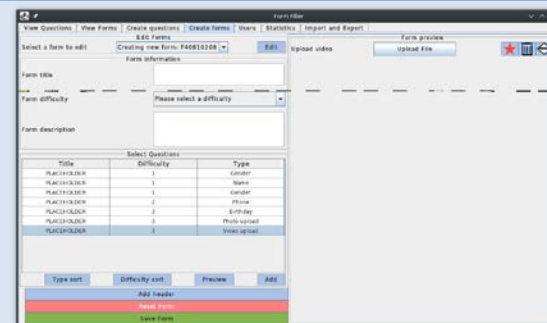
The up and down buttons have arrows as their icons as per the design.



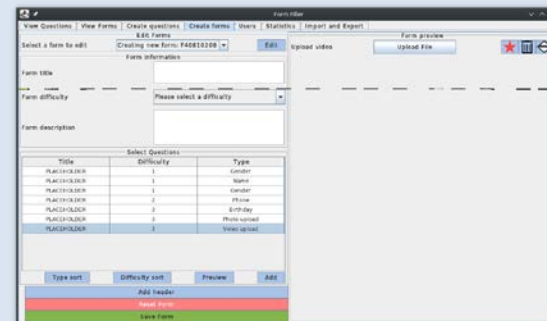
35

Soft Dev

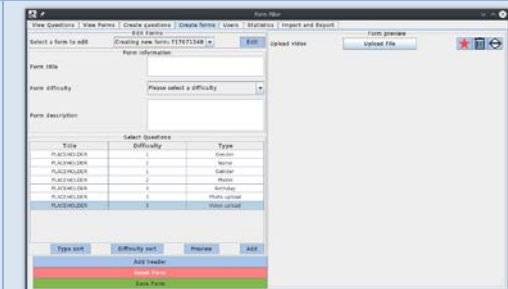
When the delete button is pressed on a question the form preview doesn't update until the window is resized.



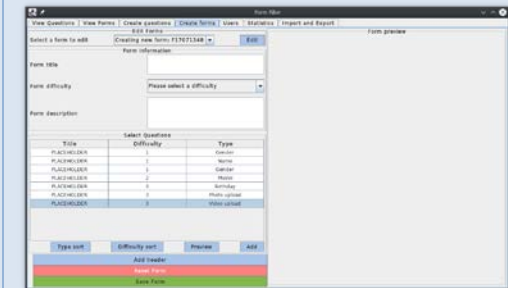
[INFO] <FORM_CREATION_PANEL> Running removeQuestionFromForm
[INFO] <FORM_BUILDER> Running remove
[INFO] <FORM_CREATION_PANEL> Running updateFormPreview



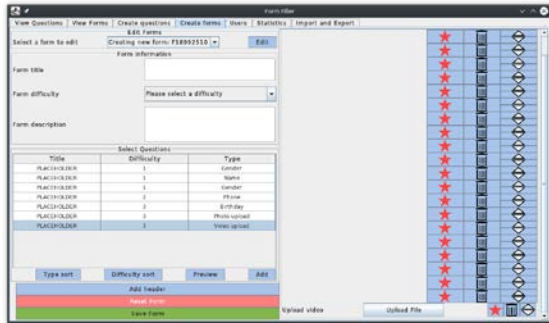
Call repaint on the form preview panel after deleting the question to cause it to update.

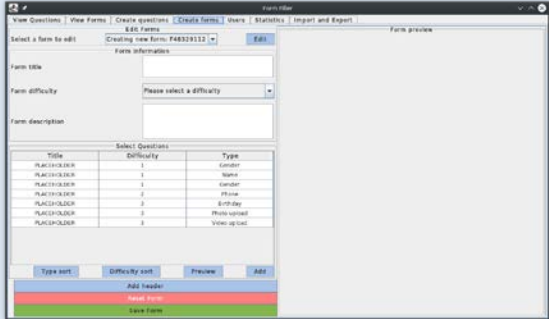
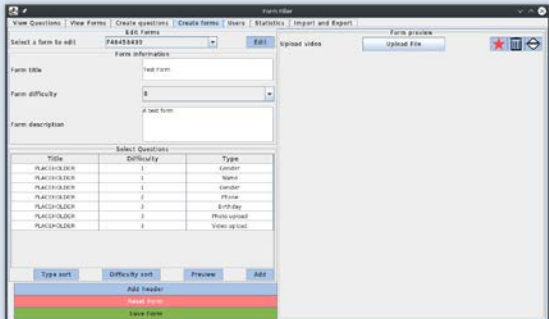
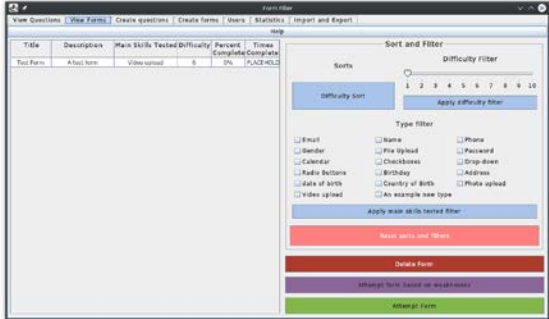


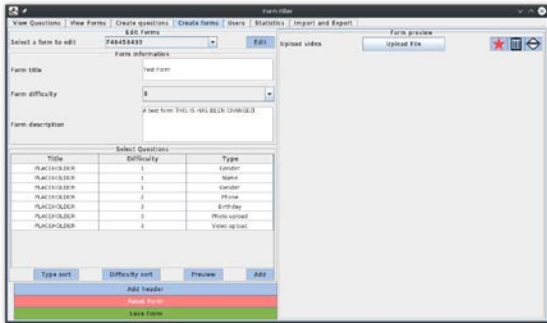
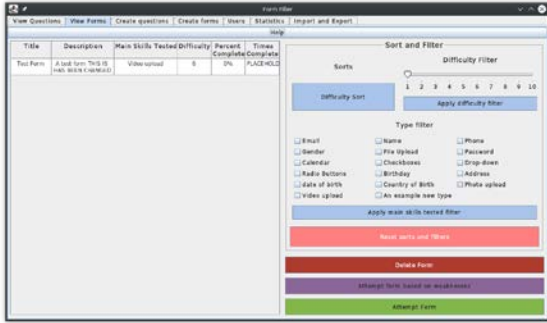
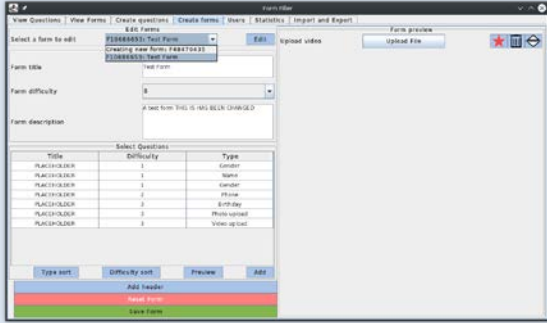
[INFO] <FORM_CREATION_PANEL> Running removeQuestionFromForm
[INFO] <FORM_BUILDER> Running remove
[INFO] <FORM_CREATION_PANEL> Running updateFormPreview

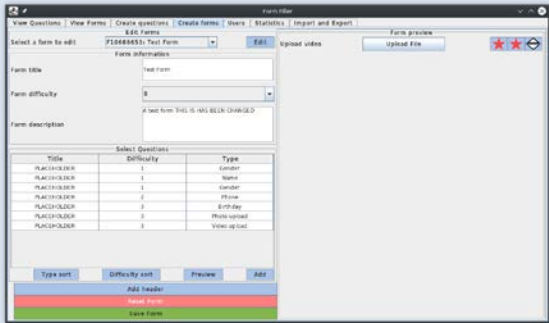
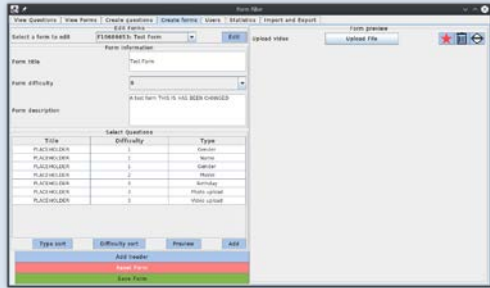
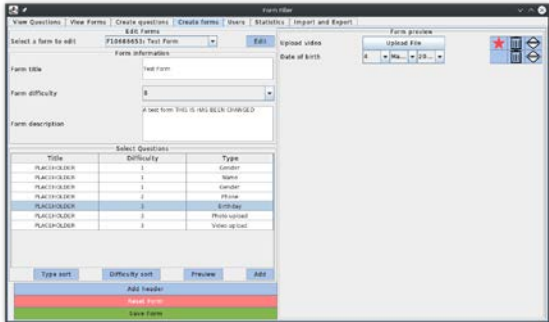


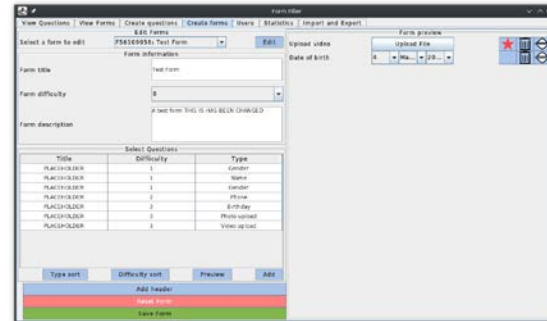
Question is removed immediately

36	Soft Dev	When there are too many questions to fit on screen a scroll bar appears allowing them all to be view. Ignore the large buttons that's just an unrelated bug due to adding the same question multiple times			
37	Soft Dev	When creating a new form and writing to the database there is a null pointer exception	<pre>[INFO] <FORM_BUILDER> Running trimArray [INFO] <FORM_LIST> Running writeDatabase [ERROR] <FORM_LIST> Error writing database to filejava.lang.NullPointerException</pre>	The null pointer exception was caused because the requiredQuestions array of the form created was empty because it wasn't loaded from the builder in the constructor of form. Storing the requiredQuestions array from the builder in the requiredQuestions variable of the form should fix this.	<pre>[INFO] <FORM_CREATION_PANEL> saveFormButton pressed [INFO] <FORM_LIST> Running getFormByID [INFO] <FORM_CREATION_PANEL> Running getFinalDetails [INFO] <FORM_BUILDER> Running build [INFO] <FORM_BUILDER> Running trimArray [INFO] <FORM_LIST> Running writeDatabase</pre>
38	Soft Dev	A large amount of null values are stored in the form database in the required questions section.		This is because the required questions array wasn't trimmed before creating a new form. Trimming the array should	<pre>F46458439,Q07218815,A test form,Video upload,6,Test Form,true</pre> The form's line in the form database doesn't have the null values

				remove all of these values.	
39	Soft Dev	When a form is selected to edit its title, difficulty, and description are loaded into the form information panel	 Before selecting a form to edit  After selecting a form.		
40	Soft Dev	When a form is loaded to be edited and changes are made and then save is pressed the changes are saved.			

			 		
41	Soft Dev	Form titles are also displayed in the edit form drop down			
42	Soft Dev	Error compiling FormCreationPanel due to a method not having public access	<pre>PCCherry@harry-pc-linux:/media/harry/HARRY_BEST/Computer Science/Component 3/SOFTDEV/Project/Forms-Editor\$./compile.sh Picked up _JAVA_OPTIONS: -Dawt.useSystemAAFontSettings=on -Dswing.aatext=true FormCreationPanel.java:401: error: isQuestionRequired(String) has private access in Form button.setIcon(formToEdit.isQuestionRequired ^ (questionID) ? requiredIcon : null; // If the question is required set the icon as the red star 1 error Picked up _JAVA_OPTIONS: -Dawt.useSystemAAFontSettings=on -Dswing.aatext=true</pre>	Change private to public so that FormCreationPanel can access the method	<pre>harry@harry-pc-linux:/media/harry/HARRY_BEST/Computer Science/Component 3/SOFTDEV/Project/Forms-Editor\$./compile.sh Picked up _JAVA_OPTIONS: -Dawt.useSystemAAFontSettings=on -Dswing.aatext=true Picked up _JAVA_OPTIONS: -Dawt.useSystemAAFontSettings=on -Dswing.aatext=true harry@harry-pc-linux:/media/harry/HARRY_BEST/Computer Science/Component 3/SOFTDEV/Project/Forms-Editor\$</pre> Successful compile

43	Soft Dev	When a form is selected to be edited all of the option buttons get the red star set as their icon		This is because the method that sets the buttons icon based on whether the question is required doesn't first check to see whether the button is actually the required button. Adding a check to ensure that only the required button's icon gets changed will fix this	 Only the required button has it's icon set
44	Soft Dev	When a form is loaded to be edited the required questions are loaded correctly (those that are required have a red star)	 Before restarting the system and reloading the form to be edited.		

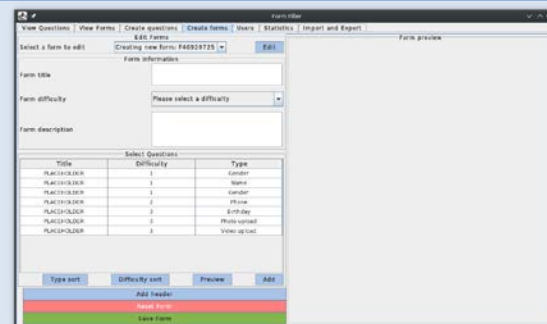


After restarting the system and loading the form to be edited, the questions that are required have red stars.

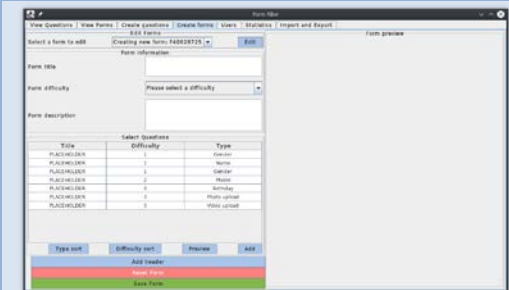
45

Soft Dev

Null pointer exception when the edit button is pressed and the "creating new form" option is pressed

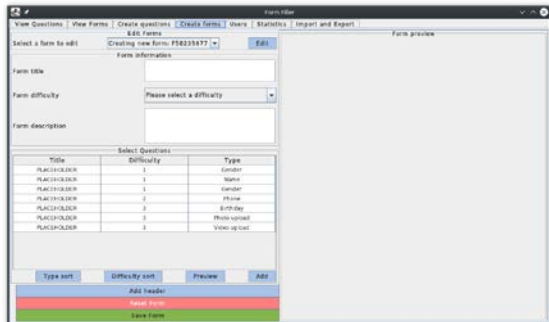


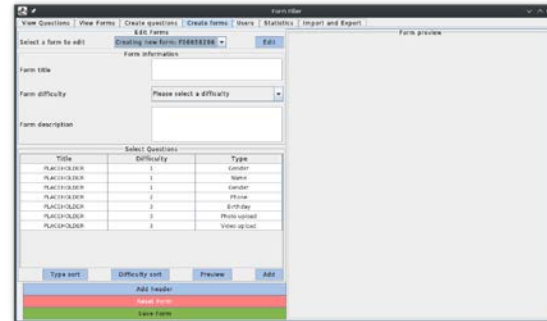
Add a check to ensure that the selected option isn't the "creating new form" option before loading the form to edit.



[INFO] <FORM_CREATION_PANEL> editFormButton pressed

No error after pressing the edit form button

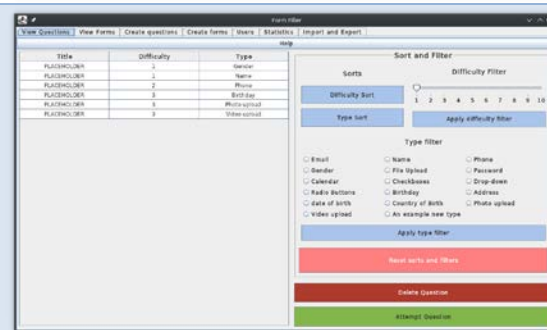
			<pre>[INFO] <FORM_LIST> Running getFormByID Exception in thread "AWT-EventQueue-1" java.lang.NullPointerException at FormCreationPanel.loadFormToBeEdited(FormCreationPanel.java:383) at FormCreationPanel.actionPerformed(FormCreationPanel.java:333) at javax.swing.AbstractButton.fireActionPerformed(AbstractButton.java:2022) at javax.swing.AbstractButton\$Handler.actionPerformed(AbstractButton.java:2348) at javax.swing.DefaultButtonModel.fireActionPerformed(DefaultButtonModel.java:402) at javax.swing.DefaultButtonModel.setPressed(DefaultButtonModel.java:259) at javax.swing.plaf.basic.BasicButtonListener.mouseReleased(BasicButtonListener.java:252) at java.awt.Component.processMouseEvent(Component.java:6539) at javax.swing.JComponent.processMouseEvent(JComponent.java:3324) at java.awt.Component.processEvent(Component.java:6304) at java.awt.Container.processEvent(Container.java:2239) at java.awt.Component.dispatchEventImpl(Component.java:4889) at java.awt.Container.dispatchEventImpl(Container.java:2297) at java.awt.Component.dispatchEvent(Component.java:4711) at java.awt.LightweightDispatcher.retargetMouseEvent(Container.java:4904) at java.awt.LightweightDispatcher.processMouseEvent(Container.java:4535) at java.awt.LightweightDispatcher.dispatchEvent(Container.java:4476) at java.awt.Container.dispatchEventImpl(Container.java:2283) at java.awt.Window.dispatchEventImpl(Window.java:2746) at java.awt.Component.dispatchEvent(Component.java:4711) at java.awt.EventQueue.dispatchEventImpl(EventQueue.java:760) at java.awt.EventQueue.access\$500(EventQueue.java:97) at java.awt.EventQueue\$3.run(EventQueue.java:709) at java.awt.EventQueue\$3.run(EventQueue.java:703) at java.security.AccessController.doPrivileged(Native Method) at java.security.ProtectionDomain\$JavaSecurityAccessImpl.doIntersectionPrivilege(ProtectionDomain.java:74) at java.security.ProtectionDomain\$JavaSecurityAccessImpl.doIntersectionPrivilege(ProtectionDomain.java:84) at java.awt.EventQueue\$4.run(EventQueue.java:733) at java.awt.EventQueue\$4.run(EventQueue.java:731) at java.security.AccessController.doPrivileged(Native Method) at java.security.ProtectionDomain\$JavaSecurityAccessImpl.doIntersectionPrivilege(ProtectionDomain.java:74) at java.awt.EventQueue.dispatchEvent(EventQueue.java:730) at org.gnome.Accessibility.AtkWrapper\$6.dispatchEvent(AtkWrapper.java:715) at java.awt.EventDispatchThread.pumpOneEventForFilters(EventDispatchThread.java:205) at java.awt.EventDispatchThread.pumpEventsForFilter(EventDispatchThread.java:116) at java.awt.EventDispatchThread.pumpEventsForHierarchy(EventDispatchThread.java:105) at java.awt.EventDispatchThread.pumpEvents(EventDispatchThread.java:101) at java.awt.EventDispatchThread.pumpEvents(EventDispatchThread.java:93) at java.awt.EventDispatchThread.run(EventDispatchThread.java:82)</pre>		
46	Soft Dev	When the reset form button is pressed the create forms tab is reset. This can be seen as the id of the form being edited in the edit forms combobox changes	 <pre>[INFO] <FORM_CREATION_PANEL> resetFormButton pressed [INFO] <FORM_LIST> Running getFreeID [INFO] <FORM_LIST> Running generateId [INFO] <FORM_LIST> Running getFormByID [INFO] <FORM_CREATION_PANEL> Running prepareGUI [INFO] <SELECT_QUESTIONS_PANEL> Running populateTable</pre>		



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Soft Dev

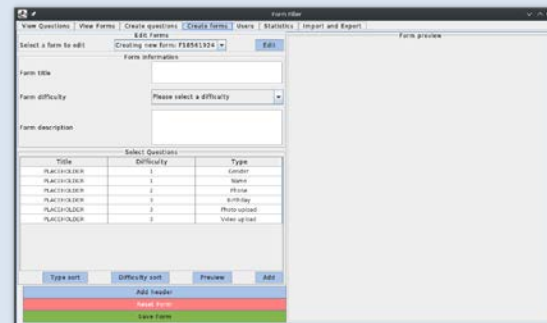
When the user opens the form creation panel the table is refreshed.



The view questions screen is selected.

```
[INFO] <FORM_SELECTION_PANEL> Running prepareGUI
[INFO] <FORM_SELECTION_PANEL> Running populateTable
[]
```

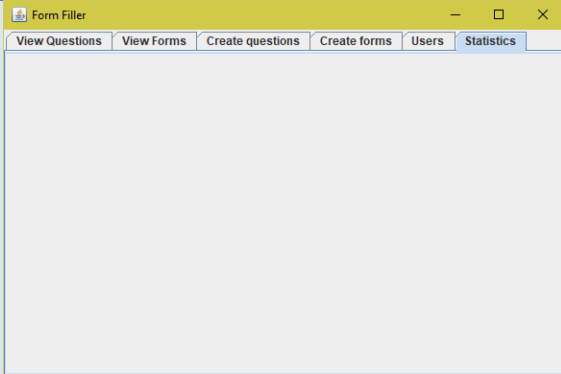
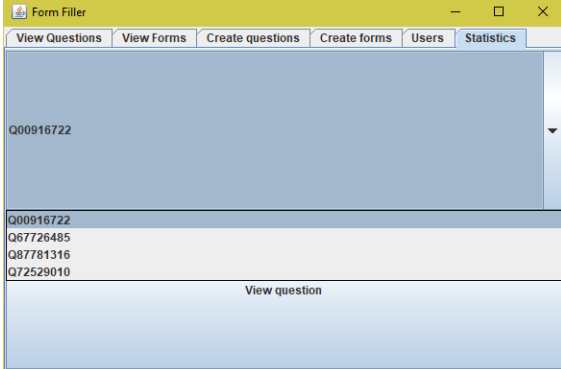
Most recent command line output.

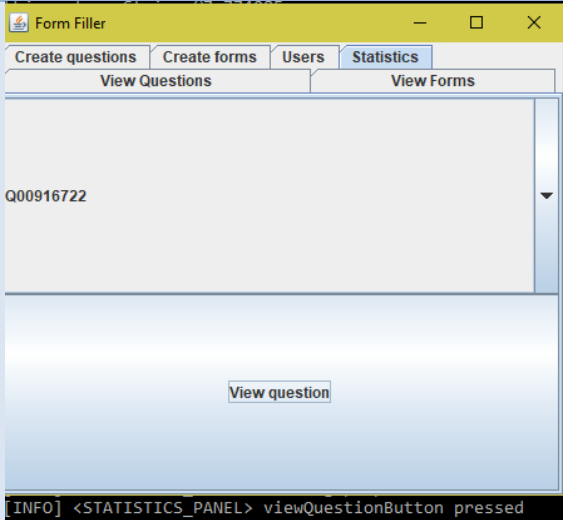
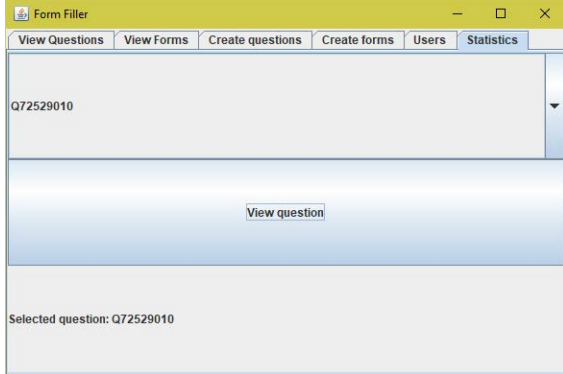


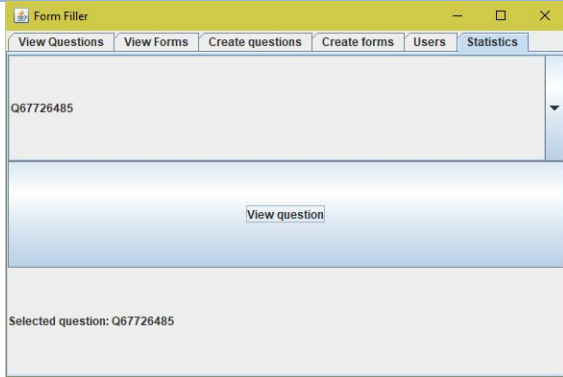
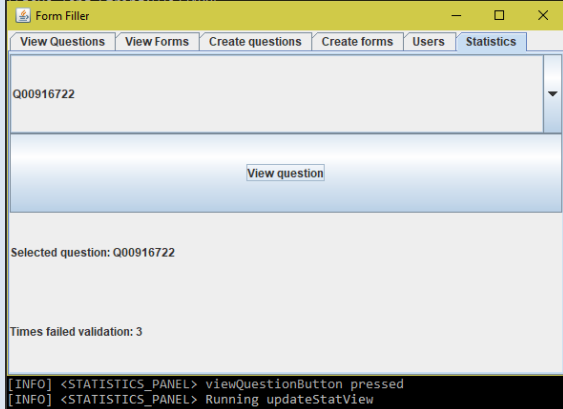
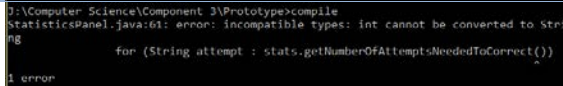
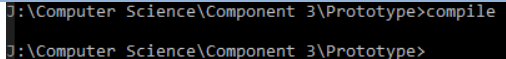
The form creation screen is selected.

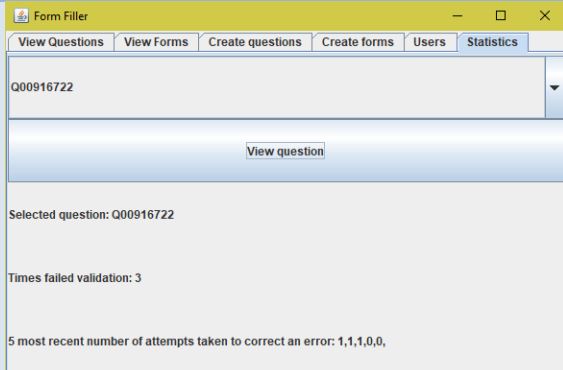
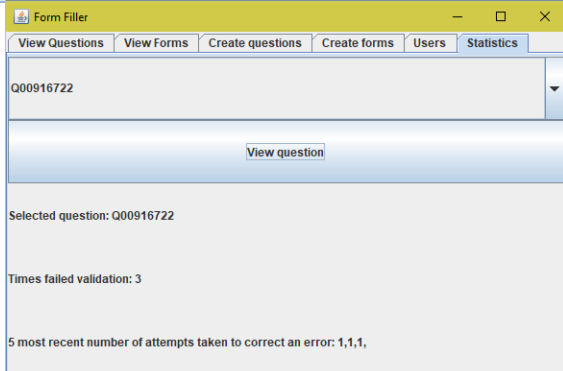
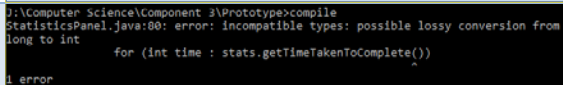
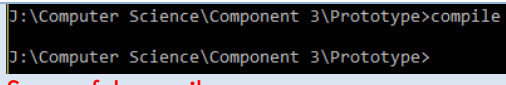
			<pre>[INFO] <FORM_SELECTION_PANEL> Running prepareGUI [INFO] <FORM_SELECTION_PANEL> Running populateTable [INFO] <SELECT_QUESTIONS_PANEL> Running populateTable []</pre> The question selection panel has been refreshed.		
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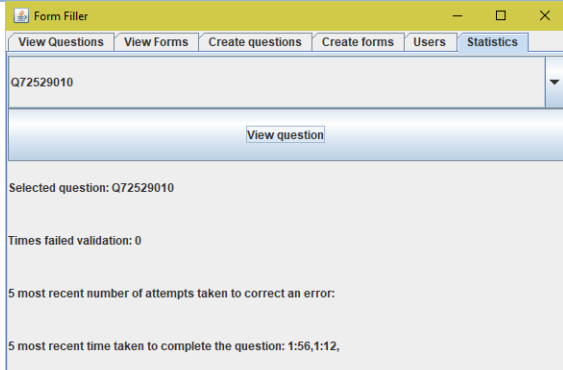
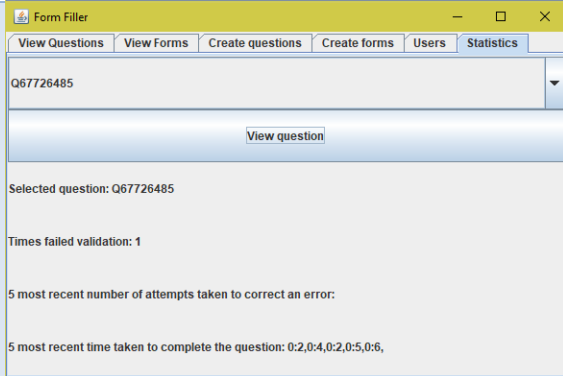
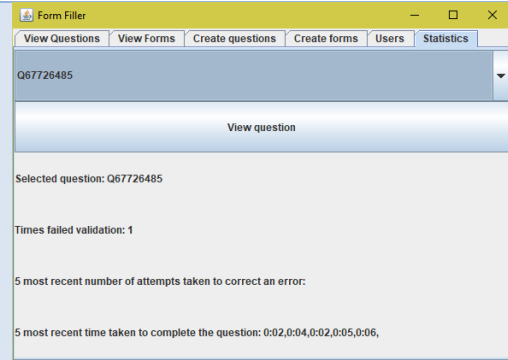
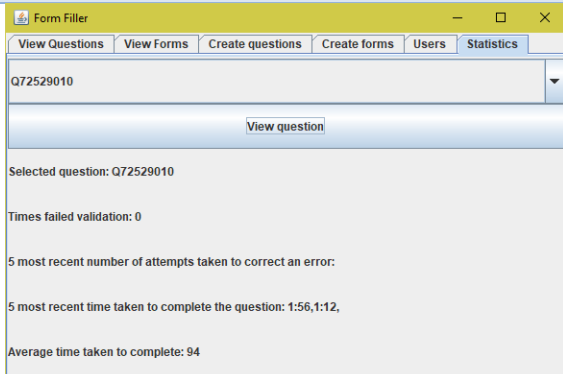
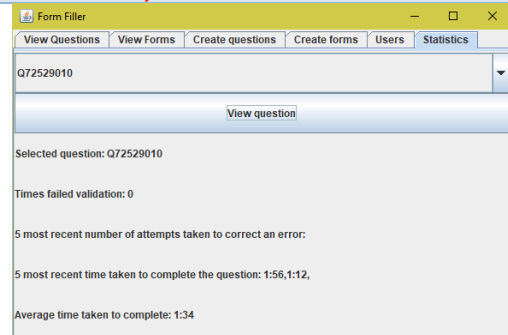
58: There should be a statistics screen which shows the user's progress

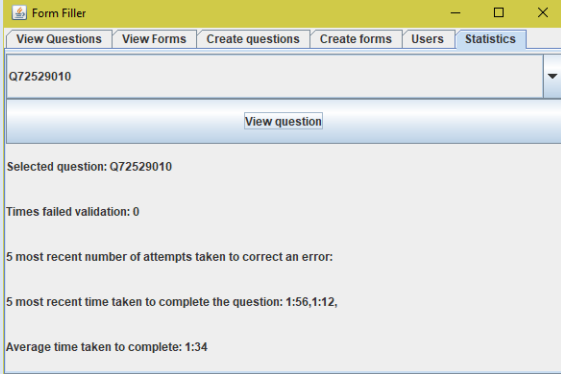
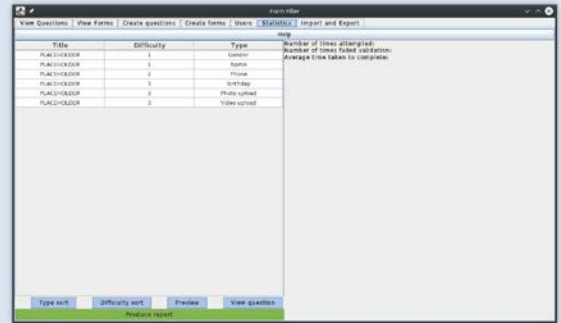
Test	Stage	Result Problem	Evidence	Action and Justification (If Needed)	Evidence of Action (If Needed)
1	Proto	The program has a statistics tab.			
2	Proto	There is a drop down that displays the question ids of all the questions that the user has attempted.			

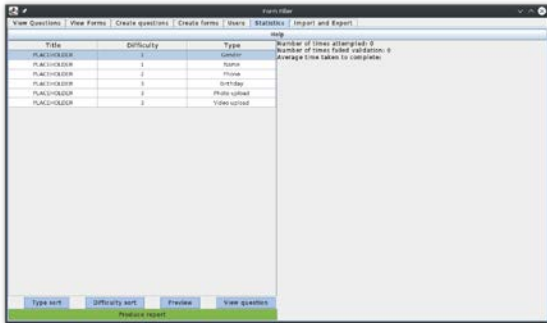
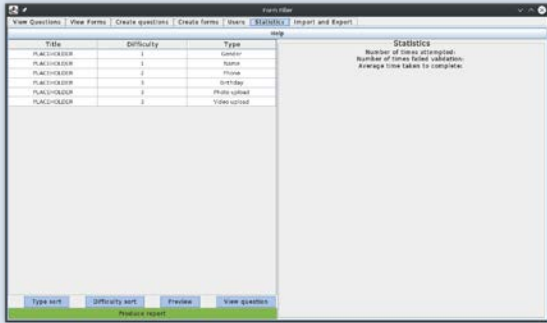
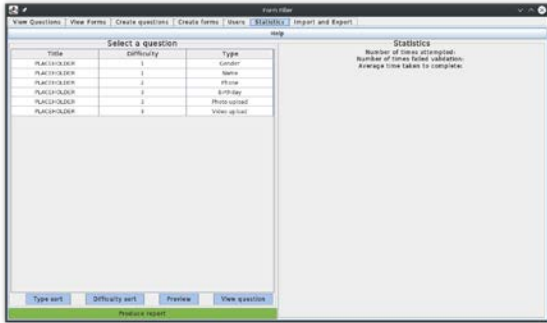
3	Proto	There is a view question button and presses to it are detected.	 The screenshot shows the 'Form Filler' application window. The 'Statistics' tab is active, and the 'View Questions' sub-tab is selected. A list of questions is displayed, with 'Q00916722' selected. Below the list is a 'View question' button. The bottom status bar shows the log entry: '[INFO] <STATISTICS_PANEL> viewQuestionButton pressed'.		
4	Proto	When the view question button is pressed the label at the bottom updates so that it shows the currently selected question	 The screenshot shows the 'Form Filler' application window. The 'Statistics' tab is active, and the 'View Questions' sub-tab is selected. A list of questions is displayed, with 'Q72529010' selected. Below the list is a 'View question' button. The bottom status bar shows the log entry: '[INFO] <STATISTICS_PANEL> viewQuestionButton pressed'. Below the main content area, a label reads 'Selected question: Q72529010'.		

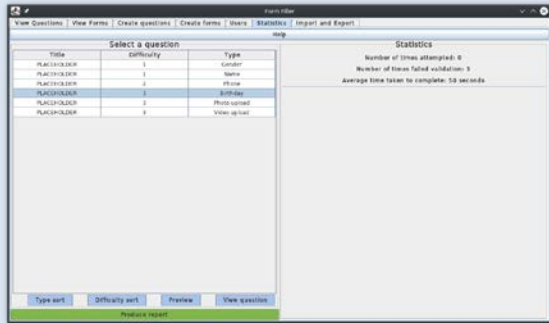
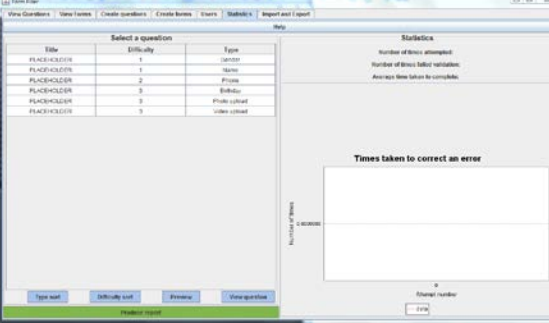
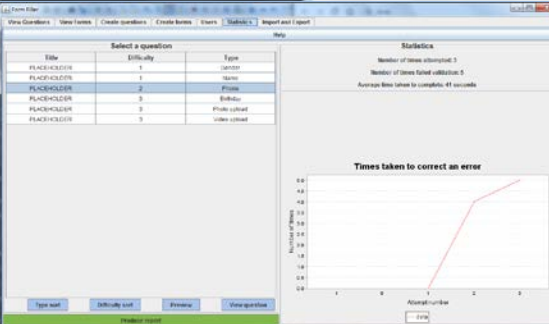
					
5	Proto	The number of times that the user has failed the validation check of the question is displayed.			
6	Proto	Type error when compiling StatisticsPanel.		Change “String attempt” to “int attempt” because integers are stored in the numberOfAttemptsNeededToCorrect array - not Strings.	

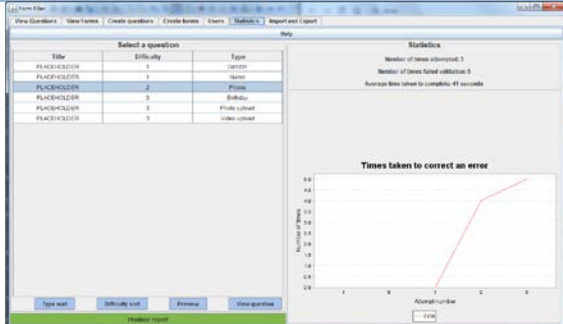
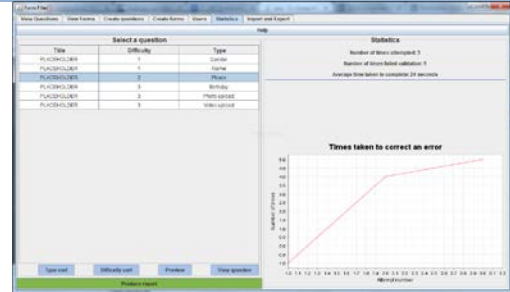
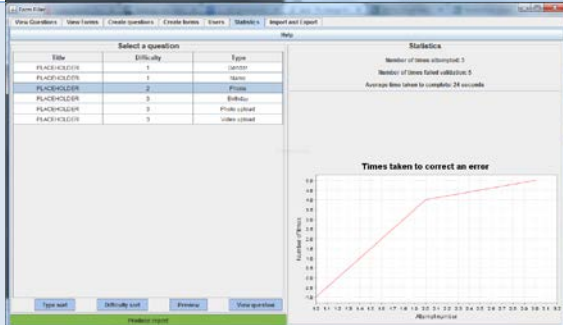
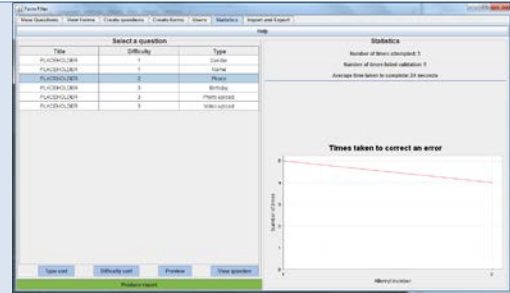
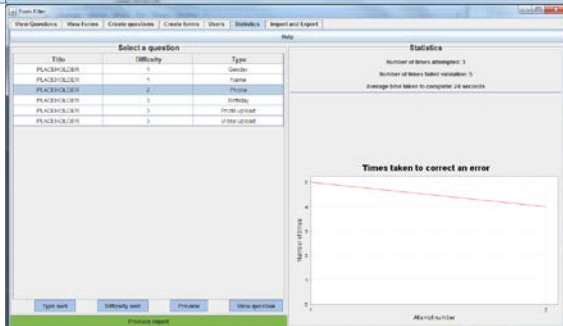
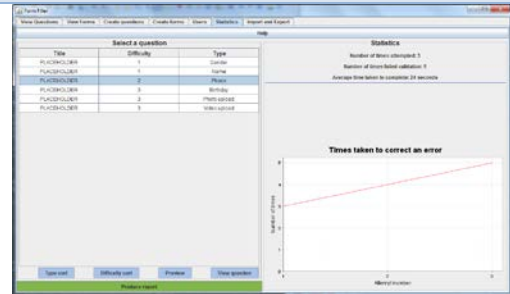
7	Proto	The 5 most recent number of attempts that it took the user to correct an error in their response to a question are outputted.			
8	Proto	The number of attempts that are 0 are not displayed because they are null data.			
9	Proto	Error compiling StatisticsPanel.java because of attempting to store a long in an int		Change the type of time from int to long	

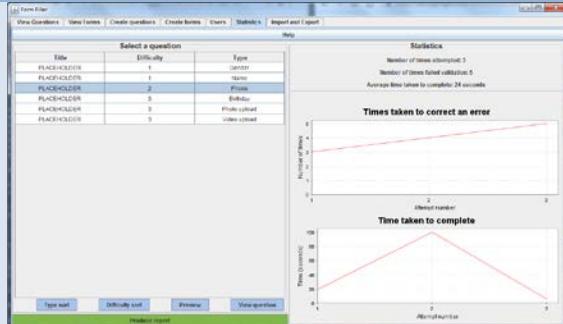
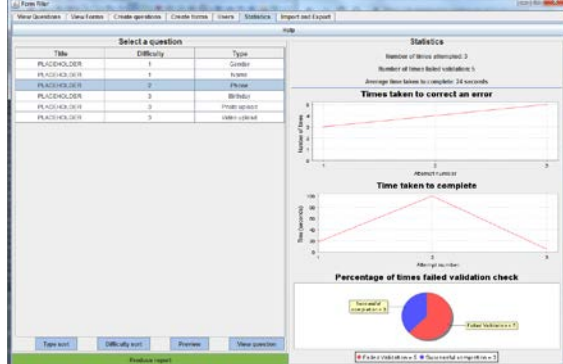
10	Proto	The most recent times taken to complete the question are displayed.			
11	Proto	Seconds aren't padded with leading zeros if they're only 1 digit		Check the size of the seconds and if it's less than 10 pad with a leading zero.	
12	Proto	The average time taken to complete is displayed however it's in seconds and not formatted in minutes and seconds.		Divide the time by 60 to get the number of minutes and mod it by 60 to get the remainder number of seconds and output that.	

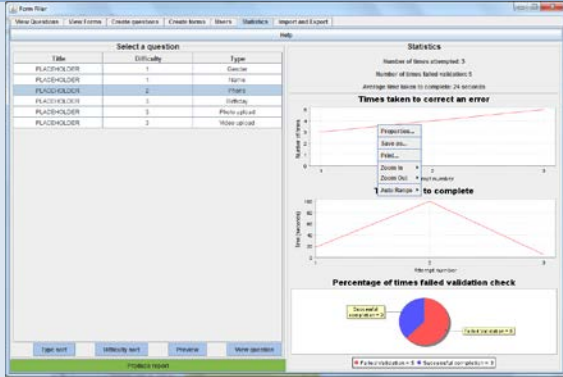
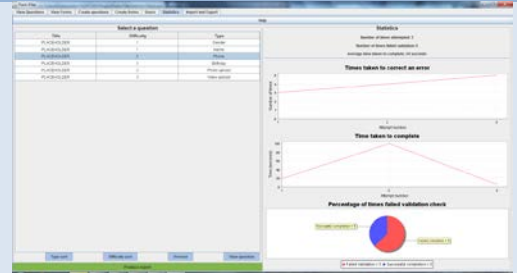
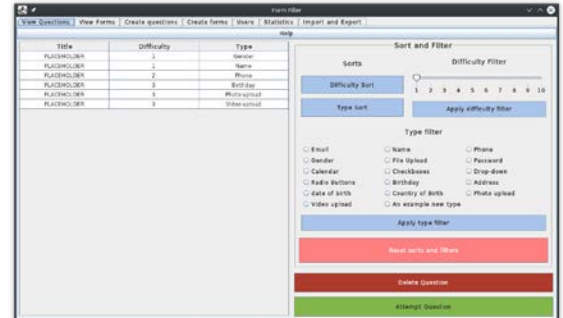
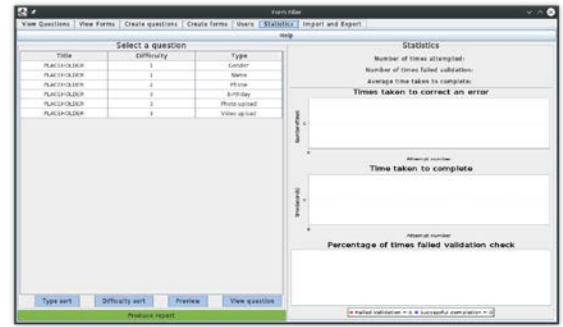
13	Proto	The average of the past times that it's taken the user to complete the question is displayed			
14	Soft Dev	Multiple symbol not found errors when compiling StatisticsPanel.java	<pre> harry@harry-pc-linux:/media/harry/HARRY_BEST/Computer Science/Component 3/SOFTDEV/Project/For- m-Filler\$./compile.sh Picked up JAVA_OPTIONS: -Dawt.useSystemAAFontSettings=on -Dswing.aatext=true StatisticsPanel.java:77: error: cannot find symbol mainPanel.add(selectedQuestionLabel); ^ symbol: variable selectedQuestionLabel location: class StatisticsPanel StatisticsPanel.java:79: error: cannot find symbol mainPanel.add(mostRecentAttemptsNeededToCorrectLabel); ^ symbol: variable mostRecentAttemptsNeededToCorrectLabel location: class StatisticsPanel StatisticsPanel.java:80: error: cannot find symbol mainPanel.add(timeTakenToCompleteLabel); ^ symbol: variable timeTakenToCompleteLabel location: class StatisticsPanel StatisticsPanel.java:127: error: cannot find symbol numberOfAttemptsLabel.setTest(numberOfAttemptsLabel.getText().split(":")[0] + " " + stats.getNumberOfAttempts()); ^ symbol: method setTest(String) location: variable numberOfAttemptsLabel of type JLabel Note: Register.java uses or overrides a deprecated API. Note: Recompile with -Xlint:deprecation for details. 4 errors Picked up JAVA_OPTIONS: -Dawt.useSystemAAFontSettings=on -Dswing.aatext=true </pre>	This is because the labels referenced in the error do not exist anymore as I removed them. There are a few references left to the labels, however, removing these will fix these errors	<pre> harry@harry-pc-linux:/media/harry/HARRY_BEST/Computer Science/Component 3/SOFTDEV/Project/For- m-Filler\$./compile.sh Picked up JAVA_OPTIONS: -Dawt.useSystemAAFontSettings=on -Dswing.aatext=true Note: Register.java uses or overrides a deprecated API. Note: Recompile with -Xlint:deprecation for details. Picked up JAVA_OPTIONS: -Dawt.useSystemAAFontSettings=on -Dswing.aatext=true </pre> Successful compile
15	Soft Dev	The statistics panel uses the new QuestionSletionPanel. There are also labels on the right which show the statistics detailed in the design.			

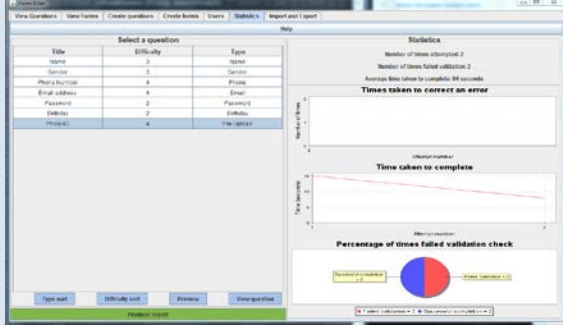
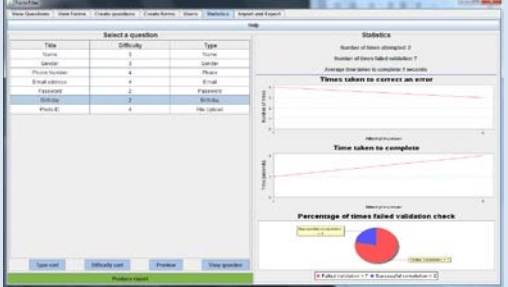
16	Soft Dev	There are the three labels that are detailed in the design.			
17	Soft Dev	There is a border around the statistics labels and they are centered.			
18	Soft Dev	There is a border around the select questions area.			

19	Soft Dev	There is padding between each label and there is a horizontal separator bar as detailed in the design.			
20	Soft Dev	There is a blank chart which shows the number of attempts the user took to correct an error over attempt number.			
21	Soft Dev	When the user selects a question the chart's data is updated to show the number of times that it has taken them to correct an error based on their most recent number of attempts.			

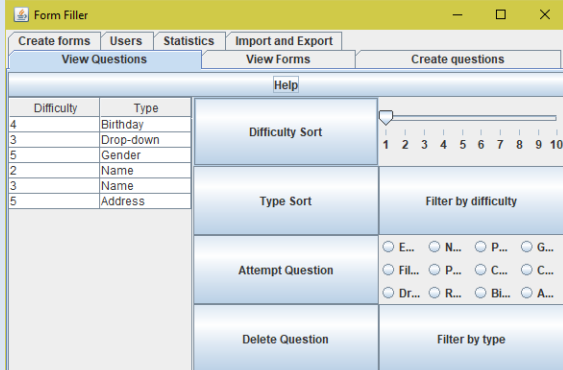
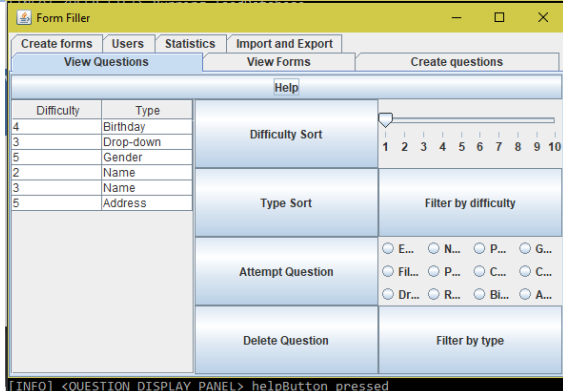
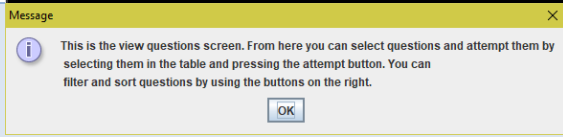
22	Soft Dev	The chart doesn't start at the first data point.		Count the number of valid data points and then subtract that from the total number of attempts to find the attempt number that the chart has to start on.	
23	Soft Dev	The x and y axis show decimal ticks. It's preferable for them to show integer ticks.		Use <code>axis.setStandardTickUnits(NumberAxis.createIntegerTickUnits())</code> on the X and Y axis to make them only display integer ticks.	
24	Soft Dev	The times taken to correct an error chart displays the data in the incorrect order.		The calculation to find the x coordinate to display the data is <code>totalNumberOfAttempts - validNumberOfAttempts + 1 + i</code> ; This is incorrect. It should be <code>totalNumberOfAttempts - i</code> ; This is because when i is 0 the x coordinate should be equal to <code>totalNumberOfAttempts</code>	

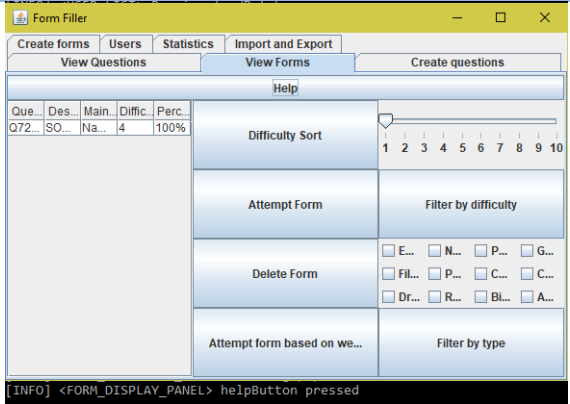
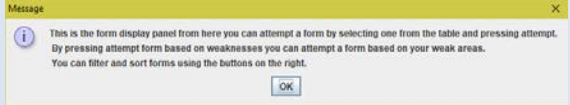
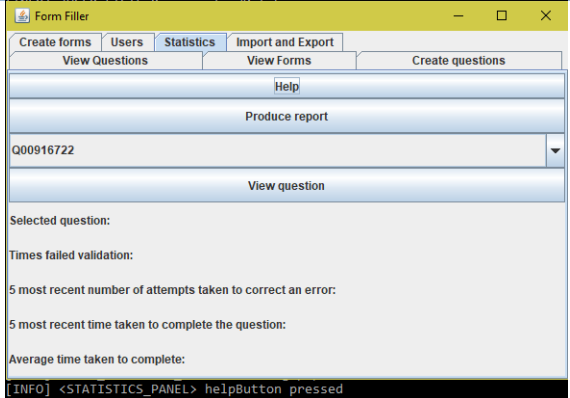
				empt when x is zero and equal to totalNumberOfAttempts - validNumberOfAttempts - 1 when i = (validNumberOfAttempts - 1). With the original calculation the inverse was true.	
25	Soft Dev	There is a time taken to complete chart.			
26	Soft Dev	There is a pie chart that displays information about the number of times the user has attempted the question compared to the number of times that they have failed the validation check.			

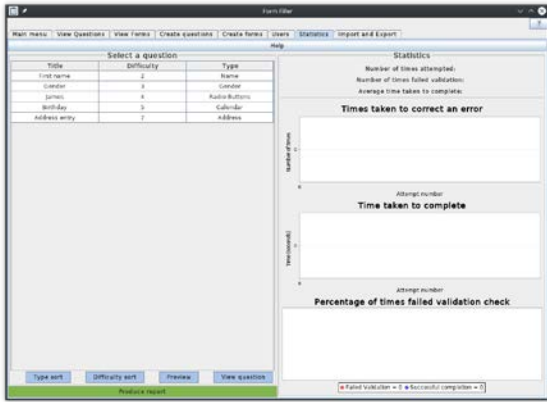
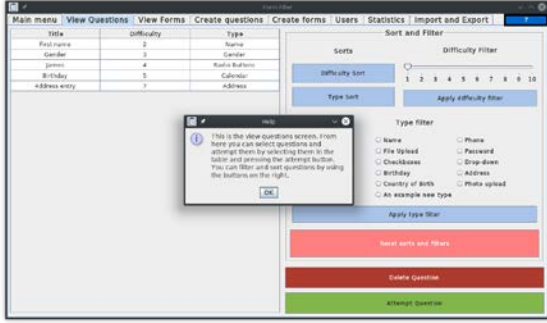
27	Soft Dev	When a chart is right clicked some chart options appear. This is undesired and should be removed.		Use <code>.setPopupMenu(null)</code> on the <code>ChartPanel</code> to disable the panel.	 The right click menu is disabled.
28	Soft Dev	When the statistics panel is selected the question selection panel refreshes.	 The view questions screen is selected. <pre>[INFO] <FORM_SELECTION_PANEL> Running prepareGUI [INFO] <FORM_SELECTION_PANEL> Running populateTable</pre> Most recent command line output. 		

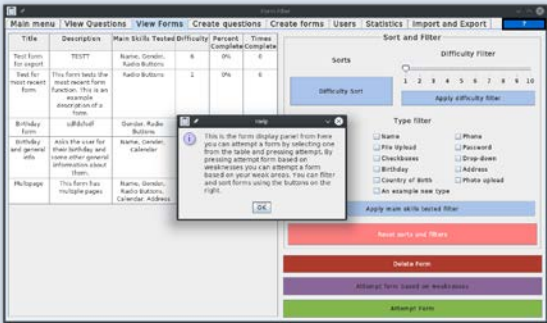
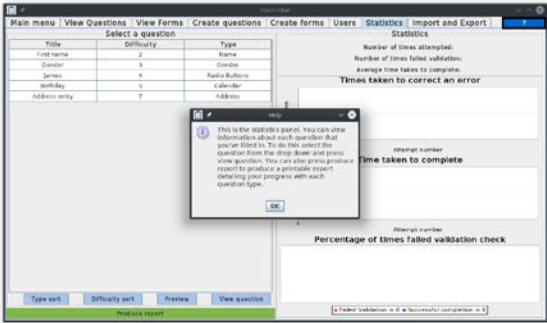
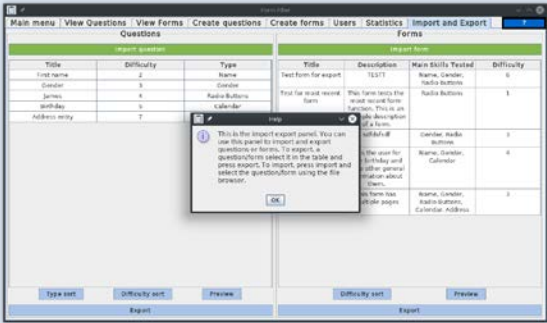
			The statistics screen is selected. <pre>[INFO] <FORM_SELECTION_PANEL> Running prepareGUI [INFO] <FORM_SELECTION_PANEL> Running populateTable [INFO] <SELECT_QUESTIONS_PANEL> Running populateTable []</pre> The question selection panel has been refreshed.		
29	Soft Dev	The average time taken to complete isn't correct, the average is too low.		This is because the time taken to complete array contains values of -1 for attempts that haven't been done yet. However these are still included in the average. Adding a check so that the average is only calculated with the non-rogue values will mean that the average is correct.	 The average time is correct.

59: There should be a context sensitive help button available in each window and panel of the application accessible by a regular user

Test	Stage	Result Problem	Evidence	Action and Justification (If Needed)	Evidence of Action (If Needed)
1	Proto	There is a help button in the view Questions window.			
2	Proto	Presses to the help button are detected			
3	Proto	When the help button is pressed a help dialog appears explaining how to use the screen.			

4	Proto	There is a help button in the form display panel and presses to it are detected			
5	Proto	When the help button is pressed a help dialog appears explaining how to use the screen.			
6	Proto	There is a help button in the Statistics panel and presses to it are detected			
7	Proto	When the help button is pressed a help dialog appears.			

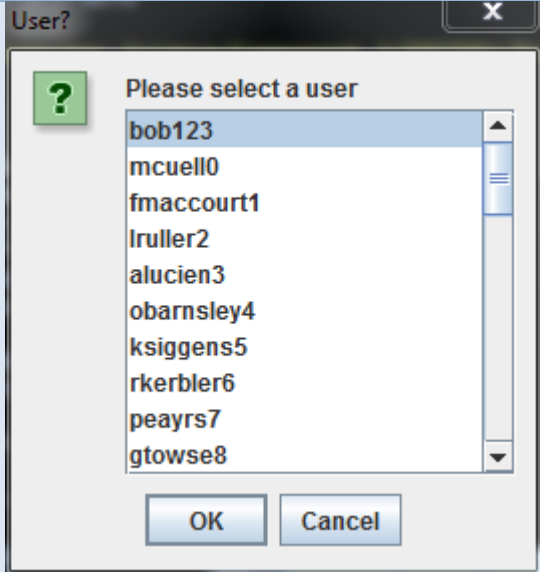
8	Soft Dev	There is a help button in the top right of the application			
9	Soft Dev	The help button is in a slightly better place and has the correct styling.			
10	Soft Dev	When the help button is pressed in the view questions panel a relevant help message appears.			

11	Soft Dev	When the help button is pressed in the view forms panel a relevant help message appears.			
12	Soft Dev	When the help button is pressed in the statistics panel a relevant help message appears.			
13	Soft Dev	When the help button is pressed in the import export panel a relevant help message appears.			

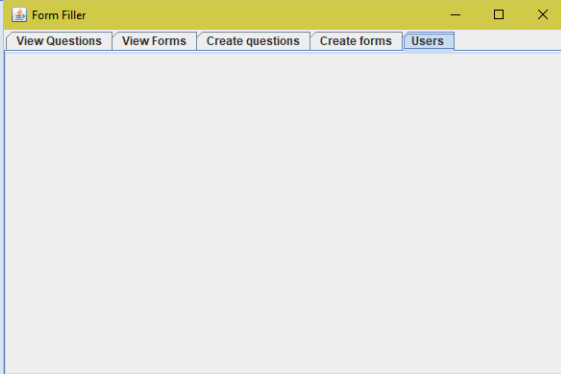
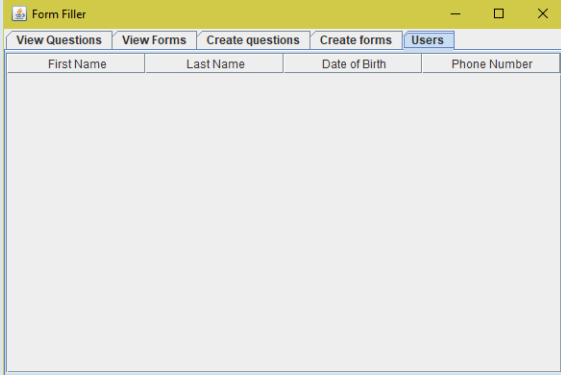
14	Soft Dev	When the help button is pressed in the main menu a relevant help message appears.	 The screenshot shows a software application window titled 'Welcome efenty12'. The window has a menu bar with 'Main menu', 'View Questions', 'View Forms', 'Statistics', and 'Import and Export'. On the left, there is a 'Navigation' panel with buttons for 'View questions', 'View forms', 'Import/Export', and 'View statistics'. On the right, there is an 'Attendance' section showing '40%'. A small help dialog box is open in the center, displaying a message: 'This is the main menu. You can navigate to other areas of the system from here. You can also resume attempting the most recent form that you attempted. Your attendance is displayed on the right.' with an 'OK' button.		
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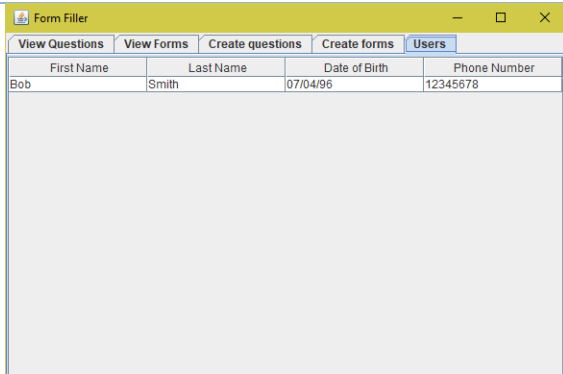
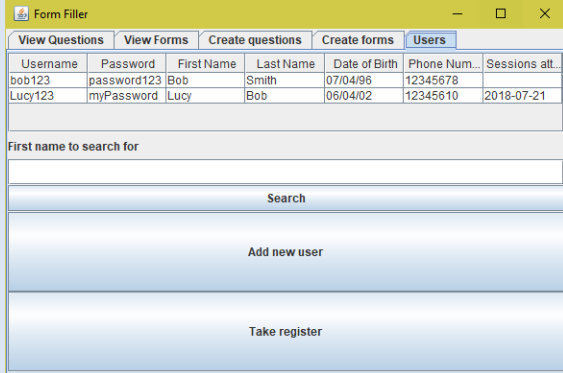
60: The admin user should have a different main menu

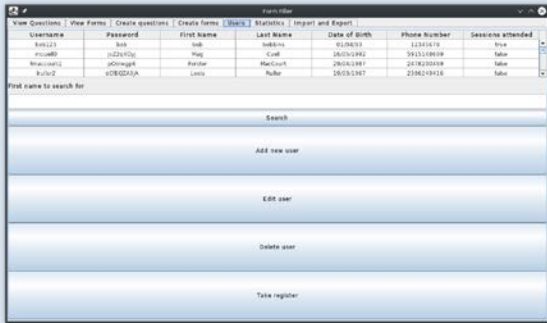
Test	Stage	Result	Problem	Evidence	Action and Justification (If Needed)	Evidence of Action (If Needed)
1	Soft Dev	Multiple compilation errors after adding admin attribute to user class. These are due to admin being parsed as a string rather than a boolean.		<pre> harry@harry-pc-linux:~/Documents/SoftDev/form-filler\$./compile.sh Picked up _JAVA_OPTIONS: -Dawt.useSystemAAFontSettings=on -Dswing.aatext=true UserList.java:213: error: incompatible types: boolean cannot be converted to String String admin = Boolean.parseBoolean(splitUserData[7]); ^ UserList.java:230: error: incompatible types: String cannot be converted to boolean addUser(new UserId, username, password, firstName, lastName, dateOfBirth, phone Number, admin, sessionsPresentAt, questionState)); ^ UserPanel.java:111: error: constructor User in class User cannot be applied to given types; users.addUser(new UserId,username,password,firstName,lastName,dateOfBirth,phone Number, new String[0], new QuestionStatList()); ^ required: String,String,String,String,String,String,String,boolean,String[],QuestionStatList found: String,String,String,String,String,String,String,String[],QuestionStatList reason: actual and formal argument lists differ in length Note: Some messages have been simplified; recompile with -Xdiags:verbose to get full output 3 errors Picked up _JAVA_OPTIONS: -Dawt.useSystemAAFontSettings=on -Dswing.aatext=true harry@harry-pc-linux:~/Documents/SoftDev/form-filler\$ </pre>	Change the type of admin in the read from database method to boolean	<pre> harry@harry-pc-linux:~/Documents/SoftDev/form-filler\$./compile.sh Picked up _JAVA_OPTIONS: -Dawt.useSystemAAFontSettings=on -Dswing.aatext=true UserPanel.java:111: error: constructor User in class User cannot be applied to given types; users.addUser(new UserId,username,password,firstName,lastName,dateOfBirth,phone Number, new String[0], new QuestionStatList()); ^ required: String,String,String,String,String,String,String,boolean,String[],QuestionStatList found: String,String,String,String,String,String,String,String[],QuestionStatList reason: actual and formal argument lists differ in length 1 error Picked up _JAVA_OPTIONS: -Dawt.useSystemAAFontSettings=on -Dswing.aatext=true harry@harry-pc-linux:~/Documents/SoftDev/form-filler\$ </pre> Only the error about incorrect argument lengths remain
2	Soft Dev	There is an error when compiling UserPanel.java because there is a line where a user is created with the incorrect number of arguments.		<pre> harry@harry-pc-linux:~/Documents/SoftDev/form-filler\$./compile.sh Picked up _JAVA_OPTIONS: -Dawt.useSystemAAFontSettings=on -Dswing.aatext=true UserPanel.java:111: error: constructor User in class User cannot be applied to given types; users.addUser(new UserId,username,password,firstName,lastName,dateOfBirth,phone Number, new String[0], new QuestionStatList()); ^ required: String,String,String,String,String,String,String,boolean,String[],QuestionStatList found: String,String,String,String,String,String,String,String[],QuestionStatList reason: actual and formal argument lists differ in length 1 error Picked up _JAVA_OPTIONS: -Dawt.useSystemAAFontSettings=on -Dswing.aatext=true harry@harry-pc-linux:~/Documents/SoftDev/form-filler\$ </pre>	Add a false to make the created user a regular user. This satisfies the admin parameter of the user class.	<pre> harry@harry-pc-linux:~/Documents/SoftDev/form-filler\$./compile.sh Picked up _JAVA_OPTIONS: -Dawt.useSystemAAFontSettings=on -Dswing.aatext=true Picked up _JAVA_OPTIONS: -Dawt.useSystemAAFontSettings=on -Dswing.aatext=true harry@harry-pc-linux:~/Documents/SoftDev/form-filler\$ </pre> Successful compile

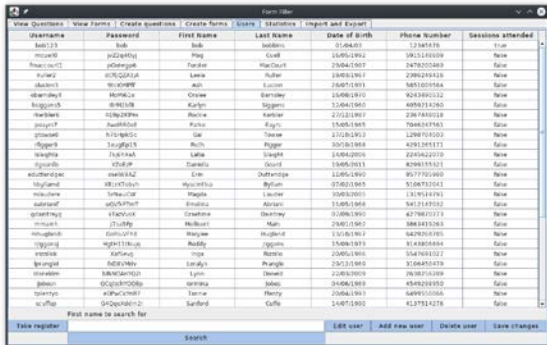
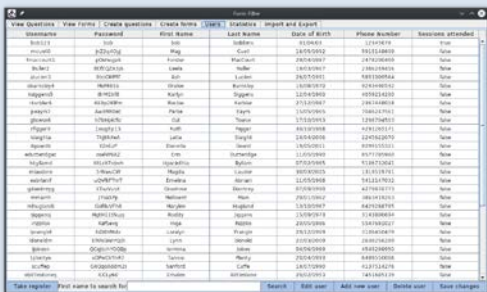
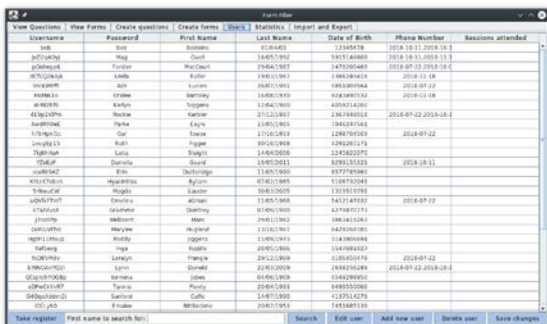
3	Soft Dev	When an admin user is logged in the main menu that is displayed is the admin main menu.	 bob123 is an admin 		
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61: The admin user should be able to view information about each user

Test	Stage	Result Problem	Evidence	Action and Justification (If Needed)	Evidence of Action (If Needed)
1	Proto	There is a user tab in the application			
2	Proto	UserPanel fails to compile due to Symbol not found errors.	<pre>D:\Computer Science\Component 3\Prototype>compile UserPanel.java:12: error: cannot find symbol private DefaultTableModel userTableModel = new DefaultTableModel(userData, tableHeaders); ^ symbol: class DefaultTableModel location: class UserPanel UserPanel.java:12: error: cannot find symbol private DefaultTableModel userTableModel = new DefaultTableModel(userData, tableHeaders); ^ symbol: class DefaultTableModel location: class UserPanel 2 errors</pre>	These errors are due to javax.swing.table.*; not being imported. So importing this should fix the errors.	<pre>D:\Computer Science\Component 3\Prototype>compile D:\Computer Science\Component 3\Prototype></pre> Successful compile
3	Proto	There is an empty table to display the user data			

4	Proto	The user data is displayed in the table			
5	Proto	The sessions that each user was present at are displayed			

6	Soft Dev	Error when compiling user panel because TableColumnModelListener, and ChangeEvent cannot be found.	<pre> harryharry-pc-linux:/media/harry/HARRY_BEST/Computer Science/Component 3/50FTDEV/Project/For- rm-Filler\$./compile.sh Picked up _JAVA_OPTIONS: -Dawt.useSystemAAFontSettings=on -Dswing.aatext=true UserPanel.java:5: error: cannot find symbol public class UserPanel extends JPanel implements ActionListener, TableColumnModelListener ^ symbol: class TableColumnModelListener UserPanel.java:246: error: cannot find symbol public void columnMarginChanged(ChangeEvent e) ^ symbol: class ChangeEvent location: class UserPanel UserPanel.java:250: error: cannot find symbol public void columnAdded(TableColumnModelEvent e) {} ^ symbol: class TableColumnModelEvent location: class UserPanel UserPanel.java:252: error: cannot find symbol public void columnRemoved(TableColumnModelEvent e) {} ^ symbol: class TableColumnModelEvent location: class UserPanel UserPanel.java:254: error: cannot find symbol public void columnMoved(TableColumnModelEvent e) {} ^ symbol: class TableColumnModelEvent location: class UserPanel UserPanel.java:256: error: cannot find symbol public void columnSelectionChanged(ListSelectionEvent e) {} ^ symbol: class ListSelectionEvent location: class UserPanel UserPanel.java:55: error: incompatible types: UserPanel cannot be converted to TableColumnMo delListener tcm.addColumnModelListener(this); ^ Note: Some messages have been simplified; recompile with -Xdiags:verbose to get full output 7 errors Picked up _JAVA_OPTIONS: -Dawt.useSystemAAFontSettings=on -Dswing.aatext=true </pre>	This is because those classes are in the javax.swing.event package and that package hasn't been imported. Importing the package will fix the errors.	<pre> harryharry-pc-linux:/media/harry/HARRY_BEST/Computer Science/Component 3/50FTDEV/Project/For- rm-Filler\$./compile.sh Picked up _JAVA_OPTIONS: -Dawt.useSystemAAFontSettings=on -Dswing.aatext=true Picked up _JAVA_OPTIONS: -Dawt.useSystemAAFontSettings=on -Dswing.aatext=true </pre> Successful compile
7	Soft Dev	The view users table has the updated table formatting (this also means that the rows will automatically resize)			

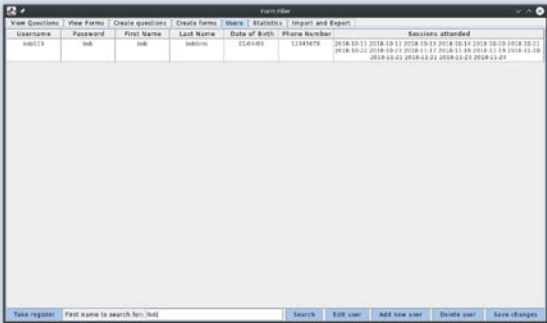
8	Soft Dev	The panel has the layout that was designed in the design.		The search section is arranged wrong. The label, text field, and button should be in a line. Changing the layout of the search panel from a gridlayout to a horizontal box layout should fix this because it'll put all of the components in a line.	
9	Soft Dev	The buttons at the bottom are all bunched up with no gaps		Add a small rigid area of width 5 between each component to force a gap between them.	
10	Soft Dev	All of the data in the table is one column too far to the left		This is because the ID of the user hasn't been added to the table. Adding it to the table will shift all of the columns to the right one as required.	

The search panel has a horizontal layout

There are now spaces between the buttons

The data is in the correct columns now

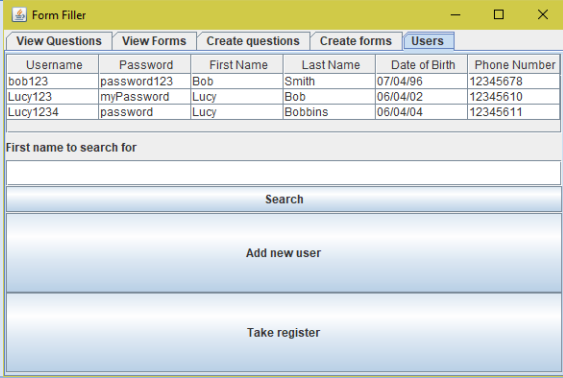
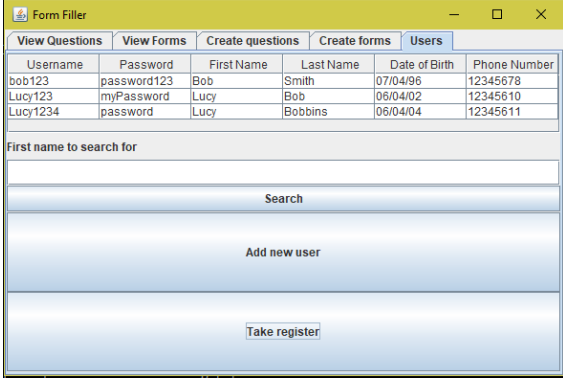
11	Soft Dev	The sessions attended now wrap		
12	Soft Dev	The table has better sizing		
13	Soft Dev	When the enter key is pressed in the search by first name text field the search is applied.		Enter pressed

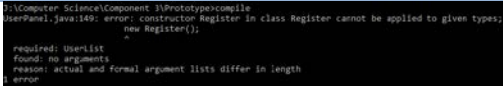
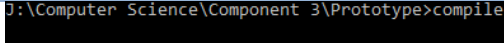
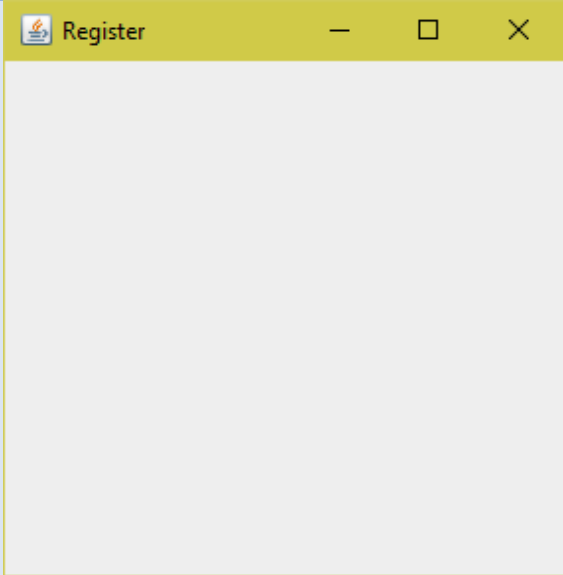
					
			Search has been applied		

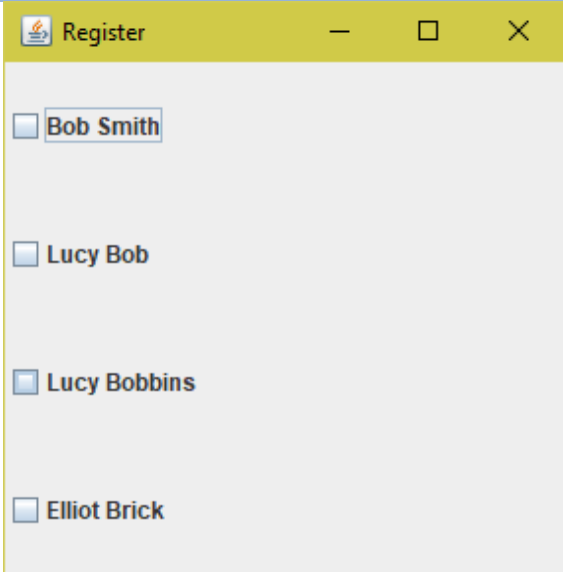
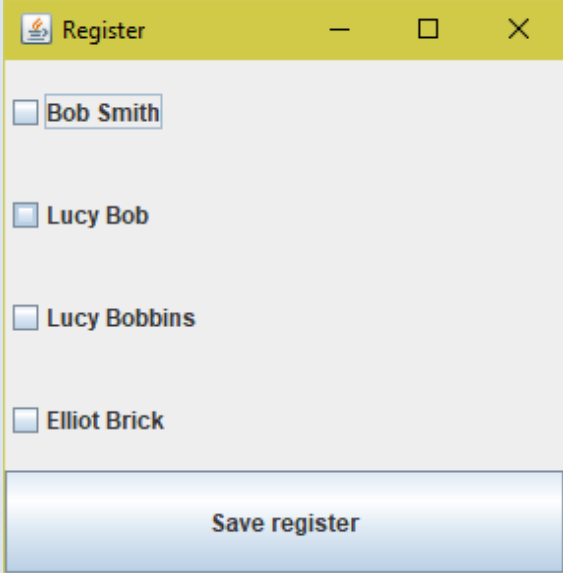
62: The admin user should be able to edit information about each user

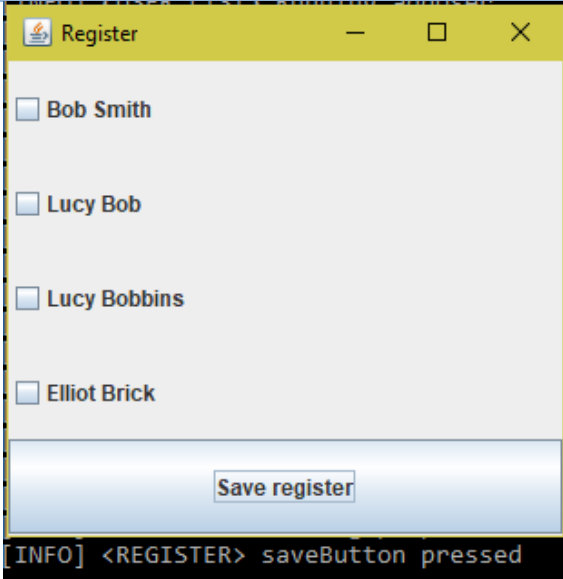
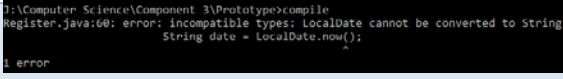
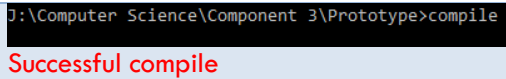
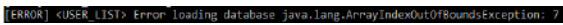
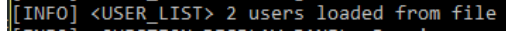
Test	Stage	Result	Problem	Evidence	Action and Justification (If Needed)	Evidence of Action (If Needed)
				See “Ability to edit users”		

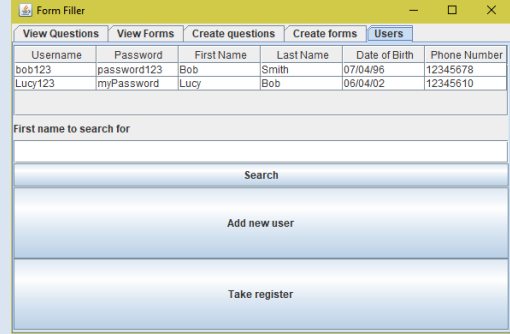
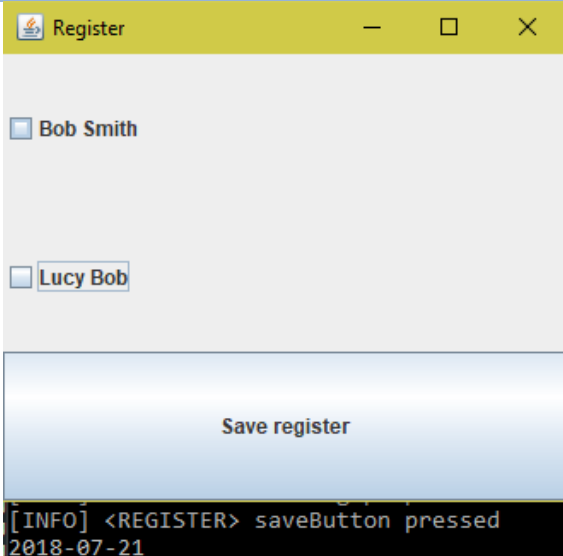
63: The admin user should be able to take a register

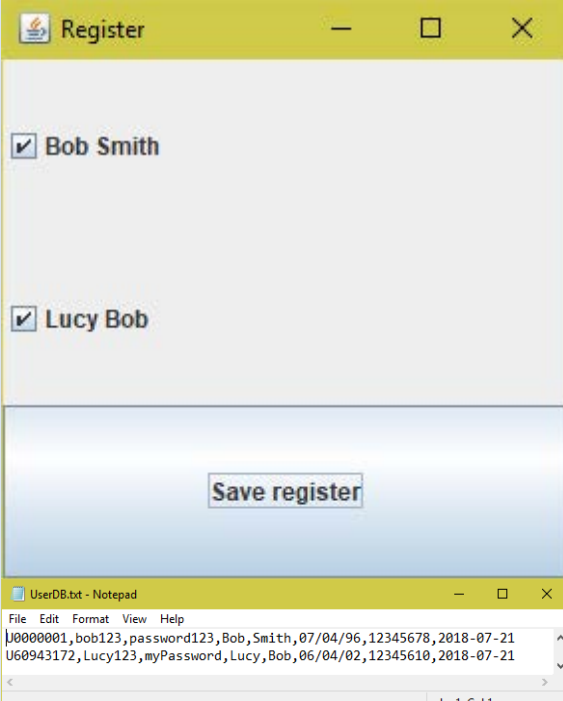
Test	Stage	Result Problem	Evidence	Action and Justification (If Needed)	Evidence of Action (If Needed)
1	Proto	There is a take register button			
2	Proto	Presses to the take register button are detected			
3	Proto	Register.java fails to compile because the symbol GridLayout can't be found	<pre>Register.java:19: error: cannot find symbol this.setLayout(new GridLayout(0,2)); ^ symbol: class GridLayout location: class Register</pre>	Import java.awt.*; as the GridLayout class is part of the java.awt package.	<pre>J:\Computer Science\Component 3\Prototypes\compile UserPanel.java:149: error: constructor Register in class Register cannot be applied to given types; new Register(); ^ required: UserList found: no arguments reason: actual and formal argument lists differ in length 1 error</pre>

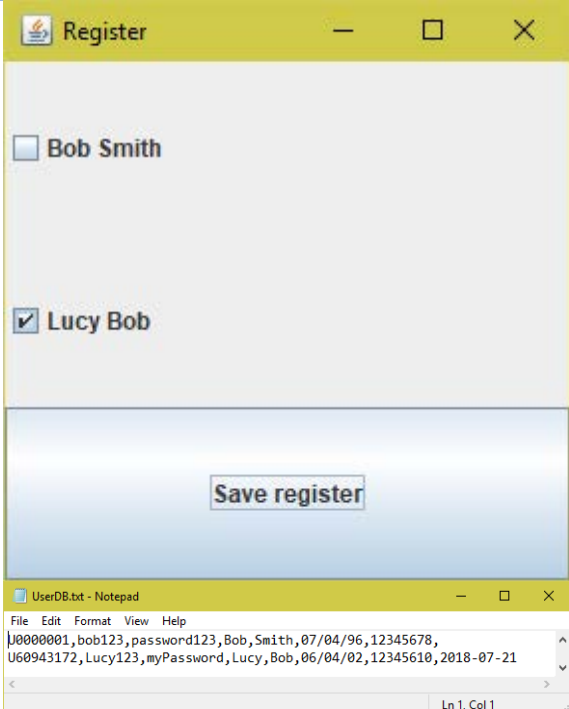
4	Proto	Error when compiling UserPanel.java because Register wasn't correctly called		When creating a new instance of register pass it the instance of UserList.	 Successful compile
5	Proto	An empty register window opens when pressing take register			

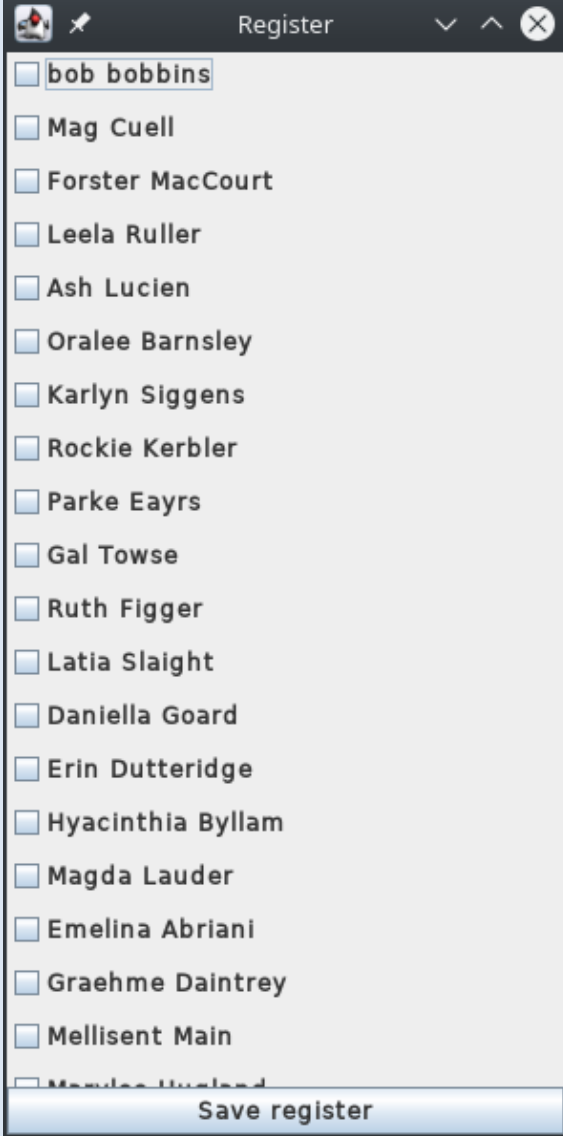
6	Proto	There is a check box for every user	 The screenshot shows a window titled 'Register' with a yellow header bar. Inside the window, there is a list of four names, each preceded by an unchecked checkbox: 'Bob Smith', 'Lucy Bob', 'Lucy Bobbins', and 'Elliot Brick'.		
7	Proto	There is a save register button	 This screenshot is identical to the one in row 6, but it includes a 'Save register' button at the bottom of the window, which has a blue gradient background.		

8	Proto	Presses to the save register button are detected			
9	Proto	Error compiling register because of a type error.		Call toString() on the date to convert it to a string.	
10	Proto	Array out of bounds exception when loading the user database.		This is because the user has been present at no sessions therefore the last part of the saved data is empty (this is where the sessions attended would be stored). Making the database loader check the length of the array containing the data before attempting to access the 7th element should fix this. If the user hasn't been present at any sessions an empty array	 The users are successfully loaded from file

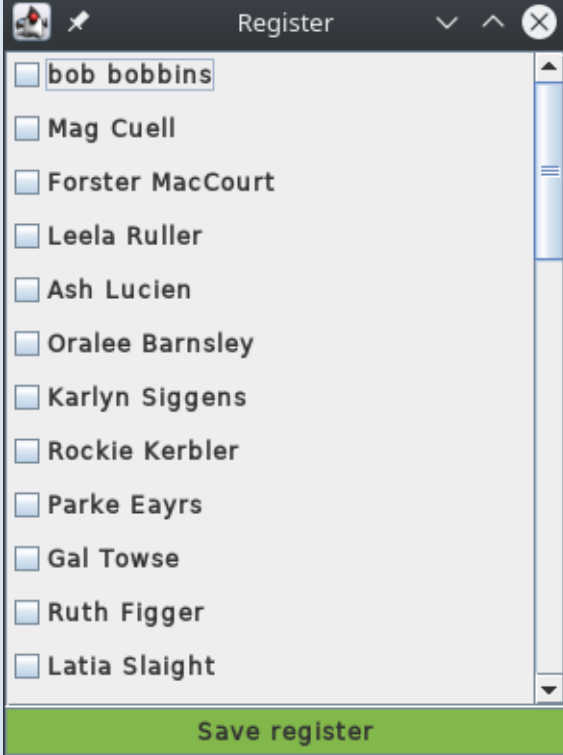
				will be passed to the user constructor rather than the data loaded from file in place of the sessions present array.																			
11	Proto	Array index out of bounds exception when converting the sessionsPresent AtArray to string if it's empty.	<pre>Exception in thread "main" java.lang.ArrayIndexOutOfBoundsException: 0 at User.sessionsToString(User.java:86) at User.toString(User.java:51) at User.toStringArray(User.java:56) at UserPanel.populateTable(UserPanel.java:81) at UserPanel.prepareGUI(UserPanel.java:46) at UserPanel.<init>(UserPanel.java:31) at GUI.prepareGUI(GUI.java:49) at GUI.setup(GUI.java:30) at GUI.<init>(GUI.java:16) at Main.main(Main.java:5)</pre>	When converting the sessionsPresentAtArray to string check whether it's empty and if it's empty return an empty string.	 Form Filler <table border="1"> <thead> <tr> <th>Username</th> <th>Password</th> <th>First Name</th> <th>Last Name</th> <th>Date of Birth</th> <th>Phone Number</th> </tr> </thead> <tbody> <tr> <td>bob123</td> <td>password123</td> <td>Bob</td> <td>Smith</td> <td>07/04/96</td> <td>12345678</td> </tr> <tr> <td>Lucy123</td> <td>myPassword</td> <td>Lucy</td> <td>Bob</td> <td>06/04/02</td> <td>12345610</td> </tr> </tbody> </table> First name to search for Search Add new user Take register Users successfully loaded from file with no errors.	Username	Password	First Name	Last Name	Date of Birth	Phone Number	bob123	password123	Bob	Smith	07/04/96	12345678	Lucy123	myPassword	Lucy	Bob	06/04/02	12345610
Username	Password	First Name	Last Name	Date of Birth	Phone Number																		
bob123	password123	Bob	Smith	07/04/96	12345678																		
Lucy123	myPassword	Lucy	Bob	06/04/02	12345610																		
12	Proto	When the save register button is pressed the date is outputted.	 Register ☐ Bob Smith ☐ Lucy Bob Save register [INFO] <REGISTER> saveButton pressed 2018-07-21																				

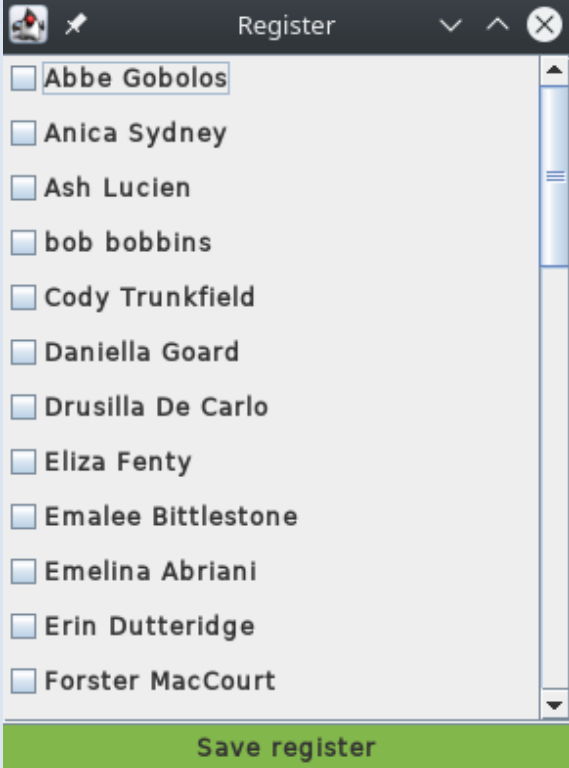
13	Proto	If a user is selected and the save register button is pressed then today's date is added to their sessionsPresent AtArray and saved to file.	 The screenshot displays two windows. The top window, titled 'Register', has a yellow title bar and contains a list of two items, each with a checked checkbox: 'Bob Smith' and 'Lucy Bob'. Below the list is a button labeled 'Save register'. The bottom window, titled 'UserDB.txt - Notepad', has a yellow title bar and a menu bar (File, Edit, Format, View, Help). The text area contains two lines of data: 'U0000001,bob123,password123,Bob,Smith,07/04/96,12345678,2018-07-21' and 'U60943172, Lucy123,myPassword,Lucy,Bob,06/04/02,12345610,2018-07-21'. The status bar at the bottom indicates 'Ln 1, Col 1'.		
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14	Proto	If only 1 user is selected only they are marked as present.	 The screenshot shows two windows. The top window, titled 'Register', has a yellow title bar and contains two checkboxes: 'Bob Smith' (unchecked) and 'Lucy Bob' (checked). Below the checkboxes is a 'Save register' button. The bottom window, titled 'UserDB.txt - Notepad', has a yellow title bar and a menu bar (File, Edit, Format, View, Help). The text area contains two lines of text: 'U0000001,bob123,password123,Bob,Smith,07/04/96,12345678,' and 'U60943172, Lucy123,myPassword, Lucy,Bob,06/04/02,12345610,2018-07-21'. The status bar at the bottom indicates 'Ln 1, Col 1'.		
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15	Soft Dev	The register better resembles the one in the design			
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16	Soft Dev	The register has a scroll bar	 The screenshot shows a window titled "Register" with a list of names, each preceded by a checkbox. The names are: bob bobbins, Mag Cuell, Forster MacCourt, Leela Ruller, Ash Lucien, Oralee Barnsley, Karlyn Siggens, Rockie Kerbler, Parke Eayrs, Gal Towse, Ruth Figger, and Latia Slaight. A vertical scrollbar is on the right side of the list. At the bottom of the window is a button labeled "Save register".		
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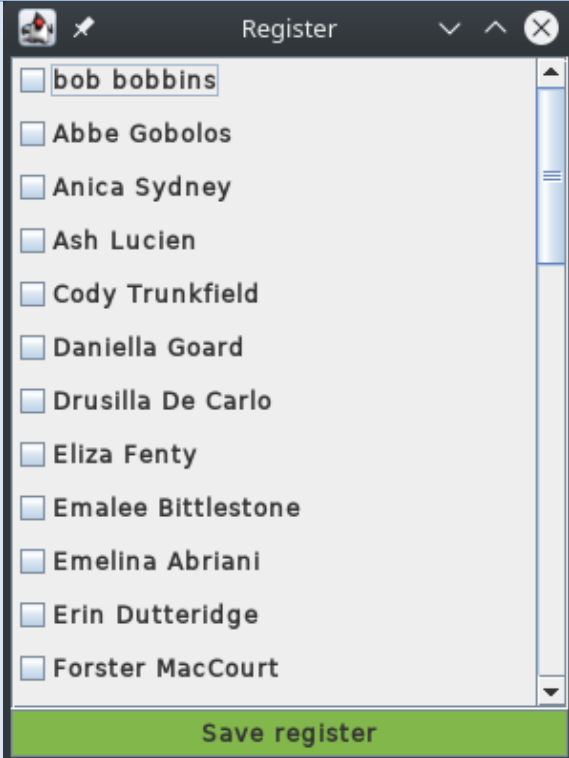
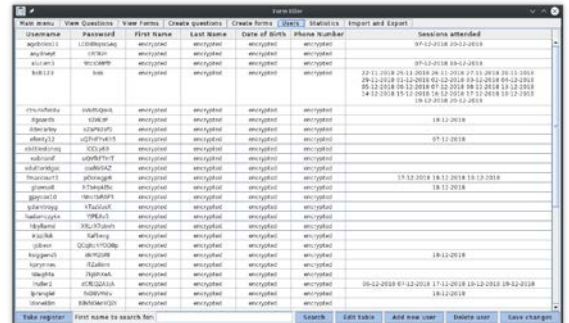
17	Soft Dev	The save register button is the correct colour	 The screenshot shows a window titled 'Register' with a list of names, each preceded by a checkbox. The names are: bob bobbins, Mag Cuell, Forster MacCourt, Leela Ruller, Ash Lucien, Oralee Barnsley, Karlyn Siggens, Rockie Kerbler, Parke Eayrs, Gal Towse, Ruth Figger, and Latia Slaight. At the bottom of the window is a green button labeled 'Save register'.		
18	Soft Dev	Infinte loop when UserList runs sort by FirstName	<pre>[INFO] <REGISTER> Running prepareGUI [INFO] <USER_LIST> Running sortByFirstName</pre>	This is due to a fault terminating condition in the isBeforeInDictionary method. The result of the recursive call isn't being saved to the beforeInDictionaryVariable and so the function is always returning false. This means that the sortByFirstName method never stops swapping values hence teh infinite	<pre>[INFO] <USER_LIST> Running sortByFirstName</pre> No infinite loop

19	Soft Dev	The user array is in alphabetical order by first name.		loop.	

20	Soft Dev	Multiple errors compiling Register.java	<pre> harry@harry-pc:~\$ cd /media/harry/HARRY_BEST/Computer_Science/Component_3/SoftDev/Project/For-Filler/; ./compile.sh Picked up _JAVA_OPTIONS: -Dawt.useSystemAAFontSettings=on -Dswing.aatext=true Register.java:79: error: cannot find symbol Date twoWeeksAgo = new Date(System.currentTimeMillis() - (15 * DAY_IN_MS)); ^ // Calculate the date two weeks ago symbol: class Date location: class Register Register.java:79: error: cannot find symbol Date twoWeeksAgo = new Date(System.currentTimeMillis() - (15 * DAY_IN_MS)); ^ // Calculate the date two weeks ago symbol: class Date location: class Register Register.java:79: error: cannot find symbol Date twoWeeksAgo = new Date(System.currentTimeMillis() - (15 * DAY_IN_MS)); ^ // Calculate the date two weeks ago symbol: variable DAY_IN_MS location: class Register Register.java:97: error: incompatible types: String[] cannot be converted to String String splitSession = session.split("-"); ^ Register.java:98: error: array required, but String found int day = Integer.parseInt(splitSession[0]); ^ Register.java:99: error: array required, but String found int month = Integer.parseInt(splitSession[1]); ^ Register.java:100: error: array required, but String found int year = Integer.parseInt(splitSession[2]); ^ Register.java:101: error: cannot find symbol Date sessionDate = new Date(year, month, day); ^ symbol: class Date location: class Register Register.java:101: error: cannot find symbol Date sessionDate = new Date(year, month, day); ^ symbol: class Date location: class Register 9 errors Picked up _JAVA_OPTIONS: -Dawt.useSystemAAFontSettings=on -Dswing.aatext=true Picked up _JAVA_OPTIONS: -Dawt.useSystemAAFontSettings=on -Dswing.aatext=true Register.java:79: error: cannot find symbol Date twoWeeksAgo = new Date(System.currentTimeMillis() - (15 * DAY_IN_MS)); ^ // Calculate the date two weeks ago symbol: class Date location: class Register Register.java:79: error: cannot find symbol Date twoWeeksAgo = new Date(System.currentTimeMillis() - (15 * DAY_IN_MS)); ^ // Calculate the date two weeks ago symbol: class Date location: class Register Register.java:79: error: cannot find symbol Date twoWeeksAgo = new Date(System.currentTimeMillis() - (15 * DAY_IN_MS)); ^ // Calculate the date two weeks ago symbol: variable DAY_IN_MS location: class Register </pre>	
21	Soft Dev	Symbol not found exception for Date	<pre> harry@harry-pc:~\$ cd /media/harry/HARRY_BEST/Computer_Science/Component_3/SoftDev/Project/For-Filler/; ./compile.sh Picked up _JAVA_OPTIONS: -Dawt.useSystemAAFontSettings=on -Dswing.aatext=true Register.java:99: error: incompatible types: String[] cannot be converted to String String splitSession = session.split("-"); ^ Register.java:100: error: array required, but String found int day = Integer.parseInt(splitSession[0]); ^ Register.java:101: error: array required, but String found int month = Integer.parseInt(splitSession[1]); ^ Register.java:102: error: array required, but String found int year = Integer.parseInt(splitSession[2]); ^ Note: Register.java uses or overrides a deprecated API. Note: Recompile with -Xlint:deprecation for details. 4 errors Picked up _JAVA_OPTIONS: -Dawt.useSystemAAFontSettings=on -Dswing.aatext=true </pre>	This is because the java.util.Date package hasn't been imported, importing it should fix this error.
22	Soft Dev	Incompatible types error when compiling	<pre> harry@harry-pc:~\$ cd /media/harry/HARRY_BEST/Computer_Science/Component_3/SoftDev/Project/For-Filler/; ./compile.sh Picked up _JAVA_OPTIONS: -Dawt.useSystemAAFontSettings=on -Dswing.aatext=true Register.java:99: error: incompatible types: String[] cannot be converted to String String splitSession = session.split("-"); ^ Register.java:100: error: array required, but String found int day = Integer.parseInt(splitSession[0]); ^ Register.java:101: error: array required, but String found int month = Integer.parseInt(splitSession[1]); ^ Register.java:102: error: array required, but String found int year = Integer.parseInt(splitSession[2]); ^ Note: Register.java uses or overrides a deprecated API. Note: Recompile with -Xlint:deprecation for details. 4 errors Picked up _JAVA_OPTIONS: -Dawt.useSystemAAFontSettings=on -Dswing.aatext=true </pre>	This is because the splitSession variable is declared as a String whereas it should be declared as a String[] as it's declaring a string array.

Error no longer present when compiling

Successful compile

33	Soft Dev	The register is in the order detailed in the PPROD. The users that have attended a session in the past 2 weeks are at the top of the register. In this case, bob.			
34	Soft Dev	If the userdatabase is encrypted the user is required to decrypt it before the register is opened			

The image displays two sequential screenshots of a Windows 'Input' dialog box. The dialog has a dark title bar with the word 'Input' in the center, a minimize button, a maximize button, and a close button. On the left side of the title bar is a small icon of a person with a red exclamation mark. The main area of the dialog contains a green square icon with a white question mark on the left, followed by the text 'Please enter a key to decrypt the database'. Below this text is a white text input field. In the bottom right corner, there are two buttons: 'OK' and 'Cancel'. The top screenshot shows the input field as empty. The bottom screenshot shows the input field containing the text 'ANINCORRECT KEY'.

An incorrect key is entered

[illegible]

The database is not decrypted and the register is not opened.

The correct key is entered

